

17-Hydroxyprogesterone

<https://medlineplus.gov/lab-tests/17-hydroxyprogesterone/>

17-HYDROXYPROGESTERONE

WHAT IS A 17-HYDROXYPROGESTERONE (17-OHP) TEST?

This test measures the amount of 17-hydroxyprogesterone (17-OHP) in a sample of your blood. 17-OHP is a substance that your adrenal glands make.

Your adrenal glands are two small organs that sit on top of each kidney. They make different types of hormones you need to stay alive and healthy. Hormones are chemical messengers in your bloodstream that control the actions of certain cells or organs.

Normally, your adrenal glands use 17-OHP to make a hormone called cortisol. Cortisol helps control your energy levels, blood glucose (sugar), blood pressure, and your body's response to stress, illness and injury.

A 17-OHP test helps diagnose a group of uncommon, inherited genetic disorders that affect how well your adrenal glands make cortisol. These genetic adrenal gland disorders are called congenital adrenal hyperplasia (CAH).

People with CAH have a change in a gene that causes the lack of an enzyme that your adrenal glands need to make hormones. The most common type of CAH is caused by a lack of an enzyme called 21-hydroxylase. 21-hydroxylase helps your adrenal glands use 17-OHP to make cortisol.

Without enough 21-hydroxylase, the adrenal glands struggle to make enough cortisol. As the adrenal glands work harder to do their job, they become enlarged and make abnormally high levels of 17-OHP. These high levels of 17-OHP are a sign of CAH with 21-hydroxylase deficiency.

The adrenal glands use the extra 17-OHP to make larger than normal amounts of male sex hormones. This happens in children and adults of both sexes. So, tests for male hormones, including testosterone, are often used with 17-OHP tests to help diagnose CAH.

The symptoms of CAH depend on how much cortisol the adrenal glands are able to make. CAH is often organized into two groups based on how severe the symptoms are:

Classic CAH includes the most severe forms of the disorder. It's usually diagnosed at birth. Without diagnosis and treatment, severe CAH can be life-threatening. Serious symptoms may show up shortly after birth. They may include:

- Dehydration
- Low blood pressure
- An irregular heartbeat (arrhythmia)
- Shock

Newborn girls with severe CAH may have external genitals that are not clearly female or male. This is caused by high levels of male hormones. In newborn boys, the genitals usually appear normal, but sometimes the penis may be larger than usual.

Classic CAH that is caused by a 21-hydroxylase deficiency can be diagnosed with 17-OHP testing. Less often, CAH is caused by a lack of a different enzyme.

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Nonclassic CAH includes mild forms of the disorder. It's also called late-onset CAH because symptoms show up between early childhood and early adulthood. It is almost always caused by a lack of 21-hydroxylase and can be found with a 17-OHP test.

Treatment is only needed if a person has symptoms. The symptoms depend on the person's age and sex. For example:

Children may have signs of early puberty, which means having signs of sexual development before age 8 in girls and before age 9 in boys.

Teenage girls and women may have more facial and body hair than usual and/or a deep voice.

Adults may have infertility.

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Adults may have infertility.

Other names: 17-OH progesterone, 17-OHP, 17 alphahydroxyprogesterone, 17 hydroxy progesterone: serum, progesterone: 17-hydroxy

WHAT IS IT USED FOR?

A 17-OHP test is used to help find and monitor the most common type of CAH, which is sometimes called 21-hydroxylase deficiency. The test is used to:

Check all newborns for severe CAH (classic CAH). Newborn screening tests can't check for mild CAH (nonclassic).

Help diagnose mild CAH in children, teens, and adults who have symptoms that could be caused by the disorder.

Find out if treatment for CAH is working.

Help rule out CAH in women who have certain symptoms such as large amounts of facial hair and irregular menstrual periods. These symptoms can also be caused by other conditions, such as polycystic ovarian syndrome (PCOS).

WHY DO I NEED A 17-OHP TEST?

Babies need a 17-OHP test as part of routine newborn screening tests to check for severe CAH:

If you give birth in a hospital, your baby will usually have a 17-OHP test before going home. The test is most accurate when it's done more than 24 hours after birth. If your baby is tested sooner, ask your child's provider if a second test should be done at a later time. Some states require two tests for all babies.

If you give birth at home, make sure your baby has all the necessary screening tests.

Symptoms of severe CAH usually begin within days or weeks after birth, but in certain cases, they may not show up until early childhood. These symptoms may be serious and include:

Lack of energy

Vomiting and other feeding problems

Diarrhea

Dehydration

Failure to thrive (not gaining weight and growing as expected)

Children, teens, and adults may need testing if they develop symptoms of mild CAH (nonclassic CAH):

In children and teens, symptoms of mild CAH may include:

Rapid growth that stops at a younger age than usual. As a result, older teens and adults with CAH may be shorter than average.

Signs of early puberty.

Severe acne, which may continue into adulthood.

In girls and women, symptoms of mild CAH may include:

More facial and/or body hair than usual

Irregular menstrual periods or no periods at all
Thinning hair on the head, especially near the temples (male pattern baldness)
Enlarged clitoris
Infertility
In boys and men, symptoms of mild CAH may include:
Enlarged penis with small testicles (testes)
Low sperm count
Infertility

WHAT HAPPENS DURING A 17-OHP TEST?

For a newborn screening, a health care professional will clean your baby's heel with alcohol. Then the provider will stick the baby's heel with a small needle to collect a few drops of blood. The heel will then be bandaged.
For older children and adults, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

There are no special preparations needed for a 17-OHP test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk from a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly. Your baby may feel a little pinch when the heel is poked. A small bruise may form on the heel, but it should go away quickly.

WHAT DO THE RESULTS MEAN?

The results of a 17-OHP test help show whether you or your child have the most common type of CAH, which is caused by a lack of 21-hydroxylase.
If test results show a normal amount of 17-OHP, it means that you or your child probably do not have CAH with 21-hydroxylase deficiency. If you or your child has symptoms, the provider may order other tests to figure out the cause.
If test results show higher than normal levels of 17-OHP, you or your child probably has CAH. The provider may order other tests to learn more about how CAH is affecting you or your child.
If you or your child is being treated for CAH, decreasing levels of 17-OHP over time means that the treatment is working. Treatment for CAH may include medicines to replace the cortisol your body can't make. Other medicines may also be needed, depending on the type of CAH. If a girl is born with atypical external genitals, surgery may also be done.
If you, your child, or another family member has been diagnosed with CAH, you may want to talk with a genetic counselor. Genetic counselors are trained in genetics and genetic testing. They can help you learn how likely you are to pass CAH to future children.
If you have questions about the results of a 17-OHP test, talk with your provider.
Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A 17-OHP TEST?

If the results of your 17-OHP test aren't clear, your provider may order another test to measure your 17-OHP. This is called an ACTH stimulation test. It involves measuring your cortisol and 17-OHP levels before and after you have an injection (shot) of ACTH (adrenocorticotropic hormone).

ACTH is a hormone your body makes to trigger your adrenal glands to make cortisol. If you have CAH, the ACTH injection will make your 17-OHP levels increase a lot, but it will not increase your cortisol levels.

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Acetaminophen Level

<https://medlineplus.gov/lab-tests/acetaminophen-level/>

ACETAMINOPHEN LEVEL

WHAT IS AN ACETAMINOPHEN LEVEL TEST?

This test measures the amount of acetaminophen in your blood. Acetaminophen is a medicine used in many over-the-counter (OTC) and prescription medicines. It is used to relieve pain and reduce fever. It comes in liquid, chewable, capsule, or pill form. It is found in many brand name medicines, including Tylenol and Excedrin. Outside the U.S., it's called paracetamol. Acetaminophen can also be combined with other active ingredients in medicines that treat allergy, cough, colds, flu, and sleeplessness.

Acetaminophen is safe when taken at the proper dose. However, taking too much of it can cause an acetaminophen overdose or toxicity. This may cause serious and sometimes deadly liver damage.

Unfortunately, it's common for people to accidentally take too much and have an acetaminophen overdose. It can happen if you:

Take more than one medicine that contains acetaminophen. For example, you may not realize that a cold, flu, allergy, or other medicine you are taking contains acetaminophen. If you take that medicine plus another medicine that contains acetaminophen, you may end up taking an unsafe dose.

Aren't following the dose recommendations in the instructions. For example, you might take more than the maximum dose or multiple doses more often than you are supposed to. The adult maximum dose is generally 4000 mgs in 24 hours. But that may be too much for some people. Check with your health care provider to see how much medicine is safe for you. Children's dosing recommendations depend on their weight, age, whether the medicine is in a liquid or solid form, if they are taking additional medicines that also have acetaminophen, and if they are taking any other medicines.

Give a child an adult version of the medicine, rather than the children's version.

Check with your provider before taking medicines containing acetaminophen for more than ten days, or five days for your child. When you take most medicines, including acetaminophen, your liver breaks them down and filters out the toxins. It may be hard for your liver to keep up if you're taking medicines containing acetaminophen for an extended time.

Other names: acetaminophen drug test, acetaminophen blood test, Paracetamol test, Tylenol drug test

WHAT IS IT USED FOR?

The test is used to find out if you or your child has taken too much acetaminophen. It may also be used to monitor how well the treatment of an overdose is working.

WHY DO I NEED AN ACETAMINOPHEN LEVEL TEST?

Your provider may order this test if you or your child has symptoms of an overdose. When you take too much of the medicine, your liver cannot process the excessive dose and it can build up in your body. Symptoms may happen as soon as two to three hours after taking the medicine but can take as long as 12 hours to appear.

Symptoms in adults and children are similar and may include:

Nausea and vomiting

Diarrhea

Abdominal (belly) pain

Loss of appetite

Trouble breathing

Fatigue

Irritability

Sweating

Jaundice, a condition that causes your skin and eyes to turn yellow

Acetaminophen overdose can be a life-threatening emergency if not treated quickly. If you think you or your child has taken too much acetaminophen, or you notice early symptoms of an acetaminophen overdose, call 911 or go to the nearest emergency room.

WHAT HAPPENS DURING AN ACETAMINOPHEN LEVEL TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an acetaminophen level test.

ARE THERE ANY RISKS TO AN ACETAMINOPHEN LEVEL TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If results show a high level of acetaminophen, you or your child may be at risk for liver damage and may need immediate treatment. The type of treatment will depend on how much excess acetaminophen is in your system. After you get your results, your provider may recommend other tests or treatment. They may repeat this test every four to six hours to make sure you are out of danger.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ACETAMINOPHEN LEVEL TEST?

Before you or your child take any medicine, read the label carefully. Make sure that you only use the recommended dose. Check the ingredient list to see whether the medicines contain acetaminophen so that you don't take too much. Some common medicines that may contain acetaminophen include:

Nyquil

Dayquil

Dristan

Contact

Theraflu

Actifed

Mucinex

Sudafed

Also, if you drink three or more alcoholic beverages a day, ask your provider if it is safe to take acetaminophen. Drinking alcohol while taking acetaminophen can increase your risk of liver damage.

Keep medicines out of reach of children and don't take medicines with acetaminophen for more than ten days without talking to your provider.

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Acid-Fast Bacillus (AFB) Tests

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ACID-FAST BACILLUS (AFB) TESTS

WHAT ARE ACID-FAST BACILLUS (AFB) TESTS?

Acid-fast bacillus (AFB) is a type of bacteria that causes tuberculosis and other types of mycobacterial infections, such as leprosy (Hansen's disease). AFB tests are mainly ordered for people with:

A positive result on a TB screening test. An AFB can then confirm this result.

Abnormal chest x-rays.

Symptoms of active tuberculosis, commonly known as TB

An AFB test can also sometimes detect other similar infections.

TB is a serious bacterial infection caused by *Mycobacterium tuberculosis*. It mainly affects the lungs but can also affect other parts of the body, including the brain, spine, and kidneys. TB can be latent or active. If you have latent TB, you'll have TB bacteria in your body but won't feel sick and can't spread the disease to others through coughing or sneezing. If you have active TB, you'll have symptoms of the disease and could spread the infection to others.

AFB tests look for AFB bacteria in your sputum. Sputum is a thick mucus that is coughed up from the lungs. It is different from spit or saliva. There are four tests used to identify AFB:

AFB smear. In this test, your sputum sample is "smear" on a glass slide and looked at under a microscope. It can provide results in 1-2 days. These results can show a possible or likely infection but cannot be used to provide a diagnosis.

AFB culture. In this test, your sample is sent to a lab to allow the cells to grow until there are enough to test. An AFB culture can positively confirm a diagnosis of TB or other mycobacterial infection. But it takes 6-8 weeks to grow enough bacteria to detect an infection.

Molecular test. In this test, your sample is tested using nucleic acid amplification (NAAT) with AFB smear and culture. The NAAT test is done directly on sputum samples and can detect TB bacteria. If there is TB bacteria, the test can also check whether the bacteria are resistant to rifampicin, a common treatment for TB. This test can be done in less than two hours.

Susceptibility test. This test is usually ordered along with an AFB culture. It is used to figure out the most effective antibiotic to treat your infection. *Mycobacterium tuberculosis* may be resistant to one or more of the medicines that are used to treat TB.

Other names: AFB smear and culture, TB culture and sensitivity, mycobacteria smear and culture, TB NAAT, Acid-Fast Bacillus Smear and Culture and Sensitivity, *Mycobacterium tuberculosis* Nucleic Acid Amplification Test

WHAT ARE THEY USED FOR?

AFB tests are most often used to diagnose an active tuberculosis (TB) infection, especially after a positive result in a TB screening test. A TB screening test can't tell the difference between an inactive and active TB infection. So, if the screening test comes back positive, your provider may want to verify the results by using an AFB test. AFB tests may also be used to help diagnose other types of AFB infections. These include:

Leprosy, a once feared, but rare and easily treatable disease caused by *Mycobacterium leprae* and *Mycobacterium lepromatosis* that affects the nerves, eyes, and skin. The skin often becomes red and flaky, with a loss of feeling.

An infection similar to TB, such as *Mycobacterium avium*, that mostly affects people with HIV and others with weakened immune systems.

AFB tests may also be used for people who have already been diagnosed with TB. The tests can show if the treatment is working and whether the infection can still be spread to others.

WHY DO I NEED AN AFB TEST?

You may need an AFB test if you have symptoms of active TB (TB disease). Which symptoms you have will depend on where in the body the TB bacteria are growing.

If TB disease is in the lungs (pulmonary), symptoms may include:

Cough that lasts for three weeks or more

Coughing up blood and/or sputum

Chest pain

Other symptoms of TB disease may include:

Chills and fever

Fatigue

Night sweats (heavy sweating during sleep)

Losing weight without trying

Back pain and paralysis (for spinal TB)

Blood in your urine

Headache and coma (for TB meningitis)

Joint pain

Abdominal (belly) pain

Generally, certain people are more likely to develop TB disease. They include:

People who have recently been infected with TB bacteria

People with medical conditions that weaken the immune system

Babies and young children often have weakened immune systems. Others can have weakened immune systems, too, especially people with any of these conditions:

HIV

Substance use

Silicosis, a lung disease caused by breathing in silica dust

Diabetes

Severe kidney disease

Low body weight

Organ transplants

Head and neck cancer

People can also have weakened immune systems due to certain treatments:

Medical treatments such as corticosteroids

Specialized treatment for rheumatoid arthritis or Crohn's disease

You may also need testing if you are at higher risk of getting TB because you:

Have been in close contact with someone who has been diagnosed with TB

Have HIV or another disease that weakens your immune system

Live or work in a place with a high rate of TB infection, such as a homeless shelter, nursing home, or prison

If you are currently being treated for TB, you might need testing to see if your treatment is working and if you could still spread the TB bacteria to others. Lastly, you may need an AFB test if you've had a positive result on a TB screening test.

WHAT HAPPENS DURING AFB TESTING?

Your health care provider will need a sample of your sputum for all AFB tests. The AFB smear and AFB culture are usually done at the same time. To get sputum samples:

You will be asked to breathe deeply and then cough and spit into a sterile container.

If you have trouble coughing up enough sputum, your provider may ask you to breathe in a sterile saline (salt) mist that can help you cough more deeply.

If you still can't cough up enough sputum, your provider may perform a procedure called a bronchoscopy. In this procedure, you will first get medicine so you won't feel any pain. Then, a thin, lighted tube will be put through your mouth or nose and into your airways. The sample may be collected by suction or with a small brush.

If your provider wants to test you for leprosy (Hansen's disease), they may take a sample from your skin. Your provider will use a special tool to take a small sample from your skin, or in some cases from the nerves around a sore caused by the condition. This procedure is called a biopsy.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

There are no special preparations needed for AFB testing.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to providing a sputum sample by coughing into a container. If you have a bronchoscopy, your throat may feel sore after the procedure. For a skin/nerve biopsy, there is also a small risk of infection and bleeding at the site where the sample is taken.

WHAT DO THE RESULTS MEAN?

If your results on an AFB smear or culture were negative, you probably don't have active TB. But it could also mean there weren't enough bacteria in the sample for your provider to make a diagnosis.

If your AFB smear was positive, it means you probably have TB or another mycobacterial infection, but an AFB culture is needed to confirm the diagnosis. Culture results can take several weeks, so your provider may decide to treat your infection in the meantime.

If your AFB culture was positive, it means you have active TB or another type of AFB infection. The culture can identify which type of infection you have.

If your molecular (NAAT) test was positive, it means you probably have TB or another kind of AFB infection.

Once you have been diagnosed, your provider may order a "susceptibility test" on your sample. A susceptibility test is used to help determine which antibiotic will provide the most effective treatment.

If you have the signs and symptoms of an active AFB infection, your provider will consider both your AFB smear and NAAT molecule test results when deciding whether to treat you for TB or any other mycobacterial infection while you wait for the culture results. If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AFB TESTING?

If not treated, TB can be deadly. But most cases of TB can be cured if you take antibiotics as directed by your provider. Treating TB takes much longer than treating other types of bacterial infections. After a few weeks on antibiotics, you will no longer be contagious, but you will still have TB.

To cure active TB, you usually need to take antibiotics for six to twelve months. How long you need to take it will depend on your overall health, age, and other factors. It's important to take the antibiotics for as long as your provider tells you, even if you feel better. Stopping early can cause the infection to come back.

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ADHD Screening

<https://medlineplus.gov/lab-tests/adhd-screening/>

ADHD SCREENING

WHAT IS ADHD SCREENING?

ADHD screening is also called ADHD testing. It helps find out if a child, teen, or adult has ADHD. ADHD stands for attention deficit hyperactivity disorder. It used to be called ADD (attention-deficit disorder).

A person who has ADHD has a hard time paying attention and focusing on tasks. ADHD is a common developmental disorder that begins in childhood and often lasts into adulthood. Although it's usually diagnosed in childhood, some people with ADHD don't get diagnosed with ADHD until they are adults.

There are three types of ADHD:

Mostly Inattentive ADHD. People with this type of ADHD have trouble paying attention and are easily distracted. It's hard for them to organize or finish tasks. They may have trouble following instructions or conversations.

Mostly Hyperactive-Impulsive ADHD. People with this type of ADHD have symptoms of both hyperactivity and impulsivity:

With hyperactivity, people feel a need to always be moving. They have trouble sitting still and may fidget and/or talk too much.

With impulsivity, people have trouble controlling their actions and words. They tend to act on sudden ideas or feelings without thinking about the possible results. They may interrupt others a lot or have trouble waiting their turn.

With hyperactivity, people feel a need to always be moving. They have trouble sitting still and may fidget and/or talk too much.

With impulsivity, people have trouble controlling their actions and words. They tend to act on sudden ideas or feelings without thinking about the possible results. They may interrupt others a lot or have trouble waiting their turn.

Combined ADHD. People with this type of ADHD have a mix of inattentive and hyperactive-impulsive symptoms. Combined ADHD is the most common type.

It's normal to sometimes have trouble paying attention or sitting still, especially for children. But people with ADHD have more severe symptoms that cause serious problems, for example, failing grades for a child or a job loss for an adult. The symptoms are ongoing and may affect family and social life, too.

There's no single test to show whether a person has ADHD. Instead, a health care provider will follow a set of professional guidelines for gathering information, doing tests, and diagnosing the cause of a person's symptoms. Treatment is available to help people of all ages who have been diagnosed with ADHD.

Other names: ADHD test

WHAT IS IT USED FOR?

ADHD screening is used to check for ADHD in people who have frequent, serious problems with attention, hyperactivity, and/or impulsive behavior. The screening also checks for other mental disorders that often happen with ADHD. Sometimes symptoms of these other disorders are mistaken for ADHD.

ADHD screening is used with children as young as age four. If you have a younger child with symptoms that could be ADHD, ask your child's provider which type of testing might be best.

WHY DO I NEED ADHD SCREENING?

Your child may need screening if you, a teacher, or another caregiver thinks your child has symptoms of ADHD.

Adults who have had challenges since childhood (before age 12) with focusing, impulsive behavior, restlessness, and/or being organized can ask their providers if they need to be screened for ADHD. The symptoms depend on the type of

ADHD a person has.

People with symptoms of attention problems may often:

Miss details or make careless mistakes in schoolwork, at work, or in other activities

Have trouble staying focused on play activities or work tasks

Not seem to listen when spoken to directly

Find it hard to follow instructions or finish tasks, or may start but get easily sidetracked

Have trouble being organized, keeping belongings in order, and managing time

Avoid doing tasks that require long periods of mental effort

Lose important items, such as books, wallets, keys, eyeglasses, and cellphones

Forget about doing daily activities

People with symptoms of hyperactivity-impulsivity may often:

Fidget and squirm while seated

Get up when staying seated is expected, such as at school or work

Run around or climb when it's not appropriate (children) or feel restless (teens and adults)

Have trouble doing quiet activities

Be constantly moving or "on the go"

Talk much more than is normal

Blurt out answers before questions are completed

Have trouble waiting for their turn

Interrupt others for example during conversations or games

People with combined ADHD show a mix of inattention and hyperactivity-impulsivity symptoms.

WHAT HAPPENS DURING AN ADHD SCREENING?

The general screening process is similar for people of all ages. It involves several steps. A primary care provider may do the entire screening or refer the person to a mental health provider. A mental health provider is a health care professional who specializes in diagnosing and treating mental health problems.

During an ADHD screening, the provider will:

Do a physical exam. An exam by a primary care provider checks for certain health conditions that can cause symptoms similar to ADHD. The exam may include vision and hearing tests (for children and adults), thyroid tests, and other tests.

Review medical history, family health history, and other records. The cause of ADHD isn't known, but it often runs in families. So, family history is important. The results of any past tests and school reports provide valuable information, too. For adults, old school records can help a provider find out if symptoms began in childhood.

Gather information using standardized ADHD symptom checklists, questionnaires, and/or interview questions. A provider will use one or more tools to gather, organize, and evaluate information during an ADHD screening. These tools have rating scales (scoring systems) that help the provider see if a person's symptoms and history fit a diagnosis of ADHD. The provider will use these tools to:

Interview the person about their symptoms. For children, questions and activities are geared toward their age. Parents are usually interviewed, too.

Gather information from others. To make a diagnosis, it's necessary to know how the person acts during different activities and situations. So, the provider may request permission to ask others for information. These people may include:

For children and teens: Teachers, babysitters, coaches, and other family members who spend time with the child. Parents may also complete checklists and/or questionnaires.

For adults: Family members who know the person from childhood, and spouses, friends, and co-workers.

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For adults: Family members who know the person from childhood, and spouses, friends, and co-workers.

Screen for other mental health disorders. People who have ADHD often have other mental health disorders. It's also possible that another disorder, and not ADHD, is causing symptoms. So, mental health screening tests are often done as needed to check for conditions such as, anxiety, depression, and bipolar disorder. Children and may also be tested for learning disabilities, autism spectrum disorder, or behavior disorders.

WILL I NEED TO DO ANYTHING TO PREPARE FOR ADHD SCREENING?

You usually don't need any special preparations for ADHD screening. Your provider will let you know what health and school records you may need to gather.

ARE THERE ANY RISKS TO SCREENING?

There is no risk to a physical exam, written test, or questionnaire.

WHAT DO THE RESULTS MEAN?

The provider will usually write up the results of an ADHD screening in a report that includes a diagnosis. To diagnose ADHD, the screening must find all of these things:

Several symptoms of ADHD that began before age 12.

Several symptoms of inattention and/or hyperactivity-impulsivity that have lasted for at least 6 months and cause serious problems. To be diagnosed with ADHD:

Children up to age 16 must have at least 6 ongoing symptoms.

People 17 and older must have at least 5 ongoing symptoms.

Children up to age 16 must have at least 6 ongoing symptoms.

People 17 and older must have at least 5 ongoing symptoms.

Symptoms that happen in 2 or more settings, for example, at home and at work or school.

Symptoms that clearly get in the way of functioning well at school, work, and/or in social situations.

Symptoms aren't caused by another mental health disorder.

If there is a diagnosis of ADHD, the report will include the type of ADHD (inattentive, hyperactive-impulsive, or combined). It will also say whether ADHD is mild, moderate, or severe. Any other disorders that may have been found will be included in the report.

There's no cure for ADHD, but treatment may help reduce symptoms. Treatment may include medicine, talk therapy, behavioral therapy, and lifestyle changes. Usually, two or more treatments are combined. Parents of children with ADHD may also receive training to help their children manage ADHD.

Learn more about laboratory tests, reference ranges, and understanding results.

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Adrenocorticotrophic Hormone (ACTH)

<https://medlineplus.gov/lab-tests/adrenocorticotrophic-hormone-acth/>

ADRENOCORTICOTROPIC HORMONE (ACTH)

WHAT IS AN ADRENOCORTICOTROPIC HORMONE (ACTH) TEST?

This test measures the level of adrenocorticotrophic hormone (ACTH) in a sample of your blood. A hormone is a chemical messenger in your bloodstream that controls the actions of certain cells or organs. ACTH tells your adrenal glands, two small glands that sit above your kidneys, to make another hormone called cortisol.

Cortisol affects almost every organ and tissue in your body. It helps your body:

Respond to stress (cortisol is sometimes called the "stress hormone")

Reduce inflammation

Regulate blood glucose, also called blood sugar, and metabolism (how your body uses food for energy)

Control blood pressure

Because ACTH controls how much cortisol you make, ACTH testing is usually done with a blood test to measure your cortisol levels. Normally, if cortisol levels drop too low, your ACTH levels will increase to trigger your adrenal glands to make more cortisol. And if cortisol levels are high, ACTH levels will decrease so your adrenal glands will make less cortisol.

Your ACTH and cortisol levels are controlled by a complex feedback system of hormones made in different parts of your body. These include your:

Pituitary gland. This small gland is in the base of your brain. It makes ACTH and other hormones.

Hypothalamus. This is a part of your brain. It makes a hormone that tells your pituitary gland how much ACTH to make.

Adrenal glands. These glands respond to ACTH by making cortisol and releasing it into your bloodstream. The increased amount of cortisol in your blood signals your hypothalamus to stop making the hormone that tells your pituitary to make ACTH.

Conditions that affect any part of this system can affect the amount of cortisol in your body. Too much or too little cortisol can lead to serious health problems.

Other names: Adrenocorticotrophic hormone blood test, corticotropin

WHAT IS IT USED FOR?

An ACTH test is used to help diagnose and monitor conditions that affect cortisol levels. These conditions include disorders of the pituitary and adrenal glands.

High cortisol levels may be from:

Cushing's disease, a tumor in the pituitary gland that's usually benign (not cancer). The tumor makes too much ACTH, which triggers the adrenal glands to make too much cortisol. The tumor keeps making ACTH even when cortisol levels are high.

Cushing's syndrome, a disorder in which your body is exposed to too much cortisol over a long period of time. The extra cortisol may be caused by:

Long-term use of high doses of steroid medicine. Steroids can have side effects that increase cortisol levels.

Tumors in the adrenal glands that make extra cortisol, even when blood cortisol levels are high.

Tumors outside of the pituitary gland that make extra ACTH. This triggers the adrenal glands to make too much cortisol. These tumors are often found in the lungs, but they aren't common.

Long-term use of high doses of steroid medicine. Steroids can have side effects that increase cortisol levels.

Tumors in the adrenal glands that make extra cortisol, even when blood cortisol levels are high.

Tumors outside of the pituitary gland that make extra ACTH. This triggers the adrenal glands to make too much cortisol. These tumors are often found in the lungs, but they aren't common.

Low cortisol levels may be from:

Addison disease, a condition in which the adrenal glands can't make enough cortisol because they are damaged. The damage is usually caused by autoimmune disease.

Hypopituitarism, a condition in which the pituitary gland either stops making one or more hormones or can't make enough hormones, including ACTH. This rare disorder may be caused by pituitary tumors, damage to the pituitary gland, or a problem with the hypothalamus. If the pituitary gland can't make enough ACTH, the adrenal glands won't make enough cortisol.

ACTH and cortisol tests are used together to help find out which type of condition may be causing abnormal cortisol levels. Other tests are often needed to help make a final diagnosis.

WHY DO I NEED AN ACTH TEST?

You may need this test if you had abnormal results on a cortisol test or have symptoms of too much or too little cortisol.

Symptoms of too much cortisol may vary. They may include:

Weight gain

Thin arms and legs

Round face

Increased fat around the base of the neck or between the shoulder blades

Easy bruising

Wide purple streaks on the stomach, breasts, hips, and under the arms

Muscle weakness

Symptoms of too little cortisol include:

Long-lasting fatigue

Muscle weakness

Loss of appetite

Weight loss

Abdominal (belly) pain

You may also need this test if you have symptoms of hypopituitarism. Symptoms will vary depending on which pituitary hormones are affected and how low the levels are. If ACTH is decreased, you may have symptoms from low cortisol levels. If a tumor is causing hypopituitarism, symptoms may also include headaches and vision problems.

WHAT HAPPENS DURING AN ACTH TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your health care provider will tell you how to prepare for your test. You may need to fast (not eat or drink) for several hours before your test. Blood samples for ACTH testing are often taken early in the morning when ACTH levels are at their highest for the day.

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand what the results of your ACTH test mean, your provider will usually compare your ACTH level with your cortisol level. If your results are abnormal, you may need other tests to confirm the specific cause. Ask your provider to explain what your test results say about your health.

In general:

High ACTH and high cortisol levels are a sign of pituitary tumor that makes ACTH (Cushing's disease). The extra ACTH triggers your adrenal glands to make too much cortisol.

You'll likely need more tests, such as a CT or MRI scan, to confirm whether you have a pituitary tumor. The high level of ACTH could also be from other causes including:

A rare type of tumor in your lung, pancreas, or thyroid gland

A condition called multiple endocrine neoplasia

High ACTH and high cortisol levels are a sign of pituitary tumor that makes ACTH (Cushing's disease). The extra ACTH triggers your adrenal glands to make too much cortisol.

A rare type of tumor in your lung, pancreas, or thyroid gland

A condition called multiple endocrine neoplasia

High ACTH and low cortisol levels are a sign of damage to your adrenal glands (Addison disease). If your adrenal glands can't make enough cortisol, your pituitary gland usually makes more ACTH to try to get your adrenal glands to work better.

Low ACTH and high cortisol levels are a sign of Cushing's syndrome. This means that high cortisol levels may be caused by long-term use of steroid medicines or an adrenal gland tumor that makes extra cortisol. The high cortisol levels signal your hypothalamus and pituitary gland to decrease ACTH levels.

Low or normal ACTH and low cortisol levels are a sign of hypopituitarism. This means that a pituitary tumor or damage to your pituitary gland may prevent it from making enough ACTH. Without enough ACTH, your adrenal glands aren't stimulated to make enough cortisol. Over time, they may shrink.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ACTH TEST?

Several other tests may be used to help diagnose the specific cause of abnormal cortisol levels. These tests measure changes in your ACTH or cortisol levels after you are given medicines that either trigger or stop the production of the hormones involved with making cortisol.

For example, an ACTH stimulation test is commonly used to help find out if a problem with your adrenal glands is causing low cortisol levels. This blood test measures your cortisol levels before and after you get an injection (shot) of ACTH. The results show whether your adrenal glands are able to respond normally to ACTH by making cortisol.

If your adrenal glands make a normal amount of cortisol after the shot, it suggests that a problem with your hypothalamus or pituitary gland is causing low cortisol levels. If your adrenal glands make little or no cortisol after the shot, you most likely have a problem that affects the health of your adrenal glands.

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RELATED HEALTH TOPICS

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Albumin Blood Test

<https://medlineplus.gov/lab-tests/albumin-blood-test/>

ALBUMIN BLOOD TEST

WHAT IS AN ALBUMIN BLOOD TEST?

An albumin blood test measures the amount of albumin in your blood. Albumin is a protein made by your liver. Low albumin levels can be a sign of liver or kidney disease or another medical condition. High levels may be a sign of dehydration.

Blood is made of fluid and blood cells. Albumin keeps the fluid part of your blood from leaking out of your blood vessels (the tubes your blood flows through) and into other tissues. If you don't have enough albumin, fluid can leak out of your blood and build up in your lungs, abdomen (belly), or other parts of your body.

Albumin also helps carry important substances throughout your body. These include hormones, vitamins, and enzymes (proteins that speed up certain chemical reactions in your body).

Other names: ALB, serum albumin test

WHAT IS IT USED FOR?

An albumin blood test is used to check your general health and nutrition, and to see how well your liver and kidneys are working. If your liver is damaged or you're not well nourished, your liver may not make enough albumin. If your kidneys are damaged, they may let albumin pass from your blood into your urine (pee).

An albumin test may also be used to monitor treatment for liver or kidney conditions, or other conditions that may affect albumin levels.

An albumin blood test is often done as part of a group of blood tests that measure different enzymes, proteins, and other substances made in your liver. These tests are called liver function tests or a liver panel. An albumin test may also be part of a comprehensive metabolic panel (CMP), a group of routine blood tests that measures 14 substances in a sample of your blood.

WHY DO I NEED AN ALBUMIN BLOOD TEST?

Your health care provider may order an albumin test as part of your routine checkup. In this case, the test is usually ordered as part of a group of liver function tests or a comprehensive metabolic panel.

You may also need an albumin test if you have symptoms of liver or kidney disease. Symptoms of liver disease include:

Nausea and vomiting

Lack of appetite

Fatigue

Weakness

Jaundice, a condition that causes your skin and eyes to turn yellow

Swelling and/or pain in your abdomen (belly)

Swelling in your ankles and legs

Dark-colored urine (pee) and/or light-colored stool (poop)

Frequent itching

Symptoms of kidney disease include:

Swelling in your hands and feet or puffy eyelids

Dry skin, itching, or numbness

Fatigue

Increased or decreased urination

Urine that is bloody or foamy

Loss of appetite and weight loss

Muscle cramps
Nausea and vomiting
Shortness of breath
Sleep problems
Trouble thinking clearly

With some types of kidney disease, such as chronic kidney disease, you may not have symptoms until the later stages.

WHAT HAPPENS DURING AN ALBUMIN BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations to test for albumin in blood. If your provider ordered other blood tests, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow. Certain medicines may affect your test results, so tell your provider what you are taking. But don't stop taking any medicines without talking with your provider first.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

An albumin blood test alone cannot diagnose a condition. Your provider will usually consider your albumin test results with the results of other tests to make a diagnosis.

Lower than normal albumin levels in your blood may be a sign of:

Liver disease, including severe cirrhosis, hepatitis, and fatty liver disease

Kidney disease

Malnutrition, especially a lack of protein in your diet

Infection

Digestive diseases that involve problems using protein from food, such as Crohn's disease and malabsorption disorders

Burns over a large area of your body

Thyroid disease

Higher than normal albumin levels in your blood may be a sign of dehydration, which may be caused by severe diarrhea or other conditions.

If your albumin levels are not in the normal range, it doesn't always mean you have a medical condition that needs treatment. Certain medicines, including steroids, insulin, and hormones, can increase albumin levels. Your albumin levels may be lower than normal if you haven't eaten for 24 to 48 hours, or take certain medicines, including birth control pills. Albumin levels are also lower during pregnancy.

If you have questions about your test results, ask your provider to explain what they mean.

Learn more about laboratory tests, reference ranges, and understanding results.

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Total Protein and Albumin/Globulin (A/G) Ratio
What You Need to Know About Blood Testing

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Alcohol Use Screening Tests

<https://medlineplus.gov/lab-tests/alcohol-use-screening-tests/>

ALCOHOL USE SCREENING TESTS

WHAT ARE ALCOHOL USE SCREENING TESTS?

Alcohol use screening tests are questionnaires you answer about your alcohol use. Your responses are used to check for unhealthy alcohol use. Unhealthy alcohol use, also called alcohol misuse, use can cause health problems and may lead to alcohol use disorder (AUD). AUD is a brain disorder in which you can't stop or control your drinking, even though it causes distress and harm. It can cause problems in your work, relationships, and health. AUD can be mild, moderate, or severe, depending on your symptoms. A severe AUD used to be called alcohol dependence and alcoholism.

Unhealthy alcohol use can also include binge drinking, which is drinking so much at once that your blood alcohol concentration (BAC) level is 0.08% or more. For a man, this usually happens after having five or more drinks within a few hours. For a woman, it is after about four or more drinks within a few hours. Not everyone who binge drinks has an AUD, but they are at higher risk of getting one.

Besides checking for unhealthy alcohol use, alcohol screening tests can let you know if you are at risk for AUD. This risk may depend on how much, how often, and how quickly you drink alcohol. Knowing that you are at risk may help you to cut back on your drinking.

If the alcohol screening test finds that you do have AUD, you and your health care provider can make an effective treatment plan for you.

Other names: Alcohol Use Disorders Identification Test (AUDIT), AUDIT-C, Cut down Annoyed, Guilty, Eye-opener (CAGE) tool, Tolerance, Annoyed Cut down, Eye-opener (T-ACE) tool, Screening with Brief Intervention (SBI) Tool

WHAT ARE THEY USED FOR?

Alcohol use screening tests are questionnaires used to check for unhealthy alcohol use and your risk for AUD. Some tests can also help show if your disorder is mild, moderate, or severe.

WHY DO I NEED AN ALCOHOL USE SCREENING TEST?

Your provider may do a screening test as part of a routine health care visit. You may also have a screening test if a provider thinks that you could have a problem with alcohol. The screening may include questions to help identify if you have symptoms of alcohol use disorder (AUD). These can include:

Drinking more or longer than you planned.

Trying to cut down or quit but being unable to do it.

Hiding liquor bottles and glasses so others don't see evidence of drinking.

Having strong cravings for alcohol.

Continuing to drink even if it causes problems in your personal relationships, work, school, and/or health.

Building up a tolerance to alcohol. That means you need more and more alcohol to feel its effects.

Using alcohol when it's not safe, such as driving.

Having symptoms of withdrawal such as tremors, headache, weakness, sweating, or nausea when you try to stop drinking.

Some people with AUD don't know or want to admit they have a problem. If your family, friends, or co-workers express concerns about your drinking, talk to your provider about getting a screening. Your provider may also recommend a screening if they notice signs and symptoms of an AUD.

WHAT HAPPENS DURING AN ALCOHOL USE SCREENING TEST?

Your primary care provider or a mental health provider may do an alcohol use screening. A mental health provider is a health care professional who specializes in diagnosing and treating mental health problems. Some mental health providers specialize in treating AUD and other substance use disorders.

There are different types of alcohol use screening tests. But they each include questions about your drinking habits and how they may affect your life. The most commonly used alcohol screening tests are:

Alcohol Use Disorders Identification Test (AUDIT). This test contains 10 multiple-choice questions about how much and how often you drink alcohol and if you have any alcohol-related problems or reactions. The answers are scored on a point system. A score of 8 or more may indicate AUD.

AUDIT-C is a shortened version of the Alcohol Use Disorders Identification Test (AUDIT). It contains three multiple-choice questions. Each answer is valued from 0 to 4 points. The higher the score, the more likely it is you have AUD.

Cut down Annoyed, Guilty, Eye-opener (CAGE). This test is still often used, but it's no longer recommended because there are better tests available. It contains the following yes or no questions:

Have you ever felt you should cut down on your drinking?

Have people annoyed you by criticizing your drinking?

Have you ever felt bad or guilty about your drinking?

Eye-opener: have you ever had a drink first thing in the morning to steady your nerves?

Two or more "yes" answers may be a sign of AUD.

Have you ever felt you should cut down on your drinking?

Have people annoyed you by criticizing your drinking?

Have you ever felt bad or guilty about your drinking?

Eye-opener: have you ever had a drink first thing in the morning to steady your nerves?

The NIAAA Single Alcohol Screening Question (SASQ). This test isn't scored and may be part of a routine health care visit. Your provider may first ask how often you have had an alcoholic drink in the past year, but the SASQ test only contains one question.

How many times in the past year have you had (4 for women, or 5 for men) or more drinks in a day?

If your answer is one or more, then your provider may want to ask additional questions to help determine if you have, or are at risk, for AUD. Two or more "yes" answers to those questions may be a sign of AUD.

How many times in the past year have you had (4 for women, or 5 for men) or more drinks in a day?

Screening with Brief Intervention (SBI) Tools. These tests contain a series of yes or no questions to help identify, reduce, and prevent alcohol misuse. They can help identify brief interventions and the best treatment for you. Brief interventions are short-term counseling sessions and treatment strategies. These are meant to help you change your drinking behaviors and habits. If your test shows you have or are at risk for AUD, a long-term treatment plan may be recommended.

Some questionnaires, including the Alcohol Use Disorders Identification Test (AUDIT), are available online for self-testing. You can ask your provider how to access these tests.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an alcohol use screening test.

ARE THERE ANY RISKS TO THE TEST?

There is no risk in taking a questionnaire.

WHAT DO THE RESULTS MEAN?

Some results may show whether you have AUD. Others may show you how severe your AUD may be. For those tests, your results may be in one of the following or similarly named categories:

Mild. You have two to three symptoms of AUD. This pattern of drinking puts your health at risk and/or has harmful social effects on you or others. If you have symptoms now, you may be at risk for more serious problems in the future. Your provider may recommend steps to help you change your drinking habits.

Moderate. You have four or five symptoms of AUD. Your provider may recommend a brief intervention or other treatments.

Severe. You have six or more symptoms of AUD. You may need more intensive treatment. This can include seeing a specialist, medicines, support groups, and/or addiction recovery programs.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ALCOHOL USE SCREENING TEST?

Some people with AUD can change their behaviors to cut back on drinking. Other people will need to stop completely. This is known as abstinence. Depending on the severity of your disorder, your provider may recommend one or more of the following treatments:

Brief or long-term counseling from a mental health provider who specializes in treating AUD. This may help you learn ways to change your drinking behavior and not start drinking again.

Support groups. There are many different approaches and formats to alcohol use disorder support groups. Groups may help provide peer support to stop or cut back on your drinking. Talk with your provider to find a group and approach that's right for you.

Medicines. Certain medicines may reduce your cravings for alcohol. These medicines are to help you stop, cut back, quit, or not start drinking again.

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RELATED MEDICAL TESTS

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Aldosterone Test

<https://medlineplus.gov/lab-tests/aldosterone-test/>

ALDOSTERONE TEST

WHAT IS AN ALDOSTERONE (ALD) TEST?

An aldosterone test measures the amount of aldosterone (ALD) in your blood or urine (pee). ALD is a hormone made by your adrenal glands, two small glands that sit above your kidneys. ALD helps keep your blood pressure stable. It does this by helping to balance the levels of two electrolytes in your blood, sodium and potassium.

If your blood pressure gets too low, ALD signals your kidneys to release more sodium into your blood and to get rid of potassium through urine. This helps keep more water in your bloodstream, which makes your blood pressure increase. When your blood pressure returns to normal, your ALD levels decrease.

Abnormal ALD levels can cause serious health conditions, including:

- High blood pressure

- Low blood pressure

Complications from abnormal levels of sodium and potassium, which can affect your heart, kidneys and other parts of your body

ALD testing may be used with other tests to help diagnose adrenal gland disorders. It can also help find out if other medical conditions are affecting the way your adrenal glands work.

An ALD test is commonly combined with a test that measures renin levels in your blood. This is called an aldosterone-renin ratio test. Renin is a hormone that your kidneys make when your blood pressure gets too low. It's part of a complex system that triggers your adrenal glands to make ALD.

Comparing renin and ALD levels helps show whether abnormal levels of ALD are caused by a problem with your adrenal glands or a problem with your kidneys or other parts of the system that triggers the production of ALD.

Other names: aldosterone, serum; aldosterone urine; aldosterone-renin ratio test; aldosterone-plasma renin activity

WHAT IS IT USED FOR?

An aldosterone (ALD) test is mainly used with a renin test to find out whether too much ALD is the cause of high blood pressure that:

- Happens with low potassium levels

- Doesn't improve with usual blood pressure medicines

- Develops at a young age

An ALD test is also used to help:

- Diagnose adrenal glands disorders

- Find the cause of an abnormal sodium or potassium levels on an electrolyte test

- Find the cause of orthostatic hypotension (low blood pressure that makes you dizzy when you get up from sitting or lying down)

WHY DO I NEED AN ALDOSTERONE TEST?

You may need this test if your health care provider thinks you may have too much or too little aldosterone (ALD).

Signs and symptoms of too much ALD (called hyperaldosteronism or aldosteronism) include:

- High blood pressure

- Low potassium test results and/or high sodium test results

- Muscle weakness, cramps, and/or spasms

- Fatigue

- Headache

- Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Increased thirst and urination

Signs and symptoms of too little ALD (called hypoaldosteronism) include:

Low blood pressure

Test results that show high potassium and/or low sodium levels

Fatigue

Muscle weakness

Loss of appetite

Weight loss

Abdominal (belly) pain

WHAT HAPPENS DURING AN ALDOSTERONE TEST?

An aldosterone (ALD) test uses a sample of blood or urine. A test that compares your ALD and renin levels uses a blood sample.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

The amount of ALD in your blood changes depending on whether you are standing up or lying down. So, you may be asked to lie down or stand for a period of time before the test. You may also be tested in both positions.

For an ALD urine test, your provider may ask you to collect all urine during a 24-hour period. You will be given a special container to collect your urine and instructions on how to collect and store your samples. Your provider will tell you what time to start. The test generally includes the following steps:

Urinate (pee) in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Follow the instructions your provider gives you. You may need to stop taking certain medicines for a period of time before the test. This includes certain over-the-counter medicines. So, be sure to tell your provider everything you're taking. But don't stop taking any prescribed medicine unless your provider tells you to stop.

You may also be asked to limit the amount of salt you eat each day for about two weeks before your test. Your provider will explain how to limit salt in your diet.

You should not eat natural black licorice for at least two weeks before your test, because it can affect your results. Most licorice products in the United States don't contain natural licorice. But if you eat licorice, check the ingredients label to be sure.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There are no known risks to having a urine test.

WHAT DO THE RESULTS MEAN?

To understand what your test results mean, your provider will consider your aldosterone (ALD) levels along with the results of other tests. Ask your provider to explain what your results say about your health.

In general, higher than normal amounts of ALD may mean you have:

Primary aldosteronism (also called Conn syndrome). This disorder is usually caused by a benign (not cancerous) tumor on an adrenal gland that produces extra ALD.

Secondary aldosteronism. This means that your adrenal glands are healthy, but another medical condition is causing them to make too much ALD. Conditions that can cause high ALD levels include:

Kidney disease

Cirrhosis of the liver

Heart failure

Preeclampsia, a type of high blood pressure that that happens in pregnancy

Dehydration

Kidney disease

Cirrhosis of the liver

Heart failure

Preeclampsia, a type of high blood pressure that that happens in pregnancy

Dehydration

In general, lower than normal amounts of ALD may mean you have:

Addison disease (also called adrenal insufficiency). With this condition, damage to your adrenal glands causes low ALD.

The damage may be from conditions, such as:

An immune disorder (most common cause)

Certain infections

Cancer cells in the adrenal glands

An immune disorder (most common cause)

Certain infections

Cancer cells in the adrenal glands

Congenital adrenal hyperplasia (CAH). This is a group of genetic disorders you're born with.

Secondary adrenal insufficiency. This condition starts with a problem in your pituitary, a gland at the base of your brain that controls your adrenal glands.

Tertiary adrenal insufficiency. With this condition, low ALD levels are caused by a problem in your hypothalamus, a gland that tells your pituitary when to signal your adrenal glands to make hormones. The most common cause is suddenly stopping steroid medicines after taking them for a long time. Steroids can cause hormonal changes that make your adrenal glands shut down. When you stop taking the steroids, your adrenal glands may be slow to start working again.

Liddle syndrome. This is an inherited condition that causes high blood pressure.

Low levels of ALD may also be linked to:

Certain other health conditions, such as diabetes and lead poisoning

Certain medicines, including non-steroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen, and some heart failure medicines

If you have questions about your results and treatment options, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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RELATED HEALTH TOPICS

Addison Disease
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RELATED MEDICAL TESTS

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Alkaline Phosphatase

<https://medlineplus.gov/lab-tests/alkaline-phosphatase/>

ALKALINE PHOSPHATASE

WHAT IS AN ALKALINE PHOSPHATASE TEST?

An alkaline phosphatase (ALP) test measures the amount of ALP in your blood. ALP is an enzyme, a protein that speeds up certain chemical reactions in your body. ALP is found in all your body tissues, but higher amounts can be found in your liver, bile ducts, and bones. Each part of your body makes a different type of ALP.

Abnormal levels of ALP in your blood may be a sign of a wide range of health conditions, including liver disease, bone disorders, and chronic kidney disease. The amount of ALP released into your blood can be affected by liver damage, conditions that affect bone growth, and certain gastrointestinal diseases. But an alkaline phosphatase test alone can't identify the source of ALP in your blood, so other tests are usually needed to make a diagnosis.

Other names: ALP, ALK, PHOS, Alkp, ALK PHOS

WHAT IS IT USED FOR?

An alkaline phosphatase test is often used to screen for or help diagnose diseases of the liver or bones. The test may also help diagnose or monitor other health conditions.

WHY DO I NEED AN ALKALINE PHOSPHATASE TEST?

Your health care provider may order an alkaline phosphatase test as part of a routine checkup. Many conditions may affect ALP levels, so the test is often done with other blood tests. These other tests can include a comprehensive metabolic panel (CMP) or liver function tests, which check how well your liver is working.

An alkaline phosphatase test may also be ordered if you have symptoms of liver damage or a bone disorder.

Symptoms of liver disease can include:

Nausea and vomiting

Lack of appetite

Fatigue

Weakness

Jaundice, a condition that causes your skin and eyes to turn yellow

Swelling and/or pain in your abdomen (belly)

Swelling in your ankles and legs

Dark-colored urine (pee) and/or light-colored stool (poop)

Frequent itching

Symptoms of bone disorders can include:

Bone pain

Inflammation and arthritis in your joints

Bones that are unusually shaped and/or too large

Broken bones

WHAT HAPPENS DURING AN ALKALINE PHOSPHATASE TEST?

An alkaline phosphatase test is a type of blood test. During the test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know if there are any special instructions. An ALP test is usually ordered with other blood tests. You usually need to fast (not eat or drink) for several hours before these tests.

You may need to stop taking certain medicines before the test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

High alkaline phosphatase (ALP) levels may be a sign of a liver problem or a bone disorder. Liver problems and bone disorders create different types of ALP. But your ALP test results can't tell which type of ALP is high.

If your test results show high ALP levels, your provider may order other tests to help figure out what's causing the problem. These tests may include:

An ALP isoenzyme test. This test can tell which part of your body is making the ALP. But this test may not be available everywhere.

Liver function tests. If the results of these tests are also high, then your high ALP level is likely from a problem in your liver.

High alkaline phosphatase levels from your liver may be a sign of:

Blockages in your bile ducts

Cirrhosis

Hepatitis

Mononucleosis, which can sometimes cause swelling in the liver

If alkaline phosphatase levels are high and the results of liver tests are normal, the problem may be a bone disorder, such as Paget's disease of bone. This disease makes your bones unusually large and weak, causing them to break more easily.

Moderately high levels of ALP may be a sign of many different types of conditions, including:

Hodgkin lymphoma

Heart failure

Certain infections

It's possible to have higher than normal levels of ALP and not have a medical condition that needs treatment. Your provider will consider your symptoms, medical history, and other test results to make a diagnosis.

Low levels of ALP are less common. They may be a sign of:

Zinc deficiency

Protein deficiency

Malnutrition

Pernicious anemia

Thyroid disease

Wilson disease

Hypophosphatasia, a rare genetic disease that affects bones and teeth

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT THE ALKALINE PHOSPHATASE TEST?

Many things can affect ALP levels. For example:

Pregnancy can cause higher than normal ALP levels.

Children and teens may have high levels of ALP because their bones are growing.
Birth control pills and certain medicines may lower ALP levels, while other medicines can cause the levels to increase.
Eating a fatty meal before an alkaline phosphatase test may also cause a small increase in ALP.
To learn what your results mean, talk with your provider.

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RELATED HEALTH TOPICS

Bile Duct Diseases
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Paget's Disease of Bone

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Allergy Blood Test

<https://medlineplus.gov/lab-tests/allergy-blood-test/>

ALLERGY BLOOD TEST

WHAT IS AN ALLERGY BLOOD TEST?

An allergy blood test measures the amount of allergy-causing antibodies called immunoglobulin E (IgE). Antibodies are proteins that your immune system makes to fight foreign substances, such as viruses and bacteria. But your immune system may also make them in response to allergens. These are harmless substances, such as pollen and peanuts, that may cause an allergic reaction in some people.

It's normal to have small amounts of IgE in your blood. But when you have an allergic reaction, your immune system sees the allergens as a threat and makes IgE antibodies. This increases the amount of IgE in your blood and causes your allergy symptoms.

Common allergens include:

Pollen

Dust

Mold

Animal dander

Certain foods, including nuts and shellfish

Certain medicines, such as penicillin

Allergy symptoms can be mild to severe, depending on the type of allergy you have. They can range from itching and sneezing to asthma or even a life-threatening condition called anaphylactic shock.

Other names: IgE allergy test, Quantitative IgE, Immunoglobulin E, Total IgE, Specific IgE, RAST, CAP, ELISA, Allergy Screen

WHAT IS IT USED FOR?

Allergy blood tests are used to help find out if you have an allergy. There are two general types of allergy blood tests:

A total IgE test measures the total amount of IgE antibodies in your blood.

A specific IgE test measures how much IgE your body makes in response to a single allergen. A separate test is done for each allergen that may be causing your allergies.

WHY DO I NEED AN ALLERGY BLOOD TEST?

Your health care provider may order allergy testing if you have symptoms of an allergy. These include:

Stuffy or runny nose

Sneezing

Itchy, watery eyes

Hives (itchy raised red patches on your skin)

Diarrhea

Vomiting

Shortness of breath

Coughing

Wheezing

Your provider may order an allergy blood test if you can't have allergy skin testing. Skin testing involves putting allergens directly on or into your skin. You may not be able to have skin testing if you:

Have certain skin conditions

Take certain medicines that may affect the results of the test

Are likely to have a serious allergic reaction to the allergens used in skin testing

In certain cases, providers may order allergy blood tests for young children because skin testing may be too uncomfortable for them.

WHAT HAPPENS DURING AN ALLERGY BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an allergy blood test. You may need to stop taking certain medicines before this test, such as antihistamines, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having an allergy blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A total IgE test result that is high means that you may have some kind of allergy. But the results of a total IgE test don't show what you're allergic to or how serious your allergy may be.

A specific IgE test result that is high means that you may be allergic to the allergen that was tested. But the amount of IgE measured doesn't predict how serious your allergy may be.

If the results from either type of test show that you could have an allergy, your provider may refer you to an allergy specialist or recommend a treatment plan. Your treatment plan will depend on what you are allergic to and how serious your symptoms are.

If you're at risk for anaphylactic shock, you'll need to be very careful to avoid the things you are allergic to. You may need to carry an emergency epinephrine treatment (an epinephrine auto-injector) with you at all times. Anaphylactic shock is most common with allergies to certain foods, medicines, insect bites and stings, and latex.

Ask your provider if you are at risk for anaphylactic shock and discuss any questions you have about your test results or your allergy treatment plan.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ALLERGY BLOOD TEST?

Allergy blood tests may not always be accurate. Sometimes, the results may say you have an allergy when you actually don't (also known as a false positive). This may happen if your body is having a slight reaction to substances in certain foods that you may have eaten before the test. Higher amounts of IgE may also be caused by smoking or a parasitic infection.

It's uncommon for a blood test to show that you don't have an allergy when you actually do (also known as a false negative).

Depending on your medical history and symptoms, your provider may order an allergy skin test with an allergy blood test, or you may have a skin test alone.

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Allergy Skin Test

<https://medlineplus.gov/lab-tests/allergy-skin-test/>

ALLERGY SKIN TEST

WHAT IS AN ALLERGY SKIN TEST?

An allergy is an overreaction by your immune system to something that doesn't bother most people. An allergy skin test helps find out what is causing your allergic reaction by exposing your skin to allergy-causing substances known as allergens.

Your immune system is your body's defense system. It usually fights germs. But when you have an allergy, your immune system sees a harmless substance, like dust or pollen, as a threat. Your immune system responds to this "threat" and causes an allergic reaction. Symptoms of an allergic reaction can range from sneezing and a stuffy nose to a life-threatening condition known as anaphylactic shock.

During an allergy skin test, your health care provider will expose your skin to a suspected allergen to see if you have an allergic reaction. Substances that could cause an allergic reaction can include:

- Pollen
- Dust mites
- Molds
- Pet dander
- Food
- Insect bites and stings
- Medicines
- Household chemicals

The overreaction an allergy causes is also known as a hypersensitivity of your immune system. There are four main types of overreactions, known as Type 1 through Type IV hypersensitivities. Type 1 hypersensitivity causes some of the most common allergies. These include dust mites, pollens, foods, and animal dander. Other types of hypersensitivities cause different immune system overreactions. These range from mild skin rashes to serious autoimmune disorders.

An allergy skin test usually checks for allergies caused by Type 1 hypersensitivity. The test looks for reactions to specific allergens that are placed on the skin.

Other names: type 1 hypersensitivity skin test, hypersensitivity test allergy scratch test, allergy patch test, intradermal test, prick test, puncture test

WHAT IS IT USED FOR?

An allergy skin test is used to diagnose certain allergies. The test can show which substances (allergens) are causing your allergic reaction. This can help your provider plan how to treat your allergy.

WHY DO I NEED AN ALLERGY SKIN TEST?

Your provider may order allergy testing if you have symptoms of an allergy. These include:

- Stuffy or runny nose
- Sneezing
- Itchy, watery eyes
- Hives, a rash with raised red patches
- Diarrhea
- Vomiting
- Shortness of breath
- Coughing
- Wheezing

WHAT HAPPENS DURING AN ALLERGY SKIN TEST?

You will most likely get tested by an allergist or a dermatologist. You may get one or more of the following allergy skin tests:

An allergy scratch test, also known as a skin prick test, is the most common allergy skin test.

During the test:

Your provider will place small drops of specific allergens on different areas on your skin, such as your forearm or back.

Your provider will then lightly scratch or prick your skin through each drop.

If you are allergic to any allergens, you will develop a hive or a small red bump, known as a wheal, at that area or areas within about 15 to 20 minutes.

An intradermal test may be used to see if you are allergic to insect stings and the medicine penicillin. Food allergies are not tested with this method. You need to use more of the allergen to test for food allergies, and that raises the risk of severe reactions such as anaphylactic shock.

During the test:

Your provider will inject a small amount of allergen just below the skin surface, using a tiny, thin needle.

Your provider will watch the spot for a reaction. If you are allergic to any allergens, you will develop a hive or a small red bump, known as a wheal, at the area or areas within about 60 to 90 minutes.

This test is sometimes used if your allergy scratch test is negative, but your provider still thinks you might have an allergy.

An allergy patch test checks to see if you have an allergic reaction to allergens that come in contact with your skin. Your provider may ask what soap, lotion, or makeup you use.

During the test:

Your provider will place small patches on your skin. The patches look like adhesive bandages. They contain small amounts of specific allergens. You'll be asked to avoid getting the testing area wet.

You'll wear the patches for 48 to 96 hours and then return to your provider's office.

Your provider will remove the patches and check for rashes, a condition called contact dermatitis, or other reactions.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines before the test. These could include antihistamines and certain antidepressants. Your provider will tell you which medicines to avoid before your test and for how long. They may also ask about any products you use on your skin and if you are normally around animals.

If you are having a patch test, your provider may want you to avoid heavy sun exposure and tanning for at least two weeks before the test.

If your child is being tested, the provider may apply a numbing cream to their skin before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having allergy skin tests. The test itself is not painful. The most common side effect is red, itchy skin at the test areas.

In very rare cases, an allergy skin test may cause anaphylactic shock. This is why skin tests must be done in a provider's office where emergency equipment is available. If you've had a patch test and feel intense itching or pain under the patches once you are home, remove the patches and call your provider.

WHAT DO THE RESULTS MEAN?

Test results are usually reported as positive or negative:

If you have red bumps or swelling at any testing areas, the test is positive. This means you are probably allergic to those substances. Usually, the larger the reaction, the more likely you are to be allergic.

If you do not have red bumps or swelling at any of the testing areas, the test is negative. This means that you are probably not allergic to those substances.

If you are diagnosed with an allergy, your provider will recommend a treatment plan. The plan may include:

Avoiding the allergen when possible

Medicines

Lifestyle changes such as reducing dust in your home

Your provider will consider your symptoms and medical history to understand the results of an allergy skin test. They may also want to do a physical exam.

Learn more about laboratory tests, reference ranges, and understanding results.

If you are at risk for anaphylactic shock, you may need to carry an emergency epinephrine treatment with you at all times. Epinephrine is a drug used to treat severe allergies. It comes in a device called an epinephrine auto-injector that contains a premeasured amount of epinephrine. If you experience symptoms of anaphylactic shock, you should inject the device into your skin and call 911.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ALLERGY SKIN TEST?

Your provider may recommend an allergy blood test^Â instead if you:

Have a skin condition such as severe eczema or psoriasis^Â on your arms and back where the skin is usually tested

Are taking medicines that might interfere with test results

Have had a severe allergic reaction in the past

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Alpha-1 Antitrypsin Testing

<https://medlineplus.gov/lab-tests/alpha-1-antitrypsin-testing/>

ALPHA-1 ANTITRYPSIN TESTING

WHAT IS ALPHA-1 ANTITRYPSIN (AAT) TESTING?

Alpha-1 antitrypsin (AAT) testing uses a sample of blood or a cheek swab to diagnose a condition called alpha-1 antitrypsin deficiency (AAT deficiency). This condition is sometimes known as "alpha-1" or AATD. If you have AAT deficiency, your body doesn't make enough AAT.

AAT is made by your liver. It helps protect your lungs from inflammation and irritating substances you might breathe in, such as smoke. If your liver doesn't make enough AAT, your lungs may be more easily damaged from smoking, pollution, or dust from the environment. This can lead to a serious lung condition called chronic obstructive pulmonary disease (COPD). AAT deficiency may also cause a liver disease called cirrhosis. This is more common in children who have AAT deficiency.

AAT deficiency is a genetic disorder. That means it's caused by changes in your genes, which may also be called gene variants or mutations.

Genes carry information that controls what you look like and how your body works. AAT deficiency is caused by changes in the SERPINA1 gene, which carries instructions for making the AAT protein. These gene changes are inherited from your parents, so AAT deficiency tends to run in families.

If you have two mutated copies of the gene, it means you have a condition called AAT deficiency. People with this disorder have a higher risk of getting lung disease or liver damage before the age of 45.

If you have one mutated copy of the gene, you are a carrier of AAT deficiency. In these cases, this means you are at slightly higher risk of developing lung disease, especially if you have other risk factors, such as being a smoker. You could pass the mutated gene on to your children.

There are a few gene changes that cause AAT deficiency. These gene changes can:

- Decrease the amount of AAT protein your liver makes.

- Prevent your liver from making any AAT.

- Affect the shape of the AAT protein so that it can't move out of your liver to protect your lungs. Over time, AAT builds up in your liver and causes damage.

A genetic test can help you find out whether you have the gene change that increases your risk for lung and liver disease.

There are three types of testing to help diagnose AAT deficiency:

The AAT blood level test measures the amount of protein found in your blood to see if your AAT levels are abnormal.

If your AAT levels are abnormally low, a genetic test (either a genotype test or a phenotype test) is needed to confirm a diagnosis of AAT deficiency.

If your AAT levels are abnormally low, a genetic test (either a genotype test or a phenotype test) is needed to confirm a diagnosis of AAT deficiency.

The genotype test looks for the more common types of gene changes that can cause AAT deficiency.

The phenotype test checks for changes in the AAT protein that change how it would normally work.

Other names: A1AT, AAT, alpha-1-antiprotease deficiency, α_1 -antitrypsin, serum AAT test, AAT phenotyping, AAT genotyping, AAT deficiency test, AAT DNA sequencing test, AAT isoelectric focusing test, A1AT test, AATD test, alpha-1 protease inhibitor deficiency test

WHAT IS IT USED FOR?

There are three main uses for AAT testing:

To diagnose or rule out AAT deficiency in people who have symptoms of lung and/or liver disease that may be caused by AAT deficiency

To screen people who don't have symptoms, but have family members with AAT deficiency
To guide treatment choices for lung disease that may be caused by AAT deficiency

WHY DO I NEED AAT TESTING?

Diagnostic AAT testing may be recommended if you have the signs and symptoms of AAT deficiency or if you have conditions that could be caused by AAT deficiency.

Signs and symptoms of AAT deficiency may include:

Wheezing

Shortness of breath after exercise

Chronic cough with phlegm (mucus)

Chest pain

Fatigue

Faster-than-normal heartbeat when you stand up

Vision problems

Conditions that could be caused by AAT deficiency can include:

COPD (chronic obstructive pulmonary disease)

Repeated respiratory infections, such as colds and bronchitis

Asthma that doesn't respond well to treatment

Panniculitis, a skin condition that causes hardened skin with painful lumps or patches

Liver disease without a known cause

Jaundice (yellowing of the skin and eyes)

You may also get this test if you have a family history of AAT deficiency, emphysema, or unexplained cirrhosis.

AAT deficiency in babies often affects the liver. Your baby may need AAT testing if he or she has signs of liver disease such as jaundice or abnormal liver enzyme tests.

WHAT HAPPENS DURING AAT TESTING?

AAT blood testing may involve a blood test with blood from your arm or a finger prick. Some genetics tests involve a cheek swab.

For a blood test from your arm, a health care professional will take the blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a finger prick test, a health care professional will prick your fingertip to obtain a small amount of blood.

For a cheek swab, a health care professional will wipe the inside of your cheek with a small tool to remove some cells. You may have the option of doing it yourself.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an AAT test.

ARE THERE ANY RISKS TO THE TEST?

There is very little physical risk to a blood test. If the blood was taken from your arm, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There are no risks to having a cheek swab.

WHAT DO THE RESULTS MEAN?

If your AAT blood levels are low, it may mean you have AAT deficiency. To confirm the diagnosis, you'll need a genotype or phenotype test.

If that testing shows that you have AAT deficiency, your provider may recommend additional testing to evaluate your lung and liver function.

There is no cure for AAT deficiency. But treatments can help prevent your condition from getting worse. Your exact treatment will depend on your symptoms.

You can also take steps to stay as healthy as possible, including:

Not smoking. If you are a smoker, quit smoking. If you don't smoke, don't start. Smoking is the leading risk factor for life-threatening lung disease in people with AAT deficiency.

Not drinking alcohol. Alcohol can damage your liver.

Following a healthy diet.

Getting regular exercise.

Seeing your provider regularly.

Taking medicines as prescribed by your provider.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AAT TESTING?

If you're thinking about having AAT testing or have received abnormal results, it may be helpful to speak with a genetic counselor. A genetic counselor is a specially trained professional in genetics and genetic testing. A counselor can help you understand the risks and benefits of testing for you and your family. After testing, a counselor can help you understand your results. If you have AAT deficiency, the counselor can provide information on the condition, including your risk of passing the disease to your children.

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Alpha-Fetoprotein (AFP) Test

<https://medlineplus.gov/lab-tests/alpha-fetoprotein-afp-test/>

ALPHA-FETOPROTEIN (AFP) TEST

WHAT IS AN ALPHA-FETOPROTEIN (AFP) TEST?

An AFP test is a blood test that measures the level of alpha-fetoprotein (AFP) in your blood when you are pregnant. The test checks your baby's risk of having certain birth defects and genetic problems, but it does not diagnose any health conditions. The test is usually done between 15 and 20 weeks of pregnancy.

AFP is a protein mostly made in your developing baby's liver. Usually, some AFP passes from the baby into your blood. But certain medical conditions can make the baby's body release more or less AFP than usual. When this happens, your AFP blood levels will be higher or lower than normal. Abnormal AFP levels may be a sign that your baby has an increased risk of a health problem, such as:

A neural tube defect, a serious condition that happens when a baby's brain, spine and/or spinal cord do not form properly.

Down syndrome, a genetic disorder that changes the way a baby's body and brain develop. It can cause both mental and physical health challenges. These problems may range from mild to severe.

Edwards syndrome (trisomy 18), an uncommon genetic condition that causes an abnormal head shape and many organ defects. Most babies with this condition die in the first year of life.

For people who aren't pregnant, an AFP blood test may be used to help diagnose certain cancers, especially liver cancer. When the test is used this way, it's called an AFP tumor marker test.

Other names: AFP Maternal; Maternal Serum AFP; msAFP screen

WHAT IS IT USED FOR?

An AFP blood test is used during pregnancy to check your baby's risk of birth defects and genetic disorders, such as neural tube defects or Down syndrome. This information helps you and your health care provider decide if you need to have other tests to learn more about your baby's health.

AFP tests are often done as part of a group of prenatal tests, including:

A multiple marker test, also called a triple screen

A quad screen

WHY DO I NEED AN AFP TEST?

An AFP test is routinely offered, usually between week 15 and 20 of your pregnancy. Your provider may especially recommend the test if your risk for having a baby with a birth defect or a genetic disorder is higher than usual because you:

Have a family history of birth defects

Are 35 years or older

Have diabetes

Have used certain medicines or drugs during pregnancy that could harm your baby

WHAT HAPPENS DURING AN AFP TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparation for an AFP test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to you or your baby with an AFP blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The results of your AFP test tell you whether your baby is more or less likely to have certain birth defects and genetic disorders. The test can't diagnose any conditions.

Normal AFP test results mean that it's unlikely that your baby has a problem. But it's possible for a test result to be normal and miss a problem that does exist.

Abnormal AFP results may be a sign that your baby has a problem:

Lower than normal AFP levels may mean your baby has an increased risk of a genetic disorder such as Down syndrome or Edwards syndrome.

Higher than normal AFP levels may mean your baby has an increased risk of:

A neural tube defect, such as:

Spina bifida, a condition in which the bones of the baby's spine don't close around part of the spinal cord. It often causes damage to the spinal cord and nerves.

Anencephaly, a condition in which most of the baby's brain and skull don't develop. Almost all babies with this condition die shortly after birth.

A defect in the abdomen (belly) or esophagus (the tube that carries food from the baby's mouth to the stomach).

A neural tube defect, such as:

Spina bifida, a condition in which the bones of the baby's spine don't close around part of the spinal cord. It often causes damage to the spinal cord and nerves.

Anencephaly, a condition in which most of the baby's brain and skull don't develop. Almost all babies with this condition die shortly after birth.

Spina bifida, a condition in which the bones of the baby's spine don't close around part of the spinal cord. It often causes damage to the spinal cord and nerves.

Anencephaly, a condition in which most of the baby's brain and skull don't develop. Almost all babies with this condition die shortly after birth.

A defect in the abdomen (belly) or esophagus (the tube that carries food from the baby's mouth to the stomach).

Abnormal AFP results don't always mean that your baby has an increased risk of a serious problem. High or low AFP levels may be caused by:

An incorrect due date. AFP levels rise and fall at set times during pregnancy as the baby develops. That means normal levels of AFP are different at different times of your pregnancy. If your due date is wrong, your AFP blood level may seem abnormal when it's just right for how far along you really are in your pregnancy.

Being pregnant with more than one baby. Each baby makes AFP, so your AFP blood levels will be higher with two or more babies.

A false positive result. This means that your AFP results show an increased risk of a problem when there's really nothing wrong with your baby.

If your AFP test results aren't normal, you will likely have more tests to help find out why. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Alpha Fetoprotein (AFP) Tumor Marker Test

<https://medlineplus.gov/lab-tests/alpha-fetoprotein-afp-tumor-marker-test/>

ALPHA FETOPROTEIN (AFP) TUMOR MARKER TEST

WHAT IS AN AFP (ALPHA-FETOPROTEIN) TUMOR MARKER TEST?

An AFP tumor marker test is a blood test that measures the level of AFP (alpha-fetoprotein) in a sample of your blood. It's usually used to help diagnose certain types of cancer and to check how well treatment is working.

AFP is a protein that the liver makes when its cells are growing and dividing to make new cells. AFP is normally high in unborn babies. After birth, AFP levels drop very low. Healthy children and adults who aren't pregnant have very little AFP in their blood.

If you aren't pregnant, AFP is mainly measured as a tumor marker. Tumor markers are substances that are often made by cancer cells or by normal cells in response to cancer. High levels of AFP can be a sign of cancer of the liver, ovaries or testicles.

An AFP tumor marker test cannot be used by itself to screen for or diagnose cancer. That's because other conditions can increase AFP levels, including liver diseases that aren't cancer. And some people who do have liver, ovarian, or testicular cancer will have normal AFP levels. So, an AFP tumor marker test can't rule out cancer for sure. But when used with other tests and exams, AFP tumor marker testing can help diagnose and monitor cancers that cause high AFP levels.

Other names: total AFP, alpha-fetoprotein-L3 Percent

WHAT IS IT USED FOR?

An AFP tumor marker test may be used during the diagnosis and/or treatment of cancer of the liver, ovaries, or testicles that make high levels of AFP. It is used to:

Help confirm or rule out a cancer diagnosis when used with other exams and tests.

Predict how cancer may behave over time.

Monitor cancer treatment. AFP levels often go up if cancer is growing and go down when treatment is working.

Check whether cancer has returned after treatment.

In certain cases, results from an AFP tumor marker test may be used to guide treatment choices for specific types of cancer. The test may also be used to monitor your health if you have chronic (long lasting) hepatitis or cirrhosis of the liver. These conditions aren't cancer, but they increase your risk of developing liver cancer.

WHY DO I NEED AN AFP TUMOR MARKER TEST?

You may need an AFP tumor marker test if:

A physical exam and/or other tests suggest that you may have cancer of the liver, ovaries, or testicles.

You are currently being treated for a cancer that causes high AFP levels. Measuring your AFP test can show how well your treatment is working.

You have completed treatment for a cancer that increased your AFP level. You may need an AFP tumor marker test from time to time to check whether your cancer is coming back.

If you have chronic hepatitis or cirrhosis, you have a higher risk of developing liver cancer. A very high level of AFP or a sudden increase can be an early sign of liver cancer. Most medical experts don't recommend measuring AFP levels to screen for cancer in these diseases. But, some health care providers may still use an AFP tumor marker test with other tests to watch for liver cancer.

WHAT HAPPENS DURING AN AFP TUMOR MARKER TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an AFP tumor marker test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If you haven't been diagnosed with cancer, test results that show:

High levels of AFP may be a sign of cancer of the liver, ovaries, or testicles. But having a high AFP level doesn't mean you have cancer or that you will get cancer. Liver injury and liver diseases that aren't cancer can also cause high AFP levels. Less often, high levels of AFP may be a sign of other cancers, including lymphoma or lung cancer. Your provider will use your medical history and other test results to make a diagnosis.

Normal levels of AFP mean you're less likely to have a cancer that causes high AFP levels. But a normal test result doesn't rule out cancer because some people with these cancers have normal AFP levels.

If you're being treated for a cancer that increased your AFP levels, you may be tested several times during treatment. Your provider will look at all your AFP test results to see how your levels have changed over time. If your results show:

Your AFP levels are increasing, it may mean that your treatment isn't working.

Your AFP levels are decreasing, it may mean your treatment is working.

Your AFP levels have stayed the same, it may mean your disease is stable and hasn't gotten better or worse.

If you've finished treatment for cancer that caused high AFP levels and your test results are:

Not normal, it may mean that you still have some cancer in your body.

Higher now than they were shortly after treatment, it may mean your cancer is growing again.

If you have a long-lasting liver disease that's not cancer, you may need other tests to check for liver cancer if your test results show a sudden increase in AFP or your level is very high.

Ask your provider to explain what your test results mean for your health.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN AFP TUMOR MARKER TEST?

In the U.S., the most common AFP tumor marker test measures all forms of AFP in your blood. A less common AFP test may be used to check the risk of liver cancer in people who have chronic liver disease. It measures a form of AFP called L3.

An AFP-L3 percent test (AFP-L3%) compares the amount of L3 to the amount of total AFP in your blood. If the portion of L3 increases, it may mean you have a high risk of developing a serious form of the most common liver cancer called hepatocellular carcinoma.

You may have heard of another type of AFP test that's used during pregnancy. This test measures total AFP levels in the blood, but it's not a tumor marker test for cancer. It's used to check for the risk of certain birth defects in an unborn baby.

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ALT Blood Test

<https://medlineplus.gov/lab-tests/alt-blood-test/>

ALT BLOOD TEST

WHAT IS AN ALT BLOOD TEST?

An ALT test measures the amount of ALT in your blood. This test is commonly used to help diagnose liver damage or disease.

ALT (alanine transaminase) is an enzyme, a protein that speeds up certain chemical reactions in your body. It is found mainly in your liver. Usually, you will have low levels of ALT in your blood. But when liver cells are damaged, they release ALT into the bloodstream. High levels of ALT in your blood may be a sign of a liver injury or disease. Some types of liver disease cause high ALT levels even before you have symptoms of the disease. So, an ALT blood test may help diagnose certain liver diseases early, when they may be easier to treat.

An ALT test is usually ordered as part of a group of liver function tests.

Other names: Alanine aminotransferase (ALT), Serum Glutamic-Pyruvic Transaminase (SGPT), GPT, Alanine Transaminase

WHAT IS IT USED FOR?

An ALT blood test is often part of a routine blood screening to check the health of your liver. The test may also help diagnose or monitor liver problems.

WHY DO I NEED AN ALT BLOOD TEST?

Your health care provider may order an ALT blood test as part of a routine checkup. An ALT test is usually done with a group of other liver function tests that check how well your liver is working. Many liver function tests are included in a common blood test called a comprehensive metabolic panel (CMP).

These tests may also be ordered if you have symptoms of liver damage, such as:

Nausea and vomiting

Lack of appetite

Fatigue

Weakness

Jaundice, a condition that causes your skin and eyes to turn yellow

Swelling and/or pain in your abdomen (belly)

Swelling in your ankles and legs

Dark-colored urine (pee) and/or light-colored stool (poop)

Frequent itching

Your provider may also order an ALT blood test if you have a high risk of liver damage because of:

A family history of liver disease

Alcohol use disorder (AUD)

Obesity

Diabetes

Taking certain medicines that can cause liver damage

Hepatitis or exposure to hepatitis

WHAT HAPPENS DURING AN ALT BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is

inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines before the test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

An ALT test is usually ordered with other blood tests. You usually need to fast (not eat or drink) for several hours before these tests. Your provider will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

An ALT blood test is often done as part of a group of liver function tests. Liver function tests measure several different proteins, substances, and enzymes that show how well your liver is working. Your provider usually compares your ALT results with the results of the other liver tests to evaluate your liver health and decide if you need other tests to make a diagnosis.

Lower-than-usual levels of ALT in your blood are not common. Your provider may do more tests since this may be a sign of:

- B6 deficiency

- Chronic kidney disease

Higher-than-usual levels of ALT in your blood may be a sign of liver damage from:

- Hepatitis

- Infection

- Cirrhosis

- Liver cancer

- Other liver diseases

- A lack of blood flow to the liver

- Certain medicines, such as cholesterol or pain medicines

- Poisons

If your results show you have a high level of ALT, it doesn't always mean that you have a medical condition that needs treatment. Many things can affect your results, such as your age, sex, weight, and certain medicines and dietary supplements. Your ALT levels can also be affected by intense exercise and where you are in your menstrual cycle.

To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ALT BLOOD TEST?

ALT test results may help tell the difference between damage from acute (sudden) liver problems and chronic (long-term) liver problems. But the amount of ALT in your blood isn't related to how much your liver may be damaged.

Your ALT is usually measured along with another liver enzyme called AST as part of a liver function panel. Since ALT is mostly found in the liver, an ALT test checks more specifically for liver damage.

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Ammonia Levels

<https://medlineplus.gov/lab-tests/ammonia-levels/>

AMMONIA LEVELS

WHAT IS AN AMMONIA LEVELS TEST?

An ammonia levels test measures the amount of ammonia in a sample of your blood. Ammonia is also called NH_3 . It is a normal waste product in your body. Healthy bacteria in your intestines make ammonia when you digest protein in the foods you eat.

Normally, your liver changes ammonia into another waste product called urea. Your kidneys get rid of urea in urine (pee). This process is called the urea cycle.

When you're healthy, the urea cycle prevents ammonia from building up in your blood. This is very important because ammonia is toxic (poisonous) to your brain. Even small increases in the level of ammonia in your blood can cause permanent brain damage, coma, and even death.

Liver disease is the most common cause of high ammonia levels. Other causes include kidney failure and genetic disorders called urea cycle disorders.

The medical term for high ammonia levels is hyperammonemia.

Other names: NH_3 test, blood ammonia test, serum ammonia, ammonia; plasma

WHAT IS IT USED FOR?

An ammonia levels test may be used to help diagnose high ammonia levels in people who have symptoms. The test alone can't diagnose conditions that increase the amount of ammonia in your blood. But it may be used with other tests to help find the cause.

Ammonia levels testing can help diagnose conditions such as:

Hepatic encephalopathy. This condition happens when toxins (poisons), including ammonia, build up in your brain because your liver is unable to break them down. It can cause memory loss, confusion, loss of consciousness, and coma.

Reye syndrome. This rare disease damages the brain and liver. Without treatment, it causes death. It mostly happens in children younger than 15 who have had a viral infection that causes a fever, such as chickenpox or the flu. Taking aspirin during a viral illness may increase the risk of Reye syndrome.

Urea cycle disorders (UCDs). This is a group of rare genetic conditions that you inherit from your parents. If you have a UCD, you lack enzymes that help change ammonia into urea. An enzyme is a protein that speeds up certain chemical reactions in your body. Symptoms may show up shortly after birth or later in life. Newborns are tested for certain UCDs as part of routine newborn screening. Your child's health care provider can explain more about which UCD screening tests are used in your state.

Ammonia levels testing is also used to monitor conditions that cause high ammonia levels and to check if treatment is working.

WHY DO I NEED AN AMMONIA LEVELS TEST?

You may need this test if you have symptoms that could be from too much ammonia in your blood. The symptoms depend on how high your ammonia levels are and your age. Symptoms in children and adults may include:

Feeling irritable

Lack of energy and mental alertness or being very sleepy

Vomiting

Headache

Loss of muscle coordination that may cause difficulties with walking or speech

Changes in behavior

Confusion, including not being sure about the time and/or where or who you are

Mood swings

Seizures

Other symptoms in children may include:

Not growing taller or gaining weight as expected

Lack of muscle tone (limp muscles)

Developmental delays in children or infants

Newborn babies may develop symptoms of high ammonia levels within three days after birth. Symptoms in newborns include the above symptoms and may also include fast breathing and/or making grunting sounds when breathing. If your baby or child has any symptoms of high ammonia levels, contact their provider right away.

If you have a family history of genetic urea cycle disorders, you or your child may need this test even if you don't have symptoms.

If you've been diagnosed with high ammonia levels, you may need this test to see how well your treatment is working.

WHAT HAPPENS DURING AN AMMONIA LEVELS TEST?

In most cases, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

In certain cases, the blood sample may be taken from an artery, which is a blood vessel that carries blood from your heart to the rest of your body. Some providers think blood from an artery may provide more accurate results. The blood is taken from an artery in your wrist, arm, or groin using a needle attached to a syringe.

Taking blood from an artery tends to be more uncomfortable than blood tests that use a vein. So, numbing medicine may be applied to your skin first. Afterwards, the site will be bandaged, and pressure will be applied for at least five minutes to stop bleeding. You may be told to avoid lifting heavy objects for 24 hours after the test.

To get a blood sample from a newborn, a provider will clean the baby's heel with alcohol. Then the provider will stick the baby's heel with a small needle to collect a few drops of blood. The heel will then be bandaged.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will give you specific instructions about how to prepare for an ammonia levels test. You should not smoke for several hours before the test. You may be asked to avoid strenuous exercise and alcohol for a period of time.

You may also need to fast (not eat or drink) for several hours. Your provider may tell you to stop taking certain medicines, including diuretics (water pills) and opioid pain medicines. But never stop taking any prescription medicines without talking with your provider first.

Babies do not need any special preparations before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

If your blood sample was taken from an artery, you may also have some bleeding where the needle was put in. Very rarely, the needle may damage a nerve or the artery.

WHAT DO THE RESULTS MEAN?

Your provider will use your test results, symptoms, and medical history to understand what your ammonia levels say about your health. Normal levels vary with age, but in general, low levels are normal and healthy.

If your test results are normal, your provider will order other tests to find out what's causing your symptoms.

If your test results show ammonia levels that are higher than normal, it usually means that your body is having trouble

getting rid of ammonia. Many types of health conditions can cause high ammonia levels, including:

Liver diseases, such as cirrhosis or hepatitis

Hepatic encephalopathy

Kidney disease or kidney failure

Certain genetic disorders, including urea cycle disorders

Reye syndrome (most common in children and teens)

Taking certain medicines

If the results of an ammonia levels test are high, more tests will usually be done to diagnose the cause. Treatment will depend on the condition that's causing the problem. If you have questions about your test results or treatment, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Amniocentesis (amniotic fluid test)

<https://medlineplus.gov/lab-tests/amniocentesis-amniotic-fluid-test/>

AMNIOCENTESIS (AMNIOTIC FLUID TEST)

WHAT IS AMNIOCENTESIS?

Amniocentesis is a test done during pregnancy. A sample of amniotic fluid is taken from your abdomen (belly). It is used to diagnose certain genetic disorders, birth defects, and other conditions in your unborn baby. The test is usually done between weeks 15 and 20 of pregnancy.

Your baby grows in an amniotic sac inside your uterus. The sac is filled with amniotic fluid, a clear or pale yellow liquid. The fluid that surrounds and protects your baby throughout pregnancy. It also helps with your baby's development and helps keep your baby at the right temperature. The fluid contains cells that can provide important information about your baby's health.

Amniocentesis is a diagnostic test. That means it will tell you whether your baby has a specific health problem, and the results are almost always accurate. Diagnostic tests are different from prenatal screening tests for genetic disorders. Prenatal screening tests have no risks for you or your baby, but they don't diagnose problems. They can only show if your baby might have a health problem. If a screening test shows your baby could have a problem, your health care provider may recommend amniocentesis or another diagnostic test to find out for sure.

Other names: amniotic fluid analysis

WHAT IS IT USED FOR?

Amniocentesis is used to diagnose certain health problems in an unborn baby. It is commonly used to find:

Genetic and chromosomal disorders, including:

Down syndrome, a disorder that causes delays in physical and mental development and other health problems.

Cystic fibrosis, a disease of the mucus and sweat glands that causes thick sticky mucus, which can lead to problems with breathing and digestion.

Sickle cell disease, a group of red blood cell disorders that can cause anemia and other health problems.

Tay-Sachs disease, a disease that destroys nerve cells, causes mental and physical problems, and often death in early childhood (uncommon).

Down syndrome, a disorder that causes delays in physical and mental development and other health problems.

Cystic fibrosis, a disease of the mucus and sweat glands that causes thick sticky mucus, which can lead to problems with breathing and digestion.

Sickle cell disease, a group of red blood cell disorders that can cause anemia and other health problems.

Tay-Sachs disease, a disease that destroys nerve cells, causes mental and physical problems, and often death in early childhood (uncommon).

Neural tube defects, severe birth defects of the baby's brain and/or spine, such as spina bifida and anencephaly.

The test may also be used to:

Check your baby's lung development if you have a risk of giving birth too soon (premature birth). In this case, amniocentesis is done later in your pregnancy.

Diagnose an infection or certain other illnesses in the baby.

WHY DO I NEED AMNIOCENTESIS?

Having amniocentesis is your choice. You may want this test if you have a higher risk of having a baby with a health problem. You may have a higher risk if:

You are age 35 or older; the risk of having a baby with a genetic disorder increases with age

You had a prenatal screening test that showed your baby might have a problem
You or your partner have a family history of a genetic disorder or neural tube defect
You or your partner had genetic testing that showed you carry a genetic disorder
You or your partner have a child with a genetic disorder or birth defect
Amniocentesis isn't right for everyone. Before you decide to get tested, think about how you might feel and what you might do after learning the results.

WHAT HAPPENS DURING AMNIOCENTESIS?

The test is usually done between 15 and 20 weeks of pregnancy. It is sometimes done later in pregnancy to check the baby's lung development or diagnose certain infections or illnesses, such as anemia in the unborn baby caused by Rh incompatibility.

During the procedure:

You'll lie on your back on an exam table.

Your provider will apply a gel to your belly.

Your provider will move an ultrasound wand-like device, called a transducer, on your belly. Ultrasound uses sound waves to show the position of your baby and placenta so your provider can see where to take a sample of amniotic fluid.

Your provider will clean your belly.

Your provider will insert a thin needle into your belly, withdraw a small amount of amniotic fluid, and then remove the needle.

When the sample is removed, your provider will check your baby's heartbeat with the ultrasound.

The sample is sent to a lab for testing. Results may take from a few days to a few weeks.

The procedure usually takes about 15 minutes. Afterward, you may be told not to exercise or have sex for a day or two.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you are early in your pregnancy, you may be asked not to urinate (pee) before the test to make sure that you have a full bladder. This helps move the uterus into a better position for the test. In later pregnancy, you need to have an empty bladder for the test.

ARE THERE ANY RISKS TO THE TEST?

You may feel mild stinging, pressure, and/or cramping during the test. Afterward, you may have mild discomfort or vaginal bleeding, or a small amount of amniotic fluid may leak from your vagina.

The risk of serious complications is extremely low. Your provider will let you know if there are any symptoms that you should tell them about. These may include ongoing vaginal bleeding, amniotic fluid leakage, severe cramping, or a fever.

Amniocentesis does have a very slight risk (less than 1 percent) of causing a miscarriage.

WHAT DO THE RESULTS MEAN?

Amniotic fluid may be tested for many different disorders. Your test results will depend on which tests your provider ordered.

Normal results are reported as "normal" or "negative." This means that it's very unlikely that your baby has the disorder that was tested, but it does not guarantee your baby will not have any health problems.

Results that are not normal are reported as "abnormal" or "positive." This means that your baby very likely has the disorder that was tested.

Your provider will explain your test results. Amniocentesis is very accurate, but in certain cases, your provider may order more tests to learn about your baby's health.

It may help to speak to a genetic counselor before testing and/or after you get your results. A genetic counselor is a specially trained professional in genetics and genetic testing who can help you understand what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AMNIOCENTESIS?

In some cases, you may want information about your unborn baby sooner than 15 to 20 weeks, when amniocentesis is usually done. In that case, you might consider another diagnostic test called a chorionic villus sampling (CVS). This test takes a small sample of tissue from the placenta.

You can have a CVS test between 10 and 13 weeks of pregnancy, which gives you more time to decide what to do if the results are abnormal. Like amniocentesis, CVS is an accurate diagnostic test that can find certain genetic disorders. But it has a slightly higher risk of miscarriage.

As you make decisions about testing, remember that prenatal genetic diagnostic tests are different from screening tests. Screening tests estimate the chances that your unborn baby may have certain problems. It's possible to have an abnormal screening test when your baby is actually healthy. But if a screening test shows that your baby has a risk for certain problems, you may have amniocentesis or CVS to find out for sure.

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Down Syndrome Tests

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Prenatal Cell-Free DNA Screening

Red Blood Cell Antibody Screen

Sweat Test for Cystic Fibrosis

Ultrasound

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Amylase Test

<https://medlineplus.gov/lab-tests/amylase-test/>

AMYLASE TEST

WHAT IS AN AMYLASE TEST?

An amylase test measures the amount of amylase in your blood or urine (pee). Amylase is an enzyme, a protein that speeds up chemical reactions in your body. Amylase helps you digest carbohydrates. Most of the amylase in your body is made by your pancreas and salivary (spit) glands.

It's normal to have a small amount of amylase in your blood and urine. But having too much amylase may be a sign of a disorder of the pancreas or salivary glands or another medical condition.

Other names: Amy test, serum amylase, urine amylase

WHAT IS IT USED FOR?

Tests for amylase in blood or urine are mainly used to diagnose problems with your pancreas, including pancreatitis, which is inflammation of the pancreas. The test is also used to monitor chronic (long-term) pancreatitis.

Changes in amylase levels show up in blood before urine, so an amylase urine test may be done with or after an amylase blood test.

One or both types of amylase test may also be used to help diagnose or monitor treatment for other disorders that may affect amylase levels, such as salivary gland disorders and certain digestive conditions.

WHY DO I NEED AN AMYLASE TEST?

Your health care provider may order an amylase blood and/or urine test if you have symptoms of a pancreatic disorder. Symptoms may begin suddenly or slowly. They may include:

Pain in your upper abdomen (belly) that may spread to your back, get worse after eating, or last for several days

Loss of appetite

Nausea and vomiting

Fever

Fast heartbeat

Jaundice (a condition that causes your skin and eyes to turn yellow)

Greasy, foul-smelling stool (poop)

Your provider may also order an amylase test to monitor an existing condition that affects your pancreas, including:

Chronic pancreatitis

Pancreatic cancer

Eating disorders

Cystic fibrosis

Alcohol use disorder (AUD)

Recovery after having gallstones removed because of a gallbladder attack

WHAT HAPPENS DURING AN AMYLASE TEST?

For an amylase blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For an amylase urine test, you will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample.

It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

Your provider may request that you collect all your urine during a 24-hour period. That's because the amount of amylase in urine can vary throughout the day. Collecting urine from an entire day can provide more accurate information about your amylase levels. For a 24-hour urine test, you will be given a container and specific instructions on how to collect your sample at home. Be sure to follow all instructions carefully.

In certain cases, your provider may order a test for amylase in peritoneal fluid. This fluid is found between the layers of tissue that line the inside of your abdomen (belly) and cover most of your organs. To get a sample, a health care professional will numb your skin and use a needle and tubing to draw some fluid into a container.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You should not drink alcohol for 24 hours before having an amylase blood or urine test. If you are having a blood test, your provider may also tell you to fast (not eat or drink) for two hours before the test. Your provider will let you know if there are any special instructions to follow.

Be sure to tell your provider about any prescription or over-the-counter medicines you are taking because they may affect your test results. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. During a blood test, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no known risk to having a urine test.

If you have a peritoneal fluid test, you may feel a little dizzy or lightheaded after the procedure. There is a small risk that the needle could damage your bowel or bladder, which may lead to bleeding or infection.

WHAT DO THE RESULTS MEAN?

High levels of amylase in blood or urine may be a sign of:

Acute (sudden) pancreatitis

A blockage in a duct (small tube) in the pancreas

Cancer or a benign (not cancerous) tumor of the pancreas

High levels of amylase in peritoneal fluid may be a sign of:

Acute (sudden) pancreatitis

A blocked intestine

Lack of blood flow to the intestines

Low levels of amylase in blood or urine are uncommon, but may be a sign of:

Chronic pancreatitis causing permanent damage to the cells that make amylase
Kidney disease
Cystic fibrosis
Liver disease
Pre-eclampsia (a type of high blood pressure in pregnancy)
To learn more about your results, talk with your provider.
Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN AMYLASE TEST?

If your provider suspects you have pancreatitis, they may order a lipase blood test, along with an amylase blood test. Lipase is another enzyme produced by the pancreas. Lipase tests are considered to be more accurate for diagnosing pancreatitis, especially when pancreatitis is caused by alcohol use disorder.

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ANA (Antinuclear Antibody) Test

<https://medlineplus.gov/lab-tests/ana-antinuclear-antibody-test/>

ANA (ANTINUCLEAR ANTIBODY) TEST

WHAT IS AN ANA (ANTINUCLEAR ANTIBODY) TEST?

An ANA test is a blood test that looks for antinuclear antibodies in your blood. Antibodies are proteins that your immune system makes to fight foreign substances, like viruses and bacteria. But an antinuclear antibody attacks your healthy cells instead. It's called "antinuclear" because it targets the nucleus of the cells, which is in charge of sending signals that are needed for important cell functions.

It's normal to have a few antinuclear antibodies in your blood. But a large number may be a sign of an autoimmune disorder. If you have an autoimmune disorder, your immune system attacks the healthy cells of your organs and tissues by mistake. These disorders can cause serious health problems.

Other names: antinuclear antibody panel, fluorescent antinuclear antibody, FANA, ANA, ANA reflexive panel

WHAT IS IT USED FOR?

An ANA test is used to help diagnose autoimmune disorders, such as:

Systemic lupus erythematosus (SLE), the most common type of lupus. Lupus is a chronic (long-lasting) disease that affects many parts of the body, including the joints, skin, heart, lungs, blood vessels, kidneys, and brain.

Rheumatoid arthritis, a condition that mostly affects joints, causing pain and swelling often in the wrists, hands, and feet.

Scleroderma, a rare disease that may affect the skin, blood vessels, and organs.

Sjögren's syndrome, a rare disease that affects the glands that make tears and saliva (spit) and other parts of the body.

Addison Disease, which affects your adrenal glands (glands on top of the kidneys), causing fatigue and weakness.

Autoimmune hepatitis, which causes swelling in your liver.

Thyroid diseases, which can cause your thyroid to make too much or too little thyroid hormones.

WHY DO I NEED AN ANA TEST?

Your health care provider may order an ANA test if you have symptoms of an autoimmune disorder. The symptoms depend on what part of your body is affected. They may include:

Fever

Rash, blisters, or skin color changes

Fatigue

Joint pain, stiffness, and swelling

Muscle pain

Swollen glands

Abdominal (belly) pain

WHAT HAPPENS DURING AN ANA TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Some medicines may affect the results of your ANA test. You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Results from an ANA test alone cannot diagnose a specific disease. Your provider will use your ANA test results along with other blood and imaging tests and information about your health to make a diagnosis.

A negative result on an ANA test means that antinuclear antibodies were not found in your blood, and you're less likely to have an autoimmune disorder. But a negative ANA test doesn't completely rule out the possibility that you could have an autoimmune disorder.

A positive result on an ANA test means that antinuclear antibodies were found in your blood. A positive result may be a sign of:

- Systemic lupus erythematosus (SLE)

- A different type of autoimmune disease

- A viral infection (antinuclear antibodies from a virus are usually only in your blood for a short time)

- Another health condition that can cause antinuclear antibodies, such as cancer

If your ANA test results are positive, your provider will likely order more tests to make a diagnosis.

Having antinuclear antibodies in your blood doesn't always mean you have a disease. Some healthy people have antinuclear antibodies in their blood, and levels tend to increase with age. Many healthy adults may have a positive ANA test result, especially women over the age of 65. Also, certain medicines can cause antinuclear antibodies.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Anion Gap Blood Test

<https://medlineplus.gov/lab-tests/anion-gap-blood-test/>

ANION GAP BLOOD TEST

WHAT IS AN ANION GAP BLOOD TEST?

An anion gap blood test gives a measurement of the acid-base balance (pH balance) in your blood. This lets you know whether your blood is too acidic or not acidic enough. The test also checks for an electrolyte imbalance in your blood. To do the measurement, an anion gap test uses the results of an electrolyte panel¹ or other blood tests that measure your electrolytes.²

Electrolytes are minerals that have an electrical charge when they are dissolved in water or body fluids. You get them from the foods you eat and the fluids you drink. You have electrolytes in your blood, urine, tissues, and other body fluids. They help control the acid-base balance of your blood. A few of your body's electrolytes that are often measured in your blood include sodium, chloride, potassium, and bicarbonate.³

Some electrolytes have a positive electric charge. Others have a negative electric charge. The anion gap measures the difference⁴ or gap⁵ between the negatively charged and positively charged electrolytes in your blood. If the anion gap is too high, your blood is more acidic than normal. If the anion gap is too low, your blood isn't acidic enough. Both high and low results may be signs of a serious disorder that needs attention.

Other names: Serum anion gap

WHAT IS IT USED FOR?

The anion gap blood test shows whether your electrolytes are out of balance or if your blood is too acidic or not acidic enough. Too much acid in the blood is called acidosis. Too little acid in your blood is called alkalosis. Both conditions can be serious.⁶

WHY DO I NEED AN ANION GAP BLOOD TEST?

Your health care provider may order an anion gap blood test if you have symptoms of acidosis or alkalosis. Symptoms may vary depending on the underlying disease or condition.

Symptoms of acidosis may include:

Shortness of breath

Vomiting

Confusion

Fatigue

Tachycardia (very fast heartbeat)

Symptoms of alkalosis may include:

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Irritability

Hand tremor

Muscle twitching

Nausea, vomiting

Numbness or tingling in the face, hands, or feet

WHAT HAPPENS DURING AN ANION GAP BLOOD TEST?

The anion gap test is a calculation that's done with the results of an electrolyte panel, which is a blood test.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle.

After the needle is inserted, a small amount of blood will be collected into a test tube. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

No special preparation is necessary for an anion gap blood test. If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having this test. You may feel slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Normal results may vary by lab. Your anion gap results may be based on the results of an electrolyte panel, a basic metabolic panel (BMP), a comprehensive metabolic panel (CMP), or a renal function panel (RFP).

Several types of medical conditions can cause abnormal anion gap test results. Your provider will use the results of this test, your medical history, and other tests to make a diagnosis.

A high anion gap test result may mean that you may have acidosis, which may be caused by:

Dehydration

Diarrhea

Too much exercise

Kidney diseases

Diabetes or diabetic ketoacidosis

Salicylate poisoning, for example from too much aspirin

Antifreeze poisoning

A low anion gap test result may mean you have alkalosis. But this result is so rare that your provider will usually have you tested again to make sure the results are accurate.

The most common cause of a low anion gap is a low level of albumin, a protein in the blood. Low albumin may be a sign of not eating enough protein, kidney problems, liver disease, heart disease, or some types of cancer.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ANION GAP BLOOD TEST?

The anion gap blood test provides information about the acid-base balance of your blood. This tells you about the health of many of your body functions. But there is a wide range of normal results and many possible causes of abnormal results. So, talk with your provider about what your test results mean and whether you need additional testing to make a diagnosis.

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Anoscopy

<https://medlineplus.gov/lab-tests/anoscopy/>

ANOSCOPY

WHAT IS AN ANOSCOPY?

An anoscopy is a procedure that uses a special lighted scope called an anoscope or anal speculum. An anoscope allows your health care provider to view the lining of your lower rectum and anus to check for signs of injury or disease. Your lower rectum is the last few inches of your large intestine. This is where your body stores stool (poop) before it passes out of your body. Your anus is the opening of the rectum to the outside of your body. This is where your digestive tract ends and your stool leaves your body.

A related procedure is called a high-resolution anoscopy. This procedure uses a special magnifying device called a colposcope along with the anoscope to view these areas. A colposcope may detect small changes in your tissues that the anoscope alone may miss.

An anoscopy can help your provider find problems in the anus and rectum, including hemorrhoids, fissures (tears), and abnormal growths.

WHAT IS IT USED FOR?

An anoscopy is most often used to diagnose:

Hemorrhoids, a condition that causes swollen, inflamed veins around the anus and lower rectum. They can be inside the anus or on the skin around the anus. Hemorrhoids are usually not serious, but they can cause bleeding and discomfort.

Anal fissures, small tears in the lining of the anus.

Anal polyps, abnormal growths on the lining of the anus.

Inflammation. The test can help find the cause of unusual redness, swelling, and/or irritation around the anus.

Cancer. High-resolution anoscopy is often used to look for cancer of the anus or rectum. The procedure can make it easier for your provider to find abnormal cells.

WHY DO I NEED AN ANOSCOPY?

You may need this test if you have symptoms of a problem in your anus or rectum. These may include:

Blood in your stool or on toilet paper after a bowel movement

Itching around your anus

Swelling or hard lumps around your anus

Discharge from your anus

Painful bowel movements

Difficulty having or holding a bowel movement

People who have HIV may also need this test as part of an anal cancer screening. This screening is done to help detect and treat precancer (which can turn into cancer) and prevent anal cancer:

Your provider may recommend an anoscopy as part of the screening if you are:

Under age 35 and have symptoms or signs of anal cancer

Under age 35 and have symptoms or signs of anal cancer

Your provider may recommend lab screening, along with a high resolution anoscopy if you have symptoms or signs of anal cancer, as part of the screening if you are:

A man who has sex with men and are age 35 or older

All other people with HIV age 45 or older
A man who has sex with men and are age 35 or older
All other people with HIV age 45 or older

WHAT HAPPENS DURING AN ANOSCOPY?

An anoscopy may be done in a provider's office or outpatient clinic. The anoscopy procedure only takes a few minutes. If a high-resolution anoscopy is done or tissue samples are taken, the procedure might take longer.

During an anoscopy:

You will put on a gown and remove your underwear.

You will lie on an exam table. Your provider will ask you to either lie on your side with your knees bent, on your stomach or to bend forward over the table with your rear end raised in the air.

Your provider will gently insert a gloved, lubricated finger into your anus to check for hemorrhoids, fissures, or other problems. This is known as a digital rectal exam. Sometimes your provider may apply a numbing medicine if you have anal pain.

Your provider will then insert a lubricated tube called an anoscope about two inches into your anus.

Most anosscopes have a light on the end to give your provider a better view of the anus and lower rectum area.

If your provider finds cells that don't look normal, they may use a swab or other tool to collect a tissue sample for testing (biopsy). High-resolution anoscopy may be better than regular anoscopy at finding abnormal cells.

During a high resolution anoscopy:

Your provider will insert a swab coated with a liquid called acetic acid through the anoscope and into the anus.

The anoscope will be removed, but the swab will remain.

The acetic acid on the swab will cause abnormal cells to turn white.

After a few minutes, your provider will remove the swab and reinsert the anoscope, along with a magnifying instrument called a colposcope.

Using the colposcope, your provider will look for any cells that have turned white.

If abnormal cells are found, your provider will take a biopsy.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may want to empty your bladder and/or have a bowel movement before the test. This may make the procedure more comfortable.

Your provider might suggest that you avoid putting anything (like medicines and other products) into your anus for 24 hours before the test. This would also include avoiding anal sex. They will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having an anoscopy or a high-resolution anoscopy. You may have some discomfort during the procedure. You may also feel a little pinch if your provider took a biopsy.

In addition, you may have a little bleeding when the anoscope is pulled out, especially if you have hemorrhoids.

After the anoscopy:

You may have some soreness or bleeding for a few days after the procedure. Your provider may encourage you to sit in a warm bath or to use ice or a topical medicine for soreness.

If you had a biopsy, your provider may tell you when to expect your results and if there are any special instructions. These may include:

Avoiding strenuous activities or lifting

Taking over-the-counter pain relievers

Avoiding constipation by drinking water, eating a high-fiber diet, and taking a supplement, if necessary

Not putting anything into your anus until bleeding and pain has stopped

WHAT DO THE RESULTS MEAN?

Your results may show a problem with your anus or rectum. These may include:

Hemorrhoids.

Anal fissure.

Anal polyp.

Infection.

Cancer. The biopsy results may confirm or rule out cancer.

Depending on the results, your provider may recommend more tests and/or treatment options.

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Anti-Müllerian Hormone Test

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ANTI-MÜLLERIAN HORMONE TEST

WHAT IS AN ANTI-MÜLLERIAN HORMONE (AMH) TEST?

An anti-müllerian hormone (AMH) test measures the amount of AMH in a blood sample. In males, AMH is made by the testicles (or testes), which are glands that make sperm and male hormones. In females, the ovaries make AMH. The ovaries are glands where eggs form and female hormones are made.

AMH plays different roles in males and females and normal levels of AMH vary with your sex and your age. Measuring AMH levels can provide information about a variety of reproductive health conditions.

In unborn babies, AMH helps form the male and female reproductive organs. The sex of unborn babies is set by the chromosomes they inherit from their parents. Male babies have XY chromosomes and female babies have XX chromosomes. But the development of their reproductive organs and genitals is affected by hormones, including AMH.

In the early weeks of pregnancy, both male and female babies have a set of ducts (tubes) called Müllerian ducts. Normally, male babies make high levels of AMH in their testicular tissue. The AMH makes the Müllerian ducts shrink and helps male organs to grow. AMH levels stay high in male children until puberty when they begin to decrease.

Unborn female babies have very low levels of AMH. This allows the Müllerian ducts to develop into the uterus, fallopian tubes, and the upper part of the vagina. AMH stays low in female children. At puberty, follicles inside the ovaries begin to make more AMH. Follicles are small sacs in the ovaries that hold immature eggs.

In healthy females of childbearing age, higher levels of AMH mean that the ovaries have a larger supply of eggs. As females age, the number of eggs decreases, which causes AMH levels to decrease. At menopause, no eggs are left, and AMH levels drop to zero.

Other names: AMH hormone test, müllerian-inhibiting hormone, MIH, müllerian inhibiting factor, MIF, müllerian-inhibiting substance, MIS

WHAT IS IT USED FOR?

AMH tests are mainly used with other tests to make decisions about treating female infertility (not being able to get pregnant). If you're having infertility treatment, AMH testing can:

Check how many eggs you have left in your ovaries. This is called your "ovarian reserve." It's normal for your ovarian reserve to decrease with age. An AMH test can tell you the size of your ovarian reserve, but it can't tell you about the health of your eggs or predict whether you'll be able to get pregnant.

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Predict how well you may respond to fertility medicine. Normally, your ovaries prepare one egg for fertilization each month. If you're using in vitro fertilization (IVF) to have a baby, your health care provider will prescribe fertility medicine to make your ovaries prepare many eggs at the same time. The eggs are removed and mixed with sperm to make embryos outside of your body. Then the embryos are either frozen or put into the uterus to start a pregnancy.

Testing your AMH level helps your provider know what dose of fertility medicine you may need to get the best response.

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Testing your AMH level helps your provider know what dose of fertility medicine you may need to get the best response.

In females, AMH testing may also be used to:

Find out if you're getting close to menopause or have already begun menopause. As you approach menopause your egg supply shrinks and AMH levels drop. AMH levels can be used to check for premature menopause (before age 40) and early menopause (before age 45). But an AMH test can't predict when you'll actually reach menopause. The

average age of menopause is 52.

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Help diagnose and monitor problems with the ovaries that cause high AMH levels.

These include:

Polycystic ovary syndrome (PCOS), a hormonal disorder that can cause infertility

Certain types of ovarian cancer

Polycystic ovary syndrome (PCOS), a hormonal disorder that can cause infertility

Certain types of ovarian cancer

For babies and children, an AMH test may be used:

To check for healthy testicles in a male baby or child that may have undescended testicles. This is a condition where the testicles fail to move from the belly, where they develop before birth, into their proper place in the scrotum. Healthy testicles in a male baby produce AMH. So, normal AMH levels mean that the baby has healthy, working testicles, but they just haven't dropped into the scrotum. Little to no AMH is a sign of other conditions that need more testing.

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To learn more about a baby born with genitals that aren't clearly male or female. This condition is called "atypical genitalia." In the past, it has been called "ambiguous genitalia" or "intersex." There are many types of atypical genitalia that have different causes. For example, problems with AMH and other hormones in a male baby can lead to the development of internal female reproductive organs and external genitals that don't look typical.

An AMH test can show whether the baby has any working testicular tissue. This information can help diagnose the cause of the problem. The test is usually done with other tests, including chromosome testing, other hormone tests, and ultrasound scans to check for sex organs and glands inside the body.

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WHY DO I NEED AN AMH TEST?

If you're female, you may need an AMH test if you:

Are having fertility problems. You may need an AMH test to:

Find out if your egg supply is normal for your age.

Plan in vitro fertilization (IVF) treatment. Higher levels of AMH mean that you're likely to respond to fertility medicine and you may only need a small dose. Low levels of AMH may mean need higher doses to respond.

Find out if your egg supply is normal for your age.

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Have symptoms of polycystic ovary syndrome (PCOS), including:

Irregular menstrual periods, or no periods at all (amenorrhea)

Acne

Too much hair on the face, chest, stomach, or thighs

Hair loss on the head (male pattern baldness)
Weight gain
Dark patches of skin
Irregular menstrual periods, or no periods at all (amenorrhea)
Acne
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Are being treated for certain types of ovarian cancer. AMH testing can show if your treatment is working. After treatment the test can show whether cancer has returned.

A male baby or child who doesn't have testicles in the scrotum may need an AMH test to help find out if there are healthy testicles inside the body.

A baby with genitals that aren't clearly male or female may need an AMH test along with other tests to help diagnose the cause of the disorder and confirm the sex.

WHAT HAPPENS DURING AN AMH TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an AMH test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

What the AMH test results mean depends on the reason the test was done:

If you're healthy and trying to get pregnant, ask your provider to explain what your AMH test results mean for your fertility based on your age and other test results. In general, a high level of AMH means you have more eggs available, and a low level means your egg supply is shrinking and your time to get pregnant may be shorter.

If you were tested for polycystic ovary syndrome (PCOS), a high level of AMH is a sign that you may have the disease. An AMH test alone cannot diagnose PCOS. There is no cure for PCOS, but symptoms can be managed with medications and/or lifestyle changes.

If you were tested for menopause, low AMH may mean you're getting close to menopause. But AMH test results can't predict how long you have until menopause. If your test result showed no AMH in your blood, it means you are in menopause. If you're younger than age 40 and have symptoms of menopause, an AMH level that's lower than average for your age may be a sign of primary ovarian insufficiency.

If you are being treated for ovarian cancer, a decrease in AMH usually mean that your treatment is working. If AMH increases, it may mean your treatment isn't working or cancer has returned.

If a male baby or child is tested for undescended testicles:

Normal AMH levels mean the baby has working testicles, but they are not in the right location. This condition can be treated with surgery and/or hormone therapy.

Little or no AMH may mean the testicles are not working or missing completely. This may be caused by a change

(mutation) in the AMH gene. The baby may have atypical internal female reproductive organs with normal male genitals. If a baby is tested because of genitals that aren't clearly male or female, the meaning of AMH test results depends on the results of other tests.

If you have questions about test results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Antibiotic Sensitivity Test

<https://medlineplus.gov/lab-tests/antibiotic-sensitivity-test/>

ANTIBIOTIC SENSITIVITY TEST

WHAT IS AN ANTIBIOTIC SENSITIVITY TEST?

Antibiotics are medicines used to fight bacterial and certain fungal infections. They are not effective for viral infections. There are different antibiotics, and each one only works well against certain types of bacteria or fungi. An antibiotic sensitivity test can help find out which antibiotic will be most effective in treating your infection.

The test can also help find a treatment for antibiotic-resistant infections. Antibiotic resistance happens when standard antibiotics become less effective or ineffective against certain bacteria. That means that the bacteria change and resist the effect of the antibiotic. The antibiotic no longer works well against the bacteria causing your infection. This is not the same as antibiotic misuse (not taking all of your medicine or taking someone else's medicine), which can contribute to antibiotic resistance. With antibiotic resistance, some diseases that were once easily treated can become serious or even life-threatening.

Other names: antibiotic susceptibility test, sensitivity testing, antimicrobial susceptibility test, drug resistance testing, culture and sensitivity, C & S

WHAT IS IT USED FOR?

An antibiotic sensitivity test is used to help find the best treatment for a bacterial infection and certain fungal infections.

WHY DO I NEED AN ANTIBIOTIC SENSITIVITY TEST?

You may need this test if you have a bacterial or fungal infection that has been shown to have antibiotic resistance. These could include tuberculosis, MRSA, and C. diff.

WHAT HAPPENS DURING AN ANTIBIOTIC SENSITIVITY TEST?

To find out what type of bacteria or fungi you have, your provider may order a fungal or bacteria culture. The test is done by taking a sample from the infected site. The most common types of tests include:

Blood culture

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial.

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Urine culture

You will provide a sterile sample of urine in a cup, as instructed by your health care provider.

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Wound culture

Your provider will use a special swab to collect a sample from the site of your wound.

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Sputum culture

You may be asked to cough up sputum (a mucus, also called phlegm, made in the lungs) into a special cup, or a special

swab may be used to take a sample from your nose.

You may be asked to cough up sputum (a mucus, also called phlegm, made in the lungs) into a special cup, or a special swab may be used to take a sample from your nose.

Throat culture

Your provider will insert a special swab into your mouth to take a sample from the back of the throat and tonsils.

Your provider will insert a special swab into your mouth to take a sample from the back of the throat and tonsils.

Your sample will be tested against different antibiotics to see if they are effective in killing the germs causing your infection.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

There are no special preparations needed for an antibiotic sensitivity test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood culture test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no risk to having a throat culture, but it may cause slight discomfort or gagging.

There is no risk to having a urine, sputum, or wound culture.

WHAT DO THE RESULTS MEAN?

Your results will show how effective different antibiotics were in killing the bacteria or fungus causing your infection. Results are usually described in one of the following ways:

Susceptible. The tested antibiotic stopped the growth or killed the bacteria or fungus causing your infection, so it may be a good choice for treatment.

Intermediate. The antibiotic may work at a higher dose, or you may need to take the dose more often.

Resistant. The antibiotic did not stop the growth or kill the bacteria or fungus causing the infection. It would not be a good choice for treatment.

If your results are resistant to all of the medicines, you may need to take a few medicines that may work better together.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ANTIBIOTIC SENSITIVITY TEST?

Incorrect use of antibiotics has played a big role in the rise in antibiotic resistance. Make sure you use antibiotics the right way:

Take them exactly as your provider tells you. Make sure to finish all of the antibiotics.

Do not use antibiotics for viral infections. They don't work on viruses like colds and flu.

Don't save them for later. Talk to your pharmacist about safely discarding leftover medicines.

Don't share your antibiotics with others.

Don't take antibiotics prescribed for someone else.

Talk with your provider or pharmacist if you have any questions about your antibiotics.

Learn more about laboratory tests, reference ranges, and understanding results.

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Antibody Serology Tests

<https://medlineplus.gov/lab-tests/antibody-serology-tests/>

ANTIBODY SEROLOGY TESTS

WHAT ARE ANTIBODY SEROLOGY TESTS?

Antibody serology tests look for specific antibodies in your blood. Antibodies are proteins your immune system makes to fight disease-causing germs (also called pathogens), such as viruses and bacteria.

Your immune system creates antibodies after you've been:

Infected with or exposed to a pathogen.

Vaccinated against a pathogen.

In both cases, your body stores the antibodies so that they can fight that pathogen if you're ever exposed to it again. A vaccine, however, teaches your body how to make antibodies before you're ever exposed to a pathogen. This way, you don't have to get an infection before developing immunity.

It may take a few weeks for your body to make antibodies after you have had an infection, been exposed to a pathogen, or had a vaccine.

Other names: antibody titer test, antibody test, antibody serum test

WHAT ARE THEY USED FOR?

Antibody serology tests are generally used to look for antibodies. However, the type of test used can vary depending on:
The type of antibody or pathogen the test is being used to detect.

What condition your provider is attempting to diagnose.

Whether or not you're receiving medical treatment.

The most common use for an antibody serology test is to look for antibodies to specific diseases. These are called endogenous antibodies, and your body makes them after being exposed to a pathogen or after being vaccinated against it. When testing for endogenous antibodies, your provider can:

See if you've had a specific infection.

Check for antibodies that may be a sign of an autoimmune disease (a type of disease that causes your immune system to attack your own cells, tissues, and/or organs by mistake).

Determine if you need another vaccination (booster shot) for a specific disease.

Antibody serology tests are not used to diagnose diseases such as:

Measles and mumps.

Chickenpox and shingles (which are both caused by the varicella-zoster virus).

Hepatitis.

Mononucleosis.

They look only at your body's response to a pathogen by checking if you have antibodies to a disease. Because of this, these tests can't show if the antibodies are from a current or past infection or from a vaccination.

A provider may use a different type of antibody serology test when monitoring your response to certain treatments, including those for:

Cancer.

Organ transplant rejection.

Inflammatory and autoimmune disorders.

Osteoporosis.

Severe viral infections.

Nervous system disorders.

These tests check your blood for monoclonal antibodies. Laboratories make monoclonal antibodies to only interact with specific cells, such as those that are cancerous. Because their targets are so specific, these artificial antibodies can make treatment much more effective. They can, for example, bind to a cancerous cell and mark it for attack by your immune system. By checking your monoclonal antibody levels, your provider can see how well your body is handling a

certain treatment.

WHY DO I NEED AN ANTIBODY SEROLOGY TEST?

You may need this test:

To find out if you've had a recent or past infection.

To check your vaccination status. If your medical records are incomplete, you may need this test to find out if you've been vaccinated.

To find out if a vaccine is effective. If you've already been vaccinated against a disease, the test can show if your vaccine is providing enough protection.

As a requirement for your school or job. Some organizations may require proof of past infection or vaccination.

To find out if you have an autoimmune disease, such as lupus.

If you're receiving any treatment that involves monoclonal antibodies.

WHAT HAPPENS DURING AN ANTIBODY SEROLOGY TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You don't need any special preparations for an antibody serology test.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will depend on which antibodies were measured. Results may be given as titers (levels of antibodies), positive (you have antibodies), or negative (you do not have antibodies).

Common results for antibody serology tests that look for endogenous antibodies include:

Antibodies to a specific pathogen were found. This may mean you had an infection before. It could also mean you've been vaccinated against a certain disease.

Low levels of certain antibodies were found. This may mean an earlier vaccination does not provide you with enough protection against a disease. It also means you may need a booster shot.

While antibody serology tests can show that you have an immune response to a pathogen, they cannot show if you have full protection from it or how long that protection lasts.

Results for antibody serology tests that look for monoclonal antibodies depend on the type of treatment you are receiving. Talk to your provider about what these results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ANTIBODY SEROLOGY TEST?

An antibody serology test may also detect autoantibodies. Autoantibodies are a type of antibody that attacks healthy cells by mistake. If they're present in your antibody serology test results, your provider may order separate autoantibody

tests to determine if you have an autoimmune disease.

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RELATED HEALTH TOPICS

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- ANA (Antinuclear Antibody) Test
- Chickenpox and Shingles Tests
- Hepatitis Testing
- HIV Screening Test
- How to Cope with Medical Test Anxiety
- How to Prepare for a Lab Test
- How to Understand Your Lab Results
- Measles and Mumps Tests
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- What You Need to Know About Blood Testing
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Antineutrophil Cytoplasmic Antibodies (ANCA) Test

<https://medlineplus.gov/lab-tests/antineutrophil-cytoplasmic-antibodies-anca-test/>

ANTINEUTROPHIL CYTOPLASMIC ANTIBODIES (ANCA) TEST

WHAT IS AN ANTINEUTROPHIL CYTOPLASMIC ANTIBODIES (ANCA) TEST?

This test looks for antineutrophil cytoplasmic antibodies (ANCA) in a sample of your blood. Antibodies are proteins that your immune system makes to fight foreign substances, such as viruses and bacteria. But sometimes antibodies attack the healthy cells of your own tissues and organs by mistake. This is called autoimmune disease.

ANCAs attack healthy white blood cells called neutrophils. This can lead to a disorder called autoimmune vasculitis. There are several types of autoimmune vasculitis, but they all cause inflammation and swelling in your blood vessels. When this happens, your blood vessels can narrow or close off. This can cause different types of serious health problems depending on which blood vessels in your body are affected.

There are two main kinds of ANCA called pANCA and cANCA. Each type targets a specific protein inside white blood cells. An ANCA test can show whether you have one or both types of antibodies. This information helps your health care provider diagnose which type of autoimmune vasculitis you have so you can get the right treatment.

Testing for pANCA can also help diagnose certain types of inflammatory bowel disease (IBD). IBD is a group of chronic (long-lasting) autoimmune diseases that cause swelling and irritation in your stomach and intestines.

Other names: ANCA antibodies, cANCA, pANCA, cytoplasmic neutrophil antibodies, serum, anticytoplasmic autoantibodies

WHAT IS IT USED FOR?

ANCA testing is used to help:

Find out if you have autoimmune vasculitis and which type.

Monitor treatment for autoimmune vasculitis.

Diagnose whether you have ulcerative colitis or Crohn's disease. Both conditions are types of inflammatory bowel disease. ANCAs are more commonly found in ulcerative colitis.

Types of autoimmune vasculitis include:

Granulomatosis with polyangiitis (GPA). This condition used to be called Wegener's disease. It most often affects blood vessels in the lungs and/or sinuses. It may also affect the nose, windpipe, or kidneys.

Microscopic polyangiitis (MPA). This disorder can affect several parts of the body, including the lungs, kidneys, nerves, skin, and joints.

Eosinophilic granulomatosis with polyangiitis (EGPA). This condition used to be called Churg Strauss syndrome. It usually affects blood vessels in the lungs and/or sinuses. It may also affect the stomach and intestines, skin, heart, and nervous system. It often causes asthma and a high level of white blood cells called eosinophils.

WHY DO I NEED AN ANCA TEST?

If you have symptoms of autoimmune vasculitis, you may need an ANCA test. Symptoms may develop slowly or quickly. They may be mild or severe.

General symptoms may include:

Fatigue

Fever

General aches and pains

Loss of appetite

Weight loss

Other symptoms depend on which type of vasculitis you have and which parts of your body are affected. Symptoms in some commonly affected areas include:

Eyes, ears, and nose

Changes in vision or loss of vision

Red and/or itching, burning eyes

Ringling in the ears (tinnitus)

Hearing loss

Dizziness

Sinus infections

Changes in vision or loss of vision

Red and/or itching, burning eyes

Ringling in the ears (tinnitus)

Hearing loss

Dizziness

Sinus infections

Skin

Rashes or hives

Itching

Bruises

Rashes or hives

Itching

Bruises

Lungs

Coughing up blood

Shortness of breath

Coughing up blood

Shortness of breath

Kidneys

Blood in the urine (pee)

Foamy urine, which is caused by protein in the urine

Blood in the urine (pee)

Foamy urine, which is caused by protein in the urine

Nervous system

Numbness, tingling, and/or weakness in different parts of the body

Shooting pains in the arms and legs

Numbness, tingling, and/or weakness in different parts of the body

Shooting pains in the arms and legs

Stomach and intestines

Open sores in the mouth

Diarrhea

Vomiting blood

Open sores in the mouth

Diarrhea

Vomiting blood

You may also need an ANCA test if you have symptoms of inflammatory bowel disease (IBD), which may include:

Abdominal (belly) pain

Diarrhea

Weight loss

Blood in stool (poop)

Fever

WHAT HAPPENS DURING AN ANCA TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You usually don't need any special preparations for an ANCA test. But ANCA testing may be done with other blood tests. Your provider will let you know if you need to prepare.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your provider will consider the results of your ANCA test along with your symptoms and the results of other tests. Ask your provider to explain what your test results mean.

If you were tested for autoimmune vasculitis:

A negative result means that no ANCAs were found in your blood sample. Your symptoms probably aren't caused by autoimmune vasculitis.

A positive result means that ANCAs were found in your blood sample. This may mean you have autoimmune vasculitis. Your test results will also show which type of ANCAs were found. This can help diagnose the type of vasculitis you have. If ANCA antibodies are found, another test may be done on your blood sample to see how much ANCA is in your blood. You will often need other blood tests and a biopsy to confirm the diagnosis. A biopsy is a procedure that removes a small sample of tissue or cells for testing. The tissue sample will be taken from a blood vessel that is swollen.

If you were tested to monitor treatment for autoimmune vasculitis, ANCA test results may show whether your treatment is working. But this test isn't always an accurate way to measure how much disease you have.

If you were tested for inflammatory bowel disease (IBD), your provider will consider your ANCA test results along with the results from another antibody blood test. This test checks for an antibody called ASCA (anti-Saccharomyces cerevisiae antibodies). ASCA are common in people with IBD. You may have:

Ulcerative colitis if ANCA was found, but ASCA was not found.

Crohn's disease if ANCA was not found, but ASCA was found.

If neither antibody was found, you may still have IBD.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Appendicitis Tests

<https://medlineplus.gov/lab-tests/appendicitis-tests/>

APPENDICITIS TESTS

WHAT ARE APPENDICITIS TESTS?

Appendicitis is an inflammation (swelling) or infection of your appendix. Your appendix is a small finger-like pouch attached to your large intestine. It's located in the lower right side of your abdomen (belly).

The cause of appendicitis isn't always clear. Your appendix may become inflamed or infected if something blocks the opening where it attaches to your intestine. If appendicitis isn't treated promptly, your appendix can burst and spread bacteria in your abdomen. This can lead to a life-threatening bacterial infection. A burst appendix can also cause a pus-filled infection in your appendix called an abscess.

Appendicitis typically causes abdominal (belly) pain in your right side. But some people may have different symptoms because their appendix isn't in the usual place. This can make appendicitis hard to diagnose. Appendicitis tests help your health care provider make an accurate diagnosis quickly so you can get proper care.

Appendicitis tests may include:

Blood tests to look for signs of infection and inflammation that may mean you have appendicitis. But blood tests can't diagnose appendicitis.

A urine (pee) test to rule out a urinary tract infection and/or a kidney stone. These conditions can cause symptoms similar to appendicitis.

Imaging tests to take pictures of your appendix. If your symptoms and other tests suggest that you have appendicitis, imaging tests can confirm the diagnosis.

Appendicitis can happen at any age, but it's more common in people in their teens and twenties. The main treatment for appendicitis is an appendectomy, which is surgery to remove the appendix. Your appendix has no known function, and you don't need it to be healthy.

WHAT ARE THEY USED FOR?

Appendicitis tests are used to confirm or rule out a problem with the appendix in people who have appendicitis symptoms.

WHY DO I NEED APPENDICITIS TESTING?

You may need testing if you have abdominal (belly) pain that usually:

Starts near your belly button and moves lower and to your right

Hurts more when you move, take deep breaths, cough, or sneeze

Gets worse within hours

If you have any severe abdominal pain, it's important to get medical help as soon as possible.

Along with pain, other appendicitis symptoms include:

Nausea and vomiting

Fever

Loss of appetite

Swelling in your abdomen

Diarrhea or constipation

Problems passing gas

WHAT HAPPENS DURING APPENDICITIS TESTING?

Before ordering appendicitis tests, your provider will ask questions about your symptoms and medical history. You'll also have an exam. If your provider thinks you could have appendicitis, you may have one or more tests to help confirm or rule out the diagnosis.

Blood tests can help find out if you have an infection or inflammation in your body, but these tests can't show whether appendicitis is the cause.

To collect a blood sample, a health care professional will insert a small needle into a vein in your arm to collect a small amount of blood into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Urine tests can help rule out urinary tract conditions that could be causing your symptoms.

To collect your urine sample, a health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method. It's important to follow these instructions so that germs from your skin don't get into the sample.

Imaging tests check to see if your appendix looks normal. These tests can confirm whether you have appendicitis. They include:

CT scan (computed tomography scan). A CT scan is the most accurate imaging test for diagnosing appendicitis. For this test:

A contrast dye may be used to make your tissues and organs show up better in the pictures. The contrast dye may be given as:

A drink that you swallow

An injection into a vein using an intravenous (IV) line

An enema, which is a tube that injects the contrast medium into your intestine through your anus

You will lie still on a table that passes slowly through the center of a large donut-shaped x-ray machine. For some scans, the x-ray machine moves around you to take pictures. You may hear whirring sounds from the machine.

A contrast dye may be used to make your tissues and organs show up better in the pictures. The contrast dye may be given as:

A drink that you swallow

An injection into a vein using an intravenous (IV) line

An enema, which is a tube that injects the contrast medium into your intestine through your anus

A drink that you swallow

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You will lie still on a table that passes slowly through the center of a large donut-shaped x-ray machine. For some scans, the x-ray machine moves around you to take pictures. You may hear whirring sounds from the machine.

An ultrasound makes pictures using sound waves. For this test:

You'll lie on an exam table and a healthcare professional will put a special gel on your abdomen.

A handheld wand-like device will be moved across your abdomen and images of your organs will show on a monitor.

You'll lie on an exam table and a healthcare professional will put a special gel on your abdomen.

A handheld wand-like device will be moved across your abdomen and images of your organs will show on a monitor.

An MRI scan makes pictures using magnets and a computer. For this test:

You may be given a dye through an intravenous (IV) line.

You'll lie very still on a table that slides into an MRI scanner, which is a tunnel-shaped machine. If you have trouble being in closed-in spaces, talk with your provider about options that may make you more comfortable.

You may be given earplugs or headphones to help block loud sounds from the scanner.

You may be given a dye through an intravenous (IV) line.

You'll lie very still on a table that slides into an MRI scanner, which is a tunnel-shaped machine. If you have trouble being in closed-in spaces, talk with your provider about options that may make you more comfortable.

You may be given earplugs or headphones to help block loud sounds from the scanner.

If there's a chance you could be pregnant, you will likely have a pregnancy test to rule out an ectopic pregnancy, which can cause pain similar to appendicitis. It's also important to know whether you're pregnant before having a CT scan. That's because the x-rays could harm an unborn baby. A pregnancy test may be done with a blood or a urine test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TESTS?

Blood and urine tests don't require any special preparations.

For imaging tests, you may have to stop eating and drinking for several hours before the test. Your provider will tell you how to prepare.

ARE THERE ANY RISKS TO THE TESTS?

A blood test has very little risk. You may have slight pain or bruising at the spot where the needle was put in, but it usually goes away quickly.

A urine test has no risk.

Imaging tests are safe. If your test includes contrast dye, there's a small chance you could have an allergic reaction to it. Reactions can often be managed with medicine. If you get the dye through an IV, it may cause a slight burning feeling.

CT scans are generally safe. They do use x-rays which are a form of radiation. The small amount of radiation in a CT scan is not considered harmful to most adults. But exposure to radiation can slightly increase the risk of developing cancer later in life. Children and unborn babies are more sensitive to radiation than adults because their bodies are still growing.

If you're concerned about radiation exposure for yourself, your unborn baby, or your child, ask about other imaging tests to diagnose appendicitis. If your child needs a CT scan, ask if the machine will be adjusted to use the lowest amount of radiation possible.

CT scans are generally safe. They do use x-rays which are a form of radiation. The small amount of radiation in a CT scan is not considered harmful to most adults. But exposure to radiation can slightly increase the risk of developing cancer later in life. Children and unborn babies are more sensitive to radiation than adults because their bodies are still growing.

If you're concerned about radiation exposure for yourself, your unborn baby, or your child, ask about other imaging tests to diagnose appendicitis. If your child needs a CT scan, ask if the machine will be adjusted to use the lowest amount of radiation possible.

An ultrasound has no risk.

MRI scans are generally very safe. But if you have certain metals and/or implanted medical devices in your body, ask your provider if you can have an MRI.

WHAT DO THE RESULTS MEAN?

To confirm or rule out appendicitis, your provider will consider your symptoms, medical history, and the results of your exam and appendicitis tests.

Blood test results:

A high white blood cell count means you probably have an infection. But the test results don't show what's causing the infection. It's also possible to have appendicitis with a normal white blood cell count.

A high level of c-reactive protein (CRP) in your blood means that you have inflammation in your body that could be from appendicitis. But this test alone can't diagnose appendicitis.

Urine test results:

A high level of white blood cells or bacteria in your urine may mean that a urinary tract infection is causing your

symptoms.

Blood, crystals or high levels of certain minerals in your urine may mean that kidney stones are causing your symptoms.

Imaging test results:

An appendix that is blocked, larger than normal, burst, or has an abscess means that you probably have appendicitis.

If your appendix looks normal, imaging tests may help find other problems that could be causing abdominal pain.

If your provider diagnoses appendicitis, you'll usually have surgery to remove your appendix as soon as possible before it can burst. If your appendix has already burst, you must have surgery right away. The surgeon will remove your appendix and clean the infection from your abdomen.

Learn more about laboratory tests, reference ranges, and understanding results.

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Arterial Blood Gas (ABG) Test

<https://medlineplus.gov/lab-tests/arterial-blood-gas-abg-test/>

ARTERIAL BLOOD GAS (ABG) TEST

WHAT IS AN ARTERIAL BLOOD GAS TEST?

An arterial blood gas (ABG) test measures the amount of oxygen and carbon dioxide (CO₂) in a sample of your blood. The blood oxygen measurement shows how well your lungs move oxygen from the air into your blood when you breathe in. The carbon dioxide measurement shows how well your lungs remove carbon dioxide from your blood when you breathe out.

Most blood tests take a sample from a vein. For this test, a health care provider will take a sample of blood from an artery (a blood vessel that carries oxygen-rich blood from your heart and lungs to your body). That's because blood from an artery has higher oxygen levels than blood from a vein.

An ABG test usually includes the following levels:

Oxygen saturation (O₂ Sat). This measures how much oxygen your red blood cells are carrying.

Partial pressure of oxygen (PaO₂). This measures the pressure of oxygen that's dissolved in your blood. It helps show how well your lungs move oxygen from the air into your blood when you breathe in.

Partial pressure of carbon dioxide (PaCO₂). This measures the amount of carbon dioxide (an acidic waste product that your body makes naturally) in your blood. It also shows how well your lungs remove carbon dioxide from your blood when you breathe out.

Acid-base balance (pH level). This measures the acidity of your blood. Too much acid is called acidosis. Too much base (alkaline) is called alkalosis. If your blood and tissues become even slightly too acidic or too basic (alkaline), it can affect many of your organs and even become life-threatening.

An ABG test may also measure your bicarbonate (HCO₃) concentration. Bicarbonate is an electrolyte. Electrolytes are electrically charged minerals that help control the amount of fluids and the balance of acids and bases (pH balance) in your body. Most of the carbon dioxide in your blood is stored as bicarbonate. So, this measurement can help show your carbon dioxide levels.

Other names: blood gas test, arterial blood gases, ABG, oxygen saturation test

WHAT IS IT USED FOR?

An ABG test is used to help:

Diagnose serious problems with your lungs and breathing. Your lungs are responsible for getting oxygen into your blood, CO₂ out of it, and for helping to keep your pH balance normal.

Diagnose kidney disorders. Like the lungs, your kidneys help to keep your pH balance normal. They also help your body remove CO₂.

Find out whether treatment is working for breathing disorders, kidney disease, or other conditions that may affect your acid-base balance.

WHY DO I NEED AN ARTERIAL BLOOD GAS (ABG) TEST?

There are many reasons why you may need this test. For example, you may need an ABG test if you:

Have symptoms of acidosis (too much acid in your blood), such as:

Uncontrolled rapid or deep breathing, which may be a sign that your lungs are trying to adjust acids or bases by changing the amount of oxygen or carbon dioxide in your blood.

Arrhythmia. (a problem with the rate or rhythm of your heartbeat).

Confusion.

Fatigue.

Uncontrolled rapid or deep breathing, which may be a sign that your lungs are trying to adjust acids or bases by changing the amount of oxygen or carbon dioxide in your blood.

Arrhythmia. (a problem with the rate or rhythm of your heartbeat).

Confusion.

Fatigue.

Have symptoms of alkalosis (too little acid in your blood), such as:

Confusion.

Nausea and vomiting.

Muscle twitching and/or cramps.

Numbness in hands, feet, or face.

Arrhythmia.

Confusion.

Nausea and vomiting.

Muscle twitching and/or cramps.

Numbness in hands, feet, or face.

Arrhythmia.

Are being treated for a lung disease or a condition that affects your breathing, such as:

Asthma.

Chronic obstructive pulmonary disease (COPD).

Cystic fibrosis.

Myasthenia gravis.

Asthma.

Chronic obstructive pulmonary disease (COPD).

Cystic fibrosis.

Myasthenia gravis.

Have symptoms of:

Carbon monoxide poisoning.

An inhalation injury (breathing in smoke, hot air, and/or harmful chemicals).

A recent head or neck injury that could affect your breathing.

Carbon monoxide poisoning.

An inhalation injury (breathing in smoke, hot air, and/or harmful chemicals).

A recent head or neck injury that could affect your breathing.

Are receiving oxygen therapy in the hospital.

WHAT HAPPENS DURING AN ARTERIAL BLOOD GAS TEST?

The sample is usually taken from an artery on the inside of your wrist, but it may be taken from an artery in your arm or groin. For a newborn, the sample may be taken from the baby's heel or the umbilical cord shortly after birth.

If your blood sample is taken from your wrist, the provider will first test your blood circulation by applying pressure to the arteries in your wrist in order to cut off blood flow to your hand for several seconds. Then the provider will let go of your wrist to check how quickly blood flow returns to your hand. If your blood flow is normal, the provider will collect a blood sample.

A blood sample taken from an artery tends to be more uncomfortable than most blood tests, which use a vein. So, the provider may apply some numbing medicine to your skin after testing your circulation. The provider will insert a needle with a syringe into the artery to remove some blood.

When the syringe is full, the provider will bandage the puncture site. Pressure will be applied to the site for at least 5 minutes to stop the bleeding.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines before this test, so tell your provider about all the medicines and supplements you take. But don't stop taking any medicines unless your provider tells you to.

If you take blood thinners, including aspirin, ask your provider whether you should stop taking them before your test.

Certain medicines may also upset your acid-base balance and lead to abnormal ABG test results.

If you are on oxygen therapy, your oxygen may be turned off for about 20 minutes before the test. This will be done only if you can breathe without oxygen therapy.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having an arterial blood gas test. You may have some bleeding, bruising, or soreness at the spot where the needle was put in. Very rarely, the needle may damage a nerve or the artery. You may be told to avoid lifting heavy objects for 24 hours after the test.

WHAT DO THE RESULTS MEAN?

ABG test results involve many body systems that affect each other. There are also many health conditions that can cause abnormal results. For these reasons, it's best to have your provider explain what your results mean for your health.

Generally, for each of the following measurements, your ABG test results will also list their normal ranges:

Oxygen saturation (O₂Sat). 95-100%.

Partial pressure of oxygen (PaO₂). 75-100 millimeters of mercury (mmHg).

Partial pressure of carbon dioxide (PaCO₂). 35-45 mmHg.

Acid-base balance (pH level). 7.35-7.45.

Bicarbonate (HCO₃). 22-26 milliequivalents per liter (mEq/L).

An ABG test alone usually can't provide a diagnosis. So, if your results are not normal, your provider will likely order more tests. In general, abnormal results may mean you have a problem with your lungs or kidneys or a metabolic disorder. Metabolic disorders affect how your body uses food for energy.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ARTERIAL BLOOD GAS TEST?

Another type of test, called pulse oximetry, can check your blood oxygen saturation levels. A small clip-like device, called a pulse oximeter, is usually attached to your finger. The device tells you the percentage of red blood cells that are full of oxygen. Pulse oximetry may be useful if blood oxygen levels are the only concern. Ask your provider if this test is right for you.

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RELATED HEALTH TOPICS

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RELATED MEDICAL TESTS

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Lung Function Tests

Pulse Oximetry

RDW (Red Cell Distribution Width)

Sleep Study

What You Need to Know About Blood Testing

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AST Test

<https://medlineplus.gov/lab-tests/ast-test/>

AST TEST

WHAT IS AN AST TEST?

An AST blood test measures the amount of AST in your blood. This test is commonly used to help diagnose liver damage or disease.

AST (aspartate aminotransferase) is an enzyme, a protein that speeds up certain chemical reactions in your body. It is found mainly in your liver but also in your heart, muscles, and other tissues. Usually, you will have low levels of AST in your blood. When your liver or other cells that contain AST are damaged, they release AST into your blood. This can lead to high AST levels in your blood.

An AST test may be ordered by itself but is often ordered as part of a group of liver function tests.

Other names: SGOT test, serum glutamic oxaloacetic transaminase test; aspartate transaminase test

WHAT IS IT USED FOR?

An AST blood test is often part of a routine blood screening to check the health of your liver. The test may help diagnose or monitor liver problems. It may also help diagnose other health conditions.

WHY DO I NEED AN AST BLOOD TEST?

You may get an AST blood test as part of your routine checkup or to monitor the treatment of a liver problem. You may also need a test if you have symptoms of liver damage, which may include:

Nausea and vomiting

Lack of appetite

Fatigue

Weakness

Jaundice, a condition that causes your skin and eyes to turn yellow

Swelling and/or pain in your abdomen (belly)

Swelling in your ankles and legs

Dark-colored urine (pee) and/or light-colored stool (poop)

Frequent itching

Even if you don't have symptoms, your health care provider may order an AST blood test if you're more likely to develop liver disease because of:

A family history of liver disease

Alcohol use disorder (AUD)

Obesity

Diabetes

Taking certain medicines that can cause liver damage

Hepatitis or exposure to hepatitis

WHAT HAPPENS DURING AN AST BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines before the test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

An AST test is usually ordered with other blood tests. You usually need to fast[^] (not eat or drink) for up to 12 hours before these tests. Your provider will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Low levels of AST in the blood are usually considered normal.

High levels of AST in the blood may be due to liver disease or other liver problems. Other reasons you may have high levels of AST can include:

Hepatitis

Cirrhosis

Mononucleosis

Heart procedures or problems

Pancreatitis

Hemochromatosis (too much iron in the body)

Deep burns

Seizures

Use of drugs that are toxic to the liver, especially alcohol use

Surgery

If your results are not in the normal range, it doesn't always mean that you have a medical condition that needs treatment. Many things can affect your results, such as pregnancy, exercise, certain medicines, and your age or sex. Talk with your provider to learn what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN AST BLOOD TEST?

Your health care provider may order an ALT blood test[^] along with your AST blood test. ALT stands for alanine transaminase, which is another type of liver enzyme. If you have high levels of AST and/or ALT, it may mean that you have some type of liver damage.

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RELATED HEALTH TOPICS

Liver Diseases

RELATED MEDICAL TESTS

ALT Blood Test

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What You Need to Know About Blood Testing

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At-Home Medical Tests

<https://medlineplus.gov/lab-tests/at-home-medical-tests/>

AT-HOME MEDICAL TESTS

WHAT ARE AT-HOME MEDICAL TESTS?

At-home medical tests, also known as home use tests, are kits you can buy online or at your local pharmacy or supermarket. These kits allow you to test for, screen, or monitor certain diseases and conditions in the privacy of your own home. Common at-home tests can include:

Pregnancy tests.

Blood glucose (blood sugar) tests, which may be used regularly to help you manage your diabetes.

Fecal occult blood tests, which screen for colon cancer.

Tests that check for signs of infectious diseases, such as hepatitis, HIV, and COVID-19.

Genetic tests, which may show if you are at higher risk for certain diseases.

There are two types of at-home tests:

Self-tests involve taking a sample of a body fluid, such as blood, urine, or saliva, and applying it to the kit as directed.

These tests provide immediate results.

Self-collection tests also allow you to take your sample at home, but then you'll need to package it and mail it to a lab.

You can get the results from your health care provider, or you may be able to get them in an online portal with a unique login and password. How long it takes to get your results can vary, based on the type of test.

Many kits are available without a prescription, but you should still ask your provider for advice on which kits to use.

Other names: home use tests, home test kits

HOW ARE THEY USED?

At-home testing is frequently used to:

Find or screen for disorders, such as high cholesterol or certain infections, before you have any symptoms. This may allow you to get treatment earlier and avoid serious complications. It may also help prevent the spread of infectious diseases.

Help diagnose pregnancy early so you can take steps to care for your health and your baby's by scheduling an appointment with your provider.

Monitor chronic diseases, such as diabetes and high blood pressure. This may allow you to make more quickly begin making treatment and/or lifestyle changes to improve your condition.

HOW DO I USE AN AT-HOME TEST?

The specific instructions will vary, depending on the type and brand of the test kit. Below are examples of some common at-home tests and how they are performed.

Pregnancy Test

Used to diagnose pregnancy by checking for a hormone called human chorionic gonadotropin (HCG). HCG is normally only produced during pregnancy.

Test procedure:

If possible, plan to do the test in the morning. Most pregnancy tests recommend using your first morning urine for more accurate results.

You will hold a test strip in your urine stream or collect your urine in a cup and place the test strip into the cup.

If you are pregnant, the strip will display a colored line, a double line, or a symbol such as a plus sign. The type of display depends on the brand.

If possible, plan to do the test in the morning. Most pregnancy tests recommend using your first morning urine for more accurate results.

You will hold a test strip in your urine stream or collect your urine in a cup and place the test strip into the cup.

If you are pregnant, the strip will display a colored line, a double line, or a symbol such as a plus sign. The type of display depends on the brand.

Follow up with your provider if the test is positive or if the test was negative but you still think you might be pregnant.

Glucose Test

Used to monitor blood glucose (blood sugar) levels. This test is essential for some people with diabetes. If your blood glucose gets too high or too low for you, it can cause dangerous complications.

Test procedure:

You will prick your finger with a device called a lancet to get a drop of blood.

You place the blood on a test strip and insert it into a monitor.

The meter will show your glucose level.

There are different types of glucose meters, and they work in slightly different ways. Some may not even require you to prick your finger. Be sure to read and follow all the instructions carefully.

You will prick your finger with a device called a lancet to get a drop of blood.

You place the blood on a test strip and insert it into a monitor.

The meter will show your glucose level.

There are different types of glucose meters, and they work in slightly different ways. Some may not even require you to prick your finger. Be sure to read and follow all the instructions carefully.

Fecal Occult Blood Test

Used to check stool for signs of colon cancer. Your provider will give you a test kit.

Test procedure:

You will collect small stool samples and place them on a special card or container.

Seal and mail the card or container to your provider or a lab.

The sample will be checked for blood. Blood in the stool can be a sign of colon cancer or other conditions.

The testing laboratory or your provider will contact you with the results.

You will collect small stool samples and place them on a special card or container.

Seal and mail the card or container to your provider or a lab.

The sample will be checked for blood. Blood in the stool can be a sign of colon cancer or other conditions.

The testing laboratory or your provider will contact you with the results.

Hepatitis C Test

Used to find out if you have hepatitis C now or were infected in the past.

Test procedure:

You will prick your finger with a device called a lancet to get a drop of blood.

You place the blood sample on a special piece of paper.

You will seal and mail the paper to a laboratory.

The testing laboratory or your provider will contact you with the results.

You will prick your finger with a device called a lancet to get a drop of blood.

You place the blood sample on a special piece of paper.

You will seal and mail the paper to a laboratory.

The testing laboratory or your provider will contact you with the results.

Genetic Test

Used to find out if you may be at higher risk of cancer or other diseases by looking for certain genetic changes. Genetic tests may also be used to find out information about your ancestors. These tests work by comparing your genetic data with genetic data from large groups of people.

Test procedure:

You can order many of these tests yourself. Others require a provider's prescription.
Most test kits will instruct you to provide a saliva sample or to take a swab from the inside of your cheek.
You will mail the sample to the test company or as directed.
The testing company or your provider will contact you with the results.

You can order many of these tests yourself. Others require a provider's prescription.
Most test kits will instruct you to provide a saliva sample or to take a swab from the inside of your cheek.
You will mail the sample to the test company or as directed.
The testing company or your provider will contact you with the results.

If you are using a test to find out your risk of disease, be sure to talk to your provider. These tests can't diagnose diseases and are not as accurate as traditional lab-based genetic tests. But if a home test shows you may be at higher risk, your provider can monitor your health more closely.

COVID-19 Test

Used to find out if you have COVID-19. A viral test tells if you have a current infection. Viral tests can be either rapid tests, which you can do at home, or laboratory tests.

Test procedure:

Buy a self-test online or at a store.

Most test kits will instruct you to take a nasal swab and swirl it inside your nostril.

You will place your sample swab in a vial and then place that in the test unit.

You will have results in 30 minutes or less displaying whether you have the COVID-19 virus. The type of display may depend on the brand of the test.

Follow the test kit instructions.

Buy a self-test online or at a store.

Most test kits will instruct you to take a nasal swab and swirl it inside your nostril.

You will place your sample swab in a vial and then place that in the test unit.

You will have results in 30 minutes or less displaying whether you have the COVID-19 virus. The type of display may depend on the brand of the test.

Follow the test kit instructions.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AT-HOME MEDICAL TESTS?

At-home medical tests can provide helpful information, but they are not a replacement for guidance and treatment by your provider. After at-home testing, you should follow up with your provider regardless of the result.

In addition, when purchasing and using an at-home test kit:

Only use FDA-approved or FDA-authorized tests. The FDA has strict requirements for the quality and accuracy of home test kits.

Follow the test instructions exactly. Even minor changes can affect your results.

Don't buy or use expired tests. Chemicals in the test may lose their effectiveness over time.

Don't change your medicines because of a test result without talking with your provider. Let your provider know about your results. They can tell you if you need to change your medicine or the dose you take based on your at-home test results.

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RELATED HEALTH TOPICS

Bacterial Infections
Colorectal Cancer
Diabetes
Genetic Testing
Hepatitis
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HIV
Infectious Diseases
Pregnancy
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RELATED MEDICAL TESTS

Bacterial Vaginosis Test
Bilirubin in Urine
Blood Glucose Test
BRCA Genetic Test
Cholesterol Levels
Colorectal Cancer Screening Tests
Drug Testing
Estrogen Levels Test
Fecal Occult Blood Test (FOBT)
Follicle-Stimulating Hormone (FSH) Levels Test
Gonorrhea Test
Hepatitis Testing
Herpes (HSV) Test
HIV Screening Test
How to Understand Your Lab Results
Human Papillomavirus (HPV) Test
Luteinizing Hormone (LH) Levels Test
Measuring Blood Pressure
Nitrites in Urine
Opioid Testing
Parainfluenza Tests
Pharmacogenetic Tests
Pregnancy Test
Prothrombin Time Test and INR (PT/INR)
Pulse Oximetry
Rapid Tests
Rheumatoid Factor (RF) Test
Semen Analysis
Sexually Transmitted Infection (STI) Tests
Sleep Study
Testosterone Levels Test
Trichomoniasis Test

MEDLINEPLUS GENETICS

Direct-to-Consumer Genetic Testing

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Autism Spectrum Disorder (ASD) Screening

<https://medlineplus.gov/lab-tests/autism-spectrum-disorder-asd-screening/>

AUTISM SPECTRUM DISORDER (ASD) SCREENING

WHAT IS AUTISM SPECTRUM DISORDER SCREENING?

Autism spectrum disorder (ASD) is a developmental disability caused by differences in a person's brain. ASD affects how people interact with others, communicate, learn, and behave. It can be diagnosed at any age, but symptoms usually show up in the first two years of life and can be life-long.

ASD is called a "spectrum disorder" because there is a wide range of symptoms and strengths in people with ASD. For example, some people with ASD may not talk at all, while others have strong language skills. The learning, thinking, and problem-solving abilities of people with ASD may range from severely challenged to gifted.

ASD screening is mainly used for young children. It helps find out whether a child has any early signs that could be ASD. But screening cannot diagnose ASD. If a screening shows that a child may have the disorder, more testing will be needed to find out for sure.

ASD screening in early childhood is important because early treatment for ASD can help reduce autism symptoms and improve quality of life. For this reason, health care providers routinely screen children for ASD before age two. Older children and adults may also be screened if they have symptoms of ASD but have never been diagnosed with the disorder.

Providers used to group ASD into different types of autism based on a person's symptoms. These were called Asperger syndrome, autistic disorder, and pervasive developmental disorder not otherwise specified (PDD-NOS). These terms are no longer used to diagnose ASD.

Other names: ASD screening

WHAT IS IT USED FOR?

Autism spectrum disorder screening is most often used to check for signs of autism spectrum disorder (ASD) in children age 2 and under. It helps find out which children need more testing to see if they have ASD.

ASD can be hard to diagnose, and screening may not find milder cases in early childhood. So, ASD screening may also be used in older children and adults who have certain challenges with social life and/or behavior that could be signs of ASD.

WHY DOES A CHILD OR AN ADULT NEED AN AUTISM SPECTRUM DISORDER SCREENING?

The American Academy of Pediatrics (AAP) recommends ASD screening for all children at their 18-month and 24-month well-child checkups.

More ASD screening may be needed if your child has a higher risk for ASD. The risk for ASD is higher for children who:

Have a sibling or other family member who has ASD

Have older parents

Have certain genetic disorders or chromosomal conditions, such as Down syndrome, Fragile X, Rett syndrome, or tuberous sclerosis

Had problems during childbirth and/or were born with a low birth weight

Research studies have found no link between vaccines and ASD.

Your child may need ASD screening at any age if you, a teacher, or other caregiver notices possible autism symptoms.

Some of the more common ASD symptoms in children and teens include:

Problems with communication and social behavior, such as:

Making little or no eye contact with others

Not responding when people smile or talk to them

Rarely sharing their interests with others (for example, rarely showing you their favorite toy)

Trouble having conversations

Talking a lot about a favorite subject without noticing others aren't interested

Having an unusual tone of voice that's robot-like or sing-song

Having trouble making friends

Not understanding humor or sarcasm

Making little or no eye contact with others

Not responding when people smile or talk to them

Rarely sharing their interests with others (for example, rarely showing you their favorite toy)

Trouble having conversations

Talking a lot about a favorite subject without noticing others aren't interested

Having an unusual tone of voice that's robot-like or sing-song

Having trouble making friends

Not understanding humor or sarcasm

Having unusual behaviors and/or narrow interests, called "restrictive/repetitive behaviors." For example:

Repeating words or phrases (called echolalia), or hand flapping and/or body rocking

Having a lasting, intense interest in specific topics, such as numbers, details, or facts

Showing unusual attachment to toys, objects

Becoming upset by slight changes in a routine and having difficulty changing from one activity to another

Being more sensitive or less sensitive than other people to sensory input, such as light, sound, the feel of clothing, or temperature

Repeating words or phrases (called echolalia), or hand flapping and/or body rocking

Having a lasting, intense interest in specific topics, such as numbers, details, or facts

Showing unusual attachment to toys, objects

Becoming upset by slight changes in a routine and having difficulty changing from one activity to another

Being more sensitive or less sensitive than other people to sensory input, such as light, sound, the feel of clothing, or temperature

If you're concerned about your child or teen, talk with your child's provider even if your child had a normal ASD screening in the past.

Adults may need ASD screening if they have problems that could be signs of autism, but they were never diagnosed with ASD. Usually, their symptoms are mild, which is why they were not diagnosed earlier. Their challenges may include:

Problems communicating and interacting with others, such as having trouble understanding other people's emotions

Restricted interests and/or repetitive behaviors

Trouble understanding what behavior is expected in school, work, or other areas of life

Adults with ASD may also have difficulty keeping a schedule and setting long-term goals. They often have other conditions such as anxiety, depression, or attention deficit hyperactivity disorder (ADHD).

WHAT HAPPENS DURING AN AUTISM SPECTRUM DISORDER SCREENING?

For children: Screening for ASD is usually done by your child's pediatrician (a doctor who specializes in treating children) or nurse. Screening may also be done in school by trained professionals. The screening may have one or more parts, including:

Questionnaires. You'll usually complete one or more questionnaires. The questions ask about your child's development and behavior, including speech, movement, thinking, and emotions. ASD often runs in families, so you may also be asked about your family health history.

Observation. The provider will watch how your child plays and interacts with you and/or others. For example, the provider will check if your child responds to your laugh or looks at a person who tries to get their attention. Not responding may be a sign of ASD.

Interactive screening tests. These tests are play activities, such as playing make-believe with dolls or other toys. These tests are designed to check your child's communication skills, social behavior, and other abilities.

For adults, screening tools for ASD are still being developed and tested. Your primary care provider may refer you to a

specialist, such a psychologist or psychiatrist. The specialist may:

- Talk with you about the challenges you face in your day-to-day life
- Ask you to complete a questionnaire about your symptoms
- Ask to talk with family members who remember what you were like as a young child
- Do screening tests for depression, ADHD, and/or anxiety, which are common in people who have ASD

WILL I NEED TO DO ANYTHING TO PREPARE FOR AN AUTISM SPECTRUM DISORDER SCREENING?

There are no special preparations needed for this screening.

ARE THERE ANY RISKS TO SCREENING?

There is no risk to having autism spectrum disorder screening.

WHAT DO THE RESULTS MEAN?

The results of an ASD screening may be given as a score. The score may be described as low, medium, or high risk for ASD. If screening results do not find signs of ASD, but you have concerns, discuss them with your provider.

If the screening shows signs of ASD, it may mean that more testing is needed to confirm whether they are caused by ASD.

Children who show signs of ASD often need to see a specialist for more testing. Several types of specialists may be trained to diagnose ASD, including:

- Developmental pediatricians, doctors who have training in child development and treating children with special needs
- Child neurologists, doctors who specialize in diagnosing and treating conditions involving the brain, spine, and nerves
- Child psychologists or psychiatrists, providers who diagnose and treat children who have mental health, behavioral and/or developmental conditions. Psychologists usually have doctoral degrees (PhDs), and psychiatrists are medical doctors (MDs).

There is no one test that can diagnose ASD. So, a specialist will use a combination of methods, including:

- Detailed questionnaires and/or interviews with parents, teachers, or caregivers

- Watching the child's behavior

- Tests to evaluate the child's thinking, learning, and language abilities

- Exams to check for other conditions that can cause behavior and/or communication problems. These exams may check:

- The nervous system

- Hearing

- Vision

- The nervous system

- Hearing

- Vision

Genetic testing to look for inherited disorders that can cause ASD. If ASD runs in your family, these tests may be done to help make a diagnosis. Otherwise, they may be done after ASD is diagnosed to help guide treatment.

If your child is diagnosed with ASD, get treatment as soon as possible. ASD treatment involves a variety of services and support depending on your child's needs.

Adults who show signs of ASD may see a psychologist, psychiatrist, neuropsychologist, or other specialist with experience diagnosing ASD in adults. But it can be difficult to find a specialist with this experience. That's because adult testing to diagnose ASD is fairly new. If you would like to find out if you have ASD, ask your primary care provider to help you find a specialist. You might also try contacting an organization in your community that supports people with ASD.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AUTISM SPECTRUM DISORDER SCREENING?

In early childhood, pediatricians do routine developmental and behavioral screening tests designed to catch other types of developmental problems. These tests may miss ASD, so make sure your child's provider does ASD screening, too.

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RELATED HEALTH TOPICS

Autism Spectrum Disorder

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Autoantibody Testing

<https://medlineplus.gov/lab-tests/autoantibody-testing/>

AUTOANTIBODY TESTING

WHAT IS AUTOANTIBODY TESTING?

Autoantibody testing checks a sample of your blood for autoantibodies, proteins made by the immune system.

Normally, your immune system makes proteins called antibodies to attack germs that could make you sick. Autoantibodies, however, do not attack germs. They instead attack your own healthy tissue and organs, believing them to be foreign substances. This can lead to the development of an autoimmune disease. There are more than 80 types of autoimmune diseases, and they can affect almost any part of your body.

Your immune system makes autoantibodies by mistake. Researchers aren't entirely sure why, though your body is more likely to make them if you:

- Have a family history of autoimmune diseases.

- Have a viral or bacterial infection, especially if it's chronic (long-term). This can cause your immune system to be overactive, which can lead to the production of autoantibodies.

- Are exposed to certain environmental toxins.

There are many different types of autoantibodies, and they can affect your body in different ways. Some affect only specific organs and/or parts of your body. These are called organ-specific antibodies. For example:

- Thyroid antibodies, such as anti-TPO, only affect your thyroid.

- Smooth muscle antibodies (SMAs) attack smooth muscle, usually in the liver.

- Islet cell autoantibodies attack cells in your pancreas.

Other antibodies, however, affect multiple organs and can cause a variety of symptoms. These are called systemic antibodies. This type can include:

- Antinuclear Antibody (ANA).

- Antineutrophil Cytoplasmic Antibodies (ANCA).

- Anti-Double Stranded DNA (anti-dsDNA).

- Antihistone Antibodies.

- Rheumatoid Factor (RF).

Other names: antibody serology test, antinuclear antibody (ANA) test, extractable nuclear antigen (ENA) test, anti-dsDNA test, rheumatoid factor (RF) test, antineutrophil cytoplasmic antibodies (ANCA) test, smooth muscle antibody (SMA) test, thyroid antibody test

WHAT IS AUTOANTIBODY TESTING USED FOR?

Every autoantibody test is used to help diagnose or monitor an autoimmune disease. However, because the types of autoantibodies can vary, there are different types of autoantibody tests. Some of these can be used to diagnose autoimmune conditions that affect only one organ. For example:

- Thyroid antibody tests look for thyroid-specific antibodies and can be used to help diagnose Grave's Disease and Hashimoto's Disease.

- Smooth muscle antibody (SMA) tests look for smooth muscle antibodies and can be used to help diagnose autoimmune hepatitis.

- Tissue transglutaminase antibody (tTG) tests look for the tTG antibody and can be used to help diagnose celiac disease.

Other tests can be used to diagnose autoimmune conditions that affect multiple organs. For example:

- Anti-dsDNA tests look for the anti-dsDNA autoantibody and can be used to help diagnose lupus.

- Rheumatoid factor (RF) tests look for rheumatoid factor and can be used to help diagnose rheumatoid arthritis.

- Antineutrophil cytoplasmic antibodies (ANCA) tests look for ANCA in your blood and can be used to help diagnose autoimmune vasculitis.

However, some autoantibody tests look for a wider range of autoantibodies. The most common of these are the antinuclear antibody (ANA) test and the extractable nuclear antigen (ENA) test. These tests can be used to help

diagnose many of the conditions listed above, as well as:

Sjogren's syndrome.

Polymyositis and dermatomyositis.

Addison Disease.

Mixed connective tissue disease.

Scleroderma.

Type 1 diabetes.

Psoriasis and psoriatic arthritis.

Raynaud's Phenomenon.

If your provider is using autoantibody tests to help make a diagnosis, they may also consider:

The results of any imaging scans, such as x-rays.

Biopsy results.

Your physical symptoms.

Your family and health history.

The results of an autoantibody test alone are usually not enough to make a diagnosis.

WHY DO I NEED AUTOANTIBODY TESTING?

You may need autoantibody testing if you have symptoms commonly associated with some autoimmune diseases. The symptoms of an autoimmune disease depend on the part of your body that's affected. These can include:

Fever.

Fatigue.

Joint symptoms, including pain and swelling.

Muscle pain or weakness.

Swollen glands.

Abdominal (belly) pain and/or cramps.

Rashes that can't be explained.

Unexpected weight loss or gain.

Palpitations (racing or pounding heart).

Anxiety and depression.

Seizures.

Other illnesses can, however, cause the same symptoms. Symptoms for many autoimmune conditions may also come and go.

You may also need an autoantibody test if you've been diagnosed with an autoimmune disease, and your provider wants to monitor your condition. For example, a provider may order an islet autoantibodies test if they want to monitor your Type 1 diabetes, which is a type of organ-specific autoimmune disease.

WHAT HAPPENS DURING AUTOANTIBODY TESTING?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR AUTOANTIBODY TESTING?

You don't need any special preparations for autoantibody testing.

ARE THERE ANY RISKS TO AUTOANTIBODY TESTING?

There is very little risk to a blood test. You may have slight pain or bruising at the spot where the needle was put in, but

most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results may depend on the type of autoantibody test you had. But generally, your results will be either positive or negative.

A negative (normal) result means that you have little or no autoantibodies in your blood. But that doesn't rule out an autoimmune disease. People with rheumatoid arthritis, for example, may have little to no RF in their blood. If you have symptoms of an autoimmune disease, but your results are normal, your provider may order more tests to make a diagnosis.

A positive (abnormal) result means that some amount of autoantibodies were found in your blood. This doesn't necessarily mean that you have an autoimmune disease. Some healthy people have autoantibodies in their blood. To make a diagnosis, your provider will consider other test results along with your symptoms and medical history.

Some results may give a number for how many autoantibodies were found. This is called a titer. If a titer is provided, your results will also have a reference range. A reference range is a set of numbers that form the high and low ends of the range of results that's considered normal. So, a titer value that is higher than the reference range may be a sign of an autoimmune disease.

The results for autoantibody panels, such as ANA and ENA tests, might also go into detail for each of the autoantibodies tested. This means you could have positive results for some autoantibody tests and negative results for others.

Because autoantibody test results can mean so many different things, it's important to talk to your provider about what your results mean for you.

Learn more about laboratory tests, reference ranges, and understanding results.

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Autonomic Testing

<https://medlineplus.gov/lab-tests/autonomic-testing/>

AUTONOMIC TESTING

WHAT IS AUTONOMIC TESTING?

Autonomic testing helps find out if your autonomic nervous system (ANS) is working as well as it should. Your ANS controls body functions that happen automatically (without you thinking about them). These functions include your breathing, heartbeat, body temperature, and digestion. If there are problems with the function of your ANS, it's known as autonomic neuropathy.

Autonomic neuropathy damages the nerves that control your internal organs. This can lead to problems with your:

Heart rate and blood pressure

Digestive system

Bladder

Sex organs

Sweat glands

Eyes

Blood glucose

The most common causes of autonomic neuropathy include diabetes, autoimmune diseases, some types of medicines, and certain infections, such as HIV.

Autonomic testing can show which part of your ANS is affected and how seriously it's affected. This can help your health care provider develop the most effective treatment strategy for you.

Other names: tilt table test, quantitative sudomotor axon reflex test (QSART), thermoregulatory sweat test (TST), passive head-up tilt test, head-upright tilt test

WHAT IS IT USED FOR?

Autonomic testing is used to check for problems with your autonomic nervous system (ANS).

WHY DO I NEED AUTONOMIC TESTING?

You may need this test if you have symptoms of autonomic neuropathy, sometimes referred to as dysautonomia. Symptoms will vary depending on which part of your nervous system is affected.

Stomach and gastrointestinal symptoms can include:

Constipation

Diarrhea

Nausea and vomiting

Bladder and urinary symptoms can include:

Difficulty urinating

Leaking urine

Urinating more often than usual

Heart and lung symptoms can include:

Rapid heart rate

Shortness of breath with activity or exercise

Dizziness

Fainting

Weakness

Other symptoms can include:

Too much or not enough sweating

Sexual problems; such as erectile dysfunction in men and vaginal dryness in women
The pupils of your eyes being slow to adjust to changes in light and darkness
Getting low blood glucose and not knowing it (since you may not have the symptoms)

WHAT HAPPENS DURING AUTONOMIC TESTING?

There are different types of autonomic tests. Depending on your symptoms, your provider may perform one or more of the following tests:

Deep Breathing Test and Valsalva Maneuver. These tests measure your heart rate and blood pressure while doing certain types of breathing.

For both tests, your provider will place electrodes on your chest and a small blood pressure cuff on your finger.

For a deep breathing test, you will take slow, deep breaths for one minute.

For the Valsalva maneuver, you will breathe out forcefully through a mouthpiece while your nose is pinched shut.

The electrodes and cuff will record your heart rate and blood pressure.

Tilt Table Test. This test measures your blood pressure and heart rate as you change your posture and position to find why you have had lightheadedness or fainted. During the first part of the test:

You will lie flat on a motorized table that has a footrest.

Soft safety straps will be placed across your body.

Your provider will put electrodes on your chest and blood pressure cuffs on one of your arms and a finger. An intravenous (IV) line may also be placed in your arm or hand to inject medicine or fluids if needed.

You will remain flat on the table for about 15 minutes.

Your provider will tilt the table to 30 degrees for two to three minutes, then 45 degrees for two to three minutes, and then to an almost standing position for up to 45 minutes.

Your provider will check your blood pressure and heart rate.

If the test causes you to faint, the table will be quickly returned to the flat position.

You may have to do the second part of the test if your blood pressure doesn't lower during the first part of the tilt table test. During the second part of the test:

Your provider will give you medicine that makes your heart beat faster to see how your body reacts.

Your provider will tilt the table to 60 degrees for up to 15 minutes and monitor your blood pressure.

If your blood pressure lowers, the test will end sooner than 15 minutes.

The test takes from 30 to 90 minutes.

Quantitative Sudomotor Axon Reflex Test (QSART). This test measures the function of the nerves that control sweating.

During the test:

You will remove your shoes and socks.

You will sit or lie flat on an exam table.

Your provider will clean and dry your skin and then place electrodes on the part of the body where they are doing the test.

This is usually on your foot, wrist, and leg. The electrodes will contain a substance that stimulates sweating.

A mild electrical current will be sent through the electrodes. You might feel a warm, tingling sensation as this is done.

A computer will analyze how your nerves and sweat glands react to the electrical current.

Your provider may move the electrodes and repeat these steps on different body parts.

The test takes about 45 minutes to an hour to complete.

Thermoregulatory Sweat Test (TST). This test measures how well your body sweats in a warm environment. During the test:

A special powder will be applied to your skin.

You will lie in a special room that is slowly heated up to stimulate sweating. The heat and humidity may be a little uncomfortable.

The powder will change color as you sweat.

The pattern of your sweat can show if you are sweating normally.

The test usually takes 40-65 minutes to complete.

Bladder Ultrasound. This test is used to check for problems with your bladder, which is controlled by autonomic nerves.

The ultrasound is done after you urinate to see how much urine is left in the bladder. During the test:

You will lie on a table with your lower abdomen exposed.

Your provider will spread a special gel on the skin over that area.

Your provider will move a wand-like device, called a transducer, over the area.
The device sends sound waves into your body.
The waves are recorded and turned into images on a monitor.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will give you specific instructions on how to prepare, but many autonomic tests require the following preparations:

Don't eat or drink anything with caffeine for eight hours before your test.

Keep well hydrated with noncaffeinated beverages the day before your test.

Don't drink alcohol for 12 hours before your test.

Don't smoke or use tobacco products for four hours before your test.

Check with your provider about how to prepare for your test. They will also let you know if you need to stop taking any medicines before your test.

ARE THERE ANY RISKS TO THE TEST?

Some people faint during a tilt table test. If that happens, you will immediately be tilted back to a normal position. During and after a tilt table test, you may experience nausea, vomiting, or weakness. Weakness may last a few hours, but other symptoms usually go away quickly.

There are no known risks to having an ultrasound.

You may have some skin irritation from the electrodes used in a sweat test.

WHAT DO THE RESULTS MEAN?

If any of your autonomic test results were not normal, it likely means that there is a problem with your ANS. To make a diagnosis, your provider will also consider your medical history, symptoms, and results from your physical exam. Sometimes they will order more tests, especially if the cause of your ANS problem is unknown.

If another disease is causing your ANS problems, treating that disease may help. Otherwise, treatment focuses on helping with your symptoms. This could include medicines and/or lifestyle changes.

If you have questions about your results, talk to your provider.

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Bacteria Culture Test

<https://medlineplus.gov/lab-tests/bacteria-culture-test/>

BACTERIA CULTURE TEST

WHAT IS A BACTERIA CULTURE TEST?

A bacteria culture test can determine if you have a bacterial infection and, if so, which type of bacteria is causing it. To do a bacterial culture test, you will need to give a sample of your blood, urine, skin, or other tissue. The type of sample depends on where the infection is located.

Bacteria are one-celled organisms. There are many different kinds of bacteria. They live just about everywhere in your body and on your skin. Some types of bacteria are harmless or even helpful, while other bacteria can cause infections and disease.

Knowing what type of bacteria is causing your infection helps determine the best treatment. Your health care provider will need to examine many bacteria cells to find out the type of bacteria. Your sample will be sent to a lab where the bacteria cells will be put in a dish with a special substance that allows bacteria to grow. It normally takes one to two days to grow enough bacteria for the test. So your test results are often ready within a few days. But some types of bacteria grow slowly, so sometimes your results may take several days or longer.

WHAT IS IT USED FOR?

Bacteria culture tests are used to help diagnose certain types of infections. The most common types of bacteria tests and their uses are:

Throat Culture

Used to diagnose or rule out strep throat.

Test procedure:

Your provider uses a special swab to take a sample from the back of your throat and tonsils.

Your provider uses a special swab to take a sample from the back of your throat and tonsils.

Urine Culture

Used to diagnose a urinary tract infection and identify the bacteria causing the infection.

Test procedure:

You provide a sterile urine sample in a container, following special instructions.

You provide a sterile urine sample in a container, following special instructions.

Sputum Culture

Sputum, also known as phlegm, is a thick type of mucus that is made in your lungs and can be coughed up. It is different from spit or saliva.

Used to help diagnose bacterial or fungal infections in your respiratory tract, such as bacterial pneumonia, tuberculosis, bronchiectasis, and histoplasmosis.

Test procedure:

You provide a sputum sample, usually by coughing it up into a special cup. In certain cases, you may need to have a bronchoscopy to get a sample. A bronchoscopy involves inserting a flexible tube through your nose or mouth and into your lungs.

You provide a sputum sample, usually by coughing it up into a special cup. In certain cases, you may need to have a bronchoscopy to get a sample. A bronchoscopy involves inserting a flexible tube through your nose or mouth and into your lungs.

Blood Culture

Used to look for bacterial or fungal infections in your blood.

Test procedure:

A health care professional takes a blood sample, usually from a vein in your arm.

A health care professional takes a blood sample, usually from a vein in your arm.

Stool Culture

Another name for stool (poop) is feces.

Used to detect bacterial infections in your digestive system, including food poisoning. Because many things can cause digestive illness, this test is often done with other tests to look for viruses and parasites that may be causing symptoms.

Test procedure:

You provide a sample of your feces in a clean container.

You provide a sample of your feces in a clean container.

Wound Culture

Used to find infections in open wounds or on burn injuries.

Test procedure:

A health care professional uses a special swab to collect a sample of cells or pus from your wound. For deeper wounds, a syringe may be used to draw out fluid, or you may have a biopsy to remove a piece of tissue from the wound.

A health care professional uses a special swab to collect a sample of cells or pus from your wound. For deeper wounds, a syringe may be used to draw out fluid, or you may have a biopsy to remove a piece of tissue from the wound.

WHY DO I NEED A BACTERIA CULTURE TEST?

Your provider may order a bacteria culture test if you have symptoms of a bacterial infection. The symptoms vary depending on the type of infection.

WHY DO I HAVE TO WAIT SO LONG FOR MY RESULTS?

Bacteria culture tests require a large number of cells to accurately identify what type of bacteria may be causing an infection. Most test samples don't include enough cells for that. So, your sample is sent to a lab and put in a dish with a special substance that allows bacteria cells to grow until there are enough to test. Most disease-causing bacteria will be ready for testing within one to two days, but some types of bacteria take five days or longer to grow enough cells.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

There are many different types of bacteria culture tests. Ask your provider if you need to do anything to prepare for your test.

ARE THERE ANY RISKS TO THE TEST?

There are no known risks to having a swab or blood test or to providing a urine, sputum, wound, or stool sample.

WHAT DO THE RESULTS MEAN?

If harmful bacteria are found in your sample, it usually means you have a bacterial infection. The test should also identify what type of bacteria is causing the infection. Your provider may order more tests to confirm the diagnosis or to see how serious the infection is.

Your provider may also order a test to determine which medicine will best treat the type of bacteria causing your infection. This test is called an antibiotic sensitivity test or a susceptibility test. It checks to see how sensitive the bacteria are to different antibiotic medicines. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I SHOULD KNOW ABOUT A BACTERIA CULTURE TEST?

If your results show you don't have a bacterial infection, you should not take antibiotic medicines. Antibiotics only treat bacterial infections. Taking antibiotics when you don't need them won't help you feel better and may lead to a serious problem known as antibiotic resistance.

Antibiotic resistance allows harmful bacteria to change in a way that makes antibiotics less effective or not effective at all. Resistant bacteria may continue to grow and multiply, making infections difficult and sometimes impossible to treat. These infections can spread to other people.

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Bacterial Vaginosis Test

<https://medlineplus.gov/lab-tests/bacterial-vaginosis-test/>

BACTERIAL VAGINOSIS TEST

WHAT IS A BACTERIAL VAGINOSIS (BV) TEST?

Bacterial vaginosis (BV) is a bacterial infection in the vagina. A healthy vagina contains a balance of both "good" (healthy) and "harmful" (unhealthy) bacteria. Normally, the good bacteria keep the harmful type under control.

A BV infection happens when the normal balance is upset and more harmful bacteria grow than good bacteria. A BV test checks a sample of vaginal discharge (fluid that comes out of the vagina) to look for harmful bacteria.

Any woman can get BV, even women who have never had sex. But it's most common in women who are sexually active. Sometimes a BV infection doesn't cause symptoms, and it can go away on its own. But serious health problems are linked to untreated BV. If you have symptoms, testing and treatment can help avoid the increased risk of these problems:

Getting a sexually transmitted disease (STD). BV makes your vagina more acidic than normal, which increases your chance of getting STDs, such as chlamydia, genital herpes, gonorrhea, or HIV. And if you have HIV and BV, you're more likely to pass HIV to your sex partner.

Preterm birth. If you have a BV infection during pregnancy, it can increase the chance that your baby will be born too early (premature baby) or have a low birth weight (less than 5 and a half pounds). Low birth weight can cause serious health problems for your baby, including difficulty breathing, gaining weight, and fighting infections.

There are different ways to check a sample of vaginal discharge for signs of BV:

Using a microscope. This is called a "vaginal smear" or a "wet mount." The sample is prepared on a glass slide and checked under a microscope for bacteria, white blood cells, and other cells that are signs of infection.

Checking for odor. This is called a "whiff test." The sample is mixed with a chemical to see if it makes a strong, fishy odor, which is a sign of BV.

Testing vaginal pH. This test checks the acidity of the sample. High acidity is a sign of BV. But a pH test alone can't diagnose BV, so it's used with other tests.

PCR test. This test checks your sample for genetic material from harmful bacteria. It's not used as much as the other three tests, because it's more expensive.

Other names: vaginal pH test, KOH test, wet mount test

WHAT IS IT USED FOR?

A BV test is used to find out if a BV infection is the cause of symptoms that appear to be from a vaginal infection.

WHY DO I NEED A BV TEST?

You may need testing if you have symptoms of BV. These include:

An unusual vaginal discharge that can be gray or white. It may also be foamy or watery.

A strong, fish-like odor, especially after sex

Pain and/or itching in the vagina

Itching around the outside of the vagina

Burning feeling when urinating (peeing)

You may also need a test if you have a female sex partner who has BV or symptoms that could be BV.

WHAT HAPPENS DURING A BV TEST?

A BV test is done much like a pelvic exam:

You'll take off your clothes below your waist and wear gown or use a sheet as a cover.

You'll lie on an exam table with your feet in stirrups.

Your health care provider will use a plastic or metal instrument called a speculum to gently widen your vagina so the cervix can be seen.

Your provider will use a cotton swab or wooden stick to collect a sample of your vaginal discharge.

Your provider may examine the sample or send it to a lab for testing.

At-home test kits are available to test for the bacteria that cause BV. The kits include everything you need to collect a sample of vaginal fluid and send it a lab for testing. To get proper care, share your results with your provider.

There are also at-home vaginal pH test kits. These kits use a swab or paper strip that changes color after you moisten it with vaginal fluid. A chart shows you what the color means. A high pH (high acidity) is a sign that you may have BV, but it doesn't mean you have BV for sure. And a normal pH doesn't mean you don't have BV. So, discuss your results with your provider.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Follow the instructions your provider gives you to prepare for this test. For 24 hours before your test, you'll usually need to avoid:

- Using tampons

- Douching (rinsing the vagina with water or other fluid)

- Using creams or medicines in the vagina

- Having vaginal sex

ARE THERE ANY RISKS TO THE TEST?

With a BV test from your provider, you may feel some mild discomfort when the speculum is put in your vagina.

WHAT DO THE RESULTS MEAN?

Abnormal results from a vaginal smear or a whiff test mean that you have a bacterial infection.

The results of a vaginal pH test will be a number that shows how acidic your vagina is. A pH over 4.5 is high, which means you're more likely to have BV. But a pH test can't diagnose BV.

If your results show you have a BV infection, your provider will probably prescribe antibiotic pills and/or antibiotic creams or gels that you put in your vagina. Medicines that treat BV are safe to use during pregnancy.

Sometimes a BV infection will come back after successful treatment. If this happens, your provider may prescribe different medicine or a different dose of the medicine you took before.

There are no over-the-counter treatments for BV. Treatments for vaginal yeast infections will not help BV and could make your symptoms worse. So, don't try to treat an infection yourself.

If your results show you don't have BV, your provider may do more tests to find the cause of your symptoms.

If you have questions about your results, talk with your health care provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BV TEST?

BV is not spread through sex between female and male partners. So, if you're diagnosed with BV and have a male sex partner, he will not need to be tested. But the infection can be spread between female sex partners. If you have an infection and your partner is female, she should get a BV test.

Researchers aren't sure how BV spreads, but there are steps you can take that may lower your risk of infection:

- Limit your number of sex partners or don't have sex

- Don't use douches

- Use condoms every time you have sex

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Balance Tests

<https://medlineplus.gov/lab-tests/balance-tests/>

BALANCE TESTS

WHAT ARE BALANCE TESTS?

Balance tests are a group of tests that check for balance disorders. A balance disorder is a condition that makes you feel unsteady or dizzy. Dizziness means different things to different people. It can include:

Vertigo, a feeling that you or everything around you is spinning

Feeling as if you're going to fall

Feeling lightheaded or as if you are going to faint

Balance disorders can be mild or so severe that you may have trouble walking, climbing stairs, or doing other daily activities. They can happen at any age, but they are more common in older people. Balance disorders are one of the main reasons that older adults tend to have more falls than younger people.

Good balance depends on your brain receiving signals about your position and movement from your ears, eyes, and the muscles and touch sensors in your legs. These signals also help your eyes stay focused on objects when you change position. Problems with any of the signals that are part of your sense of balance can cause dizziness and other symptoms.

The part of your ears that controls your sense of balance is called the vestibular system or "the labyrinth." It is in your inner ear. It includes special organs filled with fluid and lined with sensors that have hair-like structures. When you move your head, the fluid in your inner ear moves the hair sensors. This triggers them to send nerve signals to your brain about your head's position and which direction it's moving.

Balance disorders may be caused by conditions in your inner ear, head injuries, or certain medicines and medical conditions that affect your inner ear or brain. Other conditions can also cause problems with balance, such as vision problems, heart disease, blood vessel disorders, migraine headaches, and arthritis.

Balance tests help find the cause of balance problems so that you can get the right treatment to improve your balance and avoid falls.

Other names: vestibular balance testing, vestibular testing

WHAT ARE THEY USED FOR?

Balance tests are used to help find the cause of balance problems, so you can get the right treatment to improve your balance and avoid falls. There are many types of balance disorders, including:

Benign paroxysmal positional vertigo (BPPV). BPPV is the most common cause of vertigo in adults. With this condition, you feel a spinning sensation when you move your head in certain ways. Normally, part of your inner ear has grains of calcium that trigger signals that tell your brain about your head's position and movement. BPPV happens if these grains slip out of place and get into a part of your inner ear where they don't belong. When the grains move around in the wrong place, they trigger confusing signals about how your head is moving.

Labyrinthitis. This balance disorder can cause dizziness and temporary hearing loss. It happens if your inner ear becomes irritated and swollen, usually from a cold or flu.

Meniere's disease. This disorder causes vertigo, hearing loss, tinnitus (ringing in the ears), and a full feeling in your ears. It happens when too much fluid builds up in the vestibular system. The cause is unknown.

Vestibular neuritis. This is inflammation (swelling) of the vestibular nerve. This nerve carries signals from the vestibular system in the inner ear to your brain. It's usually caused by a virus. The main symptoms are nausea and vertigo.

Perilymph fistula. This happens when fluid in the inner ear leaks into the middle ear, where your eardrum is. It causes an unsteady feeling, dizziness, and nausea. The symptoms usually get worse with activity. Causes include head injury, ear surgery, and having many long-lasting ear infections.

Mal de Debarquement syndrome (MdDS). This condition is triggered after long periods of motion, such as being on a boat or running on a treadmill. After you stop moving, you still have a swaying or bobbing feeling that can last for hours or days. Severe cases can last months or even years. The cause is unknown.

WHY DO I NEED A BALANCE TEST?

You may need balance tests if you have symptoms of a balance disorder. Symptoms include:

Dizziness

Vertigo (a spinning feeling)

Falling or feeling as if you are going to fall

Staggering when you walk

Feeling lightheaded, like you are going to faint, or a floating sensation

Blurred vision

Confusion

Other symptoms may include nausea and vomiting, diarrhea, changes in heart rate and blood pressure, and anxiety.

If you're age 65 or older, your health care provider may use simple screening tests to check your balance even if you don't have symptoms. This helps check your risk for falls so you can make changes to reduce your risk if needed.

WHAT HAPPENS DURING A BALANCE TEST?

Balance disorders can be difficult to diagnose. If you have symptoms of a problem, your provider may suggest that you see a specialist in disorders of the ear. These include:

An audiologist, a health care provider who is trained to diagnose hearing loss and balance disorders. They also can provide certain treatments to improve these conditions.

An otolaryngologist (ENT), a doctor who specializes in treating diseases of the ears, nose, throat, and head and neck.

You may need several tests to help find out if you have a disorder in your inner ear. Common tests include:

Electronystagmography (ENG) or videonystagmography (VNG) tests. These tests measure eye movements that you can't control, called nystagmus. Normally, these eye movements happen briefly when you move your head into certain positions. If they happen at other times or don't happen when they should, you may have a problem in your inner ear.

During the test:

You will sit in an exam chair in a dark room.

For an ENG, electrodes will be placed near your eyes to record their movement. The electrodes are small sensors that either stick to your skin or are put in a headband that you wear.

For a VNG, you'll wear special goggles that record your eye movements.

For an ENG, electrodes will be placed near your eyes to record their movement. The electrodes are small sensors that either stick to your skin or are put in a headband that you wear.

For a VNG, you'll wear special goggles that record your eye movements.

You'll follow a light on a screen with your eyes.

You'll be asked to move your head and body into different positions. The provider will also move your head into certain positions.

Warm and cool water or air will be put in one ear at a time. This should make your eyes move in specific ways.

Rotary test, also called a rotary chair test. This test measures how well your eyes and inner ear are working together to help you keep your balance. During this test:

You'll sit in a computer-controlled chair with a motor that makes it move.

You'll wear special goggles that records your eye movements as the chair moves back and forth and turns at different speeds.

Posturography, also known as computerized dynamic posturography (CDP). This test measures how well you can keep your balance while standing. It can help find out whether a balance disorder is caused by a problem with your inner ear, eyes, or nerve signals from your feet and legs. During this test:

You'll stand barefoot on a platform facing a screen.

You'll wear a safety harness to catch you if you lose your balance.

Your balance will be measured when the platform is still and when it's moving while:

Your eyes are opened and closed

You are looking at a moving image on the screen in front of you
Your eyes are opened and closed
You are looking at a moving image on the screen in front of you
Other tests include:

Vestibular evoked myogenic potentials (VEMP) tests to measure how certain parts of your inner ear are working. This test involves lying back in a chair and listening to sounds through earphones while you lift and turn your head. Electrodes attached to your neck and face record your muscle movements.

Dix-Hallpike maneuver to check how your eyes move when your provider quickly moves your head in different positions. This test can show whether vertigo is caused by changes in head position or a more serious condition. A newer version of this test is called a video head impulse test (vHIT). For this test, you wear goggles that record your eye movements while your provider moves your head.

Tests for balance disorders may also include hearing tests. That's because many balance disorders are related to hearing problems, such as tinnitus. You may also have imaging tests of your head and brain.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A BALANCE TEST?

Your provider will let you know how to prepare for your tests. It's best to wear loose, comfortable clothes. Depending on the test, you may need to avoid certain foods or medicines before your test. But don't stop taking any medicine without talking with your provider first.

ARE THERE ANY RISKS TO BALANCE TESTS?

Certain tests may make you feel dizzy or sick to your stomach. But these feelings usually go away in a few minutes. You may want to plan to have someone bring you home in case the dizziness lasts longer.

WHAT DO THE RESULTS MEAN?

If your results show that you may have a balance disorder, your provider may order more tests and/or start treatment. Treatment depends on the type of disorder you have. For example:

For benign paroxysmal positional vertigo (BPPV), your provider may move your head in different positions to move calcium grains in your inner ear back where they belong. The procedure is called the Epley maneuver or canalith repositioning.

For Meniere's disease, your provider may suggest diet and lifestyle changes, such as increasing physical activity and quitting smoking. Medicines may also help.

If your balance problems are from medicine or a condition that's not in your inner ear, changing medicine or treating the condition may help.

If treatment doesn't help enough, balance retraining therapy, also called vestibular rehabilitation, may help you learn to cope with your symptoms and reduce your risk of falling.

If you have questions about the results of your balance tests, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Barium Swallow

<https://medlineplus.gov/lab-tests/barium-swallow/>

BARIUM SWALLOW

WHAT IS A BARIUM SWALLOW?

A barium swallow, also called an esophagogram, is an imaging test that checks for problems in your upper gastrointestinal (GI) tract. Your upper GI tract includes your:

Mouth

Back of the throat

Esophagus (the tube that carries food and liquids from your mouth to the stomach)

Stomach

Duodenum (first part of your small intestine)

A barium swallow test involves drinking a chalky-tasting liquid that contains barium. Barium is a substance that coats your internal organs and tissues, so they can be seen more clearly on an x-ray. This can allow your health care provider to find problems and abnormal areas in your GI tract.

The test uses a special type of x-ray called fluoroscopy. Fluoroscopy shows internal organs moving in real time as a live video. It allows your provider to see how you swallow and how some of your other internal organs look. Fluoroscopy may be used to diagnose certain health conditions or to guide treatment procedures such as surgeries.

Other names: esophagogram, esophagram, upper GI series, swallowing study, esophagography, modified barium swallow study, video fluoroscopy swallow study.

WHAT IS IT USED FOR?

A barium swallow is used to see what happens when you swallow and to help diagnose conditions that affect the throat, esophagus, stomach, and first part the small intestine. These conditions can include:

Ulcers

GERD (gastroesophageal reflux disease), a condition in which contents of the stomach leak backward into the esophagus

Hiatal hernia, a condition in which part of your stomach pushes into the diaphragm (the muscle between your stomach and chest). A hiatal hernia can cause GERD.

Structural problems in the GI tract, such as polyps (abnormal growths) and diverticula (pouches in the intestinal wall)

Tumors

WHY DO I NEED A BARIUM SWALLOW?

Your provider may order this test to check how well an upper GI organ works. You may need this test if you have symptoms of an upper GI disorder, such as:

Trouble swallowing

Abdominal (belly) pain

Vomiting

Bloating

WHAT HAPPENS DURING A BARIUM SWALLOW?

A barium swallow is most often done by a radiologist or radiology technician. A radiologist is a doctor who specializes in using imaging tests to diagnose and treat diseases and injuries.

A barium swallow usually includes the following steps:

You may need to remove your clothing. If so, you will be given a hospital gown.

You may be given a lead shield or apron to wear over your pelvic area. This protects the area from unnecessary radiation.

You will stand, sit, or lie down on an x-ray table. You may be asked to change positions during the test.

The radiologist may take x-rays of your chest and belly first.

You will swallow a drink that contains barium. The drink is thick and chalky. It's usually flavored with chocolate or strawberry to make it easier to swallow.

While you swallow, the radiologist will watch images of the barium traveling down your throat to your upper GI tract.

You may be asked to hold your breath at certain times.

The images will be recorded so they can be reviewed later.

The procedure may take from 30 to 60 minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know if there are any special instructions to follow. You will probably be asked to fast (not eat or drink) after midnight on the night before the test. You may be asked to avoid anything that might coat your throat, such as smoking, chewing gum, or sucking on hard candy.

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

You should not have this test if you are pregnant or think you may be pregnant. Radiation can be harmful to a developing baby.

For others, there is little risk to having this test. The radiation dose is very low and not considered harmful for most people. But talk to your provider about all the x-rays you've had recently. The risks from radiation exposure may be linked to the number of x-ray treatments you've had over time.

Tell your provider if you have any known allergies to the barium used in the test.

Your poop may look white for several days after the test as you pass the barium. You may have constipation if all of the barium doesn't pass out of your body after the procedure. Drinking lots of fluids and eating high-fiber foods may help. But if you continue to have constipation, contact your provider.

WHAT DO THE RESULTS MEAN?

A normal result means that no abnormalities in size, shape, and movement were found in your upper GI tract.

If your results were not normal, it may mean you have one of the following conditions:

Hiatal hernia

Ulcers

Tumors

Polyps

Diverticula, a condition in which small sacs form in the inner wall of the intestine

Esophageal stricture, a narrowing of the esophagus that can make it hard to swallow

If you have questions about your results, talk to your provider. Your provider may consider your symptoms, medical history, and the results of other tests to understand the results of your barium swallow.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BARIUM SWALLOW?

Your results may also show signs of esophageal cancer. If your provider thinks you may have it, they may do a procedure called an esophagoscopy. During an esophagoscopy, a thin, flexible tube is inserted through your mouth or

nose and down into your esophagus. The tube has a video camera so a provider can view the area. The tube may also have a tool attached that can be used to remove tissue samples for testing (biopsy).

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RELATED HEALTH TOPICS

Benign Tumors

- Cancer
- Esophageal Cancer
- Esophagus Disorders
- Hiatal Hernia
- Peptic Ulcer
- Stomach Disorders
- Throat Disorders
- X-Rays

RELATED MEDICAL TESTS

- Dysphagia Tests
- Esophageal pH Test
- Fluoroscopy
- Gastric Emptying Tests
- Helicobacter Pylori (H. Pylori) Tests
- How to Cope with Medical Test Anxiety
- How to Understand Your Lab Results

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Basic Metabolic Panel (BMP)

<https://medlineplus.gov/lab-tests/basic-metabolic-panel-bmp/>

BASIC METABOLIC PANEL (BMP)

WHAT IS A BASIC METABOLIC PANEL (BMP)?

A basic metabolic panel (BMP) measures eight different substances in your blood. It provides important information about your body's fluid balance, your metabolism (the process your body uses to make energy from food you eat), and how well your kidneys are working. This test is often used to check your general health.

A BMP test includes measurements of the following:

Glucose, a type of sugar and your body's main energy source. If your glucose level is high, it may be a sign of diabetes.

Calcium, one of the body's most important minerals. Calcium is essential for the proper functioning of your nerves, muscles, and heart. It also helps with blood clotting, an important process that prevents you from losing too much blood when you get injured.

Sodium, potassium, carbon dioxide, and chloride. These are electrolytes, which are electrically charged minerals that help control the amount of fluids and the balance of acids and bases (pH balance) in your body. They help move nutrients into your cells and waste out. They also help your nerves, muscles, heart, and brain work well.

BUN (blood urea nitrogen) and creatinine, waste products removed from your blood by your kidneys.

Abnormal levels of any of these substances or a combination of them can be a sign of a health problem.

Other names: BMP, chemistry panel, chemistry screen, chem 7, electrolyte panel

WHAT IS IT USED FOR?

A BMP can be used to screen or diagnose health conditions or monitor treatment. This test may also be used to check how well your body functions. It can provide information on your:

Kidney function

Fluid and electrolyte balance

Blood glucose levels

Acid and base balance

Metabolism

WHY DO I NEED A BMP?

A BMP is often done as part of a regular checkup. This may help your health care provider find health problems before you have any symptoms.

You may also need this test if you:

Have general symptoms such as fatigue, confusion, breathing problems, or vomiting that lasts a long time

Are being treated in the emergency room

Are being monitored for a certain chronic (long-lasting) condition, such as high blood pressure or kidney disease

WHAT HAPPENS DURING A BMP?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for eight hours before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If any one result or combination of BMP results were not normal, it could be a sign of several different health conditions. These include kidney disease, breathing problems, and complications from diabetes. You will likely need more tests to confirm or rule out a specific diagnosis.

To understand what your test results may mean, your provider may also consider your symptoms, your medical history, and the results of other tests you had. If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BMP?

There is a similar test to a BMP called a comprehensive metabolic panel (CMP). A CMP includes the same eight tests as a BMP, plus six more tests, which measure certain proteins and liver enzymes. The additional tests are:

Albumin, a protein made in the liver

Total protein, which measures the total amount of protein in the blood

ALP (alkaline phosphatase), ALT (alanine transaminase), and AST (aspartate aminotransferase). These are different enzymes (proteins that speed up certain chemical reactions in your body) made by the liver.

Bilirubin, a waste product made by the liver

Your provider may order a CMP instead of a BMP to get a more complete picture of the health of your organs or to check for liver disease or other specific conditions.

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RELATED HEALTH TOPICS

Calcium
Fluid and Electrolyte Balance
Kidney Diseases
Potassium
Sodium

RELATED MEDICAL TESTS

Blood Glucose Test
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Calcium Blood Test
Carbon Dioxide (CO₂) in Blood
Chloride Blood Test
Comprehensive Metabolic Panel (CMP)
Creatinine Test
Electrolyte Panel
Fasting for a Blood Test
How to Cope with Medical Test Anxiety
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Potassium Blood Test
Sodium Blood Test
What You Need to Know About Blood Testing

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BCR-ABL1 Genetic Test

<https://medlineplus.gov/lab-tests/bcr-abl1-genetic-test/>

BCR-ABL1 GENETIC TEST

WHAT IS A BCR-ABL1 GENETIC TEST?

A BCR-ABL1 genetic test uses a sample of blood or bone marrow to look for an abnormal gene called BCR-ABL1. This abnormal gene is involved in certain types of blood cancer. It's found in:

Most people who have chronic myeloid leukemia (CML), which is also called chronic granulocytic leukemia or chronic myeloid leukemia

Some people who have certain types of acute lymphoblastic leukemia (ALL)

Some people with acute myeloid leukemia (AML), but very rarely

The abnormal BCR-ABL1 gene is made when two separate genes merge together and become one gene. Genes are short sections of DNA. DNA is made of long strands of material that carry information that controls what you look like and how your body works. DNA is packaged into structures in your cells called chromosomes. Normally, most of your cells have the same set of 23 pairs of chromosomes.

The abnormal BCR-ABL1 gene is formed when pieces of chromosomes 9 and 22 break off and trade places. The broken piece of chromosome 9 includes part of the ABL1 gene. It attaches (fuses) to part of the BCR gene on chromosome 22. This makes the abnormal fusion gene, BCR-ABL1. The changed chromosome 22 is called a "Philadelphia chromosome" after the city where researchers discovered it.

Your genes carry special instructions for making proteins that help your cells do their work. But the abnormal BCR-ABL1 gene has instructions for an abnormal protein. The protein causes certain bone marrow cells to make large numbers of abnormal white blood cells. These abnormal white blood cells are leukemia cells. They don't protect you from infections like healthy white blood cells do. They can build up in the blood and bone marrow and crowd out healthy blood cells.

There are different types of BCR-ABL1 genetic tests. They check your blood or bone marrow for:

The abnormal BCR-ABL1 gene

Changes in chromosome 22 that show it has become a Philadelphia chromosome

An abnormal BCR-ABL1 gene is an acquired genetic change. That means you don't inherit it from your parents, and you can't pass it on to your children. Acquired changes happen after birth if cells make a mistake when dividing to make new cells or if your DNA is damaged from exposure to substances that cause cancer. The changes show up only in certain types of cells. Changes in your genes are also called mutations or variants.

Other names: BCR-ABL, BCR-ABL fusion, Philadelphia chromosome, B-cell acute lymphoblastic leukemia (B-ALL), chronic myelogenous leukemia (CML), t(9;22), BCR-ABL1 Transcript Detection by RT-PCR, Quantitative BCR-ABL1 Translocation Detection by RT-PCR

WHAT IS IT USED FOR?

BCR-ABL1 genetic testing is most often used with other tests to diagnose or rule out two specific types of blood cancer: Chronic myeloid leukemia (CML).

Philadelphia-positive acute lymphoblastic leukemia (Ph+ ALL). This is a type of acute lymphoblastic leukemia that has the Philadelphia chromosome.

Certain types of BCR-ABL1 testing may also be used to help plan treatment for these blood cancers. There are several treatment options. Some medicines target the protein made by the BCR-ABL1 gene. In general, these types of medicines cause less harm to normal cells than cancer chemotherapy or radiation therapy do.

During treatment, BCR-ABL1 tests may be used to see if cancer treatment working. And after treatment is finished, testing is often used to see if cancer is coming back.

WHY DO I NEED A BCR-ABL1 GENETIC TEST?

If your health care provider thinks you may have a type of leukemia that involves the BCR-ABL1 gene, you may need a BCR-ABL1 genetic test to make a diagnosis. Signs and symptoms of these blood cancers include:

High levels of white blood cells on a complete blood count (CBC) test

Fatigue

Fever

Weight loss

Night sweats (heavy sweating during sleep)

Joint or bone pain

Pain or fullness below your ribs on the left side

Painless lumps in your armpits, groin, neck, or belly

If you have already been diagnosed with a type of leukemia that involves a BCR-ABL1 gene, you may need this test to:

Help guide treatment decisions

See how well your treatment is working (if treatment is helping, the number of cells with BCR-ABL1 genes will decrease.)

Monitor your health after treatment to see if blood cancer is coming back

WHAT HAPPENS DURING A BCR-ABL1 GENETIC TEST?

A BCR-ABL1 genetic test usually uses a sample of blood or bone marrow.

If you are having a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

If you are having a bone marrow test, a provider usually takes a sample from the back of the hip bone. You will be given an injection (shot) of medicine to numb the area. You may also be given medicine to help you relax or sleep.

A bone marrow test has two parts that are usually done one after the other:

For a bone marrow aspiration, the provider pushes a thin needle into the bone and removes a sample of bone marrow fluid with a syringe attached to the needle.

For a bone marrow biopsy, the provider inserts a hollow needle into the bone and twists the needle to take out a small piece of bone marrow tissue. You may feel some pressure or brief pain while the sample is being taken.

The procedure takes only a few minutes. Afterwards, the area will be bandaged. You'll stay lying down for about 15 minutes to make sure the bleeding stops. You may need to stay longer if you had medicine to relax or sleep.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You usually don't need any special preparations for a blood test. If you're having a bone marrow test, your provider will tell you whether you need to fast (not eat or drink) for a few hours before the procedure.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

After a bone marrow test, you may feel stiff or sore where the sample was taken. This usually goes away in a few days. If you need pain relief, talk with your provider.

WHAT DO THE RESULTS MEAN?

If you had BCR-ABL1 genetic testing to diagnose a blood cancer, your provider will probably use the results of more than one test to make a diagnosis. Your provider will likely diagnose either chronic myeloid leukemia (CML) or Philadelphia chromosome-positive acute lymphoblastic leukemia (Ph+ ALL) if your tests show all three of these results:

Abnormal white blood cells

The BCR-ABL1 gene

The Philadelphia chromosome

In certain cases, a person may have CML without having the Philadelphia chromosome, but this isn't common.

If you had a BCR-ABL1 gene test to see if your cancer treatment is working, your provider will compare your results with past test results to look for changes over time. The way your results are reported depends on which test you had. Ask your provider to explain what your test results mean for your health and treatment plan. In general:

If the amount of BCR-ABL1 in your sample decreases, it means your treatment is working to some degree. Your test results may describe your response to treatment using words such as minor, minimal, partial, major, or complete.

If the amount of BCR-ABL1 in your sample increases, it may mean your treatment is not working. You may need to try another treatment or have more tests. An increase in BCR-ABL1 levels after successful treatment, may mean your cancer has come back.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BCR-ABL1 GENETIC TEST?

Labs use different methods to test for the BCR-ABL1 gene and Philadelphia chromosome. The test method can affect your results. So, it's important to have your tests done the same way, and usually in the same lab. This allows your provider to compare your results over time. If you have questions about how your tests are done, ask your provider.

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RELATED HEALTH TOPICS

- Acute Lymphocytic Leukemia
- Acute Myeloid Leukemia
- Bone Marrow Diseases
- Cancer
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- Genetic Disorders
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RELATED MEDICAL TESTS

- Beta 2 Microglobulin (B2M) Tumor Marker Test
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- Complete Blood Count (CBC)
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- Karyotype Genetic Test
- What You Need to Know About Blood Testing

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Beta 2 Microglobulin (B2M) Tumor Marker Test

<https://medlineplus.gov/lab-tests/beta-2-microglobulin-b2m-tumor-marker-test/>

BETA 2 MICROGLOBULIN (B2M) TUMOR MARKER TEST

WHAT IS A BETA-2 MICROGLOBULIN TUMOR MARKER TEST?

A beta-2 microglobulin tumor marker test¹ measures the amount of a protein called beta-2 microglobulin (B2M) in your body fluids. The test usually uses a sample of your blood. It may also be done using urine (pee) or in rare cases cerebrospinal fluid (CSF).

B2M is a small protein found on the surface of most cells in your body. Your cells release B2M when they are damaged or when they grow more quickly than normal. Usually, your kidneys filter B2M out of your blood. So, only small amounts of this protein are present in healthy people.

However, B2M is also a type of tumor marker. Tumor markers are substances that are often made by cancer cells or by normal cells in response to cancer. High levels of B2M in body fluids are often caused by blood and bone marrow cancers, such as:

- Multiple myeloma.

- Chronic lymphocytic leukemia (CLL).

- Certain types of lymphoma.

High B2M levels can also be caused by noncancerous conditions, such as kidney disease² and autoimmune disorders. Because high levels of B2M can be caused by such a wide variety of conditions, health care providers don't use the results from a beta-2 microglobulin tumor marker test to diagnose cancer or any other condition. Instead, it is often done after you've been diagnosed with certain cancers of the bone marrow or blood.

Other names: total beta-2 microglobulin, β_2 -microglobulin, B2M, B2MG, thymotaxin

WHAT IS IT USED FOR?

Though a beta-2 microglobulin tumor marker test cannot be used to screen for or diagnose cancer, it can provide valuable information about any bone marrow or blood cancer you already have. For example, testing your B2M levels can help your provider:

- Find out how much the cancer has grown or spread.

- Predict how the cancer might change (or progress) over time.

- Pick your treatment and check whether it's working. If B2M levels don't decrease during a treatment, your provider may need to adjust it.

- Check whether cancer has spread to your brain and/or spinal cord. This is done only if you have symptoms of these types of cancer.

WHY DO I NEED A BETA-2 MICROGLOBULIN TUMOR MARKER TEST?

You may need this test if you've been diagnosed with multiple myeloma, chronic lymphocytic leukemia, or certain types of lymphoma. A B2M test can help determine how serious your disease is, monitor your cancer over time, and check how well your treatment is working.

If you have multiple myeloma, a urine test for B2M may be used to see if your disease is affecting your kidneys.

WHAT HAPPENS DURING A BETA-2 MICROGLOBULIN TUMOR MARKER TEST?

A beta-2 microglobulin test usually uses a sample of your blood. In certain cases, you may need to provide a 24-hour urine sample, which means collecting all your urine for a full day. If your provider suspects that cancer has spread to your brain or spine, you may have a cerebrospinal fluid (CSF) analysis.

For a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a 24-hour urine sample, you will be given a special container to collect your urine and instructions on how to collect and store your samples. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate (pee) in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in the refrigerator or in a cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

For a cerebrospinal fluid (CSF) analysis, a provider will do a procedure called a spinal tap (also known as a lumbar puncture). A spinal tap is usually done in a hospital. A provider numbs your skin with medicine and uses a thin, hollow needle to remove some fluid from between the bones in your lower spine. You usually go home the same day.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

For blood and urine tests, there are no special preparations.

For a CSF analysis (spinal tap), you may be asked to empty your bladder (pee) and bowels (poop) before the test.

ARE THERE ANY RISKS TO THE TEST?

With a blood test, there is very little risk. After the test, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

With a urine test, there is no known risk.

With a CSF analysis, there is very little risk from a spinal tap. You may feel a little pinch or pressure when the needle is inserted. After the test, you may feel some pain or tenderness in your back or have some bleeding where the needle was inserted. You may also get a headache. The headache may last for several hours or up to a week or more. Your provider may suggest treatment to relieve the pain.

WHAT DO THE RESULTS MEAN?

If your B2M levels were measured to learn about your cancer:

The higher your B2M levels, the more cancer you have in your body. Higher levels are linked to cancers that tend to grow faster.

If you have multiple myeloma, higher levels of B2M are often linked to kidney problems.

If your B2M levels were measured to check how well your treatment is working, your provider may look at several test results over time to look for a trend:

Increasing B2M levels may mean your treatment is not working.

Decreasing B2M levels may mean your treatment is working.

B2M levels that don't change may mean that your disease is stable and hasn't gotten better or worse.

If you have questions about your test results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BETA-2 MICROGLOBULIN TUMOR MARKER TEST?

B2M levels may be higher than normal in many conditions that aren't cancer, including:

Kidney disease

HIV

Multiple sclerosis

Because of this, providers may test your B2M levels for reasons not related to cancer treatment. These can include:

Checking kidney function. B2M is filtered by the kidneys. So, high levels of B2M in your blood or urine may be an early sign of kidney injury.

Monitoring immune and autoimmune diseases. Increased B2M levels may show that your immune system is more active than normal. For example, people with primary Sjogren's syndrome¹ may have higher B2M levels in blood and even in tears.

Detecting transplant rejection. In liver transplant patients, high B2M levels in bile and blood may be an early sign that the body is rejecting the new organ.

Evaluating dialysis treatment. In people on long-term dialysis, B2M can build up in tissues (a condition called dialysis-related amyloidosis). Measuring B2M helps providers check how well the dialysis filter works.

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Bilirubin Blood Test

<https://medlineplus.gov/lab-tests/bilirubin-blood-test/>

BILIRUBIN BLOOD TEST

WHAT IS A BILIRUBIN BLOOD TEST?

A bilirubin blood test measures the levels of bilirubin in your blood. Bilirubin is a yellowish substance made during your body's normal process of breaking down old red blood cells. Bilirubin is found in bile, a fluid your liver makes to help you digest food.

If your liver is healthy, it will remove most of the bilirubin from your body through your bile ducts (the tubes that carry bile from your liver). If your liver is damaged or your bile ducts are blocked, bilirubin can leak out of your liver and into your blood. When too much bilirubin gets into the bloodstream, it can cause jaundice, a condition that causes your skin and eyes to turn yellow. If you have jaundice, your health care provider may order a bilirubin blood test to check for liver disease or damage.

Bilirubin is usually tested with a blood test but can also be tested in your urine.

Other names: Total serum bilirubin, TSB total bilirubin, TBIL neonatal bilirubin, direct bilirubin, conjugated bilirubin, indirect bilirubin, unconjugated bilirubin

WHAT IS IT USED FOR?

A bilirubin blood test may be used to check the health of your liver, look for the cause of jaundice, or determine how well a treatment is working.

The test is also commonly used to help diagnose newborn jaundice. Many healthy babies get jaundice because their livers haven't developed enough to get rid of the bilirubin. Newborn jaundice is usually not harmful and clears up within a few weeks. But in some cases, high bilirubin levels can lead to brain damage, so infants are often tested as a precaution.

WHY DO I NEED A BILIRUBIN BLOOD TEST?

Your health care provider may order a bilirubin blood test:

If you have symptoms such as jaundice, dark urine, clay-colored stool (poop), or stomach pain. These could indicate hepatitis, cirrhosis, or other liver diseases.

To find out if there is a blockage in your bile ducts.

To monitor an existing liver disease.

To diagnose disorders related to your body breaking down red blood cells too quickly. For example, high bilirubin levels in the bloodstream may be a sign of hemolytic anemia. This condition happens when your body destroys red blood cells faster than it makes them.

WHAT HAPPENS DURING A BILIRUBIN BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

If the test is done on your newborn, they will have a heel stick test. A health care professional will clean your baby's heel with alcohol and poke the heel with a small needle. They will collect a few drops of blood and put a bandage on the site.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider may tell you to fast (not eat or drink) for several hours before your blood test. You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. Your provider will let you know if there are any other special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The results will show if the amount of bilirubin in your blood is in the normal range. The normal ranges can vary from lab to lab.

If the results are outside your lab's reference range, you may need further testing. Normal results can vary based on the lab, your sex, and if the test is being done on an infant or child. but high bilirubin levels may mean your liver isn't working right.

High bilirubin levels (hyperbilirubinemia) may be due to:

- Hemolytic anemia

- Liver disease

- Blockage in your bile ducts

Low bilirubin levels (hypobilirubinemia) are not usually a cause for concern but may be due to:

- Medicines such as antibiotics, birth control pills, sleeping pills, and seizure medicines

Abnormal results don't always mean you have a medical condition that needs treatment. Higher-than-normal bilirubin levels can also be caused by some medicines, certain foods, or strenuous exercise. To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BILIRUBIN BLOOD TEST?

A bilirubin blood test is only one measure of your liver health. If your provider thinks you might have a liver disease or a red blood cell disorder, you may need other tests, such as:

- Liver function tests, a group of tests that measure different substances in your blood

- Liver protein tests

- Urine tests

- An ultrasound

- A liver biopsy, to get a sample of your liver tissue to examine under a microscope

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Bilirubin in Urine

<https://medlineplus.gov/lab-tests/bilirubin-in-urine/>

BILIRUBIN IN URINE

WHAT IS A BILIRUBIN IN URINE TEST?

A bilirubin in urine test measures the levels of bilirubin in your urine. Normally, bilirubin isn't found in urine. If there is bilirubin in your urine (called bilirubinuria), it may be an early sign of a liver condition.

Bilirubin is a yellow substance your body makes during the normal process of breaking down red blood cells. Your liver uses bilirubin to make bile, a fluid that helps you digest food in your intestines. Some bile flows through bile ducts (small tubes) from your liver directly into your intestines. The rest is stored in your gallbladder for when you need it.

A healthy liver removes most of the bilirubin from your body. But if there is a problem with your liver or bile ducts, bilirubin can build up in your blood and get into your urine.

Other names: urine test, urine analysis, UA, chemical urinalysis, direct bilirubin

WHAT IS IT USED FOR?

A bilirubin in urine test is often part of a urinalysis, which may include a visual check of your urine (pee) sample, tests for certain chemicals, and an examination under a microscope to look for certain types of cells. A urinalysis is often used to check your general health, including the health of your urinary tract (which includes your kidneys, ureters, bladder, and urethra). It can also help diagnose liver problems and many other medical conditions.

WHY DO I NEED A BILIRUBIN IN URINE TEST?

Your health care provider may have ordered a bilirubin in urine test as part of your routine checkup. The test may also be ordered if you have symptoms of a liver condition, gallbladder disease, or a problem with your bile ducts. These symptoms may include:

Nausea and vomiting

Lack of appetite

Fatigue

Weakness

Jaundice, a condition that causes your skin and eyes to turn yellow

Swelling and/or pain in your abdomen (belly)

Swelling in your ankles and legs

Dark-colored urine (pee) and/or light-colored stool (poop)

Frequent itching

Bilirubin in urine can be an early sign of liver damage. It may even show up before you have symptoms. So, your provider may order a bilirubin in urine test if you have a high risk for liver damage or disease. Reasons for an increased risk may include the following:

A family history of liver disease

Alcohol use disorder (AUD)

Obesity

Diabetes

Taking certain medicines that can cause liver damage

Hepatitis or exposure to hepatitis

WHAT HAPPENS DURING A BILIRUBIN IN URINE TEST?

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a bilirubin in urine test. If your provider has ordered other urine or blood tests, you may need to fast (not eat or drink) for several hours before the test.

Certain medicines and supplements may affect your results, so let your provider know what you are taking. But don't stop taking any medicines unless your provider tells you to. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a urinalysis or a bilirubin in urine test.

WHAT DO THE RESULTS MEAN?

If bilirubin is found in your urine, it may be a sign that you have:

A liver disease, such as hepatitis or cirrhosis

A blockage in your bile ducts, the small tubes that carry bile out of your liver

To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I SHOULD KNOW ABOUT A BILIRUBIN IN URINE TEST?

A bilirubin in urine test is only one measure of your liver health. If your test results aren't normal, your provider may order other tests, including a liver panel. A liver panel is a group of blood tests that measures many different enzymes, proteins, and other substances in the liver. It is often used to help find and diagnose liver disease.

Some at-home urine tests that you buy without a prescription may measure bilirubin in urine. These tests usually involve dipping a test strip in a cup of urine or placing a few drops of urine on the strip. If you do an at-home bilirubin in urine test, talk with your provider about your results.

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Blood Alcohol Level

<https://medlineplus.gov/lab-tests/blood-alcohol-level/>

BLOOD ALCOHOL LEVEL

WHAT IS A BLOOD ALCOHOL TEST?

A blood alcohol test measures the amount of alcohol (also called ethanol) in a sample of your blood. Alcohol is the ingredient in beer, wine, and distilled spirits (liquor) that makes you feel intoxicated (drunk).

When you have an alcoholic drink, your digestive tract quickly absorbs the alcohol. It then enters your bloodstream and stays there until your liver can break it down and filter it out of your blood. Your liver needs time to do this. If you consume alcohol faster than your liver can get rid of it, your blood alcohol levels will rise. The higher these levels rise, the more intoxicated you'll feel.

There are a few blood tests that can measure whether a person has been drinking alcohol. The most common is the blood alcohol concentration/content (BAC) test. This test shows the amount of alcohol in your blood for up to 12 hours after drinking.

Other blood tests measure substances that may stay in your blood for weeks after you've had alcohol. These tests are typically used to test for alcohol use disorder (AUD) or to check whether a person recovering from AUD has stayed sober.

Other names: blood alcohol level test, ethanol test, ethyl alcohol, blood alcohol content

WHAT IS IT USED FOR?

A blood alcohol test is most often used to determine not only if alcohol is present in your blood but also to what extent it's affecting you. Alcohol can change various brain functions. If you've only consumed a small amount, these changes aren't often dangerous to your immediate health. Your mood might change, and your reaction time may be reduced. Larger amounts of alcohol, however, can be much more dangerous, leading to coma or even death.

You may need to take a blood alcohol test for a number of reasons, including:

Legal evidence. Law officials may use BAC tests to see if someone drank while underage, broke a parole agreement, or drove while above the legal BAC limit. For quick results, a police officer may use a breathalyzer, a device that checks your breath for alcohol. Or they may bring you to a medical facility to get a blood test, which can provide a more accurate reading of your BAC.

Employment. Your employer may test for alcohol use before hiring you. After you're hired, they may also test you to see if you're under the influence of alcohol while on the job, especially if you're ever involved in a work accident.

Testing for alcohol poisoning. Hospital emergency staff may use a blood alcohol test to check for this life-threatening condition that can happen if your blood alcohol level gets dangerously high from drinking alcohol. Young children can also get alcohol poisoning from drinking household products that contain alcohol, such as mouthwash and certain cold medicines.

Monitoring treatment for alcohol use disorder. Some programs may require testing to see if you have been able to avoid drinking.

WHY DO I NEED A BLOOD ALCOHOL TEST?

You may need a blood alcohol test if you are suspected of drunk driving and/or have symptoms of intoxication. These include:

Difficulty with balance and coordination.

Slurred speech.

Slowed reflexes.

Nausea and vomiting.

Mood changes.

Poor judgment.

You or your child may also need this test if there are symptoms of alcohol poisoning. In addition to the above symptoms, alcohol poisoning can cause:

Confusion.

Irregular breathing.

Seizures.

Low body temperature.

WHAT HAPPENS DURING A BLOOD ALCOHOL TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This process usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a blood alcohol test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Blood alcohol level results may be given in different ways, including percentage of blood alcohol content (BAC). The legal limit for BAC is usually 0.08%, but this limit is lower in certain states and for certain driver classifications. The effects of alcohol and your level of impairment increase as your BAC increases. Typical results are below.

0.01 to 0.05% BAC: You may feel relaxed, less alert, and have a slight loss of judgement.

0.06-.15% BAC: You may have slurred speech, reduced muscle coordination and impaired judgement, memory or balance.

0.16 to 0.30% BAC: In this range, you will feel more serious signs of intoxication such as difficulty walking and speaking. You may also feel drowsy, confused, and nauseated. You may experience gaps in memory (blackouts), vomiting or loss of consciousness.

Over 0.31% BAC: Levels are especially dangerous and may even be fatal. You may lose consciousness, have trouble breathing, or go into a coma.

The timing of this test can affect the accuracy of the results. A blood alcohol test can detect alcohol in your system for up to about 12 hours after your last drink. Other factors can also affect your BAC and your level of impairment, including:

Weight.

Age.

Sex.

Type of alcohol.

How quickly you drank.

How much you ate before you drank.

Medicines you're taking.

Sensitivity (tolerance) to alcohol.

If you have questions or concerns about your results, you may want to talk to a health care provider and/or a lawyer.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BLOOD ALCOHOL TEST?

Some employers or legal officials may rely on other methods to check for alcohol use besides a breath or blood test.

Hair. This test can show past alcohol use over a longer period.

Urine. This test can check for recent alcohol use. Some urine tests can detect alcohol use over a longer time frame.

Saliva. This method of testing is less common, but it can still be used to see if you've recently had an alcoholic drink.

Sweat. This method of testing is also less common, but it can still be used to find traces of alcohol.

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Blood Differential

<https://medlineplus.gov/lab-tests/blood-differential/>

BLOOD DIFFERENTIAL

WHAT IS A BLOOD DIFFERENTIAL TEST?

A blood differential test measures the amount of each type of white blood cell (WBC) that you have in your blood. This test can help diagnose or monitor an infection or condition that affects your immune system.

White blood cells are also called leukocytes and WBCs. They are a type of blood cell that is made in your bone marrow and found in your blood and lymph tissue (which is part of your lymphatic system). White blood cells are an important part of your immune system. When you are injured or sick, your white blood cells will travel through your bloodstream and tissues to where they are needed. There, they will help your body fight off the infection or disease.

There are five different types of white blood cells:

Neutrophils are the most common type of white blood cell. They are your body's main defense against infection when bacteria, viruses, or other germs enter your body.

Lymphocytes include two main types of white blood cells: B cells and T cells. B cells fight off invading viruses, bacteria, or toxins. Certain T cells can target and destroy your body's own cells, such as cancer cells and cells that have been infected by viruses.

Monocytes kill bacteria, viruses, and other germs that may make you sick. They also boost your body's immune response and clear away dead cells.

Eosinophils defend against parasites and infections. They are also involved in allergies and help control inflammation (swelling and redness).

Basophils release enzymes during allergic reactions and asthma attacks.

A blood differential test measures the amount and/or percentage of each type of white blood cell. Another test called a white blood count (WBC) measures the total number of these cells in your blood.

Other names for a blood differential test: Complete blood count (CBC) with differential, Differential, White blood cell differential count, Leukocyte differential count, Peripheral Differential

WHAT IS IT USED FOR?

A blood differential test is often part of a routine checkup. Because the five types of white blood cell do different jobs, measuring them separately can give your health care provider important information about your health.

The test can also help diagnose a variety of medical conditions, such as:

Infections

Autoimmune diseases such as rheumatoid arthritis

Leukemia and other types of cancer

WHY DO I NEED A BLOOD DIFFERENTIAL TEST?

A blood differential test is used for many reasons. Your provider may order the test to:

Monitor your overall health or as part of a routine checkup.

Diagnose the cause of your symptoms or your abnormal WBC count from another blood test. For example, if you have symptoms such as feeling unusually tired or weak, or have unexplained bruising or other symptoms, this test may help uncover the cause.

Monitor an existing blood disorder or related condition.

WHAT HAPPENS DURING A BLOOD DIFFERENTIAL TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

No special preparation is necessary.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. After the test, some people experience mild pain, dizziness, or bruising. These symptoms usually go away quickly.

WHAT DO THE RESULTS MEAN?

Your blood differential test results may be higher or lower than normal for many reasons.

Conditions that may cause a high white blood cell count, also called leukocytosis, can include:

Infections

Autoimmune or inflammatory disorders

Leukemia or Hodgkin disease

Allergic reactions

Certain steroids

Conditions that may cause a low white blood cell count, also called leukopenia, can include:

Bone marrow diseases

Chemotherapy

Certain medicines, such as antibiotics

Cancer

HIV

But abnormal results don't always mean you have a condition that needs medical treatment. Some other factors that may affect your results can include:

Physical injury or extreme stress

Smoking

Alcohol use

If your results seem abnormal, your provider may order other tests to help figure out the cause. If you have questions about your results, talk with your provider. To understand your results, your provider will also consider your symptoms, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

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- Pneumococcal Disease Tests
- What You Need to Know About Blood Testing
- White Blood Count (WBC)

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Blood Glucose Test

<https://medlineplus.gov/lab-tests/blood-glucose-test/>

BLOOD GLUCOSE TEST

WHAT IS A BLOOD GLUCOSE TEST?

A blood glucose test measures the glucose levels in your blood. Glucose is a type of sugar. It is your body's main source of energy. It comes from the food you eat. A hormone called insulin helps move glucose from your bloodstream into your cells for use as energy. If you have too much or too little glucose in your blood, it may be a sign of a serious medical condition.

High blood glucose levels (hyperglycemia) may be a sign of prediabetes or diabetes. Prediabetes is a condition in which your blood glucose levels are higher than what is healthy for you, but not high enough to be considered diabetes. If you have diabetes, your body can't make insulin or can't use it as well as it should, or both. Too much glucose stays in your blood and doesn't reach your cells. This can cause glucose levels to get too high. If blood glucose isn't controlled, it can lead to serious, long-term health conditions, such as heart disease and nerve problems.

High blood glucose may also be caused by other conditions that can affect insulin or glucose levels in your blood, such as problems with your pancreas or adrenal glands, and side effects from certain medicines.

Low blood glucose levels (hypoglycemia) is often caused by certain diabetes medicines for type 1 diabetes or type 2 diabetes, including insulin. In people without diabetes, low blood glucose is much less common. It may be caused by certain medicines and conditions, such as some kidney or liver diseases. Without treatment, severe low blood glucose can lead to major health problems, including seizures and brain damage.

Several types of blood tests check blood glucose levels. Your health care provider will choose the test that's right for you.

Other names: blood sugar, self-monitoring of blood glucose (SMBG), fasting plasma glucose (FPG), fasting blood sugar (FBS), fasting blood glucose (FBG), random blood sugar, glucose challenge test, oral glucose tolerance test (OGTT)

WHAT IS IT USED FOR?

A blood glucose test may be used:

As part of a routine checkup to screen for prediabetes and diabetes

To help diagnose the cause of symptoms that may mean your blood glucose is too high or too low

To monitor the side effects of certain long-term medicines that may affect blood glucose

WHY DO I NEED A BLOOD GLUCOSE TEST?

Your provider may order a blood glucose test as part of your routine checkup. Routine blood tests that include a glucose test include a basic metabolic panel (BMP) and a comprehensive metabolic panel (CMP).

The most common reasons for a blood glucose test are to screen for and monitor prediabetes and diabetes. Your provider may order a blood glucose test if:

You have a high risk for developing type 2 diabetes. Your risk may be high if you:

Are over 35 years old. The American Diabetes Association recommends diabetes screening for prediabetes and diabetes beginning at age 35 years and older.

Have prediabetes.

Are overweight or have obesity.

Have a family history of diabetes.

Have high blood pressure or heart disease.

Have ever had gestational diabetes (diabetes that develops during pregnancy) or given birth to a baby who weighed

over 9 pounds.

Exercise less than three times a week.

Have non-alcoholic fatty liver disease (NAFLD).

Are an African American, Hispanic or Latino, American Indian, or an Alaska Native.

Some Pacific Islander people and Asian American people also have a higher risk for developing type 2 diabetes.

Are over 35 years old. The American Diabetes Association recommends diabetes screening for prediabetes and diabetes beginning at age 35 years and older.

Have prediabetes.

Are overweight or have obesity.

Have a family history of diabetes.

Have high blood pressure or heart disease.

Have ever had gestational diabetes (diabetes that develops during pregnancy) or given birth to a baby who weighed over 9 pounds.

Exercise less than three times a week.

Have non-alcoholic fatty liver disease (NAFLD).

Are an African American, Hispanic or Latino, American Indian, or an Alaska Native.

You have symptoms of high or low glucose levels, which could mean you have diabetes or another condition that affects blood glucose levels.

Symptoms of high blood glucose levels include:

Increased thirst

Frequent urination (peeing)

Blurred vision

Fatigue

Sores that are slow to heal

Losing weight without trying

Numbness or tingling in your feet or hands

Symptoms of low blood glucose levels include:

Feeling shaky or jittery

Hunger

Fatigue

Feeling dizzy, confused, or irritable

Headache

A fast heartbeat or arrhythmia (a problem with the rate or rhythm of your heartbeat)

Having trouble seeing or speaking clearly

Fainting or seizures

Increased thirst

Frequent urination (peeing)

Blurred vision

Fatigue

Sores that are slow to heal

Losing weight without trying

Numbness or tingling in your feet or hands

Feeling shaky or jittery

Hunger

Fatigue

Feeling dizzy, confused, or irritable

Headache

A fast heartbeat or arrhythmia (a problem with the rate or rhythm of your heartbeat)

Having trouble seeing or speaking clearly

Fainting or seizures

You have prediabetes or diabetes. Your provider may schedule blood glucose tests to monitor your condition. You may also monitor your blood glucose at home.

You are pregnant. You will likely have a blood glucose test between the 24th and 28th week of your pregnancy to check for gestational diabetes. If you have a high risk for gestational diabetes, your provider may test you earlier.

You take long-term medicines, other than diabetes medicines, that may affect blood glucose levels. Examples of medicines that may cause high blood glucose in some people include certain types of:

Steroids

Antipsychotics

Beta-blockers

Statins

Examples of medicines that may cause low blood glucose in some people include certain types of:

ACE inhibitors

Antibiotics called quinolones

Medicines to treat malaria

Steroids

Antipsychotics

Beta-blockers

Statins

ACE inhibitors

Antibiotics called quinolones

Medicines to treat malaria

WHAT HAPPENS DURING A BLOOD GLUCOSE TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

There are several types of blood glucose tests:

A fasting blood glucose test measures your glucose after you fast (not eating or drinking, except for water) for at least 8 hours. It is often used to:

Screen for or monitor prediabetes or diabetes

Help diagnose or monitor other conditions that affect blood glucose

Check for side effects from medications that can affect blood glucose

Screen for or monitor prediabetes or diabetes

Help diagnose or monitor other conditions that affect blood glucose

Check for side effects from medications that can affect blood glucose

A random blood glucose test can be done at any time, even after eating. If you have symptoms of diabetes, your provider may use this test first. That's because it can be done right away.

A glucose challenge test may be used to screen for gestational diabetes if you are pregnant. You don't need to fast for this test. You will drink a sugary drink that contains glucose. A blood sample is taken one hour later. If your blood glucose is too high, you will usually need an oral glucose tolerance test to confirm or rule out gestational diabetes.

An oral glucose tolerance test (OGTT) is used to diagnose gestational diabetes. If you aren't pregnant, it's sometimes used to confirm a diagnosis of prediabetes or type 2 diabetes. You will need to fast for this test. A blood sample will be

taken before you have a sugary drink. After you have the drink, more blood samples will be taken, usually about every hour for the next 2 or 3 hours.

A hemoglobin A1C (HbA1c) test provides information about your average blood glucose levels over the past 3 months. You don't need to fast for this test. It uses one blood sample. An A1C test can diagnose prediabetes and diabetes. Most people with diabetes have an A1C test at least twice a year to check how well they are managing their blood glucose levels.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If your provider orders a fasting blood glucose test or an oral glucose tolerance test, you will need to fast for at least eight hours before the test. If your glucose test is part of a BMP or CMP test, you may need to fast. Other blood glucose tests don't require any special preparations. Ask your provider whether you need to fast before your glucose test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly. After an oral glucose tolerance test, you may feel light-headed. Your provider may suggest that you plan to have someone take you home.

WHAT DO THE RESULTS MEAN?

The meaning of your test results will depend on which test you had, the results of other tests, and your medical history. Ask your provider what your test results say about your health.

In general:

If your results show higher than normal glucose levels, it may mean you have or are at risk for getting diabetes. Prediabetes and diabetes are the most common causes of high glucose levels, but other possible causes include:

An overactive thyroid gland (hyperthyroidism)

A problem with your pancreas, including pancreatitis

Stress from surgery, very serious illness, or trauma

A problem with your adrenal glands, including Cushing's syndrome and pheochromocytoma

Certain medicines

If you have diabetes, glucose levels that are lower than normal for you may be caused by:

Certain diabetes medicines and other medicines that may lower blood glucose

Not eating enough, especially after taking diabetes medicine

Being more physically active than usual

If you don't have diabetes, low blood glucose levels aren't common. If they happen, they may be caused by:

Liver or kidney disease

Underactive adrenal glands, for example, Addison disease

Certain pituitary disorders

An underactive thyroid gland (hypothyroidism)

Malnutrition

Certain medicines

Alcohol use disorder (AUD)

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I SHOULD KNOW ABOUT A BLOOD GLUCOSE TEST?

If you have diabetes, you may need to do blood glucose testing at home to help manage your blood glucose levels. Your provider will let you know how often you should test. There are two ways to check your blood glucose. Your provider can help you decide which one is best for you:

Blood glucose meters require you to prick your finger with a small device called a lancet. Then, you apply a drop of blood to a test strip and insert it into a small electronic glucose meter, which measures the glucose in your blood. Continuous glucose monitors (CGM) usually use a tiny sensor placed under the skin of your arm or belly. Depending on the type of CGM you use, the sensor stays in place from a week to several months. It automatically estimates your glucose level every few minutes and tracks it so you can see what your blood glucose level is at any time.

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Ketones in Blood
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What You Need to Know About Blood Testing

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Blood in Urine

<https://medlineplus.gov/lab-tests/blood-in-urine/>

BLOOD IN URINE

HOW DO YOU TEST FOR BLOOD IN URINE?

A urinalysis test^Â checks a sample of your urine (pee). It can find a small amount of blood cells in your urine and other types of cells, chemicals, and substances.^Â

Having blood in your urine is called hematuria. The blood may make your urine or the water in the toilet bowl turn pink, red, or reddish brown. But you might not always be able to see the blood without a lab test. There are two main types of hematuria, based on whether you can see the blood:

Gross hematuria is when you can see blood in your urine without having a test.^Â

Microscopic hematuria is when you can't see small amounts of blood in your urine, but it can be seen under a microscope or is found using a urinalysis.^Â

Having blood in your urine usually isn't serious. Some medicines and certain foods, such as beets or rhubarb, might also cause your urine to turn red. But in some cases, red or white blood cells in your urine may mean that you have a medical condition that needs treatment, such as a urinary tract infection (UTI), kidney disease, or liver disease.

Other names: microscopic urine analysis, microscopic examination of urine, urine test, urine analysis, UA, urine microscopy

WHAT IS IT USED FOR?

A urinalysis, which includes a test for blood in urine, is used to check your general health, including the health of your urinary tract, kidneys, and liver. The test can also be used to check for other health problems besides blood in urine.

WHY DO I NEED A BLOOD IN URINE TEST?

Your health care provider may order a urinalysis as part of a routine exam. You may also need this test if you have seen blood in your urine or have other symptoms that could be caused by a problem with your kidneys or urinary tract. These symptoms include:

Painful urination

Frequent urination

Back pain

Abdominal (belly) pain

WHAT HAPPENS DURING A BLOOD IN URINE TEST?

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test. Outside blood could get into your urine sample and affect your test results.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations before getting a test for blood in your urine. If your provider has ordered other urine or blood tests, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. They will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a urinalysis or a blood in urine test.

WHAT DO THE RESULTS MEAN?

Many things can cause blood in urine, and most of them aren't serious. Blood may be caused by certain medicines, intense exercise, sexual activity, or menstruation.Â

But blood in your urine can sometimes be a sign of a more serious problem, such as:

Infection in your bladder, kidney, or prostate

Bladder or kidney stones

Kidney injury from an accident or sports

A viral infection, including hepatitis (a disease of the liver causing inflammation)

Cancer of the bladder, kidney, or prostate

Enlarged prostate (BPH)

Inflammation of the kidney, urethra, or bladder

A blood disorder

Polycystic kidney disease

Disorders of the ureters (tubes that connect your kidneys to your bladder)

If your test result shows blood in your urine, you may need more tests to find out why. To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BLOOD IN URINE TEST?

A blood in urine test is usually part of a routine urinalysis. A urinalysis also measures other substances in the urine, including proteins, acid and sugar levels, cell fragments, and crystals.

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RELATED HEALTH TOPICS

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Mucus in Urine

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Blood Smear

<https://medlineplus.gov/lab-tests/blood-smear/>

BLOOD SMEAR

WHAT IS A BLOOD SMEAR?

A blood smear is a test that's used to help diagnose and monitor a wide variety of blood disorders and other conditions that affect your blood. The test checks a sample of your blood under a microscope to see whether your blood cells look normal. It looks at your:

Red blood cells (RBCs), which carry oxygen from your lungs to the rest of your body

White blood cells (WBCs), which fight infection and are part of your immune system

Platelets, which help your blood clot to stop bleeding when you have a cut or wound

To do a blood smear, a sample of blood is spread on a glass slide. The slide is treated with a special stain that makes the cells easier to see. Then a laboratory professional looks at the slide under a microscope to check the size, shape, and number of cells in your blood sample. Automatic computer programs may be used to help examine your blood smears.

Other names: peripheral smear, peripheral blood film, smear, blood film, manual differential, differential slide, blood cell morphology, blood smear analysis, leukocyte differential

WHAT IS IT USED FOR?

A blood smear alone can't diagnose a disease. But it can be used with other tests to help diagnose and monitor a variety of conditions, such as:

Blood and bone marrow disorders

Certain types of cancer, including those that affect your blood, bone marrow, and lymphatic system (a part of your immune system)

Infections

Certain parasites

WHY DO I NEED A BLOOD SMEAR?

You may need a blood smear if:

You had a complete blood count (CBC) with abnormal results. A CBC is a group of blood tests that measures the number and size of the different cells in your blood. It's often done as part of a routine checkup. A blood smear can confirm abnormal CBC results by checking the size and number of your cells. It can also give your health care provider more information about what your blood cells look like, including their color.

You have symptoms of a condition that affects your blood cells. Symptoms may include:

Fatigue

Jaundice, a condition that causes your skin and eyes to turn yellow

Nosebleeds or other unusual bleeding

Easy bruising

Fever

Bone pain

Swollen lymph nodes, also called "swollen glands"

You are being treated for a condition that affects your blood cells, such as anemia or blood cancer. A blood smear can help find out if your treatment is working.

You have symptoms of certain diseases caused by parasites, and you live in or travel to areas where those diseases are common. These diseases include:

Malaria, which is spread by bites from infected mosquitos

Babesiosis, which is spread mainly by bite from infected ticks

Symptoms of these parasitic diseases can be like having the flu. They include, fever, chills, body aches, nausea, and fatigue.

WHAT HAPPENS DURING A BLOOD SMEAR?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Sometimes, the blood sample will be taken by pricking your finger with a needle. When testing a baby, the heel is pricked.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparation for a blood smear. If your provider has ordered other blood tests, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The results of a blood smear alone usually can't diagnose a medical condition. Your provider will use your test results combined with your medical history, symptoms, and the results of other tests to make a diagnosis.

Your blood smear results will describe what your blood cells look like and how many there are. The results will also show anything else that's unusual about your blood, for example, if any parasites were found.

Red blood cell results that aren't normal, may be a sign of conditions, such as:

Anemia

Sickle cell disease

Hemolytic anemia, a type of anemia that causes your body to destroy red blood cells faster than it makes them

Thalassemia

Bone marrow disorders

White blood cell results that aren't normal may be a sign of conditions, such as:

Infection

Leukemia

Bone marrow disorders

Platelet results that aren't normal may be a sign of conditions, such as:

Thrombocytopenia, a lower-than-normal number of platelets. This may increase your risk of bleeding.

Thrombocytosis, a higher-than-normal number of platelets that is caused by another disease or condition, such as anemia, or an infection. Having too many platelets can cause blood clots to form in your blood vessels (the tubes that carry blood in your body).

Thrombocythemia, a higher-than-normal number of platelets that is caused by a problem with the cells in your bone marrow that make platelets.

Other conditions can also cause abnormal blood smear results, including:

Severe illness

Stress

Blood transfusions

If your blood smear results are not normal, your provider may order more tests to help find the cause. Talk with your provider to learn more about your results.

Learn more about laboratory tests, reference ranges, and understanding results.

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Bone Density Scan

<https://medlineplus.gov/lab-tests/bone-density-scan/>

BONE DENSITY SCAN

WHAT IS A BONE DENSITY SCAN?

A bone density scan, also known as a DEXA or DXA scan, is a low-dose x-ray that measures calcium and other minerals in your bones. Bones containing more minerals are denser and tend to be stronger and less likely to break. The measurement helps show your bone's strength and thickness (known as bone density or mass).

Your bones usually become thinner as you get older or if you develop certain medical conditions. When bones become thinner than usual, it's known as osteopenia. Osteopenia puts you at risk for a more serious condition called osteoporosis. Osteoporosis is a disease that causes bones to become very thin and brittle. It gets worse over time. It usually affects older people and is most common in women over the age of 65. People with osteoporosis are at higher risk for fractures (broken bones), especially in their hips, spine, and wrists.

Other names: bone mineral density test, BMD test, DEXA scan, DXA; Dual-energy x-ray absorptiometry scan, Bone densitometry, Dual x-ray absorptiometry, DXA

WHAT IS IT USED FOR?

A bone density scan is used to:

Diagnose osteopenia (low bone mass)

Diagnose osteoporosis

Predict your risk of future fractures

See if treatment for osteoporosis is working

WHY DO I NEED A BONE DENSITY SCAN?

Most women age 65 or older should have a bone density scan. Women in this age group are at high risk of losing bone density, which can lead to fractures. Women of other ages and men may also be at risk for losing bone density if they:

Have a very low body weight

Have had one or more fractures after the age of 50

Have lost a half inch or more in height within one year

Have a family history of osteoporosis

Other risk factors for losing bone density include:

Lack of physical activity

Long-time use of tobacco or alcohol

Not getting enough calcium and vitamin D in your diet

Having medical conditions such as diabetes or rheumatoid arthritis

Being an older male with risk factors for weak bones or osteoporosis

WHAT HAPPENS DURING A BONE DENSITY SCAN?

There are different ways to measure bone density. The most common and accurate method is with dual-energy x-ray absorptiometry, also known as a central DEXA scan. The scan checks the bone density of your lower spine, hip, wrist, or entire body. It is usually done in a hospital or clinic.

During a central DEXA scan:

You will lie on your back on a padded table. You will probably be able to leave your clothes on.

You may need to lie with your legs straight, or you may be asked to rest your legs on a padded platform.

The arm of the scanning machine will pass over your hips and spine. At the same time, another scanning machine will pass under you. The images from the two machines will be combined and sent to a computer. A health care provider will view the images on the computer screen.

While the machines are scanning, you need to stay very still. You may be asked to hold your breath to keep the images from blurring.

The test usually takes 10 to 30 minutes. Your provider will usually get the results in a few days.

Sometimes, a bone density scan is only done on your wrist, hand, or foot bones. This is known as a peripheral DEXA (p-DEXA) scan. The portable p-DEXA device has a box-like structure. This kind of scan might be done in a provider's office, mobile health van, or drugstore.

During a peripheral DEXA (p-DEXA) scan:

You will be asked to place your finger, hand, forearm, or foot in the space for imaging.

This test takes a few minutes, and your provider should receive the results shortly after.

This scan can provide good information about your fracture risk but cannot give as much detail as the central DEXA scan.

If you are found to be at risk of bone fractures, your provider may recommend a central DEXA scan.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may be told to stop taking calcium supplements 24 to 48 hours before your test. You should also avoid wearing metal jewelry or clothes with metal parts, such as buttons or buckles.

ARE THERE ANY RISKS TO THE TEST?

A bone density scan uses very low doses of radiation. It is safe for most people. But it is not recommended during pregnancy. Even low doses of radiation could harm a developing baby. Be sure to tell your provider if you are pregnant or think you may be pregnant.

WHAT DO THE RESULTS MEAN?

Bone density results are often given as T-scores and Z-scores. A higher number means that you have greater bone density. The lower your score, the higher your risk of bone fracture.

T-score may be used if you are a postmenopausal woman or a man age 50 or older. This measurement compares your bone density with a healthy young adult of the same sex.

Z-score may be used if you are premenopausal, a male under age 50, or a child. This measurement compares your bone density measurement with other people your age, weight, sex, and ethnic or racial origin.

Your T-score results may show one of the following:

A T-score of -1.0 or higher. This is considered normal bone density.

A T-score between -1.0 and -2.4. This means you have low bone density (osteopenia) and may be at risk for developing osteoporosis.

A T-score of -2.5 or less. This means you probably have osteoporosis.

If your results show you have low bone density, your provider will recommend steps to prevent further bone loss. These may include:

Getting more exercise, with activities such as walking, dancing, and using weight machines.

Adding calcium and vitamin D to your diet.

Taking prescription medicines to increase bone density.

Your provider may recommend how often you need to have repeat scans, based on your risk of bone fractures. The recommendations are to get a central DEXA scan every two years for high risk, every 3 to 5 years for moderate risk, and every 10 to 15 years for low risk.

If you have questions about your results and/or treatments for bone loss, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BONE DENSITY SCAN?

A DEXA scan is the most common way to measure bone density. But your provider may order more tests to confirm a diagnosis or to find out if bone loss treatment is working. These include a calcium blood test, a vitamin D test, and/or tests for certain hormones.

DEXA scans are not as accurate if you have a history of spinal surgery or spinal deformity, fractures, or arthritis in your spine.

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Bone Marrow Tests

<https://medlineplus.gov/lab-tests/bone-marrow-tests/>

BONE MARROW TESTS

WHAT ARE BONE MARROW TESTS?

Bone marrow is a soft, spongy tissue found in the center of your bones. Bone marrow makes different types of blood cells, including:

Red blood cells (also called erythrocytes), which carry oxygen from your lungs to every cell in your body.

White blood cells (also called leukocytes), which help you fight infections.

Platelets, which help with blood clotting.

Bone marrow tests remove a sample of your bone marrow and bone. The sample is usually taken from the back of your hip. It is examined under a microscope and tested to look for signs of disease.

Bone marrow tests help find out whether your bone marrow is working well to make normal amounts of healthy blood cells. Bone marrow tests can help diagnose and monitor bone marrow disorders, blood disorders, and certain types of cancer.

There are two types of procedures used to collect bone marrow samples for testing:

Bone marrow aspiration removes a small amount of fluid and cells from your bone marrow.

Bone marrow biopsy removes a small piece of bone and bone marrow.

Bone marrow aspiration and bone marrow biopsy are usually done at the same time.

Other names: bone marrow examination

WHAT ARE THEY USED FOR?

Bone marrow tests are used to help find the cause of too many or too few blood cells, or blood cells that don't look normal. The results can help:

Diagnose blood and bone marrow disorders, such as:

Anemia (when the cause is unknown)

Hemochromatosis

Myelodysplastic syndromes

Polycythemia vera

Platelet disorders

Certain cancers involving blood and bone marrow, including leukemia, multiple myeloma, and lymphoma

Anemia (when the cause is unknown)

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Certain cancers involving blood and bone marrow, including leukemia, multiple myeloma, and lymphoma

Plan and monitor treatment for conditions that involve your blood or bone marrow

Find out if certain types of cancer, such as breast, lung, or prostate cancer, have spread to your bone marrow

Find the cause of an unexplained fever, which could be from a bone marrow infection

WHY DO I NEED BONE MARROW TESTS?

You may need a bone marrow aspiration and a bone marrow biopsy if you:

Had abnormal results on other blood tests, such as a complete blood count (CBC) or a blood smear

Have signs and symptoms, and a medical history that could mean you have a bone marrow disease

Have certain types of cancers that involve your blood and bone marrow, and your health care provider needs to know how much cancer is in your body (This is called "cancer staging.")

Have a type of cancer in another part of your body that commonly spreads to bone marrow

Are being treated for a condition that affects your blood or bone marrow, and your provider needs to find out how well your treatment is working

WHAT HAPPENS DURING A BONE MARROW TEST?

Bone marrow aspiration and bone marrow biopsy procedures are usually done at the same time. A provider will collect the marrow samples for testing. Usually, the samples can be collected in about ten minutes. The procedure may be done in a hospital, provider's office, or clinic.

Before the procedure, you may be asked to put on a hospital gown. Your blood pressure, heart rate, and temperature will be checked.

You may choose to have a mild sedative, which is medicine to help you relax. You may also have the choice to use stronger medicine that will make you sleep. Your provider can help you decide which option is best for you.

During the procedure:

You'll lie down on your side or your stomach, depending on which bone will be used for the samples. Most bone marrow samples are taken from a bone in the back of your hip, called the iliac crest. But other bones may be used.

The skin over the bone will be cleaned with an antiseptic.

You will get an injection (shot) of medicine to numb your skin and the bone underneath. It may sting.

When the area is numb, your provider will make a very small incision (cut) in your skin and insert a hollow needle. You will need to lie very still during the procedure:

For the bone marrow aspiration, your provider pushes a thin needle into the bone and removes a sample of bone marrow fluid with a syringe attached to the needle. You may feel a brief some pressure or brief pain. The aspiration takes only a few minutes.

For the bone marrow biopsy, your provider inserts a hollow needle into the bone and twists the needle to take out a small piece of bone marrow tissue. You may feel some pressure or brief pain while the sample is being taken.

For the bone marrow aspiration, your provider pushes a thin needle into the bone and removes a sample of bone marrow fluid with a syringe attached to the needle. You may feel a brief some pressure or brief pain. The aspiration takes only a few minutes.

For the bone marrow biopsy, your provider inserts a hollow needle into the bone and twists the needle to take out a small piece of bone marrow tissue. You may feel some pressure or brief pain while the sample is being taken.

After the test, your provider will press on your skin to stop the bleeding and then cover the area with a bandage.

If you didn't use medicine to relax or sleep, you'll usually need to stay lying down for about 15 minutes to make sure that the bleeding has stopped. If you used medicine to relax or sleep, you'll need to stay longer before you can go home.

Your provider will tell you how to care for yourself after your test. You may need to avoid hard physical activity and keep the bandage dry for at least a day. You may feel sore for a week or more. Your provider can tell you which pain relievers are safe to use.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You'll receive complete instructions for how to prepare.

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

Tell your provider if you have a bleeding disorder.

Your provider will tell you whether you need to fast (not eat or drink) for a few hours before the procedure.

Plan to have someone take you home after the test, because you may be drowsy if you are given medicine to help you relax or sleep during the procedure.

Be sure to ask your provider any questions you may have about the procedure.

ARE THERE ANY RISKS TO THE TEST?

After a bone marrow aspiration and bone marrow biopsy you may feel stiff or sore where the sample was taken. This usually goes away in a few days. Your provider may recommend or prescribe a pain reliever to help. Don't take any pain medicine your provider hasn't approved. Certain pain relievers, such as aspirin, could increase your risk of bleeding.

Serious symptoms are very rare, but may include:

Increased pain or discomfort where the sample was taken

Redness, swelling, bleeding, or other fluids leaking from the area where the needles were inserted

Fever

If you have any of these symptoms, call your provider.

WHAT DO THE RESULTS MEAN?

It may take several days or even weeks to get your bone marrow test results. Your provider may have ordered many different types of tests on your sample, so your results often include a lot of complex information. Your provider can explain what your results mean.

In certain cases, if your test results are not normal, you may need to have more tests to confirm a diagnosis or to decide which treatment would be best for you.

If you have already been diagnosed with a disease that involves your bone marrow, your test results may show:

How well your treatment is working

Whether your treatment is causing side effects that affect your bone marrow

Learn more about laboratory tests, reference ranges, and understanding results.

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BRAF Genetic Test

<https://medlineplus.gov/lab-tests/braf-genetic-test/>

BRAF GENETIC TEST

WHAT IS A BRAF GENETIC TEST?

A BRAF genetic test uses a sample of tumor tissue, bone marrow, or blood to look for changes in a gene called BRAF. Changes in your genes are also called gene variants or mutations. The test is usually done to look for changes in the BRAF gene that may be involved in cancer.

Genes are parts of DNA in your cells that you inherit from your parents. Genes carry information that controls what you look like and how your body works. The BRAF gene provides instructions for making a protein that tells your cells when to grow and divide to make new cells.

Certain changes in your BRAF gene may change the gene's instructions and lead to BRAF proteins that are too active. These abnormal BRAF proteins trigger too much cell growth. As a result, cells that have the changed BRAF gene may grow out of control and become cancer.

Changed BRAF genes and their proteins are commonly found in melanoma, the most serious form of skin cancer. Changes are also found in several other types of cancer, including colon, thyroid, ovarian, lung, and certain blood cancers.

Cancers with changes in the BRAF gene tend to grow more quickly than cancers that don't have this gene change. Knowing whether your cancer has a changed BRAF gene can help your health care provider choose the most effective treatment for you.

Changes in your BRAF genes can be acquired or inherited from your parents:

Acquired changes in your BRAF genes develop sometime after you're born. Most changes in BRAF genes are acquired. These changes can happen if your cells make a mistake when they divide to make new cells. Changes can also happen if you're exposed to substances that cause cancer. Acquired changes in genes show up only in certain types of cells. They're usually not in sperm or egg cells, so you can't pass them down to children.

Inherited changes in your genes are changes that you're born with. All the cells in your body have the gene change, including your egg or sperm cells. That means you can pass the change down to your children. Inherited changes in the BRAF gene are very rare, but they can increase your risk of cancer or cause different types of genetic conditions that are present from birth.

Other names: BRAF gene mutation analysis, Melanoma, BRAF V600 mutation

WHAT IS IT USED FOR?

BRAF genetic testing may be used to:

Look for changes in the BRAF gene in certain types of cancer. This information may be used to diagnose certain types of cancer and/or guide treatment decisions.

Check for inherited changes in the BRAF gene. This is not a routine screening test. But it may be used if your personal or family health history suggests that you may have a high risk for cancer or another condition that's caused by inherited changes in the BRAF gene. Your risk of cancer from inherited gene change may be high if:

- You or a family member have had cancer at an unusually young age
- You have several close family members who have the same type of cancer
- You or a family member have two or more types of cancer
- You or a family member have had cancer at an unusually young age
- You have several close family members who have the same type of cancer
- You or a family member have two or more types of cancer

WHY DO I NEED A BRAF GENETIC TEST

You may need a BRAF genetic test if you:

Have a cancer that could be caused by changes in your BRAF gene. The results from a BRAF genetic test can help your provider:

Predict how fast your cancer may grow.

Know whether targeted treatment is likely to help you. Targeted treatment for cancer with changes in BRAF genes blocks or slows the abnormal proteins that make cancer grow. These medicines could be harmful if they are prescribed for cancers that don't have changed BRAF genes. So BRAF testing is needed to match you with the best treatment for your cancer.

Understand how your cancer may respond to treatments, such as chemotherapy or radiation therapy.

Predict how fast your cancer may grow.

Know whether targeted treatment is likely to help you. Targeted treatment for cancer with changes in BRAF genes blocks or slows the abnormal proteins that make cancer grow. These medicines could be harmful if they are prescribed for cancers that don't have changed BRAF genes. So BRAF testing is needed to match you with the best treatment for your cancer.

Understand how your cancer may respond to treatments, such as chemotherapy or radiation therapy.

Have a personal or family health history that increases your risk of cancer involving inherited changes in your BRAF gene. A BRAF genetic test can help you estimate and manage your cancer risk. The test can also show whether you could pass a changed BRAF gene to your children.

Have a family health history of a genetic condition caused by BRAF gene changes. BRAF genetic testing can help you understand if you and your partner have BRAF gene changes that you could pass to your children.

If you're considering a BRAF gene test to look for an inherited gene change, a genetic counselor can help you consider the pros and cons of this test.

WHAT HAPPENS DURING A BRAF GENETIC TEST?

If you're having a BRAF genetic test because you have cancer, the test will depend what type of cancer you have:

For solid tumors, the test often uses a sample of cells from the tumor. The sample may be taken:

During surgery to remove the entire tumor.

With a biopsy to remove a small amount of tissue. The type of biopsy you have depends on where your cancer is located. A biopsy may be done with:

A hollow needle inserted through your skin. Imaging tests, such as ultrasound may be used to guide the needle.

Minor surgery to remove a sample of tissue (an incisional biopsy).

With a blood test. If surgery or a biopsy isn't possible, you may have a blood test to look for DNA that the tumor has shed in your bloodstream. This is called a "liquid biopsy." To do a blood test, a health care professional will use a small needle to take a blood sample from a vein in your arm.

During surgery to remove the entire tumor.

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For blood cancer, a blood test or a bone marrow test may be used. Marrow is soft, spongy tissue found inside of bones.

Most bone marrow samples are taken from the back of the hip bone.

If you're having a BRAF genetic test to check for inherited gene changes, you will usually have a blood test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you're having a blood test, you usually don't need any special preparations. Ask your provider about how to prepare for other types of BRAF gene tests.

ARE THERE ANY RISKS TO THE TEST?

The risks of a BRAF gene test depend on how your sample is collected. Talk with your provider about your risks.

In general, with a biopsy, you may have soreness, pain, or bleeding where the tissue was removed.

A blood test has very little risk. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

After a bone marrow test, you may feel stiff or sore where the sample was taken. This usually goes away in a few days. Serious symptoms are very rare, but may include swelling, bleeding at the site, or fever.

WHAT DO THE RESULTS MEAN?

If you have cancer and your test results show:

You have a change in your BRAF gene, your provider may prescribe medicines that are designed to target the changed BRAF proteins that are making your cancer grow. You may also need other treatments.

You don't have a change in your BRAF gene, your provider will prescribe another type of treatment for your cancer.

If you don't have cancer and your results show you have a change in your BRAF gene, it does not mean you'll get cancer. But, depending on the change, you may have a higher risk of developing cancer.

You may be able to reduce your risk by having more frequent cancer screenings. For example, to reduce or risk of melanoma, you can have more skin exams to check your whole body for moles and other suspicious growths. Cancer screening can help catch cancer early when it's easier to treat.

If you have an inherited change in your BRAF gene, talk with your provider or a genetic counselor to learn about the risks of passing the change to your children.

Learn more about laboratory tests, reference ranges, and understanding results.

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BRCA Genetic Test

<https://medlineplus.gov/lab-tests/brca-test/>

BRCA GENETIC TEST

WHAT IS A BRCA GENETIC TEST?

BRCA is short for "breast cancer gene." A BRCA genetic test uses a sample of your blood, saliva (spit), or cells from inside of your cheek to look for changes in your BRCA1 and BRCA2 genes. Genes are parts of DNA in your cells that you inherit from your parents. You inherit two copies of each gene, one from each parent. Genes carry information that controls what you look like and how your body works.

BRCA is a type of gene called a "tumor suppressor." Tumor suppressor genes provide instructions for making proteins that help prevent or slow the growth of cells that may become tumors. The proteins work by:

Preventing cells from dividing too quickly to make new cells.

Repairing damaged DNA in cells.

Starting the normal process of cell death in cells that can't be repaired.

If you have certain changes (or variations) in a BRCA gene, the gene or the protein it makes may not work well or may stop working completely. As a result, your cells may grow and divide out of control, which can lead to tumors and cancer.

Other names: BRCA gene test, BRCA gene 1, BRCA gene 2, breast cancer susceptibility gene1, breast cancer susceptibility gene 2

WHAT IS IT USED FOR?

This test is used to find out if you have harmful changes in your BRCA1 or BRCA2 genes that increase your risk of getting:

Breast cancer. This is cancer of the breast tissue. It mostly affects females, but harmful BRCA variations can also increase the breast cancer risk for males.

Ovarian cancer. This is cancer of the female reproductive glands where eggs form.

Prostate cancer. This is cancer of the male reproductive gland that makes fluid for semen.

Pancreatic cancer. This is cancer of the pancreas, an organ that helps you digest food and makes important hormones.

WHY DO I NEED A BRCA GENETIC TEST?

Harmful variations in the BRCA gene are rare. They affect only about 0.2% of the U.S. population. So, BRCA testing is not recommended for most people.

You and your family members are more likely to have a BRCA1 or BRCA2 variation if either side of your family has a strong history of:

Female breast cancer diagnosed under the age of 50.

Triple negative breast cancer (an especially aggressive form less likely to respond to treatment) diagnosed under the age of 60.

Male breast cancer.

Cancer of the ovaries, fallopian tubes (tubes connecting the ovaries to the uterus), or peritoneum (tissue that covers your belly organs).

Prostate cancer that spreads to other parts of the body (metastatic cancer).

Pancreatic cancer.

If you're concerned that you may have a harmful variant in the BRCA1 or BRCA2 gene, your health care provider or a genetic counselor can review your personal and family health history to see if you need this test.

You should also talk to your provider about testing if you've had any of these cancers and/or are of Ashkenazi (Eastern European) Jewish descent. BRCA variations are much more common in those with Ashkenazi Jewish ancestry.

compared with the general population.

WHAT HAPPENS DURING A BRCA GENETIC TEST?

A BRCA test can use either a sample of your blood, saliva, or a cheek swab.

For a blood test: A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a saliva test: You'll spit into a container or use a cotton pad to soak up some saliva.

For a cheek swab: A health care professional will use a special swab or brush to take a sample from your cheek. You may have the option of doing it yourself.

You may be able to use an at-home saliva test kit to check for the three most common harmful variations in BRCA genes. But there are more than 1000 known variations. A home test can't rule out all of them. If you decide to use an at-home test kit, be sure to follow all your provider's or the kit's instructions for collecting a sample and getting that sample to a lab.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

For a blood test: You don't need any special preparations.

For a saliva test: A half hour before the test, you may need to stop eating, drinking, or smoking. Follow all the instructions your provider gives you or the instructions in an at-home kit.

For a cheek swab: You may be asked to rinse your mouth before the test.

With any type of genetic test, you may want to meet with a genetic counselor first to see if testing is right for you. Your counselor can explain the pros and cons of learning more about your cancer risk. After your test, a counselor can help with the medical and emotional impact that your results may have on you and your family.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There are no risks linked to providing a saliva sample or having a cheek swab.

WHAT DO THE RESULTS MEAN?

Your provider or genetic counselor will need to explain how your results affect your risk for certain cancers. That's because the meaning of your test results depends on the exact type of BRCA variant you have and your personal and family health history.

In general, your results may be reported using these terms:

A negative (normal) result means that the test didn't find any harmful changes in your BRCA genes. How this affects your cancer risk depends on whether you've already had cancer and whether a member of your family has a harmful BRCA variant.

An uncertain result may also be called a "variant of uncertain significance (VUS)". It means that a variant in your BRCA genes was found, but researchers don't know whether that variant causes cancer.

A positive result may also be called a "likely pathogenic variant." It means that you have a harmful gene variant that is known to increase the risk of certain cancers. But the test cannot tell whether you will develop cancer.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BRCA GENETIC TEST?

Not everyone who has a harmful variant in BRCA1 or BRCA2 will get cancer. If your results show that you have a harmful variant in your BRCA genes, talk with your provider about ways to lower your cancer risk. You may discuss:

- Your schedule for cancer screening tests. Your provider may recommend that you get tested sooner and more often than usual.
- Taking certain medicines and/or having surgery to reduce the risk of cancer. Surgery may include removing both breasts and/or the ovaries and fallopian tubes. Medicines such as tamoxifen and raloxifene can also reduce your cancer risk by controlling the levels of estrogen (a hormone that plays a key role in female reproductive health) in your body.

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Breast Biopsy

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BREAST BIOPSY

WHAT IS A BREAST BIOPSY?

A breast biopsy is a procedure that removes a sample of breast tissue so that a pathologist (a type of doctor) can look at it under a microscope for signs of disease and cancer.

There are three different ways to do a breast biopsy.

A fine needle aspiration biopsy uses a very thin needle to remove a sample of breast cells or fluid.

A core needle biopsy uses a larger needle and sometimes a vacuum probe to remove a sample. This sample is about the size of a grain of rice.

A surgical biopsy is surgery to remove a sample of tissue (an incisional biopsy). In certain cases, the entire tumor will be removed (an excisional biopsy).

Your health care provider will decide which of these biopsy types is best for you depending on the results of a recent clinical breast exam or mammogram (an x-ray picture of the breast).

Other names: core needle biopsy; core biopsy, breast; fine-needle aspiration; open surgery biopsy

WHAT IS IT USED FOR?

A breast biopsy is the only way to find out whether a suspicious change in your breast is cancer. Breast cancer is cancer that forms in the milk ducts (tubes that carry milk to the nipple) and the lobules (the small lobes of breast tissue that make milk). Breast cancer is much more common in women, but men can get it, too.

WHY DO I NEED A BREAST BIOPSY?

You may need a breast biopsy if:

You or your provider felt a lump, thickening, or other change in your breast.

A mammogram, ultrasound, or MRI test shows a lump, calcium deposits, or other signs that might mean cancer.

The skin on your breast or nipple is red, scaly, or swollen, or your nipple is pulled inward.

You have a discharge of abnormal fluid coming from your nipple.

If your provider suggests that you have a breast biopsy, it doesn't mean you have breast cancer. Most breast lumps and other changes that are checked with biopsies turn out to be benign, which means they are not cancer.

WHAT HAPPENS DURING A BREAST BIOPSY?

Biopsies are often done using mammography, ultrasound, MRI, or x-rays to help see exactly where to take the tissue sample. Your procedure will vary depending on which method is used to guide the biopsy, but the general steps are usually the same.

For a fine needle aspiration biopsy or a core needle biopsy:

Your provider will clean the skin on your breast and give you a shot of medicine to numb the area, so you won't feel any pain. The shot may sting briefly.

You may be sitting or lying down. If images are used to guide the biopsy, you may lie on your side, back, or belly with your breast in an opening on the table.

For a fine needle aspiration biopsy, your provider will insert the needle into the biopsy site and remove a sample of cells or fluid. For a core biopsy, a tiny cut may be made to insert a wide needle or a vacuum device. You may feel a little pressure when the sample or samples are removed.

Pressure will be applied to the area until the bleeding stops.

Your provider will cover the biopsy site with a sterile bandage. If you had a small incision, small strips of medical tape may be used to close the wound.

For a surgical biopsy:

You'll lie on an operating table.

The skin over the biopsy area will be cleaned.

Your provider will give you medicine to make sure you don't feel any pain during the procedure. They may give you a single injection (shot) to numb your breast. Or they may give you general anesthesia, which is medicine given through an IV (intravenous line) in your arm or hand to make you sleep.

When you are numb or asleep, the surgeon will make a small cut in your breast tissue to remove part or all of the lump. In certain cases, tissue around the lump may also be removed. This may help avoid the need for more surgery if cancer cells are found in the lump.

The cut in your skin will be closed with small strips of medical tape or stitches and covered with a sterile bandage.

If you're getting a surgical biopsy, your provider may also perform a wire localization procedure to guide them to the lump. They will insert the wire either the day before or the day of the biopsy. To do this, your provider will clean the skin on your breast and give you a shot of medicine to numb the area. Then they will:

Place a thin, hollow needle into the biopsy area.

Insert a sterile wire through the center of the needle.

Remove the needle, leaving the wire in your breast until the biopsy is done. Once the biopsy is done, your provider will remove the wire.

The type of biopsy you have will depend on:

The size and location of the suspicious tissue in your breast.

How many areas of your breast are involved.

How abnormal the tissue looks on a mammogram or other image.

Your general health and preferences.

Ask your provider about why you need a biopsy and which type is right for you.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines, such as blood thinners—like aspirin, before this test, so tell your provider about all the medicines and supplements you take. Don't stop or start taking anything unless your provider tells you to.

Before you get any procedure, you should tell your provider about any recent surgeries or known allergies. You should also tell them if you're pregnant.

If you're having general anesthesia, you will probably need to fast—(not eat or drink) for several hours before surgery. If you have general anesthesia or medicine to relax, you may be groggy after the procedure, so plan to have someone take you home.

ARE THERE ANY RISKS TO THE TEST?

It's common to have some bruising and temporary discomfort after a breast biopsy. Possible risks include:

Infection, which can be treated with antibiotics

Bleeding at the biopsy site

Your provider will give you instructions on how to care for the biopsy area and manage any discomfort. If you're having general anesthesia, talk with your provider about how it may affect you.

WHAT DO THE RESULTS MEAN?

It may take several days to a week to get your biopsy results (often called a pathology report). The report is written for your provider and will include a lot of medical terms. But the most important part of your report will be the diagnosis, which will be one of these categories:

Normal. No cancer or abnormal cells were found.

Abnormal breast changes that aren't cancer and don't increase your risk for breast cancer. Some of these conditions often get better on their own, while others may need treatment.

Abnormal breast changes that increase your risk for breast cancer. These cells are not cancer, but if you have them, you are more likely to develop cancer. You may need a surgical biopsy to remove any additional abnormal tissue. In addition, your provider may refer you to a doctor who specializes in breast cancer.

Breast cancer. If your provider finds cancer cells, your report will include details about how fast the cancer cells are growing, how much they look like normal cells, and other information to help plan the most effective treatment. Usually, a doctor who specializes in breast cancer will provide your care.

Your test results may also tell you about your tumor type and grade and whether or not you have certain hormones or proteins that make you more likely to develop cancer. Ask your provider to explain any additional information in your report.

Learn more about laboratory tests, reference ranges, and understanding results.

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Bronchoscopy and Bronchoalveolar Lavage (BAL)

<https://medlineplus.gov/lab-tests/bronchoscopy-and-bronchoalveolar-lavage-bal/>

BRONCHOSCOPY AND BRONCHOALVEOLAR LAVAGE (BAL)

WHAT ARE BRONCHOSCOPY AND BRONCHOALVEOLAR LAVAGE (BAL)?

A bronchoscopy is a procedure that's used to check for the cause of a lung problem. It can also be used to treat some lung diseases.

The procedure uses a bronchoscope, which is a thin, lighted tube with a tiny camera. This allows your health care provider to view images of your airways. The tube is put through your mouth or nose, down your throat, and into your airways. The tube is usually flexible but can also be rigid:

A flexible bronchoscope may be used to keep your airway open, suck up secretions (mucus made in your airways), or take a tissue sample (biopsy).

A rigid bronchoscope may be used to treat a tumor or bleeding or to remove something large stuck in your airway. It may also be used to insert a stent, which is a tiny tube placed in your airway to help you breathe.

Bronchoalveolar lavage (BAL) is a procedure that is sometimes done during a bronchoscopy. It's done to collect a sample from your lungs. It is also called bronchoalveolar washing. During the procedure, a saline solution is put through the bronchoscope to wash your airways and capture a fluid sample. The cells or bacteria in the fluid sample may be tested for infections or certain lung conditions.

Other names: flexible bronchoscopy, bronchoalveolar washing

WHAT ARE THEY USED FOR?

Bronchoscopy may be used to:

Find and treat growths or other blockages in your airways

Remove lung tumors

Control bleeding in your airway

Help find the cause of certain lung symptoms such as a cough that doesn't go away

Help guide placement of a breathing tube

Place medicine in your lungs to treat certain conditions

Help determine how severe lung cancer is (if you have already been diagnosed with it)

Bronchoscopy with BAL is used to collect tissue for testing, to help diagnose certain lung diseases such as:

Bacterial infections such as tuberculosis and bacterial pneumonia

Fungal infections

Lung cancer

WHY DO I NEED BRONCHOSCOPY AND BAL?

You may need one or both tests if you have symptoms of a lung disease, such as:

A cough that doesn't go away

Trouble breathing

Coughing up blood

You may also have one or both tests if a chest x-ray or another imaging test shows a potential problem with your lungs.

If you have an immune system disorder, you may be more likely to get certain lung problems. For example, HIV or an organ transplant can put you at higher risk for some lung infections. If your provider thinks that you may have a lung infection or other lung problem, they can use bronchoscopy, often with a BAL, to make a diagnosis. Getting an early diagnosis helps you get the correct treatment as soon as possible.

WHAT HAPPENS DURING BRONCHOSCOPY AND BAL?

Bronchoscopy and BAL are often done by a pulmonologist. A pulmonologist is a doctor who specializes in diagnosing and treating lung diseases.

A bronchoscopy takes about 30 to 90 minutes and usually includes the following steps:

You may need to remove some or all of your clothing. If so, you will be given a hospital gown.

You may also need to remove dentures and any removable dental appliances.

You will lie on a bed or table with your head raised.

You may get medicine (sedative) to help you relax. The medicine will be injected into a vein or given through an IV (intravenous) line placed in your arm or hand.

Your provider will spray a numbing medicine in your mouth and throat so you won't feel any pain during the procedure.

Your provider will insert the bronchoscope down your throat and into your airways. The camera on the bronchoscope will take photos and connect to a video screen.

As the bronchoscope is moved down, your provider can look into your airways and lungs through the bronchoscope and on the video screen.

Your provider may perform other treatments at this time, such as removing a tumor or clearing a blockage.

At this point, you may also get a BAL.

During a BAL:

Your provider will put a small amount of saline through the bronchoscope.

After washing the airways, the saline is sucked up into the bronchoscope.

The saline solution will contain cells and other substances, such as bacteria, which will be taken to a lab for testing.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for several hours before your procedure. Your provider will let you know how long you need to avoid food and drink.

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

WHAT TO EXPECT AFTER THE TEST:

Your mouth and throat may be numb for a few hours.

You will have to wait to eat or drink until the numbness is gone.

If you've been given a sedative, you may be drowsy for a few hours after your procedure and should arrange to have someone take you home.

You may have a sore throat, cough, or hoarseness for a day or more.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a bronchoscopy or a BAL. Serious complications are rare, but they may include bleeding in the airways, infection, or a pneumothorax (collapse of part of your lung).

WHAT DO THE RESULTS MEAN?

If your bronchoscopy results were not normal, it may be a sign of lung disorder such as:

A blockage, growth, or tumor in the airways

Narrowing of part of the airways

Lung damage due to an immune disorder such as rheumatoid arthritis

If you had BAL and your lung sample results were not normal, it may be a sign of lung cancer or a type of infection such

as:

Tuberculosis

Bacterial pneumonia

Fungal infection

If you have questions about your results, talk to your provider. Your provider may consider your symptoms, medical history, and the results of other tests to understand the results of your procedure.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT BRONCHOSCOPY AND BAL?

In addition to BAL, other procedures can be done during a bronchoscopy. These include:

Sputum culture. Sputum is a thick type of mucus made in your lungs. It is different than spit or saliva. A sputum culture checks for certain types of infections.

Laser therapy or radiation to treat tumors or cancer

Treatment to control bleeding in the lungs

If you had a tissue sample taken, you may also have a chest x-ray done to check for any issues.

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BUN (Blood Urea Nitrogen)

<https://medlineplus.gov/lab-tests/bun-blood-urea-nitrogen/>

BUN (BLOOD UREA NITROGEN)

WHAT IS A BUN (BLOOD UREA NITROGEN) TEST?

A BUN, or blood urea nitrogen test, measures the amount of urea nitrogen in your blood. Urea nitrogen is a waste product that forms as your body breaks down proteins. It's carried in your blood and then removed by your kidneys when you urinate (pee).

Having a small amount of urea nitrogen in your blood is normal. But higher-than-normal BUN levels may be a sign that your kidneys aren't working well. If you have kidney disease, this waste material can build up in your blood and may lead to serious health problems, including high blood pressure, anemia, and heart disease.

People with early kidney disease may not have any symptoms. A BUN test can help uncover kidney problems early, when treatment can be more effective.

Other names for a BUN test: Urea nitrogen test, serum BUN, BUN to Creatinine Ratio

WHAT IS IT USED FOR?

A BUN test is often part of a series of tests called a comprehensive metabolic panel (CMP). It can help diagnose or monitor kidney diseases.

WHY DO I NEED A BUN TEST?

Your health care provider may order a BUN test:

- As part of a routine checkup

- To monitor the treatment of a kidney disease

- For screening if you are at risk of a kidney problem, since early kidney disease usually does not have any signs or symptoms. You may be at risk of developing a kidney problem because of:

 - A family history of kidney problems

 - Diabetes

 - High blood pressure

 - Heart disease

 - Obesity

 - Being over the age of 60

 - Smoking or using other tobacco products

 - A family history of kidney problems

 - Diabetes

 - High blood pressure

 - Heart disease

 - Obesity

 - Being over the age of 60

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Your provider may also check your BUN levels if you are having symptoms of later-stage kidney disease, which can include:

- Needing to go to the bathroom (urinate) frequently or infrequently

- Itching

- Fatigue

Swelling in your arms, legs, or feet
Muscle cramps
Trouble sleeping

WHAT HAPPENS DURING A BUN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Usually, no special preparation is necessary for a BUN test. But if your provider has ordered other tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Normal BUN levels can vary, but generally, a high blood urea nitrogen level is a sign that your kidneys are not working well. But abnormal results don't always mean that you have a medical condition that needs treatment. BUN levels can vary according to your sex. They also usually increase with age.

A few other reasons your BUN level might be higher than normal include:

Dehydration (too little fluid in your body)

Burns

Certain medicines

A high-protein diet

A recent heart attack

A few other reasons your BUN levels might be lower than normal include:

A lack of protein in your diet

Malnutrition

Liver disease

To learn what your results mean, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BUN TEST?

A BUN test is only one type of measurement of kidney function. If your provider thinks you may have kidney disease, you may need other tests. These may include tests to measure your:

Creatinine, which is another waste product that your kidneys remove from your body

GFR (Glomerular Filtration Rate), which gives an estimate of how well your kidneys are filtering blood

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Burn Evaluation

<https://medlineplus.gov/lab-tests/burn-evaluation/>

BURN EVALUATION

WHAT IS A BURN EVALUATION?

A burn is damage to your skin and/or other tissues that may be caused by:

Heat, such as fire, steam, and hot surfaces or liquids. These are called thermal burns. They are the most common type of burn.

Chemicals, such as acids or certain cleaning products.

Electricity, including lightning.

Radiation, including:

Sunburn.

Burns from radiation therapy for cancer.

Sunburn.

Burns from radiation therapy for cancer.

Friction. A friction burn happens when skin is scraped on a rough surface. The skin is torn and often burned from the heat caused by friction. Common causes of friction burns include bicycle and motorcycle accidents when skin rubs against the pavement. These friction burns are sometimes called "road rash." Carpet burns and rope burns are also types of friction burns.

A burn evaluation is an exam that health care providers do to find out how serious a burn is. The evaluation is done to estimate:

How deep into the skin the burn goes. The burn depth may be called the degree of the burn or the thickness of the burn.

How much of the skin is burned. This is called the extent of the burn.

This information helps providers decide how to treat a burn. You can often treat minor burns at home. But if you've been badly burned, you may need care in a hospital or a special burn center.

Other names: burn assessment

WHAT IS A BURN EVALUATION USED FOR?

Burn evaluations are used to find out how serious burn injuries may be. Providers use information from burn evaluations to make a treatment plan and predict how the burn will heal. This includes:

Deciding if a burn victim needs to receive intravenous (IV) fluids, and if so, how much fluid is needed. Serious burns may cause fluid to leak from tissues. This can lead to dehydration and other complications.

Deciding whether a burn victim needs care in the hospital or a special burn center.

Preparing for the risk of complications from burns, such as:

Breathing problems

Shock, a life-threatening drop in blood pressure

Infections and sepsis

Breathing problems

Shock, a life-threatening drop in blood pressure

Infections and sepsis

WHAT HAPPENS DURING A BURN EVALUATION?

In the case of serious burn injuries, the burn victim is first checked for life-threatening conditions that must be managed

right away. These include breathing and blood pressure problems, and other injuries. Clothing will be removed to make sure hot substances and/or chemicals don't continue to cause burns.

When a burn victim is in stable condition, a provider will begin a burn evaluation, which includes two steps.

The first step of a burn evaluation is figuring out the degree of the burn. This means checking to see how far into the skin the burn goes. Burns are grouped into three main types depending on how deep they are:

First-degree burns are also called superficial burns. These burns are the least serious type. They damage only the outer layer of the skin, called the epidermis. A sunburn is an example of a common first-degree burn. First-degree burns can often be treated at home, at your provider's office, or in an emergency room. They generally heal within a week.

Signs and symptoms of first-degree burns include:

Pain

Redness

Mild swelling of the skin, but usually no blisters

First-degree burns are also called superficial burns. These burns are the least serious type. They damage only the outer layer of the skin, called the epidermis. A sunburn is an example of a common first-degree burn. First-degree burns can often be treated at home, at your provider's office, or in an emergency room. They generally heal within a week.

Pain

Redness

Mild swelling of the skin, but usually no blisters

Second-degree burns are also called partial thickness burns. These burns damage the outer layer of skin and the layer underneath, called the dermis. Burns that involve the top part of the dermis are called superficial partial thickness burns. Burns that go deep into the dermis are called deep partial thickness burns.

Signs and symptoms include:

Pain

Deep redness of the skin

Blisters

Wet or glossy look from leaking fluid

Possible loss of some skin

Second-degree burns are also called partial thickness burns. These burns damage the outer layer of skin and the layer underneath, called the dermis. Burns that involve the top part of the dermis are called superficial partial thickness burns. Burns that go deep into the dermis are called deep partial thickness burns.

Pain

Deep redness of the skin

Blisters

Wet or glossy look from leaking fluid

Possible loss of some skin

Third-degree burns are also called full-thickness burns. They require emergency medical attention. They damage or completely destroy the entire thickness of the skin.

Third-degree burns usually require surgery to heal. Signs and symptoms include:

Loss of skin layers

No pain if the nerves have been destroyed (but you may feel pain from any first or second-degree burns near the third-degree burns)

Dry, leathery skin

Charred skin or patches that appear white, brown, or black

Third-degree burns are also called full-thickness burns. They require emergency medical attention. They damage or completely destroy the entire thickness of the skin.

Loss of skin layers

No pain if the nerves have been destroyed (but you may feel pain from any first or second-degree burns near the

third-degree burns)

Dry, leathery skin

Charred skin or patches that appear white, brown, or black

Burns that go into the fat layer under the skin or deeper into the muscle or bone may be called fourth or fifth-degree burns.

If a burn victim has burns that are second-degree (partial thickness) or deeper, the provider will continue with the next step of the evaluation.

The second step of a burn evaluation estimates how much of the body is burned. This estimate is a percentage of the person's total body surface area (TBSA) that has second-degree or deeper burns. First-degree burns are not included in this estimate.

There are different ways to estimate the percentage of the body that is burned:

The "rule of nines" divides the body into sections that equal about 9% or 18% of the total body surface area. For example, an arm equals 9% of a person's total body surface. This method helps providers quickly estimate how much of the body is burned in adults.

For children, the rule of nines has different surface area estimates. That's because children's heads and bodies have different proportions than adults.

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For children, the rule of nines has different surface area estimates. That's because children's heads and bodies have different proportions than adults.

The Lund-Browder chart is the most accurate way to estimate the percentage of the body that's burned, especially for children. Children's body proportions change as they grow, so the chart has different percentages for the head, neck, and body areas based on a child's age.

The Lund-Browder chart is the most accurate way to estimate the percentage of the body that's burned, especially for children. Children's body proportions change as they grow, so the chart has different percentages for the head, neck, and body areas based on a child's age.

The palmar surface or palm method is used in adults and children. It's helpful for estimating the extent of smaller burn areas. With this method, the surface of a person's hand (palm and fingers) equals about 1% of their total body surface area. For example, if you have a burn that appears to be the size of two of your hands, the extent of the burn is about 2% of your total body surface area.

WHAT DO THE RESULTS MEAN?

To understand the results of a burn evaluation, a provider will also consider the part of the body that's burned, the cause of the burn, and the health of the person with the burn. All of this information helps decide whether a burn is minor, moderate, or severe.

The seriousness of the burn guides treatment decisions. Your provider can explain whether a burn is minor, moderate, or severe and why.

In general:

Minor burns usually don't require hospitalization and they heal well. They involve only first and/or second-degree burns. But they may still be very painful. They can also make it hard to do normal activities for a few days.

Moderate burns may require a stay in the hospital, but they usually don't need treatment in a burn center. They involve only first and/or second-degree burns that cover a larger area than minor burns. With treatment, they usually heal well.

Severe burns require special care at a burn center. Treatment at a burn center may be recommended for:

Second-degree burns that cover more than 10% of the body

Third-degree burns

Second and third-degree burns of the face, feet, hands, genitals, or over the joints, such as the knees

Burns that happen with other injuries, such as broken bones

Burns in people who have other medical conditions that could affect their recovery, such as diabetes or heart disease

Chemical burns and burns from electricity, including lightning

Inhalation injuries that may be caused by breathing in smoke, hot gasses from a fire, or other substances that can burn

your mouth, airway, or lungs.

Second-degree burns that cover more than 10% of the body

Third-degree burns

Second and third-degree burns of the face, feet, hands, genitals, or over the joints, such as the knees

Burns that happen with other injuries, such as broken bones

Burns in people who have other medical conditions that could affect their recovery, such as diabetes or heart disease

Chemical burns and burns from electricity, including lightning

Inhalation injuries that may be caused by breathing in smoke, hot gasses from a fire, or other substances that can burn your mouth, airway, or lungs.

If you have questions about the results of a burn evaluation and/or burn treatment, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I SHOULD KNOW ABOUT A BURN EVALUATION?

You can often treat minor burns at home, but it's important to get medical help if a burn has any of these signs:

Appears to go deep into the skin

Is larger than 3 inches across

Looks leathery and dry

Has patches of white, brown, or black

Swells very fast

Gets more painful over time, has pus, or causes a fever, which may be signs of an infection

Also, get medical help for all burns that:

Are caused by electricity or chemicals

Are on the face, hands, genitals, buttocks, over a joint, or go all the way around a part of the body, such as a leg or the chest

Happen with inhalation (breathing in) of smoke, hot gasses, or chemicals

Happen to a child under age 5 or an adult older than age 70

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C. diff Testing

<https://medlineplus.gov/lab-tests/c-diff-testing/>

C. DIFF TESTING

WHAT IS C. DIFF TESTING?

C. diff testing checks a sample of your stool (poop) for signs of an infection with a bacteria called C. diff. C. diff is also called C. difficile or *Clostridioides difficile*. It used to be called *Clostridium difficile*.

A C. diff infection affects your large intestine. It can cause symptoms that range from mild diarrhea to serious dehydration and colitis (inflammation in your colon, which is the biggest part of your large intestine). In rare cases, a C. diff infection can become life-threatening.

C. diff bacteria are common in the environment, and it's in the stool of people who have the bacteria. If a person with a C. diff infection doesn't wash their hands well after using the bathroom, tiny amounts of contaminated stool can get on anything or anyone that they touch.

You can get the bacteria if you touch a contaminated surface and then touch your mouth. But if you're healthy, you usually won't get sick. Good bacteria in your digestive system will prevent the C. diff from growing out of control and causing an infection.

Most C. diff infections happen while you're taking certain antibiotics or within weeks after you finish taking them. That's because the antibiotics that kill bad bacteria may also get rid of good bacteria in your digestive system that help prevent infections. If your good bacteria are gone, C. diff can grow out of control and make you sick.

When C. diff grows, it can release toxins that irritate the lining of your colon and cause symptoms. Some C. diff tests check your stool sample for these toxins. Other tests check for genes from the types of C. diff that make toxins. Some types of C. diff don't make toxins or cause illness.

There are also tests that look for certain proteins from C. diff bacteria called GDH antigens. GDH antigen tests can show if you have C. diff in your body. But they can't tell if it's making you sick. To make an accurate diagnosis, your health care provider may order more than one type of C. diff test.

Other names: C. difficile, *Clostridium difficile*, Glutamate dehydrogenase test GDH *Clostridioides difficile*, C. difficile toxin test, C. difficile culture, antibiotic-associated pseudomembranous colitis, C. difficile toxin, PCR

WHAT IS IT USED FOR?

C. diff testing is most often used to find out if C. diff is causing diarrhea. But diarrhea can have many causes, so the test may be used with tests for parasites and other bacteria that also cause diarrhea.

WHY DO I NEED C. DIFF TESTING?

You may need C. diff testing if you have symptoms of an infection. The main symptoms of a C. diff infection include:

Watery diarrhea three or more times a day with no other known cause

Abdominal (belly) pain

Nausea

Loss of appetite

Fever

It is especially important to be tested if you have symptoms and you have a high risk of getting a C. diff infection. Your risk is high if you:

Have taken antibiotics within the last six to eight weeks.

Are aged 65 or older.

Have recently stayed in or worked at a hospital, nursing home, or other health care facility. C. diff bacteria are more common in health care facilities.

Have a weakened immune system from a health condition such as HIV or certain medicines or treatments, including

chemotherapy for cancer.

Had C. diff in the past. After you recover from a C. diff infection, you may still have the bacteria in your colon. It could begin to grow again and cause a new infection.

Know you were exposed to C. diff.

WHAT HAPPENS DURING C. DIFF TESTING?

You will need to provide a fresh sample of loose or liquid stool for your test. Your provider will probably give you a container or kit with instructions on how to collect the sample. There are different ways to collect the stool samples so follow the instructions carefully. In general, you'll need to:

Label the container with your name, the collection date, and time.

Collect a stool sample as instructed. This usually involves using a clean, dry container, or special paper or plastic wrap placed over the toilet to catch the stool.

Make sure the stool doesn't mix with any urine, toilet paper, or water from the toilet.

Close the container tightly.

Wash your hands well with soap and water.

Return the container according to the instructions. This is important to get an accurate test result. If you are unable to return the container right away, you should refrigerate your sample until you're ready to deliver it.

If you're collecting a sample from diapers, you'll get special instructions for using plastic wrap inside of a clean diaper. In certain cases, a provider may use a swab to take a stool sample from the rectum.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for C. diff testing.

ARE THERE ANY RISKS TO TESTING?

There is no known risk to having C. diff testing.

WHAT DO THE RESULTS MEAN?

There are several tests for C. diff that check for different signs of infection in your stool sample. Your provider may order a combination or two or more types of tests along with tests for other conditions that can cause diarrhea. Your provider will consider all your test results, your symptoms, and your medical history to figure out whether C. diff is causing your illness.

In general:

A positive (abnormal) test result may mean that toxins from C. diff bacteria are causing your symptoms. You may be diagnosed with C. diff if your tests show that your sample contained:

C. diff toxin genes. These are genes from the types of C. diff that can cause illness.

C. diff toxins. These tests check for two types of toxins called A and B.

A negative (normal) test result generally means that no signs of a C. diff infection were found in your sample. Your symptoms are probably caused by something else.

But a negative result on a C. diff toxin test alone isn't enough to rule out a C. diff infection. That's because C. diff toxins can break down quickly if your sample isn't kept cool. If this happens, a toxin test may not find toxins in your sample, even when they're making you sick. For this reason, toxin tests are usually done with tests for C. diff toxin genes and/or C. diff GDH antigens.

If you're diagnosed with a C. diff infection while you're taking an antibiotic for another infection, your provider may tell you to stop taking it. Your treatment may include:

Taking a different type of antibiotic that targets C. diff bacteria

Fluids if you are dehydrated

Some people may take probiotics, a type of supplement that might support good bacteria in your digestive system. But researchers aren't sure if these supplements are helpful.

If C. diff infections keep returning after treatment, talk with your provider. There are treatments that can help reduce the chance that C. diff will come back. These treatments use good bacteria from the stool of healthy people to help control the growth of C. diff.

If you have questions about your results and/or treatment, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT C. DIFF TESTING?

After treatment for a C. diff infection, you won't be tested again to see if you're cured. That's because it's common to still have C. diff in your body after you get better. A test would only show that the bacteria is still there, but not whether you're likely to get sick again.

To lower your chance of getting or spreading C. diff:

Avoid taking antibiotics that you don't really need.

Let all your providers know if you've had a C. diff infection, including your dentist. This information will help them make decisions if you need antibiotics in the future.

Wash your hands with soap and water every time you use the bathroom and before you eat anything. C. diff is contagious. People who have it spread it by not washing their hands before touching people and things that other people touch.

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RELATED HEALTH TOPICS

- Antibiotics
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- Bowel Movement
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- Ova and Parasite Test
- White Blood Cell (WBC) in Stool

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C-Peptide Test

<https://medlineplus.gov/lab-tests/c-peptide-test/>

C-PEPTIDE TEST

WHAT IS A C-PEPTIDE TEST?

This test measures the level of C-peptide in a sample of your blood or urine (pee). C-peptide is a chain of amino acids (the building blocks of proteins) that your pancreas makes in the process of making insulin. Insulin is a hormone that your pancreas makes to help blood glucose (blood sugar) get into your cells, where it can be used for energy. This helps keep your blood glucose at healthy levels.

C-peptide and insulin enter your bloodstream at the same time and in equal amounts, but C-peptide stays in your blood longer. It is also unaffected by outside sources of insulin such as the medicines involved in diabetes treatment. In other words, C-peptide can give an accurate picture of how much insulin your body is making naturally.

Because of this, health care providers may rely on a C-peptide test to measure insulin levels rather than testing insulin levels directly.

Other names: insulin C-peptide, connecting peptide insulin, proinsulin C-peptide

WHAT IS IT USED FOR?

A C-peptide test can provide important information to help:

Find the cause of hypoglycemia (low blood glucose). Some types of hypoglycemia are linked to high insulin levels. By checking your C-peptide levels, your provider can tell whether this is the case. Some causes of this kind of hypoglycemia include:

Liver or kidney disease.

Malnutrition.

Drinking too much alcohol without eating.

A tumor in your pancreas (uncommon).

Side effects from certain diabetes medicines, including insulin.

Liver or kidney disease.

Malnutrition.

Drinking too much alcohol without eating.

A tumor in your pancreas (uncommon).

Side effects from certain diabetes medicines, including insulin.

Manage diabetes treatment. If you have type 1 or type 2 diabetes, treatment decisions may depend on knowing how much insulin your pancreas is making. A C-peptide test can provide an accurate measurement, even if you take insulin for diabetes. That's because your C-peptide levels depend only on how much insulin your pancreas makes.

Monitor treatment for a tumor in your pancreas, called an insulinoma (uncommon). These tumors make too much insulin and cause low blood glucose. They are almost always benign (not cancer) and can usually be removed with surgery.

Find out whether you have type 1 or type 2 diabetes if your diagnosis is uncertain. Usually, your health care provider can diagnose which type of diabetes you have without this test. But in certain cases, your provider may need to use a C-peptide test to make their diagnosis.

WHY DO I NEED A C-PEPTIDE TEST?

You may need a C-peptide test if you have been diagnosed with a pancreatic tumor (insulinoma) or hypoglycemia (low blood glucose) without a clear cause.

Hypoglycemia is a condition in which your blood glucose is lower than what is healthy for you. If you have hypoglycemia, you may experience:

Shakiness

Sweating

Headache

Hunger or nausea

An irregular or fast heartbeat (called arrhythmia)

Dizziness

Fatigue

In more extreme cases, hypoglycemia can cause confusion, coma, and seizures.

You may also need the test if you have diabetes and:

Take insulin, and your provider is considering changing your treatment.

Have type 2 diabetes, and your provider wants to see if you need to start taking insulin.

Have hypoglycemia, which could be caused by taking too much diabetes medicines.

WHAT HAPPENS DURING A C-PEPTIDE TEST?

A C-peptide test usually uses a sample of your blood. But the test may also be done on a sample of all your urine collected over a 24-hour period.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a 24-hour urine sample test, you will be given a special container to collect your urine and instructions on how to collect and store your samples. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or in a cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

For a C-peptide blood test, you may need to fast¹ (not eat or drink) for 8-12 hours before the test. Your provider may also want to test C-peptide after you've eaten something. This way, they can see what your C-peptide levels are when your pancreas is actively working. Ask your provider if there are any specific instructions you need to follow before either a blood or a urine test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly. There are no known risks to a urine test.

WHAT DO THE RESULTS MEAN?

The meaning of your results depends on your age, health, the medicines you take, and the results of other tests, such as a blood glucose test. In general:

A high level of C-peptide usually means that your body is making too much insulin. Conditions that cause insulin levels that are high for you include:

Type 2 diabetes.

Cushing's syndrome.

Insulinoma, a tumor in your pancreas.

Kidney failure.

A low level of potassium in your blood.

Type 2 diabetes.

Cushing's syndrome.

Insulinoma, a tumor in your pancreas.

Kidney failure.

A low level of potassium in your blood.

A low level of C-peptide may mean your body isn't making enough insulin. Conditions that cause insulin levels that are low for you include:

Type 1 diabetes and, in some cases, type 2 diabetes.

Taking too much insulin to treat diabetes, which may block your pancreas from making insulin on its own.

A severe infection.

Addison disease.

Liver disease.

Type 1 diabetes and, in some cases, type 2 diabetes.

Taking too much insulin to treat diabetes, which may block your pancreas from making insulin on its own.

A severe infection.

Addison disease.

Liver disease.

If you have been treated for an insulin-producing tumor in your pancreas, a decrease in your C-peptide levels means your treatment is working. An increase in your C-peptide may mean that your tumor is back.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A C-PEPTIDE TEST?

A C-peptide test is not used to diagnose diabetes. Other tests, such as blood glucose and urine glucose, are used to screen for and diagnose diabetes.

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RELATED HEALTH TOPICS

Blood Glucose
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What You Need to Know About Blood Testing

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C-Reactive Protein (CRP) Test

<https://medlineplus.gov/lab-tests/c-reactive-protein-crp-test/>

C-REACTIVE PROTEIN (CRP) TEST

WHAT IS A C-REACTIVE (CRP) PROTEIN TEST?

A c-reactive protein test measures the level of c-reactive protein (CRP) in a sample of your blood. Your liver makes CRP in response to inflammation.

Inflammation is your body's way of protecting your tissues and helping them heal from an injury, infection, or other disease. When it's acute (sudden) and for a short time, inflammation is helpful. For example, when you get a cut on your skin, the area around the cut may turn red and swell. These are signs that inflammation is working to stop infection. Inflammation acts in a similar way when responding to germs or harmful substances (toxins) inside your body.

If inflammation lasts too long, however, it can damage healthy tissues. This is called chronic (long-term) inflammation, and it is most often caused by certain autoimmune disorders. Chronic inflammation can also happen if your tissues are repeatedly injured or irritated, for example from smoking or chemicals in the environment.

Other names: c-reactive protein, serum

WHAT IS IT USED FOR?

By measuring the levels of c-reactive protein in your blood, a CRP test can tell your health care provider how much inflammation you have in your body. High CRP levels may mean you have an acute or chronic health condition, such as: Infections from bacteria or viruses.

Inflammatory bowel disease, disorders of the intestines that include Crohn's disease and ulcerative colitis.

Autoimmune disorders, such as lupus, rheumatoid arthritis, and vasculitis.

Lung diseases, such as asthma.

Smoking and exposure to environmental toxins such as polluted air and hazardous waste can also cause high CRP levels.

WHY DO I NEED A CRP TEST?

You may need this test if you have symptoms of a bacterial infection, such as:

Fever or chills

Rapid heart rate

Rapid breathing

Nausea and vomiting

You may also need a CRP test if your provider wants to:

Check for a chronic condition that causes inflammation.

Monitor for infection after surgery.

Monitor your condition and treatment if you have an infection or a chronic condition.

Make treatment decisions if you have sepsis. Sepsis is your body's extreme response to an infection that spreads to your blood. It's a life-threatening medical emergency.

CRP levels rise and fall depending on how much inflammation is in your body. So, if your CRP levels fall, it's a sign that your treatment for inflammation is working or that you're healing on your own.

WHAT HAPPENS DURING A CRP TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle

goes in or out. This process usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Some medicines may affect your results. So, tell your provider about any supplements or medicines that you take, including magnesium and non-steroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen and aspirin. Don't stop taking any prescription medicines without talking with your provider first.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

In general, healthy people have very low amounts of CRP in their blood. A value of 0.8-1.0 milligrams per deciliter (mg/dL) or lower is thought to be a healthy amount. Any increases above normal may mean you have inflammation in your body. However, increases can also be linked to:

Insomnia.

Depression.

Hormone replacement therapy.

Obesity.

Sex. Females often have slightly higher CRP levels than males.

Your CRP test results tell you how much inflammation you have in your body, but not what's causing it or where it is. To make a diagnosis, your provider will look at your CRP results along with the results of other tests, your symptoms, and medical history.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CRP TEST?

A CRP test is sometimes confused with a high-sensitivity-(hs) CRP test. They both measure CRP, but they are used for different conditions. An hs-CRP test measures very tiny increases in your CRP levels. It is used to estimate your risk of heart disease.

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CA-125 Blood Test (Ovarian Cancer)

<https://medlineplus.gov/lab-tests/ca-125-blood-test-ovarian-cancer/>

CA-125 BLOOD TEST (OVARIAN CANCER)

WHAT IS A CA-125 BLOOD TEST?

This test measures the amount of a protein called CA-125 (cancer antigen 125) in a sample of your blood. CA-125 is a type of tumor marker. Tumor markers are substances that are often made by cancer cells or by normal cells in response to cancer.

Organs in the female reproductive system (such as the ovaries and fallopian tubes) may make CA-125 when those organs are inflamed. This may happen in response to normal bodily functions, such as a menstrual period. However, your body may also produce high levels of CA-125 in response to ovarian cancer.

The ovaries are a pair of female reproductive glands that store ova (eggs) and make female hormones. Ovarian cancer happens when the cells in an ovary begin to grow out of control.

Other names: cancer antigen 125, glycoprotein antigen, ovarian cancer antigen, CA-125 tumor marker

WHAT IS IT USED FOR?

Measuring certain tumor markers often help health care providers treat cancer. So, because high levels of CA-125 can be a sign of ovarian cancer, a CA-125 blood test can help your provider:

See if ovarian cancer treatment is working or to check if ovarian cancer has come back. This is the most common use of CA-125 blood testing.

To learn more about a growth or lump in your pelvis (the area below your belly). If a suspicious lump shows up on imaging, such as an ultrasound, your provider may check your CA-125 levels along with other tests to find out whether the lump could be ovarian cancer.

To screen for ovarian cancer if your risk is very high. If your family health history includes ovarian cancer, your provider may suggest a CA-125 blood test and other tests to look for signs of cancer.

Your provider cannot use a CA-125 to screen for ovarian cancer if you aren't at high risk. That's because other conditions that aren't cancer can also cause high CA-125 levels. These include:

Endometriosis

Pelvic inflammatory disease (PID)

Uterine fibroids

Liver disease

Natural bodily functions such as your menstrual period and pregnancy can also cause high CA-125 levels.

WHY DO I NEED A CA-125 BLOOD TEST?

If you've been diagnosed with ovarian cancer, you may have several CA-125 blood tests:

During your treatment to see if your cancer is going away.

After your treatment, to check if your cancer has returned.

If you have a lump in your pelvis that could be ovarian cancer, your provider may use a CA-125 test to help find out if it could be ovarian cancer. But only a biopsy can diagnose ovarian cancer.

If you have a very high risk of getting ovarian cancer, your provider may suggest checking your CA-125 levels. If they're high, you'll probably need more tests to find out if you have cancer. You're more likely to get ovarian cancer if you:

Have a mother or sister, or two or more other relatives who've had ovarian cancer.

Have family members who've had breast cancer or colorectal cancer (colon cancer).

Have inherited certain gene changes or conditions that increase your risk of ovarian cancer, such as:

Changes in your BRCA 1 or BRCA 2 genes.

Lynch syndrome (also called hereditary non-polyposis colorectal cancer), an inherited disorder that increases the risk for many types of cancer.

Changes in your BRCA 1 or BRCA 2 genes.

Lynch syndrome (also called hereditary non-polyposis colorectal cancer), an inherited disorder that increases the risk for many types of cancer.

Have had breast, uterine (uterus), or colorectal cancer.

If you're concerned about getting ovarian cancer, talk with your provider about your risk.

WHAT HAPPENS DURING A CA-125 BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a CA-125 blood test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your provider will review your CA-125 test results along with other information about your condition. Together, you can discuss how your results affect your diagnosis, treatment, and need for more testing.

In general:

If you are being treated for ovarian cancer, and the results of several tests show that your CA-125 levels are going down, it usually means that the treatment is helping. If your levels go up or stay the same over time, it may mean that the treatment isn't working.

If you have finished your treatment for ovarian cancer and your CA-125 levels begin to increase over time, your cancer may be coming back.

If you have a high risk for ovarian cancer or have a suspicious pelvic lump, high CA-125 levels could be a sign of cancer. Your provider will usually order more tests to make a diagnosis.

While high CA-125 levels don't always mean ovarian cancer, a normal CA-125 test result doesn't rule it out. That's because CA-125 levels may be low in the early stages of cancer. And not everyone with ovarian cancer makes high levels of CA-125.

Talk with your provider if you have questions about your results.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CA-125 BLOOD TEST?

The most common type of ovarian cancer is epithelial ovarian cancer. If you have been treated for this type of cancer, you may be tested for a tumor marker called HE4 along with CA-125. Some studies show that measuring both tumor markers provides more accurate information to check whether treatment is working and to look for the return of this type of cancer.

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CA 19-9 Blood Test (Pancreatic Cancer)

<https://medlineplus.gov/lab-tests/ca-19-9-blood-test-pancreatic-cancer/>

CA 19-9 BLOOD TEST (PANCREATIC CANCER)

WHAT IS A CA 19-9 BLOOD TEST?

A CA 19-9 test measures the amount of a protein called CA 19-9 (cancer antigen 19-9) in a sample of your blood. CA 19-9 is a type of tumor marker. Tumor markers are substances made by cancer cells or by normal cells in response to cancer.

High levels of CA 19-9 can be a sign of certain cancers of the digestive tract. These include cancers of the:

Pancreas.

Bile duct.

Colon and rectum.

Stomach.

Ovaries.

Bladder.

However, high CA 19-9 levels can also be a sign of other conditions, including:

Gallstones, which can block your bile ducts.

Cholangitis, which is an infection of the bile ducts.

Pancreatitis, which is inflammation of the pancreas.

Cirrhosis (or scarring) of the liver.

Bile duct disease, including a blockage or infection.

Liver disease.

Cystic fibrosis.

Because high levels of CA 19-9 can mean different things, health care providers don't use the results from a CA 19-9 test alone to screen for or diagnose cancer or other diseases. To do this, your provider will have to consider the results of a CA19-9 test as well as the results of other tests.

Other names: cancer antigen 19-9, carbohydrate antigen 19-9, CA 19-9 measurement, CA 19-9 radioimmunoassay (RIA)

WHAT IS IT USED FOR?

CA 19-9 blood tests may be used to:

Monitor certain types of cancer and cancer treatment. CA 19-9 levels often go up as cancer grows and go down as tumors shrink. So, your provider may order this test throughout your treatment to see how well it's working.

Help make a prognosis. A prognosis is a health care professional's prediction for how an illness will behave over time. Your provider may use this test to help determine your cancer's stage (how advanced it is) and how likely it is to respond to treatment.

Check whether cancer has returned after treatment. Because high CA 19-9 levels will often show up before you get any other symptoms of cancer, your provider may use this test along with several others to see if your cancer has returned.

Check for the presence of certain cancers, but only when used with other tests.

Some people do not make CA 19-9 even when they have a cancer that usually produces high levels of CA 19-9. For these people, a CA 19-9 test is not useful.

If your provider is using a CA 19-9 blood test to help check for signs of cancer, they will perform additional tests, including:

A physical examination. Some cancers associated with CA 19-9 can have noticeable symptoms. For example, pancreatic cancer can cause your liver or gallbladder to swell.

Imaging tests, such as x-rays, computer tomography (CT) scans, and magnetic resonance imaging (MRI).

Cholangiopancreatography. Your provider can use this to look at the pancreatic ducts and bile ducts to see if they're blocked or narrowed. If they are, it could be a sign of a pancreatic tumor.

WHY DO I NEED A CA 19-9 TEST?

You may need a CA 19-9 blood test if you:

Have been diagnosed with any of the types of cancer associated with CA 19-9.

Are going through cancer treatment. Your provider may test you on a regular basis to see if your treatment is working.

Have completed cancer treatment. Your provider may test you to see if your CA 19-9 levels stay low. If they rise, it could be a sign that your cancer has returned.

Your provider may also order this test if you have symptoms of the cancers associated with high CA 19-9 levels. These can include:

Jaundice (yellowing of the skin).

Weight loss.

Abdominal (belly) pain.

Weakness.

Swollen gallbladder or liver.

WHAT HAPPENS DURING A CA 19-9 BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Usually, you don't need any special preparations for a CA 19-9 blood test. But ask your provider to be sure. If you take certain vitamins, you may need to stop them before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If you're being treated for cancer, you may be tested several times during your treatment. Your provider will look at all your test results to see how your CA 19-9 levels have changed. Your results may show:

Your levels of CA 19-9 are increasing. This may mean your tumor is growing, and/or your treatment is not working. More tests are usually needed to find out for sure.

Your levels of CA 19-9 are decreasing. This may mean your tumor is shrinking and your treatment is working.

Your levels of CA 19-9 have stayed the same. This may mean your disease is stable and hasn't gotten better or worse.

Your CA 19-9 levels decreased after treatment but increased later. This may mean your cancer has come back or grown. You'll need more tests to check whether cancer is really causing the higher level of CA 19-9.

If you don't have cancer but your test results show a high level of CA 19-9, you may not have a health problem that needs treatment. Healthy people can have high CA 19-9 levels. But high levels may be a sign of a condition, such as pancreatitis or gallstones.

Talk with your provider about what your results may mean.

If your provider thinks you may have one of these conditions, you will probably need other tests to find out for sure.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CA 19-9 TEST?

Labs use different methods to measure the amount of CA 19-9 in your blood sample. The test method can affect your results. If you're having regular tests to monitor cancer, it's important to have your tests done the same way, and usually in the same lab. This allows your provider to compare your results over time. Your test report usually says what method was used to measure your CA 19-9 levels. If you have any questions about how your tests are done, talk with your provider.

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Calcitonin Test

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CALCITONIN TEST

WHAT IS A CALCITONIN TEST?

A calcitonin test measures the level of calcitonin in a sample of your blood. Calcitonin is a hormone that helps control the level of calcium in your blood. Calcitonin is made in your thyroid gland by cells called "C cells." Your thyroid is a small, butterfly-shaped gland in your neck.

Normally, you have a small amount of calcitonin in your blood. But if you have too much, it may be a sign of:

Medullary thyroid cancer (MTC), a rare cancer that happens when C cells in your thyroid grow out of control. C cells grow in the inside part of your thyroid, which is called the medulla.

C-cell hyperplasia, a rare condition that causes abnormal growth of C cells in your thyroid. C-cell hyperplasia can turn into medullary thyroid carcinoma.

Your health care provider may order a calcitonin test if you have symptoms of these conditions or if they run in your family.

Other names: thyrocalcitonin, CT, human calcitonin, hCT

WHAT IS IT USED FOR?

A calcitonin test is most often used to:

Help diagnose C-cell hyperplasia and medullary thyroid cancer in people who have symptoms

Find out if treatment for medullary thyroid cancer is working

Check whether medullary thyroid cancer has returned after treatment

Screen people who have a high risk of medullary thyroid cancer. The risk is high for people who have a family history of a rare genetic disorder called multiple endocrine neoplasia type 2 (MEN 2).

There are different types of MEN 2 that often cause tumors in certain glands that make hormones, including your thyroid. All types of MEN 2 increase your risk of medullary thyroid cancer. MEN 2 is caused by changes in a gene called RET. Changes in genes are also called gene variants or mutations.

You can inherit MEN 2 - and an increased risk for medullary thyroid cancer - if either of your parents passed a changed RET gene down to you. If MEN 2 runs in your family, regular calcitonin tests can help find cancer early when it may be easier to treat.

WHY DO I NEED A CALCITONIN TEST?

You may need a calcitonin test to help check for medullary thyroid cancer or C-cell hyperplasia if:

You have symptoms of either condition which may include:

A lump in the front of your neck

Swollen lymph nodes in your neck (also called "swollen glands")

Pain in your throat

Trouble swallowing or breathing

A change in your voice, such as hoarseness

A long-lasting cough when you don't have a cold

Diarrhea

A lump in the front of your neck

Swollen lymph nodes in your neck (also called "swollen glands")

Pain in your throat

Trouble swallowing or breathing

A change in your voice, such as hoarseness

A long-lasting cough when you don't have a cold

Diarrhea

You have symptoms of multiple endocrine neoplasia type 2 (MEN 2). MEN 2 can affect your thyroid, adrenal, and parathyroid glands. Symptoms depend on which glands are affected. They may include:

Fatigue

Depression

Aches and pains in bones and joints

Nausea and vomiting

Increased thirst and urination (peeing)

High blood pressure

Headaches

Arrhythmia (a problem with the rate or rhythm of your heart)

Sweating

Fatigue

Depression

Aches and pains in bones and joints

Nausea and vomiting

Increased thirst and urination (peeing)

High blood pressure

Headaches

Arrhythmia (a problem with the rate or rhythm of your heart)

Sweating

You have a family history of MEN 2, which increases your risk of developing medullary thyroid cancer. If MEN 2 runs in your family, you and your family members, including children, may need regular calcitonin tests to help find cancer before you have symptoms. Your provider may also suggest a blood test to check for changes in your RET genes that cause MEN 2.

If you've been diagnosed with medullary thyroid cancer, you'll need calcitonin testing to:

Find out how well your treatment is working.

Check whether the cancer has come back after you finish treatment.

WHAT HAPPENS DURING A CALCITONIN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if you need to fast and if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The meaning of your calcitonin test results depends on the reason you were tested. If you had a calcitonin test:

To help diagnose or screen for medullary thyroid cancer or C-cell hyperplasia:

A normal calcitonin level may mean that you don't have either condition, but it doesn't rule them out. That's because medullary thyroid cancer and C-cell hyperplasia don't always cause high calcitonin. Your provider may order other tests to learn more.

A high calcitonin level means that the C cells in your thyroid are making too much calcitonin. The higher the level, the more likely it is that you have medullary thyroid cancer or C-cell hyperplasia. But high calcitonin levels may be caused by other conditions, including certain other cancers, certain medicines, and pregnancy. To make a diagnosis, your provider will consider other information, including other test results.

A low calcitonin level is not known to be a medical problem.

To monitor treatment for medullary thyroid cancer or to check whether cancer has come back after treatment: The meaning of your test results depends on the type of treatment you're having or had. Treatment usually starts with surgery to remove your thyroid gland. Other treatments include radiation therapy, chemotherapy, and/or medicines.

In general:

Calcitonin levels that decrease and stay low over time usually mean that your treatment is helping.

A calcitonin level that decreases but is still higher than normal may mean that you still have some cancer tissue left in your body. You may need more tests and treatment.

A calcitonin level that increases over time usually means that your cancer has returned or spread to other parts of your body.

Ask your provider to explain what your test results mean for your health.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CALCITONIN TEST?

In certain cases, your provider may order a more accurate type of calcitonin test called a "calcitonin stimulation test." For this test, a sample of your blood will be taken as usual. Then you'll be given medicine to make your body produce calcitonin. The medicine is injected into a vein intravenously (by IV). Afterwards, more blood samples will be taken a few minutes apart. If your calcitonin increases by a lot in each blood sample, it may mean you have C-cell hyperplasia or medullary thyroid carcinoma, but you'll need other tests to find out for sure.

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Calcium Blood Test

<https://medlineplus.gov/lab-tests/calcium-blood-test/>

CALCIUM BLOOD TEST

WHAT IS A CALCIUM BLOOD TEST?

A calcium blood test measures the amount of calcium in your blood. Too much or too little calcium in your blood may be a sign of a wide range of medical conditions, such as bone disease, thyroid disease, parathyroid disorders, kidney disease, and other conditions.

Calcium is one of the most important minerals in your body. About 1% of the calcium in your body is in your blood. The rest is stored in your bones and teeth. Having the right amount of calcium in your blood is necessary for your nerves, muscles, and heart to work properly. It also helps blood vessels move blood throughout your body and helps release hormones that affect many body functions.

Other names: total calcium, ionized calcium

WHAT IS IT USED FOR?

A blood calcium test is used to check your general health. It's also used to help diagnose or monitor many medical conditions, including those that affect your bones, kidneys, digestive system, thyroid, and parathyroid glands.

There are two types of calcium blood tests that measure different forms of blood calcium:

Total calcium test measures all the calcium in your blood. You have two types of blood calcium that are normally present in about equal amounts:

"Bound calcium" is attached to proteins in your blood.

"Free calcium" is not attached to proteins. It's also called ionized calcium. This form of blood calcium is active in many body functions.

Normally, your body tightly controls the balance of bound and ionized calcium, so a total calcium test gives a good estimate of how much ionized calcium you have.

A total calcium test is the most common test for blood calcium. It's often part of a basic metabolic panel (BMP) and a comprehensive metabolic panel (CMP), which are both routine screening tests.

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An ionized calcium test measures only the "free calcium" in your blood that isn't attached to proteins. An ionized calcium test is more difficult to do, so it's usually ordered if the results of a total calcium test aren't normal. You may also have this test if you have a condition that affects your body's ability to balance the amounts of ionized and bound calcium in your blood or if you are seriously ill or having surgery.

WHY DO I NEED A CALCIUM BLOOD TEST?

Your health care provider may have ordered a basic or comprehensive metabolic panel, which includes a calcium blood test, as part of your routine checkup. You may also have this test to diagnose or monitor conditions that can affect your blood calcium or if you have symptoms of abnormal calcium levels.

Symptoms of high calcium levels may include:

Digestive symptoms, such as:

Constipation

Nausea and vomiting

Abdominal (belly) pain

Loss of appetite

Constipation

Nausea and vomiting

Abdominal (belly) pain

Loss of appetite

Increased thirst

Urinating (peeing) more than usual

Kidney stones

Bone, joint, and muscle symptoms, such as weakness, aches, and pains

Fatigue

A change in mental health, such as depression or confusion

Symptoms of low calcium levels may include:

Dry skin, coarse hair, and nails that easily break (after a long period of low levels)

Muscle cramps, spasms, or stiffness

Tingling in the lips, tongue, fingers, and feet

Irregular heartbeat (arrhythmia)

Seizures, if calcium levels are extremely low

Many people with high or low calcium levels don't have any symptoms. So your provider may order a calcium test if you have a known condition that may affect your calcium levels, such as:

Kidney disease

Thyroid or parathyroid disease

Malnutrition

Problems absorbing calcium

Certain types of cancer

WHAT HAPPENS DURING A CALCIUM BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You usually don't need any special preparations for a calcium blood test or a basic or comprehensive metabolic panel. You may need to stop taking certain medicines or supplements, such as vitamin D, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Results from a total calcium test that are higher than normal (hypercalcemia) may be a sign of many types of conditions, such as:

Overactive parathyroid glands (hyperparathyroidism), a condition in which your parathyroid glands produce too much parathyroid hormone

Certain types of cancer, including cancer that spreads to the bone

Bone disorders, including Paget's disease of bone

Taking too much vitamin D over a long period of time

Results from a total calcium test that are lower than normal (hypocalcemia) may be a sign of:

Low blood protein levels, which may be caused by liver disease or malnutrition

Underactive parathyroid glands (hypoparathyroidism), a condition in which your parathyroid glands produce too little parathyroid hormone

Too little calcium in your diet

Too little vitamin D or magnesium

Pancreatitis

Kidney disease

If your results from a total calcium blood test are not in the normal range, it doesn't always mean that you have a medical condition that needs treatment. Your diet and certain medicines can affect your calcium levels. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CALCIUM BLOOD TEST?

A calcium blood test does not tell you how much calcium is in your bones. Bone health can be measured with a type of x-ray called a bone density scan, or DEXA scan. A DEXA scan measures the mineral content, including calcium, and other aspects of your bones.

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Phosphate in Urine

Uric Acid Test

What You Need to Know About Blood Testing

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Calcium in Urine Test

<https://medlineplus.gov/lab-tests/calcium-in-urine-test/>

CALCIUM IN URINE TEST

WHAT IS A CALCIUM IN URINE TEST?

A calcium in urine test measures the amount of calcium in your urine (pee). If your urine calcium levels aren't normal, it may be a sign of kidney disease, a parathyroid disorder, bone disease, or another condition that affects how your body uses calcium.

Calcium is one of the most important minerals in your body. Most of your calcium is stored in your bones and teeth. About 1% of the calcium in your body is in your blood. If there's too much calcium in your blood, the extra calcium may be stored in your bones or passed in your urine and stool (poop).

Having just the right amount of calcium in your blood is necessary for your nerves, muscles, and heart to work properly. Your blood calcium levels are carefully controlled by your parathyroid glands, which are 4 pea-sized glands in your neck. If your blood calcium levels get too low, these glands make parathyroid hormones (PTH). The hormones are chemical messengers in your bloodstream that work to increase your blood calcium levels until they are normal again.

PTH raises blood calcium levels by:

Causing your bones to release calcium into your blood

Helping your intestines absorb calcium from food

Helping your kidneys hold on to calcium and return it to your blood instead of flushing it out in urine

Other names: urinalysis (calcium), urinary Ca^{+2}

WHAT IS IT USED FOR?

A calcium in urine test may be used with other tests to help:

Diagnose kidney stones if you have symptoms. Kidney stones form when you have high levels of certain minerals in your urine. Most kidney stones are made of calcium. So, a calcium in urine test helps your health care provider figure out if a kidney stone could be causing your symptoms and whether it's likely made of calcium. This information also helps decide what treatment you may need to prevent future stones.

Diagnose or monitor problems with how well your kidneys are working.

Learn more about how your parathyroid glands are working if you:

Have symptoms of underactive parathyroid glands

Have been diagnosed with overactive parathyroid glands

Have symptoms of underactive parathyroid glands

Have been diagnosed with overactive parathyroid glands

Learn more about the cause of a bone disease you have, such as osteoporosis.

Learn more about the cause of abnormal results on a calcium blood test.

WHY DO I NEED A CALCIUM IN URINE TEST?

You may need a calcium in urine test if you have symptoms of a kidney stone. These symptoms include:

Sharp pain in your lower abdomen (belly), back, side, or groin

Blood in your urine

Frequent need to urinate (pee)

Pain while urinating

Inability to urinate or urinating only small amounts

Cloudy or bad-smelling urine

Nausea and vomiting

You may also need a calcium in urine test if:

You have been diagnosed with overactive parathyroid glands (hyperparathyroidism). This means your glands make too much parathyroid hormone (PTH). The test can help find out how severe your condition is. It can also help show if a kidney problem may be causing high PTH.

You have symptoms of underactive parathyroid glands (hypoparathyroidism). This means your glands make too little PTH. Without enough PTH, your kidneys may filter too much calcium out of your blood and into your urine. Symptoms of hypoparathyroidism include:

Tingling fingers and toes

Muscle cramps and spasms in the face, hands, legs, and feet

Muscle weakness and fatigue

Seizures

Tingling fingers and toes

Muscle cramps and spasms in the face, hands, legs, and feet

Muscle weakness and fatigue

Seizures

Your provider may also order a calcium in urine test if you had a routine calcium blood test with abnormal results.

WHAT HAPPENS DURING A CALCIUM IN URINE TEST?

You'll need to collect all your urine during a 24-hour period. This is called a 24-hour urine sample test. You will be given a special container to collect your urine and instructions on how to collect and store your samples. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or in a cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know if there are any special instructions to follow. You may be asked to avoid certain foods and medicines for several days before the test. But don't stop taking any medicines without talking with your provider first.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a calcium in urine test.

WHAT DO THE RESULTS MEAN?

Higher than normal calcium levels in urine may be a sign of many conditions, including:

A kidney stone

Overactive parathyroid glands (hyperparathyroidism)

Certain types of cancer, including cancer that spreads to the bones

Bone disorders, including Paget's disease of bone and osteoporosis

Sarcoidosis

Lower than normal calcium levels in urine may be a sign of many conditions, including:

Kidney disease

Underactive parathyroid glands (hypoparathyroidism)

Too little vitamin D or magnesium

Malabsorption disorder

Malnutrition

If your calcium levels aren't normal, it doesn't always mean you have a medical condition that needs treatment. Urine calcium levels can be affected by:

Your diet, for example, eating foods with a lot of sodium (salt)

Certain supplements, such as taking too much vitamin D for a long period of time

Certain medicines, including antacids and some diuretics (water pills)

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CALCIUM IN URINE TEST?

A calcium in urine test does not tell you how much calcium is in your bones. Bone health can be measured with a type of x-ray called a bone density scan, or DEXA scan. A DEXA scan measures calcium and other minerals in your bones.

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RELATED HEALTH TOPICS

Calcium
Kidney Stones
Paget's Disease of Bone
Parathyroid Disorders
Vitamin D Deficiency

RELATED MEDICAL TESTS

Calcium Blood Test
Crystals in Urine
How to Cope with Medical Test Anxiety
How to Understand Your Lab Results
Intravenous Pyelogram (IVP)
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Phosphate in Blood
Phosphate in Urine
Uric Acid Test

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Calprotectin Stool Test

<https://medlineplus.gov/lab-tests/calprotectin-stool-test/>

CALPROTECTIN STOOL TEST

WHAT IS A CALPROTECTIN STOOL TEST?

A calprotectin stool test measures a protein called calprotectin in a sample of your stool (poop). The test is also called a fecal calprotectin test. It's used to check for inflammation (swelling and irritation) in your intestines.

It's normal to have a small amount of calprotectin in your stool. But high levels are a sign that your intestines are inflamed. When you have inflammation in your intestines, your immune system sends certain types of white blood cells (neutrophils) to the inflamed area. These white blood cells release calprotectin into your intestines where it mixes with your stool.

Inflammation in your intestines can cause severe watery or bloody diarrhea with abdominal (belly) pain and/or cramping that last for more than a few days. Other types of digestive conditions can also cause these symptoms. If these symptoms last a long time or come and go, it's important to know if your intestines are inflamed. That's because inflammation can damage the lining of your intestines and lead to other serious health conditions over time.

A calprotectin test can find out whether your intestinal condition involves inflammation. But it can't diagnose the specific cause. Knowing whether your intestines are inflamed helps your health care provider decide what other tests you may need and what treatments are best for you.

Other names: Fecal calprotectin

WHAT IS IT USED FOR?

Calprotectin stool testing is used to check for inflammation in the intestines. In most cases, providers use calprotectin stool testing to help tell the difference between inflammatory bowel disease (IBD) and irritable bowel syndrome (IBS).

IBD and IBS are both chronic (long-lasting) conditions that affect adults and children. They cause similar symptoms that can come and go, including diarrhea, belly pain, and cramps. But only IBD involves inflammation:

IBD is a group of digestive diseases. It may cause watery and/or bloody diarrhea. Over time, inflammation from IBD damages the lining of the intestines. Symptoms may disappear for weeks or years and then come back. Experts think IBD may have many causes, including autoimmune disorders and genes. The main types of IBD are:

Crohn's disease

Ulcerative colitis

Crohn's disease

Ulcerative colitis

IBS is a group of symptoms that happen together. They include bloating, pain and/or cramps when you have bowel movements, and diarrhea, constipation, or both. Researchers think IBS may be caused by a problem with how your brain and intestines (gut) work together. IBS does not damage your intestines and tests do not show inflammation.

Depending on your symptoms and medical history, a calprotectin in stool test may be used with other stool tests that can help find the cause of long-lasting diarrhea. These include tests for fecal occult blood, bacteria in stool, parasites and eggs, and/or a white blood cell (WBC) in stool test.

If a person has been diagnosed with IBD, a calprotectin stool test may be used to help:

Find out how severe the inflammation is

Guide treatment choices

Check to see if IBD is getting better or worse

Predict the chance that IBD symptoms will return, including after surgery to treat IBD

WHY DO I NEED A CALPROTECTIN STOOL TEST?

Most watery diarrhea goes away on its own after a day or two, and your provider doesn't need to do tests. But if your symptoms don't improve after a few days or keep coming back, you may need a calprotectin stool test to check for inflammation in your intestines.

Along with diarrhea, belly pain and cramping, other symptoms of inflammation in your intestines may include:

Blood, mucus, and/or pus in your stool. Contact your provider right away if you see blood or pus in your stool, or if your stool is black and tarry.

Always feeling like you need to move your bowels (poop) even when there's no stool. This condition may be painful and cause cramps.

Feeling an urgent need to move your bowels.

Weight loss when you're not trying to lose weight.

If you already have IBD, you may need calprotectin testing to monitor your condition. This information helps your provider choose treatment and check whether it's working.

WHAT HAPPENS DURING A CALPROTECTIN STOOL TEST?

You will need to provide a stool sample for your test. Your provider will probably give you a container or kit with instructions on how to collect the sample. There are different ways to collect stool samples, so follow the instructions carefully. In general, you'll need to:

Label the container with your name, the collection date, and time.

Collect a stool sample as instructed. This usually involves using a clean, dry container, or special paper or plastic wrap placed over the toilet to catch the stool.

Make sure the stool doesn't mix with any urine (pee), toilet paper, or water from the toilet.

Close the container tightly.

Wash your hands well with soap and water.

Return the container according to the instructions.

If you're collecting a sample from diapers, you'll get special instructions for using plastic wrap inside of a clean diaper. In certain cases, a provider may use a swab to take a stool sample from the rectum.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Ask your provider if you need to stop taking any medicines before your test. But never stop taking any medicine without talking with your provider first. Medicines that may affect your test results include certain over-the-counter medicines, including:

Nonsteroidal anti-inflammatory drugs (NSAIDs), which are used to relieve pain. Examples include aspirin, ibuprofen, and naproxen.

Proton pump inhibitors (PPIs), which are used to control stomach acid and relieve symptoms of GERD. Examples include omeprazole, esomeprazole, and lansoprazole.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a calprotectin stool test.

WHAT DO THE RESULTS MEAN?

Normal or low levels of calprotectin usually mean that your intestines are not inflamed, which means you don't have inflammatory bowel disease (IBD). Your symptoms are probably caused by a type of non-inflammatory condition. Your provider may order other tests to figure out which condition is causing your symptoms.

But it's possible to have normal calprotectin test results even when your intestines are inflamed. This is called a "false negative." False negative calprotectin results mostly happen in children.

High levels of calprotectin usually mean that your intestines are inflamed. The higher the amount of calprotectin in your

stool, the more inflammation you have. If your calprotectin levels are high, your provider may retest you in a few weeks to see if they change.

Several conditions can cause high calprotectin levels, so you may need other tests to find the cause. In general:

Very high levels of calprotectin are often linked to:

Inflammatory bowel disease (IBD)

Food poisoning caused by certain types of bacteria

Inflammatory bowel disease (IBD)

Food poisoning caused by certain types of bacteria

Somewhat high levels of calprotectin are often linked to:

Intestinal or colorectal cancer

Celiac disease

Certain bacterial infections in your intestines, such as *Clostridium difficile* (C. diff)

Taking NSAIDs

IBD that has been treated

Intestinal or colorectal cancer

Celiac disease

Certain bacterial infections in your intestines, such as *Clostridium difficile* (C. diff)

Taking NSAIDs

IBD that has been treated

If your calprotectin test results show that your intestines are inflamed, your provider may order a colonoscopy, which uses a tiny camera to look inside your intestines. This helps your provider learn whether IBD or another condition is causing the inflammation. But if your calprotectin test results show that your intestines probably aren't inflamed, a colonoscopy is unlikely to help make a diagnosis. In this way, a calprotectin stool test can help avoid unnecessary colonoscopies.

If you have questions about your test results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CALPROTECTIN STOOL TEST?

If your provider thinks you could have IBD, you may first have a blood test to check for inflammation. The blood test may be a C-reactive protein (CRP) test or an erythrocyte sedimentation rate (ESR) test. But a calprotectin stool test is more accurate at finding inflammation in the intestines.

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RELATED HEALTH TOPICS

- Bacterial Infections
- Bowel Movement
- C. diff Infections
- Crohn's Disease
- Diarrhea
- Digestive Diseases
- Foodborne Illness
- Gastrointestinal Bleeding
- Irritable Bowel Syndrome
- Ulcerative Colitis

RELATED MEDICAL TESTS

- Antineutrophil Cytoplasmic Antibodies (ANCA) Test
- Bacteria Culture Test
- C-Reactive Protein (CRP) Test
- C. diff Testing
- Colorectal Cancer Screening Tests
- Erythrocyte Sedimentation Rate (ESR)
- Fecal Occult Blood Test (FOBT)
- How to Cope with Medical Test Anxiety

How to Understand Your Lab Results

Ova and Parasite Test

Platelet Tests

White Blood Cell (WBC) in Stool

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Carbon Dioxide (CO₂) in Blood

<https://medlineplus.gov/lab-tests/carbon-dioxide-co2-in-blood/>

CARBON DIOXIDE (CO₂) IN BLOOD

WHAT IS A CARBON DIOXIDE (CO₂) BLOOD TEST?

A carbon dioxide (CO₂) blood test measures the amount of carbon dioxide in your blood. Carbon dioxide is an odorless, colorless gas. It is a waste product from metabolism, the process your body uses to make energy from food you eat. Most of the carbon dioxide in your body is in the form of bicarbonate (HCO₃), which is a type of electrolyte. Electrolytes are electrically charged minerals that help control the amount of fluid and the balance of acids and bases (pH balance) in your body.

Your blood carries carbon dioxide to your lungs, and then you breathe it out. Too much or too little carbon dioxide in your blood can be a sign of a health problem.

Other names: carbon dioxide content, CO₂ content, carbon dioxide blood test, bicarbonate blood test, bicarbonate test, total CO₂; TCO₂; carbon dioxide content; CO₂ content; bicarb; HCO₃

WHAT IS IT USED FOR?

A CO₂ blood test is often part of a group of tests called an electrolyte panel, which may be part of a routine checkup. The test may also help diagnose or monitor conditions related to an electrolyte imbalance. These may include high blood pressure and diseases of the kidneys, lungs, or liver.

WHY DO I NEED A CO₂ IN BLOOD TEST?

Your health care provider may order a CO₂ blood test as part of your routine checkup or if you have symptoms of an electrolyte imbalance. These symptoms may include:

Trouble breathing

Confusion

Weakness

Fatigue

Vomiting and/or diarrhea over a long period

Your provider may also order a CO₂ blood test to check for side effects of certain medicines that can cause electrolyte imbalances.

WHAT HAPPENS DURING A CO₂ BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Some prescription and over-the-counter medicines can increase or decrease the amount of carbon dioxide in your blood. You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Abnormal test results may be a sign that your body is having a hard time keeping the right acid-base balance (pH balance). This may be because your lungs or kidneys are having a problem removing carbon dioxide or because you have an electrolyte imbalance. Many different types of conditions can cause these problems.

Too much CO₂ in the blood (called hypercapnia or hypercarbia) can be a sign of many conditions, including:

Lung diseases.

Cushing's syndrome.

Kidney failure.

Metabolic alkalosis, a condition in which your blood is not acidic enough. You may lose acid from conditions such as vomiting, dehydration, and anorexia.

Too little CO₂ in the blood (hypocarbica) may be a sign of:

Addison disease.

Diabetic ketoacidosis, a complication of type 1 and type 2 diabetes.

Shock.

Metabolic acidosis, a condition in which your blood is too acidic. It can have many causes, including kidney or liver disease or long-lasting diarrhea.

Respiratory alkalosis, a condition in which your blood is not acidic enough because of a lung or breathing problem, including hyperventilation (rapid, deep breathing).

If your test results are not in the normal range, it doesn't necessarily mean you have a medical condition that needs treatment. Your provider may order other tests to confirm a diagnosis. Other factors, including certain medicines, can affect the level of CO₂ in your blood. To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CO₂ BLOOD TEST?

A CO₂ blood test may also be completed as part of a basic or comprehensive metabolic panel. These tests measure different substances in your blood to learn about your body's chemical balance and metabolism.

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RELATED HEALTH TOPICS

- Adrenal Gland Disorders
- Fluid and Electrolyte Balance
- Kidney Diseases
- Kidney Tests
- Liver Diseases
- Lung Diseases

RELATED MEDICAL TESTS

- Arterial Blood Gas (ABG) Test
- Basic Metabolic Panel (BMP)
- Comprehensive Metabolic Panel (CMP)
- Electrolyte Panel
- Fasting for a Blood Test
- How to Cope with Medical Test Anxiety
- How to Understand Your Lab Results
- Lung Function Tests
- What You Need to Know About Blood Testing

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Catecholamine Tests

<https://medlineplus.gov/lab-tests/catecholamine-tests/>

CATECHOLAMINE TESTS

WHAT ARE CATECHOLAMINE TESTS?

Catecholamines are a group of hormones made by your adrenal glands, two small glands located above your kidneys. Hormones act as chemical messengers in your bloodstream. They carry information that controls the actions of certain cells or organs. The main types of catecholamines are dopamine, norepinephrine, and epinephrine. Epinephrine is also known as adrenaline.

These hormones are released into the body in response to physical or emotional stress. This response is sometimes referred to as your body's "fight or flight" response to stress. When catecholamines enter your bloodstream:

Your blood flow increases to your muscles and brain

You become more alert

Your heart rate and blood pressure increase

You get more adrenaline for energy

Catecholamine tests measure the amount of catecholamines in your urine or blood. Usually, these hormone levels are low and increase during and just after a stressful situation. Then these levels return to a normal level once the stressful situation has ended. Blood or urine levels of dopamine, norepinephrine, and/or epinephrine that stay higher than normal can be a sign of certain rare tumors.

Other names: dopamine, norepinephrine, epinephrine tests, free catecholamines, fractionated catecholamines

WHAT ARE THEY USED FOR?

Catecholamine tests are most often used to diagnose or rule out certain types of rare tumors, including:

Pheochromocytoma, a tumor of the adrenal glands. This type of tumor is usually benign (not cancerous). But it can be fatal if left untreated.

Neuroblastoma, a cancerous tumor that develops from nerve tissue. It mostly affects infants and children.

Paraganglioma, a type of tumor that forms near the adrenal glands and can cause high blood pressure. This type of tumor is sometimes cancerous, but it usually grows very slowly.

The tests may also be used to see if treatments for these tumors are working.

WHY DO I NEED A CATECHOLAMINE TEST?

You or your child may need this test if you have symptoms of a tumor that affects catecholamine levels. Symptoms in adults include:

High blood pressure, especially if it isn't getting better with treatment

Severe headaches

Sweating

Rapid heartbeat

Anxiety

Tingling in your arms and legs

Symptoms in children include:

Bone pain or tenderness

An abnormal lump in the abdomen

Weight loss for no known reason

Uncontrolled eye movements

Treatment of these rare tumors may help to control or reduce your high blood pressure. But if the tumors are not treated, your symptoms may get worse. The tumors may continue to grow and could cause more severe health problems.

WHAT HAPPENS DURING A CATECHOLAMINE TEST?

A catecholamine test may be done in urine or blood. Urine testing is done more often because catecholamine blood levels can change quickly and may also be affected by the stress of testing.

But blood testing can be useful in helping to diagnose a pheochromocytoma tumor. This type of tumor may make certain substances and release them into your bloodstream. These substances may include extra catecholamines.

For a catecholamine urine test, your health care provider will ask you to collect all your urine during a 24-hour period. This is called a 24-hour urine sample test. For 24-hour urine sample test, you'll be given a special container to collect your urine and instructions on how to collect and store your samples. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may be asked to avoid certain foods for two to three days before the test. These could include:

Caffeinated foods and drinks, such as coffee, tea, energy drinks, and chocolate

Alcohol and tobacco

Bananas

Citrus fruits

Foods that contain vanilla

You may also be asked to avoid stress and vigorous exercise before your test, as these can affect catecholamine levels. Certain medicines may also affect levels. Be sure to tell your provider about all the medicines you are taking. But don't stop taking any medicines unless your provider tells you to.

If you are having a blood test, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to having a urine test.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results show high levels of catecholamines in your urine or blood, it may be a sign of a pheochromocytoma, neuroblastoma, or paraganglioma tumor. If you are being treated for one of these tumors, high levels may mean your treatment is not working.

High levels of these hormones don't always mean you have a tumor. Your levels of dopamine, norepinephrine, and/or epinephrine can be affected by stress, vigorous exercise, caffeine, smoking, and alcohol.

To understand the results of your catecholamine test, your provider may consider your symptoms, medical history, and the results of other tests.

If you have questions about your results or your child's results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT CATECHOLAMINE TESTS?

These tests can help diagnose certain tumors, but they can't tell whether the tumor is cancerous. Most tumors are not. If your results showed high levels of these hormones, your provider will probably order more tests. These include imaging tests such as a CT scan or an MRI, which can help your provider get more information about a suspected tumor. Learn more about laboratory tests, reference ranges, and understanding results.

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CCP Antibody Test

<https://medlineplus.gov/lab-tests/ccp-antibody-test/>

CCP ANTIBODY TEST

WHAT IS A CCP ANTIBODY TEST?

This test looks for CCP (cyclic citrullinated peptide) antibodies in your blood. CCP antibodies, also called anti-CCP antibodies, are a type of antibody called autoantibodies. Antibodies and autoantibodies are proteins made by the immune system:

Antibodies protect you from disease by fighting foreign substances like viruses and bacteria.

Autoantibodies like CCP antibodies are abnormal proteins. Instead of protecting your body, they attack your body's healthy cells by mistake.

CCP antibodies target healthy tissues in the joints. Joints are places in your body where two bones meet, such as your elbows, wrists, and knees. If CCP antibodies are found in your blood, it can be a sign of rheumatoid arthritis. Rheumatoid arthritis is an autoimmune disease that gets worse over time. It causes pain, swelling, stiffness, and a loss of function in your joints.

CCP antibodies are found in most people who have rheumatoid arthritis. They are almost never found in people who don't have the disease.

Other names: Cyclic citrullinated peptide antibody, anticitrullinated peptide antibody, citrulline antibody, anti-cyclic citrullinated peptide, anti-CCP antibody, ACPA

WHAT IS IT USED FOR?

A CCP antibody test is used to help diagnose or rule out rheumatoid arthritis.

It's often done along with or after a rheumatoid factor (RF) test. Rheumatoid factors are another type of autoantibody. RF tests used to be the main test to help diagnose rheumatoid arthritis. However, RF factors can be found in people with other autoimmune diseases and even in some healthy people. And some people with RA have little to no RF factors. Doing both tests provides a more accurate diagnosis than only using RF testing results.

WHY DO I NEED A CCP ANTIBODY TEST?

You may need this test if you have symptoms of rheumatoid arthritis. These include:

Joint pain

Joint stiffness, especially in the morning

Joint swelling

Fatigue

Low-grade fever

You may also need this test if other tests couldn't confirm or rule out a diagnosis of rheumatoid arthritis.

WHAT HAPPENS DURING A CCP ANTIBODY TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Be sure to tell your provider about all medicines, vitamins, and dietary supplements you are taking. You may need to stop taking certain substances for 8 hours before your test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your CCP antibody results were positive, it means these antibodies were found in your blood. A negative result means no CCP antibodies were found. The meaning of these results may depend on the results of a rheumatoid factor (RF) test as well as a physical exam.

If you have symptoms of rheumatoid arthritis, and your results show:

Positive CCP antibodies and positive RF, it means that you probably have rheumatoid arthritis.

Positive CCP antibodies and negative RF, it could mean you are in the early stages of rheumatoid arthritis or will develop it in the future.

Negative CCP antibodies and negative RF, it means you are less likely to have rheumatoid arthritis. Your provider may need to do more tests to help find out what is causing your symptoms.

Sometimes, CCP antibodies may be found in other conditions such as:

Other autoimmune disorders such as Sjogren's syndrome or lupus

Active tuberculosis (TB)

Chronic lung disease

To understand the results of a CCP antibody test, your provider will consider your symptoms, medical history, physical exam, and the results of other tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CCP ANTIBODY TEST?

Rheumatoid arthritis can be difficult to diagnose, especially in its early stages. Your provider may order one or more tests in addition to CCP antibody and RF tests. These include x-rays of your joints and the following blood tests:

Erythrocyte sedimentation rate (ESR)

Synovial fluid analysis

C-reactive protein

Antinuclear antibody

These blood tests can show increased protein in the blood, which may be a sign of having a medical condition that causes inflammation. Inflammation can be a symptom of rheumatoid arthritis.

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CD4 Lymphocyte Count

<https://medlineplus.gov/lab-tests/cd4-lymphocyte-count/>

CD4 LYMPHOCYTE COUNT

WHAT IS A CD4 COUNT?

A CD4 count is a blood test that measures the number of CD4 cells in a sample of your blood. CD4 cells are a type of white blood cell that circulates throughout the body in the blood. They're also called CD4 T lymphocytes or "helper T cells." That's because they help fight infection by triggering your immune system to destroy viruses, bacteria, and other germs that may make you sick.

A CD4 count is mostly used to check the health of your immune system if you are infected with HIV (human immunodeficiency virus).

HIV attacks and destroys CD4 cells. Without treatment, HIV may destroy so many CD4 cells that your immune system will have trouble fighting off infections. HIV is the virus that causes AIDS (acquired immunodeficiency syndrome). AIDS is the most serious stage of an HIV infection. If you have AIDS, your CD4 count is so low that you may develop serious infections from viruses, bacteria, or fungi that usually don't cause problems in healthy people. These are called "opportunistic infections," and they can become life-threatening. AIDS increases your risk of developing certain cancers, too.

Most people with HIV don't have AIDS. And if they take their HIV medicine as prescribed, they may never develop AIDS.

If you have HIV, a CD4 count can help your health care provider check your risk for serious infections. A CD4 count may also be used to help diagnose and monitor certain other conditions that affect your immune system.

Other names: CD4 lymphocyte count CD4+ count, T4 count, T-helper cell count, CD4 percent

WHAT IS IT USED FOR?

If you have HIV, a CD4 count may be used to:

See how HIV is affecting your immune system. CD4 counts can help monitor your risk for developing opportunistic infections or certain cancers. If your risk increases, your provider may give you treatment to help prevent infections.

Help check how well HIV treatment is working. A CD4 count is used with a test called an HIV viral load test to see if HIV medicines are working. A viral load test measures how much HIV is in your blood.

Help check the stage of HIV and diagnose AIDS. CD4 count decreases over time as the HIV disease progresses. Without treatment, HIV can lead to a very low CD4 count, which means you have AIDS.

A CD4 count may also be used to:

Monitor treatment after an organ transplant. If you've had an organ transplantation, you'll need to take medicine to prevent your immune system from attacking the new organ. These medicines are called "anti-rejection" drugs or immunosuppressants. You may also take these medicines to treat certain autoimmune diseases. A low CD4 count means the medicine is working.

Help diagnose different types of lymphoma. Lymphoma is a cancer of a part of the immune system called the lymph system. A CD4 count may be used with other tests to find out which type of immune cells are causing lymphoma. The test results help choose the right treatment.

Help diagnose DiGeorge syndrome. This is an uncommon disorder that is inherited (passed down through families). It often causes immune problems and other health conditions that start at birth.

WHY DO I NEED A CD4 COUNT?

Your provider may order a CD4 count when you are first diagnosed with HIV. You will probably be tested again every few months to see if your counts have changed since your first test.

If you are taking medicine for HIV, your provider may order regular CD4 counts with an HIV viral load test to see how

well your medicines are working.

WHAT HAPPENS DURING A CD4 COUNT?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a CD4 count.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

CD4 results are usually given as a number of cells per cubic millimeter of blood. But labs may have different ways of describing "normal" CD4 counts. What's normal for you depends on your age, other health conditions, and the medicines you may take. That's why it's important to talk with your provider about what your test results mean.

CD4 counts may change even when your health has not changed. So, your provider will usually look at a few test results over time to see if there's a trend in your CD4 counts.

If you're diagnosed with HIV, it's important to start treatment as soon as possible. If your CD4 count continues to decrease for many months, your provider may change your medicine and also may start preventive treatment for opportunistic infections. The goal of this is to increase or stabilize your CD4 count.

In general, ranges for CD4 counts are:

Normal CD4 count for healthy adults and teens: 500 to 1,200 cells per cubic millimeter

Low CD4 count: Below 500 cells per cubic millimeter

If you have HIV, a low CD4 count means that HIV has weakened your immune system. A CD4 count of 200 or fewer cells per cubic millimeter means that you have AIDS. With AIDS you have a high risk of developing life-threatening infections or cancers.

If you don't have HIV, a low CD4 count may be caused by an infection. Cancer chemotherapy¹ and medicines that weaken your immune system may also cause low CD4 counts. In certain cases, the cause of a low CD4 count is unknown, but this is uncommon.

If you have HIV, a low CD4 count means that HIV has weakened your immune system. A CD4 count of 200 or fewer cells per cubic millimeter means that you have AIDS. With AIDS you have a high risk of developing life-threatening infections or cancers.

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High CD4 Count: Counts above the normal range may be a sign of an infection or a blood cancer.

While there is no cure for HIV, there are different medicines that you can take to control the virus and protect your immune system. In fact, CD4 counts usually increase with these medicines, which helps keep you healthy. If you are living with HIV, it's important to take your medicine as prescribed and get regular blood tests, including CD4 counts.

Learn more about laboratory tests, reference ranges, and understanding results.

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CEA Test

<https://medlineplus.gov/lab-tests/cea-test/>

CEA TEST

WHAT IS A CEA TEST?

CEA stands for carcinoembryonic antigen. CEA is a protein that is a type of "tumor marker." Tumor markers are substances that are often made by cancer cells or by normal cells in response to cancer.

High levels of CEA are normal in healthy, unborn babies. After birth, CEA levels become very low or disappear completely. So, healthy adults should have little or no CEA in their bodies.

Cancers that may cause high levels of CEA include cancers of the:

Colon and rectum (colorectal or bowel cancer)

Prostate

Ovary

Lung

Thyroid

Liver

Pancreas

Breast

If you've been diagnosed with a cancer that can cause high CEA levels, CEA testing may help your health care provider learn more about your cancer and chance of recovery. The test is often used with other tests to check if cancer treatment is working.

CEA tests are not used to screen for or diagnose cancer. That's because:

Cancers that often cause high CEA levels don't always cause high levels. You can have normal CEA test results even though you have one of these cancers.

Many other health problems that aren't cancer can make CEA levels rise, including certain conditions that affect your liver, digestion, or breathing.

Most CEA tests use a sample of your blood to measure your CEA level. Sometimes the test uses a sample of fluid from your spine, chest, or abdomen (belly), but this is less common.

Other names: CEA assay, CEA blood test, carcinoembryonic antigen test

WHAT IS IT USED FOR?

A CEA test is mostly used in people who have been diagnosed with cancers that often increase CEA levels. It's most commonly used in colorectal cancer. A provider may order a CEA test along with other tests to:

Learn more about a cancer soon after it has been diagnosed. CEA levels can help predict the likelihood of recovery and/or the chance that cancer will come back after treatment.

Check how well cancer treatment is working.

See if cancer has returned after treatment.

WHY DO I NEED A CEA TEST?

If you've been diagnosed with a type of cancer that can cause high CEA levels, you may need CEA testing:

Before treatment to:

Help understand how serious your cancer is.

Help guide treatment decisions.

Get a measurement of your CEA level before treatment to compare with CEA levels after treatment.

Help understand how serious your cancer is.

Help guide treatment decisions.

Get a measurement of your CEA level before treatment to compare with CEA levels after treatment.

During cancer treatment to see if the treatment is working.

After treatment to check whether cancer has come back. You may need to have tests on a regular schedule.

WHAT HAPPENS DURING A CEA TEST?

CEA is usually measured with a blood sample. During a CEA blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Sometimes, CEA is tested in other body fluids. For these tests, your provider will remove a small sample of fluid using a thin needle.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

For a CEA blood test, you usually don't need to prepare. If you smoke, you may need to stop for a while before your test. That's because smoking increases CEA levels.

ARE THERE ANY RISKS TO THE TEST?

A CEA blood test has very little risk. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand your CEA test results, your provider will consider the results of other tests and exams. Ask your provider to explain what your test results mean for your health and treatment.

If you had a test before starting cancer treatment, in general:

A low level of CEA may mean your tumor is small and the cancer has not spread to other parts of your body. But some cancers don't make much CEA, so your provider may use other tests to learn more about how much cancer you have and if it's spread.

A high level of CEA may mean you have a larger tumor and/or your cancer has spread. You'll need other tests to confirm how serious your cancer is.

If you had a test to monitor your cancer during or after treatment, your provider will compare your current CEA test results with your past test results. In general:

Decreases in CEA over time often mean that treatment is working.

Increases in CEA or high levels that stay high may mean that treatment isn't working. For example, if you had surgery for colorectal cancer, these levels may mean that the entire tumor wasn't removed or the cancer is growing back.

Decreases in CEA after treatment followed by increases may mean cancer has come back.

CEA levels that remain high or increase after treatment don't always mean that treatment isn't working, or cancer is growing. So, if your CEA levels don't come down, your provider will likely order other tests to find out why.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CEA TEST?

Labs use different methods to measure CEA. The test method can affect your results. So, it's best to have your tests done the same way, and usually in the same lab. This allows your provider to compare your results over time. If you

have questions about how your tests are done, ask your provider.

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Celiac Disease Screening

<https://medlineplus.gov/lab-tests/celiac-disease-screening/>

CELIAC DISEASE SCREENING

WHAT IS A CELIAC DISEASE SCREENING?

A celiac disease screening is a blood test that looks for signs that you may have celiac disease, an autoimmune disorder that causes a serious allergic reaction to gluten. Gluten is a protein found in wheat, barley, rye and other grains. It may also be found in other products like vitamins and supplements, hair and skin products, toothpaste, and lip balm.

This test can tell if you have celiac disease by looking for antibodies to gluten in your blood. Antibodies are disease-fighting substances made by the immune system.

Normally, your immune system makes antibodies to fight foreign substances like viruses and bacteria. But if you have celiac disease, your immune system does this whenever you eat or drink gluten. This can cause your antibodies to attack the lining of the small intestine, where cells called villi absorb nutrients from the food that you eat. Repeated attacks will damage these cells making nutrient-absorption much more difficult.

Other names: celiac disease antibody test, anti-tissue transglutaminase antibody (anti-tTG), deaminated gliadin peptide antibodies, anti-endomysial antibodies

WHAT IS IT USED FOR?

A celiac disease test is used to:

Help diagnose celiac disease.

Monitor celiac disease.

See if a gluten-free diet is relieving symptoms of celiac disease.

Your health care provider may use a different type of test to measure different antibodies in your blood. The most common tests include:

Tissue transglutaminase antibody (tTG), IgA test. This type looks for the tTG antibody and for immunoglobulin A, another antibody your body makes to fight germs. It is also the most commonly used test type.

Immunoglobulin A (IgA) test. This type specifically checks to see if you have a deficiency in the IgA antibody, which can happen in those with celiac disease.

Immunoglobulin G (IgG) Test. This type specifically checks to see if you have high levels of IgG, which can be a sign of celiac disease.

Deamidated gliadin peptide (DGP) antibodies test. This type is often used if you have low IgA and for children younger than 2 years old.

WHY DO I NEED A CELIAC DISEASE SCREENING TEST?

You may need a celiac disease test if you have symptoms of celiac disease. Symptoms are different for children and adults.

Symptoms of celiac disease in children include:

Nausea and vomiting

Abdominal (belly) bloating

Constipation

Chronic diarrhea and foul-smelling stool (poop)

Weight loss and/or failure to gain weight

Delayed puberty

Irritable behavior

Because celiac disease damages the digestive system over time, symptoms of celiac disease in adults often include digestive problems such as:

Nausea and vomiting
Chronic diarrhea
Unexplained weight loss
Decreased appetite
Abdominal (belly) pain
Bloating and gas

Digestive symptoms are also common in people with gluten sensitivity and wheat allergies. But celiac disease is different than these conditions. They can cause similar symptoms, but celiac disease is the only one that damages the small intestine.

Many adults with celiac disease may also have symptoms that are not related to digestion. These include:

Iron-deficient anemia
An itchy rash called dermatitis herpetiformis
Mouth sores
Bone loss
Depression or anxiety
Fatigue
Headaches
Missed menstrual periods
Tingling in the hands and/or feet

You may need a celiac test if you are at higher risk of having the disease, even if you don't have symptoms. You are more likely to have celiac disease if a close family member has celiac disease. You may also be at higher risk if you have another autoimmune disorder, such as type 1 diabetes.

WHAT HAPPENS DURING A CELIAC DISEASE SCREENING TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If the test is being used to diagnose celiac disease, you'll need to continue to eat foods with gluten for a few weeks before testing. Your provider will give you specific instructions about how to prepare for the test.
If the test is being used to monitor celiac disease, you don't need any special preparations.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Typical results may show one of the following:

Negative: You probably don't have celiac disease.

Positive: You probably do have celiac disease.

Uncertain or indeterminate: It's unclear whether you have celiac disease.

If your results were positive or uncertain, your provider may order other tests to confirm that you have celiac disease and/or to see how much damage the disease may have caused. These tests may include:

A biopsy to examine a tissue sample from your small intestine or your skin if you have a rash that could be from celiac disease. An endoscopy is used to take a sample of tissue from your small intestine.

Capsule endoscopy to look at your small intestine. For this test, you swallow a tiny camera in a capsule. As it passes through your small intestine, it records pictures. The camera leaves your body during a bowel movement (poop).

Genetic testing to see if you have a gene linked to celiac disease.

Tests for other health problems that celiac disease may cause such as anemia, osteoporosis, or a lack of certain vitamins.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CELIAC DISEASE TEST?

Most people with celiac disease can reduce and often eliminate symptoms if they keep a strict gluten-free diet. Although many gluten-free products are available today, it can still be challenging to completely avoid gluten. Your provider may refer you to a dietician (a nutrition expert) who can help you enjoy a healthy diet without gluten.

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Cerebrospinal Fluid (CSF) Analysis

<https://medlineplus.gov/lab-tests/cerebrospinal-fluid-csf-analysis/>

CEREBROSPINAL FLUID (CSF) ANALYSIS

WHAT IS A CEREBROSPINAL FLUID (CSF) ANALYSIS?

A cerebrospinal fluid (CSF) analysis tests a sample of your CSF for diseases that affect the brain and spinal cord, which make up your central nervous system. Your central nervous system controls and coordinates everything you do, including your ability to move, breathe, see, think, and more.

CSF is a clear, colorless, watery fluid that flows in and around your brain and spinal cord. CSF acts like a cushion that helps protect them from sudden impact or injury. It also removes waste products from the brain and helps your central nervous system work properly.

The CSF normally has substances like water, vitamins, and nutrients in it and is free of cells, proteins, and toxins (a type of poison). Certain diseases of the brain and spinal cord and other conditions that affect the central nervous system can cause a change in the amount of CSF you have or what substances are in your CSF. A CSF analysis is a group of tests that use a sample of your CSF to help diagnose these types of diseases.

Other names: Spinal Fluid Analysis, CSF Analysis

WHAT IS IT USED FOR?

A CSF analysis is used to measure different substances in your CSF. It may include tests to diagnose:

Infectious diseases of the brain and spinal cord, including meningitis (inflammation of the brain and spinal cord) and encephalitis (inflammation of the brain). CSF tests for infections look at white blood cells, bacteria, and other substances in the CSF.

Autoimmune disorders, such as Guillain-Barré Syndrome (a rare disorder where the immune system attacks the nerves that connect the brain and spinal cord to the body) and multiple sclerosis (MS; a nervous system disease of the brain and spinal cord). CSF tests for these disorders look for high levels of certain proteins in the CSF.

Bleeding in the brain.

Brain tumors, including from cancers in other parts of the body that may spread to the central nervous system.

Alzheimer's disease, the most common form of dementia, which includes memory loss, confusion, and changes in behavior.

WHY DO I NEED A CSF ANALYSIS?

Your health care provider may order a CSF analysis if you have:

Symptoms of an infection or bleeding in the brain or spinal cord

Symptoms of an autoimmune disorder, such as multiple sclerosis (MS)

Had a brain injury or an injury to your spinal cord

Have cancer that may have spread to your central nervous system

Symptoms that may be from another central nervous system condition, such as headaches

Symptoms of a brain or spinal cord infection include:

Fever

Severe headache

Seizures

Stiff neck

Nausea and vomiting

Sensitivity to light

Double vision

Changes in behavior

Confusion

Symptoms of MS often vary and come and go, or they may steadily get worse. They may include:

Blurred or double vision

Tingling, numbness, or pain in the arms, legs, body, or face

Painful muscle spasms

Weak muscles, often in the hands and legs

Dizziness, balance problems, or clumsy movement when walking

Bladder control problems

Fatigue

Problems thinking or learning new things

WHAT HAPPENS DURING A CSF ANALYSIS?

To get a sample of CSF, a provider will do a procedure called a spinal tap, also known as a lumbar puncture. A spinal tap is usually done in a hospital. During the procedure:

You will lie on your side or sit on an exam table.

A provider will clean your back to disinfect it and inject an anesthetic into your skin, so you won't feel pain during the procedure. Your provider may put a numbing cream on your back before this injection.

When the area on your back is completely numb, your provider will insert a thin, hollow needle between two vertebrae in your lower spine. Vertebrae are the small backbones that make up your spine.

Your provider will withdraw a small amount of CSF for testing. This will take about five minutes.

You'll need to stay very still while the fluid is being withdrawn.

Your provider may ask you to lie on your back for an hour or two after the procedure. This may prevent you from getting a headache afterward.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a CSF analysis, but you may be asked to empty your bladder (pee) and bowels (poop) before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a spinal tap. You may feel a little pinch or pressure when the needle is inserted. After the test, you may feel some pain or tenderness in your back at the site where the needle was inserted.

You may also have some bleeding at the site or get a headache. The headache may last for several hours or up to a week or more, but your provider may suggest treatment to help relieve the pain.

WHAT DO THE RESULTS MEAN?

A CSF analysis may include a variety of different tests on your sample. So, the measurements on your test results will depend on which tests were done. Your provider can explain what your results mean.

In general, your CSF analysis results may show that you have an infection, an autoimmune disorder, such as multiple sclerosis (MS), or another disease or injury of the brain or spinal cord. Your provider will likely order more tests to confirm your diagnosis.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CSF ANALYSIS?

Some infections, such as meningitis caused by bacteria, are life-threatening emergencies. If your provider suspects you

have bacterial meningitis or another serious infection, you may need to start medicine before you have a final diagnosis.

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RELATED HEALTH TOPICS

- Alzheimer's Disease
- Brain Diseases
- Encephalitis
- Guillain-Barre Syndrome
- Head Injuries
- Meningitis
- Multiple Sclerosis
- Spinal Cord Diseases
- Spinal Cord Injuries

RELATED MEDICAL TESTS

- Bacteria Culture Test
- Beta 2 Microglobulin (B2M) Tumor Marker Test
- CEA Test
- Complete Blood Count (CBC)
- CSF Immunoglobulin G (IgG) Index
- Cytomegalovirus (CMV) Tests
- Electromyography (EMG) and Nerve Conduction Studies
- Herpes (HSV) Test
- How to Cope with Medical Test Anxiety
- How to Understand Your Lab Results
- Immunoglobulins Blood Test
- Lactate Dehydrogenase (LDH) Test
- Lactate Test
- Lyme Disease Tests
- Measles and Mumps Tests
- Myelography
- Neurological Exam
- Pneumococcal Disease Tests
- Strep B Test
- Syphilis Tests

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Ceruloplasmin Test

<https://medlineplus.gov/lab-tests/ceruloplasmin-test/>

CERULOPLASMIN TEST

WHAT IS A CERULOPLASMIN TEST?

A ceruloplasmin test measures the amount of ceruloplasmin in a sample of your blood. Your liver makes ceruloplasmin by attaching copper to a protein. Your liver releases the ceruloplasmin into your bloodstream so it can carry copper to the parts of your body that need it.

Copper is a mineral that your body uses for many important processes, including making energy, blood vessels, and melanin (the substance that gives your skin and eyes their color). Copper helps your body use iron, and it supports brain development. It helps keep your nervous and immune systems healthy, too.

You need only small amounts of copper to stay healthy. Most people get enough copper from the foods they eat. Good sources of copper include shellfish, nuts and seeds, beef liver and other organ meats, wheat-bran cereals, and chocolate.

Normally, most of the copper in your blood is part of ceruloplasmin. So, ceruloplasmin testing can help diagnose conditions that involve having too much or too little copper in your body.

Other names: CP, ceruloplasmin blood test, ceruloplasmin, serum, copper oxidase, ferroxidase

WHAT IS IT USED FOR?

A ceruloplasmin test is most often used with other tests to help diagnose Wilson disease. Wilson disease is a rare genetic disorder that prevents the body from removing extra copper.

Wilson disease can lead to a harmful buildup of copper in the liver, brain, eyes, and other organs. The high levels of copper can severely damage organs and become life-threatening.

Ceruloplasmin tests may also be used to help diagnose conditions that cause low levels of copper. Although low levels of copper are uncommon, they may be caused by conditions such as:

Malnutrition, a condition that happens if your diet doesn't include enough vitamins, minerals, and other key nutrients.

Malabsorption, a condition in which your body has trouble absorbing nutrients from the food you eat. Many diseases can cause malabsorption, including celiac disease.

Menkes syndrome, also called Menkes disease, a rare, incurable genetic disease that affects the body's ability to use copper. Symptoms usually show up in the first few months of life.

Taking too many zinc supplements, which can affect how well your body can use copper

If a person has been diagnosed with too much or too little copper in their blood, regular ceruloplasmin testing may be used to monitor their condition and to check whether treatment is working.

Ceruloplasmin tests are mostly used with tests that measure copper levels in blood and/or urine (pee).

WHY DO I NEED A CERULOPLASMIN TEST?

Your health care provider may order a ceruloplasmin test if you have symptoms of Wilson disease. Even though people are born with the disease, the symptoms don't appear until copper builds up in your liver, brain, or other organs. This usually happens between ages 4 and 40, but symptoms can appear earlier or later in life.

Wilson disease can cause many types of symptoms depending on which parts of your body are affected. For example:

Nausea and vomiting

Pain in your upper abdomen (belly)

Urine that's a darker color than usual and/or stool (poop) that's a lighter color

Jaundice, a condition that causes your skin and eyes to turn yellow

A type of anemia, called hemolytic anemia, that destroys red blood cells faster than your body can make them

Swelling from fluid buildup in your belly, lower legs, or feet

Itchy skin

Trouble with speech, swallowing, or physical coordination

Tremors

Stiff muscles and/or sudden jerky movements you can't control

Mental health disorders, including anxiety, depression, or psychosis

Green, gold, or brown rings around your irises (the colored part of your eyes)

If you have a family history of Wilson disease, you may need a test even if you don't have symptoms.

You may also need this test if you have symptoms that could be a sign of too little copper, such as:

Abnormally low levels of white blood cells called neutrophils

Osteoporosis, a condition that causes bones to become weak and break easily

Fatigue

Tingling in your hands and feet

Pale skin

Your baby may need a ceruloplasmin test if they have symptoms of Menkes syndrome. Symptoms usually show up when babies are about two or three months old. They may include:

Hair that is colorless, thin, breaks easily, and becomes unusually tangled

Floppy muscles

Seizures

Slow weight gain and growth

Without treatment, most children with Menkes syndrome die within the first few years of life. If treatment is given in the first month after birth, some children may live longer. But even with treatment, the disease still tends to get worse over time.

If you have been diagnosed with abnormal copper levels, you may need regular ceruloplasmin testing to check your condition and to see how well your treatment is working.

WHAT HAPPENS DURING A CERULOPLASMIN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a ceruloplasmin test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The results of a ceruloplasmin test alone cannot diagnose any conditions. The meaning of your test results will depend on the results of other tests, including tests that measure the amount of copper in your blood and/or urine.

To make a diagnosis, your provider will consider all your test results along with your symptoms, medical history, and family history. Ask your provider to explain what your ceruloplasmin levels say about your health.

In general, lower than normal ceruloplasmin levels may mean your body is not able to use copper properly. Low ceruloplasmin may be a sign of:

Wilson disease, although ceruloplasmin levels are sometimes normal with this disease

Menkes syndrome

Severe liver disease

Kidney disease

A lack of copper caused by malnutrition and/or malabsorption

Higher than normal ceruloplasmin levels may be linked to:

Heart disease

Rheumatoid arthritis

Leukemia

Hodgkin lymphoma

Infection, inflammation, or injury

But high levels of ceruloplasmin don't always mean you have a condition that needs medical treatment. Pregnancy, birth control pills, and certain medicines, including estrogen therapy to treat menopause, may cause high blood ceruloplasmin. Too much copper in your diet rarely causes high ceruloplasmin.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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RELATED HEALTH TOPICS

- Genetic Counseling
- Genetic Disorders
- Genetic Testing
- Liver Diseases
- Minerals
- Wilson Disease

RELATED MEDICAL TESTS

- How to Cope with Medical Test Anxiety
- How to Understand Your Lab Results
- What You Need to Know About Blood Testing

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Chickenpox and Shingles Tests

<https://medlineplus.gov/lab-tests/chickenpox-and-shingles-tests/>

CHICKENPOX AND SHINGLES TESTS

WHAT ARE CHICKENPOX AND SHINGLES TESTS?

Chickenpox and shingles tests check to see if you are or have ever been infected with the varicella zoster virus (VZV). VZV is a type of herpes virus that causes chickenpox and shingles. This virus is also known as human herpesvirus 3 (HHV-3) and the herpes zoster virus. The first time you are infected with this virus, you get chickenpox. Once you have had chickenpox, you can't get it again. The virus stays in some of your nerve cells but is dormant (inactive). Later in life, the virus can become active again and cause shingles. Unlike chickenpox, you can get shingles more than once, but it is rare. A shingles infection is sometimes also called herpes zoster.

Testing for VZV involves checking for VZV in a blood, fluid, or tissue sample. There are different types of tests to do this: A PCR (polymerase chain reaction) test. This test looks for genetic material from VZV in a sample. It is most often done on a sample of fluid taken by swabbing open blisters (a swab test). In some cases, other fluids may be used, but the results may not always be as accurate. The PCR test can get results quickly and is the most useful test for confirming a VZV infection.

A viral culture test. For this test, cells from a swab test sample are grown in a lab and then checked for VZV. This test is not as accurate as a PCR test.

A direct fluorescent antibody (DFA) test. This test is also done on a swab test sample that is sent to a lab. It involves using a fluorescent dye and a special microscope to look for VZV in the sample. It is a rapid test, but it is not as accurate as a PCR.

An antibody test. This is a blood test that looks for VZV antibodies. Antibodies are proteins that your immune system makes to fight foreign substances, such as viruses and bacteria. Testing for one type of antibody (IgM) can show if you have a current or recent VZV infection. Testing for another type of antibody (IgG) can show if you have ever been exposed to VZV.

Which tests and samples are used depends on the person and their symptoms.

Other names: varicella zoster virus antibody, serum varicella immunoglobulin G antibody level, VZV antibodies IgG and IgM, herpes zoster

WHAT ARE THEY USED FOR?

Health care providers can usually diagnose chickenpox or shingles with a visual examination. But in some cases, your provider may also order a chickenpox or shingles test to confirm the diagnosis.

These tests are also sometimes ordered to check for immunity to the varicella zoster virus (VZV). You have immunity if you've had chickenpox before or have had the chickenpox vaccine. When you have immunity, it means you can't get chickenpox. But you can still get shingles later in life.

These tests may be done on people who are at higher risk of complications from VZV and either don't have immunity or don't know if they have immunity. People who are at higher risk of complications include:

During pregnancy

Infants

Teen and adults with symptoms of chickenpox

People who have weakened immune systems, for example, because they:

Have HIV, cancer, or another condition that weakens their immune system

Have had an organ transplant

Are taking medicines that weaken the immune system, such as chemotherapy and steroids

Have HIV, cancer, or another condition that weakens their immune system

Have had an organ transplant

Are taking medicines that weaken the immune system, such as chemotherapy and steroids

WHY DO I NEED A CHICKENPOX OR SHINGLES TEST?

You may need a chickenpox or shingles test if you are at higher risk of complications, are not immune to VZV, and/or have symptoms of infection.

Contact your provider right away if you have symptoms of chickenpox or shingles. Some of the symptoms of chickenpox and shingles are similar, but they are not the same illness. Common symptoms for both include:

A red, blistering rash

Fever

Headache

Sore throat

With chickenpox, the rash first appears on the chest, back, and face. Then it spreads over the entire body, including inside the mouth, eyelids, or genital area. It usually takes about 1 week for all of the blisters to become scabs.

One to two days before you get the rash, you may have other typical symptoms, such as:

Fever

Tiredness

Loss of appetite

Headache

Some people who have been vaccinated against chickenpox can still get the disease. But they often have milder symptoms and are sick for a shorter period of time than people who are not vaccinated.

Before getting a rash, people with shingles may first have pain, itching, or tingling in the area where the rash later develops. This can happen several days before the rash appears. They may also have a fever at this time.

The rash that develops is painful and itchy, and it often develops on just one side of the body or face, and in a small area. The most common place for shingles to occur is in a band around one side of the waistline. Shingles on the face can affect the eye and cause vision loss. Rarely, the rash can be more widespread on the body and look similar to a chickenpox rash. This usually happens in people with weakened immune systems.

The rash consists of blisters that typically scab over in 7 to 10 days and fully clear up within 2 to 4 weeks.

Other symptoms of shingles can include:

Headache

Chills

Upset stomach

You may also need this test if you are at higher risk of complications and were recently exposed to chickenpox or shingles. You can't catch shingles from another person. But the shingles virus (VZV) can spread and cause chickenpox in someone who doesn't have immunity.

WHAT HAPPENS DURING CHICKENPOX AND SHINGLES TESTING?

You will need to provide a sample of blood from your vein or from the fluid in one of your blisters. Blood tests check for antibodies to the VZV. Blister tests check for the virus itself.

For a blood test from a vein, a health care professional will take a blood sample from a vein in your arm using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

For a blister test, a provider will gently press a cotton swab on a blister to collect a sample of fluid for testing.

Both tests are quick, each usually taking less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a blood or blister test.

ARE THERE ANY RISKS TO THE TEST?

After a blood test, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly. There is no risk to having a blister test.

WHAT DO THE RESULTS MEAN?

If you have symptoms and your test results show VZV antibodies or the virus itself, it's likely you have chickenpox or shingles. Your diagnosis of either chickenpox or shingles will depend on your age and specific symptoms.

If you don't have symptoms and your results show antibodies to the virus or the virus itself, you either had chickenpox at one point or received the chickenpox vaccine.

If you are diagnosed with an infection and are in a high-risk group, your provider may prescribe antiviral medicines. Early treatment can prevent serious and painful complications.

Most healthy children and adults with chickenpox will recover from chickenpox within a week or two. Home treatment can help relieve symptoms. More serious cases may be treated with antiviral medicines. Shingles may also be treated with antiviral medicines as well as pain relievers.

If you have questions about your results or your child's results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT CHICKENPOX AND SHINGLES TESTS?

The best way to prevent chickenpox is to get the chickenpox vaccine. The Centers for Disease Control and Prevention (CDC) recommends the chickenpox vaccine for children, teens, and adults who never had chickenpox or the chickenpox vaccine:

For children:

The first dose is recommended at 12 through 15 months old

The second dose is recommended at 4 through 6 years old

The first dose is recommended at 12 through 15 months old

The second dose is recommended at 4 through 6 years old

People 13 years of age and older who have never had chickenpox and haven't had a chickenpox vaccine should get two doses, at least 28 days apart

There is also a vaccine for shingles. It's called Shingrix. The CDC recommends that healthy adults age 50 and older get two doses of Shingrix, separated by 2 to 6 months. Shingrix is also recommended for adults 19 years and older who have weakened immune systems due to disease or medicines. The vaccine provides strong protection against shingles and complications from the disease.

Since you can get shingles more than once, you should get the vaccine even if you've already had shingles.

A different shingles vaccine called Zostavax is no longer available in the United States. If you had a Zostavax vaccine in the past, you should still get vaccinated with Shingrix. Talk to your provider about the best time to get Shingrix.

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Chlamydia Test

<https://medlineplus.gov/lab-tests/chlamydia-test/>

CHLAMYDIA TEST

WHAT IS A CHLAMYDIA TEST?

Chlamydia is a common bacterial sexually transmitted infection (STI), especially in sexually active people younger than age 25. You can get a chlamydia infection through sexual contact. This includes having vaginal, oral, or anal sex without a condom with a partner who has chlamydia.

A chlamydia test checks a sample of fluid from your body to look for the bacteria that causes chlamydia. The type of fluid depends on the part of your body that may be infected. Your sample may be:

Urine (pee)

Fluid that's swabbed from your:

Urethra (the tube that carries urine out of your body)

Rectum (the last part of your intestine that connects to the anus)

Vagina

Urethra (the tube that carries urine out of your body)

Rectum (the last part of your intestine that connects to the anus)

Vagina

Chlamydia often has no symptoms, so you may spread the disease to others without even knowing you have it. If you are pregnant and have chlamydia, you can give the infection to your baby during childbirth. Chlamydia can cause pneumonia and eye problems in babies.

Chlamydia can be cured with antibiotics. But if it's not treated, it can cause serious, lasting health problems, even if you don't have symptoms:

In women, untreated chlamydia can cause pelvic inflammatory disease (PID). PID can permanently damage the female reproductive system and may cause infertility (problems getting pregnant).

In men, untreated chlamydia can lead to swelling and pain in the tubes that carry sperm out of the testicles (male sex glands that produce sperm, also called testes). It can also cause infertility (problems getting someone pregnant), but this rarely happens.

A chlamydia test helps find the disease early so you can get treatment before you develop lasting health problems or spread the disease to others.

Other names: Chlamydia NAAT or NAT, Chlamydia/GC STD Panel, Chlamydia Culture

WHAT IS IT USED FOR?

If you're sexually active, a chlamydia test is used to:

Check for a chlamydia infection when you don't have symptoms

Diagnose or rule out chlamydia when you do have symptoms

Symptoms of chlamydia may include:

An unusual fluid discharge from your genitals (penis or vagina) or rectum

Irritation or itching around your genitals

Pain or burning when you urinate (pee)

Pain or bleeding in your rectum

In men, pain and swelling in one or both testicles

Chlamydia tests are often done with other STI tests because people often have more than one STI at the same time. For example, many people who have chlamydia also have gonorrhea.

If you have chlamydia, you'll be tested again three months after you finish treatment. That's because it's common to get chlamydia again.

WHY DO I NEED A CHLAMYDIA TEST?

If your sex partner has been diagnosed with chlamydia or if you have symptoms that could mean you have a chlamydia infection, you will need to be tested. But chlamydia usually doesn't cause symptoms. So, getting tested is the only way to know for sure if you have an infection. That's why medical experts recommend routine screening tests if you have a high risk of getting chlamydia:

If you are a sexually active woman, you should get tested for chlamydia at least every year if you:

Are younger than age 25

Are age 25 or older and have a high risk of infection because you have:

A new sex partner

More than one sex partner

A sex partner who has sex with other people

A sex partner who has an STI

Have HIV (You should be tested for chlamydia the first time you see your provider for HIV care. After that you should be tested at least once a year. But you may need to get tested more often if your sexual activity increases your risk of getting chlamydia.)

Are younger than age 25

Are age 25 or older and have a high risk of infection because you have:

A new sex partner

More than one sex partner

A sex partner who has sex with other people

A sex partner who has an STI

A new sex partner

More than one sex partner

A sex partner who has sex with other people

A sex partner who has an STI

Have HIV (You should be tested for chlamydia the first time you see your provider for HIV care. After that you should be tested at least once a year. But you may need to get tested more often if your sexual activity increases your risk of getting chlamydia.)

If you are pregnant, ask your health care provider about chlamydia testing at your first prenatal visit. The test is often included with routine prenatal tests. You should be tested if you are younger than age 25. If you are older than age 25, you should be tested if you have a high risk of getting chlamydia.

If you are a sexually active gay or bisexual man, you should get tested for chlamydia at least every year. But you will need to get tested more frequently (every 3 to 6 months) if you have:

More than one sex partner

A sex partner who has sex with other people

Anonymous sex (having sex with people you don't know)

HIV or take PrEP medicine to prevent HIV

More than one sex partner

A sex partner who has sex with other people

Anonymous sex (having sex with people you don't know)

HIV or take PrEP medicine to prevent HIV

If you are a heterosexual (straight) man:

Medical experts don't suggest routine testing for chlamydia if you have no symptoms and a low risk of infection.

If you have HIV, you should be tested for chlamydia the first time you see your provider for HIV care. After that you

should be tested at least once a year. But you may need to be tested more often if your sexual activity increases your risk of getting chlamydia.

Medical experts don't suggest routine testing for chlamydia if you have no symptoms and a low risk of infection.

If you have HIV, you should be tested for chlamydia the first time you see your provider for HIV care. After that you should be tested at least once a year. But you may need to be tested more often if your sexual activity increases your risk of getting chlamydia.

The best testing schedule for you may be different than the recommendations. So have an honest talk with your provider about your sexual activity to help them figure out how often you should get tested.

WHAT HAPPENS DURING A CHLAMYDIA TEST?

To do a chlamydia test, you will need to provide a sample of fluid from the part of your body that may be infected. There are two ways to gather the sample:

For a first-catch urine sample, you'll use a sterile cup to collect urine from the very first part of your urine stream. To get an accurate test result, you will need to avoid urinating for two hours before the test.

For a swab sample, your provider will use a special swab or brush to collect cells, usually from your genital area. You may feel some brief discomfort. Your provider may give you the choice to swab yourself. In certain cases, fluid may be swabbed from your rectum, throat, eyes, or cervix (the lower part of the uterus).

Test results are usually ready in a day. There are some rapid chlamydia tests that can provide results in 90 minutes or less.

You can also buy at-home tests¹ for chlamydia and other STIs. With these tests, you collect a swab or urine sample at home and send it to a lab for testing. It's important to follow all the instructions carefully and discuss your test results with your provider.

You should not use an at-home test if you have symptoms of chlamydia or if your partner has chlamydia. In that case, it's important to talk with your provider right away so you don't delay treatment. Ask your provider whether at-home chlamydia testing is right for you.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Preparations for a chlamydia test depend on the type of sample you will be providing. You may need to avoid antibiotic medicines and vaginal douches and creams for 24 hours before your test. But don't stop taking any medicines without talking with your provider first. Ask your provider if there are any special instructions for your test.

ARE THERE ANY RISKS TO THE TEST?

There are no known risks to having a chlamydia test.

WHAT DO THE RESULTS MEAN?

A negative test result means that you did not have a chlamydia infection when the sample was collected.

A positive test result means you have been infected with chlamydia.

If you have chlamydia, you will need to:

Take antibiotics to treat your infection. Your provider will tell you how to take your medicine. To make sure your infection is cured, follow the instructions carefully and finish all the medicine.

Tell your sexual partner/s that you have chlamydia so that they can be tested and treated if necessary.

Get retested for chlamydia three months after you finish treatment. That's because repeat infections are very common.

Ask your provider to tell you how often to get tested after that.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CHLAMYDIA TEST?

The most reliable way to prevent chlamydia is not to have vaginal, oral, or anal sex. But if you are sexually active, you can reduce your risk of getting chlamydia and other STIs by:

Having sex with only one partner who has sex only with you (mutual monogamy). But first, both of you should get tested for chlamydia and other STIs.

Using latex condoms correctly every time you have sex. If you or your partner is allergic to latex, you can use polyurethane condoms.

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Chloride Blood Test

<https://medlineplus.gov/lab-tests/chloride-blood-test/>

CHLORIDE BLOOD TEST

WHAT IS A CHLORIDE BLOOD TEST?

A chloride blood test measures the amount of chloride in your blood. Chloride is a type of electrolyte. Electrolytes are electrically charged minerals that help control the amount of fluids and the balance of acids and bases (pH balance) in your body. You have electrolytes in your blood, urine, tissues, and other body fluids.

You get chloride through the food you eat. It is mainly found in salt. Your body gets rid of the extra chloride in your urine (pee). Chloride helps control the amount of fluid in your body. In addition, it helps maintain healthy blood volume and blood pressure.

Many diseases and other factors can cause abnormal chloride levels. Chloride is often measured with other electrolytes to diagnose or monitor conditions such as kidney disease, heart failure, liver disease, and high blood pressure.

Other names: Cl, Serum chloride

WHAT IS IT USED FOR?

You usually get a chloride test as part of a routine blood screening to check your general health. It's also used to help monitor or diagnose conditions related to an imbalance of acids or fluids in your body.

WHY DO I NEED A CHLORIDE BLOOD TEST?

Your health care provider may have ordered a chloride blood test as part of an electrolyte panel, a routine blood test that measures chloride and other electrolytes, such as potassium, sodium, and bicarbonate.

You may also need a chloride blood test if you have symptoms of an acid or fluid imbalance, including:

Vomiting over a long period of time

Diarrhea

Fatigue

Weakness

Dehydration

Trouble breathing

WHAT HAPPENS DURING A CHLORIDE BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

If your provider has ordered other blood tests, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

There are many reasons why your chloride levels may not be in the normal range.

High levels of chloride (hyperchloremia) may be a sign of:

Dehydration.

Kidney disease.

Metabolic acidosis, a condition in which you have too much acid in your blood. It can cause nausea, vomiting, and fatigue.

Low levels of chloride (hypochloremia) may be a sign of:

Heart failure.

Lung disease.

Addison disease, a condition in which your body's adrenal glands don't produce enough of certain types of hormones. It can cause various symptoms, including weakness, dizziness, weight loss, and dehydration.

Metabolic alkalosis, a condition in which you have too much base in your blood. It can cause irritability, muscle twitching, and tingling in the fingers and toes.

If your chloride levels are not in the normal range, it doesn't always mean you have a medical problem that needs treatment. Many things can affect your chloride levels, such as drinking too much fluid or losing fluid because of vomiting or diarrhea. Also, certain medicines, such as antacids, can cause abnormal results. To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CHLORIDE BLOOD TEST?

Urine also contains some chloride. Your provider may also recommend a urine chloride test to get more information about your chloride levels.

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- What You Need to Know About Blood Testing

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Cholesterol Levels

<https://medlineplus.gov/lab-tests/cholesterol-levels/>

CHOLESTEROL LEVELS

WHAT IS A CHOLESTEROL TEST?

A cholesterol test is a blood test that measures the amount of cholesterol and triglycerides (a type of fat) in your blood. Cholesterol is a waxy, fat-like substance that's found in your blood and every cell of your body. Your body needs some cholesterol to make hormones, vitamin D, and substances that help you digest foods.

Your liver makes all the cholesterol your body needs and removes excess amounts. Cholesterol is also found in foods from animal sources, such as meat, egg yolks, poultry, and dairy products. Foods high in dietary fat can increase the cholesterol in your blood. If there's too much cholesterol in your blood, your liver can't remove it all.

There are two main types of cholesterol: LDL (low-density lipoprotein), or "bad" cholesterol, and HDL (high-density lipoprotein), or "good" cholesterol.

Too much LDL cholesterol in your blood increases your risk for coronary artery disease and other heart diseases. High LDL levels can cause the buildup of a sticky substance called plaque in your arteries. This buildup of plaque is known as atherosclerosis. Over time, plaque can narrow your arteries or fully block them. When this happens, parts of your body may not get enough blood:

If the blood flow to your heart is blocked, it can cause a heart attack.

If the blood flow to your brain is blocked, it can cause a stroke.

If the blood flow to your arms or legs is blocked, it can cause peripheral artery disease.

Other names for a cholesterol test: Lipid profile, Lipid panel

WHAT IS IT USED FOR?

A cholesterol test gives you and your health care provider important information about your risk of developing heart disease. If your test shows you have high cholesterol, you can take steps to lower it. This may decrease your risk of developing heart problems in the future.

A cholesterol test measures your:

LDL level. LDL ("bad") cholesterol is the main source of blockages in the arteries.

HDL level. HDL ("good") cholesterol helps get rid of "bad" LDL cholesterol. A higher HDL level may help reduce your risk of heart attack or stroke.

Total cholesterol. This is a measure of the total amount of cholesterol in your blood. It includes HDL and LDL cholesterol.

Triglyceride level. Triglycerides are a type of fat found in your blood. High levels of triglycerides may increase the risk of heart disease, especially in women.

There is another type of cholesterol called VLDL (very low-density lipoprotein). Some people also call VLDL a "bad" cholesterol because it contributes to the buildup of plaque in your arteries. But VLDL and LDL are different; VLDL mainly carries triglycerides, and LDL mainly carries cholesterol. VLDL isn't usually included in routine cholesterol tests because it's difficult to measure. Because VLDL contains a certain percentage of triglycerides, a lab can use your triglycerides level to estimate your VLDL level.

WHY DO I NEED A CHOLESTEROL TEST?

Your provider may order a cholesterol test as part of a routine exam. When and how often you should get a cholesterol test depends on your age, risk factors, and family history. The general recommendations are:

For people who are age 19 or younger:

The first test should be between ages 9 to 11

Children should have the test again every 5 years

Some children may have this test starting at age 2 if there is a family history of high cholesterol, heart attack, or stroke

For people who are ages 20 to 65:

Younger adults should have the test every 5 years

Men ages 45 to 65 and women ages 55 to 65 should have it every 1 to 2 years

For people older than 65:

They should be tested every year

You may also have a cholesterol test more often if you are at high risk of heart problems because of:

- A family history of heart disease
- High blood pressure
- Type 2 diabetes
- Smoking
- Excess weight or obesity
- Lack of physical activity
- A diet high in saturated fat

WHAT HAPPENS DURING A CHOLESTEROL TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

You may be able to use an at-home kit^Â to check your cholesterol levels. Your kit will include a device to prick your finger to collect a drop of blood for testing. Be sure to follow the kit instructions carefully. Also, be sure to tell your provider if your at-home test shows that your total cholesterol level is higher than 200 mg/dl.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for 9 to 12 hours before your blood cholesterol test. That's why the tests are often done in the morning. Your provider will let you know if you need to fast and if there are any other special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Cholesterol is usually measured in milligrams (mg) of cholesterol per deciliter (dL) of blood. The information below will help you understand what your test results mean. In general, low LDL levels and high HDL cholesterol levels are good for heart health.

Anyone age 19 or younger:

Men age 20 or older:

Women age 20 or older:

The LDL listed on your results may say "calculated." This means that your LDL level is an estimate based on your total cholesterol, HDL, and triglycerides. Your LDL level may also be measured "directly" from your blood sample. Either way, you want your LDL number to be low.

A healthy cholesterol level for you may depend on your age, family history, lifestyle, and other risk factors for heart disease, such as high triglyceride levels. Your provider can explain what's right for you.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT MY CHOLESTEROL LEVELS?

High cholesterol can lead to heart disease, the number one cause of death in the United States. You can't change some risk factors for high cholesterol, such as age and your genes. But there are actions you can take to lower your LDL levels and reduce your risk, including:

Eat a healthy diet. Reduce or avoid foods high in saturated fat and cholesterol to help lower the cholesterol levels in your blood.

Manage your weight. Being overweight can increase your cholesterol and risk for heart disease.

Stay active. Regular physical activity may help lower your LDL (bad) cholesterol levels and raise your HDL (good) cholesterol levels. It may also help you lose weight.

Quit smoking. Smoking lowers your HDL (good) cholesterol, especially in women. Smoking also raises your LDL (bad) cholesterol.

Reduce stress. Stress may raise levels of certain hormones such as corticosteroids. These can cause your body to make more cholesterol.

Avoid drinking too much alcohol. Alcohol can raise your total cholesterol level.

Talk to your provider before making any major change in your diet or exercise routine.

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Chorionic Villus Sampling (CVS)

<https://medlineplus.gov/lab-tests/chorionic-villus-sampling-cvs/>

CHORIONIC VILLUS SAMPLING (CVS)

WHAT IS CHORIONIC VILLUS SAMPLING (CVS)?

Chorionic villus sampling (CVS) is a genetic test done during pregnancy. This test checks chorionic villus cells to see if your baby has certain health conditions. Chorionic villus cells are found in the placenta (the organ that brings oxygen and nutrients to the baby).

The test looks for abnormalities in your baby's chromosomes. Chromosomes are tiny "packages" in your cells that contain your genes. Genes carry information, called DNA, that controls what you look like and how your body works. Normally, you're born with a set of 46 chromosomes, with one set coming from each parent.

The chorionic villi cells in the placenta contain the same chromosomes as your developing baby. So, a CVS test can show if your baby has an extra chromosome or a missing or damaged chromosome. These chromosome differences can cause serious health problems.

CVS is a prenatal diagnostic test, which means it can almost always tell for sure whether a developing baby has a chromosome disorder. This is different than a prenatal screening test, which only shows whether a baby has an increased risk of a serious health problem.

Other names: CVS test

WHAT IS IT USED FOR?

CVS testing is used to diagnose chromosome problems or other genetic diseases in developing baby. These include:

Down syndrome (also called Trisomy 21), a condition in which a person has an extra chromosome or a piece of a chromosome. This extra copy changes how a baby's body and brain develop and can cause mental and physical challenges during their lifetime.

Cystic fibrosis (CF), a disease that's inherited (passed down through families). It causes mucus buildup in the lungs and other organs, making it hard to breathe.

Sickle cell disease (SCD), a group of inherited red blood cell disorders. SCD can cause pain, infections, organ damage, and strokes.

Tay-Sachs disease, a disorder that causes fatty proteins to build up in the brain. It affects sight, hearing, and mental development. Most children with Tay-Sachs die by the age of 5.

Trisomy 18 (Edward syndrome), a condition that causes the baby to have growth delays while developing. It also causes physical and mental abnormalities after they are born. Most babies with Trisomy 18 die before birth or within their first month.

CVS testing is very accurate and can be done early in pregnancy, between the 10th and 13th week. But it can only diagnose certain genetic diseases. A CVS test does not diagnose or screen for birth defects such as neural tube defects. These are conditions that cause abnormal development of a developing baby's brain and/or spine. Different tests, including an alpha-fetoprotein (AFP) blood test, are used to screen for or diagnose these and other birth defects.

WHY DO I NEED A CVS TEST?

A CVS test is not a routine pregnancy test. But your health care provider may recommend CVS testing if you are at higher risk for having a baby with a chromosome disorder. The risk factors include:

Your age. If you are age 35 and older, you have a higher risk of having a baby with Down syndrome or another genetic disorder.

A family history of a genetic disorder.

Having another child with a genetic disorder.

You may also need CVS testing if you had abnormal results on a prenatal screening test.

WHAT HAPPENS DURING A CVS TEST?

Your provider will use an ultrasound to check your baby's position and guide the procedure. An ultrasound is an imaging test that uses sound waves to look at your tissues and organs. The test usually takes about 30 to 45 minutes.

There are two types of CVS tests:

Transabdominal (through your abdomen (belly)). A long, thin needle is guided through your abdomen (belly) to take a sample.

Transcervical (through your vagina). A thin catheter is guided through your cervix, (the lower part of the uterus), to your placenta to take a sample. This is the most common method.

During a transabdominal CVS, your provider will:

Clean your abdomen with an antiseptic.

Apply a numbing medicine to your abdomen.

Insert a long, thin needle through your abdomen and uterus and into the placenta. You may feel a cramping or stinging sensation as the needle enters the uterus.

Use the needle to withdraw a sample of tissue (villi) from the placenta.

Remove the needle.

During a transcervical CVS, your provider will:

Clean your vagina and cervix with an antiseptic.

Use an instrument called a speculum to gently spread apart the sides of your vagina. This is the same instrument used for a pap test.

Insert a thin tube through your vagina and cervix and up to the placenta. You may feel a slight twinge or cramping as this is done.

Use the tube to gently suck in a sample of tissue from the placenta.

Remove the tube.

You may have light vaginal spotting after the procedure.

A CVS test isn't recommended if you have a sexually transmitted infection (STI) or have had vaginal bleeding during the pregnancy.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

On the morning of the test, you may be asked to drink extra fluids and not urinate. This will fill your bladder, which may help move the uterus into a better position for the procedure.

ARE THERE ANY RISKS TO THE TEST?

CVS is generally considered to be a safe procedure, but it does have some risks. These include:

Miscarriage (this is a small risk).

Infection.

Bleeding.

Rh sensitization. This is a condition in which your body makes antibodies (proteins made by the immune system) that attack your baby's red blood cells. If diagnosed during pregnancy, it is easily treatable.

Limb defects in the baby (this is very rare).

Call your provider if you have chills, contractions, a fever, or amniotic fluid leaking after the procedure.

WHAT DO THE RESULTS MEAN?

CVS test results are usually available within two weeks.

If your CVS test results were normal, it is unlikely that your baby has the disorder that was tested. But it does not guarantee that your baby won't have any health problems.

If your results were not normal, it may mean your baby has a chromosome or genetic disorder, such as Down syndrome or cystic fibrosis.

Occasionally, CVS test results are unclear, and your provider may recommend amniocentesis. Amniocentesis is another prenatal diagnostic test that checks the amniotic fluid (the liquid in the sac surrounding the baby). It is performed between the 15th and 20th week of pregnancy.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT CVS TESTING?

Some chromosome and genetic disorders require your baby to have extensive medical treatments. Other disorders may not be treatable.

If your CVS test results were not normal, you should talk with your provider about your options and the decisions you may need to make. It may also help to speak to a genetic counselor—both before and after you get your results. A genetic counselor is a specially trained professional in genetics and can help you understand what your results mean.

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Chymotrypsin in Stool

<https://medlineplus.gov/lab-tests/chymotrypsin-in-stool/>

CHYMOTRYPSIN IN STOOL

WHAT IS A CHYMOTRYPSIN IN STOOL TEST?

This test measures the amount of chymotrypsin in a stool (poop) sample. Chymotrypsin is an enzyme, a protein that speeds up certain chemical reactions in your body. It is released by your pancreas (an organ behind your stomach) to help break down the protein in your food during digestion. If your pancreas is working as it should, there will be some chymotrypsin in your stool sample.

However, if your pancreas is releasing lower-than-normal amounts of chymotrypsin, you may not have enough enzymes to digest your food properly. This is a condition known as exocrine pancreatic insufficiency. It's often called pancreatic insufficiency or EPI for short.

Pancreatic insufficiency is most often caused by chronic pancreatitis, inflammation of the pancreas. It may also be caused by cystic fibrosis (CF), a genetic disorder that causes mucus to build up in your lungs and other organs, including the pancreas. People with CF often have trouble absorbing nutrients from food.

Your stool sample may also be tested for trypsin. This is another enzyme that is released by the pancreas and helps with digestion.

Other names: fecal chymotrypsin, trypsin and chymotrypsin stool test

WHAT IS IT USED FOR?

A chymotrypsin in stool test is used to check for pancreatic insufficiency.

Knowing whether or not you have pancreatic insufficiency can help your health care provider diagnose certain conditions that can cause it, such as pancreatitis and cystic fibrosis. It can also help in diagnosing the cause of malnutrition, a condition in which your body does not get the calories, vitamins, and/or minerals needed for good health.

WHY DO I NEED A CHYMOTRYPSIN IN STOOL TEST?

You may need this test if you have symptoms of a pancreatic insufficiency. These include:

Chronic diarrhea

Abdominal (belly) pain

Bloating and gas

Loose, foul-smelling, greasy stools

Weight loss

WHAT HAPPENS DURING A CHYMOTRYPSIN IN STOOL TEST?

You will need to provide a fresh sample of loose or liquid stool for your test. Your provider will probably give you a container or kit with instructions on how to collect the sample. There are different ways to collect the stool samples so follow the instructions carefully. In general, you'll need to:

Label the container with your name, the collection date, and time.

Collect a stool sample as instructed. This usually involves using a clean, dry container, or special paper or plastic wrap placed over the toilet to catch the stool.

Make sure the stool doesn't mix with any urine, toilet paper or water from the toilet.

Close the container tightly.

Wash your hands well with soap and water.

Return the container according to the instructions. This is important to get an accurate test result. If you are unable to

return the container right away, you should refrigerate your sample until you're ready to deliver it.

If you're collecting a sample from diapers, you'll get special instructions for using plastic wrap inside of a clean diaper. In certain cases, a provider may use a swab to take a stool sample from the rectum

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You may need to stop taking certain medicines before this test, such as if you are taking pancreatic enzymes, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THIS TEST?

There is no known risk to having a stool test.

WHAT DO THE RESULTS MEAN?

Your results will show the amount of chymotrypsin in your stool. Your results may also include the amount of trypsin found. Â Â

A positive result means that your stool sample shows a normal amount of chymotrypsin and, if it was tested for, trypsin.

A negative result means that the amount of chymotrypsin and/or trypsin in your stool sample is below normal. It may mean that you have pancreatic insufficiency.

If you have a negative result, this doesn't diagnose any specific disorders by itself. Your provider will need to order more tests. These may include the following, which measure different pancreatic enzymes:

Stool elastase

Lipase blood test

Amylase blood test

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Coagulation Factor Tests

<https://medlineplus.gov/lab-tests/coagulation-factor-tests/>

COAGULATION FACTOR TESTS

WHAT ARE COAGULATION FACTOR TESTS?

Coagulation factors are proteins in your blood. They help form blood clots to stop bleeding when you have an injury. These proteins are also called clotting factors. You have several different types of clotting factors that are all important for making blood clots.

Coagulation factor tests are blood tests that check one or more of your clotting factors to see if you:

Have too much or too little of a clotting factor

Are missing a clotting factor

Have a clotting factor that isn't working right

Your liver makes most of your clotting factors. But normally, clotting factors are turned off, so you don't form abnormal blood clots. When you have an injury that causes bleeding, blood cells called platelets become stickier, so they can stick to each other to make a clot and stop the bleeding.

The platelets release molecules into your blood that begin to turn on the clotting factors. The clotting factors work together in a chain reaction to form a blood clot that will stay firmly in place.

Problems with any one of your clotting factors may mean that:

Your blood clots too easily, even without an injury. This condition may lead to clots that block your blood flow and cause serious conditions, such as heart attack, stroke, or clots in the lungs.

Your blood doesn't clot enough after an injury or surgery. If this happens, you are likely to have a bleeding disorder. Bleeding disorders can lead to serious blood loss after an injury.

Clotting factors have names, such as fibrinogen and prothrombin. Each clotting factor also has a Roman numeral name, such as "clotting factor II."

Other names: blood clotting factors, factor assays, factor assay by number (Factor I, Factor II, Factor VIII, etc.) or by name (fibrinogen, prothrombin, hemophilia A, hemophilia B, etc.), coagulation panel

WHAT IS IT USED FOR?

A coagulation factor test is used to find out if you have a problem with any of your clotting factors that may cause too little or too much blood clotting.

Coagulation factor tests are also used to check on people who have a known problem with clotting factors or who take medicine called blood thinners to lower the risk of blood clots.

You may have tests for one or more factors at a time.

WHY DO I NEED A COAGULATION FACTOR TEST?

You may need this test if you have:

An abnormal result on a blood test that checks how long it takes your blood to clot. These tests include a prothrombin time test and INR (PT/INR) and/or a partial thromboplastin time (PTT) test.

A family health history of problems with clotting factors. Some conditions that affect clotting factors, such as hemophilia (a rare bleeding disorder where the blood doesn't clot properly), are inherited. That means that your parents passed the gene for the disease to you. These conditions are not common.

A health condition that may affect clotting factors in your blood:

Conditions that may cause a bleeding disorder include:

Severe liver disease

A lack of vitamin K, a vitamin that makes proteins for healthy bones and tissues
Blood transfusions, a procedure where blood is transferred from a donor to you
Cancer
Immune disorders

Conditions that may cause a problem with blood clots include:

Autoimmune diseases, such as lupus, where your immune system attacks healthy cells and tissues
Cancer
Obesity
Certain infections, such as sepsis and COVID-19
Not moving for long periods of time, such as after surgery
Lack of vitamins B6, B12, and folate, vitamins that help the process your body uses to get or make energy from the food you eat
Severe liver disease
A lack of vitamin K, a vitamin that makes proteins for healthy bones and tissues
Blood transfusions, a procedure where blood is transferred from a donor to you
Cancer
Immune disorders
Autoimmune diseases, such as lupus, where your immune system attacks healthy cells and tissues
Cancer
Obesity
Certain infections, such as sepsis and COVID-19
Not moving for long periods of time, such as after surgery
Lack of vitamins B6, B12, and folate, vitamins that help the process your body uses to get or make energy from the food you eat
Symptoms that may be from a problem with clotting factors:

Symptoms of bleeding disorders may include:

Heavy bleeding that doesn't stop with pressure after an injury, dental procedure, or surgery
Frequent nosebleeds that start on their own
Blood in urine (pee) or stool (poop)
Frequent, large bruises or tiny red or brown spots under the skin
Redness, swelling, pain, or stiffness from bleeding into muscles or joints
Heavy menstrual periods

Symptoms of too much blood clotting may include:

Swelling, redness, warmth, and pain in your arms or legs which may be from a clot
Trouble breathing from a clot that's traveled to your lung
Nausea
Heavy bleeding that doesn't stop with pressure after an injury, dental procedure, or surgery
Frequent nosebleeds that start on their own
Blood in urine (pee) or stool (poop)
Frequent, large bruises or tiny red or brown spots under the skin
Redness, swelling, pain, or stiffness from bleeding into muscles or joints
Heavy menstrual periods
Swelling, redness, warmth, and pain in your arms or legs which may be from a clot
Trouble breathing from a clot that's traveled to your lung
Nausea

WHAT HAPPENS DURING A COAGULATION FACTOR TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a coagulation factor test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your provider may need to order other tests to find the cause of a problem with your clotting factors.

Lower than normal levels of one or more clotting factors or a missing clotting factor may mean you have a bleeding disorder. Depending on which clotting factors were tested, your results may show the type of bleeding disorder you have and how serious it is.

Bleeding disorders that you inherit usually involve only one clotting factor. There is no cure for inherited bleeding disorders, but treatment can help manage your condition.

Bleeding disorders caused by other conditions usually involve low levels of two or more clotting factors. Treatment depends on the cause of your bleeding disorder.

Higher than normal levels of one or more clotting factors may mean you have a disorder that makes your blood clot more than it should. Your provider may recommend medicine and heart-healthy lifestyle changes to help prevent clots. You may also need to avoid hormone replacement therapy^Â for menopause and birth control pills with estrogen, because they may increase the risk of blood clots.

Talk with your provider to find out what your tests results mean and what treatment is best for you.

Learn more about laboratory tests, reference ranges, and understanding results.

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Cognitive Testing

<https://medlineplus.gov/lab-tests/cognitive-testing/>

COGNITIVE TESTING

WHAT IS COGNITIVE TESTING?

Cognitive testing checks for problems with certain brain functions called "cognition." Cognition includes thinking, learning, remembering, and using judgment and language. Problems with cognition are called "cognitive impairment." There are many different cognitive tests that check for cognitive impairment. They generally involve answering questions and doing simple tasks, such as repeating lists of words or spelling words backwards. The most commonly used tests usually take 15 minutes or less.

Cognitive impairment is more common among older people, but it's not a normal part of aging. It can be caused by many medical and mental health conditions. Some of these conditions may be treatable, such as urinary tract infections (UTIs), depression, and side effects from medicine.

Treating these conditions may improve cognitive impairment or cure it completely. But cognitive impairment caused by dementia, such as Alzheimer's disease, has no cure and gets worse over time.

Cognitive testing alone can't diagnose dementia or any other cause of cognitive impairment. But the test results can show whether you have a problem with how your brain is functioning that needs more testing. That helps your health care provider order the right tests to learn more about your cognitive impairment and whether it can be treated. Even if the cause of cognitive impairment can't be treated, the information you gain from cognitive testing can help you and your family plan for your future care.

Other names: cognitive assessment, cognitive screening, Montreal Cognitive Assessment, MoCA test, Mini-Mental State Exam (MMSE), and Mini-Cog

WHAT IS IT USED FOR?

Cognitive testing is used if a person shows signs of a problem with memory, thinking, or other brain functions. The test shows if a person has a problem that requires more testing.

Cognitive testing is often used to screen older adults for a condition called mild cognitive impairment (MCI). People with MCI may notice that they have more trouble with memory than other people their age. They may lose things more often or have more trouble coming up with words for what they want to say. But they're still able to do their usual daily activities.

There's no cure for MCI. If you have the condition, your provider may give you cognitive tests as part of your routine checkup to see if your brain function has changed. That's because researchers have found that people with MCI have a higher risk of developing dementia than those without MCI. But in many cases, the symptoms of MCI stay the same or even improve over time.

WHY DO I NEED COGNITIVE TESTING?

You may need cognitive testing if you show signs of cognitive impairment. You may notice these changes yourself or your family or friends may notice them. Signs of cognitive impairment include:

Forgetting appointments and important events

Losing things often

Having trouble coming up with words that you usually know

Losing your train of thought in conversations, movies, or books

Feeling increased irritability and/or anxiety

WHAT HAPPENS DURING A COGNITIVE TEST?

There are different types of cognitive tests. They all involve answering questions and usually doing some simple tasks. Commonly used tests include:

Montreal Cognitive Assessment (MoCA) test. This test takes about 15 minutes. It includes memorizing a short list of words, copying a drawing of a shape or object, and looking at pictures of animals and saying which animals you see.

Mini-Mental State Exam (MMSE). This test takes about 10 minutes. It includes knowing what date it is, counting backward, and identifying everyday objects, such as a pencil or watch

Mini-Cog. This test usually takes about 3 minutes. It includes recalling a three-word list of objects and drawing a clock with hands that shows a specific time.

WILL I NEED TO DO ANYTHING TO PREPARE FOR COGNITIVE TESTING?

You don't need any special preparations for a cognitive test.

ARE THERE ANY RISKS TO TESTING?

There is no risk to having cognitive testing.

WHAT DO THE RESULTS MEAN?

Your cognitive test result will be a score.

If your test score is normal, you could still have some cognitive impairment that the test may not show. If you or your family are concerned about your brain function, but your test results were normal, talk with your provider about having another type of cognitive test.

If your score is lower than normal, it usually means you have some level of cognitive impairment. But your provider can't make a diagnosis on these test results alone.

Depending on your score, your provider may have you see a neurologist (a doctor with special training diagnosing and treating problems with the brain and nervous system). The neurologist may do longer tests called "neuropsychological testing" to learn more about how your brain is working. These detailed tests check your ability to plan, solve problems, and make decisions.

Your provider may also order tests to confirm or rule out treatable conditions that could be causing cognitive impairment. Which tests you have will depend on your medical history, a physical exam, and the results of cognitive testing. You may be tested for treatable conditions such as:

Blood vessel disorders

Sleep disorders

Hypothyroidism

Lack of certain vitamins, such as vitamin B12, or minerals

Mental health conditions, such as depression, anxiety, or stress

Concussion or other head injury from an accident or fall

Stroke

Urinary tract infection (UTI)

If you have cognitive impairment that can't be cured, medicines and healthy lifestyle changes may help slow the loss of brain function over time.

If you have questions or are concerned about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT COGNITIVE TESTING?

You may have heard about cognitive tests that you can take yourself online. If you or someone you know is concerned about cognitive impairment, doing one of these tests may be helpful. But it's important to follow up with your provider to

discuss the test results and any other tests you may need to properly diagnose your condition.

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Cold Stimulation Test

<https://medlineplus.gov/lab-tests/cold-stimulation-test/>

COLD STIMULATION TEST

WHAT IS A COLD STIMULATION TEST?

A cold stimulation test involves placing your fingers in ice water and checking to see how long it takes your skin to return to a normal temperature. This test is used to find out if you have Raynaud phenomenon. Raynaud phenomenon, also known as Raynaud disease or Raynaud syndrome, is a condition that causes brief episodes or "attacks" where your blood vessels (the tubes that your blood flows through) narrow. This decreases the blood flow to certain parts of your body. The attacks usually happen after exposure to cold or stress. They mostly affect your fingers and toes. But in rare cases, the attacks can affect other areas of your body, such as your ears, nose, lips, or nipples.

When you are exposed to cold, your body tries to slow the loss of heat and maintain its temperature. To do this, the blood vessels in the top layer of your skin constrict (narrow). This moves blood from those vessels near the surface to vessels deeper in the body. But when you have Raynaud phenomenon, the blood vessels in your hands and feet react to cold and stress by narrowing quickly. They also stay narrowed for a long period of time.

A Raynaud attack can be triggered by very mild and limited exposure to cold, such as taking food out of the freezer or holding a cold drink. When an attack happens, your skin may feel cold and numb. Your skin may also turn white or blue due to a lack of oxygen. As blood flow returns, your skin may tingle, throb, or turn red. An attack may last a few minutes or a few hours.Â

There are two types of Raynaud phenomenon:

Primary Raynaud phenomenon:

Is the more common type and is often milder.Â

Has no known cause.

A family history of the disease and being a woman may increase your risk for this type.

Symptoms usually start before age 30, often during the teenage years.Â

Most people can manage their mild symptoms with lifestyle changes.

Secondary Raynaud Â phenomenon:

Is usually caused by another health condition, such as a rheumatic disease like lupus or scleroderma, carpal tunnel syndrome, or disorders of the immune system, connective tissue, or arteries.

Other causes may include repeated use of vibrating machinery (such as a jackhammer), or exposure to cold or certain chemicals.

Medicines to treat high blood pressure, migraines, or attention deficit hyperactivity disorder (ADHD) may cause similar symptoms or make your symptoms worse.

Is less common than primary Raynaud but is more serious because it can damage your blood vessels.Â

Symptoms may include skin ulcers (open sores caused by poor blood flow) or skin infections.

Symptoms usually start between the ages of 35 and 40.Â

Other names: cold challenge test

WHAT IS IT USED FOR?

Your provider will usually diagnose Raynaud phenomenon by doing a physical exam and reviewing your symptoms and medical history. Some providers also use a cold stimulation test to help confirm your diagnosis of Raynaud phenomenon.

WHY DO I NEED A COLD STIMULATION TEST?

You may need this test if you have symptoms of Raynaud phenomenon. Its symptoms usually show up after exposure to cold or emotional stress. You may also have them after smoking cigarettes or vaping. These symptoms mostly affect

fingers and toes and can include:

Skin that turns white or blue (although this can be harder to see if you have darker skin)

Skin that is numb, painful, or stinging once it starts to warm up

Skin that takes a long time to warm up after exposure

WHAT HAPPENS DURING A COLD STIMULATION TEST?

A small device that measures temperature will be taped to your fingers.

Your provider will record the temperature.

You will then place your hands into an ice water bath for 20 seconds.

You will remove your hands from the ice water.

Your provider will record your finger temperature every five minutes for up to 20 minutes, or until your temperature returns to the same as it was before the ice water bath.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a cold stimulation test.

ARE THERE ANY RISKS TO THE TEST?

You may feel some discomfort while your hands are in the ice water.

WHAT DO THE RESULTS MEAN?

If it takes 20 minutes or more for your finger temperature to return to normal after the ice water bath, it probably means you have Raynaud phenomenon. If your symptoms are severe and/or you are age 35 or older, you may have secondary Raynaud phenomenon.

If your health care provider thinks you have secondary Raynaud phenomenon, they may order more tests to help diagnose the disorder that is causing it. These tests can include:

Nailfold capillaroscopy. For this test, your provider will place a drop of oil on the base of one of your fingernails. The nail area is then examined under a microscope. If abnormal arteries are seen, it can be a sign of scleroderma or other connective tissue disease.

Blood tests to check for immune system disorders. These include an antinuclear antibody (ANA) test, erythrocyte sedimentation rate (ESR), and C-reactive protein.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A COLD STIMULATION TEST?

There are steps you can take that may help you avoid or reduce the symptoms of Raynaud phenomenon, including:

Dressing extra warmly for cold weather. This includes wearing a hat, mittens, scarf, and heavy socks.

Wearing gloves when you are holding cold items.

Using battery- or chemical-operated hand and/or foot warmers.

Turning down air conditioning and wearing a sweater or jacket when in an air-conditioned space.

Getting regular exercise, which increases blood flow.

Limiting caffeine and alcohol.

Avoiding stress.

Quitting smoking (if you smoke). Nicotine in cigarettes can cause your blood vessels to narrow, increasing the chance of an episode.

Talking to your provider about the medicines you take, since some medicines can bring on episodes.

These changes should be enough to manage the symptoms of primary Raynaud phenomenon. If your symptoms are causing too much discomfort, or if you have secondary Raynaud phenomenon, your provider may also prescribe medicines that can help open your blood vessels. For severe cases, your provider may recommend a procedure to destroy the nerves that trigger blood vessel narrowing in the affected areas of your body.Â

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Colorectal Cancer Screening Tests

<https://medlineplus.gov/lab-tests/colorectal-cancer-screening-tests/>

COLORECTAL CANCER SCREENING TESTS

WHAT ARE COLORECTAL CANCER SCREENING TESTS?

Colorectal cancer screening tests check for colorectal cancer before you have any symptoms. Colorectal cancer is a type of cancer that affects the colon or rectum. The colon makes up most of the large intestine. The rectum is at the end of the large intestine and connects to the anus.

Colorectal cancer screening can help to:

Find colorectal cancer early, when treatment is most effective.

Prevent cancer from developing. Some screening tests let your health care provider find and remove abnormal growths, known as colorectal polyps. Cancer in the colon or rectum often starts as a polyp. Most of these polyps are benign (not cancerous) but could become cancerous if not removed.

Find bleeding that you can't see, which may mean you have a polyp.

Other names: colon cancer screening

WHAT ARE THEY USED FOR?

Colorectal cancer screening tests are used to check for signs of colorectal cancer and remove any colorectal polyps.

WHY DO I NEED A COLORECTAL CANCER SCREENING TEST?

Regular screening can detect colon polyps and colorectal cancer. If you aren't at higher risk for colorectal cancer, your provider will likely recommend you start screenings at age 45. If you are at higher risk, you may need to start getting screened for colorectal cancer earlier. Talk with your provider about when to start, how often to get screened, and which type of test you should have.

You may need to be screened at a younger age, or more often, if you have certain risk factors, including:

A family history of colorectal cancer

A personal or family history of ovarian cancer

An inflammatory bowel disease such as ulcerative colitis or Crohn's disease

Being overweight or having obesity

Smoking tobacco products

Alcohol misuse

Some of the tests used to screen for colorectal cancer may also be used as diagnostic tests. They can help your provider make a diagnosis when you are having symptoms of a colonic disease.

WHAT HAPPENS DURING A COLORECTAL CANCER SCREENING?

There are two main types of colorectal cancer screening tests: visual exams and stool tests. There are different options for both types of tests.

Visual exams

Visual exams are procedures that allow your provider to look inside your colon and rectum. They are done using a thin tube with a camera attached or a special imaging device. They usually require some type of bowel prep to clear the stool out of your colon. If you need a bowel prep, your provider will give you instructions on how to do it.

Types of visual exams include:

Colonoscopy. This test lets your provider view your entire rectum and colon (large intestine). It is often done in the outpatient department of a hospital or medical center. The procedure takes about 30 to 60 minutes.

Before the test, you will need to empty your colon (large bowel) in a procedure called bowel prep.

During the test:

You will lie on an exam table.

You will be given a medicine called a sedative to help you relax and prevent you from feeling pain during the procedure. You may be awake for the test but probably won't remember anything.

A colonoscope, a thin, lighted tube with a camera attached, will be inserted into your rectum and up into your colon.

The scope inflates your large intestine with air for a better view. The camera sends a video image to a monitor so your provider can see your colon.

Your provider will look for colorectal polyps or other abnormal areas.

If you have polyps, your provider may remove them or tissue samples using special tools inserted through the scope.

Polyps or samples may be sent to a lab for testing. Most polyps aren't cancer, but removing them can prevent them from becoming cancer later on.

If you have abnormal tissue, your provider may do a biopsy.

The sedative may make you drowsy for several hours. You should arrange for someone to take you home.

Sigmoidoscopy. This test allows your provider to look at your rectum and the lower colon (sigmoid colon). During the test:

You will lie on an exam table with your knees drawn up to your chest.

A sigmoidoscope, a flexible, lighted tube with a camera, will be inserted into your rectum and up into the lower part of your colon.

Air will be pumped through the sigmoidoscope to make your colon bigger and easier to see.

The air may make you feel like you need to have a bowel movement or pass gas.

Your provider may remove polyps or tissue samples using special tools inserted through the scope.

Polyps or samples may be sent to a lab for testing.

Virtual Colonoscopy. This test is also called a computerized tomography (CT) colonography. It uses x-rays and a computer to take detailed, 3-dimensional pictures of the entire colon and rectum. Since this is an x-ray, this test is not recommended if you are pregnant. Tell your provider if you have any implanted medical devices, such as a pacemaker.

This test is usually done in the radiology department of a hospital or medical center and takes about 10 to 15 minutes. A virtual colonoscopy might be done if you can't have a colonoscopy due to medical reasons. This procedure also requires bowel prep beforehand. During the test:

You will lie on a narrow table.

You may be asked to drink a liquid containing contrast dye. Contrast dye is a substance that makes parts of your body show up more clearly on an x-ray.

A provider will insert a small tube into your rectum.

Air will be pumped through the tube to make your colon bigger and easier to see.

The table will slide into the CT machine. This is a tunnel-shaped device where the technician takes the x-ray images.

You may be asked to hold your breath a few times during the test.

You will turn over on your side or stomach to get more images taken.

A series of images will be taken.

You may have some cramping or bloating for about an hour after the test.

Stool tests

Stool tests look at stool samples for blood or other signs of cancer you can't see. But you can also have blood in your stool if you have a health condition that isn't cancer, such as hemorrhoids. Your provider usually prescribes the test, which can be done at home. You take a stool sample and send it to a lab.

Types of stool tests include:

Fecal occult blood tests (FOBT). These tests look for hidden (known as occult) blood in your stool. Blood in the stool may be a sign of polyps, cancer, or other conditions. This test can check if there is blood in your stool, but the test can't tell your provider what is causing the bleeding. Your provider will give you a kit that includes instructions on how to do the test.

There are two types of fecal occult blood tests: the fecal immunochemical test (FIT) and the guaiac smear method (gFOBT). Below are typical instructions for each test. Your instructions may vary slightly, depending on the manufacturer of the test kit.

For a fecal immunochemical test (FIT), you will most likely need to:

Collect multiple samples from one bowel movement.

Collect the sample from the toilet using the special brush or other device included in your kit.

For each sample, use the brush or device to take the sample from the stool's surface.

Brush the sample onto a test card.

Label and seal all your samples as directed.

Mail the samples to your provider or lab.

For a guaiac smear test (gFOBT), you will most likely need to:

Collect samples from three separate bowel movements.

For each sample, collect the stool and store it in a clean container. Make sure the sample does not mix in with urine or water from the toilet.

Use the applicator from your test kit to smear some of the stool on the test card or slide, which is also included in your kit.

Label and seal all your samples as directed.

Mail the samples to your provider or lab.

Stool DNA test (FIT-DNA test). This test checks for genetic changes in your stool that may be signs of colorectal polyps or cancer. Your provider will let you know to get a test kit. The kit will include instructions on how to do the test. The instructions will likely include the following steps:

Put a special container inside your toilet.

Collect an entire bowel movement.

Close and seal the container.

Mail the container with the stool sample to a lab for testing.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

There are no special preparations needed for a FIT or stool DNA test. The typical preparations for other colorectal cancer screening tests are below.

For a colonoscopy and virtual colonoscopy, you will need to do a bowel prep. Your provider will give you specific instructions on how to do your bowel prep, but steps for bowel prep may include:

Following a liquid diet for one to three days before the test.

Drinking plenty of clear liquids one to three days before the test. Clear liquids include water, black coffee or tea, fat-free broth, and sports drinks without added color. Avoid red or purple drinks or gelatin; the dye in those can look like blood in the colon.

Drinking a strong liquid laxative (a medicine used to cause a bowel movement), taking laxative pills, and/or using an enema on the evening before your test.

The laxative or enema will cause diarrhea. Be prepared to stay close to a bathroom. Bowel prep can be inconvenient and uncomfortable, but if the colon is not thoroughly cleaned out, your provider may not be able to get a complete picture of your colon and rectum. They might not be able to see if you have polyps and other abnormal areas.

Also, you may need to stop taking certain medicines before these tests, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

For a sigmoidoscopy, before the test, you may need to:

Follow a clear liquid diet one or more days before the procedure. This isn't always required, so be sure to talk to your provider before the test.

You may be asked to fast (not eat or drink) on the morning of the procedure.

You may need to take a laxative (a medicine used to cause a bowel movement) the night before the test.

You should empty your bowels with an enema about one hour before the procedure. A second enema is sometimes needed.

Also, you may need to stop taking certain medicines before these tests, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

For a g(FOBT) test, you may need to avoid the following foods and medicines for several days before your test:

Nonsteroidal, anti-inflammatory drugs (NSAIDs), such as ibuprofen, naproxen, or aspirin

Vitamin C daily from supplements, fruit juices, or fruit

Certain vegetables, such as broccoli and turnips

Red meat, such as beef, lamb, or pork

Your provider will give you full instructions on how to prepare for your screening test.

ARE THERE ANY RISKS TO THE TEST?

There are no risks to having a fecal occult blood test or a stool DNA test.

Serious risks of colonoscopy are rare but can include:

Tears in the colon or rectum wall

Bleeding

Reaction to the sedative

There are fewer risks to sigmoidoscopy than with a colonoscopy. Tears and bleeding can happen in the colon, but they are even less common than with a colonoscopy.

In a virtual colonoscopy, risks include:

Exposure to a small dose of radiation from the scan

Rarely, a tear in the colon or rectum can happen when it is inflated with air

Seek medical care immediately if you have severe abdominal pain, blood in your stool or bleeding from your anus, dizziness, or weakness.

WHAT DO THE RESULTS MEAN?

If your fecal occult blood test or stool DNA test had abnormal results, it doesn't necessarily mean you have cancer. But your provider will probably order more tests, such as a colonoscopy, to help make a diagnosis.

Abnormal results for the other types of screening tests may include the following:

Colonoscopy

Colorectal polyps or abnormal tissue were found and removed.

Samples were sent to a lab for testing.

Most polyps aren't cancerous but can turn into cancer if not removed. Depending on the size and number of polyps, your provider may recommend more frequent colonoscopies.

Sigmoidoscopy

Colorectal polyps or abnormal tissue were found. They may or may not be removed depending on their location.

Whether or not your polyps were removed, you will probably need a colonoscopy so your provider can view your entire colon and rectum.

Virtual colonoscopy

Colorectal polyps or abnormal tissue were seen on the CT scan.

You may need a traditional colonoscopy so your provider can get a tissue sample. Sometimes this procedure can be done on the same day as your virtual colonoscopy.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT COLORECTAL CANCER SCREENING?

Each type of screening test has advantages and disadvantages. You and your provider should discuss the benefits and risks of each option. The following may help guide your decision:

The most sensitive colon cancer screening test

Looks at the entire colon

Can remove polyps and test for cancer

May help find other diseases

Needed only every 10 years (if no polyps were found in previous tests)

Small risk of tears and bleeding

Bowel prep needed

Sedatives are needed for the procedure

Takes at least one day for bowel prep and recovery

Usually doesn't require bowel prep

Sedatives are not usually needed

Needed every five years (if no polyps were found in previous tests)

Only looks at a third of the colon
 May miss small polyps
 Can't remove all polyps
 Very small risk of tears or bleeding
 Not widely used as a screening test
 Colonoscopy will be needed if results are abnormal
 Looks at the entire colon
 Sedatives not needed
 Needed every five years (if no polyps were found in previous tests)
 Bowel prep needed
 May miss small polyps
 Polyps cannot be removed during the procedure
 Exposure to a small amount of radiation
 Traditional colonoscopy will be needed if results are abnormal
 Fairly new test; may not be covered by insurance
 No risk to colon
 No bowel prep
 Sample can be collected at home
 Can have false positives, which means your test results show that blood is in your stool, but there isn't any
 Cannot detect precancerous changes (cells that are at risk for turning into cancer)
 May miss some signs of polyps and/or cancer
 Colonoscopy will be needed if results are abnormal
 Should be done every year
 No risk to colon
 No bowel prep
 Sample can be collected at home
 Can have false positives, which means your test results show that blood is in your stool, but there isn't any
 Cannot detect precancerous changes
 May miss some signs of polyps and/or cancer
 Dietary restrictions required before testing
 Colonoscopy will be needed if results are abnormal
 Should be done every year
 No risk to colon
 No bowel prep
 Sample can be collected at home
 Can have false positives
 Cannot detect precancerous changes
 May miss some cancers
 Colonoscopy will be needed if results are abnormal
 Should be done every three years

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Colposcopy

<https://medlineplus.gov/lab-tests/colposcopy/>

COLPOSCOPY

WHAT IS A COLPOSCOPY?

A colposcopy is a procedure to examine your cervix, vagina, and vulva:

The vagina connects your womb (uterus) and cervix to the outside of your body.

The cervix is the lower part of your uterus (the uterus is the place where a baby grows during pregnancy).

The vulva includes your genital parts that are outside your body.

During the procedure, your health care provider will be looking for abnormal cells, which could be cancer or precancer. Precancer means that the cells could turn into cancer over time.

To do the procedure, your provider will use a lighted, magnifying device called a colposcope. They will place the device at the opening of your vagina. The device magnifies the normal view, allowing your provider to see problems that can't be seen by the eyes alone.

If your provider sees a problem, they may take a tissue sample for testing (biopsy). The sample is most often taken from the cervix. This procedure is known as a cervical biopsy. Biopsies may also be taken from the vagina or vulva. The results of the biopsy will show if you have cells that are cancer or precancer.

Other names: colposcopy with directed biopsy

WHAT IS IT USED FOR?

A colposcopy is most often used to find abnormal cells in the cervix, vagina, or vulva. It may also be used to:

Check for genital warts, which may be a sign of an HPV (human papillomavirus) infection. Having HPV may put you at higher risk for developing cervical, vaginal, or vulvar cancer.

Look for noncancerous growths called polyps.

Check for irritation or inflammation of the cervix.

Check for the cause of abnormal vaginal bleeding or vulvar itching.

If you've already been diagnosed and treated for HPV, the test may be used to monitor cell changes in the cervix. Sometimes, abnormal cells return after treatment.

WHY DO I NEED A COLPOSCOPY?

You may need this test if you had abnormal results on a Pap smear. A Pap smear is a test that involves collecting a sample of cells from the cervix. It can show if there are abnormal cells but can't provide a diagnosis. A colposcopy provides a more detailed look at the cells. It may help your provider confirm a diagnosis and/or find other potential problems. It could allow your provider to find precancer cells early enough that they can be treated and might not turn into cancer.

You may also need this test if you have:

Been diagnosed with HPV

Abnormal areas on your cervix (which were seen during a routine pelvic exam)

Bleeding after sex

WHAT HAPPENS DURING A COLPOSCOPY?

A colposcopy may be done by your primary care provider or a gynecologist, a doctor who specializes in diagnosing and treating diseases of the female reproductive system. The test is usually done in the provider's office and takes 10 to 20 minutes. If the provider finds abnormal tissue, you may also get a biopsy, which may take about 10 minutes longer.

During a colposcopy:

You will remove your clothing and put on a hospital gown.

You will lie on your back on an exam table with your feet in stirrups.

Your provider will insert an instrument called a speculum into your vagina. A speculum is the same instrument used in a Pap smear. It is used to spread open your vaginal walls so that your cervix is visible.

Your provider will place the colposcope outside your vagina.

Your provider will shine a light into your vagina and look through the colposcope, which provides a magnified view of the cervix, vagina, and vulva.

Your provider will gently swab your cervix and vagina with a vinegar or iodine solution. The solution makes abnormal tissues easier to see.

If any areas of tissue look abnormal, your provider may perform a cervical, vaginal, or vulvar biopsy.

During a biopsy:

A vaginal biopsy may cause mild to moderate pain, and your provider may recommend you take an over-the-counter pain reliever before the procedure or give you medicine to numb the area.

Once the area is numb, your provider will use a small tool to remove a tissue sample for testing. Sometimes, many samples are taken.

Your provider may also do a procedure called an endocervical curettage (ECC). This procedure involves taking a tissue sample from the inside of the opening of the cervix. This area can't be seen during a colposcopy. An ECC is done with a special tool called a curette. You may feel a pinch or cramp as the tissue is removed.

Your provider may apply a topical medicine to the biopsy site to treat any bleeding you may have.

After a biopsy, you should not douche, use tampons, or have sex for a week after your procedure or for as long as your provider advises.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Do not douche, use tampons or vaginal medicines, or have any kind of vaginal penetration for 48 hours before the test. Also, it's best to schedule your colposcopy when you are not having your menstrual period. Your provider may recommend taking an over-the-counter pain reliever before your appointment.

Tell your provider if you are pregnant or think you may be pregnant. Colposcopy is generally safe during pregnancy, but being pregnant raises your risk of bleeding after a biopsy.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a colposcopy. You may have some discomfort when the speculum is inserted into the vagina, and the vinegar or iodine solution may sting, tingle, or burn.

A biopsy is also a safe procedure. You may feel a pinch or some pain when the tissue sample is taken. After the procedure, you may be sore for a day or two. You may have some cramping and slight bleeding. It's normal to have a little bleeding and discharge for up to a week after the biopsy.

Serious complications from a biopsy are rare, but call your provider if you have any of the following symptoms:

Heavy bleeding

Abdominal (belly) pain

Signs of infection, such as fever, chills, and/or bad smelling vaginal discharge

WHAT DO THE RESULTS MEAN?

During your colposcopy, your provider may find one or more of the following conditions:

Genital warts

Polyps

Swelling or irritation of the cervix

Abnormal tissue

If your provider also performed a biopsy, your results may show you have:

Precancerous cells in the cervix, vagina, or vulva

An HPV infection

Cancer of the cervix, vagina, or vulva

If your biopsy results were normal, it's unlikely that you have cells in your cervix, vagina, or vulva that are at risk for turning into cancer. But that can change. So, your provider may want to monitor you for cell changes with more frequent Pap smears and/or additional colposcopies.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A COLPOSCOPY?

If your results show that you have precancerous cells, you may need to have a procedure to remove them. This may prevent them from developing into cancer.

If your results show that you have cancer, you may be referred to a gynecologic oncologist. This is a provider who specializes in treating cancers of the female reproductive system.

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Complement Blood Test

<https://medlineplus.gov/lab-tests/complement-blood-test/>

COMPLEMENT BLOOD TEST

WHAT IS A COMPLEMENT BLOOD TEST?

A complement blood test measures the levels and/or activity of a group of proteins that are part of the complement system. This system "complements" or enhances your body's defenses against infections and other diseases. Your immune system makes antibodies, which are proteins to fight disease-causing substances like viruses and bacteria. The proteins in the complement system will also help fight the infection or disease. These proteins work together in a sequence, with one protein activating the next.

There are nine major complement proteins. They are labeled C1 through C9. Complement proteins may be measured individually or together:

C3 and C4 proteins are the most commonly tested individual complement proteins.

A CH50 test (sometimes called CH100 or a total complement test) measures the total activity and levels of all the major complement proteins.

If the CH50 level is not normal, each of the nine complement levels can be measured individually to find out which one is abnormal.

If the test shows that your complement protein levels are not normal or that the proteins aren't working with the immune system as well as they should, it can be a sign of an autoimmune disease or other health condition. If you have an autoimmune disease your immune system attacks the healthy cells of your organs and tissues by mistake.

Other names: complement antigen, complement activity C3, C4, CH50, CH100, C1 C1q, C2, total complement test

WHAT IS IT USED FOR?

A complement blood test is most often used to diagnose or monitor autoimmune diseases such as:

Lupus, a chronic (long-lasting) disease that affects many parts of the body, including the joints, blood vessels, kidneys, and brain

Rheumatoid arthritis, a condition that causes pain and swelling of the joints, often in the wrist, hands, and feet

It may also be used to help diagnose certain bacterial, viral, or fungal infections.

WHY DO I NEED A COMPLEMENT BLOOD TEST?

Your health care provider may order complement blood test if you have an autoimmune disease such as lupus. The test can show how well the treatment is working.

Your provider may also order this test if you have the symptoms of an autoimmune disease. The symptoms of autoimmune disease can depend on the part of your body that's affected. The symptoms can come and go and may include:

Infections

Unexplained inflammation, such as redness, swelling, heat and pain

Fatigue

Joint pain or swelling

Muscle pain

WHAT HAPPENS DURING A COMPLEMENT BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle

goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a complement blood test.

ARE THERE ANY RISKS TO A COMPLEMENT BLOOD TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results show decreased levels or activity of complement proteins, it may mean that your body is trying to get rid of foreign substances or attacking its own tissues.

Decreased levels may be a sign of the following conditions:

Lupus.

Rheumatoid arthritis.

Cirrhosis, scarring of the liver.

Certain types of kidney disease.

Hereditary angioedema, a rare but serious disorder of the immune system. It can cause swelling of the face and airways.

Malnutrition.

A recurring infection (usually bacterial).

If your results show increased levels or activity of complement proteins, it may mean that your body has acute or chronic inflammation.

Increased levels may be a sign of the following conditions:

Certain types of cancer, such as leukemia or non-Hodgkin lymphoma

Ulcerative colitis, a condition in which the lining of the large intestine and rectum become inflamed

If you are being treated for lupus or another autoimmune disease, increased levels or activity of complement proteins may mean your treatment is working.

Increased or decreased complement levels can also be a sign of a condition affecting your immune system. If you have questions about your results, talk to your provider. They will consider your symptoms, medical history, and the results of other blood tests to determine if additional tests are needed to make a diagnosis.

Learn more about laboratory tests, reference ranges, and understanding results.

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Complete Blood Count (CBC)

<https://medlineplus.gov/lab-tests/complete-blood-count-cbc/>

COMPLETE BLOOD COUNT (CBC)

WHAT IS A COMPLETE BLOOD COUNT?

A complete blood count, or CBC, is a group of blood tests that measure the number and size of the different cells in your blood. A CBC measures:

Red blood cells, which carry oxygen from your lungs to the rest of your body.

White blood cells, which fight infections and other diseases. There are five major types of white blood cells. A CBC test measures the total number of white cells in your blood. A different test called a CBC with differential measures the number of each type of these white blood cells.

Platelets, which stop bleeding by helping your blood to clot.

Hemoglobin, an iron-rich protein in red blood cells that carries oxygen from your lungs to the rest of your body.

Hematocrit, the amount of whole blood that is made up of red blood cells.

Mean corpuscular volume (MCV), the average size of your red blood cells.

Other names for a complete blood count: CBC, full blood count, blood cell count

WHAT IS IT USED FOR?

A complete blood count is often part of a routine checkup. It is also used to monitor a condition or treatment that may affect your blood cell counts such as infections, anemia, immune system disorders, and blood cancers.

WHY DO I NEED A COMPLETE BLOOD COUNT?

Your health care provider may have ordered a complete blood count as part of your checkup or to monitor your overall health. The test may also be used to:

Help diagnose blood diseases, infections, immune system disorders, or other medical conditions

Check for changes in an existing blood disorder

WHAT HAPPENS DURING A COMPLETE BLOOD COUNT?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Usually, no special preparation is necessary for a complete blood count. But if your provider ordered other tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle went in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A CBC counts the cells in your blood. There are many reasons your levels may not be in the normal range. For example: Abnormal levels of red blood cells, hemoglobin, or hematocrit may be a sign of dehydration, anemia, heart disease, or too little iron in your body.

A low white cell count may be a sign of an autoimmune disorder, bone marrow disorder, or cancer.

A high white cell count may be a sign of an infection or a reaction to a medicine.

If any of your levels are abnormal, it doesn't always mean you have a medical condition that needs treatment. Diet, activity level, medicines, having a menstrual period, not drinking enough water, and other factors can affect the results. Talk with your provider to learn what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A COMPLETE BLOOD COUNT?

A complete blood count is only one tool your provider uses to learn about your health. Your provider will consider your medical history, symptoms, and other factors to make a diagnosis. You may also need additional tests.

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- What You Need to Know About Blood Testing
- White Blood Count (WBC)

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Comprehensive Metabolic Panel (CMP)

<https://medlineplus.gov/lab-tests/comprehensive-metabolic-panel-cmp/>

COMPREHENSIVE METABOLIC PANEL (CMP)

WHAT IS A COMPREHENSIVE METABOLIC PANEL (CMP)?

A comprehensive metabolic panel (CMP) is a routine blood test that measures 14 different substances in a sample of your blood. It provides important information about your metabolism (how your body uses food and energy) and the balance of certain chemicals in your body.

A CMP includes tests that measure your blood levels of:

Glucose, also called "blood sugar." Glucose is your body's main source of energy.

Calcium. This is one of the most important minerals in your body. Most of it is stored in your bones and teeth. Having the right amount of calcium in your blood is necessary for your nerves, muscles, and heart to work properly.

Sodium, potassium, bicarbonate, and chloride. These are electrolytes, electrically charged minerals that help control the amount of fluid in your body. They also help control the balance of acids and bases (pH balance) in your body.

Albumin. This is the main protein in your blood. It's made in your liver.

Total protein. This is the total amount of protein in your blood. It includes albumin and globulins, which are mainly made in your liver.

ALP (alkaline phosphatase), ALT (alanine transaminase), and AST (aspartate aminotransferase). These are types of enzymes that are mainly made in your liver. Enzymes are proteins that speed up certain chemical reactions in your body.

Bilirubin. This is a waste product your body makes when it breaks down old red blood cells. Your liver removes most of the bilirubin from your body.

BUN (blood urea nitrogen) and creatinine. These are waste products that your kidneys filter out of your blood and remove from your body through urine (pee).

Other names: chem 14, chemistry panel, chemistry screen, metabolic panel

WHAT IS IT USED FOR?

A CMP is commonly used as part of a routine checkup. It can provide information about your overall health and help find certain conditions before you have symptoms. For example, a CMP can check your:

Liver and kidney health

Blood glucose

Protein levels

Fluid and electrolyte balance, which can affect the acid-base balance in your blood

Metabolism

A CMP may also be used to help:

Diagnose the cause of certain symptoms you may have

Guide treatment for many diseases

Check to see if treatment is working and/or if it's causing certain side effects

WHY DO I NEED A CMP?

Your health care provider may order a CMP as part of a routine checkup. You may also need this test if you:

Have signs or symptoms that could be caused by liver or kidney disease, or metabolic disorders

Have a general symptom, such as fatigue, that could be caused by many different types of conditions

Had an abnormal CMP result in the past

Are being treated for certain medical conditions and your provider wants to find out if your treatment is working and/or affecting your liver or kidneys

WHAT HAPPENS DURING A CMP?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for several hours before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The results of your CMP test will usually list the levels of each of the 14 substances that were measured. To understand what your results mean, your provider will compare all the results. Your provider will also consider your health history, any medicines you take, and other things that could affect your test results. Ask your provider to explain what your CMP results say about your health.

In general, if you have one or more results that aren't normal, it may be a sign of a health condition. For example, high blood glucose may be a sign of diabetes. You will likely need more tests to confirm or rule out a specific diagnosis.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CMP?

Another test called a basic metabolic panel (BMP) is similar to a CMP. A BMP includes 8 of the 14 tests that are part of a CMP. A BMP does not include a measurement of:

Total protein

Albumin

Bilirubin

Liver enzymes (ALP, ALT, and AST)

Your provider may choose a CMP or a BMP, depending on your health history and needs.

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RELATED HEALTH TOPICS

- Blood Glucose
- Calcium
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RELATED MEDICAL TESTS

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- AST Test
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- BUN (Blood Urea Nitrogen)
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- Carbon Dioxide (CO₂) in Blood
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Concussion Tests

<https://medlineplus.gov/lab-tests/concussion-tests/>

CONCUSSION TESTS

WHAT ARE CONCUSSION TESTS?

Concussion tests can help find out if an adult or child has had a concussion after a head injury. A concussion is also called a mild traumatic brain injury (mTBI).

A concussion happens when sudden movement causes the brain to bounce around or twist inside the skull. It can be caused by a bump or blow to the head. It may also happen if a hit to the body causes the head and brain to move rapidly back and forth. This jolting movement can create chemical changes in the brain. And it can stretch and damage brain cells.

Even though a concussion is a "mild" brain injury, it can seriously affect the brain. Most people don't lose consciousness when they have a concussion. But some signs and symptoms usually show up right away after the injury and others may appear hours or days later. Symptoms are different for each person, but often include headaches, problems with memory, and changes in how a person thinks, acts, feels, and sleeps.

The symptoms of a concussion are usually temporary. With treatment most people feel better in a couple weeks. The main treatment is physical and mental rest. Without proper rest, recovery from a concussion may take longer.

There are many types of concussion tests that health care providers use to help diagnose concussions. They usually include questionnaires that you fill out and tests of mental and physical abilities. Some tests have special versions for children. All concussion tests have scoring systems that help show how much an injury has affected the brain. The tests check:

How well your brain processes information. This includes testing memory, problem solving, and other brain functions. These tests are often done on a computer or on paper.

How well your nervous system is working. These tests check balance, muscle coordination, and other physical abilities. A provider often does these tests as part of a physical exam.

A concussion will not show up on imaging tests that take pictures of the brain. So, these tests usually aren't needed to diagnose a concussion. But a CT or MRI scan of the brain may be done if there are signs of a more serious brain injury, such as bleeding in the brain.

Some concussion tests are designed for sports coaches or trainers to use with athletes of all ages who have hurt their heads playing sports. These tests are sometimes called "sideline assessments." They are not a substitute for a medical examination after a blow to the head.

Other names: concussion assessment

WHAT ARE THEY USED FOR?

Concussion tests are used after a head injury to see if the injury is affecting brain function. The tests are also be used to monitor recovery after a concussion.

Athletes, including student athletes, may have a routine concussion test at the start of the sports season even when they haven't had a brain injury. This is called a "baseline test," and it's usually done by a trained health care professional.

The purpose of baseline testing is to have a record of the athlete's normal brain function. Later on, if the athlete has a head injury, a provider can compare the baseline test results with the results of a concussion test done after the injury. This helps to:

Confirm or rule out a concussion

Find out how severe a concussion is

Know when an athlete is fully recovered and ready to return to normal activities

Baseline concussion tests are mostly done on athletes who play contact sports that have a high risk for concussion, such as football and hockey.

WHY DO I NEED CONCUSSION TESTING?

You may need concussion testing if you have had a head injury or a blow to the body that jolts your head. Testing may be necessary even if you think the injury isn't serious. That's because it's possible to have a concussion and not even know it.

Concussion symptoms usually show up soon after a head injury. But sometimes, they don't show up for hours or days. So, it's important to keep watching for signs and symptoms of a concussion for a day or two after a head injury. Signs and symptoms of a concussion include:

- Appearing dazed or stunned
- Clumsy movements
- Answering questions slowly
- Headache
- Nausea or vomiting
- Fatigue
- Confusion
- Dizziness
- Sensitivity to light and/or blurry or double vision
- Changes in sleep and/or mood
- Difficulty paying attention
- Memory problems

In rare cases, a dangerous blood clot can form on the brain a day or two after a head injury. This is a medical emergency. Call 911 or go to an emergency department right away if a child or adult has one or more of these symptoms:

- A headache that gets worse or doesn't go away
- Weakness, numbness, decreased coordination, or seizures
- Repeated vomiting
- Slurred speech and/or unusual behavior
- One pupil (the black part in the middle of the eye) that's larger than the other
- Confusion
- Not being able to wake up, a loss of consciousness, or looking very drowsy

For infants and toddlers, the symptoms of a medical emergency are the same as for adults and children, but also include:

- Crying that won't stop
- Not eating or nursing

WHAT HAPPENS DURING CONCUSSION TESTING?

To do concussion testing a provider will usually use one or more standard concussion tests to learn about the injury and how it's affecting the brain. During the testing you'll usually:

Answer questions about the injury, such as:

- How the head injury happened
- Whether the injured person was unconscious and for how long
- Signs and symptoms after the injury
- How the head injury happened
- Whether the injured person was unconscious and for how long
- Signs and symptoms after the injury

Take a test, often on a computer, to check for problems with:

- Learning
- Memory
- Thinking and reasoning
- Ability to pay attention

Learning
Memory
Thinking and reasoning
Ability to pay attention
Have a physical exam that checks for problems with:

Vision and eye movement
Hearing
Balance
Coordination
Reflexes
Vision and eye movement
Hearing
Balance
Coordination
Reflexes

In certain cases, your provider may order a blood test to help decide if a brain scan is needed to check for a more serious injury. The test checks a blood sample for certain proteins that may show up in the bloodstream after a mild concussion. To collect a blood sample, a health care professional inserts a small needle into a vein in your arm and collects a small amount of blood into a test tube or vial.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A CONCUSSION TEST?

You don't need any special preparations for concussion testing.

ARE THERE ANY RISKS TO THE TESTS?

There's little risk to having concussion testing. If you have a blood test, you may have slight pain or bruising at the spot where the needle was put in, but it usually goes away quickly.

WHAT DO THE RESULTS MEAN?

If concussion testing shows that a head injury has caused a concussion, the treatment is physical and mental rest to help the brain heal. This often includes limiting time spent on electronic screens (televisions, computers, phones, and tablets). It may also be necessary to take two or three days off from school or work.

Ask your provider about a recovery plan, including how to relieve symptoms. As symptoms improve, physical and mental activities can gradually be increased. But it's important to make sure that the concussion is fully healed before returning to all usual activities. That's because having a second concussion before the first has healed can cause serious brain damage. Your provider may do another concussion test to see if recovery is complete.

Complete healing is very important for athletes who want to return to their sport after a concussion. Different sports organizations may have "concussion protocols" that set guidelines for when an athlete can participate again after having a concussion.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT CONCUSSION TESTING?

Concussions are common among children and young adults. Adults age 75 and up also have a higher-than-average risk of concussion because they have a higher risk for falls. But there are steps you can take to prevent concussions at any age, including:

Wearing the correct helmet and making sure it fits when you or your child play sports such as, biking, skiing, skating,

hockey, football, and batting in baseball or softball

Wearing seat belts

Preventing falls by keeping your home well-lit, removing tripping hazards, and using nonslip mats and grab bars in the bathroom

Making home and play areas safe for young children:

Making sure playgrounds have shock-absorbing material, such as mulch or sand

Installing window guards

Using safety gates on stairs

Making sure playgrounds have shock-absorbing material, such as mulch or sand

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Cord Blood Testing and Banking

<https://medlineplus.gov/lab-tests/cord-blood-testing-and-banking/>

CORD BLOOD TESTING AND BANKING

WHAT ARE CORD BLOOD TESTING AND CORD BLOOD BANKING?

Cord blood is the blood left in the umbilical cord after a baby is born. The umbilical cord connects a mother to her unborn baby during pregnancy. It contains blood vessels that bring nourishment to the baby and remove waste products.

After a baby is born, the cord is clamped shut and cut. The blood that remains in the cord may be collected and used in two ways:

Cord blood testing. This means doing a variety of tests on cord blood to check the baby's health. Tests may also be done on cord blood if a health care provider is concerned that the baby may have a health condition, such as an infection or an acid-base (pH) imbalance. Some hospitals may routinely collect cord blood samples to do tests to check a baby's general health.

Cord blood banking. This means saving blood from the baby's umbilical cord in a special facility called a cord blood bank. The blood is saved so it can be used to treat certain diseases. Umbilical cord blood is full of special cells called stem cells. Stem cells can develop into many different types of cells found in the body, such as blood, brain, muscle, and organ cells.

Stem cells from cord blood can be used to treat certain blood disorders, including leukemia, Hodgkin disease, and some types of anemia. They can also be used to treat certain inherited metabolic and immune system disorders. Researchers are studying whether stem cell treatment can also help treat other types of diseases.

If you choose to save your baby's cord blood, you have two main options:

You can donate the blood to a public cord blood bank. Cord blood in public banks may be used by anyone who needs a cord blood transplant. This treatment replaces diseased blood cells with healthy cells from cord blood. In certain cases, donated cord blood may be used for research. If you donate your baby's cord blood, there's no cost to you. But only certain hospitals can collect cord blood for storage in public cord blood banks.

You can save the blood in a family (private) cord blood bank. This allows you to use the blood for a family member who may benefit from treatment with cord blood. You may be interested in this option if you have a family history of health conditions that can be treated with stem cells from cord blood. You will usually have to pay a fee for the blood collection and storage at a private cord blood bank. But if you or a close family member already has a disease that needs treatment with cord blood, you may qualify for free or low-cost cord blood banking.

WHAT IS CORD BLOOD TESTING USED FOR?

Cord blood tests may be done after your baby is born to:

Measure blood gases. This helps to see if a baby's blood has a healthy level of oxygen and acid-base (pH) balance.

Measure bilirubin levels. Bilirubin is a waste product made by the liver. High bilirubin levels can cause jaundice, a condition that causes the skin and eyes to turn yellow. Jaundice is common in healthy newborns, and it usually clears up in a few weeks. But it can sometimes be a sign of liver disease, a blood disorder, or another type of condition.

Do a blood culture test. This test may be done if a provider thinks a baby may have an infection.

Do a complete blood count (CBC). This test measures many different parts of the blood. It may be done to check for an infection.

Check blood glucose (sugar) levels. Blood glucose may be high if the mother has diabetes.

Check to see if a baby was exposed to illegal drugs or certain prescription drugs that the mother may have misused during pregnancy. Umbilical cord blood testing can find many drugs, including sedatives, cocaine, and opioids such as heroin and fentanyl. If harmful drugs are found in cord blood, a provider can treat the baby to help reduce the chance of problems, such as developmental delays.

WHAT IS CORD BLOOD BANKING USED FOR?

Cord blood banking is used to save cord blood so that the stem cells can be used to treat certain blood disorders and cancers.

You may want to consider donating cord blood if you want to help others. A public blood cord bank provides lifesaving stem cells to people who need treatment now.

You may want to consider family cord blood banking if:

You have a family history of a genetic disorder or cancer that can be treated with stem cells.

You or a family member currently has a disorder that may be treated with stem cells.

If neither of these situations applies to you, providers don't recommend saving your baby's cord blood in a family cord blood bank. There are two main reasons:

The conditions that are currently treated with stem cells from cord blood aren't common. That means your child and other family members are unlikely to ever need the cord blood. Most cord blood that's saved in case of a future illness is never used. But as research continues, it is possible that stem cell treatment may be developed for more common diseases.

If your child should someday need stem cell treatment for a genetic disorder or cancer, they won't be able to use their own stem cells for treatment. That's because their stem cells will have the same genes or pre-cancerous cells that caused the disease in the first place.

HOW IS CORD BLOOD COLLECTED?

Collecting cord blood does not change anything about your birth experience. After your baby is born, the umbilical cord will be pinched closed with a clamp and cut. Then the provider will either use a needle to remove blood from the cord or let the blood drip out of the cord. The blood is collected in a special bag and sent to a lab for testing or to a cord blood bank for long-term storage.

Sometimes, the cord doesn't contain enough blood to use for cord blood banking.

IS THERE ANY PREPARATION NEEDED FOR CORD BLOOD TESTING OR BANKING?

For cord blood testing, you don't need to prepare anything.

For cord blood banking, you need to plan in advance. If you're considering donating cord blood or privately banking it, talk with your provider at least three months before your baby is due. This will give you time to:

Decide whether to donate the blood to a public bank or save it in a family cord blood bank.

Get information about your options for collecting the blood. Not all hospitals can collect blood for public cord blood banks. If you choose to use a family cord blood bank, you'll need to select which bank to use.

Finish paperwork about your medical history and your family health history.

Receive a cord blood collection kit if you are using a family cord blood bank. Your provider will use this kit to collect and send the blood to the bank.

ARE THERE ANY RISKS TO CORD BLOOD TESTING OR BANKING?

Cord blood testing has no risks.

With cord blood banking, the mother will need to have a blood test to check for infections to help make sure the cord blood is safe to use. There is very little risk to having a blood test. You may have slight pain or bruising where the needle was put in, but any discomfort or bruises usually go away quickly.

Some cord blood banks may require testing of the mother's blood for genetic disorders, too. If you find out you have a gene that could cause a serious disease, you or your family members may feel upset.

If you choose family cord blood banking, the costs can be very high.

WHAT DO CORD BLOOD TEST RESULTS MEAN?

Cord blood test results will depend on what was measured. If the results were not normal, ask your baby's provider to explain what they mean for your baby's health.

Learn more about laboratory tests, reference ranges, and understanding results.

For more information on cord blood testing and banking, talk with your provider.

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Cortisol Test

<https://medlineplus.gov/lab-tests/cortisol-test/>

CORTISOL TEST

WHAT IS A CORTISOL TEST?

A cortisol test measures the level of cortisol in your blood, urine (pee), or saliva (spit). Cortisol is a hormone made by your adrenal glands, two small glands that sit above your kidneys. Hormones are chemical messengers in your bloodstream that control the actions of certain cells or organs. Cortisol affects almost every organ and tissue in your body. It helps your body:

- Respond to stress (cortisol is sometimes called the "stress hormone.")

- Reduce inflammation (swelling)

- Control blood glucose (also called "blood sugar") and metabolism (how your body uses food for energy)

- Control your blood pressure

Your pituitary gland, a gland in your brain, makes another hormone, called ACTH (adrenocorticotropic hormone). ACTH tells your adrenal glands how much cortisol to make. If your cortisol levels are too high or too low, it may mean you have a problem with your adrenal glands or your pituitary gland.

High cortisol levels may also be caused by a tumor that makes cortisol, or taking large doses of certain steroid medicines, such as prednisone, for a long time. Low levels may happen if you stop the medicine suddenly. Without treatment, cortisol levels that are too high or too low can be very serious.

Other names: urinary cortisol, salivary cortisol, free cortisol, blood cortisol, plasma cortisol, cortisol serum, 24-hour urinary cortisol excretion, 24-hour urinary free cortisol (UFC), late-night salivary cortisol, cortisol competitive protein binding assay, cortisol fluorometric assay

WHAT IS IT USED FOR?

A cortisol test is used to help diagnose or rule out medical conditions that cause too much or too little cortisol, such as: Cushing's syndrome, a disorder that causes high cortisol levels. It happens when your body has been exposed to too much cortisol for a long period of time. It may be caused by steroid medicines or certain tumors that cause your body to make extra cortisol.

Addison disease, a condition that causes low cortisol levels. It happens when your adrenal glands can't make enough cortisol because certain infections or diseases have damaged them.

Secondary adrenal insufficiency, a condition that causes low cortisol levels. It happens when your pituitary gland can't make enough ACTH to tell your adrenal glands to make cortisol. Over time, your adrenal glands may shrink and stop working.

Cortisol testing is also used to monitor treatment for these conditions.

WHY DO I NEED A CORTISOL TEST?

You may need a cortisol test if you have symptoms of a condition that affects cortisol levels.

Symptoms of Cushing's syndrome (too much cortisol) may include:

- Weight gain

- Thin arms and legs

- Round face

- Increased fat around the base of your neck or between your shoulder blades

- Easy bruising

- Wide purple streaks on your stomach, breasts, hips, and under your arms

- Muscle weakness

Weight gain
Thin arms and legs
Round face
Increased fat around the base of your neck or between your shoulder blades
Easy bruising
Wide purple streaks on your stomach, breasts, hips, and under your arms
Muscle weakness
Common symptoms of Addison disease and adrenal insufficiency (too little cortisol) may include:

Long-lasting fatigue
Muscle weakness
Loss of appetite
Weight loss
Abdominal (belly) pain
Long-lasting fatigue
Muscle weakness
Loss of appetite
Weight loss
Abdominal (belly) pain

WHAT HAPPENS DURING A CORTISOL TEST?

Most of your cortisol is in your blood, so cortisol tests often use a sample of blood drawn at a lab. But some cortisol passes into your urine and saliva. Measuring cortisol in these fluids is also an accurate way to estimate how much cortisol is in your blood.

Because cortisol levels vary during the day, your provider may ask you to collect a few saliva samples at home at different times. This is a convenient way to gather information about when your cortisol levels rise and fall. To learn more about your cortisol levels, your health care provider may order more than one type of test.

For a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Blood samples are usually taken twice during the day--once in the morning when cortisol levels are normally at their highest, and again around 4 p.m., when levels are normally much lower.

For a cortisol urine test, your provider may ask you to collect all your urine during a 24-hour period. This is called a "24-hour urine sample test." For this test, you'll be given a special container and instructions for how to collect and store your urine sample. Your provider will tell you what time to start. The test usually includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

Store the urine container in a refrigerator or in a cooler with ice during the collection period.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

In certain cases, a urine test for cortisol may be done on one sample of urine collected in the morning.

A cortisol saliva test may be done at your provider's office or at home with a kit you use to collect a saliva sample. Your sample may be collected with a swab, or you may spit into a tube.

If you're collecting your own sample, your provider will tell you what time to collect it. You may need to collect more than one sample. Many kits use a swab. Be careful to follow the instructions carefully. They usually include these general steps:

Do not eat, drink, brush, or floss your teeth for 30 minutes before the test.

Wash and dry your hands.

Open the tube that holds the swab and let the swab fall into your mouth without touching it with your hands.

Roll the swab in your mouth or hold it under your tongue for about 2 minutes until it is soaked with saliva.

Spit the swab back into the tube without touching it and close the tube.

Label the tube with the time you collected the sample.

Take your saliva sample to your doctor's office or the lab as instructed.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

The preparations for a cortisol test will depend on the type of test you are having. Be sure to follow all the instructions that your provider gives you.

Stress and exercise can raise your cortisol levels, so you may need to rest before your test. For a blood test, you'll need two appointments at different times on the same day. Before a saliva test, you may need to stop using certain medicines. Let your provider know about all the medicines you use, including skin creams. But don't stop using any medicines without talking with your provider first.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There are no known risks to a urine or saliva test.

WHAT DO THE RESULTS MEAN?

A cortisol test alone can't diagnose the cause of abnormal cortisol levels. If your cortisol level isn't normal, you will usually have more tests to find out what is causing the problem.

High levels of cortisol may be a sign that you have Cushing's syndrome. It may be caused by:

Taking high doses of certain steroid medicines for a long time to treat conditions, such as asthma, rheumatoid arthritis, or lupus.

Tumors that make extra ACTH, which tells your adrenal glands to make cortisol. These tumors may be in your pituitary gland or other parts of your body, including your lungs.

Tumors in your adrenal glands that make extra cortisol.

Low levels of cortisol may mean you have Addison disease or secondary adrenal insufficiency:

Common causes of Addison disease include damage to your adrenal glands from conditions, such as:

Autoimmune diseases

Certain infections, including tuberculosis (TB) and HIV

Autoimmune diseases

Certain infections, including tuberculosis (TB) and HIV

Common causes of secondary adrenal insufficiency include:

Autoimmune diseases

Problems with your pituitary gland

Traumatic brain injury

Surgical removal of your pituitary gland to treat other conditions

Autoimmune diseases

Problems with your pituitary gland

Traumatic brain injury

Surgical removal of your pituitary gland to treat other conditions

If your cortisol results aren't normal, it doesn't always mean you have a medical condition that needs treatment. The most common cause of abnormal cortisol levels is taking steroid medicines for a long time or suddenly stopping them.

Cortisol levels can also be affected by:

Stress

Pregnancy

Exercise

Hot and cold temperatures

Certain medicines, such as birth control pills

To learn what your test results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CORTISOL TEST?

High cortisol levels may come and go in some people. This is sometimes called pseudo-Cushing's syndrome, which means false Cushing's syndrome. It's also called "non-neoplastic hypercortisolism."

It may be caused by depression, anxiety, alcohol use disorder, poorly controlled diabetes, obesity, and other health problems. Tests to find the cause of high cortisol levels may help diagnose non-neoplastic hypercortisolism.

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RELATED HEALTH TOPICS

Addison Disease
Adrenal Gland Disorders
Cushing's Syndrome
Hormones
Stress

RELATED MEDICAL TESTS

Adrenocorticotrophic Hormone (ACTH)
Aldosterone Test
How to Cope with Medical Test Anxiety
How to Understand Your Lab Results
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Creatine Kinase

<https://medlineplus.gov/lab-tests/creatine-kinase/>

CREATINE KINASE

WHAT IS A CREATINE KINASE (CK) TEST?

This test measures the amount of creatine kinase (CK) in a sample of your blood. CK is also called creatine phosphokinase (CPK).

CK is an enzyme. An enzyme is a protein that speeds up certain chemical reactions in your body. CK helps make energy. Most of the CK in your body is in your skeletal muscles. These are the muscles you use to move. You also have CK in your heart muscle and small amounts in your brain.

Normally, a small amount of CK gets into your blood from the usual wear and tear on your muscles. But if your muscles, heart, or brain tissues are damaged, larger amounts of CK leak out of your cells into your bloodstream.

A CK test is mainly used to help diagnose and monitor injuries and diseases that damage skeletal muscles and cause high levels of CK in your blood. But it may also be used for conditions that damage the heart muscle and the brain.

Other names: CK, total CK, creatine phosphokinase, CPK

WHAT IS IT USED FOR?

A CK test is most often used to help diagnose and monitor:

Muscle injuries, including injuries from accidents, serious burns, or extreme exercise.

Muscle diseases, including:

Muscular dystrophy, a group of inherited muscle diseases that weaken muscles over time.

Rhabdomyolysis, or "rhabdo," a rapid breakdown of muscle tissue which releases proteins and electrolytes into the blood. This uncommon condition can damage the heart and lead to sudden kidney failure. Rhabdo has many causes, including serious injuries, working in very hot places, overusing muscles, and certain medicines and medical conditions that affect the muscles.

Myositis, a group of rare diseases that involve long-term muscle inflammation (swelling), weakness, and sometimes pain.

Muscular dystrophy, a group of inherited muscle diseases that weaken muscles over time.

Rhabdomyolysis, or "rhabdo," a rapid breakdown of muscle tissue which releases proteins and electrolytes into the blood. This uncommon condition can damage the heart and lead to sudden kidney failure. Rhabdo has many causes, including serious injuries, working in very hot places, overusing muscles, and certain medicines and medical conditions that affect the muscles.

Myositis, a group of rare diseases that involve long-term muscle inflammation (swelling), weakness, and sometimes pain.

In certain cases, a CK test may be used to help diagnose a heart attack. Another blood test, called a troponin test is used more often. That's because troponin testing is better at finding damage to the heart muscle from a heart attack.

If you've had a stroke, a CK test may be done to find out how severe it was. The test may also help predict the chance that you may have another stroke.

WHY DO I NEED A CK TEST?

You may need a CK test if you have symptoms of a muscular disorder. These include:

Muscle pain and/or cramps

Muscle weakness

Balance problems and/or falling a lot

Numbness or tingling
Dark colored urine (pee)
Swollen legs or feet

You may also need this test if you:

Have had an injury that crushed or tore your muscles

Have had serious burns

Take certain medicines that may cause muscle damage, such as statins to lower blood cholesterol

Have had a stroke or traumatic brain injury (TBI)

WHAT HAPPENS DURING A CK TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a CK test. To help make sure your results are accurate, your provider may ask you to avoid intense exercise and drinking alcohol for a few days before your test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Normal CK levels for you will depend on:

Your age, sex, and race

How much muscle you have

How physically active you are

If your CK level is higher than normal, it usually means you have some type of muscle damage. CK levels may not reach their highest until up to two days after certain injuries. So, you may have more than one CK test to see if your levels go up or down. CK levels that stay high or increase may mean that muscle damage is continuing to happen.

But a CK test result can't show where the damage is or what's causing it. Your provider will consider your symptoms and medical history to understand what your test results mean.

If the source of high or increasing CK levels isn't clear, you may need a more specific type of CK test to find out if the CK is coming from your muscles, heart, or brain. This more specific test is called a CK isoenzymes test.

A CK isoenzymes test measures three different types of CK:

High levels of CK-MM enzymes are most likely from damage to your skeletal muscles. The damage may be from disease, sudden injury, certain medicines, or recent intense exercise.

High levels of CK-MB enzymes are most likely from damage to your heart muscle. The damage is often caused by heart attack or inflammation in your heart muscle (myocarditis). But other conditions can also cause high levels of CK-MB, such as:

A blood clot in the lung

Heart surgery

Congestive heart failure

Hypothyroidism

Kidney failure

A blood clot in the lung
Heart surgery
Congestive heart failure
Hypothyroidism
Kidney failure

Higher than normal CK-BB enzymes may mean you had a stroke or brain injury. But conditions that affect smooth muscles can also increase CK-BB levels. Smooth muscles are muscles you can't control, for example, the muscles in your digestive system.

To diagnose the condition causing high CK levels, you may need other types of tests, too. The tests you have will depend on which types of CK enzymes are high as well as your symptoms and medical history.

If you have questions about any of your test results or diagnosis, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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RELATED HEALTH TOPICS

Heart Attack

Muscle Disorders

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RELATED MEDICAL TESTS

BUN (Blood Urea Nitrogen)

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Creatinine Test

<https://medlineplus.gov/lab-tests/creatinine-test/>

CREATININE TEST

WHAT IS A CREATININE TEST?

This test measures creatinine levels in a sample of your blood and/or urine (pee). Creatinine is a normal waste product in your body. It's made when you use your muscles and some of the muscle tissue breaks down.

Normally, your kidneys filter creatinine from your blood and remove it from your body in your urine. If there is a problem with your kidneys, creatinine can build up in your blood and less may be released in urine. If blood and/or urine creatinine levels are not normal, it may be a sign of kidney disease.

Creatinine in blood may be measured by itself or as part of a group of tests called a comprehensive metabolic panel (CMP) or a basic metabolic panel (BMP). Your health care provider may order these tests as part of a routine checkup.

Other names: blood creatinine, serum creatinine, urine creatinine, kidney function, renal function

WHAT IS IT USED FOR?

A creatinine test is used to help:

Check the health of your kidneys

Diagnose kidney disease

Monitor known kidney problems and see how well treatment is working

Check for side effects from medicines that may affect your kidneys

Creatinine testing alone isn't the best way to check how well your kidneys are working. That's because people make different amounts creatinine depending on how much muscle they have, the foods they eat, their age, and how active they are. So, the results from creatinine testing are often used in calculations or compared with other substances to help get more information:

Creatinine levels in blood are often used to calculate how fast your kidneys filter waste out of your blood. This is called an estimated glomerular filtration rate (eGFR). The calculation includes information about your age, weight, height, and sex. An eGFR is a more accurate way to measure kidney health than creatinine levels in blood or urine alone. An eGFR can also help show how serious kidney disease may be.

Creatinine levels in blood and urine maybe be compared with each other. This is called a creatinine clearance calculation. Like an eGFR, it estimates how fast your kidneys filter waste. But an eGFR is more accurate. A creatinine clearance may still be useful to help identify the cause of high or low levels of blood creatinine in people who have very high muscle mass or a loss of muscle mass from age, illness, or the loss of an arm and/or leg.

Creatinine levels in urine may be used to calculate a urine albumin to creatinine ratio (UACR), which is sometimes called a microalbumin creatinine ratio. Albumin is the main protein found in blood. Normally your kidneys filter out just a trace of albumin, or none at all. If larger amounts of albumin pass into your urine, it may be a sign of kidney damage. A UACR compares the amounts of creatinine and albumin in your urine to get a more accurate measurement of how much albumin is in your urine.

Creatinine blood levels measured as part of a CMP or a BMP may be compared with the level of BUN (blood urea nitrogen) that's measured in the same test. This can help find out the cause of a kidney problem.

WHY DO I NEED A CREATININE TEST?

Your provider may order a creatinine test:

To check your kidney health as part of a routine checkup.

If you have symptoms of kidney disease. Symptoms may include:

Swelling in the hands and feet or puffy eyelids

Dry skin, itching, or numbness

Fatigue

Increased or decreased need to urinate (pee)

Urine that is bloody or foamy

Loss of appetite and weight loss

Muscle cramps

Nausea and vomiting

Shortness of breath

Sleep problems

Trouble thinking clearly

Swelling in the hands and feet or puffy eyelids

Dry skin, itching, or numbness

Fatigue

Increased or decreased need to urinate (pee)

Urine that is bloody or foamy

Loss of appetite and weight loss

Muscle cramps

Nausea and vomiting

Shortness of breath

Sleep problems

Trouble thinking clearly

If you have a high risk for developing kidney disease, even if you don't have symptoms now. Chronic (long-term) kidney disease (CKD) often doesn't have symptoms in the early stages. Your risk for kidney disease is increased if you:

Have diabetes

Have high blood pressure

Have a family health history of kidney disease, diabetes, or high blood pressure

Have heart disease

Are over 50 years old

Smoke

Have obesity

Have diabetes

Have high blood pressure

Have a family health history of kidney disease, diabetes, or high blood pressure

Have heart disease

Are over 50 years old

Smoke

Have obesity

If you have kidney problems or possible kidney problems because you have:

Had an abnormal result on a kidney test in the past

Been diagnosed with kidney disease

Taken certain medicines that could affect your kidneys

Had a kidney transplant

Had an abnormal result on a kidney test in the past

Been diagnosed with kidney disease

Taken certain medicines that could affect your kidneys

Had a kidney transplant

WHAT HAPPENS DURING A CREATININE TEST?

For a creatinine blood test:

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a creatinine urine test:

You may need to provide all the urine you pass over 24 hours. That's because creatinine levels vary through the day. But your provider may use a urine sample from a shorter period of time.

For a 24-hour urine sample, you will be given a special container to collect your urine over a full day and instructions on how to collect and store your sample. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or in a cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know how to prepare for your test. You may be told to not eat meat for 24 hours before your test. That's because meat can temporarily increase creatinine levels.

If creatinine is being measured as part of a CMP or a BMP test, you may need to fast (not eat or drink) for up to 12 hours before your test.

Certain medicines and supplements can affect your test results. So be sure to tell your provider everything you're taking. But don't stop taking any medicine unless your provider tells you to stop.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no risk to having a urine test.

WHAT DO THE RESULTS MEAN?

Creatinine levels that are considered normal for you will depend on how much muscle you have, what you eat, your age, and how active you are. If you're healthy, your levels are usually fairly stable over time.

But a normal creatinine test result doesn't always mean that your kidneys are healthy. Sometimes, blood creatinine levels remain in a normal range during the early stages of kidney disease. They rise as the condition of your kidneys becomes more serious. If your provider suspects you have a kidney condition, you will likely have other kidney tests even if your creatinine results seem normal.

If your results are abnormal, a single high creatinine test can't diagnose a specific condition. You will likely need to be retested and/or have other tests, too.

In general, if your blood creatinine level is:

High for you, it may be a sign of:

Kidney disease or injury, including infection, poor blood flow to the kidneys, a blockage in the urinary system, or kidney failure

A condition that affects your kidneys, such as heart failure or diabetes

High blood creatinine levels don't always mean you have a kidney problem. They may be caused by dehydration, muscle disorders and injuries, muscular dystrophy, intense exercise, or a diet high in meat. Certain health problems in pregnancy can also cause increases in creatinine.

Kidney disease or injury, including infection, poor blood flow to the kidneys, a blockage in the urinary system, or kidney failure

A condition that affects your kidneys, such as heart failure or diabetes

High blood creatinine levels don't always mean you have a kidney problem. They may be caused by dehydration, muscle disorders and injuries, muscular dystrophy, intense exercise, or a diet high in meat. Certain health problems in pregnancy can also cause increases in creatinine.

Low for you, it may be a sign of malnutrition or a condition that causes your muscles to get smaller, such as a long illness, a nerve disorder, or muscle loss from aging. Serious liver disease can also lead to low levels. Low levels of blood creatinine aren't common.

If your results from creatinine blood or urine tests were used to calculate an eGFR, creatinine clearance, and/or a urine albumin to creatinine ratio, ask your provider to explain what these measurements say about your kidney health.

If you have questions about any your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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RELATED HEALTH TOPICS

- Chronic Kidney Disease
- Kidney Diseases
- Kidney Failure
- Kidney Tests

RELATED MEDICAL TESTS

- Basic Metabolic Panel (BMP)
- BUN (Blood Urea Nitrogen)
- Comprehensive Metabolic Panel (CMP)
- Fasting for a Blood Test
- Glomerular Filtration Rate (GFR) Test
- How to Cope with Medical Test Anxiety
- How to Understand Your Lab Results
- Microalbumin Creatinine Ratio

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Crystals in Urine

<https://medlineplus.gov/lab-tests/crystals-in-urine/>

CRYSTALS IN URINE

WHAT IS A CRYSTALS IN URINE TEST?

A crystals in urine test checks a sample of your urine (pee) under a microscope to look for crystals. The test shows what the crystals are made of, how large they are, and how many are in your urine.

Urine crystals are formed in your kidneys. Your kidneys are two organs located below your rib cage, one on each side of your spine. They make urine when they filter your blood to remove waste products and extra water from your body.

Urine contains many dissolved substances, including extra minerals and acids your body doesn't need. Certain minerals in your urine may clump together with other substances and form solid crystals. This usually happens if your urine contains high levels of those minerals or has too much or too little acid.

Having a few small crystals in your urine is normal, and they usually don't cause problems. But large amounts of certain types of crystals may be a sign of a health condition that needs treatment.

Some types of crystals may stick together and make kidney stones. These are hard, pebble-like pieces of material that form in the kidneys. Small kidney stones may pass out of your body in your urine with little or no pain.

But without treatment, a large kidney stone may get stuck and block your urine flow. This can cause pain, bleeding, and serious damage to your kidneys and other parts of your urinary system. A crystals in urine test can help find the cause of kidney stones so you can get the right treatment to help prevent them.

Other names: urinalysis (crystals) microscopic urine analysis, microscopic examination of urine

WHAT IS IT USED FOR?

A crystals in urine test may be used to help diagnose or monitor several types of medical conditions linked to crystals in urine. These conditions include:

Kidney stones and other kidney diseases

Urinary tract infections (UTIs)

Certain problems with metabolism (the process your body uses to make energy from the food you eat)

Crystals in urine testing is often part of a urinalysis, a test that measures different substances in your urine. It may be used to check your general health or to learn more about a possible health problem. A urinalysis may include a visual check of your urine sample, tests for certain chemicals, and an examination under a microscope to look for certain types of cells and crystals.

WHY DO I NEED A CRYSTALS IN URINE TEST?

You may have a crystals in urine test as part of a urinalysis to check your general health. You may also need this test if:

You have symptoms of a kidney stone or urinary tract infection, which include:

Sharp pains in your lower abdomen (belly), side, groin, or back

Blood in your urine

Frequent need to urinate (pee)

Not being able to urinate at all or only urinating a little bit

Pain when urinating

Cloudy or bad-smelling urine

Nausea and vomiting

Fever and chills

Sharp pains in your lower abdomen (belly), side, groin, or back

Blood in your urine

Frequent need to urinate (pee)
Not being able to urinate at all or only urinating a little bit
Pain when urinating
Cloudy or bad-smelling urine
Nausea and vomiting
Fever and chills
You have a condition that can cause kidney stones.
You have had a kidney stone, and your health care provider needs to know your risk for developing another one.

WHAT HAPPENS DURING A CRYSTALS IN URINE TEST?

You will need to give a urine sample for the test. Your provider may ask you to collect your sample using the clean catch method or a 24-hour urine test.

Clean Catch

A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

24-hour Urine Test

Your provider may ask you to collect all your urine during a 24-hour period. This test may provide more accurate results because the amount of crystals and other substances in your urine can vary throughout the day.

If you need to have a 24-hour urine sample test, you'll be given a special container and instructions for how to collect and store your urine sample. Your provider will tell you what time to start. The test usually includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

Store the urine container in a refrigerator or in a cooler with ice.

24 hours after starting the test, try to urinate if you can. This is your last urine collection.

Return the container with your urine to your provider's office or the laboratory as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a crystals in urine test. To make sure your test results are accurate, be sure to carefully follow all the instructions for providing your urine sample.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a crystals in urine test.

WHAT DO THE RESULTS MEAN?

The test results will usually show if you have any crystals in your urine, the amount, and what they are made of. It's best to ask your provider to explain what your results mean. That's because different types of crystals are linked to different medical conditions, and your provider may need to order other tests to make a diagnosis.

Some common types of crystals that may form stones in some people include:

Calcium oxalate and calcium phosphate. These crystals are the most common cause of kidney stones.

Uric acid and urates. Large amounts of these crystals may form when metabolic problems, such as gout and type 2 diabetes, make your urine too acidic.

Struvite. Repeated urinary tract infections may cause large amounts of struvite crystals.

Having crystals in your urine doesn't always mean that you have a medical condition that needs treatment. Large amounts of some types of crystals may be caused by:

Certain medicines, such as calcium-based antacids, some types of antibiotics, diuretics (water pills), and antiviral medicines (medicines that treat viruses)

Your diet, especially eating too much animal protein, such as fish, shellfish, and organ meats

Not drinking enough fluids

If you have questions about your test results and how you can prevent crystals in urine, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

CSF Immunoglobulin G (IgG) Index

<https://medlineplus.gov/lab-tests/csf-immunoglobulin-g-igg-index/>

CSF IMMUNOGLOBULIN G (IGG) INDEX

WHAT IS A CSF IGG INDEX?

CSF stands for cerebrospinal fluid. IgG stands for immunoglobulin G. It is a type of antibody. Antibodies are proteins that your immune system makes to fight germs. A CSF IgG index measures the levels of IgG in your cerebrospinal fluid.

CSF is a clear, colorless fluid that flows in and around your brain and spinal cord. Your brain and spinal cord make up your central nervous system. Cerebrospinal fluid cushions your central nervous system from impact or injury. It also removes waste products from the brain and helps your central nervous system work right.

Normally you have a tiny amount of IgG in your cerebrospinal fluid. High levels of IgG may mean you have an infection or an inflammatory or autoimmune disease that involves your central nervous system. An autoimmune disorder causes your immune system to attack your own healthy cells by mistake, which can lead to serious health problems.

Other names: cerebrospinal fluid IgG level, cerebrospinal fluid IgG measurement, CSF IgG level, IgG (Immunoglobulin G) spinal fluid, IgG synthesis rate

WHAT IS IT USED FOR?

A CSF IgG index is used to check for diseases of the central nervous system. It is often used to help diagnose multiple sclerosis (MS). MS is a chronic (long-lasting) autoimmune disorder that attacks the outside covering of your nerves. Most people with MS have higher than normal levels of IgG. But a CSF IgG test alone can't diagnose MS.

WHY DO I NEED A CSF IGG INDEX?

You may need a CSF IgG index if you have symptoms of a central nervous system disorder, especially multiple sclerosis (MS). The symptoms of MS often vary. They may come and go or steadily get worse. They include:

Blurred or double vision

Tingling, numbness, or pain in the arms, legs, body, or face

Painful muscle spasms

Weak muscles, often in the hands and legs

Dizziness

Problems with balance and walking

Bladder control problems

Fatigue

Problems with thinking or memory

WHAT HAPPENS DURING A CSF IGG INDEX?

To get a sample of cerebrospinal fluid, a provider will do a procedure called a spinal tap, also known as a lumbar puncture. A spinal tap is usually done in a hospital. During the procedure:

You will lie on your side or sit on an exam table.

A provider will clean your back and inject an anesthetic into your skin, so you won't feel pain during the procedure. Your provider may put a numbing cream on your back before this injection.

When the area on your back is completely numb, your provider will insert a thin, hollow needle between two vertebrae in your lower spine. Vertebrae are the small backbones that make up your spine.

Your provider will withdraw a small amount of cerebrospinal fluid for testing. This will take about five minutes.

You'll need to stay very still while the fluid is being withdrawn.

Your provider may ask you to lie on your back for an hour or two after the procedure. This may prevent you from getting a headache afterward.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a CSF IgG index, but you may be asked to empty your bladder (pee) and bowels (poop) before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a spinal tap. You may feel a little pinch or pressure when the needle is inserted. After the test, you may feel some pain or tenderness in your back at the site where the needle was inserted.

You may also have some bleeding at the site or get a headache. The headache may last for several hours or up to a week or more. Your provider may suggest treatment to help relieve the pain.

WHAT DO THE RESULTS MEAN?

A CSF IgG index that's higher than normal may be a sign of an infection or an inflammatory or autoimmune condition that affects your central nervous system, such as:

Multiple sclerosis

Transverse myelitis (TM)

Syphilis that infects your nervous system (neurosyphilis)

Connective tissue disorders

Guillain-Barre syndrome

Encephalitis caused by herpes simplex

You may need more tests before your provider can diagnose your condition.

A CSF IgG index that's lower than normal may be a sign that your body is unable to make enough IgG antibodies, and you are likely to get sick with infections. Experts don't know what causes this problem. Normally, a test of IgG levels in blood^Â is used to check whether you may lack IgG antibodies.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A CSF IGG INDEX?

The CSF IgG index is often used to help diagnose multiple sclerosis (MS), but there is no specific test for MS. If your provider thinks you have MS, you will probably have other tests to help diagnose or rule out the disease.

While there is no cure for MS, there are many treatments available that can relieve symptoms and help keep it from getting worse.

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Cytomegalovirus (CMV) Tests

<https://medlineplus.gov/lab-tests/cytomegalovirus-cmv-tests/>

CYTOMEGALOVIRUS (CMV) TESTS

WHAT ARE CYTOMEGALOVIRUS (CMV) TESTS?

Cytomegalovirus (CMV) tests check for signs of CMV in your blood, sputum, or other body fluids. CMV is a virus in the herpes family. Other types of herpes viruses include the viruses that cause chickenpox and mononucleosis (mono).

CMV is a common virus that infects people of all ages. Most people have been exposed to CMV during childhood. Once you get a CMV infection, the virus stays in your body for the rest of your life. Most of the time, the virus stays dormant (inactive). However, it can become active again (reactivated) in certain situations, such as when you are under a lot of stress or if you have an immune system problem.

If the CMV virus is active, it can be spread through body fluids such as:

Saliva

Urine

Blood

Tears

Semen

Breast milk

Most healthy people don't have any signs or symptoms when they get CMV. They usually aren't even aware that they have the infection. But sometimes a CMV infection may cause mild, flu-like symptoms in healthy people.

However, certain people are at higher risk of health problems from a CMV infection:

People with weakened immune systems due to conditions such as HIV or cancer. For them, CMV can cause vision problems and other serious health problems.

If you are pregnant and have an active CMV infection you can pass the virus to your fetus. In severe cases, this could lead to the loss of the baby through a miscarriage or stillbirth.

Newborns who got a CMV infection before they were born. This type of CMV is known as congenital CMV. Most babies born with CMV do not have health problems from the virus. Of the babies that do, the most common long-term health problem is hearing loss. Congenital CMV can also cause vision problems, intellectual disabilities, and other serious disorders.

Very low birth weight and premature infants. They can get a CMV infection during childbirth or from breastfeeding. They are less able than healthy babies to fight the infection.

CMV testing can help those at risk for complications get the treatment they need. While there is no cure for CMV, antiviral medicines and other treatments may reduce your symptoms and improve outcomes.

Other names: CMV IgG and IgM, cytomegalovirus antibody

WHAT ARE THEY USED FOR?

CMV tests help diagnose a current, reactivated, or past CMV infection in people at risk for health complications. Risk groups include:

People with weakened immune systems due to certain infections or diseases

People who have recently received an organ transplant

You are pregnant and have symptoms of a CMV infection.

Newborns with signs or symptoms of infection

WHY DO I NEED A CMV TEST?

You may need testing if you have a weakened immune system or are pregnant and have the following symptoms:

Swollen lymph nodes ("swollen glands")

Sore throat

Fever

Fatigue

Weakness

Muscle aches

Headache

Your baby may need this test if they have the following symptoms:

Jaundice, a condition that causes your skin and eyes to turn yellow

Low birth weight

Small head

Hearing and/or vision problems

Seizures

Rash

WHAT HAPPENS DURING A CMV TEST?

There are several types of CMV tests, including:

Blood tests

This is the most common way to test adults for CMV.

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Cerebrospinal fluid (CSF) analysis (spinal tap)

To get a sample of cerebrospinal fluid, a provider will do a procedure called a spinal tap, also known as a lumbar puncture. A spinal tap is usually done in a hospital. During the procedure:

You will lie on your side or sit on an exam table.

A provider will clean your back and inject an anesthetic into your skin, so you won't feel pain during the procedure. Your provider may put a numbing cream on your back before this injection.

When the area on your back is completely numb, your provider will insert a thin, hollow needle between two vertebrae in your lower spine. Vertebrae are the small backbones that make up your spine.

Your provider will withdraw a small amount of cerebrospinal fluid for testing. This will take about five minutes.

You'll need to stay very still while the fluid is being withdrawn.

Your provider may ask you to lie on your back for an hour or two after the procedure. This may prevent you from getting a headache afterward.

Sputum test

Sputum is a thick mucus made in your lungs.

You may be asked to rinse your mouth out with water before the sample is taken.

Your provider will ask you to breathe deeply and then cough deeply into a special cup.

A health care provider will ask you to breathe deeply and then cough deeply into a special cup.

Your provider may tap you on the chest to help loosen sputum from your lungs.

If you have trouble coughing up enough sputum, your provider may ask you to breathe in a salty mist to help you cough more deeply.

If you still can't cough up enough sputum, your provider may perform a procedure called a bronchoscopy. In this procedure, you'll first get a medicine to help you relax, and then a numbing medicine so you won't feel any pain.

Then a camera attached to a thin, lighted tube will be put through your mouth or nose and into your airway. This helps your provider see the inside of your lungs and airway.

Your provider will collect a sample from your airway using a small brush or suction

Biopsy

During the test, your provider will remove a small sample of tissue for testing. There are many ways to do a biopsy. The type of biopsy you have depends on the type of cancer you have and where the tumor is located. In general, a biopsy may be done using:

A hollow needle inserted through your skin. Imaging tests, such as ultrasound, may be used to guide the needle:

A fine needle aspiration biopsy uses a very thin needle to remove a sample of cells and/or fluid.

A core needle biopsy uses a larger needle to remove a sample.

A fine needle aspiration biopsy uses a very thin needle to remove a sample of cells and/or fluid.

A core needle biopsy uses a larger needle to remove a sample.

Amniocentesis

This procedure is done on pregnant women to see if an unborn baby has been infected with CMV.

You'll lie on your back on an exam table.

Your provider will apply a gel to your belly.

Your provider will move an ultrasound wand-like device, called a transducer, on your belly. Ultrasound uses sound waves to show the position of your baby and placenta so your provider can see where to take a sample of amniotic fluid.

Your provider will clean your belly.

Your provider will insert a thin needle into your belly, withdraw a small amount of amniotic fluid, and then remove the needle.

When the sample is removed, your provider will check your baby's heartbeat with the ultrasound.

The sample is sent to a lab for testing. Results may take from a few days to a few weeks.

Newborns are usually given a saliva or urine test.

During an infant saliva test:

A saliva specimen is collected within three weeks of your baby's birth and at least an hour after breastfeeding.

Your provider will insert a sterile swab into your baby's cheek and swirl for several seconds.

The swab will be placed in a special solution for testing.

During an infant urine test:

You will be given a special plastic bag that will fit over your baby's genital area.

You will place a diaper over the bag.

After your baby has urinated, you will remove the bag from the diaper and empty the urine into a container given to you by your provider.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a blood or sputum test. There are also no preparations needed for an infant saliva or urine test.

You may need to empty your bladder before a spinal tap.

You may be asked to fast (not eat or drink) for several hours before a biopsy.

For an amniocentesis, you may be asked to not urinate (pee) before the test to make sure you have a full bladder. This helps move the uterus into a better position for the test. In later pregnancy, you need to have an empty bladder for the test.

Your provider or your child's provider will let you know if you need to make any other preparations.

ARE THERE ANY RISKS TO THE TEST?

There are no known risks to having a sputum test, infant saliva test, or infant urine test.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is very little risk to having a spinal tap. You may feel a little pinch or pressure when the needle is inserted. After the test, you feel some pain or tenderness in your back, or have some bleeding where the needle was inserted. You may also get a headache. The headache may last for several hours or up to a week or more, but your provider may suggest treatment to relieve the pain.

If you had a biopsy, you may have a little bruising, bleeding, or soreness at the biopsy site. This usually goes away after a few days.

If you had amniocentesis, you may have some stinging, pressure, and/or cramping during the test. Afterward, you may have mild discomfort or vaginal bleeding, or a small amount of amniotic fluid may leak from your vagina. The risk of

serious complications is extremely low. Your provider will let you know if there are any symptoms that you should tell them about. These may include ongoing vaginal bleeding, amniotic fluid leakage, severe cramping, or a fever.

WHAT DO THE RESULTS MEAN?

The test result can show whether you have ever had a CMV infection. But the tests can't show if it's a current, past, or reactivated infection.

If you have symptoms and/or risk factors such as an immune system disorder, your provider may do additional tests to help make a diagnosis and create a treatment plan.

If you are pregnant and your amniocentesis shows your baby has CMV, your child's provider may test and treat your baby soon after birth to help prevent complications.

If you have questions about your results, talk to your provider or your child's provider.

Learn more about laboratory tests, references ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT CMV TESTING?

CMV tests are included as part of a TORCH panel, a group of blood tests used to screen newborns and sometimes during pregnancy for the following infections:

Toxoplasmosis

Rubella

Cytomegalovirus

Herpes simplex virus

These infections can cause birth defects if a mother gets any of these infections during pregnancy. Early diagnosis and treatment may help prevent a baby from getting a serious health problem.

You can't eliminate your risk for CMV, but you can lower your chance of getting infected or spreading the infection by:

Not sharing food or dishes with a child or putting their pacifier in your mouth.

Washing your hands well after changing a diaper.

Wearing gloves and washing your hands well if you are caring for someone who is sick and have contact with their body fluids.

Using latex condoms correctly every time you have sex. If you or your partner is allergic to latex, you can use polyurethane condoms.

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D-Dimer Test

<https://medlineplus.gov/lab-tests/d-dimer-test/>

D-DIMER TEST

WHAT IS A D-DIMER TEST?

A D-dimer test looks for D-dimer in blood. D-dimer is a protein fragment (small piece) that's made when a blood clot dissolves in your body. D-dimer isn't usually found in your blood unless your body is making or breaking up blood clots. A blood clot is a mass of blood that forms when platelets, proteins, and cells in the blood stick together. When you get hurt, your body forms a blood clot to stop the bleeding. Blood clotting is an important process that prevents you from losing too much blood when you are injured.

Normally, your body will dissolve the clot once your injury has healed. With a blood clotting disorder, clots can form when you don't have an injury, or they don't dissolve when they should. These disorders can be serious and even life-threatening.

Other names: fragment D-dimer, fibrin degradation fragment

WHAT IS IT USED FOR?

A D-dimer test is used to check if you may have a blood clot. The test helps to find out if you need more tests to check for blood clotting disorders. A D-dimer test may also be used to monitor how well treatments for a blood clotting disorder like disseminated intravascular coagulation (DIC) are working.

Blood clotting disorders that the D-dimer test may be used for include:

Deep vein thrombosis (DVT), a blood clot that forms in a vein deep in your body. These clots usually affect the lower legs, but they can also happen in other parts of your body.

Pulmonary embolism (PE), a blockage in a lung artery. It usually happens when a blood clot in another part of your body breaks loose and travels to your lungs. DVT clots are a common cause of PE.

Disseminated intravascular coagulation (DIC), a condition that causes too many blood clots to form. They can form throughout the body, causing organ damage and other serious health issues. DIC may be caused by inflammation, infection, or cancer.

Stroke, which can happen when blood flow to a part of the brain is blocked.

WHY DO I NEED A D-DIMER TEST?

You may need this test if you have symptoms of a blood clotting disorder, such as deep vein thrombosis (DVT) or a pulmonary embolism (PE).

Symptoms of DVT in your leg include:

Leg pain or tenderness

Leg swelling

Redness or darkening of the skin on the leg

Skin that feels warm

Symptoms of PE include:

Trouble breathing

Cough

Chest pain

Rapid heartbeat and breathing

Many people do not have any symptoms of PE.

Both DVT and PE require quick medical treatment. If you have DVT symptoms and are not in a health care setting, call your health care provider. If you have symptoms of PE, seek immediate medical attention.

WHAT HAPPENS DURING A D-DIMER TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a D-dimer test.

ARE THERE ANY RISKS TO A D-DIMER TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results show low or normal D-dimer levels in the blood, it means you probably don't have a clotting disorder. But if you are being treated with blood thinner medicines, they may cause a false negative D-dimer test. This means your test results show that you don't have a clotting disorder, but you really do.

If your results show higher than normal levels of D-dimer, it may be a sign of a clotting disorder. Additional tests will be needed to show where the blood clot is located or what type of clotting disorder you have.

High D-dimer levels are not always caused by clotting disorders. A few other possible reasons why you may have high D-dimer levels include pregnancy, heart disease, rheumatoid arthritis, and recent surgery. Being older or immobilized (unable to move) may also cause a high D-dimer level.

If your D-dimer results were not normal, your provider will probably order more tests to make a diagnosis. To understand the results of a D-dimer test, your provider will consider your symptoms, medical history, and the results of other tests.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A D-DIMER TEST?

If your D-dimer test results were not normal, your provider may order one or more imaging tests to find out if you have a clotting disorder. These include:

Doppler ultrasound, a test that uses sound waves to create images of your veins. This may help to find out if the blood in your veins is flowing as it should.

CT angiography. In this test, you are injected with a special dye that helps your blood vessels show up on a special type of x-ray machine. This may help to find blood clots or check if your blood vessels are damaged.

Ventilation-perfusion (V/Q) scan. These are two tests that look for certain lung problems. These tests may be done separately or together. The ventilation scan measures how air moves in and out of your lungs. The perfusion scan measures how blood flows in the lungs. They both use small amounts of radioactive tracer to help a scanning machine check how well your lungs are working.

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Dengue Fever Test

<https://medlineplus.gov/lab-tests/dengue-fever-test/>

DENGUE FEVER TEST

WHAT IS A DENGUE FEVER TEST?

Dengue fever is a viral infection spread to people through bites from infected mosquitos. You can't catch the virus from a person who is sick with dengue fever.

A dengue fever test checks a sample of your blood to see if you have the virus. The test is used if you have symptoms of dengue fever and have been in an area where mosquitos carry the virus.

The dengue virus is most common in parts of the world with tropical and subtropical climates, including:

South and Central America

The Caribbean, including Puerto Rico and the U.S. Virgin Islands

Asia

The Middle East

The Pacific Islands

Africa

Outbreaks of dengue fever are rare in the U.S. mainland. But small outbreaks have been reported in parts of Florida and Texas.

Most people who get the dengue virus won't become sick. People who do get sick may have mild to severe symptoms that usually begin four to ten days after a bite from an infected mosquito. The most common symptom is a sudden high fever. There is no specific medicine that can cure or treat dengue fever, but most people will recover after about a week. Some people who get sick with dengue will develop severe dengue disease. Severe dengue can cause internal bleeding and shock, which can be life-threatening.

Infants and pregnant women have a higher risk for developing severe dengue. The risk is also higher for people who have had dengue and then get it again. That's why it's so important to get a dengue fever test if you have symptoms after being in an area where the virus is common.

There are two main types of dengue fever blood tests:

Antibody tests check your blood sample for proteins, called antibodies, that your immune system makes to fight viruses and other germs. Your body needs time to make specific antibodies to fight dengue. So, these tests are most accurate when they are done four days or more after your symptoms begin.

Molecular tests look for genetic material from the dengue virus in your blood sample. A PCR test (polymerase chain reaction) is a type of molecular test that's used to check for dengue. These tests can be done during the first seven days after your symptoms begin. Tests done later than that may not have accurate results.

One type of PCR test can check for dengue along with two other viruses that mosquitos carry. These viruses, chikungunya and zika, are common in the same places as the dengue virus, and they cause similar symptoms. Your provider can order this test from your state or local public health department.

Other names: dengue virus antibody, dengue virus by PCR

WHAT IS IT USED FOR?

A dengue fever test is used to find out if you have the dengue virus. It is mostly used for people who have symptoms related to dengue fever and have recently traveled to an area where dengue infections are common.

WHY DO I NEED A DENGUE FEVER TEST?

You may need to be tested for the dengue virus if you have symptoms of dengue fever and have been in an area where the virus is common. Symptoms usually show up four to ten days after being bitten by an infected mosquito.

Symptoms of dengue fever include a sudden high fever (104°F or higher) with any of these other symptoms:

Nausea and vomiting
Rash
Eye pain, typically behind the eyes
Headache
Joint, bone or muscle pain

Symptoms of severe dengue may begin 24 to 48 hours after the fever has gone away. Get medical help right away if you or a family member has any of these warning signs of severe dengue:

Abdominal (belly) pain or tenderness
Vomiting (at least 3 times in 24 hours)
Bleeding from the nose or gums
Vomiting blood or blood in stool (poop)
Feeling tired, restlessness, or irritable

WHAT HAPPENS DURING A DENGUE FEVER TEST?

Your health care provider will ask about your symptoms and your recent travels. If an infection is suspected, you will get a blood test to check for the dengue virus.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a dengue fever test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If you had a molecular test (a PCR test):

A positive test result means that the test found genetic material from the virus in your blood sample. You most likely have a dengue infection.

A negative test result means that the dengue virus wasn't found in your blood sample. You probably don't have dengue. But it's possible that there wasn't enough virus in your blood sample for the test to find. That might happen if the test was done more than seven days after your symptoms began. If your molecular test result is negative, your provider will usually order an antibody test to confirm that you don't have dengue.

If you had an antibody test, your test results may be reported as positive or negative. Or they may be given as a measurement of the dengue antibodies found in your blood sample.

A positive test result means that antibodies for the dengue virus were found. You probably have a dengue infection, but you may need another test to confirm the results. That's because the antibody test may have found antibodies that fight other similar viruses that mosquitos carry. Your provider may order another type of blood test to find out exactly which type of viral infection you have.

A negative test result means that dengue antibodies weren't found in your blood sample. Dengue may not be causing your symptoms. But a negative result could mean that you were tested before your body had time to make enough antibodies to show on the test. So, your provider may order another antibody test to check again.

If you have dengue fever, your provider can tell you how to treat your symptoms. Always follow your provider's instructions. In general, care for dengue fever includes:

Getting plenty of rest.

Drinking lots of fluids.

Taking acetaminophen to control fever and relieve pain. Do not take aspirin, ibuprofen, or naproxen sodium, because they can increase the risk of bleeding.

Watching carefully for symptoms of severe dengue. Dengue fever can become severe within hours. Get medical help right away if you develop the symptoms of severe dengue.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A DENGUE FEVER TEST?

A vaccine is available for children ages 9 to 16 who:

Have already had dengue fever. The vaccine helps protect against future dengue infections or from getting severe dengue disease. Before getting the vaccine, a child must have a blood test to confirm a past dengue infection. That's because the vaccine can be harmful to a child who has never had dengue fever.

Live in areas where dengue is common. These areas include Puerto Rico, the U.S. Virgin Islands, the U.S. territories of American Samoa, the Federated States of Micronesia, the Republic of the Marshall Islands, and the Republic of Palau. The vaccine isn't approved for children traveling to these areas to visit.

If you travel to an area where dengue is common, you can reduce the risk of infection by preventing mosquito bites.

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Dental Exam

<https://medlineplus.gov/lab-tests/dental-exam/>

DENTAL EXAM

WHAT IS A DENTAL EXAM?

A dental exam is part of a checkup of your teeth and gums. When you see your dentist for a checkup, the visit usually includes three parts:

A dental cleaning to remove the buildup of plaque, a sticky film of bacteria that coats your teeth

X-rays of your teeth on certain visits, depending on the health of your teeth

An exam of your teeth and mouth by a dentist

Dental exams are important for protecting your oral health. Oral health includes your teeth, gums, and all the muscles and bones that let you smile, speak, and chew. Common oral health problems, such as include tooth decay (cavities) and gum disease (periodontal disease), can lead to pain, infection, and tooth loss.

Poor oral health can affect your general health, too. For example, losing teeth can cause problems with eating and nutrition, which can lead to other health problems. You can help prevent oral health problems with regular visits to the dentist and good toothbrushing and flossing habits at home.

At a dental checkup, you'll usually see both a dentist and a dental hygienist. A dentist is a doctor who has special training to care for teeth and gums. A dental hygienist is a health care professional trained to clean teeth and teach you how to take good care of your teeth and gums.

Dentists can treat people of all ages. There are also pediatric dentists who have received additional training on dental care for children.

Other names: dental checkup, oral exam, oral hygiene, dental hygiene, dental prophylaxis

WHAT IS IT USED FOR?

Dental exams are used to:

Clean teeth to help prevent tooth decay, gum disease, and other oral health problems

Find problems early when they're easier to treat

Help people learn the best ways to care for their oral health

WHY DO I NEED A DENTAL EXAM?

Ask your dentist or hygienist how often you should have exams. Most adults and children need to have a routine dental exam every six months. You may need more frequent dental exams if you:

Have gum disease

Get a lot of cavities

Have a family history of oral health problems

Have a health condition that may increase your risk for oral health problems, such as:

Diabetes

Heart disease

HIV

Conditions that affect your immune system

Diabetes

Heart disease

HIV

Conditions that affect your immune system

Take medicines that make your mouth dry

Smoke

Are pregnant, which increases your risk for gum disease

If you have swollen, dark red, and/or bleeding gums, tooth pain, or other discomfort in your mouth, don't wait for your routine dental exam. Contact your dentist as soon as possible.

Babies should have their first dental appointment within six months of getting their first tooth, or by their first birthday. Ask your baby's dentist how often your baby should have a dental exam. Most babies should be checked every six months.

WHAT HAPPENS DURING A DENTAL EXAM?

At a typical dental exam, a hygienist will clean your teeth. Then the dentist will examine your mouth. At certain visits, you may have x-rays of your teeth before the dentist examines you.

Dental x-rays can show hidden cavities, gum disease, bone loss in your jaw, and other problems that can't be seen by looking in your mouth. Ask your dentist how often you need x-rays.

For a dental cleaning:

The first step is called "scaling." The hygienist will use small, metal tools to scrape plaque and tartar from your teeth:

Plaque is a film of bacteria that begins to coat your teeth soon after you eat. Brushing and flossing removes plaque, but not all of it. If you let plaque build up on your teeth, it can cause cavities and destroy the tissue and bone around your teeth, leading to tooth loss. Scaling gets rid of the plaque that tooth brushing leaves behind.

Tartar is a hard substance that can form over time from plaque that's not removed from your teeth. Tartar can build up along your gumline and cause red, swollen gums (gingivitis). This can lead to more serious gum infections and tooth loss. Only a professional dental scaling can remove tartar. And regular professional dental cleanings can prevent it from forming.

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Next the hygienist will polish your teeth, using a device like an electric toothbrush with a rubber tip. The polishing is done with a special paste that's slightly gritty.

After the polishing, the hygienist will floss your teeth.

When needed, the hygienist may apply a fluoride gel or foam, or dental sealants to your teeth:

Fluoride is a mineral that prevents tooth decay (cavities). Fluoride treatments are given to children more often than adults.

Sealants are thin coatings painted on the chewing surfaces of the back teeth. They can help prevent cavities for many years by keeping food and bacteria out of the grooves in the teeth. Sealants are mostly used in children and teens. But adults who have never had cavities in their back teeth may also choose to protect those teeth with dental sealants.

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The hygienist or dentist may give you tips on how to care for your teeth and gums, including the right way to brush and floss.

If dental x-rays are needed:

Your upper body will be covered with a heavy lead apron to protect you from radiation.

You'll bite down on a piece of plastic and an x-ray scanner will be positioned near your cheek.

The hygienist will leave the room or stand behind a protective shield to take the x-ray, which only takes a few seconds. You'll repeat these steps a few times to get x-rays of all your teeth.

For the dentist's exam:

If you had x-rays, your dentist will check them for problems.

Next, your dentist will:

Examine your teeth and gums.

Check your bite, which is the way your top and bottom teeth fit together. If you have a bite problem, you may be referred to an orthodontist.

Check you for oral cancer. This involves looking and feeling for unusual lumps in your mouth, including on your lips and tongue. Your dentist may also check your face and neck.

Examine your teeth and gums.

Check your bite, which is the way your top and bottom teeth fit together. If you have a bite problem, you may be referred to an orthodontist.

Check you for oral cancer. This involves looking and feeling for unusual lumps in your mouth, including on your lips and tongue. Your dentist may also check your face and neck.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A DENTAL EXAM?

If you have certain health conditions, you may need to take antibiotics before your exam to prevent infections. Ask your dentist and/or your health care provider whether you need antibiotics, especially if you have heart problems or an immune system disorder.

Some people feel anxious about going to the dentist. If you or your child feels this way, tell your dentist before your visit. Your dentist can discuss ways to help you feel more relaxed about getting dental care.

ARE THERE ANY RISKS TO A DENTAL EXAM?

There is very little risk to having a dental exam. The cleaning may be uncomfortable, but it's usually not painful.

Dental x-rays are safe for most people. The dose of radiation in an x-ray is very low. But x-rays are not recommended during pregnancy, unless it's an emergency. Be sure to tell your dentist if you are pregnant or think you may be pregnant.

WHAT DO THE RESULTS MEAN?

After your dental exam, your dentist will discuss any oral health problems you may have and your treatment options. For certain problems, improving your brushing and flossing habits may be all that's needed.

If you or your child has a cavity or a more serious problem, you'll probably need to make another appointment with the dentist to treat it.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A DENTAL EXAM?

You can help keep your teeth for your lifetime by getting regular dental exams and taking good care of your teeth and gums. For a healthy mouth and strong teeth:

Brush your teeth twice a day using a soft-bristled brush and toothpaste with fluoride.

Clean between your teeth regularly using dental floss, a special brush, or a plastic pick. Your dentist can suggest the best way to do this.

Replace your toothbrush every three or four months.

Eat a healthy diet and limit sweets and sugary drinks. If you do have sweets, brush your teeth soon afterwards.

Drink fluoridated water.

Don't smoke. Smoking increases your risk for gum disease.

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Depression Screening

<https://medlineplus.gov/lab-tests/depression-screening/>

DEPRESSION SCREENING

WHAT IS DEPRESSION SCREENING?

A depression screening is also called a depression test. It's a standard set of questions that you answer to help your health care provider find out whether you have depression.

Depression is a common, serious mental health condition. Everyone feels sad at times, but depression is different than normal sadness or grief. Depression can affect how you think, feel, and behave. It makes it hard to function at home and work. You may lose interest in activities you once enjoyed. Some people with depression may feel worthless and might even think about harming themselves. Depression may also be part of other mental health conditions, such as bipolar disorder.

Because depression is a common mental health problem, depression screening is often done as part of a routine health checkup. Medical experts recommend that depression screening should be done for everyone starting at age 12. Screening can help find depression early. And treating depression early may make recovery faster. Most people with depression will get better.

Other names: depression test, Patient Health Questionnaire-9 (PHQ-9), Beck Depression Inventory (BDI), Hamilton Depression Rating Scale (HAM-D), Zung Self-Rating Depression Scale (SDS), Geriatric Depression Scale (GDS and GDS-SF)

WHAT IS IT USED FOR?

A depression screening is used to:

- Help diagnose depression

- Understand how severe depression may be

- Help figure out what type of depression you have

There are different types of depression. The most common types are:

Major depression, also called major depressive disorder. The symptoms typically make it difficult to work, sleep, study and eat. With major depression, you have symptoms most of the time for at least two weeks.

Persistent depressive disorder, also called dysthymia. The symptoms are less severe than major depression, but they last much longer, usually for at least two years.

Seasonal affective disorder (SAD). This form of depression usually happens in winter when there's less sunlight. Most people with SAD tend to feel better in the spring and summer.

Postpartum depression. This is major depression that happens after giving birth. It's more severe and lasts longer than mild unhappiness and other mood changes that are often called the "baby blues." Major depression may also begin during pregnancy. Depression that happens during or shortly after pregnancy is called "perinatal depression." Medical experts recommend routine depression screening during pregnancy and after birth.

WHY DO I NEED DEPRESSION SCREENING?

Depression screening is often part of a routine checkup. You may also need depression screening if you show signs of depression, which may include:

- Loss of interest or pleasure in activities you used to enjoy

- Feeling sad or anxious

- Feelings of guilt, worthlessness or helplessness

- Trouble sleeping (insomnia) or sleeping too much

- Fatigue and lack of energy

- Trouble concentrating, remembering details, or making decisions

Changes in your weight

Thoughts of hurting yourself or suicide

If you are thinking about suicide or hurting yourself, get help right away:

Call 911 or go to your local emergency room

Contact a suicide hotline. In the United States, you can reach the National Suicide and Crisis Lifeline at any time:

Call or text 988

Chat online with Lifeline Chat

TTY users: Use your preferred relay service or dial 711 then 988

Call or text 988

Chat online with Lifeline Chat

TTY users: Use your preferred relay service or dial 711 then 988

Veterans can contact the Veterans Crisis Line:

Call 988 then press 1

Text 838255

Chat online

Call 988 then press 1

Text 838255

Chat online

Call your mental health provider or other provider

Reach out to a loved one or close friend

WHAT HAPPENS DURING DEPRESSION SCREENING?

During a depression screening, you'll answer a standard set of questions. Your provider may ask the questions, or you may fill out a questionnaire form to discuss with your provider later.

In general, the questions ask you about:

Changes you've noticed in your:

Mood

Sleep habits

Appetite or weight

Energy levels

Ability to focus your attention

Stress levels

Mood

Sleep habits

Appetite or weight

Energy levels

Ability to focus your attention

Stress levels

Medicines you take

Alcohol and drug use

Your personal and family history of depression and other mental health conditions

You may also have a physical exam. There's no lab test that can diagnose depression. But your provider may order blood tests to find out if another health condition, such as anemia or thyroid disease, may be causing depression.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

If you are being tested by a mental health provider, he or she may ask you more detailed questions about your feelings

and behaviors. You may also be asked to fill out a questionnaire about these issues.

WILL I NEED TO DO ANYTHING TO PREPARE FOR DEPRESSION SCREENING?

You usually don't need any special preparations for a depression test or a blood test.

ARE THERE ANY RISKS TO DEPRESSION SCREENING?

There is no risk to answering questions or having a physical exam.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If you are diagnosed with depression, your provider will discuss your treatment options. Starting treatment as soon as possible may improve your chance of recovery. Treatment for depression may take time to work, but it can help reduce symptoms and shorten how long depression lasts.

Your provider may suggest that you see a mental health provider for your care. A mental health provider is a health care professional who specializes in diagnosing and treating mental health problems. If you are already seeing a mental health provider, a depression test may help guide your treatment.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT DEPRESSION SCREENING?

There are many types of mental health providers who treat depression. Your primary health care provider can help you find the right support.

These are some of the professionals who have training to diagnose and treat depression:

Psychiatrists are medical doctors who specialize in mental health. Psychiatrists can prescribe medicine.

Psychologists generally have doctoral degrees, but they do not have medical degrees. They can't prescribe medicine unless they have a special license. Some psychologists work with providers who can prescribe medicine. Psychologists may use one-on-one counseling and/or group therapy sessions.

Psychiatric or mental health nurses are nurses with special training in mental health problems. Nurses who may have a master's or doctoral degree in psychiatric-mental health nursing include, advanced practice registered nurses (APRNs), certified nurse practitioners (CNPs), and clinical nurse specialists (CNSs). In some states, certain nurses can prescribe medicines.

Licensed clinical social workers have at least a master's degree in social work with special training in mental health. They can't prescribe medicine, but they may work with providers who can prescribe medicine. Providers who are licensed clinical social workers usually have LCSW or LICSW after their names.

Licensed professional counselors (LPC) may also be called clinicians or therapists. States have different names of these licenses, such as LMFT (licensed marriage and family therapist). These professionals usually have a master's degree in a field related to mental health. They can't prescribe medicine, but may work with providers who can prescribe.

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RELATED HEALTH TOPICS

- Antidepressants
- Anxiety
- Bipolar Disorder
- Depression
- Mental Health
- Postpartum Depression
- Psychotic Disorders
- Seasonal Affective Disorder
- Stress
- Teen Depression

RELATED MEDICAL TESTS

ADHD Screening

Mental Health Screening

Obsessive Compulsive Disorder (OCD) Test

Panic Disorder Test

Postpartum Depression Screening

Suicide Risk Screening

Tricyclic Antidepressant (TCA) Screen

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Developmental and Behavioral Screening Tests

<https://medlineplus.gov/lab-tests/developmental-and-behavioral-screening-tests/>

DEVELOPMENTAL AND BEHAVIORAL SCREENING TESTS

WHAT ARE DEVELOPMENTAL AND BEHAVIORAL SCREENING TESTS?

Developmental and behavioral screening tests look at how your child is developing. The screenings are made up of checklists and questionnaires for parents. They include questions about your child's language, movement, thinking, behavior, and emotions.

Many of the questions are based on developmental milestones. Developmental milestones are skills and behaviors that show up in babies and children reach at certain ages as they grow. They can include smiling for the first time, rolling over, and walking. The screenings compare your child's milestones to those of other children of the same age. If a screening shows your child is developing at a slower rate, it may be a sign of a developmental disability.

Developmental disabilities are conditions that begin during a child's developmental period and can cause physical, learning, language, or behavior issues. These conditions may cause challenges in completing daily tasks and last through the person's lifetime. They include:

Intellectual disabilities (IDs). These conditions cause limitations in mental abilities. People with IDs often need help with learning and daily living skills.

Autism spectrum disorder (ASD). This is a disorder that affects behavior, communication, and social skills.

Cerebral palsy (CP). This is a condition that affects movement, coordination, and balance.

Deafness or other hearing problems

Developmental and behavioral screening tests don't diagnose these conditions, but they can show if your child is not developing on schedule. If the screening tool finds an area of concern, a trained specialist may do a formal developmental evaluation. This could involve observation of your child, tests, and/or having you complete a questionnaire.

The results of a formal developmental evaluation help determine if your child needs treatment. When developmental disabilities are found and treated early, it can help to improve your child's life. Early treatment, known as early intervention, helps children learn important skills and make the most of their strengths.

Other names: Ages and Stages Questionnaire, Parents' Evaluation of Developmental Status (PEDS), Modified Checklist for Autism in Toddlers (M-CHAT), Child Development Inventories (CDI)

WHAT ARE THEY USED FOR?

Developmental and behavioral screening tests are used to see if a child is meeting their developmental milestones. Children grow at different rates, and there is a range in what is considered normal development. But looking at milestones can help identify potential problems early. Below are some examples of milestones for infants and toddlers:

By 4 months Includes smiling, cooing, and bringing their hands to their mouth.

By 6 months. Includes playing with others, looking at themselves in the mirror, sitting without support, and rolling over.

By 9 months. Includes making sounds like "mama" and "dada," understanding the word "no," crawling, and pulling themselves up to standing.

By 1 year. Includes playing peekaboo, following simple directions, and walking while holding on to furniture.

By 18 months. Includes speaking and understanding several words, eating with a spoon, walking, and walking, and going up and down stairs.

By 2 years. Includes noticing when others are upset, saying at least two words together, using more gestures such as nodding yes, and kicking a ball.

By 3 years. Includes recognizing labels and colors, naming pictures of common objects, getting dressed and undressed, and walking and running easily.

Developmental milestones are general guidelines. If your child reaches one of the developmental milestones later than other children of the same age, they may have a developmental delay.

If developmental delays don't go away on their own, early treatment may help your child catch up with other children

their age. But if a provider identifies that your child has a developmental disability, early treatment may help, but these are usually lifelong conditions.

WHY DOES MY CHILD NEED A DEVELOPMENTAL AND BEHAVIORAL SCREENING TEST?

The American Academy of Pediatrics (AAP) recommends developmental and behavioral screenings for all children during regular well-child checkups at the following ages:

9 months

18 months

30 months

The AAP also recommends that all children should be screened for autism spectrum disorder during regular checkups at:

18 months

24 months

Your child's provider may want to do additional screening if your child is at a higher risk for developmental concerns, for example because of:

A premature birth

A low birth weight

Being one of a multiple birth

Exposure to lead or other environmental risks

An infection during pregnancy

As a parent, you will probably be the first to notice any problems with how your child interacts, learns, speaks, or moves. Talk with your provider about a screening if you think there is a problem with your child's development.

WHAT HAPPENS DURING A DEVELOPMENT AND BEHAVIORAL SCREENING TEST?

Screenings may be done by your child's provider and/or by you with guidance from your child's provider.

There are many different types of developmental and behavioral screening tests. Each asks questions about a child's development. You may be asked about your child's social interactions, language skills, and/or gross and fine motor skills. Motor skills are the ability to move muscles.

Gross motor skills involve moving large muscles, for example when walking and jumping.

Fine motor skills involve the ability to move small muscles with precision, such as when picking up a toy or using a fork.

There are many screening tests. Some of the common ones include:

Ages and Stages Questionnaire. This test is designed for children between the ages of 1 month and 5 1/2 years. It contains a series of questions with answer choices of "yes," "sometimes," and "not yet."

Parents' Evaluation of Developmental Status (PEDS). This test is designed for children from birth to age 8. It is a brief test that can be completed in about five minutes.

Child Development Inventories (CDI). These are three different tests geared to age groups from infancy to preschool. Each contains 60 yes or no questions.

Modified Checklist for Autism in Toddlers (M-CHAT). This test is for toddlers between the ages of 16 and 30 months. It includes a series of 23 yes or no questions.

ARE THERE ANY SPECIAL PREPARATIONS NEEDED FOR THIS SCREENING?

There are no special preparations needed for this screening.

ARE THERE ANY RISKS TO THE SCREENING?

There is no risk in taking a questionnaire.

WHAT DO THE RESULTS MEAN?

If the results show that your child is not developing at the same rate as other children of the same age, it doesn't necessarily mean there is a problem. But your child may be referred to a specialist for further testing and treatment.

If you have questions about the results, talk to your child's provider. When deciding on the next steps, your provider may consider the results of the screening tests in combination with your child's health and family history.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT DEVELOPMENTAL AND BEHAVIORAL SCREENING TESTS?

If screening tests show a problem with your child's development, your child's provider may recommend at-home developmental monitoring. Developmental monitoring is a way to look at how your child grows and changes over time. To do it, parents and other caregivers use brief checklists to track the child's development.

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RELATED HEALTH TOPICS

Autism Spectrum Disorder

Birth Defects
Cerebral Palsy
Child Development
Developmental Disabilities
Down Syndrome
Hearing Disorders and Deafness
Infant and Newborn Development
Speech and Communication Disorders
Toddler Development

RELATED MEDICAL TESTS

ADHD Screening
Autism Spectrum Disorder (ASD) Screening
Down Syndrome Tests
Fetal Alcohol Spectrum Disorders (FASD) Screening
Hearing Tests for Children

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DHEA Sulfate Test

<https://medlineplus.gov/lab-tests/dhea-sulfate-test/>

DHEA SULFATE TEST

WHAT IS A DHEA SULFATE TEST?

This test measures the levels of DHEA sulfate (DHEAS) in your blood. DHEAS stands for dehydroepiandrosterone sulfate. DHEAS is mostly made in the adrenal glands, two small glands located above your kidneys. Your adrenal glands help control your heart rate, blood pressure, and other body functions. They make different types of hormones you need to stay healthy. This includes cortisol, a hormone that helps with your immune system, and DHEAS, a steroid hormone.

DHEAS is a type of androgen (male sex hormone), but it's found in both men and women. Smaller amounts of DHEAS are made in the testicles and ovaries.

DHEAS plays an important role in making the male sex hormone testosterone and the female sex hormone estrogen. DHEAS helps with the development of male sexual characteristics at puberty and reproduction for both sexes. Your DHEAS levels peak around puberty and then get lower as you age.

If your DHEAS levels are not normal, it may mean there is a problem with your adrenal glands or sex hormones (estrogen or testosterone).

Other names: DHEAS, DHEA-S, DHEA, DHEA-SO₄, dehydroepiandrosterone sulfate

WHAT IS IT USED FOR?

A DHEA sulfate (DHEAS) test is most often used to:

- Find out if your adrenal glands are working well

- Diagnose tumors of the adrenal glands

- Diagnose disorders of the testicles or ovaries

- Find out the cause of early or late puberty in boys

- Look for the cause of the development of masculine features in women and girls, including excess body and facial hair growth and deepening of voice

A DHEAS test is often done along with other sex hormone tests. These include testosterone tests for men and estrogen tests for women.

WHY DO I NEED A DHEA SULFATE TEST?

You may need this test if you have symptoms of high or low levels of DHEA sulfate (DHEAS). Men may not have any symptoms of high levels of DHEAS.

Symptoms of high levels of DHEAS in women and girls may include:

- Excess body and facial hair growth

- Deepening of voice

- Menstrual irregularities

- Acne

- Becoming very muscular

- Hair loss at the top of the head

Baby girls may also need testing if they have external (outer) genitals that do not look clearly male or female (ambiguous genitalia). Boys may need this test if they have signs of early puberty.

Low levels of DHEAS may have different causes. If the cause of low levels of DHEAS is an adrenal gland disorder, the symptoms can include:

- Unexplained weight loss

- Nausea and vomiting

Dizziness

Dehydration

Craving for salt

If the cause of low levels of DHEAS is related to aging, the symptoms can include:

Decreased sex drive

Erectile dysfunction in men

Thinning of vaginal tissues in women

WHAT HAPPENS DURING A DHEA SULFATE TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a DHEA sulfate test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results show high levels of DHEAS, it may be a sign of the following conditions:

Congenital adrenal hyperplasia, an inherited disorder of the adrenal glands in which the adrenal glands produce too many male sex hormones.

A tumor of the adrenal gland. It may be benign (noncancerous) or cancerous.

Polycystic ovary syndrome (PCOS). PCOS is a common hormone disorder affecting childbearing women. It is one of the leading causes of female infertility.

In rare cases, a high level of DHEAS could be a sign of an ovarian tumor.

If your results show low levels of DHEAS, it may be a sign of the following conditions:

Addison disease. Addison disease is a disorder in which the adrenal glands are not able to make enough of certain hormones.

Hypopituitarism, a condition in which the pituitary gland does not make enough of one or more of the pituitary hormones. The possible symptoms will vary, depending on which hormones are lacking.

If you have questions about your results, talk to your provider. To understand the results of a DHEA sulfate test, your provider will consider your symptoms, age, sex, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A DHEA SULFATE TEST?

Over-the-counter DHEA sulfate supplements are available and are sometimes promoted as an anti-aging therapy. However, there is no reliable evidence to support these anti-aging claims. In fact, these supplements may cause serious side effects. If you have questions about DHEA supplements, talk to your provider.

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Diabetes Tests

<https://medlineplus.gov/lab-tests/diabetes-tests/>

DIABETES TESTS

WHAT ARE DIABETES TESTS?

Diabetes, also known as diabetes mellitus, is a disease that affects how your body uses glucose (blood sugar). Glucose is your body's main source of energy. A hormone (a chemical messenger in your bloodstream that controls the actions of certain cells or organs) called insulin helps move glucose from your bloodstream into your cells.

If you have diabetes, your body can't make insulin or can't use it as well as it should, or both. Too much glucose stays in your blood and doesn't reach your cells. This can cause glucose levels to get too high, which can lead to serious health conditions. These may include heart disease, nerve damage, eye problems, and kidney disease.

Your blood glucose can also drop below what is healthy for you. If it drops very low, you could lose consciousness or have a seizure.

Diabetes tests most often measure glucose levels in your blood to monitor your blood glucose level or see if you have or are at risk for diabetes. Urine tests for diabetes are not used as much since they are not as accurate, and you would need a blood test to diagnose diabetes.

Other names: blood glucose, fasting plasma glucose, FPG, oral glucose tolerance test, OGTT, glucose screening test, glucose in urine test, hemoglobin A1c, random blood sugar, blood sugar, fasting blood sugar (FBS)

WHAT ARE THEY USED FOR?

Diabetes tests are used to screen, monitor, and/or diagnose the following:

Type 1 diabetes. If you have type 1 diabetes, your body makes little or no insulin. This disease happens when your immune system attacks and destroys cells that produce insulin. It can develop at any age but most often starts in childhood. People with type 1 diabetes must take daily doses of insulin, either by injection or a special pump.

Type 2 diabetes. This is the most common form of diabetes. If you have type 2 diabetes, your body may still be able to make insulin, but your cells don't respond well to insulin and can't easily take up enough glucose from your blood. Type 2 diabetes may be caused by your genes and lifestyle factors, such as being overweight or having obesity. The disease most often occurs in adulthood but is becoming more common in children and teens.

Gestational diabetes. This is a form of diabetes that develops during pregnancy.

Prediabetes. This is a condition in which your blood glucose levels are higher than what is healthy for you but not high enough to be considered diabetes. But having prediabetes raises your risk of getting type 2 diabetes later on.

WHY DO I NEED A DIABETES TEST?

You may need testing to monitor your blood glucose level or if you have symptoms of diabetes, such as:

Increased thirst

Frequent urination

Increased hunger

Fatigue

Blurred vision

Unexplained weight loss

Sores that are slow to heal

Numbness or tingling in your feet

Symptoms of type 1 diabetes usually come on quickly and can be severe. Symptoms of type 2 diabetes and prediabetes often develop slowly, even over years.

Gestational diabetes doesn't usually cause symptoms in the early stages of your pregnancy, but you will most likely be screened for the condition between 24 and 28 weeks of pregnancy. If testing shows your glucose levels are too high,

you will be tested again to confirm the diagnosis.

You may also need testing if you have certain risk factors for type 2 diabetes. You may be at higher risk for developing it if you:

Are over 35 years old. The American Diabetes Association recommends diabetes screening for prediabetes and diabetes beginning at age 35 years and older.

Have prediabetes.

Are overweight or have obesity.

Have a family history of diabetes.

Have high blood pressure or heart disease.

Previously had gestational diabetes.

Exercise less than three times a week.

Have non-alcoholic fatty liver disease (NAFLD).

Are an African American, Hispanic or Latino, American Indian, or an Alaska Native.

WHAT HAPPENS DURING A DIABETES TEST?

There are several ways to screen for, monitor, and diagnose diabetes. Most tests involve measuring your blood glucose levels.

To get a blood sample, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

The different types of glucose blood tests include:

Blood glucose test, also known as fasting blood glucose. Before the test, you will need to fast (not eat or drink) for eight hours before the test. This test is usually done in the morning and is often used as a screening test for diabetes. It may be repeated to confirm a diagnosis.

Oral glucose tolerance test (OGTT). This test also requires fasting. When you arrive for your test, a blood sample will be taken. You will then drink a sugary liquid that contains glucose. About two hours later, another blood sample will be taken. This test tells your health care provider how well your body processes sugar. This test may be used to test for gestational diabetes. If you are having this test done during pregnancy, you will have your blood drawn every 30 minutes for two to three hours.

Glucose challenge test, also known as the glucose screening test. This test may be used to test for gestational diabetes. You will drink a sugary liquid that contains glucose. A blood sample will be taken one hour later. You do not need to fast for this test. If your blood glucose is too high, your provider may order an oral glucose tolerance test.

Random blood sugar. This test can be taken at any time. No fasting is required. This test may be used if you have symptoms of diabetes.

Hemoglobin A1c (HbA1c). This test measures the average amount of glucose attached to hemoglobin over the past three months. Hemoglobin is the part of your red blood cells that carries oxygen from your lungs to the rest of your body. No fasting is required for this test.

Glucose may also be measured in your urine.

Urine tests are not often used to diagnose or monitor diabetes since blood tests are more accurate. But a urine test may show if you are at risk for getting the disease. If your glucose in urine levels are too high, then you will probably need a blood test to confirm the diagnosis.

For a glucose in urine test, your provider may recommend an at-home test kit. The kit will include a test strip that you hold under your stream of urine. The test strip will change colors to show different levels of glucose.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You will need to fast (not eat or drink) for a blood glucose and an oral glucose tolerance test.

You don't need any special preparations for a random blood sugar test, glucose challenge test, hemoglobin A1c test, or glucose in urine test.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no risk to having a urine test.

WHAT DO THE RESULTS MEAN?

Depending on the type of test or tests you had, your results may show one of the following:

Your glucose is at a healthy level for you. This means you probably are not at risk for or do not have diabetes.

Prediabetes. This means you have glucose levels that are higher than what is healthy for you and may be at risk of getting diabetes.

Type 1 or type 2 diabetes.

Gestational diabetes.

If you have been diagnosed with type 1 diabetes, talk to your provider about how to best manage the disease. There is no cure for type 1 diabetes, but it may be controlled with regular glucose monitoring and taking insulin.

If you've been diagnosed with prediabetes or type 2 diabetes, your condition may be managed or even reversed by taking diabetes medicines and making lifestyle changes. These include eating a healthy diet, losing weight, and increasing exercise.

If you've been diagnosed with gestational diabetes, you may be able to lower your glucose levels by eating a healthy diet and getting regular exercise. But be sure to talk to your provider about treatment options. Most of the time, gestational diabetes goes away after giving birth. But you will be at higher risk for developing type 2 diabetes in the future.

If you have other questions about your diabetes diagnosis or treatment, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT DIABETES TESTING?

If you've been diagnosed with type 1 diabetes, you will need to monitor your blood glucose levels daily, often several times a day. Your provider can recommend a kit you can use at home. Most kits include a lancet, a device that pricks your finger. You will use this to collect a drop of blood for testing. There are some newer kits available that don't require pricking your finger. If you are pregnant and have gestational diabetes, you may also need to monitor your glucose levels in this way.

People with type 2 diabetes must also check their blood sugar regularly. If you have type 2 diabetes, talk to your provider about how often it should be checked.

People with type 2 diabetes may also need to have their insulin levels checked regularly. Insulin plays a key role in keeping glucose at the right levels. An insulin in blood test is done at a provider's office.

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Diabetic Foot Exam

<https://medlineplus.gov/lab-tests/diabetic-foot-exam/>

DIABETIC FOOT EXAM

WHAT IS A DIABETIC FOOT EXAM?

People with diabetes have a high risk for many types of foot health problems. A diabetic foot exam can help find problems that can lead to serious infection and amputation (surgery to remove all or part of your foot) if they're not found and treated early.

Diabetic foot problems can happen over time if high blood glucose (blood sugar) damages the nerves and blood vessels in your feet. Nerve damage from diabetes is called diabetic neuropathy. It can make your feet feel numb or tingly. It can also cause you to lose feeling in your feet.

If you don't feel pain in your feet, you may not notice a callus, blister, or cut on your foot. Without treatment, these types of common foot wounds can become ulcers (open sores) that can get infected.

If you have diabetes, a foot sore or infection may not heal well, even with treatment. That's because diabetes can damage blood vessels and cause poor blood flow in your feet. If an infection doesn't get better, tissue in your foot may die. This is called gangrene.

If you have gangrene or an infection that won't heal, your toe, foot, or part of your leg may need to be amputated to prevent infection from spreading and to save your life. Fortunately, you can help keep your feet healthy by:

Getting regular diabetic foot exams

Caring for your feet at home

Keeping your blood sugar within a range that's healthy for you

Other names: comprehensive foot exam

WHAT IS IT USED FOR?

A diabetic foot exam is used to find foot health problems in people with diabetes before they cause serious infection. The exam is used to check the feet for:

Problems with the nerves and blood flow.

Signs of damage or injury.

Changes in shape. Foot changes, such as bunions and hammer toes (toes that curve upwards with the top part curled under), can make shoes rub and cause foot ulcers.

WHY DO I NEED A DIABETIC FOOT EXAM?

People with diabetes usually need to have a complete diabetic foot exam at least once a year to help prevent serious infections. But if you've had foot problems from diabetes, you may need to have a complete foot exam more often.

A complete foot exam may be done by your primary care provider or a foot doctor (podiatrist). A foot doctor specializes in keeping feet healthy and treating foot diseases. Ask the provider you see for foot care to tell you how often you need a complete foot exam.

It's also a good idea to ask your primary care provider to do a quick check of your feet at every health care visit you have during the year. If you're meeting your diabetes treatment goals, your provider will probably want to see you at least twice a year to check your health and review your self-care plan. Make sure to have your feet checked, too.

If you have any of these foot symptoms, contact your provider right away:

A blister, cut, bruise, or other foot injury that doesn't start to heal after a few days.

Skin on your foot that is red, warm, or painful. These are signs of a possible infection.

A callus with dried blood inside it. This may be the first sign of a wound under the callus.

A foot infection that becomes black and smelly. This may be a sign of gangrene.

WHAT HAPPENS DURING A DIABETIC FOOT EXAM?

A complete diabetic foot exam usually includes these steps:

Information about you. Your provider will ask questions about:

Your medical history. This includes how you're managing your diabetes, other conditions you may have, and your symptoms.

The medicines you take.

Your shoes. Your provider may check your shoes for proper fit and ask questions about your other footwear. Shoes that don't fit well can lead to blisters, calluses, and ulcers.

Skin exam. Your provider will check:

The skin of your feet to look for dryness, cracking, calluses, blisters, ulcers, and other damage or abnormal areas

Your toenails for cracks and fungal infection

The temperature of your feet to see if they are the same

Nerve exam. Your provider will do a few tests to check your nerves and the feeling in your feet:

Monofilament test. You will close your eyes while your provider brushes a soft strand of nylon (a monofilament) over your foot and toes. You'll tell your provider when you feel the strand touching your foot.

Tuning fork and vibration perception threshold tests (VPT). Your provider will place a tuning fork or other device that vibrates on different parts of your foot and toes to see if you can feel the vibrations.

Pinprick test. Your provider will gently press a small pin against your big toe to see if you can feel it. The pin will not break through your skin.

Ankle reflexes. Your provider will tap a special, small hammer on your Achilles tendon, the thick band of tissue that connects your calf muscle to your heel bone. If your nerves are working properly, your foot will jerk slightly on its own.

Musculoskeletal (muscle and bone) exam. Your provider will look for problems such as:

Bent or overlapping toes

Bunions

A rocker shape on the bottom of your foot (Charcot Foot)

Vascular (blood vessel) exam. To check blood flow to your feet, your provider may:

Feel the pulses in your foot and ankle

Compare blood pressure measurements in your ankles and arms. If blood pressure in your ankle is lower than in your arm, you may have reduced blood flow to your foot. This is called an ankle-brachial index test. Your provider may do this test if you have signs and symptoms of a blood flow problem.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a diabetic foot exam.

ARE THERE ANY RISKS TO THE TEST?

There are no known risks to having a diabetic foot exam.

WHAT DO THE RESULTS MEAN?

Your provider will use the results of your exam to estimate your risk of developing foot ulcers and infections that could lead to amputation. If your risk is high, your provider will probably recommend that you have a complete diabetic foot exam more than once a year.

If a problem is found that needs medical care, your treatment will depend on the type of problem and how serious it is. If you have questions about your diabetic foot exam, ask your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A DIABETIC FOOT EXAM?

You can help keep your feet healthy by controlling your blood sugar levels every day and taking good care of your feet. This includes:

Checking your feet every day. Look for cuts, redness, and other changes in the skin and toenails, including warts or other spots that your shoes could rub. Make sure to check the bottoms of your feet too.

Washing your feet every day. Use warm water and soap. Don't soak your feet because that can dry out your skin. After you dry your feet, you can use talcum powder or cornstarch between your toes. They soak up moisture that can cause infection. If you use lotion, don't apply it between your toes.

Asking your doctor how to remove corns and calluses safely. Thick skin on your feet can rub and lead to sores. But removing it the wrong way could damage your skin. So, you don't want to cut the skin or use medicated pads or liquid removers.

Trimming your toenails straight across with a clipper. If it's hard for you to trim your own toenails, or if they're thick or curve into the skin, have a podiatrist (foot doctor) do it for you.

Always wearing well-fitting shoes and socks or slippers to protect your feet when walking. You don't want to walk barefoot, even indoors. And be sure your shoes are smooth inside. A seam or pebble in your shoe could rub your skin raw.

Protecting your feet from heat and cold. Use sunscreen on exposed skin and don't walk barefoot at the beach. In cold weather, wear warm socks instead of warming your feet near a heater or fireplace.

Keeping the blood flowing in your feet. Put your feet up when you're sitting. Wiggle your toes and circle your feet throughout the day. Don't wear tight socks. And get plenty of activity that's not too hard on the feet, such as walking.

Getting your feet checked at your health care visits. Even if you haven't noticed a problem, it's good to have your provider look at your feet.

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Differential Diagnosis

<https://medlineplus.gov/lab-tests/differential-diagnosis/>

DIFFERENTIAL DIAGNOSIS

WHAT IS A DIFFERENTIAL DIAGNOSIS?

Different health conditions often cause similar symptoms. For example, fatigue is a symptom that can be caused by anemia, depression, heart disease, thyroid disease, sleep disorders, and many other conditions.

When symptoms could have more than one cause, your provider will follow a careful, step-by-step process to rule out conditions that share your symptoms and identify the most likely cause. This process is called making a differential diagnosis.

To make a diagnosis, your provider first creates a list of all the possible conditions that could explain your symptoms. This is your differential diagnosis list. Your provider will base the list on your specific:

Symptoms

Medical history

Family health history

Medicines and supplements

Lifestyle

Test results

A differential diagnosis list helps your provider decide which tests will help confirm or rule out some of the possible conditions. As test findings narrow down the list, the most likely condition will become clear. This is your final diagnosis.

The process of making a differential diagnosis may take time, but it helps make sure your provider doesn't miss a possible cause of your symptoms. That means your final diagnosis is more likely to be accurate, and you'll get the right treatment.

HOW IS IT USED?

A differential diagnosis process is used to help make a final diagnosis when you have symptoms that don't have one clear cause. It can be used to help diagnose:

Mental disorders.

Neurological conditions.

Infections.

Hormonal or metabolic conditions.

Autoimmune diseases.

Differential diagnosis is especially useful when symptoms overlap across many conditions, such as fatigue, headaches, chest pain, shortness of breath, dizziness, or unexplained weight changes.

WHAT HAPPENS DURING A DIFFERENTIAL DIAGNOSIS?

To make a differential diagnosis, your provider has to think like a detective and follow several steps:

The first step is to gather clues by asking you about your health.

Your provider will ask detailed questions about your:

Symptoms. Your provider may want to know what symptoms you have, how long you've had them, what makes them better or worse, and how they affect you.

Medical history. Your provider will consider whether your symptoms could be related to health problems you already have. They may also ask if you've ever experienced your current symptoms before.

Medicines and supplements. Sometimes symptoms can be side effects from medicines or supplements, so your provider will want to know about everything you take.

Family health history. If a certain condition runs in your family, your provider will consider whether your symptoms are a

sign of that condition.

Lifestyle and habits. Changes or problems with your sleep, diet, exercise, work, mood, and relationships can be causes or symptoms of many conditions. So, to help narrow the differential diagnosis list, your provider may ask you if there have been any changes in your life.

The second step of the differential diagnosis process is a physical exam. Information from an exam gives your provider more clues about your health. It may help rule out certain conditions or make others seem more likely.

The third step is to put together a differential diagnosis list. Your provider will make a list of possible conditions ("suspects") based on information about your symptoms, health, and lifestyle.

The fourth step is to order tests. Tests can help confirm or rule out the conditions on your differential diagnosis list. The tests you have will depend on the conditions on your list. If your list includes a serious condition that may need urgent treatment, you'll have tests for that condition first.

Common tests to help narrow your differential diagnosis list may include:

Lab tests, such as blood and urine (pee) tests.

A biopsy.

Imaging tests, such as an MRI Scan, ultrasound, or x-ray.

A mental health screeningÂ if your symptoms could be caused by a mental disorder.

As test results rule out some of the possible causes of your symptoms, your final diagnosis should become clear. Then your provider can recommend treatment.

WHAT DO MY RESULTS MEAN?

The conditions on your differential diagnosis list are not your final diagnosis. They are only the conditions that could be causing your symptoms. If, for example, you have abdominal (belly) pain, then your differential diagnosis list may include conditions such as:

Appendicitis.

Gastritis.

Inflammatory bowel disease.

Intestinal or bowel blockage.

Pancreatitis.

Your provider will then use the differential diagnosis list to choose which tests to order. Each time you get test results, ask your provider how they affect your differential diagnosis. Test results may:

Narrow the list.

Rule out certain conditions.

Suggest additional tests.

Point toward a likely diagnosis.

Guide early treatment choices.

If you have questions about the conditions in your differential diagnosis list, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A DIFFERENTIAL DIAGNOSIS?

Your provider may recommend starting treatment before your final diagnosis is certain. This usually happens when:

A condition on your list needs urgent treatment.

Your symptoms are severe.

Starting treatment could help clarify the diagnosis.

In certain cases, your provider may learn more about your condition by seeing whether a treatment improves your symptoms. This approach can be especially useful for gastrointestinal disorders and autoimmune diseases.

If you have questions about your diagnosis or next steps, talk with your provider.

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Doppler Ultrasound

<https://medlineplus.gov/lab-tests/doppler-ultrasound/>

DOPPLER ULTRASOUND

WHAT IS A DOPPLER ULTRASOUND?

A Doppler ultrasound is a type of ultrasound imaging test that uses sound waves to show how well your blood flows through your blood vessels. It can be used to examine blood flow in many parts of your body, including many of your organs and your neck, arms, and legs.

A Doppler ultrasound can examine your arteries (blood vessels that carry oxygen-rich blood from your heart and lungs to your body) and veins (blood vessels that carry blood from your body back to your heart and lungs to get more oxygen).

Doppler ultrasound works by bouncing sound waves off red blood cells flowing through your blood vessels. The ultrasound device measures the echoes that bounce back from the cells. Cells that are moving away from the sound waves make different echoes than cells that are moving closer to the sound waves. This is called the Doppler effect, named after the scientist who discovered it.

Doppler ultrasound can check for blood vessel conditions that affect the amount, speed, and direction of blood flow in different parts of your body. It can help find narrowed arteries, blood clots, and other conditions.

There are different types of Doppler ultrasound tests that can be used to gather different information. They include:

Color Doppler. This test uses a computer to change sound wave measurements into different colors. The colors show the speed and direction of blood flow.

Power Doppler. This is a newer type of color Doppler. It can show much smaller blood vessels and slower blood flow than standard color Doppler. But it can't show the direction of blood flow.

Spectral Doppler. This test shows blood flow information in a graph. It can show how much of a blood vessel is blocked.

There are two types of spectral Doppler ultrasound:

A "pulsed wave" Doppler can show where blood is flowing at different speeds, but it can't record blood flowing at high speeds.

A "continuous wave" Doppler can record very fast blood flow, but it can't show the exact location in the blood vessel.

A "pulsed wave" Doppler can show where blood is flowing at different speeds, but it can't record blood flowing at high speeds.

A "continuous wave" Doppler can record very fast blood flow, but it can't show the exact location in the blood vessel.

Duplex Doppler. This test uses standard ultrasound to take images showing blood flow. Then a computer turns the images into a graph, as in spectral Doppler.

Other names: Doppler ultrasonography, vascular ultrasound

WHAT IS IT USED FOR?

Doppler ultrasound tests are used to help diagnose problems that affect the flow of blood in many parts of the body, including:

Your heart. A Doppler ultrasound is part of an echocardiogram, which is a test that shows the size and shape of your heart and how well it's pumping blood. The test can help diagnose many heart conditions, including:

Narrowed and blocked arteries

Aneurysm, a bulge in the wall of artery

Heart valve diseases

Congenital heart defects, structural problems in the heart from birth

Narrowed and blocked arteries

Aneurysm, a bulge in the wall of artery

Heart valve diseases

Congenital heart defects, structural problems in the heart from birth

Your arms and legs. The test can check for conditions that reduce blood flow including:

Deep vein thrombosis (DVT), blood clots, usually in your legs

Peripheral arterial disease (PAD), narrow or blocked arteries in your legs or arms

Varicose veins, weak valves in your veins that let blood flow backwards, usually in your legs

Deep vein thrombosis (DVT), blood clots, usually in your legs

Peripheral arterial disease (PAD), narrow or blocked arteries in your legs or arms

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Your neck and brain. The test can look for conditions that affect blood flow to your brain and cause stroke. These include narrowed arteries and blood clots in your neck (carotid artery disease) and narrowed arteries in your brain.

Your abdomen (belly). The test can be used to check blood flow in your organs and to look for aneurysms.

A new organ after transplantation surgery. The test can show whether blood is flowing through the organ normally.

A Doppler ultrasound can also help diagnose blood vessel tumors and problems with the structure of your blood vessels that you were born with.

If you are pregnant and there are concerns about blood flow to your fetus, a Doppler ultrasound is used. It can also check the baby's blood flow.

WHY DO I NEED A DOPPLER ULTRASOUND?

You may need a Doppler ultrasound if you have symptoms of abnormal blood flow. Your symptoms will depend on the part of your body that's affected and the cause of abnormal blood flow.

You may also need a Doppler ultrasound if you:

Have a high risk of having a stroke or have had a stroke or a transient ischemic attack (a stroke that lasts a few minutes). An ultrasound of your brain, called a transcranial Doppler ultrasound, can diagnose the cause of a stroke and/or help estimate your risk for having a stroke in the future.

Had an injury to your blood vessels.

Have high blood pressure that could be caused by blocked arteries in your kidneys.

Are being treated for a condition that affects blood flow. A Doppler ultrasound can show how much the treatment is helping.

Are pregnant. You may need this test if your unborn baby may be smaller than normal for your stage of pregnancy. You may also need this test if you have certain health problems such as high blood pressure in pregnancy (preeclampsia) or sickle cell disease.

WHAT HAPPENS DURING A DOPPLER ULTRASOUND?

A Doppler ultrasound is often done by a sonographer, a health care professional who has special training to do ultrasound exams. The test may be done in different ways depending on the part of your body that's being examined.

Most Doppler ultrasound exams include these general steps:

You will remove clothing from the area that will be examined and lie on a table.

The sonographer will spread a special gel on your skin over the area that will be examined.

The sonographer will move a wand-like device, called a transducer, across your skin. The device sends sound waves into your body, but you can't feel them.

The sound waves bounce off red blood cells in your bloodstream.

You may hear whooshing sounds as the ultrasound device picks up the echoes and turns them into images or graphs on a computer screen.

After the exam is over, the sonographer will wipe the gel off your body.

For certain ultrasound exams, the ultrasound device is placed inside a body opening. Depending on the organs being checked, the device may be placed in the:

Vagina. This is called a transvaginal ultrasound. It helps view the uterus and ovaries.

Rectum. This is called a transrectal ultrasound. It's usually done to view the prostate gland.

Esophagus (the tube that connects your mouth and stomach). This is called a transesophageal echocardiogram (TEE). It's done to get clear images of the heart.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Preparations for a Doppler ultrasound depend on what part of your body is being checked. Your health care provider will tell you if you need to prepare and what to do. You may need to:

Fast (not eat or drink) for several hours before the test.

Stop smoking or using products that have nicotine for two hours or more before your test. Nicotine causes blood vessels to narrow, which could change your results.

ARE THERE ANY RISKS TO THE TEST?

Doppler ultrasound has not been linked to any health harms. It's generally considered safe when trained sonographers use it correctly. Doppler ultrasound doesn't use ionizing radiation like x-rays use, which makes it safer than x-ray. That's why ultrasound is the most widely used medical imaging method for viewing an unborn baby during pregnancy.

But in certain cases, ultrasound can affect fluids and tissues in the body. That's why most medical experts recommend using ultrasound only when it's needed to provide important medical information.

Learn more about laboratory tests, reference ranges, and understanding results.

WHAT DO THE RESULTS MEAN?

If your Doppler ultrasound results are normal, it means that the blood vessels that were checked looked healthy. They have no narrowing or blood clots. The blood flow was normal.

If your results aren't normal, the meaning will depend on what area of your body was tested and the type of Doppler ultrasound that you had. Talk with your provider about what your results say about your health and need for treatment.

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Down Syndrome Tests

<https://medlineplus.gov/lab-tests/down-syndrome-tests/>

DOWN SYNDROME TESTS

WHAT ARE DOWN SYNDROME TESTS?

Down syndrome tests check whether a developing fetus may have Down syndrome, a chromosome disorder caused by having an extra copy or piece of chromosome 21.

Chromosomes are tiny "packages" in your cells that contain your genes. Genes carry information, called DNA, that controls what you look like and how your body works. Normally, you're born with a set of 46 chromosomes in each cell. The chromosomes are grouped into 23 pairs. Each pair has two chromosomes, one coming from each parent.

Most people with Down syndrome have an extra copy of chromosome 21 in all their cells. This common form of Down syndrome is called trisomy 21. "Trisomy" means having three copies of a chromosome in your cells instead of the usual two. With less common types of Down syndrome, a person may have:

An extra piece of chromosome 21 attached to another chromosome (translocation Down syndrome).

An extra chromosome 21 in some, but not all, of their cells (mosaic Down syndrome).

The extra chromosome in Down syndrome affects the way a fetus's brain and body develop. This can cause mild or severe problems with development, thinking, learning, and physical health. Children with Down syndrome may also start talking later than other children and have:

A flat face.

Eyes that slant up.

A short neck.

Small hands and feet.

Poor muscle tone.

Loose joints.

Hearing loss.

Sleep apnea.

Ear infections.

Eye diseases

Heart defects.

There are two basic types of tests that help find Down syndrome during pregnancy:

Screening tests, which estimate the chance that a fetus has Down syndrome.

Diagnostic tests, which confirm whether Down syndrome is actually present by examining the fetus's cells or DNA.

Screening tests are simple and safe, using blood samples and ultrasounds. Diagnostic tests involve collecting cells from the amniotic fluid, umbilical cord, or placenta (the organ that grows in the uterus to provide oxygen and nutrients to the fetus). This makes diagnostic tests more accurate but also slightly riskier.

WHAT ARE THE TESTS USED FOR?

Down syndrome tests are used for several purposes during and after pregnancy. Each test works in a different way and is done at specific times.

Prenatal screening tests estimate the chance that a fetus has Down syndrome but can't diagnose the condition. There are several types of these screening tests, including:

First trimester screening (weeks 10-13). This test combines a blood test and an ultrasound.

The blood test checks for markers, such as certain proteins, that may be linked to a higher risk of Down syndrome. These can include pregnancy-associated plasma protein A (PAPP-A) and human chorionic gonadotropin (hCG).

The ultrasound measures the thickness of the fluid behind the fetus' neck. Increased fluid thickness can mean a higher chance of Down syndrome.

The blood test checks for markers, such as certain proteins, that may be linked to a higher risk of Down syndrome. These can include pregnancy-associated plasma protein A (PAPP-A) and human chorionic gonadotropin (hCG).

The ultrasound measures the thickness of the fluid behind the fetus' neck. Increased fluid thickness can mean a higher chance of Down syndrome.

Second trimester screening (weeks 15-20). Often called the quad screen (or quadruple screen test), this screening checks the levels of four substances in your blood. These are: alpha-fetoprotein (AFP), hCG, estriol (a type of estrogen), and inhibin A. Unusual levels of any of these can suggest a higher chance of Down syndrome.

Combined or integrated screening. This test combines results from both the first and second-trimester screenings. By doing so, this screening test can more clearly estimate your fetus' risk for Down syndrome.

Cell-free DNA (cfDNA) or noninvasive prenatal screening (NIPS). This test checks small pieces of your fetus' DNA found in your blood for unusually large amounts of material from chromosome 21. It is the most accurate type of test and can be done as early as 10 weeks. Medical experts recommend this test for everyone, regardless of age or risk level.

Prenatal diagnostic tests confirm whether a fetus has Down syndrome. They are usually done after the results of a screening test come back positive and your health care provider needs to confirm a diagnosis. To do this, your provider may order one of the following tests:

Amniocentesis. This test is usually performed between 15 and 20 weeks of pregnancy. Your provider will collect a small amount of amniotic fluid, which is the fluid that surrounds the fetus. This fluid contains some of your fetus's cells. By collecting a sample of your amniotic fluid, your provider can check the fetal cells for chromosome changes.

Chorionic Villus Sampling (CVS). This test is usually done between 10 and 13 weeks of pregnancy. Your provider will collect tissue from parts of the placenta called the chorionic villi. Cells from the chorionic villi usually have the same chromosomes as the fetus. So, a provider can tell from the sample if the fetus has any chromosome abnormality.

Percutaneous Umbilical Blood Sampling (PUBS). This type of test removes a sample of the blood from the umbilical cord of the fetus. It provides very accurate results but carries a higher risk of health complications for the fetus, so it's used only when other tests can't provide clear answers.

Your provider may also test for Down syndrome:

After birth. Your provider will only test for Down syndrome if your baby shows physical signs of this condition after birth. To do this, they may use a sample of your blood to perform a karyotype test, which can diagnose Down syndrome by checking the number and structure of the baby's chromosomes.

During in vitro fertilization (IVF). Your provider will test embryos for Down syndrome before they are placed in the uterus.

WHY DO I NEED A DOWN SYNDROME TEST?

Medical researchers recommend that everyone, regardless of age and risk, be offered screening for Down syndrome while pregnant. You may talk with your provider about which one of the screening test options works best for you. These options can include first and second trimester screening, cell-free DNA (cfDNA), or noninvasive prenatal screening (NIPS).

You may need additional Down syndrome testing, including diagnostic testing, if you have a high risk for having a baby with Down syndrome. Your risk may be higher if:

You were age 35 or older when you got pregnant. The risk of having a baby with Down syndrome increases with age.

You have a child with Down syndrome or had a diagnosis of Down syndrome in a past pregnancy.

You have a family history of Down syndrome.

An ultrasound of your fetus showed a possible sign of Down syndrome.

You may choose to get a diagnostic test even if you don't have a high risk for having a baby with Down syndrome. Your provider or a genetic counselor can help you consider the pros and cons of these tests.

WHAT HAPPENS DURING DOWN SYNDROME TESTING?

For a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For the first trimester ultrasound, a health care professional will move an ultrasound wand-like device on your belly. Ultrasound uses sound waves to make an image of your fetus.

For amniocentesis, a provider inserts a hollow needle through your belly and into your uterus to withdraw a sample of amniotic fluid.

For chorionic villus sampling (CVS), a provider collects a sample of tissue from the placenta with either a needle that's inserted through your belly or a tube that's inserted through your vagina.

For percutaneous umbilical blood sampling (PUBS), a provider inserts a hollow needle through your belly and into the umbilical cord in your uterus. A small blood sample is removed from a vein in the cord.

For a newborn karyotype, a provider collects a blood sample from the baby's heel or vein. They will then use that sample to check the number and structure of your baby's chromosomes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TESTS?

Blood tests and ultrasounds usually don't require any preparation.

For a CVS or amniocentesis, you may need to drink extra fluid and not urinate (pee) before your test so that your bladder is full.

If you're having a PUBS test after 23 weeks of pregnancy, you may need to fast (not eat or drink) for several hours before the test.

Your newborn will not need any preparation for a karyotype test.

ARE THERE ANY RISKS TO THE TESTS?

Blood tests and ultrasounds have very little risk. After a blood test, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

During amniocentesis or CVS, you may feel mild stinging or cramping. Both tests are safe, but they do have a slight risk of causing a miscarriage.

PUBS may cause brief discomfort. This test has a slightly higher risk of causing serious problems than amniocentesis or CVS. Talk with your provider about the risks and benefits of testing for Down syndrome.

There is no risk for a newborn karyotype test.

WHAT DO THE RESULTS MEAN?

Results from Down syndrome screening tests show your risk of having a baby with Down syndrome:

A positive screening test result means that your baby's chance of having Down syndrome is higher than average. Your test results may include a number that describes how high the risk is. But a high risk doesn't mean your baby will have Down syndrome. You could still deliver a healthy baby with no chromosomal defects or disorders.

A negative/normal screening test result means that your baby is unlikely to have Down syndrome. But a screening test can't guarantee that your baby won't have Down syndrome.

Whether or not your screening tests were normal, you may choose to have one or more diagnostic tests. Diagnostic and karyotype tests can usually tell you whether or not your baby will have Down syndrome. But diagnostic tests cannot predict how much Down syndrome will affect your baby's health and abilities.

You may find it helpful to speak with a genetic counselor before testing and again after you get your results. Genetic counselors are specially trained to help you weigh the risks and benefits of testing. They can also help you understand what your results mean so you can make informed decisions.

Learn more about laboratory tests, reference ranges, and understanding results.

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Drug Testing

<https://medlineplus.gov/lab-tests/drug-testing/>

DRUG TESTING

WHAT IS A DRUG TEST?

Drugs are chemical substances that can change how your body and mind work. A drug test looks for signs of illegal drugs or certain prescription and/or over-the-counter (OTC) medicines that can be misused. The test is usually done with a sample of your urine (pee). Less commonly, drug testing uses a sample of your blood, saliva (spit), hair, sweat, fingernails, or breath. The purpose of a drug test is to look for drug use and misuse, which includes:

Using any illegal drugs, such as:

Anabolic steroids.

Cocaine.

Club drugs.

Heroin.

Inhalants.

Marijuana (cannabis, THC).

Amphetamines, such as methamphetamines and Ecstasy (MDMA).

Phencyclidine (PCP)

Anabolic steroids.

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Marijuana (cannabis, THC).

Amphetamines, such as methamphetamines and Ecstasy (MDMA).

Phencyclidine (PCP)

Misusing prescription medicines, including opioids (such as codeine, oxycodone, morphine, and fentanyl), barbiturates (such as phenobarbital), and benzodiazepines (such as alprazolam). This means taking prescription medicines in a different way than your health care provider prescribed. This includes:

Taking a medicine that was prescribed for someone else.

Taking a larger dose than you are supposed to.

Using the medicine in a different way than you are supposed to. For example, instead of swallowing your tablets, you might crush and then snort or inject them.

Using the medicine for another purpose, such as getting high.

Taking a medicine that was prescribed for someone else.

Taking a larger dose than you are supposed to.

Using the medicine in a different way than you are supposed to. For example, instead of swallowing your tablets, you might crush and then snort or inject them.

Using the medicine for another purpose, such as getting high.

Misusing over-the-counter medicines, such as dextromethorphan (DXM) and certain laxatives. This includes using them for another purpose or in a different way than you are supposed to.

A drug test can also check your urine or blood for signs of alcohol use.

Drug tests can find signs of drugs that you took hours to several days or more before the test. How long a drug lasts in your body depends on:

The type of drug

How much you used

How long you were using it before the test

How your body reacts to the drug

Other names: drug screen, drug test, drugs of abuse testing, substance abuse testing, toxicology screen, tox screen, sports doping tests

WHAT IS IT USED FOR?

Drug testing is used to find out whether you have been using or misusing one or more drugs or prescription/OTC medicines. It can't diagnose a drug use disorder (addiction).

A drug test may be used for different purposes, including:

Employment. Employers may require that you pass a drug test before they will hire you. The testing may be part of a drug-free workplace program. After you're hired, they may test you to check for on-the-job drug use. If you have a work accident, you may be tested to see whether drugs or alcohol were involved.

Sports. Professional and other athletes are often tested for drugs that are used to improve performance, such as anabolic steroids that help build muscle.

Substance use disorder treatment. Drug testing may be used to monitor treatment in programs for drug or alcohol use disorder (AUD).

Legal evidence. Testing may be part of a criminal or motor vehicle accident investigation. Drug screening may also be ordered as part of a court case.

To prevent prescription drug misuse. If you take a prescription medicine that can be addictive, such as opioids for chronic (long-term) pain, your provider may order a drug test to make sure you're taking the medicine correctly.

To detect over-the-counter (OTC) drug misuse. If your provider suspects that you are misusing an OTC medicine, such as a laxative or cough medicine, they may order a drug test to know for sure.

Drug testing may be scheduled ahead of time or done randomly without notice.

WHY DO I NEED A DRUG TEST?

You may be asked to take a drug test for a job, to participate in certain organized sports, or as part of a police investigation or court case. In a hospital emergency room, a provider may order a drug test if you have signs of a possible drug overdose. Newborns can also be tested for possible prenatal exposure to illegal drugs.

WHAT HAPPENS DURING A DRUG TEST?

Drug testing may be done in many different locations, including labs, hospitals, drug treatment centers, and workplaces. Urine is the most commonly used sample for drug testing. You will be given instructions for how to collect your sample. In certain cases, a health care professional or other person may need to be present while you provide your sample. This is to make sure the urine is yours and isn't contaminated with anything that might affect the test results.

If a blood sample is used to test for drugs, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

If a saliva sample is used to test for drugs, your provider will take a saliva sample from your mouth using a swab. They will roll the swab in your mouth or hold it under your tongue for about 2 minutes until it is soaked with saliva. They will then either place the swab in a tube and send it to a lab for testing or test it immediately using a handheld device.

In some cases, a health care professional may test your hair, fingernails, or sweat for signs of drug use, though these testing methods are not as commonly used.

A health care professional may also use a breath test to see if you've been drinking alcohol. For this test, you will blow into a tube connected to a small device that will measure your blood alcohol content (BAC).

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Be sure to tell the testing professional if you are taking any prescription drugs, over-the-counter medicines, or supplements, because these substances may affect your test results. Also, you should avoid foods with poppy seeds, which can show up as opiates in a drug test.

ARE THERE ANY RISKS TO THE TEST?

There are no known physical risks to having a drug test. But if drugs show up in your results, it may affect your job, your eligibility to play sports, the outcome of a legal matter, or other parts of your life.

Before you take a drug test, make sure you know:

What you are being tested for

Why you are being tested

How the results will be used

If you have questions or concerns about a drug test, talk with your provider or the person or organization that is asking for the test.

WHAT DO THE RESULTS MEAN?

If a drug test result is positive, it means that one or more drugs were found in amounts that suggest drug use or misuse. If you have a positive test result, you'll usually have a follow-up urine test to make sure the first test was correct. That's because the type of urine testing that's often done first can have false positive results.

A false positive means that the test showed you had a drug in your sample when it really wasn't there. False positives can happen if the test reacts with other chemicals in your body from certain over-the-counter medicines, prescriptions, and foods.

The follow-up drug test after a false positive result is called a confirmatory test. It is a more sensitive test that provides more accurate results.

If a drug test result is negative, it means that either:

The drugs that were tested were not found in your sample.

A very small amount of drugs was found, but not enough to count as a positive test result for drugs.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A DRUG TEST?

If you test positive for a legal drug prescribed by your provider, your employer can't punish you, unless the drug is affecting your ability to perform your job.

If you test positive for marijuana and live in a state where it is legal, employers may still be able to punish you in some states. Employers may do this because they want a drug-free workplace. Also, marijuana is still illegal under federal law.

You can buy home tests for many illegal and prescription drugs. There are two types of tests:

At-home tests let you do the entire test at home and get rapid results. But if the test result is positive for drugs, you'll need to have a follow-up lab test to check the accuracy of the result. Before you buy an at-home test, check to see if it includes a kit for collecting a sample, often urine or saliva, to send to a lab in case the home test is positive.

A self-collection test has a kit for gathering urine, saliva, or another type of sample to send to a lab for testing.

When using home tests, be careful to follow all the instructions. It's best to talk with your provider about using home drug testing so you can decide if it's right for your situation.

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Drug Use Screening Tests

<https://medlineplus.gov/lab-tests/drug-use-screening-tests/>

DRUG USE SCREENING TESTS

WHAT ARE DRUG USE SCREENING TESTS?

Drug use screening tests are questionnaires you answer about your drug use. Drugs are chemical substances that can change how your body and mind work. They include prescription medicines, over-the-counter medicines, tobacco, and illegal drugs.

Drug use can cause problems in your work, relationships, and health. Drug use can include taking illegal drugs or misusing prescription or over-the-counter medicines. Misusing medicines means that you are taking them for a nonmedical reason, such as to get high. It can include taking more than your prescribed dose or taking someone else's medicine. It can also mean taking medicine in a different way than you are supposed to, such as crushing and snorting tablets.

Substance use disorder (SUD) is a treatable, chronic mental disorder. The disorder may affect your brain and behavior. Symptoms of SUD include a pattern of using drugs or alcohol, even if it causes a problem in your life. The most severe form of SUD is addiction. If you have an addiction, you may continue to use the substance even though you know it can cause you harm. A drug use screening test can help your health care provider check for a substance use disorder, and if you have one, make an effective treatment plan for you.

Commonly used drugs include:

Marijuana.

Prescription opioid pain relievers, which include oxycodone, hydrocodone, morphine, and codeine.

Club drugs, which are drugs that are most often used by young adults at bars, concerts, nightclubs, and parties. Some types include:

Methamphetamines, which are stimulant drugs.

Hallucinogens, which are drugs that cause hallucinations (seeing, hearing, and/or feeling things that are not real). They include LSD and Ecstasy (MDMA).

Methamphetamines, which are stimulant drugs.

Hallucinogens, which are drugs that cause hallucinations (seeing, hearing, and/or feeling things that are not real). They include LSD and Ecstasy (MDMA).

Cocaine, another type of stimulant drug.

Heroin, an illegal opioid.

Substance use can also refer to the misuse of other substances, including alcohol and inhalants. Inhalants contain substances that people inhale (breathe in) to get high. They are often products easily bought and can be found in the home or workplace, such as paint thinner and certain types of glue.

Other names: Drug Abuse Screen Test (DAST); DAST-10; Tobacco, Alcohol, Prescription medication, and other Substance use (TAPS), substance abuse screening; Screening, Brief Intervention, and Referral to Treatment (SBIRT); Cut down Annoyed, Guilty, Eye-opener (CAGE) tool; Car, Relax, Alone, Forget, Friends, Trouble (CRAFT) Screening Tool

WHAT ARE THEY USED FOR?

Drug use screening tests are used to check your risk of drug use or misuse and determine the best treatment for you. Some tests can also show if you are at low, moderate, or high risk for complications from drug use.

WHY DO I NEED A DRUG USE SCREENING TEST?

You may need a screening test if you have symptoms or behaviors that may be a sign that you are at risk for a substance use disorder. These include:

An urgent need to take the drug on a daily or regular basis.

Hiding your drug use from others.

Making excuses to take the drug.

Continuing to take the drug even if it causes problems in your personal relationships, work, school, and/or health.

Building up a tolerance to the drug. That means you need more and more of the drug to feel its effects.

Many people who use or misuse drugs don't know or want to admit they have a problem. If your family, friends, or co-workers express concerns about your drug use, talk to your provider about getting a screening. Your provider may also recommend a screening if they notice signs and symptoms of a substance use disorder.

WHAT HAPPENS DURING A DRUG USE SCREENING TEST?

Your primary care provider or a mental health provider may do a drug use screening. A mental health provider is a health care professional who specializes in diagnosing and treating mental health problems. Some mental health providers specialize in treating substance use disorders.

There are different types of drug use screening tests. Most of these questionnaires are also available online for self-testing. Each includes questions about your drug use and how it affects your life. The most commonly used drug use screening tests include:

Drug Abuse Screen Test (DAST, also known as DAST-10). This test contains 10 yes or no questions about how much and how often you take drugs. It also asks if the drugs are causing problems in your life and health. The answers are scored on a point system. A score of 8 or more may indicate that you have a problem with drug use.

Tobacco, Alcohol, Prescription medication, and other Substance use (TAPS). This test has two parts. The first section asks questions about your tobacco, alcohol, and other substance use in the past year. If you answer yes to using any of these substances, you will have additional questions in the second part. Your score will determine a risk level per substance.

Cut-down Annoyed, Guilty, Eye-opener (CAGE). This test is often used to check for alcohol misuse, but it is also used for drug use. But it's no longer recommended because there are better tests available. The test contains the following yes or no questions:

Have you ever felt the need to cut-down on your drug use?

Do you ever feel annoyed when people criticize your drug use?

Have you ever felt embarrassed or guilty about your drug use?

Eye-opener: Have you ever used drugs first thing in the morning?

Two or more "yes" answers may indicate signs of a substance use disorder.

Have you ever felt the need to cut-down on your drug use?

Do you ever feel annoyed when people criticize your drug use?

Have you ever felt embarrassed or guilty about your drug use?

Eye-opener: Have you ever used drugs first thing in the morning?

Car, Relax, Alone, Forget, Friends, Trouble (CRAFT). This test is designed for teens and includes yes or no questions about drug and alcohol use, such as:

Have you ever ridden in a car driven by someone (including yourself) who was high or had been using alcohol or drugs?

Do you ever use alcohol or drugs to relax, feel better about yourself, or fit in?

Do you ever use alcohol/drugs while you are alone?

Do you ever forget things you did while using alcohol or drugs?

Do your family or friends ever tell you that you should cut down on your drinking or drug use?

Have you gotten into trouble while you were using alcohol or drugs?

Two or more "yes" answers to these questions may indicate signs of a substance use disorder.

Have you ever ridden in a car driven by someone (including yourself) who was high or had been using alcohol or drugs?

Do you ever use alcohol or drugs to relax, feel better about yourself, or fit in?

Do you ever use alcohol/drugs while you are alone?

Do you ever forget things you did while using alcohol or drugs?

Do your family or friends ever tell you that you should cut down on your drinking or drug use?

Have you gotten into trouble while you were using alcohol or drugs?

These questionnaires may be used along with an approach called Screening, Brief Intervention, and Referral to Treatment (SBIRT). SBIRT helps to identify, reduce, and prevent risky substance use and find the best treatment. It focuses on helping you understand how substance use may affect your life, so you can make positive changes. If your test shows you have or are at risk for a serious drug use problem, a long-term treatment plan may be recommended.

Other screening tools may be available for specific groups, such as adolescents or pregnant women.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a drug use screening test.

ARE THERE ANY RISKS TO THE TEST?

There is no risk in taking a questionnaire.

WHAT DO THE RESULTS MEAN?

Results may show whether you have a substance use disorder and the severity of the problem. The results can help your provider develop a treatment plan that is right for you.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT DRUG USE SCREENING TESTS?

Treatment for drug use depends on the type of drug and the severity of your substance use disorder. Treatment options may include:

Brief or long-term counseling from a mental health provider who specializes in treating drug use.

Support groups. There are different approaches and formats to drug use support groups. Talk with your provider to find a group and approach that's right for you.

Medicine. Certain medicines can help establish normal brain function and decrease cravings for the drug. This may help prevent a relapse or return to the use of the drug. Medicines can also make it safer to handle the unpleasant and sometimes dangerous symptoms of withdrawal (symptoms that can happen when you cut back or stop taking drugs).

Residential treatment programs. If you have a more serious drug problem, you may be helped by a stay in a facility that specializes in treating drug use. These facilities offer a range of treatment services. Stays can last anywhere from one month to a year, depending on your condition.

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Dysphagia Tests

<https://medlineplus.gov/lab-tests/dysphagia-tests/>

DYSPHAGIA TESTS

WHAT ARE DYSPHAGIA TESTS?

Dysphagia is a term that means difficulty swallowing. Usually, when you swallow, food moves easily from your mouth, down your throat, and into your stomach. During this process, the food travels through a muscular tube called the esophagus.

It's normal to occasionally have trouble swallowing, such as if you don't chew some food well enough or if you eat too fast. But if you have dysphagia, trouble swallowing is an ongoing problem. It can take more time and effort to move food from your mouth to your stomach. When you try to swallow liquids, food, or even your saliva, you may cough or choke. Swallowing may be uncomfortable or painful, and these issues may even stop you from swallowing.

Dysphagia can happen at any age, but it's more common if you are an older adult or if you have certain neurologic diseases. Many different conditions can cause dysphagia, and some of them can be very serious.

Dysphagia tests can diagnose different types of dysphagia and may help screen for or diagnose various conditions that can cause it. It's important to get a diagnosis, because if you have dysphagia and don't treat it, you might be at risk for:

Choking

Dehydration (a loss of too much fluid)

Malnutrition (not getting enough food for your body to work properly)

Aspiration (food or liquids getting into your airway), which can lead to pneumonia

Other names: bedside swallow, clinical swallow, dysphagia screening tool, fiberoptic endoscopic evaluation of swallow (FEES), barium swallow, esophagram, videofluoroscopic swallow study (VFSS), upper endoscopy

WHAT ARE THEY USED FOR?

Dysphagia tests are used to find out why you have trouble swallowing.

WHY DO I NEED A DYSPHAGIA TEST?

You may need this test if you have symptoms of dysphagia. These can include:

Coughing or gagging when swallowing

Feeling of food being stuck in your throat

Pain when swallowing

Being unable to swallow

Food or liquids coming back up through your throat or mouth, including from GERD (gastroesophageal reflux disease)

Frequent heartburn

A hoarse or weaker voice

Weight loss

Drooling

If food has gotten stuck in your airway and you're having trouble breathing, call 911 for help right away.

WHAT HAPPENS DURING A DYSPHAGIA TEST?

There are several different types of dysphagia tests. A radiologist or a radiology technician most often does these tests. A radiologist is a doctor who specializes in using imaging tests to diagnose and treat diseases and injuries. Depending on your symptoms, you may have one or more of the following tests:

Bedside Swallow Screen

You will sit upright in a bed or chair.

A health care provider may ask you questions about your symptoms and how long you've had trouble swallowing.

You may be asked to do certain movements, such as smacking your lips together or sticking out your jaw.

You will be given different foods and drinks to swallow. These may include water, other liquids, soft foods, and solid foods.

The provider will check your teeth, lips, jaw, cheeks, and neck while you swallow.

A bedside swallow test is a common type of dysphagia screening tool. Other tools include a questionnaire or discussion about past or current swallowing difficulties and a water swallow test. During a water swallow test, your provider will observe how easily you can drink water from a cup.

Fiberoptic Endoscopic Evaluation of Swallow (FEES)

You will sit upright in a bed or chair.

A provider will slide an endoscope (a thin, flexible tube with a light and camera) into your nose and to the back of your throat.

You will eat bits of food and/or drink liquids that may be colored with a dye.

The endoscope allows your provider to watch food travel down your throat as you swallow.

Upper Endoscopy

You will lie on your side on an exam table.

An IV (intravenous) line will be placed in your arm or hand.

Medicine to relax you will be injected intravenously (by IV).

Your provider may spray a numbing medicine on the back of your throat.

Once the relaxing and numbing medicines have taken effect, your provider will insert an endoscope into your mouth and throat.

The endoscope will take pictures of your esophagus, stomach, and part of your small intestine.

The tube may also have a tool attached that can be used to take a biopsy (removal of a small sample of tissue) to examine after the procedure.

Videofluoroscopic Swallow Study (VFSS), also known as a Modified Barium Swallow

You will stand or sit on an x-ray table.

You will be given different foods and liquids covered with barium. Barium is a substance that coats your internal organs and tissues so they can be seen more clearly on an x-ray. This allows your provider to see how you swallow.

While you swallow, a special x-ray called fluoroscopy will track the barium-coated food in real-time as a live video as it moves through your mouth, throat, and esophagus.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know if there are any special instructions to follow. You may need to fast (not eat or drink) for several hours or more before your test. You may be asked to avoid doing anything that will coat your throat, such as smoking, chewing gum, or sucking on hard candy.

You may need to stop taking certain medicines before the test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

You may also want to make a list of your swallowing problems ahead of time. Then, you can share them with your provider during your test.

If you are having an upper endoscopy, you will be given a medicine that can make you drowsy, so you should arrange for someone to take you home.

ARE THERE ANY RISKS TO THE TESTS?

There is very little risk to having a bedside swallow, but there is a small chance that fluid might get into your lungs during the procedure.

During a FEES or upper endoscopy, you may feel some discomfort when the endoscope is inserted, but serious complications are rare. There is a very small risk of getting a tear in your intestine. If you had a biopsy, there is a small risk of bleeding at the site. Bleeding usually stops without treatment.

You should not have a VFSS, which is a type of x-ray, if you are pregnant or think you may be pregnant. The radiation

from these x-rays can be harmful to a developing baby. For others, there is little risk to having these tests. The radiation dose is very low and not considered harmful for most people. But talk to your provider about all the x-rays you've had in the past. The risks from radiation exposure⁵ may be linked to the number of x-ray treatments you've had over time. Tell your provider if you have any known allergies to the barium used in the test. Your poop may look white for several days after the test as you pass the barium. You may have constipation⁵ if all the barium doesn't pass out of your body.

WHAT DO THE RESULTS MEAN?

Your results may show your provider where the problem is or what type of dysphagia you might have. These can include:

Oral cavity dysphagia, a disorder of the mouth in which you have difficulty chewing or breaking down food. Causes may include:

Weakness after a stroke

Muscular or nerve problems

Oropharyngeal dysphagia, a disorder of the throat⁵ in which you have difficulty starting to swallow. Causes may include:

Certain types of cancer

Neurologic diseases such as multiple sclerosis⁵ or Parkinson's disease

Pharyngoesophageal diverticulum, a small pouch that forms and collects food particles in your throat

Esophageal dysphagia, a disorder of the esophagus⁵ in which you have difficulty moving food down your esophagus. Causes may include:

Esophageal stricture (narrowing of your esophagus)

Esophageal cancer

GERD (gastroesophageal reflux disease), a condition in which the contents of your stomach leak backward into your esophagus

A bedside swallow screen or other type of dysphagia screening tool will only show if you are at risk for one of the above disorders. If your results show you are at risk, your provider will probably order follow-up testing.

If you have questions about your results, talk to your provider. They may also consider your symptoms and medical history to understand your test results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT DYSPHAGIA TESTS?

If you are having trouble swallowing, your provider may refer you to one of the following specialists:

A speech and language pathologist, a provider that specializes in diagnosing and treating speech, language, and communication disorders.

An otolaryngologist, a doctor who specializes in diagnosing and treating disorders of the ear, nose, and throat.

A gastroenterologist, a doctor who specializes in diagnosing and treating disorders of the digestive system.

A neurologist, a doctor who specializes in diagnosing and treating disorders of the brain, spinal cord, and nervous system.

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Elastography

<https://medlineplus.gov/lab-tests/elastography/>

ELASTOGRAPHY

WHAT IS ELASTOGRAPHY?

Elastography is a type of imaging test that checks your organs to see if they are stiffer than normal. Stiff areas in your organs may be a sign of disease. Elastography is mainly used to check the liver for stiffness. Stiff areas in the liver are a sign of scar tissue (fibrosis) caused by liver disease.

Liver fibrosis happens when certain diseases damage the liver over time. When the liver tries to heal, it forms more scar tissue. Many conditions that cause ongoing or repeated liver damage can lead to liver fibrosis. Conditions that commonly cause fibrosis include:

Hepatitis C and Hepatitis B. These liver diseases are caused by viruses.

Alcohol use disorder (AUD). AUD can cause alcohol-associated liver disease, which includes the buildup of too much fat in the liver. The fat can cause liver inflammation (swelling) which damages liver tissue.

Nonalcoholic fatty liver disease (NAFLD). This type of fat buildup often happens in people who have diabetes, obesity, and/or high levels of cholesterol in their blood. It is not from drinking too much alcohol.

As fibrosis develops, scar tissue replaces healthy liver cells and prevents your liver from working normally. The scar tissue can reduce blood flow in your liver. The lack of blood causes more liver damage and more fibrosis.

Without treatment, fibrosis can become severe. Severe fibrosis is called cirrhosis. With cirrhosis, liver scarring is permanent. It can cause serious health problems, including liver failure. Cirrhosis also increases your chance of getting liver cancer.

Fibrosis and early cirrhosis don't always cause symptoms. But elastography can help find liver scarring early, so treatment can start before the damage becomes severe. There are two types of liver elastography tests:

Ultrasound elastography is also called transient elastography. An ultrasound device uses sound waves to send vibrations into your liver. The device measures how fast the vibrations move through your liver. If you have areas of stiff liver tissue, the vibrations will move faster through that area. A computer uses the measurements to make an image that shows any stiffness in your liver tissue, which is a sign of fibrosis.

MRE (magnetic resonance elastography) sends vibrations into your liver that are measured with magnetic resonance imaging (MRI). MRI is a procedure that uses powerful magnets and radio waves to create images of organs and structures inside the body. In an MRE test, a computer program makes a map that shows any stiff areas of your liver.

Other names: liver elastography, transient elastography, FibroScan, MR elastography

WHAT IS IT USED FOR?

Liver elastography is used to check for liver fibrosis (scarring). It may be used instead of a liver biopsy, which involves removing a piece of liver tissue for testing.

The information from an elastography test can help:

Diagnose liver conditions in people who have a high risk for liver disease

Show how serious liver disease is

Guide treatment choices for liver disease

Monitor how well treatment for liver disease is working

Help predict the chance that liver disease will cause certain serious problems

Check how much fat is in the liver

WHY DO I NEED AN ELASTOGRAPHY TEST?

You may need an elastography test to check for liver fibrosis (scarring) if you:

Have liver disease and your health care provider needs information to guide treatment choices.

Have had other liver tests that show you may have inflammation or damage in your liver.

Are more likely to develop cirrhosis (severe liver scarring). Conditions that increase your risk for cirrhosis include:

Alcohol use disorder (AUD)

Type 2 diabetes

Chronic (long-lasting) hepatitis B or C

Certain autoimmune diseases or genetic disorders

Long-term use of certain medicines

Alcohol use disorder (AUD)

Type 2 diabetes

Chronic (long-lasting) hepatitis B or C

Certain autoimmune diseases or genetic disorders

Long-term use of certain medicines

Have signs or symptoms of cirrhosis.

Cirrhosis may not cause symptoms until your liver is badly damaged. Symptoms of early cirrhosis may include:

Feeling tired or weak

Having a poor appetite

Losing weight without trying

Nausea and vomiting

Mild pain in the upper right side of your abdomen (belly)

As your liver condition gets worse, you may have other symptoms, including:

Bruising and bleeding easily

Confusion, memory loss, or sleep disorders

Swelling in your legs or feet

Bloating or full feeling in your abdomen from fluid buildup

Itchy skin

Dark urine (pee)

Jaundice (a condition that causes your skin and eyes to turn yellow)

WHAT HAPPENS DURING AN ELASTOGRAPHY TEST?

An ultrasound elastography is often done by a radiology technician, who is a health care professional trained to do imaging tests. Sometimes a radiologist does the test. A radiologist is a doctor who specializes in using medical imaging tests to diagnose medical conditions.

An elastography test is quick and usually includes these steps:

You'll remove clothing covering your right abdomen (belly) to expose the skin over your liver.

You'll lie on a table and the technician will spread a special gel on your skin.

The technician will hold a wand-like device, called a transducer, and move it across your skin.

You'll hold your breath for 10 to 15 seconds while the device sends a series of vibrations into your liver. You may feel the device pulse against your skin, but it doesn't hurt.

The vibrations will travel through your liver and "bounce" back to the device which sends the information to a computer.

The computer makes an image of your liver showing any areas of stiff tissue and other signs of liver disease.

MRE (magnetic resonance elastography) is done with an MRI (magnetic resonance imaging) machine. During an MRE procedure:

You'll lie on a narrow exam table.

The technician will place some small devices on your lower right abdomen. One device, called a "driver," will send painless vibrations into your liver. The other devices send and receive radio waves to a computer.

The table will slide into a large tunnel-like MRI machine. You will hear loud noises as the machine takes pictures. You may be given earplugs or headphones to block the noise.

Inside the MRI, you'll be asked to hold your breath for 10 to 15 seconds when the vibrations are going through your liver.

The computer will make images of your liver that show any scarring and/or other signs of liver disease.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will tell you how to prepare for a liver elastography. In general, it's best to wear a loose-fitting shirt or sweater that you can easily pull up. You may be told to fast (not eat or drink) for several hours before your test. If you're having an MRE, you'll need to remove all metal from your body. If you have any medical devices or metal inside your body let your provider know. The scanner can heat metal and cause burns.

ARE THERE ANY RISKS TO THE TEST?

There are no known risks to having an ultrasound elastography. There is little risk to having an MRE for most people. If being inside a small space makes you nervous, tell your provider before your test. Your provider may give you medicine to help you relax. If you are or could be pregnant, let your provider know.

WHAT DO THE RESULTS MEAN?

Your elastography images will be studied by a radiologist, who will send your results to your provider. Your results will show how much stiffness was found in your liver. This tells you how much liver scarring (fibrosis) you have. Your results will be given a fibrosis score which is based on the amount of stiffness in your liver and the cause:

F0 to F1 is normal, which means you have little or no scarring in your liver.

F2 means you have moderate scarring.

F3 means you have severe scarring.

F4 means you have cirrhosis (very severe scarring).

Your results also show how much fat buildup you have in your liver. The measurement of fat in your liver is called a "CAP score." It tells you the percentage of your liver that is fat. The amount of fat in a normal liver is 5% or less. Higher amounts mean you have fatty liver disease, which may cause fibrosis.

If you have mild to severe fibrosis and/or fatty liver disease, your condition can improve if you treat the cause and make lifestyle changes, including diet and exercise. If you have cirrhosis, treating the cause may keep liver scarring from getting worse. Successful treatment may slowly improve some of the scarring.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ELASTOGRAPHY?

Certain conditions can affect the accuracy of an elastography test. These include:

Liver inflammation (swelling) from an illness or drinking alcohol

Tumors in your liver

Obesity

Abnormal fluid buildup your abdomen (belly)

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RELATED HEALTH TOPICS

Alcohol Use Disorder (AUD)

Cirrhosis

Fatty Liver Disease

Hepatitis

Jaundice

Liver Cancer

Liver Transplantation

MRI Scans

RELATED MEDICAL TESTS

Albumin Blood Test

Alkaline Phosphatase

ALT Blood Test

AST Test

Bilirubin Blood Test

Hepatitis Testing

How to Cope with Medical Test Anxiety

Liver Function Tests

Ultrasound

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Electrocardiogram

<https://medlineplus.gov/lab-tests/electrocardiogram/>

ELECTROCARDIOGRAM

WHAT IS AN ELECTROCARDIOGRAM (EKG) TEST?

An electrocardiogram (EKG) test is a simple, painless, and quick test that records your heart's electrical activity. Each time your heart beats, an electrical signal travels through your heart. The signal triggers your heart's four chambers to contract (squeeze) in the proper rhythm so that your heart can pump blood to your body.

An EKG recording of these signals looks like wavy lines. Your provider can read these lines to look for abnormal heart activity that may be a sign of heart disease or damage.

An EKG can show:

How fast your heart is beating

Whether the rhythm of your heartbeat is steady or irregular

The strength and timing of the electrical signals passing through each part of your heart

Sometimes information from an EKG can help measure the size and position of your heart's chambers.

An EKG is often the first test you'll have if you have signs of a heart condition. It may be done in your provider's office, an outpatient clinic, in a hospital before surgery, or as part of another heart test called a stress test.

An EKG test is also called an ECG. EKG is based on the German spelling, elektrokardiogramm. EKG may be preferred over ECG to avoid confusion with an EEG, a test that measures brain waves.

Other names: ECG test, Holter monitor, ambulatory electrocardiography, continuous electrocardiograms, transtelephonic event monitors

WHAT IS IT USED FOR?

An EKG test is used to help diagnose and monitor many types of heart conditions and their treatment. These conditions include:

Arrhythmia

Cardiomyopathy

Coronary artery disease

Heart attack

Heart failure

Heart valve diseases

Congenital heart defects

EKG tests are mainly used for people who have symptoms of a heart condition or have already been diagnosed with a heart condition. They are not generally used to screen people who don't have symptoms unless they have an increased risk of developing heart disease. Your provider can explain your risk for heart disease and let you know if need to have an EKG test. In certain cases, your provider may have you see a cardiologist, a doctor who specializes in heart diseases.

WHY DO I NEED AN EKG TEST?

You may need an EKG test if you have symptoms of a heart condition, including:

Chest pain

Rapid or irregular heartbeat

Shortness of breath

Dizziness

Fatigue

A decrease in your ability to exercise

You may also need an EKG to:

Find out if you had a heart attack in the past but didn't know it

Monitor your heart if you have a known heart condition

Check how well your heart treatment is working, including medicine and/or a pacemaker

Check your heart health:

Before having surgery

If you have an increased risk for developing heart disease because:

Heart disease runs in your family

You have another condition, such as diabetes, that makes your risk higher than normal

Before having surgery

If you have an increased risk for developing heart disease because:

Heart disease runs in your family

You have another condition, such as diabetes, that makes your risk higher than normal

Heart disease runs in your family

You have another condition, such as diabetes, that makes your risk higher than normal

WHAT HAPPENS DURING AN EKG TEST?

An EKG test only takes a few minutes. It generally includes these steps:

You will lie on an exam table.

A provider will place several electrodes (small sensors that stick to your skin) on your arms, legs, and chest. The provider may need to shave body hair to make sure the electrodes stay on.

The electrodes are attached by wires to a computer or a special EKG machine

You will lie very still while your heart's electrical activity is recorded on a computer or printed on paper by an EKG machine.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an EKG test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having an EKG. You may feel a little discomfort or skin irritation after the electrodes are removed. The EKG doesn't send any electricity to your body. It only records electrical signals from your heart, so there's no risk of electric shock.

WHAT DO THE RESULTS MEAN?

Your provider will check your EKG results for a steady heartbeat and rhythm. If your results are not normal, it may be a sign of a heart condition. The specific condition depends on which part of your EKG wasn't normal.

You may need to have other heart health tests before your provider can make a diagnosis. Your provider can explain what your test results mean for your heart health and treatment.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN EKG?

An EKG is a "snapshot" of your heart's activity over a very short time. If you have heart symptoms that come and go, a regular EKG may not catch the problem. In that case, your provider may recommend that you wear a small portable EKG monitor that can record your heart for days or longer while you do your normal activities. You may also need a longer EKG recording if your provider wants to check how well your heart is working after a heart attack or to see if treatment is helping you.

There are many types of long-term EKG monitors. The two main groups are Holter monitors, which can be worn for up to two days, and event monitors, which may record your heart activity for weeks to years depending on the type.

A Holter monitor is about the size of a small camera. You usually wear it on a belt or strap around your neck for a day or two. Wires under your clothes attach to electrodes that stick to your chest. The monitor records your heart's electrical signals the whole time you're wearing it. You may be asked to keep a diary of your symptoms during the test period. After the test period, you remove the monitor and return it according to the instructions. A provider will review the recording of your heart's electrical activity from the monitor.

An event monitor records your heart's electrical activity when you press a button or when the device detects abnormal heart activity. There are two main types of event monitors:

Event monitors that you wear or carry with you. You wear some monitors on your chest or wrist. Other monitors are designed to carry. If you have symptoms, you hold the monitor to your chest. These event monitors may be used for weeks to months. Some of them wirelessly transmit information about your heart to a provider. Others must be returned so a provider can examine the recorded information.

Event monitors that are inserted under the skin of your chest. These are called implantable event monitors. They are put under your skin during minor surgery that's often done in a doctor's office. These monitors can track your heart's electrical activity for years. You may need this type of EKG monitor if you had a stroke or frequent fainting, and your provider hasn't found the cause. Implantable monitors wirelessly transmit the information they record so your provider can regularly check it.

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RELATED HEALTH TOPICS

Arrhythmia
Coronary Artery Disease
Heart Attack
Heart Diseases
Heart Failure
Heart Health Tests
Pacemakers and Implantable Defibrillators

RELATED MEDICAL TESTS

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Creatine Kinase
Doppler Ultrasound
Heart Disease Risk Assessment
How to Cope with Medical Test Anxiety
Lipoprotein (a) Blood Test
Natriuretic Peptide Tests (BNP, NT-proBNP)
Sleep Study
Stress Tests
Troponin Test

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care provider if you have questions about your health.

Electrolyte Panel

<https://medlineplus.gov/lab-tests/electrolyte-panel/>

ELECTROLYTE PANEL

WHAT IS AN ELECTROLYTE PANEL?

Electrolytes are minerals that have an electrical charge when they are dissolved in water or body fluids. You have electrolytes in your blood, urine, tissues, and other body fluids. They are important because they help:

Balance the amount of water in your body

Balance your body's acid-base (pH) level

Move nutrients into your cells

Move waste out of your cells

Support your muscle and nerve function

Keep your heart rate and rhythm steady

Keep your blood pressure stable

Keep your bones and teeth healthy

Electrolytes are in your blood, urine (pee), tissues, and other body fluids. An electrolyte panel is used to check for electrolyte, fluid, or pH imbalances.

An electrolyte panel, also known as a serum electrolyte test, is a blood test that measures levels of the body's main electrolytes:

Sodium, which helps control the amount of fluid in your body. It also helps your nerves and muscles work properly.

Chloride, which also helps control the amount of fluid in your body. In addition, it helps maintain healthy blood volume and blood pressure.

Potassium, which helps your cells, heart, and muscles work properly.

Bicarbonate, which helps maintain your body's acid- base balance (pH). It also plays an important role in moving carbon dioxide through the bloodstream.

Calcium, which helps make and keep your bones and teeth strong.

Magnesium, which helps your muscles, nerves, and heart work properly. It also helps control blood pressure and blood glucose (blood sugar) levels.

Phosphate, which works together with the mineral calcium to build strong bones and teeth.

You get these electrolytes from the foods you eat and the fluids you drink.

The levels of electrolytes in your body can become too low or too high. This can happen when the amount of water in your body changes. The amount of water that you take in should equal the amount you lose. If something upsets this balance, you may have too little water (dehydration) or too much water (overhydration).

Abnormal levels of any of these electrolytes can be a sign of a serious health problem, including kidney disease, high blood pressure, and a life-threatening type of arrhythmia (irregular heartbeat).

Other names: serum electrolyte test, electrolyte blood test, electrolyte lab test, lytes, sodium (Na), potassium (K), chloride (Cl), carbon dioxide (CO₂)

WHAT IS IT USED FOR?

An electrolyte panel is often part of a routine blood screening or a comprehensive metabolic panel. The test may also be used to find out if your body has a fluid imbalance or an imbalance in acid and base levels.

Electrolytes are usually measured together. But sometimes they are tested individually. Separate testing may be done if your health care provider thinks that there might an imbalance of a specific electrolyte.

WHY DO I NEED AN ELECTROLYTE PANEL?

You may need this test if you have symptoms indicating that your body's electrolytes may be out of balance. These

include:

Nausea and vomiting

Confusion and irritability

Weakness

Irregular or fast heartbeat (arrhythmia)

Diarrhea or constipation

Fatigue

Headaches

Muscle weakness, cramps, or spasms

Numbness or tingling in your arms, legs, fingers, and toes

WHAT HAPPENS DURING AN ELECTROLYTE PANEL?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

No special preparations are needed for an electrolyte panel.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will include measurements for each electrolyte. Abnormal electrolyte levels can be caused by many different conditions, including:

Dehydration.

Kidney disease.

Heart disease.

Diabetes.

Lung infection (such as tuberculosis).

Malabsorption, a condition in which your body is not getting enough nutrients from the foods you eat.

Liver disease.

Acidosis, a condition in which you have too much acid in your blood. It can cause nausea, vomiting, and fatigue.

Alkalosis, a condition in which you have too much base in your blood. It can cause irritability, muscle twitching, and tingling in the fingers and toes.

An electrolyte imbalance can also happen if you take certain medicines, such as:

Antibiotics

Antacids, especially if you take large amounts

Chemotherapy medicines

Diuretics ("water pills"), which are used to treat high blood pressure and other conditions

Steroid medicines

Your specific results will depend on which electrolyte or electrolytes are affected and whether levels are too low or too high. If your electrolyte levels were not in the normal range, it doesn't necessarily mean you have a medical problem needing treatment. Many factors can affect electrolyte levels. These include taking in too much fluid or losing fluid because of vomiting or diarrhea. Also, certain medicines such as antacids and blood pressure medicines may cause

abnormal results.

An electrolyte imbalance means that the level of an electrolyte is too low or too high. The names of the imbalances are:

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ELECTROLYTE PANEL?

Your provider may order another test, called an anion gap blood test, along with your electrolyte panel. Some electrolytes have a positive electric charge. Others have a negative electric charge. The anion gap blood test is a measurement of the difference between the negatively charged and positively charged electrolytes. If the anion gap is either too high or too low, it may be a sign of a serious health problem.

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RELATED HEALTH TOPICS

Calcium

Fluid and Electrolyte Balance

Minerals

Potassium

Sodium

RELATED MEDICAL TESTS

Aldosterone Test

Anion Gap Blood Test

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Electromyography (EMG) and Nerve Conduction Studies

<https://medlineplus.gov/lab-tests/electromyography-emg-and-nerve-conduction-studies/>

ELECTROMYOGRAPHY (EMG) AND NERVE CONDUCTION STUDIES

WHAT ARE ELECTROMYOGRAPHY (EMG) AND NERVE CONDUCTION STUDIES?

Electromyography (EMG) and nerve conduction studies are tests that check how well your muscles and the nerves that control them are working. These nerves control your muscles by sending out electrical signals to make your muscles move. As your muscles react by tightening (contracting), they give off electrical activity, which can then be measured.

An EMG test looks at the electrical signals your muscles make when they are at rest and when used. A healthy muscle should not give off any electrical signals when you aren't moving it. If your muscle is damaged, it may show electrical activity while at rest or activity that is not normal while using it.

A nerve conduction study measures how fast and how well your body's electrical signals move along your nerves. A damaged nerve has a slower and weaker signal. This test can help check for nerve damage.

EMG tests and nerve conduction studies can help find out if you have a health condition that has damaged your muscles or nerves or how they work together. These tests can be done separately, but they are usually done at the same time.

Other names: electrodiagnostic study, EMG test, electromyogram, NCS, nerve conduction velocity, NCV

WHAT ARE THEY USED FOR?

EMG and nerve conduction studies are used to help check for many kinds of muscle and nerve disorders. An EMG test helps find out if muscles are responding the right way to nerve signals. Nerve conduction studies help to check for nerve damage or disease. When EMG tests and nerve conduction studies are done together, it helps providers tell if your symptoms are caused by a muscle or a nerve disorder.

WHY DO I NEED AN EMG TEST AND A NERVE CONDUCTION STUDY?

You may need these tests if you have symptoms of a muscle or nerve disorder. These symptoms include:

Muscle weakness

Tingling or numbness in arms, legs, hands, feet, and/or face

Muscle cramps, spasms, and/or twitching

Paralysis of any muscles

WHAT HAPPENS DURING AN EMG TEST AND NERVE CONDUCTION STUDY?

For an EMG test:

You will sit or lie on a table or bed.

Your provider will clean the skin over the muscle being tested.

Your provider will place a small needle with an electrode into the muscle. This is connected to a machine used to record your muscle's electrical activity when you move and relax it. You may have slight pain or discomfort when the electrode is inserted.

You will stay still while the machine records your muscle activity at rest.

You will tighten (contract) the muscle slowly while the machine records this muscle activity.

The electrode may be moved to record activity in different muscles.

The electrical activity is recorded and shown on a video screen. It looks like wavy and spiky lines. The activity may also be recorded and sent to an audio speaker. The audio may make popping sounds when you're contracting your muscles.

An EMG test may take from 30 to 60 minutes.

For a nerve conduction study:

You will sit or lie on a table or bed.

Your provider will stick electrodes to your skin above a nerve they're checking. The electrodes, called stimulating electrodes, deliver a mild electrical pulse.

Your provider will attach recording electrodes to the muscles controlled by those nerves. These electrodes will record the muscles' response to the electrical stimulation from the nerve.

Your provider will send a small pulse of electricity through the stimulating electrodes. This stimulates the nerve to send a signal to the muscle.

This may cause a mild tingling feeling.

Your provider will write down how long it takes for your muscle to respond to the nerve signal.

The speed of the muscle response is called the conduction velocity.

A nerve conduction test may take from 15 minutes to over an hour. The length of time depends on how many nerves and muscles are tested.

If you are having both tests, the nerve conduction study will be done first.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THESE TESTS?

Tell your provider if you have a pacemaker or cardiac defibrillator. Special steps will need to be taken before the test if you have one of these devices.

Tell your provider if you take blood thinner medicines. If you're taking blood thinners, you may need to watch for more bleeding from the needle.

Wear loose, comfortable clothing for easy access to the test area. It can be easily removed if you need to change into a hospital gown.

Make sure your skin is clean. Don't use lotions, creams, or perfumes for a day or two before the test. They can change the results of the test.

ARE THERE ANY RISKS TO THE TESTS?

You may feel mild pain during an EMG test. Let your provider know if your pain becomes very uncomfortable because this may change your test results. You may have a tingly feeling, like mild static electricity, during a nerve conduction study.

The muscles that were tested may be sore for a few days after the test. You may also have some bruises on your skin where the needles were placed.

WHAT DO THE RESULTS MEAN?

If your results are not normal, it can be a sign of many different conditions. Depending on which muscles or nerves are affected, it could mean one of the following:

Carpal tunnel syndrome, a common condition that affects nerves in the hand and arm. It causes numbness, tingling, weakness, or pain in your wrist or hand.

Herniated disk, a condition that happens when a part of your spine, called a disk, is damaged. Part of the disk bulges or pushes out, putting pressure on the spine which causes pain and numbness.

Guillain-Barré syndrome, a rare disorder that causes your immune system to attack your nerves. It can lead to numbness, tingling, and paralysis.

Myasthenia gravis, a rare disease that causes weakness in the muscles you can control. They may include your muscles for eye movement, facial expressions, or other muscles.

Muscular dystrophy (MD), a group of genetic diseases that cause muscle weakness. This weakness can lead to trouble walking and the ability to care for yourself.

Charcot-Marie-Tooth disease (CMT), a group of nerve disorders that are inherited (passed down through families). CMT causes nerve damage and muscle weakness, mostly in the arms and legs.

Amyotrophic lateral sclerosis (ALS), also known as Lou Gehrig's disease. It is a disorder that attacks nerve cells in your brain and spinal cord. It affects all the muscles you use to move, speak, eat, and breathe. There is no cure for ALS, and

it gets worse over time.

To understand the results of an EMG test or nerve conduction study, your provider will consider your symptoms, medical history, and the results of other tests.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Epithelial Cells in Urine

<https://medlineplus.gov/lab-tests/epithelial-cells-in-urine/>

EPITHELIAL CELLS IN URINE

WHAT IS AN EPITHELIAL CELLS IN URINE TEST?

Epithelial cells cover the inside and outside of the surfaces of your body. They are found on your skin, blood vessels, and organs, including your urinary tract (which includes your kidneys, ureters, bladder, and urethra).

An epithelial cells in urine test looks at a sample of your urine (pee) under a microscope to count the number of epithelial cells. It's normal to have a small number of certain types of epithelial cells in urine. But a large number may be a sign of a urinary tract infection, kidney disease, or other serious medical condition.

Other names: microscopic urine analysis, microscopic examination of urine, urine test, urine analysis, UA

WHAT IS IT USED FOR?

An epithelial cells in urine test is a part of a urinalysis, a test that measures different substances in your urine. It may be part of a routine checkup, or your health care provider may order the test if you have signs of a urinary tract problem.

A urinalysis may include a visual check of your urine sample, tests for certain chemicals, and an examination under a microscope to look for certain types of cells. An epithelial cells in urine test is part of a microscopic exam of your urine.

WHY DO I NEED AN EPITHELIAL CELLS IN URINE TEST?

Your provider may have ordered an epithelial cells in urine test as part of your routine checkup or if your visual or chemical urine test results weren't normal. You may also need this test if you have symptoms of a urinary or kidney disorder. These symptoms may include:

Frequent and/or painful urination

Abdominal (belly) pain

Back pain

Blood in your urine

WHAT HAPPENS DURING AN EPITHELIAL CELLS IN URINE TEST?

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If your provider has ordered other urine or blood tests, you may need to fast (not eat or drink) for several hours before the test.

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. They will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having the test.

WHAT DO THE RESULTS MEAN?

Results are often reported as a general amount, such as "few," moderate," or "many" cells. A normal result is typically "few" cells.

"Moderate" or "many" cells may be a sign of a medical condition, such as:

Urinary tract infection

Yeast infection

Kidney disease

Liver disease

Certain types of cancer

If your results are not normal, it doesn't always mean that you have a medical condition that needs treatment. You may need more tests before your provider can make a diagnosis. To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN EPITHELIAL CELLS IN URINE TEST?

Three types of epithelial cells line the urinary tract. They are called transitional cells, renal (kidney) tubular cells, and squamous cells.

If squamous epithelial cells are in your urine, it may mean your sample was contaminated. This means that the sample contains cells from another part of the body. This can happen if you do not clean your genital area well enough when collecting your urine sample with the clean catch method.

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Erythrocyte Sedimentation Rate (ESR)

<https://medlineplus.gov/lab-tests/erythrocyte-sedimentation-rate-esr/>

ERYTHROCYTE SEDIMENTATION RATE (ESR)

WHAT IS AN ERYTHROCYTE SEDIMENTATION RATE (ESR)?

Erythrocytes are red blood cells. The sedimentation rate is the time it takes for your red blood cells to settle at the bottom of a test tube. An erythrocyte sedimentation rate (ESR) is a blood test that can show if you have inflammation in your body. Inflammation is your immune system's response to injury, infection, and many other conditions, including immune system disorders, certain cancers, and blood disorders.

To do an ESR test, a sample of your blood is sent to a lab. A health care professional places the sample in a tall, thin test tube and measures how quickly your red blood cells settle or sink to the bottom of the tube. Typically, red blood cells sink slowly. But inflammation makes red blood cells stick together in clumps. These clumps of cells are heavier than single cells, so they sink faster.

If an ESR test shows that your red blood cells sink faster than expected, it may mean you have a medical condition causing inflammation. How quickly the red blood cells in your sample fall to the bottom of the test tube is a sign of how much inflammation you have. Faster ESR rates mean higher levels of inflammation. But an ESR test alone cannot diagnose which condition is causing the inflammation.

Other names: ESR, SED rate sedimentation rate; Westergren sedimentation rate

WHAT IS IT USED FOR?

An ESR test can be used with other tests to help diagnose and monitor conditions that cause inflammation. Many conditions cause inflammation, including arthritis, vasculitis, infections, and inflammatory bowel diseases (ulcerative colitis and Crohn's disease). An ESR may also be used to monitor an existing condition.

WHY DO I NEED AN ESR?

Your health care provider may order an ESR if you have symptoms of a condition that causes inflammation. Your symptoms will depend on the condition you may have, but they may include:

Headaches

Unexplained fever

Unexplained weight loss

Joint stiffness

Neck or shoulder pain

Loss of appetite

Anemia

WHAT HAPPENS DURING AN ESR?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR AN ESR?

You don't need any special preparations for this test. But if your provider ordered other tests on your blood sample, you

may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having an ESR. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your provider will use your ESR test results, medical history, symptoms, and other test results to make a diagnosis. An ESR test alone cannot diagnose conditions that cause inflammation.

A high ESR test result means your red blood cells sank faster than normal. This may be from a condition that causes inflammation, such as:

Giant cell arteritis, inflammation (swelling) of your arteries, usually in your scalp, neck, and arms

Arthritis

Systemic vasculitis, inflammation of your blood vessel walls, which can affect any of your organs

Polymyalgia rheumatica

Inflammatory bowel disease

Kidney disease

Infections

Rheumatoid arthritis and other autoimmune diseases

Heart disease

Certain cancers

A low ESR test result means your red blood cells sank slower than normal. This may be caused by conditions such as:

A blood disorder, such as:

Polycythemia, an increase in your red blood cells, which causes your blood to thicken

Sickle cell disease (SCD)

Leukocytosis, a very high white blood cell count (WBC)

Polycythemia, an increase in your red blood cells, which causes your blood to thicken

Sickle cell disease (SCD)

Leukocytosis, a very high white blood cell count (WBC)

Heart failure

Certain kidney and liver problems

If your ESR results are not normal, it doesn't always mean you have a medical condition that needs treatment. Normal ESR levels will vary, depending on your age and sex. Also, pregnancy, your menstrual cycle, aging, obesity, drinking alcohol regularly, and exercise can affect ESR results. Certain medicines and supplements may also affect your results, so be sure to tell your provider about any medicines or supplements you are taking.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ESR?

Because an ESR can't diagnose a specific disease, your provider may order other tests at the same time. Also, it's possible to have a condition that causes inflammation and still have a normal ESR result. A C-reactive protein (CRP) test is commonly done with an ESR to provide more information.

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Esophageal pH Test

<https://medlineplus.gov/lab-tests/esophageal-ph-test/>

ESOPHAGEAL PH TEST

WHAT IS AN ESOPHAGEAL PH TEST?

An esophageal pH test measures how often stomach acid enters your esophagus (the muscular tube that carries food and liquids from your mouth to your stomach). It also measures how long the acid stays there. The test involves placing a catheter (a thin tube) or a special device called a pH probe into your esophagus. The catheter or device will measure your acid level (known as pH level) for 24 to 96 hours.

The test can show if you have acid reflux or GERD (gastroesophageal reflux disease):

Acid reflux, also called gastroesophageal reflux (GER), happens when stomach acid flows back into the esophagus. You might notice acid reflux after eating a big meal or if you lie down after dinner. When the acid touches the esophagus, it can cause a burning feeling in your chest or neck. This is known as heartburn.Â

GERD is a more serious and long-lasting form of reflux. It causes repeated symptoms and can lead to complications over time.Â

Â While acid reflux and GERD are not life-threatening, the symptoms can be very uncomfortable and affect your quality of life. Also, if not treated, GERD can damage the esophagus and lead to more serious health problems.Â

Other names: esophageal pH monitoring 24-hour esophageal pH test, esophageal reflux monitoring, esophageal acidity test, pH monitoring, pH probe study, 48-hour Bravo esophageal pH test

WHAT IS IT USED FOR?

An esophageal pH test is most often used to diagnose acid reflux or GERD. It may also be used to see if treatment for GERD is effective.

WHY DO I NEED AN ESOPHAGEAL PH TEST?

You may need this test if you have symptoms of acid reflux. These include:

Heartburn

Swallowed food coming back into your mouth (regurgitation)

Trouble swallowing

Feeling like there's a lump in your throat

Sore throat

Cough

Frequent burping

WHAT HAPPENS DURING AN ESOPHAGEAL PH TEST?

An esophageal pH test may be done using a thin tube called a catheter or with a wireless pH probe (a small, capsule-like device).

During a catheter test:

Your health care provider will insert a small, thin tube through your mouth or nose. You may be asked to take drinks of water to swallow the tube down to your esophagus. The tube has a sensor that measures your acid level.

The other end of the tube will be connected to a small monitor that you wear around your waist or over your shoulder.

The monitor will measure your acid levels for 24 hours.

You will keep a diary of your symptoms and the food you eat during these 24 hours.

The next day, you will see your provider, who will remove the tube and monitor. You'll also give your provider the diary of

your symptoms and food eaten.

During a wireless test, also known as a pH probe study:

Your provider will attach a pH probe to the lining of your esophagus during an upper endoscopy, a procedure that is done using an endoscope. An endoscope is a thin, tube-like instrument with a light and camera.

You will get a medicine to block the pain and make you feel relaxed and sleepy during the procedure.

The probe will measure acid levels for 48 to 96 hours.

The probe will send the information to a device you wear on your wrist or belt.

You will need to keep a diary of your symptoms and the food you eat for 48 to 96 hours.

In 4 to 10 days, the probe will leave your body through a bowel movement.

After the probe stops transmitting (after 48-96 hours), you will give the diary to your provider.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You should not eat or drink (fast) for four to six hours before the procedure.

You may be asked to stop taking certain medicines, such as antacids, for 24 hours or for as long as two weeks before your test. Your provider will let you know which medicines to avoid and for how long.

Tell your provider if you have a pacemaker or an implantable heart defibrillator.

Tell your provider if you have had problems with your esophagus before or if you have a history of bleeding problems.

ARE THERE ANY RISKS TO THE TEST?

During a catheter test, you may feel some mild discomfort in your nose and throat, but it usually doesn't last long. You may also feel like gagging when the tube is inserted. Rarely, the test can cause a nosebleed.

During a wireless test, you may have a sore throat from the endoscope. In very rare cases, the test may cause a tear in the lining of the esophagus.

WHAT DO THE RESULTS MEAN?

If your results show high acid levels, it may be a sign of one of the following conditions:

Acid reflux

GERD

Esophagitis, a condition that cause swelling and/or inflammation of the esophagus. It can cause difficulty swallowing.

Stricture, a narrowing of the esophagus that can make it hard to swallow and cause breathing problems

Barrett's esophagus, a condition in which the lining of the esophagus is damaged by stomach acid. It can raise the risk of developing cancer of the esophagus.

Esophagitis, strictures, and Barrett's esophagus can happen if GERD is left untreated or if treatment is not effective.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN ESOPHAGEAL PH TEST?

If you've been diagnosed with acid reflux or GERD, you may be able to reduce or eliminate your symptoms with diet and lifestyle changes. Diet changes include limiting:

Fatty foods

Spicy foods

Citrus fruits and juices

Tomatoes and tomato products

Alcohol

Lifestyle changes include:

Eating smaller amounts

Quitting smoking

Waiting a few hours after eating before you lie down

Losing weight if needed

Over-the-counter medicines, such as antacids, may also be helpful.

If your symptoms don't improve or are more serious, your provider may recommend treatments to reduce or block acid production. These may include over-the-counter and/or prescription medicines. Sometimes surgery is recommended for people with more severe and/or long-lasting symptoms.

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Estrogen Levels Test

<https://medlineplus.gov/lab-tests/estrogen-levels-test/>

ESTROGEN LEVELS TEST

WHAT IS AN ESTROGEN TEST?

An estrogen test measures the level of estrogens in a sample of your blood, urine (pee), or saliva (spit).

Estrogens are a group of hormones that play a key role in female reproductive health, including puberty, menstruation, pregnancy, and menopause. However, estrogens are also important for more general bodily functions in both males and females, including:

Healthy cholesterol and blood sugar levels.

Bone and muscle strength.

Brain functions such as the ability to focus.

There are many types of estrogen, but only three types are commonly tested:

Estrone, also called E1, is the weakest of the three types and is made mostly by body fat and the adrenal glands (glands that sit on top of each kidney) in males and females. In females, the ovaries (the glands that contain eggs) also make estrone. Estrone is the only type of estrogen that females continue to make after menopause.

Estradiol, also called E2, is the main estrogen in nonpregnant females of childbearing age. It's mostly made in the ovaries and makes pregnancy possible by helping to release an egg each menstrual cycle and by thickening the lining of the uterus (the place where a fetus grows). It also helps support brain and bone health. Males make small amounts of this estrogen in the testicles (the glands that make sperm).

Estriol, also called E3, is an estrogen that increases during pregnancy. It's made in the placenta (the organ that grows in the uterus to provide nutrients and oxygen to the fetus). It helps your body prepare for breastfeeding and the birthing process. Males and nonpregnant females have very low levels of this estrogen.

Though estrogen levels depend on many factors, including sex and age, levels that stay too high or too low may be a sign of a health problem.

Other names: estradiol test, estrone (E1), estradiol (E2), estriol (E3), estrogenic hormone test

WHAT IS IT USED FOR?

In females, estradiol and/or estrone testing may be used to help diagnose conditions that may cause or be caused by estrogen levels that are too high or too low. These include:

Early or late puberty.

Polycystic ovary syndrome (PCOS), a condition that affects your ovaries.

Thyroid or pituitary gland disorders such as hypothyroidism.

Infertility (problems getting pregnant).

Certain estrogen-producing tumors of the ovaries.

Perimenopause, the period leading up to menopause.

Estradiol and/or estrone testing may also be used to help monitor:

Infertility treatments.

Hormone replacement therapy for menopause.

Hormone treatment for certain cancers, such as breast cancer.

During pregnancy, an estriol hormone test may be used to:

Help diagnose certain birth defects.

Monitor a high-risk pregnancy.

In males, estradiol and/or estrone tests may be used to see whether too much estrogen is causing conditions, such as:

Late puberty.

Breast growth (gynecomastia).

Infertility.

Erectile dysfunction (ED), a condition in which you have problems having and/or keeping an erection.

Testicular or adrenal cancer.

WHY DO I NEED AN ESTROGEN TEST?

If you are female, you may need estrogen testing if you:

Have signs of early or delayed puberty such as the development of breasts, body hair, and widened hips before the age of 8 or after the age of 13.

Are of childbearing age and have abnormal periods or no periods at all.

Are unable to get pregnant after a year of trying.

Have a high-risk pregnancy.

Have symptoms of menopause, including hot flashes and/or night sweats.

Use hormone replacement therapy after menopause.

Have vaginal bleeding after menopause.

If you're pregnant, you may have an estriol test between weeks 15 and 20 of your pregnancy. The test may be part of a group of prenatal tests called a triple screen test or a quad screening test, which are used to check whether your fetus has a high risk for a genetic birth defect such as Down syndrome. You may be more likely to have a baby with a birth defect if you:

Have a family history of birth defects.

Are age 35 or older.

Have diabetes.

Have had a viral infection during pregnancy.

If you are male, you may need estrogen testing if you have:

Signs of late puberty, such as delayed development of muscle mass and growth of testicles and penis.

Enlarged breasts or other symptoms of too much estrogen.

WHAT HAPPENS DURING AN ESTROGEN TEST?

If you are tested at your provider's office or a lab, you will provide a blood or urine sample. If you use an at-home test, the sample is usually saliva (spit).

For a blood test: A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a urine test: You'll need to collect all your urine during a 24-hour period. This is called a 24-hour urine sample test. You will be given a special container to collect your urine and instructions on how to collect and store your samples. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or in a cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

For an at-home saliva test: You'll collect a sample of saliva and send it to a lab for testing. Ask your provider which saliva test is best to use. Be sure to follow all the instructions that come with your kit and discuss your results with your provider.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain supplements and medicines, such as the birth control pill and hormone replacement therapy, before this test. Tell your provider about everything you take, but don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no known risk to a urine or saliva test.

WHAT DO THE RESULTS MEAN?

A healthy level of estrogen depends on several factors, including:

Your age.

Your sex.

The reasons for having the test.

The type of estrogen that was measured.

The results of other tests you may have had.

Estrogen levels in males, for example, don't change much throughout life. Therefore, a high level is usually a sign of a health condition such as infertility or an estrogen-producing tumor.

In females, however, estrogen levels change a lot. They rise during puberty, are at their highest during pregnancy, and decline after menopause. Because estrogen levels change so often, you may need to be tested more than once to look for a trend in your estrogen levels over time. Talk with your provider to find out what your results mean for your health.

Learn more about laboratory tests, reference ranges, and understanding results.

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Estrogen Receptor, Progesterone Receptor Tests

<https://medlineplus.gov/lab-tests/estrogen-receptor-progesterone-receptor-tests/>

ESTROGEN RECEPTOR, PROGESTERONE RECEPTOR TESTS

WHAT ARE ESTROGEN RECEPTOR/PROGESTERONE RECEPTOR (ER/PR) TESTS?

Estrogen receptor/progesterone receptor (ER/PR) tests are done on samples of your tissue that contain breast cancer cells. The tests check if your cancer cells have estrogen and progesterone receptors (proteins that estrogen and progesterone attach to). The results of the tests are used to help guide breast cancer treatment.

Receptors are special proteins inside or on the surface of cells. They can attach to certain substances in your blood, including estrogen and progesterone. Estrogen and progesterone are hormones (chemical messengers in your bloodstream that control the actions of certain cells or organs). These hormones play key roles in a woman's sexual development and reproductive functions. Men also have these hormones, but in much smaller amounts.

Breast cancer can affect both women and men. If a receptor attaches to estrogen and/or progesterone, the receptor may use that hormone to trigger certain breast cancer cells to grow.

ER/PR tests will show whether there are ER and/or PR receptors on your breast cancer cells. Test results are frequently referred to as the hormone receptor status. The different statuses include:

ER-positive (ER+): Cancers that have estrogen receptors (ER). Most breast cancers are ER-positive.

PR-positive (PR+): Cancers that have progesterone receptors (PR)

Hormone receptor-positive (HR+): Cancers with one or both receptor types.

HR-negative (HR-): Cancers without ER or PR receptors.

If your hormone receptor status shows you have one or both receptors on your cancer cells, you may respond well to certain treatments.

Other names: ER/PR IHC testing, hormone receptor status, hormone receptor test, ER/PR test

WHAT ARE THEY USED FOR?

ER/PR tests are used to guide treatment for breast cancer patients.

WHY DO I NEED AN ER/PR TEST?

You may need this test if you've been diagnosed with breast cancer. Knowing your hormone receptor status will help your health care provider decide how to treat it. If you have ER-positive, PR-positive, or HR-positive cancer, medicines that lower hormone levels or stop the hormones from helping cancer to grow can be very effective. If you have HR-negative cancer, these types of medicines won't work for you.

WHAT HAPPENS DURING ER/PR TESTING?

Your provider will need to take a sample of your breast tissue in a procedure called a breast biopsy. There are many ways to do a biopsy:

There are three main types of breast biopsy procedures. They are usually done on an outpatient basis, which means you go home the same day:

Fine needle aspiration biopsy uses a very thin needle to remove a sample of breast cells or fluid. The biopsy takes about 15 minutes.

Core needle biopsy uses a wide needle to remove one or more small tissue samples about the size of a grain of rice. Sometimes a small vacuum probe is used instead of a needle. The device gently suctions some tissue and removes it with a small rotating blade. A core needle biopsy takes between 15 minutes and an hour, depending on how it's done.

Surgical biopsy (or open biopsy) is surgery to remove all or part of a lump. The biopsy usually takes about an hour.

Biopsies are often done using mammography, ultrasound, MRI, or x-rays to help see exactly where to take the tissue sample. Your procedure will vary depending on which method is used to guide the biopsy, but the general steps are usually the same.

For a fine needle aspiration biopsy or a core needle biopsy:

Your provider will clean the skin on your breast and give you a shot of medicine to numb the area, so you won't feel any pain. The shot may sting briefly.

You may be sitting or lying down. If images are used to guide the biopsy, you may lie on your side, back, or belly with your breast over an opening on the table.

For a fine needle aspiration biopsy, your provider will insert the needle into the biopsy site and remove a sample of cells or fluid. For a core biopsy, a tiny cut may be made to insert a wide needle or a vacuum device. You may feel a little pressure when the sample or samples are removed.

Pressure will be applied to area until the bleeding stops.

Your provider will cover the biopsy site with a sterile bandage. If you had a small incision, small strips of medical tape may be used to close the wound.

For a surgical biopsy:

You'll lie on an operating table. You may have an IV (intravenous line) in your arm or hand that may be used to give you medicine to relax. The skin over the biopsy area will be cleaned.

To prevent pain, you'll have either: A shot of medicine to numb your breast. The shot may sting briefly. General anesthesia, which is medicine given through an IV to make you sleep.

A shot of medicine to numb your breast. The shot may sting briefly.

General anesthesia, which is medicine given through an IV to make you sleep.

When you are numb or asleep, the surgeon will make a small cut in your breast tissue to remove part or all of the lump. In certain cases, tissue around the lump may also be removed. This may help avoid the need for more surgery if cancer cells are found in the lump.

The cut in your skin will be closed with small strips of medical tape or stitches and covered with a sterile bandage.

The type of biopsy you have will depend on:

The size and location of the suspicious tissue in your breast

How many areas of your breast are involved

How abnormal the tissue looks on a mammogram or other image

Your general health and preferences

Ask your provider about why you need a biopsy and which type is right for you.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will give you instructions for how to prepare for your biopsy. If you take any blood thinners, including aspirin, you may need to stop taking them before your biopsy. Tell your provider about all the medicines and supplements you take. Don't stop or start taking anything without talking with your provider first.

If you're having general anesthesia, you will probably need to fast (not eat or drink) for several hours before surgery. If you have general anesthesia or medicine to relax, you may be groggy after the procedure, so plan to have someone take you home.

ARE THERE ANY RISKS TO THE TEST?

It's common to have some bruising and temporary discomfort after a breast biopsy. Possible risks include:

Infection, which can be treated with antibiotics

Bleeding

Your provider will give you instructions for how to care for biopsy area and manage any discomfort. If you're having general anesthesia, talk with your provider about how it may affect you. General anesthesia is very safe even for most people with other health conditions.

WHAT DO THE RESULTS MEAN?

The results will show whether you have a hormone receptor-positive or negative type of cancer.

If your hormone receptor status is positive:

At least 1% of the cells in your sample have estrogen and/or progesterone receptors.

The cancer is using either estrogen or progesterone to grow. The higher the levels of receptors, the more effective certain medicines may be in treating your cancer.

These results may also be called hormone-sensitive, hormone receptor-positive, or HR-positive.

If your hormone receptor status is negative:

Less than 1% of the cells in your sample have receptors, so those medicines will not be effective.

These results may also be called hormone insensitive, receptor-negative, or HR-negative.

If you have questions about your results, talk to your provider. They may want to do other tests before making a diagnosis and planning the best treatment for you.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT ER/PR TESTS?

HER2 testing is often done at the same time as ER/PR testing to help guide treatment. HER2 is a protein involved in normal cell growth. It also helps certain types of cancer cells grow quickly. Cancers with large amounts of HER2 protein tend to grow quickly and spread to other parts of the body. Breast cancer cells with higher-than-normal levels of HER2 are called HER2-positive. Certain medicines, called HER2 targeted therapy, can block or slow HER2 proteins and help control these cancers. But these medicines won't help cancers that don't have high levels of HER2 proteins. So, HER2 tumor marker testing is necessary to find out if targeted therapy will be helpful.

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Fall Risk Assessment

<https://medlineplus.gov/lab-tests/fall-risk-assessment/>

FALL RISK ASSESSMENT

WHAT IS A FALL RISK ASSESSMENT?

Falls are common in adults 65 years of age and older. In the United States, about a third of older adults who live at home and about half of people living in nursing homes fall at least once a year. There are many factors that increase the risk of falling in older adults. They include:

Mobility problems (e.g., trouble walking or standing up)

Balance disorders

Chronic (long-term) illnesses

Impaired vision

Taking certain medicines

Foot problems and wearing unsafe shoes

Mild cognitive impairment or certain types of dementia

Many falls cause at least some injury. These range from mild bruising to broken bones, head injuries, and even death. In fact, falls are a leading cause of death in older adults.

Other names: fall risk evaluation, fall risk screening, assessment, and intervention

WHAT IS IT USED FOR?

A fall risk assessment is used to find out if you have a low, moderate, or high risk of falling. If the assessment shows you are at an increased risk, your health care provider and/or caregiver may recommend strategies to prevent falls and reduce the chance of injury.

WHY DO I NEED A FALL RISK ASSESSMENT?

The Centers for Disease Control and Prevention (CDC) and the American Geriatrics Society recommend yearly fall assessment screening for all adults 65 years of age and older. If the screening shows you are at risk, you may need an assessment. The assessment includes performing a series of tasks called fall assessment tools.

You also may need an assessment if you have certain symptoms. Falls often come without warning, but if you have any of the following symptoms, you may be at higher risk:

Dizziness

Irregular or rapid heartbeats (arrhythmia)

Blood pressure that drops too much when you stand up

WHAT HAPPENS DURING A FALL RISK ASSESSMENT?

A fall risk assessment checks to see how likely it is that you will fall. It is mostly done for older adults. The assessment usually includes:

An initial screening. This includes a series of questions about your overall health and if you've had previous falls or problems with balance, standing, and/or walking.

A set of tasks known as fall assessment tools. These tools test your strength, balance, and gait (the way you walk).

Many providers use an approach developed by the CDC called STEADI (Stopping Elderly Accidents, Deaths, and Injuries). STEADI includes screening, assessing, and intervention. Interventions are recommendations that may reduce your risk of falling. STEADI includes three steps:

Screening you for your risk of falling

Assessing for your risk factors that can be improved to try to prevent falls (for example, balance problems, impaired vision)

Intervening to reduce your risk of falling by using effective strategies (for example, providing education and resources)

During the screening, you may be asked several questions including:

Have you fallen in the past year?

Do you feel unsteady when standing or walking?

Are you worried about falling?

During an assessment, your provider will test your strength, balance, and gait, using the following fall assessment tools:

Timed Up-and-Go (Tug). This test checks your gait. You'll start in a chair, stand up, and then walk for about 10 feet at your regular pace. Then you'll sit down again. Your provider will check how long it takes you to do this. If it takes you 12 seconds or more, it may mean you are at higher risk for a fall.

30-Second Chair Stand Test. This test checks strength and balance. You'll sit in a chair with your arms crossed over your chest. When your provider says "go," you'll stand up and sit down again. You'll repeat this for 30 seconds. Your provider will count how many times you can do this. A lower number may mean you are at higher risk for a fall. The specific number that indicates a risk depends on your age.

4-Stage Balance Test. This test checks how well you can keep your balance. You'll stand in four different positions, holding each one for 10 seconds. The positions will get harder as you go.

Position 1: Stand with your feet side-by-side.

Position 2: Move one foot halfway forward, so the instep is touching the big toe of your other foot.

Position 3: Move one foot fully in front of the other, so the toes are touching the heel of your other foot.

Position 4: Stand on one foot.

Position 1: Stand with your feet side-by-side.

Position 2: Move one foot halfway forward, so the instep is touching the big toe of your other foot.

Position 3: Move one foot fully in front of the other, so the toes are touching the heel of your other foot.

Position 4: Stand on one foot.

If you can't hold position 2 or position 3 for 10 seconds or you can't stand on one leg for 5 seconds, it may mean you are at higher risk for a fall.

During the assessment, your provider may also:

Identify any medicines you take that could increase your risk of falling

Ask about possible safety hazards in your home, such as throw rugs you could trip on, a slippery tub floor, etc.

Measure your blood pressure when lying and standing

Do an eye test that checks your ability to see small details (the Snellen test)

Check your feet and footwear

Measure your vitamin D levels

Identify other conditions you may have that could affect your risk of falls such as depression and osteoporosis)

Do a cognitive test to check for problems with cognition thinking, learning, remembering, being aware of surroundings, and using judgment)

There are many other fall assessment tools. If your provider recommends other assessments, he or she will let you know what to expect.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A FALL RISK ASSESSMENT?

You don't need any special preparations for a fall risk assessment.

ARE THERE ANY RISKS TO A FALL RISK ASSESSMENT?

There is a small risk that you might fall while you are doing the assessment.

WHAT DO THE RESULTS MEAN?

The results may show you have a low, moderate, or high risk of falling. They also may show which areas may be increasing your risk for falls (e.g., gait, strength, and/or balance). Based on your results, your provider may make recommendations to reduce your risk of falling. They may include:

Exercising to improve your strength and balance. You may be given instructions on specific exercises or be referred to a physical therapist.

Changing or reducing the dose of medicines that may be affecting your gait or balance. Some medicines have side effects that cause dizziness, drowsiness, or confusion.

Taking vitamin D to strengthen your bones.

Getting your vision checked by an eye doctor.

Switching to safer footwear to lower your risk of falling.

Getting a referral to a podiatrist (foot doctor) if you are having problems with balance or gait.

Getting rid of any safety hazards in your home. These may include poor lighting, loose rugs, and/or cords on the floor.

Treating any medical conditions that could increase your risk of falls.

If you have questions about your results and/or recommendations, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Fasting for a Blood Test

<https://medlineplus.gov/lab-tests/fasting-for-a-blood-test/>

FASTING FOR A BLOOD TEST

WHAT IS FASTING FOR A BLOOD TEST?

If your health care provider has told you to fast before a blood test, it means you should not eat or drink anything, except plain water, for several hours before your test. While you're fasting for a blood test, you should also not:

Chew gum

Smoke

Exercise

Your provider will give you a copy of instructions to follow if they ask you to fast for a test. If you have any questions not covered in those instructions, ask your provider at least a full day before the test so that you'll have time to properly prepare.

HOW LONG DO I HAVE TO FAST BEFORE THE TEST?

You usually need to fast for 8 to 12 hours before a test. Your provider will tell you exactly how long to fast, depending on what test they've ordered. For certain tests, there may also be additional instructions. For example, you may be asked not to drink alcohol for a longer period of time.

If your blood test requires fasting, you can usually schedule the test early in the morning. That way, most of your fasting time will be while you are sleeping.

WHY DO I NEED TO FAST BEFORE MY BLOOD TEST?

When you eat and drink, your bloodstream absorbs sugar, fat, protein, and other nutrients in those foods and beverages. These nutrients can then change the levels of various substances in your blood and therefore affect the accuracy of your test results. Inaccurate results give your provider an inaccurate view of your health and may lead to a wrong diagnosis.

WHAT TYPES OF BLOOD TESTS REQUIRE FASTING?

Some examples of common blood tests that usually require fasting are:

Blood glucose, also called blood sugar, tests. These tests measure the level of blood glucose in a sample of your blood. They are used to help diagnose and monitor diabetes, prediabetes, and gestational diabetes (diabetes that starts during pregnancy). Not all blood glucose tests require fasting, so ask your provider if you need to fast for your test.

Cholesterol levels tests (lipid panel). These tests measure the level of fats, called lipids, in your bloodstream, including cholesterol and triglycerides. The test results help show your risk of getting heart disease and can be used to monitor your health after a cardiac event like a heart attack or stroke.

Basic metabolic panel (BMP). This is a group of tests that measures several natural chemicals in your blood. The results give your provider information about how well your organs are working, including your heart, kidneys, and liver.

In some cases, you may also be asked to fast for:

Liver function tests. This group of tests measures several proteins and other substances related to your liver function. Test results can give your provider information on your liver's health.

Renal function panel. This is a group of tests that measure several waste products and substances related to your kidney function. The results are often used to monitor your kidney function if your provider suspects kidney disease or if you have diabetes.

Whether or not your provider asks you to fast for these tests depends on several factors. For example, some liver

function tests only require fasting if they're ordered as part of a comprehensive metabolic panel.

Other types of tests not listed here may require fasting. Therefore, you should always ask your provider if you aren't sure whether you should fast before your test.

CAN I DRINK ANYTHING BESIDES WATER DURING A FAST?

No. Juice, coffee, soda, and other beverages can get in your bloodstream and affect your results. But you can drink plain water. Avoid lemon-flavored or flavored carbonated water, which may contain sugars or artificial sweeteners. Staying well hydrated can actually make the testing process a little easier. This is because water helps to keep more fluid in your veins, which can make it easier to get a sample of your blood.

CAN I CONTINUE TAKING MEDICINE DURING A FAST?

Ask your provider whether it's okay to take your usual prescription medicines and/or over-the-counter medicines. For some tests, they may ask you to stop taking your medicines. But don't stop taking your medicines unless your provider tells you to stop.

Let your provider know about any vitamins and supplements that you take, too. That way you'll be sure to avoid taking anything that could affect your test results.

WHAT IF I MAKE A MISTAKE AND HAVE SOMETHING TO EAT OR DRINK BESIDES WATER DURING MY FAST?

Tell your provider if you eat or drink before your test. You may need to reschedule the test for another time when you can fast to make sure you get accurate results.

WHEN CAN I EAT AND DRINK NORMALLY AGAIN?

You'll be able to eat as soon as your test is over. It is common to feel light-headed after a blood test that requires fasting. So, you may want to bring a snack that you can eat right away.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT FASTING BEFORE A BLOOD TEST?

You should not fast if your provider doesn't tell you to do so. Fasting for some blood tests may also affect your results and give your provider an inaccurate understanding of your health. If you aren't sure if you should fast for a coming blood test, ask your provider as soon as possible. Taking the right steps before a blood test helps make sure your results will be accurate.

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Fecal Occult Blood Test (FOBT)

<https://medlineplus.gov/lab-tests/fecal-occult-blood-test-fobt/>

FECAL OCCULT BLOOD TEST (FOBT)

WHAT IS A FECAL OCCULT BLOOD TEST?

A fecal occult blood test (FOBT) checks for blood in a sample of your stool (poop). It's called occult (hidden) blood because you can't see it just by looking at it. And fecal means that it is in your stool.

Blood in your stool means there is bleeding in the digestive tract. The bleeding can be caused by a variety of conditions, including:

Polyps, abnormal growths on the lining of your colon or rectum

Hemorrhoids, swollen veins in your anus or rectum

Diverticulosis, a condition in which you have small pouches in the inside wall of your colon

Ulcers, sores in the lining of your digestive tract

Ulcerative colitis, a type of inflammatory bowel disease

Colorectal cancer, a type of cancer that starts in your colon or rectum

Colorectal cancer is one of the most common types of cancer in the United States. A fecal occult blood test can screen for colorectal cancer to help find the disease early, when treatment may be most effective.

Other names: FOBT, stool occult blood, occult blood test, Hemoccult test, guaiac smear test, gFOBT, immunochemical FOBT, iFOBT; FIT

WHAT IS IT USED FOR?

A fecal occult blood test is commonly used as a screening test to help find colorectal cancer before you have symptoms. The test may also be used to check for other conditions that can cause gastrointestinal (GI) bleeding.

In certain cases, the test is used to help find the cause of anemia. It can also help tell the difference between irritable bowel syndrome (IBS), which usually doesn't cause bleeding, and inflammatory bowel disease (IBD), which is likely to cause bleeding.

But a fecal occult blood test alone cannot diagnose any condition. If your test results show blood in your stool, you will likely need other tests to find the exact cause.

WHY DO I NEED A FECAL OCCULT BLOOD TEST?

Your health care provider may order a fecal occult blood test if you have symptoms of a condition that could involve bleeding in your digestive tract. Or you may have the test to screen for colorectal cancer when you don't have any symptoms.

Regular screening tests for colorectal cancer can detect colon polyps and colorectal cancer. If you aren't at higher risk for colorectal cancer, your provider will likely recommend you start screenings at age 45. If you are at higher risk, you may need to start getting screened for colorectal cancer earlier. Talk with your provider about when to start, how often to get screened, and which type of test you should have.

A fecal occult blood test is one of several types of colorectal screening tests. Other tests include:

A stool DNA test (FIT-DNA test). This test checks for genetic changes in your stool that may be signs of colorectal polyps or cancer.

Colonoscopy or sigmoidoscopy. Both tests use a thin tube with a camera to look inside your colon. A colonoscopy allows your provider to see your entire colon. A sigmoidoscopy shows only the lower part of your colon.

CT colonography, also called "virtual colonoscopy." For this test, you usually drink a liquid containing contrast dye before having a CT scan. The CT scan uses x-rays to take detailed 3-dimensional pictures of your entire colon and rectum. The contrast dye makes your colon and rectum show up more clearly on the x-rays.

Each type of test has pros and cons. Your provider can help you decide which test is right for you.

WHAT HAPPENS DURING A FECAL OCCULT BLOOD TEST?

Usually, your provider will give you a kit to collect samples of your stool (poop) at home. The kit will include instructions on how to do the test.

There are two main types of fecal occult blood tests:

The guaiac fecal occult blood test (gFOBT) uses a chemical (guaiac) to find blood in stool. It usually requires stool samples from two or three separate bowel movements.

The fecal immunochemical test (iFOBT or FIT) uses an antibody, a type of protein that attaches to hemoglobin (blood) in stool. FIT is considered better at finding blood in your stool than gFOBT testing. A FIT test requires stool samples from one to three separate bowel movements, depending on the brand of the test.

Different test kits have different instructions, so it's very important to follow the specific instructions that come with your kit. The typical process for gathering a stool sample usually includes these general steps:

Collecting a bowel movement. Your kit may include special paper to place over your toilet to catch your bowel movement. You may also use plastic wrap or a clean, dry container. If you are doing a guaiac test, be careful not to let any urine mix in with your stool.

Taking a stool sample from the bowel movement. Your kit will include a wooden stick or applicator brush for scraping the stool sample from your bowel movement. Follow the instructions for where to gather the sample from the stool.

Preparing the stool sample. You will either smear the stool on a special test card or insert the applicator with the stool sample into a tube that came with your kit.

Labeling and sealing the sample as directed.

Repeating the test on your next bowel movement as directed if more than one sample is needed.

Returning the samples as directed. This might be mailing them or taking them to your provider's office or a lab.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You do not need any special preparations for many fecal occult blood tests. But for some guaiac fecal occult blood tests (gFOBTs), certain foods, supplements, and medicines may affect the results of the test. If you are having one of these tests, your provider will tell you what you need to avoid and for how long. For example, you might need to avoid:

Nonsteroidal, anti-inflammatory drugs (NSAIDs), such as ibuprofen, naproxen, and aspirin. If you take aspirin for heart problems, talk with your provider before stopping your medicine. You may be able to take acetaminophen during this time, but check with your provider before taking it.

Vitamin C in amounts over 250 mg a day. This includes vitamin C from supplements, fruit juices, or fruit.

Red meat, such as beef, lamb, and liver. Traces of blood from these meats may show up in your stool.

Certain raw fruits and vegetables.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a fecal occult blood test.

WHAT DO THE RESULTS MEAN?

If your results from a fecal occult blood test show that you have blood in your stool, it means you likely have bleeding somewhere in your digestive tract. But that doesn't always mean you have cancer. Other conditions that may cause blood in your stool include ulcers, hemorrhoids, polyps, and benign (not cancer) tumors.

If you have blood in your stool, your provider will likely recommend more tests to figure out the exact location and cause of your bleeding. The most common follow-up test is a colonoscopy. If you have questions about your test results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A FECAL OCCULT BLOOD TEST?

Regular colorectal cancer screenings, such as fecal occult blood tests, are an important tool in the fight against cancer. Studies show that screening tests can help find cancer early and may reduce deaths from the disease.

If you decide to use fecal occult blood testing for your colorectal cancer screening, you will need to do the test every year.

You can buy gFOBT and FIT stool collection kits without a prescription. Most of these tests require you to send a sample of your stool to a lab. But some tests can be done completely at home for quick results. If you're considering buying your own test, ask your provider which one is best for you.

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Ferritin Blood Test

<https://medlineplus.gov/lab-tests/ferritin-blood-test/>

FERRITIN BLOOD TEST

WHAT IS A FERRITIN BLOOD TEST?

A ferritin blood test measures the level of ferritin in your blood. Ferritin is a protein that binds to iron and stores it in your body. You need iron to make healthy red blood cells. Red blood cells carry oxygen from your lungs to the rest of your body. Iron is also important for healthy muscles, bone marrow, and organs, including brain development in children. You get iron from eating food or taking supplements. There are iron-only supplements, but many multivitamin/mineral supplements also contain iron.

Too little or too much iron in your body can cause serious health problems if not treated. A ferritin blood test can show how much iron you have stored in your body.

Other names: serum ferritin, serum ferritin level, ferritin serum

WHAT IS IT USED FOR?

A ferritin blood test may be ordered as part of a group of iron tests to help check your iron levels. It can help your health care provider find out if your body is storing the right amount of iron to stay healthy.

The test may also be used to:

Help diagnose or rule out conditions related to iron levels, including:

Hemochromatosis, having too much iron in your body (also called iron overload)

Iron deficiency anemia, having too few red blood cells because you lack iron

Liver diseases (much of the ferritin in your body is stored in your liver)

Restless legs syndrome, a tingling or burning feeling in your legs that may be caused by a lack of iron

Adult Still disease (also called adult-onset Still disease or AOSD), an uncommon disease that causes joint pain, fever, and rash, often with high ferritin levels

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Monitor chronic (long-lasting) conditions that may affect your iron levels, such as cancer, kidney disease, and autoimmune diseases

Check for low levels of iron before symptoms develop if you are at high risk for developing iron deficiency, including:

Being underweight

Having heavy menstrual bleeding

Being pregnant

Not being able to absorb food well (for example, if you have inflammatory bowel disease or have had a gastrointestinal surgical procedure)

Being underweight

Having heavy menstrual bleeding

Being pregnant

Not being able to absorb food well (for example, if you have inflammatory bowel disease or have had a gastrointestinal surgical procedure)

If you're being treated for problems with iron, the test may be used to check how well the treatment is working.

WHY DO I NEED A FERRITIN BLOOD TEST?

You may need this test if you have symptoms of iron levels that are too low or too high.

Symptoms of iron levels that are too low include:

Shortness of breath

Weakness or fatigue

Dizziness

Arrhythmia, a problem with the rate or rhythm of your heartbeat

Pale skin

Restless legs syndrome

Symptoms of iron levels that are too high can vary and tend to get worse over time. Symptoms may include:

Weakness or fatigue

Joint pain, commonly in the knees or hands

Loss of interest in sex or erectile dysfunction (ED)

Abdominal (belly) pain

Change in skin color, which may look gray, metallic, or bronze

You may also need this test if the results of other blood tests show that you have low levels of hematocrit (the amount of your blood that is made up of red blood cells) or hemoglobin (the protein that carries oxygen throughout the body).

WHAT HAPPENS DURING A FERRITIN BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider may ask you to fast (not eat or drink) for 12 hours before your test. The test is usually done in the morning. If you have any questions about how to prepare for your test, talk to your provider.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Lower than normal ferritin levels may mean you have iron deficiency anemia, or another condition related to low iron levels. Iron deficiency anemia is a common type of anemia where your body doesn't make enough red blood cells. Without treatment, it can lead to heart problems, infections, slow growth and brain development in children, and other health issues.

Higher than normal ferritin levels can be caused by having too much iron in your body. Other conditions that may cause increased ferritin levels include:

Inflammation (like from infection or surgery)

An autoimmune disorder

Liver disease

Cancer

Obesity

Alcohol use disorder

Hyperthyroidism

If your ferritin results are not normal, it does not always mean you have a medical condition that needs treatment. Certain medicines can decrease or increase your ferritin levels. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A FERRITIN BLOOD TEST?

Most conditions that cause too little or too much iron can be successfully treated with medicines, diet, and/or other therapies.

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Fetal Alcohol Spectrum Disorders (FASD) Screening

<https://medlineplus.gov/lab-tests/fetal-alcohol-spectrum-disorders-fasd-screening/>

FETAL ALCOHOL SPECTRUM DISORDERS (FASD) SCREENING

WHAT IS A FETAL ALCOHOL SPECTRUM DISORDERS (FASDs) SCREENING?

Fetal alcohol spectrum disorders (FASDs) are a group of conditions that can happen in a child who was exposed to alcohol before birth (called prenatal alcohol exposure). These conditions can affect each person in different ways and can range from mild to severe. People with FASDs can have lifelong effects, including physical, emotional, and behavioral problems.

When you drink alcohol during pregnancy, the alcohol passes to your developing baby through your placenta, the organ that nourishes your baby in your uterus. The alcohol can harm your baby's development, damage their organs, or cause other health problems.

No amount of alcohol is safe to drink during pregnancy. If you are pregnant or think you could be pregnant, you should not drink alcohol. This includes hard liquor, beer, and wine. Exposing your developing baby to alcohol could cause them to have an FASD or other serious health problems.

The term FASDs includes all of the following conditions:

Alcohol-Related Birth Defects (ARBD). This condition can affect organ development. People with ARBD may have problems with the function of their heart, kidneys, hearing, vision, and/or bones. They may also have other types of FASDs.

Neurobehavioral Disorder Associated with Prenatal Alcohol Exposure (ND-PAE). People with this condition can have problems with mental health, memory, communication, impulse control, and skills of daily living.

Alcohol-Related Neurodevelopment Disorder (ARND). People with ARND have problems with the brain and nervous system. It can cause intellectual disabilities, and/or behavioral and learning problems.

Fetal Alcohol Syndrome (FAS). This is the most severe type of FASD. It causes both birth defects and neurodevelopmental disorders.

Partial Fetal Alcohol Syndrome (pFAS). People with pFAS have some of the symptoms of an FASD, such as changes to their facial features. But the condition is not as severe as the other disorders.

Other names: fetal alcohol syndrome screening, fetal alcohol 4-digit diagnostic code

WHAT IS IT USED FOR?

The screening is used to find out if your child has an FASD. There is no cure for FASDs, but early diagnosis and treatment may help reduce some of the symptoms and help your child reach their full potential.

WHY DOES MY CHILD NEED AN FASDs SCREENING?

A screening may be needed if you drank alcohol during your pregnancy and/or your child has symptoms of an FASD. Symptoms may include a mix of physical problems, intellectual disabilities, and problems coping with daily living. They can range from mild to severe and will vary depending on your child's age. Symptoms of FASDs may include a combination of the following:

Physical issues, such as:

Abnormal facial features. These include small eyes, a thin upper lip, and a smooth skin surface between the nose and upper lip.

Small head size.

Lower than average height and weight.

Poor coordination.

Vision and hearing problems.

Problems with their heart, kidneys, or bones.

Learning challenges, such as:

Intellectual disability.

Delayed development.

Difficulty concentrating.

Poor memory.

Speech and language delays.

Behavioral issues, such as:

Hyperactivity

Poor social skills.

Poor reasoning and judgment

WHAT HAPPENS DURING AN FASDS SCREENING?

There is no single test for FASDs. But your provider may:

Check for intellectual disabilities and developmental delays

Look for certain physical features such as a small head, small eyes, and thin upper lip

See if your child's weight and height are below normal

Look for behavioral symptoms such as short attention span and problems with impulse control

Check your child's coordination, vision, and hearing

Some providers may also use a method called the 4-digit code when they do a physical exam on your child to check for FASDs. Your provider will rate each category on a scale of one to four, and then add the numbers. The total number will help them in making a diagnosis based on the following categories:

Lower height and weight than other children of the same age and sex

FAS facial features

Behavior and learning difficulties

Exposure to alcohol during pregnancy

WILL MY CHILD NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

There are no special preparations needed for an FASD screening.

ARE THERE ANY RISKS TO THIS TEST?

There are no risks to having an FASDs screening.

WHAT DO THE RESULTS MEAN?

If your child is diagnosed with an FASD, your provider may recommend several treatment options. There is no cure for these disorders, but the following may greatly improve a child's development and quality of life:

Working with a team of providers that can include a special education teacher, speech therapist, physical and occupational therapists, and mental health professionals

Special services in school to help with learning and behavior difficulties

Behavior modification therapy

Mental health therapy

Social skill training

Vocational and life skill therapy

Medicines to help with some symptoms

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN FASDS SCREENING?

If you are pregnant or trying to get pregnant and cannot stop drinking, it's important to get help. Even if you are pregnant, it is never too late to stop drinking. Because brain growth takes place throughout pregnancy, stopping alcohol use will improve the baby's health and well-being. Your provider can recommend counseling and treatment programs.

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Flu (Influenza) Test

<https://medlineplus.gov/lab-tests/flu-influenza-test/>

FLU (INFLUENZA) TEST

WHAT IS A FLU (INFLUENZA) TEST?

Flu is short for influenza. It is a respiratory infection caused by viruses. Respiratory infections may affect your nose, throat, and/or lungs. Most flu tests check a sample of fluid from your nose or throat to find out if you're infected with a flu virus.

The flu usually spreads from person to person through coughing or sneezing. You can also get the flu by touching a surface that has the virus on it, and then touching your nose, eyes, or mouth. You can catch the flu at any time, but it's most common during certain times of the year, known as flu season. In the United States, flu season can begin as early as October and end as late as May.

Most people who get the flu will feel sick with muscle aches, fever, a runny or stuffy nose, and other uncomfortable symptoms that may feel like a very bad cold. They usually recover on their own in less than 2 weeks. But the flu can be serious, or even deadly, especially for children under age 5, adults age 65 and over, and people with certain chronic (long-term) illnesses, such as asthma, diabetes, or heart disease.

In some cases, your health care provider may prescribe medicine to help fight the flu virus. These medicines are called flu antivirals. They may make your symptoms milder and shorten the time you're sick by about one day. They may also help prevent serious flu complications, such as pneumonia.

Flu antivirals work best when you start taking them within 2 days after feeling sick. So early testing may be important, especially if you have a high risk of getting seriously ill from the flu. But most people get better without antiviral medicine. Flu tests check for the most common types of seasonal flu viruses, influenza A and influenza B. Some tests are done in a health care setting, such as your provider's office, pharmacies, or outpatient clinics. You can also buy at-home tests.

There are two main types of tests for seasonal flu:

Rapid influenza antigen tests, or rapid influenza diagnostic tests (RIDTs), are the most common flu tests. They check your fluid sample for certain proteins (antigens) found on flu viruses. These rapid tests can provide results in less than 30 minutes, and you can do them at home. But they aren't as accurate as other types of flu tests.

Molecular flu tests are more accurate than rapid antigen tests. They check your sample for genetic material from flu viruses. There are two types of molecular flu tests:

Rapid molecular tests can provide results in under 30 minutes. You usually need to go to a health care setting to have these tests.

Molecular laboratory tests are the most accurate type of flu test, but you may have to wait up to three days for results. That's because your sample is sent to a special lab for testing. You can go to a health care setting to have your sample collected, or you can use an at-home collection kit to gather a sample and send it to the lab. A PCR (polymerase chain reaction) is a common molecular lab test.

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Tests are also available to check for seasonal flu viruses along with other respiratory viruses that cause similar symptoms. These combination or "multiplex" tests use one fluid sample to check for three or more different viruses. Most often, they can be used to check for the seasonal flu and:

COVID-19 (coronavirus disease 2019). COVID-19 is caused by the SARS-CoV-2 virus. You can get COVID-19 at any time during the year. Combination flu/COVID-19 tests are available as at-home rapid antigen tests, rapid molecular tests, and laboratory molecular tests.

Respiratory syncytial virus infections (RSV). RSV infections are usually most common between fall and spring.

Combination tests check for RSV, seasonal flu, and COVID-19 all at once. These tests are available from your provider as rapid molecular tests and molecular laboratory tests, including tests with home collection kits.

Combination tests may be especially useful during times of the year when several respiratory viruses are spreading widely. Providers treat seasonal flu, COVID-19, and RSV with different antiviral medicines. If you need antiviral medicine, a combination test helps make sure you get the right kind. It's possible to have more than one respiratory virus at the same time.

Your provider may use molecular laboratory tests to check for bird flu, also called avian influenza, if you've been exposed to this virus. Bird flu is a subgroup of uncommon influenza A viruses (called H5) that cause respiratory illness. These viruses are different from the influenza A viruses that cause seasonal flu. They usually infect birds only, but sometimes they infect other animals, including cows.

People rarely get bird flu. When they do, it's usually because they were in contact with infected birds or animals. Most bird flu infections in people have been caused by bird flu viruses called H5N1 and H7N9.

If you have flu symptoms and have been near animals that could have had bird flu, tell your provider right away and stay away from others. Your provider can order a test for H5 bird flu. The test checks a sample from your nose, throat, or eye for signs of the virus that causes bird flu.

Your provider can help you decide if you need a test that checks for more than seasonal flu. To decide which test is right for you, your provider will consider your symptoms, medical history, and how widespread different viral illnesses are in your community.

Other names: rapid flu test, influenza antigen test, rapid influenza diagnostic test, RIDT, Flu PCR, Type A and B Antigen Detection, SARS-CoV-2 & Flu A/B, Flu SC2 multiplex assay, COVID-19, Flu A+B and RSV, H5 Bird Flu test

WHAT IS IT USED FOR?

If you have symptoms of a respiratory infection, a test for common seasonal flu may be used to help find out if the influenza A or B virus is causing your illness. Knowing whether you have the flu is especially important if you have a high risk of becoming seriously ill from the flu. The test results can help your provider decide if medicine could help treat your illness and which one might help you.

Public health officials may also use flu tests to monitor flu and COVID-19 outbreaks in the community.

WHY DO I NEED A FLU TEST?

If you have flu symptoms, testing can help you decide what to do next, such as starting antiviral medicine, and taking steps to lower your chances of spreading the virus to others.

Symptoms of the flu and other respiratory illnesses may include:

- Fever
- Chills
- Muscle or body aches
- Fatigue
- Headache
- Runny or stuffy nose
- Sore throat
- Cough

Even if you have flu symptoms, you may not always need a flu test. If the flu is widespread in your area, your provider may be able to diagnose you without a test.

Your provider may be more likely to order a flu test if you have symptoms and a high risk of developing a serious illness from the flu and other respiratory infections. Testing helps your provider make sure you get the right treatment to lower your risk. The risk of developing serious complications from the flu is higher for people who:

- Are age 65 and older

- Are younger than age 5 (especially those younger than 2 years old)

- Have a chronic health condition, such as asthma, blood disorders, diabetes, lung diseases, metabolic disorders, obesity, conditions that affect the heart, kidneys, liver, or nervous system, and diseases that weaken the immune system

It's important to understand your risk of serious illness from the flu before you get sick. That way you know if you need to

contact your provider at the first sign of symptoms.

WHAT HAPPENS DURING A FLU TEST?

If you are tested at your provider's office or other health care setting, a fluid sample may be collected using:

A nasopharyngeal swab test. A health care professional will insert a long swab into your nostril until it reaches your nasopharynx (the upper part of your throat). They will then rotate the swab for 10 to 15 seconds, remove it, and test that sample for signs of influenza A and B. In some cases, they may also test for influenza A subtype H5 (bird flu), and SARS-CoV-2 (Covid-19).

A nasal aspirate or wash. A health care professional will put some saline solution (salt water) into your nose and use gentle suction to remove it.

If your provider orders an H5 bird flu test, they may use additional methods. These include:

Conjunctival swab. Your provider may use this test if you have conjunctivitis (an eye infection). For this test, your provider will:

Gently pull down the lower eyelid of your infected eye.

Swab the inside of your lower eyelid by rotating the swab over the infected areas 2-3 times. They will not touch the surface of your eye.

Gently pull down the lower eyelid of your infected eye.

Swab the inside of your lower eyelid by rotating the swab over the infected areas 2-3 times. They will not touch the surface of your eye.

Sputum culture. This test checks a sample of your sputum (the mucus that is produced in the lungs and respiratory tract) for signs of the bird flu virus. Your provider may only order this if you have serious respiratory disease and a flu infection.

For a sputum culture:

You may be asked to rinse your mouth out with water before the sample is taken.

Your provider will ask you to breathe deeply and then cough deeply into a special cup.

Your provider may tap you on the chest to help loosen sputum from your lungs.

If you have trouble coughing up enough sputum, your provider may ask you to breathe in a salty mist to help you cough more deeply.

If you can't cough up enough sputum after breathing in the salty mist, your provider may perform a bronchoscopy.

You may be asked to rinse your mouth out with water before the sample is taken.

Your provider will ask you to breathe deeply and then cough deeply into a special cup.

Your provider may tap you on the chest to help loosen sputum from your lungs.

If you have trouble coughing up enough sputum, your provider may ask you to breathe in a salty mist to help you cough more deeply.

If you can't cough up enough sputum after breathing in the salty mist, your provider may perform a bronchoscopy.

If you have severe respiratory disease and a flu infection, your provider may use a bronchoalveolar lavage (BAL) to collect a sample from your lungs. This is a procedure that can be done during a bronchoscopy (a procedure that's used to check for the cause of a lung problem).

If you use an at-home test, it's very important to follow all the instructions that come with your test kit. There are two types of at-home tests:

Rapid antigen tests. You usually swab your nose to collect the fluid sample. The instructions will tell you how to complete the test, which takes under 30 minutes.

Self-collection tests. You usually swab your nose to collect a fluid sample to send to a lab for testing, but some tests may use saliva (spit).

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a nasopharyngeal swab, a nasal aspirate or wash, or a conjunctival swab.

Because a sputum culture is often scheduled in the morning, your provider may encourage you to drink more water the night before. This is because drinking more water can increase the amount of sputum. You may also be asked not to eat for one or two hours before the test or take some antibiotics, as they may affect bacterial growth and your results. But don't stop taking any medicines unless your provider tells you to.

If you get a BAL and/or bronchoscopy, you may be asked to fast (not eat or drink) for up to twelve hours before the test. You may also be asked to stop taking certain medicines. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

If your throat or nose is swabbed, you may feel a gagging sensation or a tickle. A nasal aspirate or wash and a conjunctival swab may feel uncomfortable. These effects are temporary.

There is no risk to providing a sputum culture.

There is very little risk to having a bronchoscopy or a BAL. Serious complications are rare, but they may include bleeding in the airways, infection, or a pneumothorax (collapse of part of your lung).

WHAT DO THE RESULTS MEAN?

The results of your flu test tell you if signs of influenza A or B virus were found in your sample. Your results are usually described as "positive" or "negative."

A positive result means you most likely have the seasonal flu. It's important to limit your contact with others to keep from infecting them. False positives (when your test says you have the flu, but you really don't) are very uncommon. Most people who have the flu won't need antiviral medicine to get better, but your provider may prescribe it depending on your health and how long you've been sick.

To help ease your symptoms, get plenty of rest and drink lots of liquids. Over-the-counter pain relievers may help with fever and aches. But children and teens should never take aspirin for flu-like symptoms. That's because aspirin increases the risk of Reye Syndrome, a rare, but serious illness that can happen after a viral illness.

You can go back to your usual activities when these things have been true for at least 24 hours:

Your symptoms have been getting better, and

You haven't had a fever and you're not taking fever-reducing medicine.

A negative result means you may not have the flu. A different virus or bacteria[^] may be causing your symptoms. But the flu can't be ruled out. That's because your results could be a false negative (when your test says you don't have the flu, but you really do.)

If you have negative test results and flu-like symptoms, ask your provider what you should do next. It's possible that your test was done when you had only a small amount of virus in your body. You may need more testing, especially if your first test was a rapid antigen flu test.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A FLU TEST?

To lower your risk of getting the flu:

Avoid close contact with people who are sick.

Wash your hands often with soap and water. Use an alcohol-based hand cleaner if soap and water aren't available.

Avoid touching your eyes, nose, and mouth.

Clean frequently touched surfaces, such as doorknobs and countertops.

Your best protection against serious illness from the flu is to get a yearly flu vaccine (flu shot). Other vaccines[^] to protect you from respiratory infections include those for COVID-19, RSV and pneumonia. Talk with your provider to find out if these vaccines are right for you.

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Fluoroscopy

<https://medlineplus.gov/lab-tests/fluoroscopy/>

FLUOROSCOPY

WHAT IS FLUOROSCOPY?

Fluoroscopy is a type of x-ray that shows organs, tissues, or other internal structures moving in real time. Standard x-rays are like still photographs. Fluoroscopy is like a video. It shows your body systems in action. These include the cardiovascular (heart and blood vessels), digestive, and reproductive systems. Fluoroscopy allows your health care provider to watch your organs or structures as they function, such as your heart beating, lungs inflating, or intestines moving food. The procedure can help your provider evaluate and diagnose a variety of conditions.

WHAT IS IT USED FOR?

Fluoroscopy is used in many types of imaging procedures. It can be used to guide your provider during a procedure visually or to help diagnose a health condition. Some of the most common uses of fluoroscopy include:

Barium swallow or barium enema. In these procedures, fluoroscopy shows the movement of the gastrointestinal (digestive) tract.

Cardiac catheterization. In this procedure, fluoroscopy shows blood flowing through the arteries. It is used to diagnose and treat some heart diseases.

Placement of a catheter or stent inside the body. Catheters are thin, hollow tubes. They are used to get fluids into the body or to drain excess fluids from the body. Stents are devices that help open narrow or blocked blood vessels. Fluoroscopy helps ensure the proper placement of these devices.

Guidance in orthopedic surgery. A surgeon may use fluoroscopy may be used by a surgeon to help guide procedures such as joint replacement and fracture (broken bone) repair.

Hysterosalpingogram. In this procedure, fluoroscopy is used to provide images of female reproductive organs.

WHY DO I NEED FLUOROSCOPY?

You may need a fluoroscopy if your provider wants to check the function of a particular organ, system, or other internal part of your body. You may also need fluoroscopy for certain medical procedures that require imaging guidance. This could include surgery or the placement of a stent.

WHAT HAPPENS DURING FLUOROSCOPY?

Depending on the type of procedure, a fluoroscopy may be done at an outpatient radiology center or as part of your stay in a hospital. The procedure may include some or most of the following steps:

You may need to remove any jewelry, removable dental appliances, eyeglasses, and metal objects that may interfere with the procedure.

You may need to remove your clothing. If so, you will be given a hospital gown.

You may be given a lead shield or apron to wear over your pelvic area or another part of your body, depending on the type of fluoroscopy. The shield or apron protects from unnecessary radiation.

For certain procedures, you may be asked to drink a liquid containing contrast dye. Contrast dye is a substance that makes parts of your body show up more clearly on an x-ray.

If you are not asked to drink a liquid with the dye, you may be given the dye through an intravenous (IV) line or enema. An IV line will send the dye directly to your vein. An enema is procedure that flushes the dye into your rectum.

You will be positioned on an x-ray table. Depending on the type of procedure, you may be asked to move your body in different positions or move a certain body part. You may also be asked to hold your breath for a brief period of time.

If your procedure involves getting a catheter, your provider will insert a needle in the appropriate body part. This may be your groin, elbow, or other site.

Your provider will use a special x-ray scanner to make the fluoroscopic images. They will be able to view them on a computer screen.

If a catheter was placed, your provider will remove it.

For certain procedures, such as those involving injections into a joint or artery, you may first be given pain medicine and/or medicine to relax you. For procedures that use fluoroscopy as imaging guidance during surgery or stent placement, you may get general anesthesia to sedate you.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your preparation will depend on the type of fluoroscopy procedure. Your provider will let you know if you need to do any special preparations. For example, for certain procedures, you may be asked to fast (not eat or drink) for several hours before the test. You may also need to avoid certain medicines before the test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

You should not have a fluoroscopy procedure if you are pregnant or think you may be pregnant. Radiation can be harmful to a developing baby.

For others, there is little risk to having this test. The radiation dose depends on the procedure, but fluoroscopy is not considered harmful for most people. But talk to your provider about all the x-rays you've had in the past. The risks from radiation exposure may be linked to the number of x-ray treatments you've had over time.

If your procedure involves using contrast dye, there is a small risk of an allergic reaction. Tell your provider if you have any allergies, especially to shellfish or iodine, or if you've ever had a reaction to contrast material.

WHAT DO THE RESULTS MEAN?

Your results will depend on what type of procedure you had. Several conditions and disorders can be diagnosed by fluoroscopy. Your provider may need to send your results to a specialist or do more tests to help make a diagnosis.

If you have questions about your results, talk to your provider. To understand the results of your fluoroscopy procedure, your provider may consider your symptoms, medical history, and the results of other tests.

Learn more about laboratory tests, reference ranges, and understanding results.

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Follicle-Stimulating Hormone (FSH) Levels Test

<https://medlineplus.gov/lab-tests/follicle-stimulating-hormone-fsh-levels-test/>

FOLLICLE-STIMULATING HORMONE (FSH) LEVELS TEST

WHAT IS A FOLLICLE-STIMULATING HORMONE (FSH) LEVELS TEST?

This test measures the level of follicle-stimulating hormone (FSH) in a sample of your blood. A hormone is a chemical messenger in your bloodstream that controls the actions of certain cells or organs. FSH plays an important role in sexual development in children and fertility in adults.

In women who menstruate (have periods), FSH helps control the menstrual cycle. It triggers the growth of eggs in the ovaries and gets the eggs ready for ovulation. Ovulation is when an ovary releases an egg so it can travel down a fallopian tube where it can be fertilized by sperm.

In men, FSH helps control the amount of sperm that the testicles (testes) make. It also affects how healthy the sperm are.

In children, FSH levels are normally low until puberty starts. Then FSH levels begin to rise along with the levels of other hormones. The increase in all these hormones triggers the physical changes of puberty:

In girls, FSH tells the ovaries to start making the hormone estrogen. Estrogen is involved in the growth of breasts, body hair, and menstruation.

In boys, FSH tells the testicles to start making the hormone testosterone. Testosterone is involved in the growth of facial and body hair, changes in a boy's voice, and sperm production.

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FSH levels are controlled by a complex system of hormones made in different parts of your body. Abnormal levels of FSH may be a sign of a problem with any of these parts. They include your:

Pituitary gland, a small gland at the base of your brain. It makes FSH.

Hypothalamus, a part of your brain. It makes hormones that tell your pituitary gland how much FSH to make.

Reproductive glands, your ovaries or testicles. The amount of sex hormones they make let your hypothalamus and pituitary gland know when to start and stop making FSH.

Other names: follitropin, FSH, follicle-stimulating hormone: serum, pituitary gonadotropins

WHAT IS IT USED FOR?

FSH testing is used to help diagnose conditions that cause too much or too little FSH. The specific way the test is used depends on your sex and age.

In women, an FSH test may be used to:

Help find the cause of infertility

Check for medical conditions that affect how the ovaries work

Find the reason for irregular or stopped menstrual periods in women of childbearing age

Find out when a woman is most likely to get pregnant

Find out if menopause or perimenopause may have begun:

Menopause happens when the ovaries stop making certain hormones, and menstrual periods have stopped for 12 months in a row. This usually happens around age 50.

Perimenopause is the time leading up to menopause when hormones and periods begin to change. It usually starts

around age 45.

Menopause happens when the ovaries stop making certain hormones, and menstrual periods have stopped for 12 months in a row. This usually happens around age 50.

Perimenopause is the time leading up to menopause when hormones and periods begin to change. It usually starts around age 45.

In men, an FSH test may be used to:

Help find the cause of infertility

Find the reason for a low sperm count

Check for medical conditions that affect the testicles

In women and men, FSH testing may be used to help diagnose disorders of the:

Pituitary gland

Hypothalamus

In children and teens, FSH testing is most often used to find out if a medical disorder is causing early or delayed (late) puberty.

An FSH test is often used with a blood test for another hormone made in the pituitary gland called luteinizing hormone (LH). These two hormones work together to control sexual development and reproduction. Blood tests to check estrogen, testosterone, and progesterone levels are also commonly used with FSH testing.

WHY DO I NEED AN FSH LEVELS TEST?

For women:

You may need this test if:

You've been unable to get pregnant after 12 months of trying.

Your menstrual periods aren't regular or have stopped.

If you're age 45 or older, testing usually isn't needed. That's because high levels of FSH are a normal sign of perimenopause and menopause. As your ovaries release fewer eggs, your body makes more hormones to try to trigger ovulation. But you may need an FSH test along with other hormone tests if there is a medical reason, including:

Premature menopause (age 40 or younger)

Early menopause (before age 45)

Having symptoms that may be related to menopause, but the cause isn't clear

Premature menopause (age 40 or younger)

Early menopause (before age 45)

Having symptoms that may be related to menopause, but the cause isn't clear

You have a less interest in sex than usual.

For men:

You may need this test if:

You've been unable to get your partner pregnant after 12 months of trying.

You have less interest in sex than usual.

You have a low sperm count.

You have a loss of muscle and/or body hair.

For women and men:

You may need an FSH test if your health care provider thinks you could have a pituitary disorder. Pituitary problems can affect fertility and decrease your interest in sex. They may also cause symptoms, such as:

Fatigue

Weakness

Decreased appetite and/or weight loss

Children and teens may need FSH testing if puberty seems to be starting too early or too late:

Early puberty means showing signs of sexual development before age 8 in girls and age 9 in boys.

Delayed puberty means not showing signs of sexual development by age 13 in girls and age 14 in boys.

WHAT HAPPENS DURING AN FSH LEVELS TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Women who have menstrual periods may need to be tested at a specific time during their menstrual cycle. That's because FSH levels change throughout the month.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand the results of your FSH test, your provider will consider your sex, age, symptoms, medical history, and the results of other hormone tests, especially your level of luteinizing hormone (LH). Your provider can explain what the results of your FSH test mean.

Generally, in women and men:

Higher levels of FSH are often a sign of a condition in the reproductive glands (ovaries or testicles) that prevents them from making normal levels of sex hormones. The pituitary responds by making more FSH to try to get them to work properly. High FSH may also be a sign of a condition outside of the reproductive glands that affects how the glands work.

Conditions linked to high FSH levels include:

In women:

- Ovaries that never developed normally
- Primary ovarian insufficiency (POI)
- Polycystic ovary syndrome (PCOS)
- Menopause or perimenopause
- An ovarian tumor
- Thyroid diseases or adrenal gland disorders
- A chromosomal disorder, such as Turner syndrome

In men:

- A germ cell tumor (a tumor that starts in the cells that become sperm)
- Mumps
- A physical injury to the testicles
- A chromosomal disorder, such as Klinefelter syndrome

In women and men, damage to the ovaries or testicles, which may be caused by:

- Autoimmune disease

Radiation therapy and/or chemotherapy

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Autoimmune disease

Radiation therapy and/or chemotherapy

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Lower than normal levels of FSH in women and men are often a sign of a problem with the pituitary gland or hypothalamus. These problems may make it hard for your body to make FSH and LH.

In women, low FSH levels may also be linked to rapid weight loss, being very underweight, or extreme exercise.

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Generally, in children with signs of early sexual development (before age 8 in girls or age 9 in boys):

High levels of FSH and LH mean the child has early puberty, also called precocious puberty. In most cases, the cause is unknown. But in certain cases, the cause is a problem in the brain, such as:

A brain tumor

A past brain injury
A past brain infection, such as meningitis or encephalitis
A brain tumor
A past brain injury
A past brain infection, such as meningitis or encephalitis

Normal levels of FSH and LH mean the child has a type of early puberty called peripheral precocious puberty. This often means that a disorder in the testicles, ovaries, or adrenal glands is causing higher than normal levels of estrogen or testosterone. Severe hypothyroidism and exposure to medicines that contain sex hormones may also cause this type of early puberty.

Generally, in teens with no signs of sexual development by age 13 for girls or age 14 for boys, low or normal levels of FSH and LH usually mean delayed puberty. Many teens with delayed puberty are healthy and go through normal puberty at a later age. But delayed puberty may be caused by certain medical conditions, such as:

Poor nutrition from a long-term illness or eating disorder

Hypogonadism, which is when the ovaries or testicles make little or no hormone. It may be caused by:

Certain genetic disorders, including Kallmann syndrome
Tumors in the brain or pituitary gland
Turner syndrome in girls
Klinefelter syndrome in boys
Certain autoimmune disorders
Radiation therapy or chemotherapy
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Certain autoimmune disorders
Radiation therapy or chemotherapy

If you have questions about test results, talk with your or your child's provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN FSH LEVELS TEST?

At-home test kits that measure FSH levels in urine (pee) may help women find out if menopause or perimenopause is causing symptoms, such as irregular periods, vaginal dryness, or hot flashes.

Home tests can only find higher than normal FSH levels. They shouldn't be used to check whether you can become pregnant because they can't accurately tell whether your ovaries are making eggs. If you use a home FSH test, discuss your symptoms and test results with your provider.

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Food Allergy Testing

<https://medlineplus.gov/lab-tests/food-allergy-testing/>

FOOD ALLERGY TESTING

WHAT IS FOOD ALLERGY TESTING?

A food allergy happens when your immune system - your body's defense against germs - overreacts to a certain food as if it was harmful. If you have symptoms after eating certain foods, food allergy testing can help find out if your symptoms are caused by an allergic reaction to those foods. And it's important to find out because allergic reactions to food can be serious.

Most food allergy symptoms are mild and include skin rashes and abdominal (belly) pain. But sometimes, symptoms quickly develop into a life-threatening allergic reaction called anaphylaxis, or anaphylactic shock. Anaphylaxis is a medical emergency that may affect your whole body, including your breathing. And there's no way to predict when an allergic reaction may change from mild to serious.

If you have a food allergy, eating even tiny amounts of a certain food or ingredient can quickly trigger your immune system to react. Most food allergies are caused by:

Milk

Eggs

Peanuts

Tree nuts (including almonds, walnuts, pecans, and cashews)

Soy

Wheat

Fish

Shellfish

Food allergies are more common in children than adults. Most children will grow out of allergies to milk, egg, soy, and wheat. But if a food allergy begins in adulthood, it's likely to continue through life.

If you or your child has symptoms that could be from a food allergy, your or your child's health care provider will probably refer you to an allergist for testing. An allergist is a doctor who specializes in diagnosing and treating allergies and asthma.

The allergist will pick the type of food allergy test that's right for you or your child. Tests include skin tests, blood tests, and a test where you eat small amounts of a suspected food while a provider watches for signs of allergy.

Other names: IgE test, sIgE test, allergen-specific IgE in serum, skin prick test (SPT), oral challenge test

WHAT IS IT USED FOR?

Food allergy testing is used to help find out if you or your child has an allergy to a specific food. It is mainly used for people who have had symptoms that could be a food allergy. Testing may also be used to see if allergies have gone away over time.

Food allergy testing can help tell whether you have a true food allergy or a food intolerance or food sensitivity, which may cause similar symptoms. The key difference is that food allergies involve your immune system and can affect your whole body and become life-threatening. But food intolerance and sensitivity don't become life-threatening and generally affect only your digestion.

Examples of food intolerances, or food sensitivities, include lactose intolerance, gluten sensitivity, and irritable bowel syndrome. The symptoms may include belly pain, nausea, gas, and diarrhea, which can also happen with food allergies.

WHY DO I NEED FOOD ALLERGY TESTING?

You or your child may need food allergy testing if either of you:

Have symptoms within minutes to hours after exposure to certain foods. Symptoms may affect your digestion, skin,

and/or breathing, and may include:

Nausea and vomiting

Stomach pain

Diarrhea

Tingling in the mouth

Skin reactions, such as hives or itching.

Red, itchy skin or a rash

Coughing

Stuffy, itchy, or runny nose

Nausea and vomiting

Stomach pain

Diarrhea

Tingling in the mouth

Skin reactions, such as hives or itching.

Red, itchy skin or a rash

Coughing

Stuffy, itchy, or runny nose

Have had symptoms of anaphylaxis. Symptoms of this life-threatening reaction may appear within minutes to hours after exposure to certain foods. Symptoms include:

Trouble breathing

Dizziness

Fainting

Swelling of the tongue, lips, and/or throat

Rapid heart rate

Cold, clammy, skin that may appear grayish, bluish, or pale

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If someone has symptoms of anaphylaxis, call 911 to get medical help right away.

You may also have food allergy testing if you or your child has been diagnosed with eosinophilic esophagitis (EoE).

Testing is done to find out which foods may make this condition worse.

Your child's provider may recommend food allergy testing if your infant or child has been diagnosed with:

Moderate to severe eczema

Enterocolitis, inflammation of the colon and small intestine

Enteropathy, a disease of the intestine

Allergic proctocolitis, a disorder that involves streaks of blood and mucus in stool (poop)

WHAT HAPPENS DURING FOOD ALLERGY TESTING?

An allergist will ask about your or your child's symptoms, family health history, and medical history, including other allergies. A physical exam is usually done before allergy testing.

Next, your allergist or another provider will do one or more allergy tests. There are many types of allergy tests, and they have different risks. So, ask your provider which tests are best for you or your child. These are the most common tests:

A skin prick test is the most commonly used test. A provider will place a drop of liquid on the skin of your arm or back. The liquid contains a protein from a specific food that may trigger an immune reaction. The provider will prick the skin under the drop so the liquid gets below the surface. Then you'll wait for 15 to 30 minutes. If a red, itchy bump forms, you may have an allergy to that food. More than one food protein may be tested.

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An allergy blood test measures immunoglobulin E (IgE) antibodies in your blood. Your immune system makes specific types of IgE in response to different foods. If you have a food allergy, your levels of IgE related to that food may be higher than normal. But the amount of IgE in your blood can't confirm a food allergy or tell you how serious an allergy may be. A blood test may be done with other tests to confirm a food allergy.

To take blood sample, a health care professional will insert a small needle into a vein in your arm. A small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

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To take blood sample, a health care professional will insert a small needle into a vein in your arm. A small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

An oral food challenge test is the most accurate food allergy test. It can be used alone to diagnose food allergies. During this test, you will slowly eat increasing amounts of a food that's suspected of causing your symptoms. You'll be closely watched for an allergic reaction. If your body doesn't react to the food, you aren't allergic to it.

If you have an allergic reaction, your food allergy is confirmed, and you'll be treated right away for the reaction. The treatment will depend on how severe your reaction is.

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If you have an allergic reaction, your food allergy is confirmed, and you'll be treated right away for the reaction. The treatment will depend on how severe your reaction is.

A food elimination diet is a test you do at home following instructions from your allergist. You'll start by removing all suspected foods from your diet for about two weeks. Then you'll slowly start adding the foods back to your diet, one at a time. If you have symptoms when you start eating a food again, you may be allergic to that food. Your symptoms could also be caused by a food intolerance, so other tests may be needed.

A food elimination diet may not be safe for someone who has had a serious allergic reaction to food in the past.

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A food elimination diet may not be safe for someone who has had a serious allergic reaction to food in the past.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you are having any type of allergy test, you may need to stop taking certain medicines and/or supplements before the test. Your provider will let you know what to avoid and for how long. Don't stop taking any medicines unless your provider tells you to. Your test may require other preparation, so be sure to follow all your provider's instructions for getting ready.

ARE THERE ANY RISKS TO THE TEST?

A skin prick test can cause itchy or irritated skin. Your allergist may recommend medicine to relieve the symptoms. The test is generally safe, but a severe reaction is possible.

A blood test has very little risk. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

An oral challenge test can cause a severe allergic reaction. That's why this test is done with close supervision by an allergist who can treat you if there is an emergency.

An elimination diet may cause a serious allergic reaction. Ask your allergist what to do if you develop symptoms.

WHAT DO THE RESULTS MEAN?

A food challenge test is the only food allergy test that can confirm a diagnosis of a food allergy. If you have other tests, your or your child's allergist will use your test results along with your symptoms and medical history to make a diagnosis. If the allergist diagnoses a food allergy, the treatment is to avoid the food that trigger your symptoms. There is no cure for food allergies. Avoiding trigger foods involves carefully reading food labels. You'll also need to explain the allergy to anyone who may prepare or serve food to you or your child, including friends, waiters, babysitters, and teachers.

Your allergist will tell you how to manage a mild allergic reaction. In case you or your child has a severe allergic reaction, your allergist may prescribe an autoinjector. An autoinjector is a small device that you use to inject an emergency allergy medicine, called epinephrine. If you need to use the device, it's still very important to get medical help right away even if you feel better. Your allergist will explain when and how to use an autoinjector.

If you have questions about test results and/or how to manage allergic reactions, talk with your allergist.

Learn more about laboratory tests, reference ranges, and understanding results.

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Free Light Chains

<https://medlineplus.gov/lab-tests/free-light-chains/>

FREE LIGHT CHAINS

WHAT IS A FREE LIGHT CHAINS TEST?

A free light chains test measures the level of free light chains in your blood. Light chains are proteins made by plasma cells, a type of white blood cell. Light chains usually link up with other proteins called heavy chains. Together, the light and heavy chains make immunoglobulins, also called antibodies. Antibodies are proteins that your immune system makes to fight germs such as viruses and bacteria.

Light chains can be bound or free:

When light chains link up with heavy chains, they are known as bound light chains.

Normally, your plasma cells will make a small amount of extra light chains that don't bind with heavy chains. Instead, they are put into your bloodstream. These unlinked chains are known as free light chains.

There are two types of light chains: lambda and kappa light chains. You will usually have some of each in your blood. A free light chains test measures the amount of lambda and kappa free light chains in your blood.

If the amount of free light chains is higher or lower than normal, it may be a sign of a disorder of the plasma cells. These disorders include multiple myeloma, a cancer that begins in the plasma cells, and amyloidosis, a condition that occurs when abnormal proteins build up and collect in different organs and tissues.

Other names: free kappa/lambda ratio, kappa/lambda quantitative free light, freelite, kappa and lambda free light chains, immunoglobulin free light chains, free light chain assay test

WHAT IS IT USED FOR?

A free light chains test is used to measure the level of free light chains in the blood and compare the amount of kappa and lambda light chains. This ratio can help diagnose or monitor how well the treatment of plasma cell disorders is working.

WHY DO I NEED A FREE LIGHT CHAINS TEST?

Your health care provider may order this test if you have symptoms of a plasma cell disorder. Depending on which plasma disorder you may have and which organs are affected, your symptoms may include:

Bone pain, often in the back or ribs

Fractures (broken bones)

Weakness, numbness, or tingling in the arms or legs

Fatigue

Frequent infections and fevers

Easy bruising or bleeding

Trouble breathing

Feeling very thirsty

Frequent urination

Tongue swelling

Purple spots on the skin

WHAT HAPPENS DURING A FREE LIGHT CHAINS TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle

goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a free light chains test.

ARE THERE ANY RISKS TO A FREE LIGHT CHAINS TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will show the amounts of lambda and kappa free light chains in your blood. It will also provide a comparison between the two, called a ratio.

If your results were not normal, it may be a sign of a plasma cell disorder, such as:

Multiple myeloma.

Amyloidosis.

MGUS (monoclonal gammopathy of unknown significance). This is a condition in which you have abnormal protein levels. It often has no symptoms but sometimes can become multiple myeloma.

Waldenstrom macroglobulinemia (WM), a cancer of the white blood cells. It's a type of non-Hodgkin lymphoma.

You may also have a higher level of free light chains if you have a kidney condition. In this case, a higher level may not mean that you have a plasma cell disorder.

To understand the results of a free light chains test, your provider will consider your symptoms, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A FREE LIGHT CHAINS TEST?

A free light chains test is often ordered with other tests, including an immunofixation blood test, to help confirm or rule out a diagnosis.

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Fungal Culture Test

<https://medlineplus.gov/lab-tests/fungal-culture-test/>

FUNGAL CULTURE TEST

WHAT IS A FUNGAL CULTURE TEST?

A fungal culture test helps diagnose fungal infections. Fungal infections may happen if you are exposed to fungi (more than one fungus). Fungi are plant-like life forms, such as yeasts and molds. Fungi live everywhere:

Outdoors in air, soil and on plants

Indoors on surfaces and in the air

On your skin and inside your body

Usually, if you are healthy, fungi will not make you sick. But many types of fungi can cause infections ranging from mild to life-threatening.

Fungal culture tests use different types of samples depending on the part of your body that's infected. Samples may include skin cells, blood, urine (pee), or other body fluid.

Your sample is sent to a lab and put in a dish with a special substance that helps grow any fungi in the sample. When there is enough material, it's checked under a microscope to see if it contains any fungi. Other tests may also be done to learn more about fungi in the sample.

Fungal infections may be grouped based on where they are in your body. In general, these groups include:

Superficial fungal infections. These infections affect the surfaces of your body, including your skin, genital area, and nails. They are very common and tend to be mild. But they often cause uncomfortable symptoms, such as itchy, scaly rashes. Two common superficial fungal infections are:

Ringworm (tinea). It usually causes a ring-like rash on the skin. Ringworm on your feet is called athlete's foot, and on your groin, it's often called "jock itch." The fungus spreads from contact with infected people or animals. You can also get it from shared objects, such as towels, or surfaces, such as locker room floors.

Yeast (candida). Yeast normally lives in and on your body. Usually, your immune system and certain "good bacteria" in your body keep it in control. If yeast grows out of control it can cause vaginal yeast infections and thrush, which affects your mouth or throat. This may happen if you have a weakened immune system or take certain medicines.

Deep and systemic fungal infections. These serious infections affect tissue deeper inside your body. The fungus may grow in your blood, lungs, and other organs, including your brain. If an infection affects many parts of your body at once, it's called a "systemic infection."

Anyone can get a deep fungal infection, but they are less common in healthy people. Most serious deep fungal infections happen in people who have weakened immune systems or are very sick with other conditions. Examples include:

Aspergillosis. This infection is caused by breathing Aspergillus mold spores, which are commonly found in the air. It mostly affects your lungs but may also spread to other parts of your body. The spores usually don't cause illness in healthy people.

Histoplasmosis. This lung infection is caused by breathing Histoplasma spores found in soil, usually where there are bird and bat droppings (poop). The infection can spread to other parts of your body, including your brain.

Pneumocystis pneumonia. This lung infection spreads from person-to-person in the air. Healthy people may carry the fungus in their lungs without having symptoms.

Invasive yeast infection (invasive candidiasis). This infection mostly happens only when your immune system is too weak to control the growth of yeast that's normally found in your body. It can affect your blood and organs.

Sporotrichosis, or "rose gardener's disease." In the United States, this infection is caused by a fungus called *Sporothrix schenckii*. It lives in soil and on plants and gets into your skin through small cuts and scrapes. It usually causes infection under the skin, but it can also cause deep infections in your lungs and other organs.

WHAT IS IT USED FOR?

Fungal infections cause symptoms that can be a lot like bacterial and viral infections, but their treatment is different. So,

a fungal culture test is used to:

Find out if a fungus is causing an infection

Help identify the type of fungus that's causing the infection

Decide what treatment will work best

Check to see if treatment for the infection is working

Your provider may also order a bacteria culture test¹ at the same time as a fungal culture test.

WHY DO I NEED A FUNGAL CULTURE TEST?

Your provider is most likely to order a fungal culture test if you have symptoms that could be from a deep fungal infection. Your symptoms will depend on the type of fungus you may have and where it's growing. They may include:

Cough, chest pain, or shortness of breath

Fever

Joint and muscle aches

Headaches

Chills

Nausea and vomiting

Fatigue

Fast heartbeat

If you have symptoms of a superficial fungal infection on your skin or nails, your provider may be able to diagnose it with a physical exam and checking a sample under a microscope. But in certain cases, especially if your symptoms aren't getting better, you may need a fungal culture test.

Your symptoms will vary depending on the type of fungus you may have and where it's growing. They may include:

Red rash

Itchy, scaly, or cracked skin

Hair loss

Vaginal itching and/or discharge

White patches inside the mouth

Nails that are discolored (yellow, brown, or white), thick, or brittle

WHAT HAPPENS DURING A FUNGAL CULTURE TEST?

To do a fungal culture test you will need to provide a sample of cells or fluid from the part of your body where a fungus may be growing. The most common samples are:

Skin scrapings or nail clippings

Used to diagnose superficial skin or nail infections

How it's done:

Your provider will use a special tool to take a small sample of your skin or nails. In certain cases, a skin biopsy² may be needed.

Your provider will use a special tool to take a small sample of your skin or nails. In certain cases, a skin biopsy² may be needed.

Body fluids

Used to diagnose yeast infections in your mouth or vagina. It may also be used to diagnose skin infections that have sores.

How it's done:

Your provider will use a special swab to gather fluid and cells from your mouth, vagina, or an open skin sore.

Your provider will use a special swab to gather fluid and cells from your mouth, vagina, or an open skin sore.

Blood

Used to find fungal infections in your blood.

How it's done:

A health care professional will collect a blood sample. The sample is most often taken from a vein in your arm.

A health care professional will collect a blood sample. The sample is most often taken from a vein in your arm.

Urine

Used mainly to find the cause of a urinary tract infection (UTI).

How it's done:

You will collect a sterile sample of urine in a container, as instructed by your provider.

You will collect a sterile sample of urine in a container, as instructed by your provider.

Sputum Culture

Sputum is a thick mucus that is coughed up from your lungs. It is different from spit or saliva.

Used to help diagnose fungal infections in your lungs

How it's done:

You may be asked to cough up sputum into a special container as instructed by your provider.

You may be asked to cough up sputum into a special container as instructed by your provider.

Many types of fungi grow slowly in the lab, so your test results may not be ready for days or weeks.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations to test for a fungal infection.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having any of the fungal culture tests. If a sample of your skin is taken, you may have a little bleeding or soreness at the site. If you get a blood test, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A positive test result means a fungus was found in your sample. You most likely have a fungal infection. Your test results will usually identify the specific type of fungus causing the infection.

Sometimes another test is done on your sample to help find out which type of medicine will kill the fungus or stop it from growing.

A negative test result means that no fungi were found in your sample. Something else is probably causing your infection.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A FUNGAL CULTURE TEST?

If you are being treated for a fungal infection, be sure to take all your medicine as prescribed, even if you feel better. Many fungal infections clear up within a few days to weeks, but some infections require months or even years of treatment.

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G6PD Test

<https://medlineplus.gov/lab-tests/g6pd-test/>

G6PD TEST

WHAT IS A G6PD TEST?

This test measures the amount of G6PD in your blood. G6PD stands for glucose-6-phosphate dehydrogenase. It's an enzyme, a protein that speeds up certain chemical reactions in your body. It helps your red blood cells work properly. Red blood cells move oxygen from your lungs to every cell in your body. Your cells need oxygen to grow, reproduce, and stay healthy.

If you don't have enough G6PD, it's known as G6PD deficiency. This deficiency:

It is a genetic disorder that is inherited (passed down from parent to child).

It mainly affects men. Women can be affected, but they are more likely to be carriers of the defective gene. This means that they may not have symptoms, but they can still pass the gene on to their children.

It is more common in people of African, Mediterranean, or Asian descent.

G6PD deficiency can cause the destruction of your red blood cells. When red blood cells are destroyed faster than your body can replace them, it's called hemolytic anemia. If you have hemolytic anemia, your cells don't get all the oxygen they need.

Most people with G6PD deficiency don't have symptoms until they are exposed to certain "triggers" which set off the destruction of red blood cells. Triggers can include:

Fava beans, also called broad beans. Hemolytic anemia which happens after eating fava beans or inhaling pollen from fava plants is called favism.

Viral infections.

Bacterial infections.

Certain antibiotics.

Anti-malaria medicines.

Other medicines, including nonsteroidal, anti-inflammatory drugs (NSAIDs) such as ibuprofen and aspirin.

Other names: G6PD deficiency test, glucose-6-phosphate dehydrogenase test, G-6-PD, RBC G6PD test

WHAT IS IT USED FOR?

A G6PD test is used to check if you have an inherited G6PD deficiency.

WHY DO I NEED A G6PD TEST?

You may need a G6PD test if you have symptoms of hemolytic anemia. These include:

Fatigue

Pale skin or jaundice, a condition that causes your skin and eyes to turn yellow

Rapid heart rate

Shortness of breath

Dark or yellow-orange colored urine (pee)

Your newborn baby might need a G6PD test if they have jaundice that doesn't go away in two weeks and/or is not caused by another condition. Your baby may also be tested if you have a family history of G6PD deficiency.

WHAT HAPPENS DURING A G6PD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle

goes in or out. This usually takes less than five minutes.

If your baby is being tested, a health care provider will clean your baby's heel with alcohol and poke the heel with a small needle. The provider will collect a few drops of blood and put a bandage on the site.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

There are no special preparations needed for a G6PD test.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

Your baby may feel a little pinch when the heel is poked, and a small bruise may form at the site. This should go away quickly.

WHAT DO THE RESULTS MEAN?

If your results show you have lower than normal amounts of G6PD, it means you have a G6PD deficiency. But your symptoms and risk of getting hemolytic anemia can vary, depending on your personal health history and exposure to triggers.

If you are a woman with slightly lower than normal amounts of G6PD, you may be a carrier of a G6PD deficiency. That means you have one defective G6PD gene and one normal G6PD gene. You might not have symptoms, since your normal G6PD genes usually make enough healthy red blood cells. But you have a risk of passing on the defective gene to your children. Male children are more likely to develop G6PD symptoms than female children.

If you are a man with a normal amount of G6PD, you are unlikely to have a deficiency. If you have anemia, it may be due to another cause. Your provider may want to repeat the G6PD test in a few weeks if you had the test during an episode of hemolytic anemia.

Learn more about laboratory tests, references ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A G6PD TEST?

Most people with G6PD deficiency can manage their condition and prevent symptoms by avoiding triggering substances. Your provider can help you figure out which substances to avoid.

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RELATED HEALTH TOPICS

Anemia
G6PD Deficiency
Genetic Disorders
Jaundice

RELATED MEDICAL TESTS

Bilirubin Blood Test
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Galactosemia Tests

<https://medlineplus.gov/lab-tests/galactosemia-tests/>

GALACTOSEMIA TESTS

WHAT IS A GALACTOSEMIA TEST?

A galactosemia test is a blood test given to newborns to check for a rare genetic metabolic disorder called galactosemia. Metabolic disorders affect metabolism, the process your body uses to make energy from the food you eat. Galactosemia can be inherited (passed down through families), but to have the condition, your baby has to receive the gene from both parents. Genes are parts of DNA in your cells that you inherit from your parents. You inherit two copies of each gene, one from each parent.

Galactosemia prevents a baby's body from breaking down a simple sugar called galactose. Galactose is in many foods, including:

Milk, milk-based baby formulas, and dairy products. These foods contain lactose, which is made up of galactose and glucose, another type of sugar.

Some fruits and vegetables.

Breast milk.

Normally, certain enzymes (proteins that speed up certain chemical reactions in your body) break down galactose. Galactosemia happens when a genetic change affects the function of these enzymes. There are three types of galactosemia. The types are based on which enzyme is affected:

GALT (galactose-1 phosphate uridyl transferase) deficiency, also known as classic galactosemia. It is the most common and severe form of the disorder.

GALK (galactose kinase) deficiency.

GALE (galactose epimerase) deficiency.

If a baby with a GALT, GALK, or GALE deficiency eats food with galactose, high levels of galactose and its byproducts build up in their blood. This can lead to serious health problems. These include liver disease, kidney failure, brain damage, and even death. But with early diagnosis and treatment, children with galactosemia can live healthy lives.

Other names: galactosemia newborn screening test, GALT test

WHAT IS IT USED FOR?

A galactosemia test is used to help diagnose galactosemia in a baby. It may also help identify whether it is a GALT, GALK, or GALE deficiency.

WHY DOES MY BABY NEED A GALACTOSEMIA TEST?

Newborns in the United States are required to get this test as part of a newborn screening. A newborn screening is a simple blood test that checks for various serious diseases.

Some older infants may need testing if they were adopted from another country and have symptoms of galactosemia. These symptoms can include:

Vomiting

Poor weight gain

Refusal to eat

Irritability

Jaundice, a condition that causes your skin and eyes to turn yellow

The symptoms often go away once your child begins avoiding galactose in their diet.

WHAT HAPPENS DURING A GALACTOSEMIA TEST?

A health care provider will clean your baby's heel with alcohol and poke their heel with a small needle. The provider will collect a few drops of blood and put a bandage on the site.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

There are no special preparations needed for a galactosemia test.

ARE THERE ANY RISKS TO THIS TEST?

Your baby may feel a little pinch when the heel is poked, and a small bruise may form at the site. This should go away quickly.

WHAT DO THE RESULTS MEAN?

Your baby's results will be either positive or negative for galactosemia.

If the results are negative, your child doesn't have the disorder.

If the results are positive, your baby will probably need more testing to confirm the diagnosis. These may include other blood tests, urine tests, and genetic tests.

Learn more about laboratory tests, references ranges, understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A GALACTOSEMIA TEST?

There is no cure for galactosemia. Infants diagnosed with the disorder can be fed soy formula or certain other lactose-free formulas. Children with classic galactosemia (GALT) must keep a special diet that avoids all milk and products that contain milk. They may need to take a calcium supplement to get enough calcium. Children with GALK and GALE deficiencies may be able to include a limited amount of dairy in their diets.

If your child has one of these deficiencies, talk to your child's provider about planning a safe and healthy diet in infancy and as your child grows up. Children with these deficiencies will probably need to stay on special diets for the rest of their lives.

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Gamma-glutamyl Transferase (GGT) Test

<https://medlineplus.gov/lab-tests/gamma-glutamyl-transferase-ggt-test/>

GAMMA-GLUTAMYL TRANSFERASE (GGT) TEST

WHAT IS A GAMMA-GLUTAMYL TRANSFERASE (GGT) TEST?

A gamma-glutamyl transferase (GGT) test measures the amount of GGT in your blood. GGT is an enzyme. Enzymes are proteins that speed up certain chemical reactions in your body. Thousands of enzymes play an important role in all of your body functions. The GGT enzyme is found throughout your body, but it is mainly found in your liver.

If your liver or bile duct is damaged, GGT may leak into your bloodstream. So high levels of GGT in your blood may be a sign of liver disease or damage to the bile ducts. Bile ducts are tubes that carry bile in and out of the liver. Bile is a fluid made by the liver that is important for digestion.

A GGT test can't diagnose the specific cause of liver disease. The test can only indicate that your liver is being damaged. This test is usually done with or after other liver function tests, most often an alkaline phosphatase (ALP) test. ALP is another type of liver enzyme. While both your GGT and ALP may be elevated in diseases that affect your bile ducts and liver, only ALP will be elevated in bone disease. This helps your health care provider to understand if your issue may be due to a bone disorder instead of your liver.

Other names: gamma-glutamyl transpeptidase, GGTP, Gamma-GT, GTP

WHAT IS IT USED FOR?

A GGT test is most often used to:

- Help diagnose liver disease

- Figure out if liver damage is due to liver disease or a bone disorder

- Check for blockages in the bile ducts

- Screen for or monitor alcohol use disorder (AUD). Drinking alcohol increases GGT levels. Most people with AUD have high GGT levels.

WHY DO I NEED A GGT TEST?

You may need a GGT test if you have symptoms of liver disease. They may include:

- Fatigue

- Weakness

- Jaundice, a condition that causes your skin and eyes to turn yellow

- Loss of appetite

- Abdominal (belly) pain or swelling

- Nausea and vomiting

- Severe itching

Your provider may use this test to monitor your alcohol use if you're being treated for AUD. You may also need this test if you have abnormal results on an ALP test and/or other liver function tests.

WHAT HAPPENS DURING A GGT TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Talk to your provider to see if there are any specific instructions, since the results of your GGT test may be affected by:

- Certain medicines or supplements
- Drinking alcohol
- Smoking cigarettes

GGT levels go down after meals. Your provider may also ask you to fast¹ for a certain amount of time before testing.

ARE THERE ANY RISKS TO A GGT TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results show higher-than-normal levels of GGT, it may be a sign of liver damage. The damage may be due to one of the following conditions:

Hepatitis.

Cirrhosis.

Alcohol use disorder (AUD).

Pancreatitis.

Diabetes.

Heart failure.

Side effects of a medicine. Certain medicines can cause liver damage in some people.

The results can't tell you which condition is causing your liver damage, but they can help show how much damage there is. Usually, the higher the level of GGT, the more liver damage you have.

If your results show low or normal levels of GGT, you probably don't have liver disease.

Your results may also be compared with the results of an ALP test. ALP tests help diagnose bone disorders. Together, your results may show one of the following:

High levels of ALP and high levels of GGT mean your symptoms are likely due to a liver disorder and not a bone disorder.

High levels of ALP and low or normal GGT levels mean it's more likely you have a bone disorder.

If you have questions about your results, talk to your provider. Your provider will consider your symptoms, medical history, and the results of other blood tests to understand the results of your GGT test.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A GGT TEST?

In addition to an ALP test, your provider may order liver function tests with or after the GGT test. These may include:

Alanine aminotransferase, or ALT

Aspartate aminotransferase, or AST

Lactic dehydrogenase, or LDH

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RELATED HEALTH TOPICS

Alcohol Use Disorder (AUD)

Bile Duct Diseases

Bone Diseases

Cirrhosis

Hepatitis

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Liver Diseases

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Alkaline Phosphatase

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Gastric Emptying Tests

<https://medlineplus.gov/lab-tests/gastric-emptying-tests/>

GASTRIC EMPTYING TESTS

WHAT ARE GASTRIC EMPTYING TESTS?

Gastric emptying tests measure the time it takes for food to empty out of your stomach. After a meal, it usually takes around four hours for 90 percent of the food to move out of your stomach and into your small intestine. But if your stomach empties too fast or too slow, it could be a sign of a health issue.

When your stomach empties too slowly (longer than normal), it may be a sign of delayed gastric emptying, called gastroparesis. It's often difficult to find the cause of gastroparesis, but diabetes is the most common underlying cause since it can damage some of the nerves in your stomach. Other causes of gastroparesis can include:

Hypothyroidism

Injury to your esophagus (the muscular tube that carries food and liquids from your mouth to the stomach), stomach, or small intestine during surgery

Certain autoimmune diseases

Nervous system disorders such as Parkinson's disease and multiple sclerosis

A viral infection of your stomach

If food stays in your stomach for too long, it can harden into a solid mass called a bezoar. A bezoar may lead to a dangerous blockage that prevents food from passing into your small intestine.

Gastroparesis may also cause malnutrition (not getting enough nutrients from your food).

When your stomach empties too rapidly (faster than normal), it may be a sign of dumping syndrome. Dumping syndrome happens when your stomach empties food too quickly into your small intestine. It is usually associated with surgery on your stomach (such as weight loss surgery) or your esophagus.

If you have symptoms such as diarrhea, bloating, or nausea within 30 minutes after eating, it may be a sign of early dumping syndrome, which is more common than late dumping syndrome.

If you have symptoms such as feeling light-headed, tired, or a fast heartbeat within one to three hours after eating a meal, it may be a sign of late dumping syndrome.

Gastric emptying tests may help you get treatment for gastroparesis and other digestive system issues before they cause serious complications.

Other names: gastric emptying study, gastric emptying scan, gastric emptying scintigraphy, smart pill, wireless motility study, gastric emptying breath test, upper GI series, barium swallow

WHAT ARE THEY USED FOR?

Gastric emptying tests are most often used to diagnose gastroparesis. They may also be used to check for other problems with food passing through your stomach.

WHY DO I NEED A GASTRIC EMPTYING TEST?

You may need this test if you have symptoms of gastroparesis, which include:

Abdominal (belly) pain

Bloating

Nausea and vomiting

A feeling of fullness after just a few bites of food

Loss of appetite

Weight loss

WHAT HAPPENS DURING A GASTRIC EMPTYING TEST?

Gastric emptying tests are most often done by a radiologist or radiology technician. A radiologist is a doctor who specializes in using imaging tests to diagnose and treat diseases and injuries. There are different types of tests. You may have one or more of the following tests:

Gastric emptying study, also known as a gastric emptying scan, or gastric emptying scintigraphy. This is the most common test used to diagnose gastroparesis. During this test:

You will start by eating a light meal, often eggs and toast. The food will contain a small, harmless amount of radioactive material called a tracer.

After you finish eating, you will lie down on an x-ray table.

The radiologist will take images of your abdomen using a scanning device.

The radiologist will watch the movements of the radioactive tracer on a monitor. The tracer will show how food travels through your stomach.

Additional images will be taken over the next few hours to see how long it takes for food to move out of your stomach and into your small intestine.

You can get up and leave the exam room during this time. Your provider will let you know when you need to return for imaging. It's usually at around 1, 2, and 4 hours after the first image was taken.

A gastric emptying study may also be done using a liquid containing the radioactive tracer instead of solid food. Since liquids move through your stomach differently, your provider may want to perform a liquid gastric emptying test with your gastric emptying study or as a follow-up.

Upper GI series, also known as a barium swallow. During this test:

You will swallow a chalky-tasting liquid that contains barium. Barium is a substance that coats your internal organs and tissues so they can be seen more clearly on an x-ray.

You will lie on an x-ray table.

A special type of x-ray called a fluoroscopy will track the barium in real-time as a live video as it moves through your esophagus, stomach, and small intestine.

You may be asked to hold your breath at certain times.

You may be asked to change positions while more images are taken.

Gastric emptying breath test. During this test:

You will eat a meal or drink a liquid that contains a type of protein that your body absorbs. The substance eventually passes into your breath.

A provider will collect samples of your breath over a period of about four hours.

You may be able to leave the exam room during this period but must sit quietly. You should not smoke or eat while you wait. Your provider will let you know when you need to return for imaging.

The amount of substance in your breath samples can help show how fast your stomach has emptied.

Smart pill, also known as a wireless motility capsule. During this test:

You will swallow a smart pill, a pill-sized wireless electronic device.

You will then proceed with your normal daily activities.

As you go through your day, the smart pill will travel through your gastrointestinal system and collect data.

The data is sent to a receiver you wear, usually around your waist.

After a day or two, the smart pill will leave your body in a bowel movement (poop).

You may not feel it when the smart pill exits your body. So when you have a bowel movement, wait 3 minutes before flushing. This will ensure the receiver you wear on your body captures the data, even after the capsule leaves your body.

After 4 or 5 days, you will return the receiver to the radiologist, who will review the results.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for several hours before your test. Your provider will also let you know if you need to stop taking certain medicines before your test.

ARE THERE ANY RISKS TO THE TEST?

You should not have a gastric emptying study or an upper GI series if you are pregnant or think you may be pregnant. Radiation from these imaging tests can be harmful to a developing baby.

For others, there is little risk to having an imaging test. The dose of radiation is very low and not considered harmful for most people. But talk to your provider about all the x-rays you've had in the past. The risks from radiation exposure¹ may be linked to the number of x-ray treatments you've had over time.

If you're having an upper GI series, tell your provider if you have any known allergies to the barium used in the test. Also, your poop may look white for several days after the test as you pass the barium. You may have constipation² if all the barium doesn't pass out of your body.

There is very little risk to having a gastric breath test or taking a smart pill. In rare cases, a smart pill may get stuck in your digestive system. If the pill takes longer than several days to leave your body in a bowel movement, contact your provider.

WHAT DO THE RESULTS MEAN?

If your results show it took a longer-than-normal time for your stomach to empty, it could mean you have gastroparesis.

If your results show it took a shorter-than-normal time for your stomach to empty, it could mean you have dumping syndrome.

If you have questions about your results, talk to your provider. Your provider may also consider your symptoms and medical history to understand your results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT GASTRIC EMPTYING TESTS?

Treatment for gastroparesis may involve taking certain medicines and/or treating the condition causing the problem, such as diabetes.

Regardless of the cause, your provider will likely recommend changing your diet and eating habits. These may include:

Eating foods low in fiber and fat.

Avoiding carbonated drinks, smoking, and alcohol.

Eating smaller meals throughout the day rather than two or three large meals.

Chewing food thoroughly.

Waiting at least two hours after eating before lying down.

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Glaucoma Tests

<https://medlineplus.gov/lab-tests/glaucoma-tests/>

GLAUCOMA TESTS

WHAT ARE GLAUCOMA TESTS?

Glaucoma tests are eye tests that help diagnose glaucoma. Glaucoma is a group of eye diseases that can damage a nerve in the back of your eye called the optic nerve. This nerve connects your eyes to your brain, allowing you to see. Damage to the optic nerve can cause vision loss and blindness. Glaucoma tests can help find the disease before you know you have it, so you can start treatment to protect your eyes against serious vision loss.

Researchers aren't sure exactly what causes the most common types of glaucoma. But having high fluid pressure in one or both eyes increases your risk for the disease. Your eyes are always making fluid that nourishes the tissues inside the eye. The fluid leaves your eyes through a drainage system with openings in the space where your iris (the colored part of your eyes) and your cornea (the clear front layer of your eyes) meet.

The pressure inside of each eyes stays in a healthy range when there's a balance between how much fluid your eye makes and how much drains out of your eye. If the drainage system becomes clogged or blocked, fluid can't drain at a normal rate. If your eye makes new fluid faster than it can drain, the fluid backs up, which increases the pressure in your eye. The high pressure may damage the optic nerve and lead to serious vision problems.

There are several types of glaucoma. The main types are:

Open-angle glaucoma, also called primary open-angle glaucoma. This is the most common type. It's called "open" because the canals that lead to your eye's drainage system are open. But deeper inside the drainage system, clogs develop. These clogs slow the fluid drainage.

Open-angle glaucoma tends to affect both eyes. In the early stages, Most people don't have any symptoms or vision changes. But without treatment, peripheral (side) vision is usually lost first.

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Closed-angle glaucoma, also called angle-closure or narrow-angle glaucoma. In this uncommon type of glaucoma, the canals that drain fluid from your eye are "closed" because your iris covers them. This condition usually affects one eye at a time. There are two main types:

Acute closed-angle glaucoma is a medical emergency. No fluid can drain out of your eye, so eye pressure increases suddenly. This causes eye pain, blurry vision, and other symptoms. Without prompt treatment, it can cause blindness in a few days.

Chronic closed-angle glaucoma develops slowly over time. You may not have any symptoms. Or you may have symptoms such as eye discomfort, blurry vision, redness, or headaches. The symptoms often get better with sleep. Without treatment, vision loss develops slowly.

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WHAT ARE THEY USED FOR?

Glaucoma tests are used to screen for and diagnose glaucoma. If glaucoma is diagnosed early, treatment may help slow or prevent vision loss.

WHY DO I NEED GLAUCOMA TESTING?

Anyone can get glaucoma. To protect your vision, it's important to be tested for glaucoma before symptoms start. When to start testing and how often to get tested will depend on your risk for glaucoma.

Your risk for glaucoma may be high if you:

Are over age 60. Glaucoma is much more common in older people. The risk is especially high if you are an older Hispanic/Latino American.

Are an African American over age 40. Glaucoma is the leading cause of blindness in African Americans.

Are an Asian American. Asian Americans have a higher risk for getting a type of open-angle glaucoma called normal-pressure or normal-tension glaucoma. With this condition, optic nerve damage begins when eye pressure is normal or a little higher than normal. But treatments that lower eye pressure often help.

Have a family history of glaucoma. You can inherit certain types of glaucoma from your parents.

Have certain medical conditions. They include diabetes and high blood pressure.

If your risk of glaucoma is high, ask your health care provider or eye care specialist when you should start getting tested and how often you need to have your eyes checked. Getting an eye exam with glaucoma testing by age 40 can help catch glaucoma and other eye diseases early.

If you have symptoms of acute closed-angle glaucoma, you need to be tested for glaucoma right away to prevent blindness. Symptoms may start suddenly and be severe. They include:

Blurry vision

Severe eye pain

Red eyes

Colored halos around lights

Nausea or vomiting

Headache

If you have glaucoma, you may need regular testing to monitor your condition and check whether your treatment is working.

WHAT HAPPENS DURING A GLAUCOMA TEST?

Glaucoma tests are done as part of a comprehensive (complete) eye exam. A complete eye exam checks both your vision and your eye health. These exams can be done by:

An ophthalmologist, a medical doctor who specializes in preventing, diagnosing, and treating all types of eye disease, including doing eye surgery.

An optometrist, a health care professional who specializes in diagnosing and treating vision problems and certain eye disorders.

If you're unsure which type of eye specialist should check you for glaucoma, talk with your provider.

Glaucoma tests check different parts of your eyes and vision, including:

Your eye pressure. There are a few ways to measure eye pressure using tonometry tests:

Air-puff test. You'll rest your chin on a machine and your eye care specialist will blow a puff of air into your eye. This quick and painless test is used as part of a routine glaucoma screening. If the results show that your eye pressure is high, your provider will do other eye-pressure tests to get a more accurate measurement.

Applanation tonometry. Your provider will numb your eyes with drops before measuring your eye pressure using one of these methods:

You'll rest your chin on a special magnifying device called a slit lamp. Your eye care specialist will examine your eye through the slit lamp while gently pressing a special tool on your eye to test the pressure.

Your eye care specialist will gently press a handheld device against your eye. The device measures your eye pressure.

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The thickness of your corneas. The cornea is the clear outer layer at the front of your eye. Its thickness can affect the results of your eye-pressure measurements. If your eye pressure is high, a pachymetry test gives your eye care specialist more information about how high it may be. To do the test, your eye will be numbed with drops and your eye care specialist will place a small device on your cornea to measure the thickness.

Your optic nerves. Your eye care specialist will do a dilated eye exam to look for damage to your optic nerve. This exam is part of a routine glaucoma check-up. It's also called ophthalmoscopy. You'll be given eye drops that widen (dilate) your pupils (the openings that let light into your eyes). You'll look straight ahead while your eye care specialist looks into your eye using a device with a light and magnifying lens.

Your peripheral (side) vision. A visual field test, also called perimetry, measures how well you can see using your side vision. There are different ways to do the test. In general, you'll look straight ahead. Objects will be shown off to the sides of your vision. Without moving your eyes, you'll say when you can see the objects.

The canals where fluid drains out of your eyes. If your eye care specialist thinks you have glaucoma, you may have a gonioscopy to see if your eye's drainage system is opened (open-angle glaucoma) or closed (close-angle glaucoma). Your eyes will be numbed with drops. Then your provider will hold a special contact lens on your eye. The lens has a mirror that shows the area where fluid drains out of your eye.

Some eye care specialists may use other tests to check for glaucoma, such as optical coherence tomography (OCT). This test makes images of the back of your eye that can show damage to the optic nerve.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A GLAUCOMA TEST?

If you wear glasses or contact lenses, bring them with you. Bring sunglasses to wear after your exam in case your eyes are sensitive to light. Also, plan to have someone take you home because your vision may be blurry.

ARE THERE ANY RISKS TO THE TESTS?

There is no risk to having glaucoma testing. Some of the tests may feel a bit uncomfortable. If your eyes were dilated, you may have blurry vision and be light sensitive for a few hours.

WHAT DO THE RESULTS MEAN?

If your test results aren't normal, you may have glaucoma or a high risk for developing glaucoma.

If your eye care specialist diagnoses glaucoma, your treatment will depend on the type of condition you have. Treatment won't cure the disease or restore any vision you have already lost. But it can prevent more vision loss. Your eye care specialist may recommend one or more treatments, such as:

Medicines to lower eye pressure or reduce the amount of fluid your eye makes.

Eye surgery or laser therapy to improve fluid drainage from the eye.

If an optometrist examined your eyes, you may be referred to an ophthalmologist for more testing and/or treatment.

You will probably have more glaucoma testing to see if the treatment is helping. Ask your eye care specialist to explain your test results and be sure to find out how often you should have your eyes checked.

Learn more about laboratory tests, reference ranges, and understanding results.

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Globulin Test

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GLOBULIN TEST

WHAT IS A GLOBULIN TEST?

Globulins are a group of proteins in your blood. Some globulins are made by your liver. Others are made by your immune system. Globulins play an important role in liver and kidney function, blood clotting, and fighting infection. There are different types of globulins called alpha, beta, and gamma globulins. Tests to measure globulins include:

Total protein test. This blood test measures the total amount of protein in your blood. The two main types of blood proteins are globulins and albumin. Albumin makes up most of the protein in the blood, while the rest are called globulin. If your total protein levels are low, it can mean that you have liver or kidney disease.

Serum protein electrophoresis. This blood test measures the level of each type of protein in your blood, including the different types of globulins. The levels of specific globulins provide information about how well your immune system is working. So, this test can be used to diagnose disorders of the immune system, a type of cancer called multiple myeloma, and a variety of other conditions.

Other names for globulin tests: Serum globulin electrophoresis, total protein

WHAT IS IT USED FOR?

Globulin tests can be used to help diagnose various conditions, including:

Liver damage or disease

Kidney disease

Nutritional problems, such as malnutrition

Certain autoimmune disorders, conditions in which your immune system attacks the healthy cells of your own tissues and organs by mistake

Certain types of cancer

WHY DO I NEED A GLOBULIN TEST?

Your health care provider may order globulin tests:

As part of your routine checkup

To help diagnose specific conditions when you have symptoms

To monitor liver or kidney problems

A total protein test may be included in a comprehensive metabolic panel (CMP), which is often part of a routine checkup. A CMP measures proteins and other substances in your blood. Your provider may order the test if you have symptoms of kidney or liver disease.

If your provider thinks you may have a liver disease, you may also have other tests that check how well your liver is working (liver function tests).

Your provider may order them if you are at risk for liver disease or have symptoms of liver disease, which may include:

Nausea and vomiting

Lack of appetite

Fatigue

Weakness

Jaundice, a condition that causes your skin and eyes to turn yellow

Swelling and/or pain in your abdomen (belly)

Swelling in your ankles and legs

Dark-colored urine (pee) and/or light-colored stool (poop)

Frequent itching

A serum protein electrophoresis test measures the levels of the different types of globulins and other proteins in your blood. Your provider may order this test to diagnose disorders related to your immune system, including:

Allergies

Autoimmune diseases such as lupus and rheumatoid arthritis

Multiple myeloma, a type of cancer

WHAT HAPPENS DURING A GLOBULIN TEST?

Globulin tests are blood tests. During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

If your globulin test is part of another blood test, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Low globulin levels may be a sign of:

Liver disease

Kidney disease

Malnutrition

High globulin levels may be a sign of:

Infection

Inflammatory disease

An immune disorder

Certain cancers, such as multiple myeloma, Hodgkin lymphoma (also called Hodgkin disease), or malignant lymphoma

Tuberculosis

Certain medicines, dehydration, pregnancy, or other conditions may also cause abnormal results. Talk with your provider to learn what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

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What You Need to Know About Blood Testing

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Glomerular Filtration Rate (GFR) Test

<https://medlineplus.gov/lab-tests/glomerular-filtration-rate-gfr-test/>

GLOMERULAR FILTRATION RATE (GFR) TEST

WHAT IS A GLOMERULAR FILTRATION RATE (GFR) TEST?

A glomerular filtration rate (GFR) test is a blood test that checks how well your kidneys are working. Your kidneys are two organs on either side of your spine near your waist. They have tiny filters called glomeruli. These filters remove waste and extra water from your blood and gets rid of them through urine (pee).

If your kidneys have been damaged by kidney disease, they can't filter your blood as fast as they should. A GFR test checks for kidney disease by measuring how much blood your kidneys filter each minute.

GFR can be measured directly, but it is a complicated test to do. So health care providers usually estimate GFR based on the amount of certain waste substances in your blood. An estimated GFR is called an eGFR.

To figure out your eGFR, your provider usually uses the results of a blood test that measures your creatinine level. Creatinine is a waste product that comes from normal wear and tear on your muscles. If your kidneys aren't working well, creatinine can build up in your blood.

People make different amounts of creatinine, depending on their size, diet, and activity levels. So, to calculate your eGFR, your provider will use your creatinine levels and other information about you, such as your:

Age

Weight

Height

Sex

This information is put into a mathematical formula, called a GFR calculator, to find your eGFR.

eGFR may also be calculated using the level of cystatin C in your blood. This is a protein that many cells in your body make. Cystatin C levels are not affected by muscle size, age, or diet, so some researchers think cystatin C provides a more accurate estimate of GFR than creatinine. In certain cases, creatinine and cystatin levels are both used to calculate eGFR in adults.

Other names: estimated GFR, eGFR, calculated glomerular filtration rate, cGFR

WHAT IS IT USED FOR?

A GFR test is used to:

Screen for kidney disease in people without symptoms

Help diagnose kidney disease in a people who have symptoms

Help find out how serious kidney disease is

Monitor people who:

Have chronic kidney disease (CKD) to see if treatment is helping

Take medicines that could harm their kidneys

Have chronic kidney disease (CKD) to see if treatment is helping

Take medicines that could harm their kidneys

Check kidney health before people start certain treatments that could affect their kidneys

WHY DO I NEED A GFR TEST?

Early-stage kidney disease doesn't usually cause symptoms, so you may have an eGFR test to check your kidney health as part of a routine exam.

You may also need this test if you have a high risk of getting kidney disease. Your risk may be higher if you:

Have diabetes

Have high blood pressure
Have a family health history of kidney disease, diabetes, or high blood pressure
Have heart disease
Are over 50 years old
Smoke
Have obesity
If you have a condition that increases your risk of kidney disease, ask your provider how often you should get tested.
Later stage kidney disease does cause symptoms. So, you may need an eGFR test if you have:
Swelling in your legs, feet, ankles, or hands or face
Urinating (peeing) more often or less often than usual
Dry skin and/or itching
Fatigue
Muscle cramps
Nausea and vomiting
Loss of appetite

WHAT HAPPENS DURING A GFR TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know how to prepare for your test. You may need to fast (not eat or drink) or avoid certain foods for several hours before the test. Certain medicines can affect your results. So be sure to tell your provider everything you're taking. But don't stop taking any medicine unless your provider tells you to stop.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your eGFR results may be reported as a number that shows how much blood your kidneys filter per minute. Your test results may also give your creatinine level and/or your cystatin C level.

Ask your provider to explain what your eGFR says about your kidney health. An eGFR isn't a perfect test and may not always reflect kidney damage. It's also possible to have an abnormal result even if you don't have kidney damage.

In general:

A normal eGFR means that you probably don't have kidney disease.

An eGFR that's below normal or low may mean that you may have kidney disease.

A very low eGFR means that you may have kidney failure.

If you're diagnosed with kidney disease, your provider will explain what you can do to protect your kidneys from more damage and help prevent kidney failure.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A GFR TEST?

Your provider may order other tests that check your kidney health. These tests may be ordered with an eGFR or to find the cause of an abnormal eGFR result. They include:

A protein in urine test. Protein in urine is a sign of kidney damage.

A microalbumin creatinine ratio test. This test checks for very small amounts of a protein called albumin in a sample of your urine. Albumin in urine may be one of the first signs of kidney disease.

A BUN (blood urea nitrogen). BUN is a waste product your kidneys remove from blood. The test checks BUN levels in a sample of your blood.

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What You Need to Know About Blood Testing

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Glucagon Blood Test

<https://medlineplus.gov/lab-tests/glucagon-blood-test/>

GLUCAGON BLOOD TEST

WHAT IS A GLUCAGON BLOOD TEST?

This test measures the amount of glucagon in your blood. Glucagon is a hormone made by your pancreas. Hormones are chemical messengers in your bloodstream that control the actions of certain cells or organs. Glucagon works with other hormones, like insulin, to help maintain your body's level of blood glucose, also called blood sugar. Glucagon increases your blood glucose levels, while insulin decreases your blood glucose levels.

Glucose is your body's main source of energy. It comes from the food you eat. Your body breaks down most of that food into glucose and releases it into your bloodstream. This causes your blood glucose levels to go up, and less glucagon is released.

And when your blood glucose levels drop, your pancreas releases glucagon into the bloodstream. But if your pancreas doesn't release the right amount of glucagon, it can lead to serious health problems.

Other names: glucagon blood test, hypoglycemia-glucagon test, glucagon-plasma

WHAT IS IT USED FOR?

A glucagon test is most often used to help diagnose the following conditions:

Hypoglycemia, also called low blood glucose, occurs when the level of glucose in your blood drops below what is healthy for you. Severe hypoglycemia can be deadly if it's not treated. The condition often affects people with diabetes. It can happen when your diabetes medicines and/or food intake are not at the proper levels.

Pancreatitis, an inflammation of the pancreas. It can be acute (sudden and short term) or chronic (long term). The systems of acute pancreatitis are severe and may require immediate medical attention. Chronic pancreatitis gets worse over time and can lead to permanent damage. Its symptoms may not appear until there are complications.

Glucagonoma, a rare tumor of the pancreas. It is usually cancerous.

The test is also sometimes used to monitor people with diabetes.

WHY DO I NEED A GLUCAGON BLOOD TEST?

You may need this test if you have symptoms of hypoglycemia or a pancreatic tumor.

Symptoms of hypoglycemia can include:

Fast or irregular heartbeat (arrhythmia)

Tingling in the lips, tongue, or cheeks

Anxiety

Irritability

Sweating

Blurred vision

Fatigue

Pale skin

Hunger

Symptoms of a pancreatic tumor can include:

A rash on the face, stomach, or legs

Diarrhea

Weight loss

Sores in the mouth or tongue

WHAT HAPPENS DURING A GLUCAGON BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for several hours before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your glucagon levels were not in the normal range, it may be a sign of one of the following conditions:

Hypoglycemia

Diabetes

Acute or chronic pancreatitis

Glucagonoma (pancreatic tumor)

If you have questions about your results, talk to your health care provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A GLUCAGON BLOOD TEST?

A glucagon blood test and a glucagon stimulation test are often called glucagon tests, but they are not the same thing. A glucagon blood test is used to check conditions related to glucose levels, while a glucagon stimulation test checks for problems with growth hormones. It is most often used for infants and children.

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care provider if you have questions about your health.

Glucose in Urine Test

<https://medlineplus.gov/lab-tests/glucose-in-urine-test/>

GLUCOSE IN URINE TEST

WHAT IS A GLUCOSE IN URINE TEST?

A glucose in urine test measures the amount of glucose in your urine (pee). Glucose is a type of sugar. It is your body's main source of energy. Glucose comes from the food you eat. Your body breaks down most of that food into glucose and releases it into your bloodstream. Your blood carries glucose to your cells for energy.

Normally, urine contains very little or no glucose. That's because your kidneys hold on to glucose and send it back to your bloodstream instead of flushing it out of your body in urine. If you have more glucose than normal in your urine, it could mean that you have:

Too much glucose in your blood, which could be a sign of diabetes. Diabetes is the most common cause of high urine glucose levels.

A rare kidney condition that causes your kidneys to remove glucose from your blood even when your blood glucose levels are normal. These conditions include:

Renal glycosuria, a genetic condition that is inherited (passed down through families)

Fanconi syndrome, a genetic condition you can inherit or develop later in life (Fanconi syndrome is not related to Fanconi anemia)

Renal glycosuria, a genetic condition that is inherited (passed down through families)

Fanconi syndrome, a genetic condition you can inherit or develop later in life (Fanconi syndrome is not related to Fanconi anemia)

If your urine glucose levels are high, your health care provider will usually order a blood glucose test to help make a diagnosis.

Other names: urine sugar test; urine glucose test; glucosuria test; glycosuria test

WHAT IS IT USED FOR?

A glucose in urine test is usually part of a urinalysis, a test that measures different cells, chemicals, and other substances in your urine. A urinalysis is often used to check your general health.

In the past, urine glucose testing was commonly used to screen for and monitor diabetes. But urine glucose testing is not as accurate as blood glucose testing, so blood tests are now more commonly used to diagnose and monitor diabetes.

A urine glucose test may still be used to help diagnose and monitor diabetes if you have a condition that makes blood testing difficult. For example, you may not be able to have a blood test if your veins are scarred from past blood tests or if you have a type of bacterial infection that could spread to your blood if a needle is inserted into a vein.

A glucose in urine test may also be done if your provider thinks you could have an uncommon type of kidney disorder.

WHY DO I NEED A GLUCOSE IN URINE TEST?

You may have a glucose in urine test as part of your routine checkup. You may also need this test if you have symptoms of diabetes or need to monitor your diabetes, but you are unable to have a blood glucose test.

Symptoms of diabetes include:

Increased thirst and urination

Increased hunger

Fatigue

Blurred vision

Numbness or tingling in the hands or feet

Sores that don't heal

Losing weight without trying

If you are pregnant, you will likely have a glucose in urine test as part of a urinalysis to check your health. Some glucose in urine is common during pregnancy. But high levels over time could be a sign of gestational diabetes, which is a type of diabetes that happens only during pregnancy.

You may need a glucose in urine test if your provider thinks you could have an uncommon kidney problem that affects how your kidneys filter glucose in blood. These conditions don't always cause symptoms. Your provider may want to test you because you have a family history of one of these conditions.

WHAT HAPPENS DURING A GLUCOSE IN URINE TEST?

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

Your provider may ask you to monitor your urine glucose at home with a test kit. Your home test kit will come with a package of strips for testing a sample of your urine and instructions on how to do the test. Be sure to follow the instructions carefully. Talk with your provider about what time of day you should test your urine to get the most accurate results.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You usually don't need any special preparations for this test. Some medicines and supplements may affect your test results, so be sure to tell your provider about everything you're taking. But don't stop taking any medicine without talking with your provider first.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a glucose in urine test.

WHAT DO THE RESULTS MEAN?

If your test result is normal, it means that you had little to no glucose in your urine. But this doesn't mean that you don't have diabetes. You can have a normal urine glucose level when your blood glucose level is high.

If your test result shows you have too much glucose in your urine, it may be a sign of:

Diabetes (If you have already been diagnosed with diabetes, high urine glucose may mean that your diabetes is not well controlled)

Pregnancy

Gestational diabetes

An uncommon type of kidney disorder

A urine glucose test alone cannot diagnose any conditions. So, if you have an abnormal test result, your provider may order other tests to help make a diagnosis, including blood tests to diagnose or rule out diabetes.

Too much urine glucose doesn't always mean you have a medical condition that needs treatment. Certain medicines, vitamin C, and eating a large meal may affect your test results. Ask your provider what your test results mean and whether you need a blood glucose test for more accurate results.

Learn more about laboratory tests, reference ranges, and understanding results.

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Gonorrhea Test

<https://medlineplus.gov/lab-tests/gonorrhea-test/>

GONORRHEA TEST

WHAT IS A GONORRHEA TEST?

A gonorrhea test looks for signs of a gonorrhea bacterial infection in a sample of your urine (pee) or saliva. Gonorrhea is one of the most common sexually transmitted infections (STIs). Anyone can get gonorrhea if they have vaginal, oral, or anal sex with someone who has the infection.

The bacteria often infect the genitals, but it can infect the rectum, throat, eyes, and other parts of the body. Your health care provider will test for a gonorrhea infection using a sample of your body fluid. The fluid used depends on the type of gonorrhea test and where the infection may be. They may collect a sample from the:

Urethra (the tube that carries urine out of the body in males and females).

Cervix (the lower part of the uterus).

Vagina.

Throat.

Rectum (the last part of your intestine that connects to the anus).

Gonorrhea often has no symptoms, so you may spread it to others without even knowing you have the infection. Gonorrhea can be passed to a baby during childbirth if the mother is infected. In babies, the infection usually affects the eyes.

Gonorrhea can be cured with antibiotics. But if it's not treated, it can lead to serious health problems including blood infection, especially in babies.

In the female reproductive system, gonorrhea can cause:

Pelvic inflammatory disease

Infertility or ectopic pregnancy

Long-term pelvic pain

In the male reproductive system, gonorrhea can cause:

Pain in the testicles

Infertility (uncommon)

In babies, gonorrhea can cause:

Blindness

Joint infections

A gonorrhea test can help you catch the infection early so you can get treatment before you develop lasting health problems or spread it to others.

Other names: GC test, gonorrhea DNA probe test, gonorrhea nucleic acid amplification test (NAAT), GC culture

WHAT IS IT USED FOR?

A gonorrhea test is used to:

Screen for a gonorrhea infection when you don't have symptoms but may have been exposed to the bacteria.

Diagnose or rule out gonorrhea when you have symptoms.

Gonorrhea testing is often done with a test for chlamydia, another type of sexually transmitted infection. That's because gonorrhea and chlamydia have similar symptoms, and it's common to have both infections at the same time. You may also be tested for syphilis and/or HIV.

WHY DO I NEED A GONORRHEA TEST?

You may need a gonorrhea test if you're sexually active and have symptoms of an infection or are at high risk for one. If you have symptoms, a test can diagnose or rule out gonorrhea. Symptoms may include:

An unusual discharge (fluid) from the genitals (vagina or penis).

Pain or burning when urinating (peeing).

Rectal symptoms, such as itching, bleeding, and anal discharge.

Pain in the lower abdomen (belly).

Sore throat (uncommon).

Vaginal bleeding between menstrual periods.

Pain, tenderness, or swelling in the testicles or scrotum.

Eye symptoms, such as pain and/or itching, sensitivity to light, and a pus-like fluid that drains from one or both eyes.

However, you may have a gonorrhea infection without any symptoms. Because of this, medical experts recommend yearly screening tests for those who are at higher risk of getting gonorrhea. You have a higher risk if you're sexually active and:

Are a woman younger than 25 years old.

Are a man who has sex with men (MSM).

Have HIV. If you have HIV, you should be tested for gonorrhea the first time you see your provider for HIV care. After that, you should be tested at least once a year. But you may be tested more often if your sexual activity increases your risk for getting gonorrhea.

If you don't have any symptoms, you may also need a gonorrhea test if you:

Have a new sex partner.

Have more than one sex partner.

Have a sex partner who has sex with others.

Have a sex partner who has an STI.

Engage in sex work.

Don't use condoms correctly every time you have sex.

If you are pregnant and younger than age 25, you should have a gonorrhea test early in your pregnancy. If you are older than age 25, ask your provider if you need to be tested.

If you are a man with no symptoms and a low risk of infection, medical experts don't suggest routine testing for gonorrhea.

Your need for testing also depends on your personal risk of getting gonorrhea. Some people may need to be tested more than once a year. So, ask your provider what's best for you.

WHAT HAPPENS DURING A GONORRHEA TEST?

To do a gonorrhea test, you will need to provide a sample of fluid from the part of your body that may be infected. There are two ways to gather the sample:

A first-catch urine sample. You use a sterile cup to collect urine from the very first part of your urine stream. To get an accurate test result, you will need to stop urinating for two hours before the test.

A swab sample. Your provider will use a special swab or brush to gather fluid and cells, usually from the genital area (the urethra or the vagina). In certain cases, the rectum, cervix, throat, or eyes may be swabbed. Your provider may give you the option to swab yourself following special instructions.

Some types of gonorrhea tests can be done with an at-home STI test kit. The kit includes a container for collecting a urine sample to send to a lab for testing. Ask your provider if at-home testing is right for you.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will tell you how to prepare for your test. If you are taking antibiotics, let your provider know.

For a urine test, you may need to avoid urinating for two hours before the test.

If you have a cervical or vaginal swab, you may need to avoid vaginal douches or creams for a day before the test.

ARE THERE ANY RISKS TO THE TEST?

There are no known risks to having a gonorrhea test. You may feel brief discomfort if you have a swab sample taken

from the urethra at the tip of the penis. You may also feel discomfort from a cervical swab and/or experience vaginal bleeding or other discharge.

WHAT DO THE RESULTS MEAN?

A negative (or normal) test result means that a gonorrhea infection wasn't found in your sample. But it doesn't rule out gonorrhea. After you are infected with gonorrhea, it may take weeks before you test positive. So, it's possible that the test was taken too soon after your exposure to the bacteria.

A positive (or abnormal) test result means that you are infected with the gonorrhea bacteria. You will need treatment with antibiotics right away. You will also need to tell your sex partner or partners so that they can be tested and treated if necessary.

To make sure your infection is cured, take all your medicine following your provider's instructions. You should also avoid having sex for at least 7 days and take another test three months after completing treatment. To prevent any future gonorrhea infections, take appropriate precautions whenever having sex. These include:

Having sex with only one partner who has tested negative for STIs and has sex only with you.

Using condoms correctly every time you have sex.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A GONORRHEA TEST?

Some types of gonorrhea bacteria are becoming resistant to certain antibiotics. This means that the bacteria have "learned" how to survive those antibiotics, so the medicines no longer work. If your symptoms don't improve after treatment, your provider may order an antibiotic "susceptibility test." This test can show which antibiotic will work best to clear up your infection.

If you have questions about your results, talk with your provider.

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Gram Stain

<https://medlineplus.gov/lab-tests/gram-stain/>

GRAM STAIN

WHAT IS A GRAM STAIN?

A Gram stain is a test that checks to see if you have a bacterial infection. A sample is taken from the site of a suspected infection, such as your throat, lungs, genitals, or a skin wound. Gram stains may also be used to check for bacteria in certain body fluids, such as blood or urine.

If you have a bacterial infection, your sample will have bacteria in it. A Gram stain will group the bacteria by color and shape to help determine which kind of bacteria is causing your infection. There are two main categories of bacterial infections: Gram-positive and Gram-negative. The categories are determined by how the bacteria react to the Gram stain test.

After the sample is taken, the Gram stain is colored purple. Then, a solvent is added to the stain. When the solvent combines with bacteria in a sample, the bacteria will either stay purple or turn pink or red. If the bacteria remain purple, they are Gram-positive. If the bacteria turn pink or red, they are Gram-negative. The two categories cause different types of infections:

Gram-positive infections include MRSA (methicillin-resistant *Staphylococcus aureus*) infections, strep infections, and toxic shock syndrome.

Gram-negative infections include salmonella infections, pneumonia, urinary tract infections, bacterial meningitis, and gonorrhea.

Knowing whether bacteria is Gram-positive or Gram-negative can help your health care provider identify the type of infection you have and which antibiotics will be most effective in treating it.

Other names: Gram's stain

WHAT IS IT USED FOR?

A Gram stain is most often used to find out if you have a bacterial infection. If you do, the test will show whether the bacteria in your infection is Gram-positive or Gram-negative. This may help your provider decide what other tests are needed to help identify which bacteria are causing your infection and how to treat it.

A Gram stain may also be used to diagnose fungal infections.

WHY DO I NEED A GRAM STAIN?

You may need this test if you have symptoms of a bacterial infection. Pain, fever, and fatigue are common symptoms of many bacterial infections. Other symptoms will depend on the type of infection you have and where it's located in your body.

WHAT HAPPENS DURING A GRAM STAIN?

Your provider will take a sample from the site of a suspected infection or from certain body fluids, depending on what type of infection you may have. Samples used for Gram staining can include:

Sputum (thick mucus) coughed up from your lungs

Drainage from a wound or your genitals

Urine

If your provider needs a sample from the fluid around your joints, heart, lungs, or spinal cord, they will use a fine-needle aspiration biopsy. This kind of biopsy uses a very thin needle to remove a sample of cells and/or fluid.

The most common types of Gram stain tests are listed below.

Wound sample:

A provider will use a special swab to collect a sample from your wound.

Blood test:

A provider will take a sample of blood from a vein in your arm.

Urine test:

You will provide a sterile sample of urine in a cup, as instructed by your provider.

Throat culture:

Your provider will insert a special swab into your mouth to take a sample from the back of your throat and tonsils.

Sputum culture. Sputum is thick mucus coughed up from the lungs. It is different from spit or saliva.

Your provider will ask you to cough up sputum into a special cup, or a special swab may be used to take a sample from your nose.

After your provider collects your sample, they will be sent to a laboratory. There, a laboratory professional will prepare and examine the sample.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a Gram stain.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to having a swab, sputum, or urine test.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your sample will be placed on a slide and treated with the Gram stain. A laboratory professional will examine the slide under a microscope. If no bacteria were found, it means you probably don't have a bacterial infection, or there weren't enough bacteria in the sample.

If bacteria were found, certain qualities may help narrow down which types may be causing your illness. This will help your provider decide how to treat it.

If the bacteria was colored purple, it means you likely have a Gram-positive infection.

If the bacteria was colored pink or red, it means you likely have a Gram-negative infection.

Your results will also include information about the shape of the bacteria in your sample. Most bacteria are either round (known as cocci) or rod-shaped (known as bacilli). The shape can provide more information about the type of infection you have.

Your results may not identify the exact type of bacteria in your sample, but they can help your provider figure out what's causing your illness and how best to treat it. You may need more tests, such as a bacteria culture, to confirm what type of bacteria it is.

Gram stain results may also show whether you have a fungal infection. They may show what category of fungal infection you have: yeast or mold. But you may need more tests to find out which specific fungal infection you have.

If you have questions about your results, talk to your provider. Your provider will consider your symptoms, and medical history to understand the results of a Gram stain test. They may also want you to have a physical exam or additional tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A GRAM STAIN?

If you are diagnosed with a bacterial infection, you will probably be prescribed antibiotics. It's important to take your medicine as prescribed, even if your symptoms are mild. This can prevent your infection from getting worse and causing

serious complications.

A Gram stain cannot detect a virus.

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Growth Hormone Tests

<https://medlineplus.gov/lab-tests/growth-hormone-tests/>

GROWTH HORMONE TESTS

WHAT ARE GROWTH HORMONE TESTS?

Growth hormone (GH) tests are blood tests that check if your body is making a normal amount of GH. Hormones are chemical messengers in your bloodstream that control the actions of certain cells or organs. GH, also known as human growth hormone, controls your body's growth. It also helps control metabolism, the process your body uses to make energy from the food you eat. GH is made in the pituitary gland, a small organ at the base of your brain that controls many functions, including growth.

In children, GH plays a major role in bone growth, muscle mass development, and height. In adults, GH affects bone and muscle health. If you or your child have too much or too little GH, it can cause health problems.

GH levels in your blood change throughout the day, depending on your diet and activity levels. So, a standard blood test doesn't usually provide helpful information about GH levels. Instead, GH levels are usually checked in blood tests taken over a few hours that measure other substances, such as proteins and other hormones that are related to GH production. This will provide more accurate results about your or your child's GH levels.

Other names: GH test, human growth hormone test, somatotropin test, growth hormone stimulation test, growth hormone suppression test

WHAT ARE THEY USED FOR?

GH tests are used to detect and monitor GH disorders, including:

GH deficiency (not enough GH). In children, GH is essential for normal growth and development. A GH deficiency can cause a child to grow more slowly and be much shorter than children of the same age and sex. In adults, GH deficiency can lead to low bone density and reduced muscle mass.

Gigantism. This rare childhood disorder causes your body to produce excess (too much) GH. Children with gigantism are very tall for their age and have large hands and feet.

Acromegaly. This condition affects adults and causes your body to produce too much growth hormone. Adults with acromegaly have thicker than normal bones and enlarged hands, feet, and facial features.

The tests may also be used if other tests show an issue with your pituitary gland. Some conditions, such as pituitary tumors, can cause your body to make too much or too little GH.

WHY DO I NEED A GROWTH HORMONE TEST?

Your provider may order a GH test if you or your child has symptoms of a GH disorder or to monitor treatment for a GH disorder.

Symptoms of GH deficiency in children include:

Slowed growth rate compared with children of the same age and sex

Shorter height, arms, and legs, and lower weight than children of the same age and sex

Small penis in males

Delayed puberty

Adults with GH deficiency may have symptoms such as fatigue and decreased bone density and muscle mass. But GH testing isn't common for adults, as other disorders are much more likely to cause these symptoms.

Symptoms of GH excess (gigantism) in children include:

Excessive growth compared with children of the same age and sex

Overly large head

Larger than normal hands and feet

Mild to moderate obesity

Symptoms of GH excess (acromegaly) in adults include:

Deep, husky voice

Larger than normal facial features such as lips, nose, and tongue

Excessive sweating and body odor

Thickening of bones

Coarse, oily skin

Irregular menstrual cycles in women

Erectile dysfunction in men

WHAT HAPPENS DURING A GROWTH HORMONE TEST?

Your provider may order a GH stimulation or a GH suppression test, depending on whether symptoms show a possible GH deficiency or a GH excess.

A GH stimulation test is used to check for a GH deficiency. During the test:

A health care professional will use a small needle to take a blood sample from a vein in your arm.

After the needle is inserted, a small amount of blood will be collected into a test tube or vial.

Your child (or you) will be given a medicine through an intravenous (IV) line that stimulates (causes) the pituitary gland to make GH.

More blood samples will be taken over the next two hours.

Each sample will be tested to see if GH levels have increased.

A GH suppression test is used to check for a GH excess. During the test:

A health care professional will take a blood sample from a vein in your arm, using a small needle.

After the needle is inserted, a small amount of blood will be collected into a test tube or vial.

You or your child will drink a solution that contains water and glucose (sugar).

Blood samples will be taken every 30 minutes for two hours after drinking the solution.

The blood samples will be tested to see if GH levels have decreased.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You may need to fast (not eat or drink) for several hours before your test.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to you or your child in having a blood test or getting an IV. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly. There is no risk to drinking the glucose solution.

WHAT DO THE RESULTS MEAN?

If GH levels don't increase to a certain level during a GH stimulation test, it may mean that there is a GH deficiency.

If your child is diagnosed with GH deficiency, they may benefit from treatment with GH supplementation. GH supplementation is an injected medicine that contains manufactured human growth hormone. When GH deficiency is diagnosed and treated early, some children can grow several inches in the first year of treatment. Others grow less and more slowly but still benefit from treatment.

If GH levels don't decrease to a certain level during a GH stimulation test, it may mean that there is a GH excess. For your child, this could mean they have gigantism. For you, it may mean you have acromegaly.

Gigantism and acromegaly are most often caused by a tumor in the pituitary gland. Treatment for the tumor may include radiation therapy, surgery, and/or medicine. But if the cause is not a tumor, you or your child may need more tests to identify the cause.

Learn more about laboratory tests, reference ranges, understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A GROWTH HORMONE TEST?

Your provider may order other blood tests to help diagnose a GH disorder. These can include:

IGF-1 test. IGF-1 is a hormone that helps manage the effects of GH in your body. Unlike GH, IGF-1 levels are more stable throughout the day. So, a reliable way to track GH in your body is by measuring the level of IGF-1 in the blood.

IGBP-3 test. IGBP-3 is a protein that is the main carrier of IGF-1. This test can help diagnose a GH deficiency, GH insensitivity (when your body can't use the growth hormone it makes), or GH excess.

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Haptoglobin (HP) Test

<https://medlineplus.gov/lab-tests/haptoglobin-hp-test/>

HAPTOGLOBIN (HP) TEST

WHAT IS A HAPTOGLOBIN (HP) TEST?

This test measures the amount of haptoglobin in your blood. Haptoglobin is a protein made by your liver to get rid of the hemoglobin outside your red blood cells.

Hemoglobin is a protein in your red blood cells that carries oxygen from your lungs to the rest of your body. Most hemoglobin is found inside your red blood cells. When red blood cells die, they release hemoglobin. This results in small amounts of hemoglobin circulating in your bloodstream. Haptoglobin binds to the hemoglobin in your bloodstream, and then your liver removes it from the body.

Damaged red blood cells also release hemoglobin. So if your red blood cells are damaged or break down faster than usual, you will have more hemoglobin than usual in your bloodstream. That means haptoglobin will bind to all of this extra hemoglobin so that it can be cleared from your body. The haptoglobin may leave the body faster than your liver can make it. This causes your haptoglobin blood levels to drop. If your haptoglobin levels are too low, it may be a sign of a disorder of the red blood cells, such as anemia.

Other names: hemoglobin-binding protein, HPT, Hp

WHAT IS IT USED FOR?

A haptoglobin test is most often used to diagnose hemolytic anemia. Hemolytic anemia is a disorder that happens when your red blood cells are destroyed faster than they can be replaced. This test may also be used to see if another type of anemia or another blood disorder is causing your symptoms.

WHY DO I NEED A HAPTOGLOBIN TEST?

You may need this test if you have symptoms of hemolytic anemia or another blood disorder. These may include:

Weakness or fatigue

Pale skin

Shortness of breath

Headache

Dizziness

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Jaundice, a condition that causes your skin and eyes to turn yellow

Dark colored urine

You may also need this test if you've had a blood transfusion. A haptoglobin test may be done along with another test called direct anti-globulin. The results of these tests can show if you've had a bad reaction to the transfusion.

WHAT HAPPENS DURING A HAPTOGLOBIN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a haptoglobin test.

ARE THERE ANY RISKS TO A HAPTOGLOBIN TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A low haptoglobin level is normal for infants less than six months old.

Otherwise, a haptoglobin level that is lower than normal may be a sign of:

Hemolytic anemia

Liver disease

Reaction to a transfusion

A haptoglobin level that is higher than normal may be a sign of:

Ulcerative colitis

Acute rheumatic disease, such as arthritis^Â or other conditions that affect the joints, bones, and muscles

A severe infection

Diabetes

Your provider may order other blood tests to help make a diagnosis. These include:

Reticulocyte Count

Hemoglobin Test

Hematocrit Test

Lactate Dehydrogenase Test

Blood Smear

Complete Blood Count

These tests may be done at the same time or after your haptoglobin test.

If you have questions about your results, talk to your provider. Tell your provider if you are taking any medicines. Some medicines may cause your haptoglobin test results to be higher or lower than normal.

To understand the results of a haptoglobin test, your provider will consider your symptoms, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A HAPTOGLOBIN TEST?

High haptoglobin levels may be a sign of an inflammatory disease. Inflammatory diseases are disorders of the immune system that can cause serious health problems. However, haptoglobin testing is not usually used to diagnose or monitor conditions related to high haptoglobin levels.

Severe blood loss and conditions that worsen your kidney function may also affect haptoglobin level results.

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Hearing Tests for Adults

<https://medlineplus.gov/lab-tests/hearing-tests-for-adults/>

HEARING TESTS FOR ADULTS

WHAT ARE HEARING TESTS?

Hearing tests measure how well you are able to hear. They can help diagnose hearing loss, how severe it is, and which part of your hearing isn't working properly.

Hearing depends on a series of steps that change sound waves into electrical signals that your brain understands as sounds. Hearing loss happens where there's a problem with any of these steps:

Sound waves enter your outer ear and travel to your eardrum in your middle ear.

The sound waves vibrate your eardrum, which sends the vibrations to tiny bones that make the vibrations stronger.

The vibrating bones make tiny waves in the fluid that's inside your cochlea. The cochlea is a snail-shaped structure in your inner ear. It is lined with sensory cells that have hair-like structures. When the hair cells move with the fluid waves, they create electrical signals.

Your auditory (hearing) nerve carries the electrical signals from your inner ear to your brain, which turns them into sounds you can recognize and understand.

There are three main types of hearing loss:

Conductive hearing loss happens when sound waves can't reach your inner ear. Earwax or abnormal fluid in your ear may be blocking the path, or a hole in your eardrum may prevent it from vibrating. Medical treatment or surgery can often improve hearing.

Sensorineural hearing loss (also called nerve deafness) happens when there's damage to the inner ear or the auditory nerve. This type of hearing loss ranges from mild (difficulty hearing certain sounds) to profound (not hearing any sound).

The hearing loss is usually permanent, but it can improve with hearing aids or other devices.

Mixed hearing loss is a combination of both sensorineural and conductive hearing loss.

Hearing loss in older adults is common. About one-third of adults over the age of 65 have some hearing loss, usually the sensorineural type. If you are diagnosed with hearing loss, there are steps you can take that may help treat or manage the condition.

Other names: audiometry; audiography, audiogram, sound test, audiologic tests

WHAT ARE THEY USED FOR?

Hearing tests are used to check for hearing loss. The tests are also used to find out how severe hearing loss is and what type it is.

WHY DO I NEED A HEARING TEST?

You may need a hearing test if you have signs of hearing loss, such as:

Having trouble hearing over the phone

Having trouble understanding what other people are saying, especially in a noisy environment

Needing to turn up the TV volume so loud that others complain

Thinking that others always seem to mumble

Having trouble hearing high-pitched sounds and voices

Hearing a ringing or other unusual sound in your ears (tinnitus)

If your workplace is noisy, you may need to have your hearing checked every year.

If you use a hearing aid or had surgery for your hearing, you may need a hearing test to see whether your hearing has improved.

WHAT HAPPENS DURING A HEARING TEST?

Your primary care provider may use a general screening test to check your hearing. This is usually a whisper test to see how well you can hear whispered words in each ear.

If the test shows you may have a hearing problem or if you notice you have trouble hearing, your provider may do more hearing tests or send you a hearing specialist, such as:

An audiologist, a health care professional who is trained to test your hearing, diagnose hearing loss, and provide hearing devices and services to improve hearing.

An otolaryngologist (ENT), a doctor who specializes in treating diseases of the ears, nose, throat, and head and neck.

Hearing aid specialists can do basic hearing tests and fit you for hearing aids. If you have a hearing aid, tests can check whether it's helping you enough. If you're unsure where to have your hearing tested, talk with your primary care provider.

There are several types of hearing tests. These are some of the more common types:

Audiometry tests check your ability to hear tones or words at different pitches and volumes. For these tests, you'll put on headphones and sit in a soundproof room. The tests may include:

A pure-tone test. This is the most common screening test for hearing loss. It tests for the quietest sounds you can hear at different pitches. During the test:

A series of high and low tones will be played in your headphones. Some will be loud, and some will be very soft.

Each time you hear a tone, you'll raise your hand, push a button, or say that you hear the sound.

A series of high and low tones will be played in your headphones. Some will be loud, and some will be very soft.

Each time you hear a tone, you'll raise your hand, push a button, or say that you hear the sound.

Speech test, also called a speech discrimination test. This test checks how well you can hear people talking. It can show how much a hearing aid might help you. During the test:

You'll hear simple words at different levels of loudness. Some of the words will be spoken over noise.

You will repeat the words that you hear.

You'll hear simple words at different levels of loudness. Some of the words will be spoken over noise.

You will repeat the words that you hear.

Tuning fork tests, also called bone conduction tests. These tests can help tell the difference between nerve problems in your inner ear (sensorineural hearing loss) and fluid or wax that's making it hard to hear (conductive hearing loss). They can also show if one ear hears better than the other.

A tuning fork is a two-pronged metal device. When it's tapped, it vibrates and makes a sound. Tuning forks are mostly used by primary care providers. Audiologists and otolaryngologists may use special devices to do bone conduction testing. The general steps are the same:

The provider will place the tuning fork or device behind your ear or on your head.

The tuning fork or device will vibrate, which makes your skull bones vibrate gently.

The vibrations travel through your bones directly into your inner ear, so your outer and middle ear aren't involved in hearing the sound.

If you have trouble hearing the sound in either ear, a problem in your inner ear may be causing hearing loss.

If the provider is using a tuning fork, you may have another test to compare how well you hear the tuning fork through:

Vibrations in the air that travel through your outer ear

Vibrations through your bones that go directly into your inner ear

Normal hearing is better through air. If you hear better through bone vibrations, you may have fluid or wax blocking sound from reaching your inner ear.

Vibrations in the air that travel through your outer ear

Vibrations through your bones that go directly into your inner ear

A tympanometry test checks to see how your eardrum and bones in your middle ear are working:

A small device will be placed in your ear.

The device will send air and sound into your ear, which makes your eardrum move. You'll feel the air pressure inside your ear.

A machine records the movements of your eardrum on graphs called tympanograms.

If your eardrum doesn't move normally, you may have fluid or wax buildup in your ear, a hole in your eardrum, a tumor in

your ear, a middle ear bones that don't vibrate well, or another problem in your middle ear.

An otoacoustic emissions test (OAE) checks for damage in the hair cells in your cochlea:

A small device will be placed in your ear. It can make sound and measure sound.

Sound from the device makes the fluid in your cochlea ripple, which moves the hair cells. When the hairs move, they make vibrations that have their own sound called OAEs.

The device measures OAEs to see how well the hair cells are working. If the test shows little or no OAEs, you have sensorineural hearing loss.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A HEARING TEST?

You don't need any special preparations for a hearing test.

ARE THERE ANY RISKS TO HEARING TESTS?

There is no risk to having a hearing test.

WHAT DO THE RESULTS MEAN?

The results of hearing tests show if your hearing is normal or if you have hearing loss. You'll usually get your results right after your tests are done.

If you have hearing loss, the results will show how severe it is, which ear is affected more, and whether you have conductive, sensorineural, or both types of hearing loss:

With conductive hearing loss, your provider may be able to treat the cause and restore your hearing. Your treatment will depend on what's blocking sound from reaching your inner ear. It may include removing wax, draining fluid, or surgery to fix problems with your eardrum or ear bones.

With sensorineural hearing loss, your hearing loss can't be treated, but you may be able to use hearing aids or have surgery to implant a device to improve your hearing. If your hearing can't be improved, assistive devices are available to help with daily living. For example, there are devices to help you use the phone and to alert you to doorbells, alarms, and other sounds.

In certain cases, more tests may be needed to diagnose the cause of the hearing loss. If you have questions about your test results, talk with your health care provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT HEARING TESTS?

Even mild hearing loss can make it difficult to understand normal speech. Because of this, many older adults will avoid social situations, leading to isolation and depression. Treating hearing loss can help prevent these problems. Treatment options include:

Hearing aids. A hearing aid is a device that's worn either behind or inside the ear. It makes sounds louder. There are many styles of hearing aids available. You can get a hearing aid with or without a prescription:

Prescription hearing aids require a visit to a hearing health professional who will program your hearing aid to match your hearing loss. Prescription hearing aids may be necessary for more severe or complicated types of hearing loss, for example, if you can only hear loud sounds.

Over-the-counter hearing aids that you buy on your own are for people with hearing loss that is:

Mild, which means you can hear some speech, but soft sounds are hard to hear.

Moderate, which means you understand very little when another person talks at a normal level.

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Cochlear implants. These devices are put in your ear with surgery. They are usually used for people with severe hearing loss who don't benefit from hearing aids. Cochlear implants send sound directly to the hearing nerve. They don't restore normal hearing. But they can help with understanding speech after people have therapy to learn how to hear with their implant.

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Hearing Tests for Children

<https://medlineplus.gov/lab-tests/hearing-tests-for-children/>

HEARING TESTS FOR CHILDREN

WHAT ARE HEARING TESTS FOR CHILDREN?

Hearing tests measure how well your baby or child can hear different sounds. Some babies are born with hearing problems. Other children are born with normal hearing and begin to have hearing problems later in childhood.

Hearing loss in infancy and early childhood can delay a child's ability to talk, understand spoken language, and develop social skills. Hearing tests can find problems early so that children with hearing loss can be treated and get help learning language skills as soon as possible.

Hearing depends on a series of steps that change sound waves into electrical signals that the brain understands as sounds. Hearing loss happens when there's a problem with any of these steps:

Sound waves enter the outer ear and travel to the eardrum in the middle ear.

The soundwaves vibrate the eardrum, which sends the vibrations to tiny bones that make the vibrations stronger.

The vibrating bones make tiny waves in the fluid that's inside the cochlea. The cochlea is a snail-shaped structure in the inner ear. It is lined with sensory cells that have hair-like structures. When the hair cells move with the fluid waves, they create electrical signals.

The auditory (hearing) nerve carries the electrical signals from the inner ear to the brain, which turns them into sounds you can recognize and understand.

There are three main types of hearing loss:

Conductive hearing loss happens when sound waves can't reach the inner ear. Earwax or abnormal fluid in the ear may be blocking the path, or a hole in the eardrum may prevent it from vibrating. Ear infections are a common cause of this type of hearing loss in infants and young children. Medical treatment or surgery can often improve hearing.

Sensorineural hearing loss (also called nerve deafness) happens when there's damage to the cochlea in the inner ear or the auditory nerve. This type of hearing loss ranges from mild (difficulty hearing certain sounds) to profound (not hearing any sound). The hearing loss is usually permanent, but it can improve with hearing aids or other devices.

Mixed hearing loss is a combination of both sensorineural and conductive hearing loss.

One type of sensorineural hearing loss is called auditory neuropathy spectrum disorder (ANSD). With ANSD, there's problem with how the inner ear or the hearing nerve sends sound to the brain. This type of hearing disorder may happen if your child has health problems before or during birth or as a newborn. These problems include premature birth (being born too soon), low birth weight, and jaundice. ANSD may also be caused by a genetic disorder.

Other names: audiometry; audiography, audiogram, sound test

WHAT ARE THEY USED FOR?

Hearing screening tests are used to find out if a newborn, baby, or child may have hearing loss. These quick pass/fail tests show whether your child needs more testing. They can't diagnose hearing loss.

More complete hearing tests are used to diagnose hearing loss if your child doesn't pass a screening test, or if you have concerns about your child's hearing. These tests can show which type of hearing loss your child has and how severe it is.

WHY DOES MY CHILD NEED A HEARING TEST?

Routine hearing screening tests are recommended for all babies and children:

Newborns should have a screening test for hearing problems by 1 month of age.

If you give birth in a hospital, the screening test is usually done before your baby goes home.

If you give birth at home, make sure to get your baby tested during the first month after birth.

If you give birth in a hospital, the screening test is usually done before your baby goes home.

If you give birth at home, make sure to get your baby tested during the first month after birth.

Children should have their hearing checked before they enter school. School-aged children may have hearing screening tests as part of their regular health checkups. The tests are often done in school, too.

More complete hearing tests are needed any time there is a concern about your child's hearing:

If your baby doesn't pass the newborn hearing screening test, your baby will need complete hearing tests as soon as possible, but no later than 3 months of age.

If your baby passes the newborn hearing screening test, but has a high risk for developing hearing loss, your baby will need complete hearing tests between 24 to 30 months of age. Your baby's risk may be higher if childhood hearing loss runs in your family or if your baby had certain infections. Ask your baby's provider about your baby's risk level.

If your baby or child shows signs of hearing loss, contact your child's provider as soon as possible.

Signs of hearing loss in a baby include:

Not jumping or being startled at loud noises

After 6 months of age, not turning toward a sound

By 12 months of age, not saying a few simple words, such as "dada" or "mama"

Signs of hearing loss in children include:

Unclear speech

Not following directions

Often saying, "Huh?"

Turning the TV volume up too high

Learning problems

WHAT HAPPENS DURING A HEARING TEST?

Hearing tests depend on your child's age and ability to follow instructions:

Hearing screening tests for newborns, infants, and very young children can measure hearing without needing your child to cooperate. These tests can even be done while your child sleeps:

Auditory brainstem (ABR) tests check the inner ear, the hearing nerve, and parts of the brain that are involved with hearing. During an ABR test:

The provider will place electrodes (small sensors that stick to the skin) on the scalp. The electrodes are connected to a computer.

Tiny, soft earphones will be placed inside the ears.

The earphones will play sounds.

The electrodes measure how the brain responds to the sounds. The results are shown on the computer.

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Otoacoustic emissions (OAE) tests check for damage in the hair cells in the cochlea. During an OAE test:

A small device will be placed in the ear. It can make sound and measure sound.

Sound from the device makes the fluid in the cochlea ripple, which moves the hair cells. When the hairs move, they make vibrations that have their own sound called OAEs.

The device measures OAEs to see how well the hair cells are working. If the test shows little or no OAEs, your child may have sensorineural hearing loss.

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Sound from the device makes the fluid in the cochlea ripple, which moves the hair cells. When the hairs move, they make vibrations that have their own sound called OAEs.

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Hearing screening tests for older children are usually audiometry tests. These tests check the quietest sounds or words that your child can hear. For these tests:

Your child will wear headphones. If your child won't wear headphones, sounds may be played through speakers in a sound booth.

For a pure-tone test, each time your child hears a tone they will raise their hand, push a button, or say that they heard the sound.

For a speech test (also called speech discrimination testing), your child will hear simple words at different levels of loudness. Some words will be spoken over noise. Your child will repeat the words they hear.

Audiometry tests can be adapted for very young children and infants. For young children, the tests may include a game to help them pay attention to the sounds. For example, a child may put a block in a box each time they hear a sound.

For infants and toddlers, the provider will watch for behavior changes that show the child heard a sound. For example, an infant may start sucking a pacifier or look to see where the sound came from. When the child responds to a sound, they get to see a moving toy as a reward.

More complete hearing tests will be done if your child doesn't pass a hearing screening test. Your child's provider will usually refer you to a hearing specialist, such as:

A pediatric audiologist, a health care professional who is trained to test hearing, diagnose hearing loss, and provide hearing devices and services to improve hearing in children.

A pediatric otolaryngologist (ENT), a doctor who specializes in diagnosing and treating ear conditions in children. This type of provider may also be called a pediatric otologist.

During a complete hearing exam, your child may have several types of hearing tests that examine different parts hearing. These tests may include an ABR and/or an OAE test for children who were screened with audiometry tests. Other common tests include:

Tympanometry. This test checks how well the eardrum and bones in the middle ear are working. Your child will need to sit very still, so this test usually isn't used for infants or young children. During the test:

A small device will be placed in your child's ear.

The device will send air and sound into the ear, which makes the eardrum move. Your child will feel the air pressure.

A machine records the movements of the eardrum on graphs called tympanograms.

If the eardrum doesn't move normally, your child may have fluid or wax buildup in the ear, a hole in the eardrum, a tumor in the ear, middle ear bones that don't vibrate well, or another problem in the middle ear.

A small device will be placed in your child's ear.

The device will send air and sound into the ear, which makes the eardrum move. Your child will feel the air pressure.

A machine records the movements of the eardrum on graphs called tympanograms.

If the eardrum doesn't move normally, your child may have fluid or wax buildup in the ear, a hole in the eardrum, a tumor in the ear, middle ear bones that don't vibrate well, or another problem in the middle ear.

Acoustic Reflex Measures, also called middle ear muscle reflex (MEMR) test. This test shows how the ear responds to loud sounds. Normally, a tiny muscle inside the ear tightens to protect your ear when you hear loud noises. If unusually loud noises are needed to trigger the muscle to tighten, or if it doesn't tighten at all, your child may have hearing loss. To do the test:

A small device will be placed in your child's ear.

The device will make loud sounds.

A machine will show if the sound triggered the ear muscle to tighten.

A small device will be placed in your child's ear.

The device will make loud sounds.

A machine will show if the sound triggered the ear muscle to tighten.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A HEARING TEST?

There are no special preparations for a hearing test. But it's helpful to tell your child what to expect. Let your child know they will hear sounds. If you have a pair of headphones, you might let your child try them on to see what they feel like.

ARE THERE ANY RISKS TO HEARING TESTS?

There is no risk to having hearing tests.

WHAT DO THE RESULTS MEAN?

If your child's hearing screening test shows normal hearing, your child usually won't need more hearing tests until their next routine screening. But if you have concerns about your child's hearing or if your child has risks for hearing loss, talk with your child's provider about getting more complete hearing tests.

If the results of complete hearing tests show that your child has hearing loss, the results will also show how severe it is, which ear is affected more, and whether they have conductive sensorineural, or both types of hearing loss.

With conductive hearing loss, treatment may be able to restore your child's hearing. The treatment will depend on what's blocking sound from reaching the inner ear. It may include removing wax, draining fluid, or doing surgery to fix problems with the eardrum or ear bones.

With sensorineural hearing loss, the loss of hearing is permanent. But your child's hearing may improve with hearing aids or surgery to implant a device called a cochlear implant.

Other treatment and support options for hearing loss in children include:

Working with professionals who can help children and their families learn to communicate. This may include learning sign language, lip reading, or other ways to communicate.

Education to help your child learn communication skills.

Assistive devices, such as devices for using the telephone and alerting your child to alarms, doorbells, and other sounds.

Support groups.

If you have questions about your child's hearing and treatment, talk with your child's primary care provider or hearing specialist.

Learn more about laboratory tests, reference ranges, and understanding results.

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Heart Disease Risk Assessment

<https://medlineplus.gov/lab-tests/heart-disease-risk-assessment/>

HEART DISEASE RISK ASSESSMENT

WHAT IS A HEART DISEASE RISK ASSESSMENT?

Heart disease is the leading cause of death in the United States. A heart disease risk assessment, or cardiovascular disease (CVD) risk assessment, is a screening tool that measures your risk of heart disease or CVD. Heart disease and CVD are related, but they are not the same thing:

CVD is the term for all diseases that affect the heart and blood vessels.

Heart disease is a type of CVD. It is the term for a variety of conditions that affect the heart's structure and function, including coronary artery disease (the most common type), heart attack, heart failure, and heart valve diseases.

There are various screening tools that can check your heart disease risk, but they are meant to be used before you have a heart problem. You can complete a heart risk assessment online or at your health care provider's office. The risk assessment includes questions about certain risk factors, including your age, family history, and lifestyle habits such as diet and exercise. It then calculates your risk of developing heart disease in the future. Even if you feel healthy now, the assessment can show if you need to take steps to prevent or reduce your chances of developing heart disease later.

Other names: cardiovascular disease assessment, CVD risk factors assessment, heart disease risk calculator, ASCVD risk calculator

WHAT IS IT USED FOR?

Most heart disease risk assessments check how likely it is that you will develop heart disease in the next 10 years. Some also evaluate your 30-year or lifetime risk.

WHY DO I NEED A HEART DISEASE RISK ASSESSMENT?

Most of the assessments can be helpful if you are 40 to 75 years old and do not have heart disease. If you are younger than 40, there are a few new assessments that your provider may use if you have a family history of heart disease or certain CVD risk factors.

WHAT HAPPENS DURING A HEART DISEASE RISK ASSESSMENT?

There are different types of heart disease risk assessments. These screening tools each include general questions about your health and possible risk factors for heart disease. Some common heart disease risk assessments include: ASCVD Risk Calculator. This tool helps predict your 10-year or lifetime risk of atherosclerotic cardiovascular disease (ASCVD). ASCVD refers to conditions caused by atherosclerosis, which develops when plaque (fatty deposits) build up inside your arteries. The buildup can block blood flow and may lead to heart attack or stroke.

ACC/AHA Cardiovascular Risk Calculator. This is a shorter version of the ASCVD Risk Calculator. Both make suggestions to improve your risk score.

Pooled Cohort Equation (PCE). This is an updated version of the ASCVD Risk Calculator, which estimates the 10-year risk of heart attack and/or stroke for people ages 30 to 79.

Predicting Risk of Cardiovascular Disease Events (PREVENT) Calculator. This new tool considers other health conditions to estimate the risk of heart attack, stroke, and heart failure for people ages 30 to 79. It is used to estimate your 10- and 30-year risk.

Reynolds Risk Score. This test was originally developed for women, but now they have a test that can be used for men. It estimates the 10-year risk of having a heart attack, stroke, or death related to heart disease for people ages 45 to 80. It is only used if you don't have diabetes.

You can choose and complete a test online, or your health care provider may tell you which one might be best for you and/or review the questions with you. Your assessment may include questions about some or all of the following:

Age

Your sex

Height and weight

Blood pressure

Cholesterol levels

Whether you smoke or have smoked in the past

Whether you have diabetes

Activity level

Diet

Family history of heart conditions

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a heart disease risk assessment.

ARE THERE ANY RISKS TO THE TEST?

There is no risk in taking a questionnaire.

WHAT DO THE RESULTS MEAN?

Heart disease risk assessment results are calculated based on data from clinical studies on heart disease. Your answers will be compared with patient data from these studies. Your results will be given as a percentage.

Â A lower percentage means you have less risk of developing heart disease.Â

A higher percentage means you have a greater risk. If your percentage is high, your provider may recommend steps to lower your risk.

Some risk factors, such as age and family history, can't be controlled. But you can take steps to lower your risk for some factors, for example by:

Eating a healthy diet

Getting regular physical activity

Limiting alcoholÂ use

Quitting smoking.

Managing stressÂ and health conditions such as diabetes, high blood pressure, or high cholesterol

These steps can benefit all adults.

If you are at a high risk of heart disease, your provider may also recommend medicines.Â These medicines may be used to treat health conditions which raise your risk or to help prevent certain heart diseases: Â They may include:

Medicines to lower cholesterol, such as statins.

Blood pressure medicines, also called antihypertensives

Blood thinners, which may help prevent heart attacks and strokes. This may include taking a daily aspirin. But aspirin therapy has some risks and is only recommended for people with certain risk factors for heart disease. Talk to your provider about the risks and benefits before you start taking a daily aspirin.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A HEART DISEASE RISK ASSESSMENT?

Depending on the results of your assessment, your provider may order additional tests including:

Cholesterol levels, a test that measures cholesterol levels in your blood

Electrocardiogram, a test that measures electrical signals in your heart

Stress test, a test that measures how well your heart handles physical activity

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Heavy Metal Blood Test

<https://medlineplus.gov/lab-tests/heavy-metal-blood-test/>

HEAVY METAL BLOOD TEST

WHAT IS A HEAVY METAL BLOOD TEST?

A heavy metal blood test is a group of tests that measures the levels of possibly harmful metals (like lead or mercury) in your blood.

Heavy metals are found naturally in the environment. They're also found in certain medicines, supplements, and foods. Chemicals used in industry and farming may also contain heavy metals, which can end up in the air, soil, and water.

Heavy metals get into your body in different ways. You might breathe them in, eat them, or absorb them through your skin. Because heavy metals are just about everywhere, it's normal to have some in your body. But if too much heavy metal gets into your body, it can cause heavy metal poisoning, which can be deadly.

Heavy metal poisoning can lead to organ damage, changes in your behavior, or difficulties with thinking and memory. The symptoms of heavy metal poisoning depend on the type of metal, how much is in your body, and your age. The highest risk for serious, long-term health problems from heavy metals is during fetal growth and a child's development.

Poisoning from heavy metals can happen slowly over time or suddenly if you are exposed to a large amount of heavy metal all at once.

The most commonly tested heavy metals are:

Lead

Mercury

Arsenic

Cadmium

Chromium

Other metals that may be tested include aluminum, beryllium, cobalt, copper, iron, manganese, nickel, platinum, selenium, silicon, silver, and thallium. You may have a heavy metal blood test for one type of metal, or you may have a test for a group of metals, called a "heavy metal panel."

Depending on what metals your health care provider is checking for, testing can also be done with samples of your urine, fingernails, or hair. But blood and urine are the samples that are most commonly used for heavy metal tests.

Other names: heavy metals panel, toxic metals, heavy metal toxicity test

WHAT IS IT USED FOR?

Heavy metal testing is used to find out if you have been exposed to certain metals, and how much of the metal is in your system.

WHY DO I NEED A HEAVY METAL BLOOD TEST?

Your provider may order a heavy metal blood test if you had a possible exposure to heavy metal or if you have symptoms of heavy metal poisoning.

Symptoms of heavy metal poisoning depend on the type of metal and how much exposure you had. Your symptoms may include:

Nausea and vomiting

Abdominal (belly) pain

Diarrhea

Tingling in the hands and feet

Shortness of breath

Chills

Muscle weakness

Memory loss or changes in behavior

Arrhythmia (problem with the rate or rhythm of your heartbeat)

If your job involves heavy metals, your employer may require regular testing to help monitor workplace safety.

Children under the age of 6 may need to be tested for lead poisoning if they have a risk of exposure to lead, even when they don't have symptoms. That's because very low levels of lead can damage a child's developing brain and cause long-term growth and learning problems.

Children have a risk of lead exposure if they spend time in homes built before 1978 when lead-based paint was still used. These homes may have lead in household dust from old chipping or peeling paint. Lead from old paint and other old sources may also remain in the soil, especially in older neighborhoods of some cities.

Young children tend to be exposed when they touch surfaces or soil with lead and put their hands in their mouths. Your child's provider may recommend lead testing based on your child's possible exposure to lead and where you live. Some state and local governments require all children to be tested for lead.

WHAT HAPPENS DURING A HEAVY METAL BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Some fish and shellfish contain high levels of mercury, so you should avoid eating seafood for 48 hours before your test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Higher than normal levels of a heavy metal in your blood may mean that you have heavy metal poisoning, but not always. Even if you don't have symptoms, you may still need to avoid all exposure to that metal.

Depending on the heavy metal and your condition, you may also need treatment to help your body get rid of the heavy metal as quickly as possible. In certain cases, your provider may recommend chelation therapy. For this therapy, you'll take medicine by mouth or have an injection (shot) of a medicine that helps remove the metals from your body through urine (pee). In certain cases, chelation medicine is given through an IV (intravenous) line.

Chelation therapy can have serious side effects, so it's important to think about the risks and benefits.

Normal or low levels of heavy metal in your blood usually mean that you don't have heavy metal poisoning. However, some heavy metals leave your bloodstream quickly and are stored in your tissues. So, if you have symptoms, your provider will likely order more tests to check for heavy metals in samples of your urine, hair, skin, or fingernail clipping.

Talk with your provider to learn what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

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Helicobacter Pylori (H. Pylori) Tests

<https://medlineplus.gov/lab-tests/helicobacter-pylori-h-pylori-tests/>

HELICOBACTER PYLORI (H. PYLORI) TESTS

WHAT ARE HELICOBACTER PYLORI (H. PYLORI) TESTS?

Helicobacter pylori (H. pylori) is a common type of bacteria that infects the lining of the stomach and the first part of the small intestine. Most people with H. pylori never have symptoms. But in some people, the bacteria can cause problems with digestion.

H. pylori is the most common cause of peptic ulcers, which are sores that form mainly in the stomach and upper part of the small intestine. H. pylori can also cause gastritis (inflammation and irritation of the stomach lining). Without treatment, H. pylori gastritis can last a lifetime and increase the chance of developing stomach cancer.

Researchers aren't sure how people get the bacteria. H. pylori may be spread from person to person through contact with vomit, stool (poop), or saliva (spit) from an infected person. The bacteria may also be spread through contaminated food and water.

Anyone who is exposed to H. pylori can become sick with an H. pylori infection. But H. pylori infections are most common in children. Testing for the bacteria can find out if H. pylori is causing gastritis or other conditions that cause ongoing indigestion. If so, treatment can kill the bacteria so the lining of your stomach can heal.

There are different ways to test for an H. pylori infection. The most common tests use samples of your breath or stool. In certain cases, tests are done on a sample of tissue that's removed from inside your stomach.

Other names: H. pylori stool antigen, H. pylori breath tests, urea breath test, rapid urease test (RUT) for H. pylori, H. pylori culture

WHAT ARE THEY USED FOR?

H. pylori tests are used to:

Find out if an H. pylori infection is the cause of digestive symptoms

Find out if treatment has cured an H. pylori infection

Because there are many possible causes of digestive symptoms, providers often order tests for other digestive disorders along with an H. pylori test.

WHY DO I NEED AN H. PYLORI TEST?

You may need an H. pylori test if you have symptoms of gastritis or an ulcer. These conditions are often caused by H. pylori bacteria. The symptoms may include:

Pain in your upper abdomen (belly)

Bloating (a feeling of fullness or swelling in your belly)

Feeling full too soon while eating a meal

Nausea and vomiting

Burping or belching

Loss of appetite

Weight loss

Gastritis can lead to an ulcer, which is a more serious condition. The symptoms of an ulcer can become severe if you develop complications. Get medical help right away if you have:

Signs of stomach bleeding which include:

Black or tarry stool

Red blood in vomit or vomit that looks like coffee grounds

Black or tarry stool
Red blood in vomit or vomit that looks like coffee grounds
Sudden, sharp or severe belly pain that doesn't go away
Dizziness or fainting
A rapid pulse (heartbeat) or other symptoms of shock
A chance or worsening in your symptoms if you already know you have an ulcer
If you've been treated for an H. pylori infection, you may need testing to see if treatment worked to get rid of all the bacteria. If H. pylori remains in your body, it can grow and cause symptoms again.

WHAT HAPPENS DURING H. PYLORI TESTING?

There are different ways to test for H. pylori. Your health care provider may order one or more tests:
A breath test is also called a urea breath test. It generally includes these steps:
First, you will breathe into a container to provide a sample of your breath.
Next you will swallow a pill or liquid that contains urea. Urea is a normal waste product found in blood. The urea used for the test contains special carbon atoms. If you have H. pylori, the bacteria will change the urea into carbon dioxide that includes these carbon atoms. Your body will get rid of the carbon dioxide in the air you exhale (breathe out).
A few minutes after swallowing the urea, you'll provide another sample of your breath.
The two breath samples will be tested and compared to each other. If you have H. pylori bacteria in your stomach:

The special carbon atoms in the urea will show up in the second sample of your breath.
The second breath sample will contain more carbon dioxide than the first.
The special carbon atoms in the urea will show up in the second sample of your breath.
The second breath sample will contain more carbon dioxide than the first.
Stool tests. There are different ways to check a stool sample for H. pylori:
Antigen tests are the most common type of test for H. pylori. Antigens are a type of protein. The test checks your sample for antigens that come from H. pylori.
PCR tests check your sample for genetic material from H. pylori.
Your provider will probably give you a container or kit with instructions on how to collect a stool sample. There are different ways to collect the sample so follow the instructions carefully. In general, you'll need to:
Label the container with your name, the collection date, and time.
Collect a stool sample as instructed. This usually involves using a clean, dry container, or special paper or plastic wrap placed over the toilet to catch the stool.
Make sure the stool doesn't mix with any urine, toilet paper, or water from the toilet.
Close the container tightly.
Wash your hands well with soap and water.
Return the container according to the instructions.
If you're collecting a sample from diapers, you'll get special instructions for using plastic wrap inside of a clean diaper. In certain cases, a provider may use a swab to take a stool sample from the rectum.
Endoscopy. If other tests don't provide enough information for a diagnosis, your provider may order a procedure called an endoscopy. Your provider will use a scope to look at your esophagus (the tube that links your mouth and stomach), the lining of your stomach, and part of your small intestine. During the procedure:
You will lie down on an exam table.
You'll be given medicine to help you relax and to numb your throat so you won't feel pain.
Your provider will insert a thin tube (an endoscope) through your mouth and into your stomach. The tube has a light and camera to let your provider examine you.
Your provider may take a biopsy to remove small samples of tissue. The tissue will be sent to a lab to test for H. pylori and other conditions that may be causing symptoms.
After the procedure, you'll recover for an hour or two while the medicine wears off.

WILL I NEED TO DO ANYTHING TO PREPARE FOR TESTING?

Before you have an H. pylori test, you may need to stop taking certain medicines for two weeks to a month. This includes many medicines you may be taking for your symptoms. Be sure to talk with your provider about all medicines you are currently taking.

For an endoscopy, you may need to fast (not eat or drink) for several hours before the procedure. You may be drowsy afterwards, so plan to have someone bring you home.

ARE THERE ANY RISKS TO TESTING?

There is no known risk to having breath or stool tests.

During an endoscopy, you may feel some discomfort when the scope is inserted, but serious complications are rare. There's a very small risk of a tear in your intestine. If you have a biopsy, there's a small risk of bleeding where the sample was removed. But it usually stops on its own.

WHAT DO THE RESULTS MEAN?

A negative (normal) H. pylori test result means that you probably don't have an H. pylori infection. Your provider may order more tests to find out the cause of your symptoms.

A positive test result means that you have an H. pylori infection. Your provider will usually prescribe one or more antibiotics to treat the infection. You will usually take other medicines to relieve your symptoms and help heal your stomach. After you finish your treatment, your provider may order repeat tests to make sure all the H. pylori bacteria is gone.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT H. PYLORI TESTING?

Blood tests for H. pylori check for certain proteins (antibodies) that your immune system makes to fight off H. pylori. These tests aren't commonly used to diagnose H. pylori. That's because the test can't show whether the antibodies are from an active H. pylori infection that you have now or a past infection that your antibodies successfully fought off.

But if an H. pylori blood test shows that you don't have any H. pylori antibodies, it usually means that you didn't have an infection when your blood sample was taken.

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Hematocrit Test

<https://medlineplus.gov/lab-tests/hematocrit-test/>

HEMATOCRIT TEST

WHAT IS A HEMATOCRIT TEST?

A hematocrit test is a blood test that measures the amount (percent) of your blood that is made up of red blood cells. Red blood cells carry oxygen from your lungs to the rest of your body. The other parts of your blood include white blood cells (to help fight infection and other diseases), platelets (to help make blood clots to stop bleeding), and a liquid called plasma.

Hematocrit levels that are too high or too low can be a sign of a blood disorder, dehydration, or other medical conditions that affect your blood.

Other names: HCT, packed cell volume, PCV, Crit; H and H (Hemoglobin and Hematocrit)

WHAT IS IT USED FOR?

A hematocrit (HCT) test is often part of a complete blood count (CBC). A CBC is a common blood test that measures the number and type of cells in your blood. It is used to check your general health. It may also be used to monitor or help diagnose blood disorders. These could include anemia, a condition in which you don't have enough red blood cells, and polycythemia (erythrocytosis), an uncommon disorder in which you have too many red blood cells and your blood becomes too thick.

WHY DO I NEED A HEMATOCRIT TEST?

Your health care provider may order a hematocrit test as part of your routine checkup or to monitor your health if you are being treated for cancer or have an ongoing health condition. Your provider may also order this test if you have symptoms of a red blood cell disorder, such as anemia or polycythemia:

Symptoms of anemia (too few red blood cells) may include:

Shortness of breath

Weakness or fatigue

Headache

Dizziness

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Symptoms of polycythemia (too many red blood cells) may include:

Headache

Feeling light-headed or dizzy

Shortness of breath when lying down

Weakness or fatigue

Skin symptoms such as itching after a shower or bath, burning, or a red face

Heavy sweating, especially during sleep

Blurred or double vision and blind spots

Unusual bleeding, such as bleeding gums or nosebleeds

WHAT HAPPENS DURING A HEMATOCRIT TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a hematocrit test. If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your hematocrit test results are reported as a number. That number is the percentage of your blood that's made up of red blood cells. For example, if your hematocrit test result is 42, it means that 42% of your blood is red blood cells, and the rest is made up of white blood cells, platelets, and blood plasma.

Normal hematocrit levels will vary, depending on your sex, age, whether or not you smoke, and the altitude where you live. Ask your provider what your hematocrit level should be.

A lower-than-normal hematocrit level may be a sign that:

You have had blood loss due to a recent injury

You have a condition causing chronic (long-term) blood loss

Your body doesn't have enough red blood cells (anemia). There are many types of anemia, and they can be caused by different medical conditions.

Your body is making too many white blood cells, which may be caused by:

Bone marrow disease

Certain cancers, including leukemia, lymphoma, multiple myeloma, or cancers that spread to your bone marrow from other parts of the body

Bone marrow disease

Certain cancers, including leukemia, lymphoma, multiple myeloma, or cancers that spread to your bone marrow from other parts of the body

A higher-than-normal hematocrit level may be a sign that:

Your body is making too many red blood cells, which may be caused by:

Lung disease

Congenital heart disease

Heart failure

Polycythemia

Lung disease

Congenital heart disease

Heart failure

Polycythemia

Your blood plasma level is too low, which may be caused by:

Dehydration, the most common cause of a high hematocrit

Shock

Dehydration, the most common cause of a high hematocrit

Shock

If your results are not in the normal range, it doesn't always mean you have a medical condition that needs treatment.

Living at high altitudes with less oxygen in the air may cause a high hematocrit. That's because your body responds to

low oxygen levels by making more red blood cells to give you the oxygen you need.

Pregnancy can cause a low hematocrit. That's because your body has more fluid than usual during pregnancy, which decreases the percentage that's made of red blood cells.

To learn what your test results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Hemoglobin A1C (HbA1c) Test

<https://medlineplus.gov/lab-tests/hemoglobin-a1c-hba1c-test/>

HEMOGLOBIN A1C (HBA1C) TEST

WHAT IS A HEMOGLOBIN A1C (HBA1C) TEST?

A hemoglobin A1C (HbA1C) test is a blood test that shows your average level of blood glucose, also called blood sugar, over the past two to three months.

Glucose is a type of sugar in your blood that comes from the foods you eat. Your cells use glucose for energy. A hormone called insulin helps glucose get into your cells. If you have diabetes—your body doesn't make enough insulin, or your cells don't use it well. As a result, glucose can't get into your cells, so your blood glucose levels increase.

Glucose in your blood sticks to hemoglobin, a protein in your red blood cells. As your blood glucose levels increase, more of your hemoglobin will be coated with glucose. An A1C test measures the percentage of your red blood cells that have glucose-coated hemoglobin.

An A1C test can show your average glucose level for the past three months because:

Glucose sticks to hemoglobin for as long as the red blood cells are alive.

Red blood cells live about three months.

High A1C levels are a sign of high blood glucose from diabetes. Diabetes can cause serious health problems, including heart disease, kidney disease, and nerve damage. But with treatment and lifestyle changes, you can control your blood glucose levels.

Other names: HbA1C, A1C, glycohemoglobin, glycated hemoglobin, glycosylated hemoglobin

WHAT IS IT USED FOR?

Your blood glucose level changes all the time. If you're checking it at home, you are measuring what your blood glucose level is at that time. An A1C test gives you an average level of your blood glucose level over the past two to three months.

An A1C test may be used to screen for or diagnose:

Type 2 diabetes. With type 2 diabetes your blood glucose gets too high because your body doesn't make enough insulin or doesn't use insulin well. The glucose then stays in your blood and not enough goes into your cells.

Prediabetes. Prediabetes means that your blood glucose levels are higher than normal, but not high enough to be diagnosed as diabetes. Lifestyle changes, such as healthy eating and exercise, may help delay or prevent prediabetes from becoming type 2 diabetes.

If you have diabetes or prediabetes, an A1C test can help monitor your condition and check how well you've been able to control your blood glucose levels.

The A1C test isn't used to diagnose type 1 diabetes—but may be used for monitoring your blood glucose levels.

WHY DO I NEED AN A1C TEST?

The Centers for Disease Control (CDC) recommends A1C testing for diabetes and prediabetes if:

You are over age 45.

If your results are normal, your provider will tell you how often you should be tested based on your age and risk factors.

If your results show you have prediabetes, you will usually need to be tested every 1 to 2 years. Ask your provider how often to get tested and what you can do to reduce your risk of developing diabetes.

If your results show you have diabetes, you should get an A1C test at least twice a year to monitor your condition and treatment.

If your results are normal, your provider will tell you how often you should be tested based on your age and risk factors.

If your results show you have prediabetes, you will usually need to be tested every 1 to 2 years. Ask your provider how

often to get tested and what you can do to reduce your risk of developing diabetes.

If your results show you have diabetes, you should get an A1C test at least twice a year to monitor your condition and treatment.

You are under 45 and are more likely to develop diabetes because you:

Have prediabetes.

Are overweight or have obesity.

Have a parent or sibling with type 2 diabetes.

Have high blood pressure or high cholesterol levels.

Have heart disease or have had a stroke.

Are physically active less than 3 times a week.

Have had gestational diabetes (diabetes during pregnancy) or given birth to a baby over 9 pounds.

Are African American, Hispanic or Latino, American Indian, or an Alaska Native person. Some Pacific Islander and Asian American people also have a higher risk of developing diabetes.

Have polycystic ovarian syndrome (PCOS).

Have prediabetes.

Are overweight or have obesity.

Have a parent or sibling with type 2 diabetes.

Have high blood pressure or high cholesterol levels.

Have heart disease or have had a stroke.

Are physically active less than 3 times a week.

Have had gestational diabetes (diabetes during pregnancy) or given birth to a baby over 9 pounds.

Are African American, Hispanic or Latino, American Indian, or an Alaska Native person. Some Pacific Islander and Asian American people also have a higher risk of developing diabetes.

Have polycystic ovarian syndrome (PCOS).

You may also need an A1C test if you have symptoms of diabetes, such as:

Feeling very thirsty

Urinating (peeing) a lot

Losing weight without trying

Feeling very hungry

Blurred vision

Numb or tingling hands or feet

Fatigue

Dry skin

Sores that heal slowly

Having more infections than usual

WHAT HAPPENS DURING AN A1C TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Certain medicines such as opioids and some HIV medicines may affect your results, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. Also, let your health care provider know if you are pregnant, since this may affect your results.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A1C results tell you what percentage of your hemoglobin is coated with glucose. The ranges are just a guide to what is normal. What's normal for you depends on your health, age, and other factors. Ask your provider what A1C percentage is healthy for you.

To diagnose diabetes or prediabetes, the percentages commonly used are:

Normal: A1C below 5.7%

Prediabetes: A1C between 5.7% and 6.4%

Diabetes: A1C of 6.5% or higher

Providers often use more than one test to diagnose diabetes. So, if your test result was higher than normal, you may have another A1C test or a different type of diabetes test, usually either a fasting blood glucose test or an oral glucose tolerance test (OGTT).

If your A1C test was done to monitor your diabetes, talk with your provider about what your test results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN A1C TEST?

The A1C test is not used to diagnose gestational diabetes. Also, if you have a condition that affects your red blood cells, such as anemia or another type of blood disorder, an A1C test may not be accurate for diagnosing diabetes. Kidney failure and liver disease can also affect A1C results. In these cases, your provider may recommend different tests to diagnose diabetes and prediabetes.

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Hemoglobin Electrophoresis

<https://medlineplus.gov/lab-tests/hemoglobin-electrophoresis/>

HEMOGLOBIN ELECTROPHORESIS

WHAT IS HEMOGLOBIN ELECTROPHORESIS?

Hemoglobin electrophoresis is a test that measures the different types of hemoglobin in the blood. It also checks for hemoglobinopathy, disorders involving abnormal types of hemoglobin. Hemoglobin is a protein in your red blood cells. It carries oxygen from your lungs to the rest of your body. There are a few different types of normal and abnormal hemoglobin.

Normal types of hemoglobin include:

Hemoglobin (Hgb) A, the most common type of hemoglobin in healthy adults.

Hemoglobin (Hgb) F, fetal hemoglobin. This type of hemoglobin is found in unborn babies and newborns. HgbF is usually replaced by HgbA by age one to two years.

If levels of HgbA or HgbF are too high or too low, it can be a sign of certain types of anemia.

Normal, healthy red blood cells are flexible, so they can move easily through blood vessels. But abnormal hemoglobin can change the size and shape of your red blood cells. This can cause the cells to clump and block blood flow, making it difficult to carry oxygen in your body well. Your body may break down these cells sooner than usual, causing anemia.

Abnormal types of hemoglobin include:

Hemoglobin (Hgb) S. This type of hemoglobin causes sickle cell disease. Sickle cell disease is a group of disorders in which you have stiff, sticky, sickle-shaped red blood cells. These sickle cells can get stuck in the blood vessels. This can block the blood flow to the organs, which can cause severe and chronic pain, infections, and other complications. Sickle cell disease is inherited (passed down in families). In the United States, it is most common in Black and African American people.

Hemoglobin (Hgb) C. This type of hemoglobin does not carry oxygen well. It can cause a mild form of anemia. In the United States, it is most common in Black and African American people.

Hemoglobin (Hgb) E. This type of hemoglobin is mostly found in people of Southeast Asian descent. People with HgbE usually have no symptoms or mild symptoms of anemia.

A hemoglobin electrophoresis test applies an electric current to a blood sample. This separates the normal and abnormal types of hemoglobin. Each type of hemoglobin can then be measured individually and compared with the normal level.

Other names: Hb electrophoresis, hemoglobin evaluation, hemoglobinopathy evaluation, hemoglobin fractionation, Hb ELP, Hb IEF, sickle cell screen

WHAT IS IT USED FOR?

Hemoglobin electrophoresis measures hemoglobin levels and looks for abnormal types of hemoglobin. If your hemoglobin type levels are too high or low, it could be a sign of a hemoglobin disorder. It's most often used to help diagnose anemia, sickle cell disease, and other hemoglobin disorders.

WHY DO I NEED HEMOGLOBIN ELECTROPHORESIS?

You may need testing if you have symptoms of a hemoglobin disorder such as anemia or sickle cell disease.

Symptoms of anemia may include:

Shortness of breath

Weakness or fatigue

Headache

Dizziness

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Symptoms of sickle cell disease may include:

Painful swelling of the hands and feet

Fatigue or fussiness in infants (from the anemia caused by sickle cell disease)

Growth problems (in children)

Jaundice, a condition that causes your skin and eyes to turn yellowish

Severe pain

If you've just had a baby, your newborn will be tested as part of a newborn screening. Newborn screening is a group of tests given to most American babies shortly after birth. The screening checks for many different serious conditions. If these conditions are found early, treatment can start right away. Early treatment may help prevent serious health problems for the baby.

You may also want testing during pregnancy if you are at risk of having a child with sickle cell disease or another inherited hemoglobin disorder. Risk factors include:

Family history, including already having a child with a hemoglobin disorder.

Ethnic background:

In the United States, most people with sickle cell disease are Black or African American.

Thalassemia, another inherited hemoglobin disorder, is most common among people of Italian, Greek, Middle Eastern, Southern Asian, and African descent.

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Thalassemia, another inherited hemoglobin disorder, is most common among people of Italian, Greek, Middle Eastern, Southern Asian, and African descent.

WHAT HAPPENS DURING HEMOGLOBIN ELECTROPHORESIS?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

To test a newborn, a health care provider will clean your baby's heel with alcohol and poke the heel with a small needle. The provider will collect a few drops of blood and put a bandage on the site.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a hemoglobin electrophoresis test.

ARE THERE ANY RISKS TO HEMOGLOBIN ELECTROPHORESIS?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

Your baby may feel a little pinch when the heel is poked, and a small bruise may form at the site. This should go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will show the types of hemoglobin found and the levels of each.

Hemoglobin type levels that are too high or too low may be a sign of:

Thalassemia, a condition that affects the production of hemoglobin. In this condition, your body makes normal hemoglobin, but it doesn't make enough of it. Symptoms range from mild to severe.

Sickle cell trait. In this condition, you have one sickle cell gene and one normal gene. Most people with sickle cell trait do not have health issues. But you can pass the sickle cell gene on to their children.

Sickle cell disease.

Hemoglobin C disease, a condition that causes a mild form of anemia and sometimes an enlarged spleen and joint pain.
Hemoglobin S-C disease, a condition that causes a mild or moderate form of sickle cell disease.

Your results may also show whether a specific disorder is mild, moderate, or severe.

There are several options for treating thalassemia, anemia, sickle cell disease, and other hemoglobin disorders. Until recently, treatment options for sickle cell disease were limited. But now, new therapies have become available. If you or your child was diagnosed with a sickle cell disease or another hemoglobin disorder, talk to your provider about treatment options.

To understand the results of a hemoglobin electrophoresis test, your provider will consider your symptoms, medical history, and the results of other blood tests. Hemoglobin electrophoresis test results are often compared with other tests, including a complete blood count and a blood smear.

Let your provider know if you've had a blood transfusion in the last 12 weeks since this can affect your results.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT HEMOGLOBIN ELECTROPHORESIS?

If you are at risk of having a child with an inherited hemoglobin disorder, you may want to speak to a genetic counselor. A genetic counselor is a specially trained professional in genetics and genetic testing. He or she can help you understand the disorder and your risk of passing it along to your child.

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Hemoglobin Test

<https://medlineplus.gov/lab-tests/hemoglobin-test/>

HEMOGLOBIN TEST

WHAT IS A HEMOGLOBIN TEST?

A hemoglobin test measures the levels of hemoglobin in your blood. Hemoglobin is an iron-rich protein in your red blood cells. It carries oxygen from your lungs to the rest of your body. Your cells need oxygen to grow, reproduce, and make energy for you to function. Your hemoglobin levels provide information about the health of your red blood cells. If your hemoglobin levels are abnormal, it may be a sign that you have a blood disorder.

Other names: Hb, Hgb, H&H

WHAT IS IT USED FOR?

A hemoglobin test is a blood test often used to check for anemia, a condition in which your body does not make enough healthy red blood cells. If you have anemia, the cells in your body don't get all the oxygen they need. Hemoglobin levels are usually measured as part of a complete blood count (CBC). A CBC is a group of tests that measure the number and type of cells in your blood.

A hemoglobin test is not the same as a hemoglobin A1C test. The A1C test is used to check your average blood glucose (blood sugar).

WHY DO I NEED A HEMOGLOBIN TEST?

Your health care provider may order the test as part of a routine exam, or if you have:

Symptoms of anemia, which include weakness, dizziness, and cold hands and feet

A family history of thalassemia, sickle cell anemia, or other inherited blood disorder

A diet low in iron and other minerals

A long-term infection

Excessive blood loss from an injury or surgical procedure

You may also need this test to monitor the treatment of anemia or the other health conditions that could affect your hemoglobin level.

WHAT HAPPENS DURING A HEMOGLOBIN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparation for a hemoglobin test. If your provider has ordered other blood tests, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will compare your hemoglobin level to the normal range, which may be called the reference range. This range may vary, based on the laboratory used and your age, race, and sex. The test can't tell you what's causing an abnormal level. But knowing how high or low your hemoglobin level is can help your provider determine how severe your condition may be.

Your hemoglobin levels may not be in the reference range for many reasons.

Low hemoglobin levels may be a sign of:

Different types of anemia

Thalassemia

Iron deficiency

Liver disease

Cancer and other diseases

Blood loss from injury, a heavy menstrual period, or other problems

Lack of certain nutrients, such as iron or vitamin B-12

High hemoglobin levels may be a sign of:

Lung disease

Heart disease

Polycythemia vera, a disease which causes your bone marrow to make too many red blood cells

Sleep apnea, a disorder that causes you to repeatedly stop breathing during sleep

If any of your levels are abnormal, it doesn't always mean you have a medical condition that needs treatment. Living at a higher altitude can also cause a high hemoglobin level. Other factors that can affect your results can include:

Diet

Activity level

Certain medicines

Dehydration

Smoking

Certain genetic conditions

Talk with your provider if you have questions about your results. To understand your hemoglobin results, your provider will also consider your symptoms, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A HEMOGLOBIN TEST?

Some forms of anemia are mild, while other types of anemia can be serious and even life threatening if not treated. If you are diagnosed with anemia, talk to your provider about the best treatment plan for you.

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What You Need to Know About Blood Testing

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Hepatitis Testing

<https://medlineplus.gov/lab-tests/hepatitis-testing/>

HEPATITIS TESTING

WHAT IS HEPATITIS TESTING?

Hepatitis testing uses a sample of your blood to see if you have a current viral hepatitis infection or had one in the past. Hepatitis is a type of liver disease that causes liver inflammation (swelling) and damage. There are several types of hepatitis. Each affects the body and is spread in different ways.

Hepatitis A (HAV) is usually spread by eating or drinking food or beverages that an infected person has handled without washing their hands. This can result in swallowing tiny amounts of the virus that can be found in an infected person's stool (poop). You can also get HAV from close physical contact with an infected person. Though no specific medicine can cure HAV, most people recover from it on their own. The best way to prevent it is to get vaccinated.

Hepatitis B (HBV) is mainly spread through blood, semen, or other body fluids. For example, you can get an HBV infection if you have sex or share needles with an infected person. It can also be passed to your fetus during pregnancy or birth. As with HAV, no medicine can cure HBV, though most adults recover from it on their own. In some cases, it can develop into a chronic (long-term) infection. The best way to prevent HBV is also to get vaccinated.

Hepatitis C (HCV) is spread through blood. Most people get infected with HCV from sharing needles or other items used for injecting drugs, or through sexual contact. The virus can be passed to the fetus during pregnancy, but this is less common. There is no vaccine for this type of hepatitis, however, unlike with hepatitis A and B, medicine can cure it. HCV is more likely than the other types to develop into a chronic infection.

Hepatitis D (HDV) is spread the same way HBV is spread, through blood, semen, or other body fluids. However, HDV is unusual among hepatitis variants because you can only develop an HDV infection if you already have HBV. This means that those with HDV have a coinfection (getting HBV and HDV at the same time) or superinfection (having HBV and then getting HDV), which can lead to health complications. The best way to prevent HDV is to get vaccinated for HBV.

Hepatitis E (HEV) is spread by drinking contaminated water or eating undercooked foods. In most people, HEV causes an acute (sudden) infection that goes away. However, some people, especially those with weakened immune systems, may develop chronic infections. These can be treated with medicines. No vaccine is available for HEV.

Most cases of hepatitis are caused by types A, B, or C. Because of this, the most common type of hepatitis test is a hepatitis panel, which checks for signs of HAV, HBV, and HCV using one blood sample.

Other names: acute viral hepatitis panel, viral hepatitis screening panel, hepatitis D virus (HDV) antibody test, hepatitis B triple panel, HBsAg, anti-HBs, anti-HBc, HBV, HAV, HCV, HDV, HEV

WHAT IS IT USED FOR?

Generally, hepatitis testing is used to find out if there is any sign of hepatitis in your blood. For hepatitis A, B, D, and E, a test can tell the difference between a present and a past infection. For hepatitis C, however, a test can't tell the difference between an active infection and a past infection that has cleared up.

If testing shows that you may have hepatitis, you'll usually need other tests to confirm the diagnosis and to find out whether your infection is acute (sudden) or chronic (long-term).

WHY DO I NEED HEPATITIS TESTING?

You may need hepatitis testing if you have symptoms of hepatitis. Hepatitis doesn't always cause symptoms, but when it does, they may include:

Jaundice, a condition that causes your skin and eyes to turn yellow.

Fever.

Fatigue.

Loss of appetite.

Dark-colored urine.

Pale or clay-colored stool (poop).

Abdominal (belly) pain.

Nausea and vomiting.

Joint pain.

Diarrhea (hepatitis A only).

If left untreated, a hepatitis infection can become chronic (long-term). These kinds of infections can lead to serious liver problems such as cirrhosis or liver cancer. Because of this, your provider may order hepatitis testing if you've had liver function tests with abnormal results.

Even if you don't have symptoms, your provider may order a hepatitis test if you've been exposed to a specific hepatitis virus, or if you're at high risk of getting an infection. You may be at high risk if you:

Work in certain industries, such as health care.

Use illegal intravenous (IV) drugs and/or share needles with others.

Have certain medical conditions that weaken your immune system, such as HIV.

Have other kinds of liver disease.

Are a man who has sex with other men.

Already have another hepatitis infection. For example, your provider may test for hepatitis D if you already have hepatitis B.

Lastly, your provider may test you for hepatitis as part of a more general health screening. Medical experts recommend that all adults aged 18 and older be tested for hepatitis B and/or C at least once in their lifetime. They also recommend HBV and HCV tests during each pregnancy as part of routine prenatal testing.

WHAT HAPPENS DURING HEPATITIS TESTING?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

At-home testing kits are available for hepatitis B and C. Usually the test kit will include a sharp device (a lancet) to prick your finger so you can collect a drop of blood and send it to a lab for testing. For more information on at-home testing for hepatitis, talk to your provider.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for hepatitis testing.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Different labs report hepatitis test results in various ways, so it's best to ask your provider what your test results mean.

In general:

A negative or normal result means you probably don't have a hepatitis infection. But if you know you were exposed to hepatitis, you may need to be tested again later on.

A positive or abnormal result may mean:

You have an active hepatitis infection now. It may be an acute infection or one you've had for a long time.

You had a hepatitis infection in the past that cleared up.

You have an active hepatitis infection now. It may be an acute infection or one you've had for a long time.

You had a hepatitis infection in the past that cleared up.

An abnormal test result means that signs of hepatitis were found in your blood. These signs include:

Hepatitis antigens, which are substances in the hepatitis virus that trigger your immune system to fight the virus. If you have hepatitis antigens in your blood, it means that you may have an active hepatitis infection. It could be an acute or chronic infection.

Hepatitis antibodies, which are proteins that your immune system makes to help fight a hepatitis infection. If you have hepatitis antibodies in your blood, it may mean that you have an acute or chronic infection or that you had an infection in the past.

If your test results are abnormal, you will likely need other tests to find out if you have an active infection and whether it is an acute or chronic infection. For example, if your test results suggest a hepatitis B infection, your provider may use a type of test called a triple panel that checks your blood for:

Hepatitis B Surface Antigen (HBsAg), an antigen on the surface of the hepatitis B virus.

Hepatitis B Surface-Antibody (Anti-HBs), an antibody that your immune system makes to fight HBV after a hepatitis B infection or vaccination.

Anti-HBc, another antibody that your immune system makes to fight HBV, but only after a hepatitis B infection.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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HER2 Tumor Marker Test

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HER2 TUMOR MARKER TEST

WHAT IS HER2 TUMOR MARKER TESTING?

Tumor marker tests look for substances, called tumor markers, that are often made by cancer cells or normal cells in response to cancer. The test results can provide important information, such as how fast a cancer may grow and which treatments are most likely to work.

HER2 is a type of tumor marker. It stands for human epidermal growth factor receptor 2. It's also called HER2/neu. HER2 is a protein involved in normal cell growth. Some types of cancer cells make larger than normal amounts of HER2, including certain cancers of the breast, stomach, and esophagus.

Cancers with large amounts of HER2 protein tend to grow quickly and spread to other parts of the body. Certain medicines, called HER2 targeted therapy, can block or slow HER2 proteins and help control these cancers. But these medicines won't help cancers that don't have high levels of HER2 proteins. So, HER2 tumor marker testing is necessary to find out if targeted therapy will be helpful.

A HER2 tumor marker test usually uses a sample of tumor tissue to measure either:

The amount of HER2 protein on the surface of cancer cells

How many copies of the HER2 gene are inside the cancer cells

Genes are parts of DNA in your cells that you inherit from your parents. HER2 genes have instructions for making the HER2 protein. Normally, you have two HER2 genes in each cell, one from each parent. But certain cancers have extra copies of the HER2 gene, which leads to too much HER2 protein on the cancer cells.

A HER2 tumor marker test looks to see whether certain cancers are:

HER2 positive, which means the cancer has high levels of the HER2 protein or HER2 genes. These cancers are likely to respond to HER2 targeted therapy.

HER2 negative, which means your cancer cells have little to no HER2 protein or a normal number of HER2 genes. This type of cancer may grow more slowly than HER2-positive cancer. It is also less likely to come back after treatment or spread to other parts of the body. HER2 targeted therapy will not work for this type of cancer, but other treatment options may help.

There are different HER2 tumor marker tests for checking HER2 proteins and HER2 genes. Usually, a protein test is done first because those tests are faster and less expensive. Blood tests to measure HER2 usually aren't recommended. That's because it's not clear how blood test results can plan cancer treatment.

Other names: human epidermal growth factor receptor 2, ERBB2 amplification, HER2 overexpression, HER2/neu tests

WHAT IS IT USED FOR?

HER2 tumor marker tests are mostly used to find out whether certain cancers are HER2 positive or negative. This information shows how fast cancer may grow and whether medicine to target HER2 proteins may be helpful.

HER2 tumor marker testing is mainly used for cancers that start in the:

Breast (most common)

Stomach

Esophagus

Ovaries

Bladder

Pancreas

WHY DO I NEED HER2 TUMOR MARKER TESTING?

You may need HER2 tumor marker testing to help guide cancer treatment decisions if you have:

Breast cancer that has:

Just been diagnosed and is growing in the normal breast tissue around the tumor. This is called "invasive" cancer. Spread beyond your breast and is growing in other parts of your body. This is called metastatic cancer. The cancer growing outside of your breast should be tested to see if it's HER2 positive, even if your breast tumor was already tested.

Come back after treatment, usually after a period of time when no cancer could be found in your body. This is called recurrent cancer. The new tumor should be tested to see if it's HER2 positive even if your first tumor was tested. That's because cancer cells can change over time.

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Stomach or esophageal cancer that:

Can't be removed using surgery.

Has come back after treatment.

Can't be removed using surgery.

Has come back after treatment.

Cancer of the pancreas, bladder, ovaries, or certain other cancers that may have high levels of HER2 which may be treated with medicines that target HER2.

WHAT HAPPENS DURING A HER2 TUMOR MARKER TEST?

Most HER2 testing uses a sample of tumor tissue. The tissue will be removed during a procedure called a biopsy. There are many ways to do a biopsy. The type of biopsy you have depends on where the tumor is located. In general, a biopsy may be done using:

A hollow needle inserted through your skin. Imaging tests, such as ultrasound, may be used to guide the needle:

A fine needle aspiration biopsy uses a very thin needle to remove a sample of cells and/or fluid.

A core needle biopsy uses a larger needle to remove a sample.

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Surgery. A surgeon may remove a sample of the tumor (incisional biopsy) or the entire tumor (excisional biopsy).

Endoscopy. This procedure uses a scope to look inside your body. Special tools may be used with a scope to remove a tissue sample. An endoscopy may be used for a biopsy of a tumor in your stomach or esophagus.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Preparations for your test depend on how your tissue sample will be taken. Ask your health care provider how to prepare for your test.

If you are getting medicine to relax or sleep during the procedure, you will probably need to fast (not eat or drink) for several hours before surgery. You will also need to have someone take you home, because you may be sleepy after the procedure.

ARE THERE ANY RISKS TO THE TEST?

Risks depend on how your tissue sample is taken. In general, if you are having a biopsy, you may have a little bruising or bleeding where the tissue was removed. Ask your provider to explain any risks from the test you're having.

WHAT DO THE RESULTS MEAN?

The results of HER2 tumor marker tests are reported differently, depending on the type of test you had. The main HER2 tumor marker tests are:

IHC (immunohistochemistry staining assay) to measure the amount of HER2 protein on cancer cells

FISH (fluorescence in situ hybridization) to check for extra copies of HER2 genes in cancer cells

IHC test results for cancer of the breast, stomach, or esophagus show the amount of HER2 protein as a score:

0 (zero) or 1+ means that your cancer cells don't have too much HER2 protein. Your cancer is HER2 negative and will not respond to HER2 targeted therapy. Your provider will discuss other treatment options.

2+ means that the amount of HER2 protein is borderline. This may be reported as "equivocal." It means the test result is not clear. You may need to have a different type of HER2 tumor marker test. The test may be done on the same tissue sample, or you may need to provide a new sample.

3+ means that your cancer cells have too much HER2 protein. Your cancer is HER2 positive and is likely to respond to HER2 targeted therapy.

FISH results for cancer of the breast, stomach, or esophagus will be reported as either:

HER2 negative, which means that you don't have too many HER2 genes in your cancer cells. HER2 targeted therapy won't help your cancer, so your provider will discuss other treatment options.

HER2 positive, which means that your cancer cells have too many HER2 genes. HER2 targeted therapy is likely to help your cancer.

Talk with your provider to learn what your results mean for your treatment.

Learn more about laboratory tests, reference ranges, and understanding results.

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Herpes (HSV) Test

<https://medlineplus.gov/lab-tests/herpes-hsv-test/>

HERPES (HSV) TEST

WHAT IS A HERPES (HSV) TEST?

A herpes test looks for signs of the herpes simplex virus (HSV), a common viral infection. Many people with HSV don't have any symptoms, but those who do often have painful blisters or sores on the affected parts of the body. These sores and other symptoms only last for several weeks. After their first appearance, they will fade and then come back in the future. These outbreaks tend to be milder over time.

A herpes test can be done using fluid from one of these sores during an outbreak or by using a sample of your blood if you have no symptoms.

There are two main types of this virus.

HSV-1 usually causes oral herpes, which can result in cold sores ("fever blisters") on or around your mouth. Most people get oral herpes by the time they're young adults, usually from non-sexual contact with saliva (spit) from a person who has an HSV-1 infection. That may happen if you share forks, cups, or towels. HSV-1 can also cause genital herpes if you receive oral sex from a partner with a cold sore.

HSV-2 is the most common cause of genital herpes. Genital herpes is a sexually transmitted infection (STI). This means you get it through sexual contact with a person who has a herpes infection, even if they don't have symptoms. HSV-2 can also spread to the mouth, causing oral herpes. This may happen if you give oral sex to a person who has genital herpes from HSV-2.

There is no cure for HSV, but medicines can help manage outbreaks. HSV usually doesn't cause any major health problems. But if you're pregnant and have genital herpes, you can pass the virus to your fetus before birth or during childbirth. This can lead to a life-threatening HSV infection for your baby.

Other names: herpes culture, herpes simplex viral culture, HSV-1 antibodies, HSV-2 antibodies, HSV DNA

WHAT IS IT USED FOR?

Generally, HSV tests are used to find out whether or not you have a herpes infection. However, there are different ways your health care provider can test for HSV, depending on whether or not there are sores present.

If there are sores or blisters, your provider will collect fluid from the sore using a swab. They can then use the fluid from the sore to test for the virus in one of two ways.

A viral culture takes cells from the sample and grows them in a lab. After the culture of cells has grown, a lab technician will be able to check it for HSV.

A polymerase chain reaction (PCR) test looks for genetic material from the HSV virus in the sample. This test is faster and more accurate than a viral culture. In the rare event that you have an HSV infection in your spine or brain, your provider may use this type of test.

If there are no sores or blisters, your provider will look for HSV antibodies in a sample of your blood. Antibodies are proteins your immune system makes to fight germs such as viruses and bacteria. So, if you have HSV antibodies, that means an HSV infection might be present. This test may be used on those who are at high risk of getting HSV.

WHY DO I NEED AN HSV TEST?

The most common reason for needing an HSV test is if you have symptoms of an infection. These include:

Blisters or sores on the mouth, genitals, anus, buttocks, or other areas of skin. The sores develop a crust as they begin to heal.

Fever and flu-like symptoms.

Swollen glands.

Pain or tingling in the affected area.

Discharge from the urethra and/or vagina.

Painful urination (pee).

If you don't have symptoms, you may also need an HSV test if your partner has genital herpes or symptoms that might be genital herpes.

In general, medical experts don't recommend HSV testing for people without symptoms of HSV.

If you are being tested for other STIs, your provider may choose to include an HSV test based on your risk of exposure.

You may be at higher risk of getting an HSV infection if you have:

Multiple sexual partners.

Other STIs.

HIV.

Sex without condoms or dental dams.

Sex in exchange for money and/or drugs.

You may also be at higher risk for HSV infection if you are a man who has sex with other men.

Using condoms or dental dams can reduce your risk of getting or passing HSV. However, they can't provide complete protection. You can still get an infection from contact with areas of skin that the condom or dental dam doesn't cover.

Rare cases of HSV in the brain and spinal cord can lead to encephalitis or meningitis. You may need an HSV test if you have one of these serious conditions.

Because an HSV infection can be passed to a baby during childbirth, your provider may order an HSV test for your baby.

WHAT HAPPENS DURING AN HSV TEST?

If you get a swab test, a health care professional will use a swab to collect fluid and cells from a herpes sore that has not begun to heal yet.

If you get a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

At-home tests are available for HSV. These tests usually use a sample of blood that you collect at home and send to a lab. Talk with your provider to find out if at-home testing is right for you.

If your provider thinks you have an HSV infection in your spinal cord or brain, you may have a test of your cerebrospinal fluid (CSF) to look for signs of the virus. CSF is a fluid that flows in and around your brain and spinal cord. A thin needle is inserted into your spine to remove a small amount of CSF. This procedure is called a lumbar puncture or spinal tap.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a swab test or a blood test. For a lumbar puncture, you may be asked to empty your bladder (pee) and bowels (poop) before the test.

ARE THERE ANY RISKS TO THE TEST?

After a swab test, you may have a little bit of bleeding or discomfort where your skin was swabbed. This usually doesn't last long.

A blood test has very little risk. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

If you have a lumbar puncture, you may feel some pain or tenderness in your back where the needle was inserted. You may also have some bleeding at the site or a headache.

WHAT DO THE RESULTS MEAN?

The meaning of your test results depends on the type of HSV test you had. To understand your results, your provider will consider your health history and any symptoms you have.

In general:

Negative/Normal test results mean that signs of the herpes virus were not found in your sample, and you are unlikely to have an HSV infection. It's important to know that you can have a negative result when you really do have a herpes infection if you had a blood test for HSV antibodies too soon after an infection began. That's because your body takes up to three months to make HSV antibodies.

Positive/Abnormal test results mean that signs of HSV were found in your sample. If you had a blood test that showed you have HSV antibodies, the antibodies could be from an active infection now or a past infection.

If you have HSV, treatment can help reduce the number of outbreaks and control your symptoms. It's important to know that having an HSV infection may increase your risk of getting HIV and other STIs. So, talk with your provider about HIV testing along with testing for other STIs.

It's also important to take steps to avoid spreading HSV to others. Using condoms can help. Since outbreaks come and go, ask your provider how to know when your risk for spreading the herpes virus may be high.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN HSV TEST?

The best way to prevent genital herpes or another STI is not to have sex. If you are sexually active, you can reduce your risk of infection by:

Having sex with only one person who has sex only with you (mutual monogamy) after both of you are tested for STIs.

Using condoms correctly every time you have sex.

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HIV Screening Test

<https://medlineplus.gov/lab-tests/hiv-screening-test/>

HIV SCREENING TEST

WHAT IS AN HIV TEST?

An HIV test checks a sample of your blood, saliva (spit), or urine (pee) to see whether you are infected with HIV (human immunodeficiency virus). HIV is a virus that destroys certain cells in your immune system. These cells protect your body against diseases from germs, such as bacteria, viruses, and fungi. If you lose too many immune cells, your body will have trouble fighting off infections and other diseases.

HIV is spread through contact with blood and other body fluids from a person who has an HIV infection. This usually happens during sex or when sharing needles or other items used to inject drugs.

HIV is the virus that causes AIDS (acquired immunodeficiency syndrome). AIDS is the final, most serious stage of an HIV infection. Without treatment, HIV gradually destroys your immune system, which leads to AIDS. With AIDS, your body has trouble fighting off infections from germs that usually don't cause problems in healthy people. These are called opportunistic infections, and they can become life-threatening. AIDS increases your risk of developing certain cancers, too.

Most people with HIV don't have AIDS. If you have HIV and you take HIV medicines[^] as prescribed, you may never get AIDS. Getting tested for HIV helps catch the virus early so you can start treatment, stay healthy, and avoid spreading HIV.

If you think you were exposed to HIV, talk with your health care provider right away about emergency treatment, called PEP (post-exposure prophylaxis). PEP may prevent an HIV infection if it is started within three days after a possible exposure. The sooner you start, the better. You will need to have HIV tests during and after taking PEP.

There are three main types of HIV tests. They are all very accurate, but no test can find HIV in your body immediately after you're infected. That's because it can take weeks until the signs of HIV in your body increase enough to show up on a test. The time between infection and when a test can find HIV depends on your body's response to HIV and the type of test you have:

HIV antibody tests look for HIV antibodies in your blood, saliva, or urine. HIV antibodies are disease-fighting proteins that your immune system makes when you have an HIV infection. Some people make antibodies faster than others. An antibody test may find HIV antibodies as early as[^] 23 days after infection, but it may take as long as 90 days before your body makes enough antibodies to show up on this test.

Antibody tests can be done in different ways:

Lab tests use a blood sample taken from a vein. In general, lab tests can find antibodies sooner after infection than other HIV antibody tests. Test results are usually ready a few days after your blood sample is taken.

Rapid tests use saliva, urine, or blood from your finger. They provide results in about 30 minutes.

At-home tests include rapid self-test kits with everything you need to collect and test a sample of saliva on your own and get your results. With mail-in tests, you collect a drop of blood from your finger to send to a lab for testing.

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HIV antibody/antigen tests are the most commonly used HIV tests. They look for HIV antibodies and antigens in your blood. An antigen is the part of the HIV virus that triggers your immune system to fight the infection. After an HIV exposure, antigens will show up in your blood sooner than antibodies:

Lab tests that use a blood sample from a vein can find an HIV infection as soon as 18 to 45 days after you were infected.

Rapid tests use a drop of blood from your finger. This test can find HIV infections starting between 18 days to 90 days after you were infected.

At-home tests, or mail-in tests, allow you to collect blood from your finger at home to send to a lab for testing. Like rapid tests, these tests can start finding HIV 18 to 90 days after infection.

Lab tests that use a blood sample from a vein can find an HIV infection as soon as 18 to 45 days after you were infected.

Rapid tests use a drop of blood from your finger. This test can find HIV infections starting between 18 days to 90 days after you were infected.

At-home tests, or mail-in tests, allow you to collect blood from your finger at home to send to a lab for testing. Like rapid tests, these tests can start finding HIV 18 to 90 days after infection.

NAT tests (nucleic acid tests) look for HIV in a sample of your blood taken from a vein and tested in a lab. These tests may also be called "viral load tests." They can usually find an HIV infection as soon as 10 to 33 days after infection. But they are mostly used for monitoring HIV treatment and not for routine screening, because they are expensive. Ask your provider about having a NAT test as soon as possible if you:

May have been exposed to HIV and have symptoms, such as:

Flu-like symptoms, including fever, chills, aches

Extreme fatigue

Swollen lymph nodes (in your neck, groin, or armpit)

Rash

Sores in your mouth

Night sweats

Sore throat

Unexplained weight loss

Know you were exposed to HIV

May have been exposed to HIV and have symptoms, such as:

Flu-like symptoms, including fever, chills, aches

Extreme fatigue

Swollen lymph nodes (in your neck, groin, or armpit)

Rash

Sores in your mouth

Night sweats

Sore throat

Unexplained weight loss

Flu-like symptoms, including fever, chills, aches

Extreme fatigue

Swollen lymph nodes (in your neck, groin, or armpit)

Rash

Sores in your mouth

Night sweats

Sore throat

Unexplained weight loss

Know you were exposed to HIV

Other names: HIV antibody/antigen tests, HIV-1 and HIV-2 antibody and antigen evaluation, HIV test, human immunodeficiency virus antibody test, type 1, HIV p24 antigen test

WHAT IS IT USED FOR?

An HIV screening test is used to find out if you have been infected with HIV. It may be done as a routine test or after a possible exposure to find out if you were infected with HIV.

If HIV is found early, you can take medicines to protect your health so you don't get AIDS. Medicines can also help you avoid spreading HIV to others.

WHY DO I NEED AN HIV TEST?

The Centers for Disease Control and Prevention (CDC) recommends that everyone between the ages of 13 and 64 get tested for HIV at least once as part of routine health care.

You may need to be tested more than once if you have a higher risk for infection. If your last HIV test was more than a year ago, you should get tested as soon as possible if you have:

Had vaginal or anal sex with someone who has HIV (HIV positive) or whose HIV status you don't know

Injected drugs and shared needles, syringes, or other drug items with other people

Exchanged sex for money or drugs

Had a sexually transmitted infection (STI), such as syphilis

Had sex with anyone who has done anything listed above

If you regularly do any of the things on the above list, you should be tested for HIV at least once every year. Certain people, including men who have sex with men (MSM), may benefit from being tested more often, depending on their risk. Ask your provider how often you should be tested.

If you have had hepatitis or tuberculosis, you should also get tested for HIV at least once a year.

If you are pregnant, your doctor may order an HIV test. That's because HIV can be passed to the fetus during pregnancy and to the baby at birth, and through breast milk. There are medicines you can take during pregnancy and delivery that greatly reduce your risk of spreading HIV to your baby.

WHAT HAPPENS DURING AN HIV TEST?

There are many places to get tested for HIV, including at your health care provider's office, health clinics, testing programs in your community, and at home. If you go somewhere for your test, a staff person or counselor will explain what type of sample will be taken and how. If you do a test at home, be sure to follow all the instructions that come with your test kit.

For blood from a vein, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For blood from a finger, a health care professional will use a special tool (lancet) to prick your finger and collect a sample of blood. If you are collecting your own blood sample to send to a lab, follow the instructions that come with your collection kit.

For a saliva sample, a health care professional will wipe a special swab along your gums. If you are doing an at-home test, you will do this yourself by following the instructions that come with your collection kit.

For a urine sample, a health care professional may give you a cleansing wipe, a small container, and instructions for how to collect your urine sample. Once you've collected your sample, your provider will tell you what to do next with it.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an HIV test. If you are getting tested in a health clinic or community program, a counselor may talk with you about your risk for HIV. So, you may want to write down any questions you have ahead of time.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having any HIV screening test. If you have blood taken from a vein, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A negative test result means that no signs of an HIV infection were found in your sample. But that doesn't always mean that you don't have HIV. You could have an HIV infection, but it's too soon for the test to tell and you may need another test later. Your provider or an HIV counselor can explain your test result and let you know if you need another test.

In general, if you have a negative result on a rapid test or an at-home test and a possible HIV exposure that was:

90 days ago or longer, you can be confident that you don't have HIV

Less than 90 days ago, you will likely need another test later to check for HIV again

A positive test result means that signs of an HIV infection were found in your sample. You will need a follow-up test to confirm an HIV diagnosis unless you had a NAT test.

If you used an at-home test, see your provider for follow-up testing.

If you had your test at a medical office or community program, the testing site will arrange your follow-up test.

If your follow-up test is also positive, it means you have HIV. It's important to start medicines called antiretroviral therapy (ART) right away, even if you're still healthy. ART can't cure HIV, but it may lower the amount of virus in your blood enough so that a test can't find it. If you're living with HIV, it's important to see your provider regularly for tests to check how your treatment is working.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT HIV SCREENING?

HIV testing can be anonymous. That means you do not have to give your name when you take an HIV test. Instead, you'll receive a number. To get your test results, you'll give the number instead of your name.

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health

care provider if you have questions about your health.

HIV Viral Load

<https://medlineplus.gov/lab-tests/hiv-viral-load/>

HIV VIRAL LOAD

WHAT IS AN HIV VIRAL LOAD TEST?

An HIV viral load test measures the amount of HIV in a sample of your blood. HIV stands for human immunodeficiency virus. The test looks for genetic material from the virus in your blood. These tests are called molecular tests or nucleic acid amplification tests (NAAT or NAT). There are several types of NAAT tests. A PCR test is one type of NAAT that may be used to find HIV.

HIV is a virus that destroys certain cells in your immune system. These cells protect your body against diseases from germs, such as viruses, bacteria, and fungi. If you lose too many immune cells, your body will have trouble fighting off infections and other diseases.

You can get HIV from contact with the blood or other body fluids of a person who has an HIV infection. This usually happens through sex or sharing needles or other equipment used to inject drugs.

HIV is the virus that causes AIDS (acquired immunodeficiency syndrome). AIDS is the final, most serious stage of an HIV infection. Without treatment, the amount of HIV in your body can increase. It can gradually destroy your immune system and become AIDS. With AIDS, your body has trouble fighting off infections from germs that usually don't cause problems in healthy people. These are called opportunistic infections, and they can become life-threatening. AIDS increases your risk of developing certain cancers, too.

Most people with HIV don't have AIDS. If you have HIV, you can take HIV medicines that protect your immune system and help prevent you from getting AIDS. Medicines that treat HIV are called antiretroviral therapy (ART). ART can't get rid of HIV completely, so you'll need to take medicines for the rest of your life. But ART can control HIV and help you live a longer, healthier life.

Keeping your HIV viral load low is important for lowering your risk of complications with this type of infection. Having regular HIV viral load tests is an important part of making sure your HIV medicines are keeping your viral load low, so you stay healthy.

Other names: nucleic acid testing, NAT, nucleic acid amplification test, NAAT, HIV PCR, RNA Test, HIV quantification

WHAT IS IT USED FOR?

An HIV viral load test is mainly used after you're diagnosed with HIV to:

- Guide decisions about your treatment

- Check how well your HIV medicines are working

- Watch for any changes in your HIV infection, such as to see if the infection is getting worse, is stable, or getting better

HIV viral load testing is also used to test newborn babies when they are born to a person who has HIV. That's because HIV can be passed to a fetus during pregnancy, a baby during childbirth, and through breast milk.

In certain cases, an HIV viral load test may be used to diagnose HIV. Usually, HIV screening tests are used first. That's because HIV viral load tests are expensive. But viral load tests can find HIV sooner after an infection than screening tests. So, your health care provider may order this test if your risk of having HIV is very high.

WHY DO I NEED AN HIV VIRAL LOAD TEST?

A viral load test is needed to guide treatment decisions. If you have HIV, you'll probably start taking HIV medicines soon after your diagnosis. But first, you'll need a viral load test to find out how much virus is in your blood. This information helps your provider choose the right medicines for you. Your first test result will be compared with later test results to see if the medicines are working.

A viral load test is needed to see how well treatment is working and to monitor your HIV infection. The goal of HIV treatment is to reduce the amount of virus in your blood until there's too little to show up on a test. You'll need to have

regular viral load tests to see whether your viral load is dropping enough. HIV viral load tests are usually done:

Before you start taking medicine

About two to eight weeks after starting or changing HIV medicines

Every three to six months to monitor your infection when your treatment is working well

Your provider may order an HIV viral load test to diagnose HIV if:

You recently had a high-risk exposure to HIV. High-risk exposures include:

Having vaginal or anal sex with someone who has HIV or whose HIV status you don't know

Sharing needles, syringes, or other items used to inject drugs with other people

Exchanging sex for money or drugs

Having a sexually transmitted infection (STI),^Â such as syphilis

Having sex with anyone who has done anything listed above

Having vaginal or anal sex with someone who has HIV or whose HIV status you don't know

Sharing needles, syringes, or other items used to inject drugs with other people

Exchanging sex for money or drugs

Having a sexually transmitted infection (STI),^Â such as syphilis

Having sex with anyone who has done anything listed above

You had a recent possible exposure to HIV, and you have early symptoms of HIV infection. The early stage of HIV doesn't always cause symptoms. When it does, they may include:

Flu-like symptoms, such as fever, chills, and aches

Extreme fatigue

Swollen lymph nodes (in your neck, groin, or armpit)

Rash

Sores in your mouth

Muscle aches

Night sweats

Flu-like symptoms, such as fever, chills, and aches

Extreme fatigue

Swollen lymph nodes (in your neck, groin, or armpit)

Rash

Sores in your mouth

Muscle aches

Night sweats

If you think you were exposed to HIV, talk with your provider right away about getting tested. You may also be able to have emergency treatment to prevent HIV infection^Â within the first 3 days after a possible exposure.

WHAT HAPPENS DURING AN HIV VIRAL LOAD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an HIV viral load test. But if you are getting this test to find out if you are infected with HIV, you should talk with a counselor before or after your test so you can better understand the results and your treatment options.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If you have an HIV viral load test because you have HIV:

A negative or undetectable viral load means that you have so little HIV in your blood that the test can't find it. This means that your HIV medicines are working well to protect your immune system, and you are unlikely to spread HIV through sex. It does not mean you're cured. You must continue to take your medicines as prescribed to keep your HIV in control.

A low viral load means your medicine is stopping the virus from growing and your infection is unlikely to get worse.

A high viral load means the virus is growing and your treatment is not working well. The higher the viral load, the more risk you have for infections and diseases related to a weak immune system. It may also mean you have a higher risk for developing AIDS. If your results show a high viral load, your provider will probably change your medicines.

If you had an HIV viral load test to diagnose whether you have HIV:

A normal or negative result means that no HIV was found in your blood, and you are probably not infected with HIV.

A result that shows any amount of virus in your blood means you have an HIV infection. Your provider will likely order other tests, including a CD4 count, to see how much HIV has damaged your immune system.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN HIV VIRAL LOAD TEST?

If you are living with HIV, you will have regular viral load tests to monitor your infection and treatment. It's best to have the same type of test done at the same lab if possible. That's because labs have different ways of measuring HIV viral load. Your provider needs to compare your test results over time to see if your viral load is going up or down, and different tests from different labs can be difficult to compare.

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Homocysteine Test

<https://medlineplus.gov/lab-tests/homocysteine-test/>

HOMOCYSTEINE TEST

WHAT IS A HOMOCYSTEINE TEST?

A homocysteine test measures the amount of homocysteine in a sample of your blood. Homocysteine is an amino acid. Amino acids are molecules that your body uses to make proteins.

Normally, your homocysteine levels are low. That's because your body uses vitamin B12, vitamin B6, and folic acid (also called folate or vitamin B9) to quickly break down homocysteine and change it into other substances that your body needs. High levels of homocysteine in your blood may be a sign that this process isn't working right or that you're lacking certain B vitamins.

If homocysteine isn't broken down, it can build up in your arteries (blood vessels that carry oxygen-rich blood from your heart and lungs to your body). This can not only damage your arteries but also increase your risk of forming blood clots. An increased risk of forming blood clots may increase your risk for heart attack, stroke, and other heart diseases and blood vessel disorders.

Other names: total homocysteine, plasma total homocysteine

WHAT IS IT USED FOR?

A homocysteine test is most often used to measure your vitamin B6, B12, and folic acid levels. Because these three vitamins break down homocysteine, not having enough of any of them can cause homocysteine levels to increase. Not having enough B6, B12, and folic acid can also lead to serious health conditions such as:

Anemia (a condition where you don't have enough healthy red blood cells).

Nerve damage throughout your body.

Cognitive decline (the loss of memory, reasoning, and thinking skills over time).

A homocysteine test may also be used to:

Help diagnose homocystinuria. Homocystinuria is a rare, genetic disease that prevents your body from making an enzyme that helps to break down homocysteine. An enzyme is a protein that speeds up certain chemical reactions in your body. Symptoms usually show up in the first year of life, but they may not appear until childhood or later. Common symptoms include problems with eyesight, blood clots, and weak bones. In the U.S., most newborns have a routine screening test to check for homocystinuria.

Better understand your risk for heart attack or stroke if you already have an increased risk. Your health care provider may order a homocysteine test if you have been diagnosed with heart or blood vessel disease or if you have conditions that increase your risk for heart and blood vessel disease, such as high blood pressure, high cholesterol, and diabetes.

Routine homocysteine testing to screen for heart disease risk isn't recommended for everyone. That's because it's uncertain how much homocysteine levels affect heart and blood vessel diseases. Even though high levels of homocysteine can damage arteries, studies have shown that in most cases lowering homocysteine levels doesn't reduce the risk of heart attack or stroke.

WHY DO I NEED A HOMOCYSTEINE TEST?

You may need this test if you are at high risk for low levels of vitamin B12 or folic acid or if you have symptoms that suggest you lack either of these vitamins. The symptoms may be very mild to severe and may include:

Dizziness.

Fatigue and/or weakness.

Headache.

Heart palpitations (racing or pounding heart).

Changes in the color of your skin or fingernails.

Sores on your tongue or in your mouth.

Tingling or numbness in your hands, feet, arms, and/or legs.

You may be at high risk for low levels of vitamin B12 or folic acid if you:

Have malnutrition.

Are an older adult. Older people often can't absorb enough vitamin B12 from food.

Have alcohol use disorder or a drug addiction.

Your provider may also recommend this test if you have:

Had a heart attack or stroke.

One or more conditions that increase your risk for heart attack or stroke, such as high LDL "bad" cholesterol or high blood pressure.

Homocystinuria. Because this condition makes the breakdown of homocysteine difficult, you are more likely to have high levels of this amino acid.

WHAT HAPPENS DURING A HOMOCYSTEINE TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for 8-12 hours before a homocysteine test. Some medicines and supplements may affect your test results. So, tell your provider about all medicines and supplements you take, especially vitamin B. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A high homocysteine level may be a sign that:

You are not getting enough vitamin B12 or folic acid in your diet.

You (or your child) have homocystinuria. You will probably need more testing to rule out or confirm a diagnosis of homocystinuria.

You may have a higher risk of heart disease, stroke, or other blood vessel disorders.

Higher than normal homocysteine levels may also happen with other conditions, such as osteoporosis, chronic kidney disease, hypothyroidism, schizophrenia, or Alzheimer's disease or other types of dementia.

If your homocysteine levels are high, it doesn't always mean you have a medical condition that needs treatment. Your results may be affected by:

Your age. Homocysteine levels may get higher as you get older.

Your sex. Males usually have higher homocysteine levels than females, but levels in females increase after menopause.

Smoking.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A HOMOCYSTEINE BLOOD TEST?

If you have high homocysteine levels, your provider may suggest that you take vitamin supplements^Â or make changes in the foods you eat. Eating a balanced diet can help you get the right amount of vitamins.

Your provider may also recommend that you take certain vitamins and supplements (known as homocysteine-lowering therapy) if you have homocystinuria or signs of cognitive decline. Research shows that this may help slow down cognitive decline and reduce your risk of heart attack or stroke if you have homocystinuria.

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How to Cope with Medical Test Anxiety

<https://medlineplus.gov/lab-tests/how-to-cope-with-medical-test-anxiety/>

HOW TO COPE WITH MEDICAL TEST ANXIETY

WHAT IS MEDICAL TEST ANXIETY?

Medical test anxiety is a fear of medical tests. Medical tests are procedures that are used to diagnose, screen for, or monitor various diseases and conditions. They can also be used to check the health of organs and body systems. While many people sometimes feel nervous or uncomfortable about testing, it doesn't usually cause serious problems or symptoms.

Medical test anxiety can be serious. It can become a type of phobia. A phobia is an anxiety disorder that causes an intense, irrational fear of something that poses little or no actual danger. Phobias can also cause physical symptoms such as rapid heartbeat, shortness of breath, and trembling.

WHAT ARE THE DIFFERENT TYPES OF MEDICAL TESTS?

The most common types of medical tests are:

Tests of body fluids. Your body fluids include blood, urine, sweat, and saliva. Testing involves getting a sample of fluid.

Imaging tests. These tests look at the inside of your body. Imaging tests include x-rays, ultrasound, and magnetic resonance imaging (MRI). Another type of imaging test is endoscopy. Endoscopy uses a thin, lighted tube with a camera that is inserted into your body. It provides images of your internal organs and other systems.

Biopsy. This test takes a small sample of your tissue for testing. It is used to check for cancer and certain other conditions.

Measurement of body functions. These tests check the activity of different organs. Testing may include checking the electrical activity of the heart or brain or measuring the function of your lungs.

Genetic testing. These tests check cells from the skin, bone marrow, or other areas. They are most often used to diagnose genetic diseases or find out if you are at risk for getting a genetic disorder.

These tests and procedures can provide important information about your health. Most tests have little or no risk to you and can help your health care provider determine the best treatment. But people with medical test anxiety may be so afraid of testing that they avoid them altogether. And this can actually put your health at risk.

WHAT ARE THE TYPES OF MEDICAL TEST ANXIETY?

The most common types of medical anxieties (phobias) are:

WHAT ARE THE TYPES OF MEDICAL TEST ANXIETY?

The most common types of medical anxieties (phobias) are:

Trypanophobia, the fear of needles. Many people have some fear of needles, but people with trypanophobia have an excessive fear of injections or needles in a medical setting. This fear may stop them from getting the tests or treatment they need. It can be especially dangerous to people with chronic medical conditions that need frequent testing or treatment.

Iatrophobia, the fear of doctors and medical tests. People with iatrophobia may avoid seeing health care providers for routine care or when they have symptoms of illness. But some minor illnesses can turn serious or even deadly if left untreated.

Claustrophobia, the fear of enclosed spaces. Claustrophobia can affect people in many different ways and is only considered a phobia if it interferes with your work or other daily activities. You may experience claustrophobia if you are getting an MRI. During an MRI, you are placed inside an enclosed, tube-shaped scanning machine. The space in the

scanner is narrow and small.

HOW DO I COPE WITH MEDICAL TEST ANXIETY?

Talk to your provider about the test. Understanding why it was ordered, what to expect during the test, and what your provider hopes to learn from the results might help to reduce some of your anxiety before the test.

During the test, some relaxation techniques may reduce your medical test anxiety, including:

Deep breathing. Take three slow breaths. Count to three for each one, then repeat. Slow down if you start to feel lightheaded.

Counting. Count to 10, slowly and silently.

Imagery. Close your eyes and picture an image or a place that makes you happy.

Muscle relaxation. Concentrate on making your muscles feel relaxed and loose.

Talking. Chat with someone in the room. It may help distract you.

Tell your provider if you feel dizzy or sick to your stomach during the test.

If you have trypanophobia, iatrophobia, or claustrophobia, the following tips may help reduce your specific type of anxiety.

For trypanophobia, fear of needles:

If you don't have to limit or avoid fluids beforehand, drink lots of water the day before and the morning of your blood test. This puts more fluid in your veins and may make it easier to draw blood.

Avoid dry skin at the place where your blood will be drawn by moisturizing to decrease any discomfort.

Stay warm to increase your blood circulation, which may make it easier for your provider to find your vein.

Ask your provider if you can get a topical anesthetic to numb the skin.

If the sight of a needle bothers you, close your eyes or turn away during the test.

If you have diabetes and need regular insulin injections, you might be able to use a needle-free alternative, such as a jet injector. A jet injector delivers insulin using a high-pressure jet of mist instead of a needle.

For iatrophobia, the fear of doctors and medical tests:

Bring a friend or family member to your appointment for support.

Bring a book, magazine, or something else to distract you while you wait for your appointment.

For moderate or severe iatrophobia, you may want to consider seeking help from a mental health professional.

If you feel comfortable talking with your provider, ask about medicines that could help reduce your anxiety.

To avoid claustrophobia during an MRI:

Ask your provider for a mild sedative before the scan.

Ask your provider if you can get tested in an open MRI scanner instead of a traditional MRI. Open MRI scanners are larger and have an open side. It may make you feel less claustrophobic. The images produced may not be as good as those from a traditional MRI, but they still may be helpful in making a diagnosis.

Avoiding medical tests may be harmful to your health. If you suffer from any type of medical anxiety, talk to your provider or a mental health professional.

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How to Prepare for a Lab Test

<https://medlineplus.gov/lab-tests/how-to-prepare-for-a-lab-test/>

HOW TO PREPARE FOR A LAB TEST

HOW DO I PREPARE FOR A LAB TEST?

A laboratory (lab) test is a procedure in which a health care provider takes a sample of your blood, urine, other body fluid, or body tissue to get information about your health. Lab tests may be used to:

Help diagnose or screen for a specific disease or condition. ScreeningÂ helps diagnose diseases before you have any symptoms.

Monitor a disease or see if treatment is working.

Give you more general information about your organs and body systems. Lab tests may be ordered with your annual checkupÂ or preventative testing.

For any type of lab test, you should prepare for it by:

Following all the instructions given to you by your provider.

Telling your provider or lab professional if you didn't follow these instructions exactly. It's important to be honest. Even a minor change from the instructions can have a big effect on your results. For example, some medicines raise or lower blood glucose (blood sugar) levels. Taking the medicines too close to a blood glucose testÂ could change your results.

Telling your provider about any medicines, vitamins, or supplementsÂ that you are taking.

Taking these steps can help make sure that your results will be accurate and reliable.

WILL I NEED TO TAKE OTHER STEPS TO PREPARE FOR MY LAB TEST?

For many lab tests, you don't need to do anything other than answer questions from your provider and/or lab professional. But for others, you may need to do some specific preparations before the test.

One of the most common lab test preparations is fasting. Fasting means you should not eat or drink anything except water for up to several hours or overnight before your test. This is done because nutrients and ingredients in food are absorbed in your bloodstream. These nutrients might still be in your bloodstream when you have your test and may change the results of certain blood tests. Fasting can help to improve the accuracy of your test results. The length of time you need to fast can vary. So if you do need to fast, make sure you ask your provider how long you should do it.

Other common test preparations may include:

Avoiding specific foods and drinks such as cooked meats, herbal tea, or alcohol.

Making sure not to overeat the day before a test.

Not smoking.

Avoiding specific behaviors such as strenuous exercise or sexual activity.

Avoiding certain medicines and/or supplements. Be sure to talk to your provider about what you are currently taking, including over-the-counter medicines, vitamins, and supplements. But don't stop taking any medicines unless your provider tells you to.

For some blood tests, you may be asked to drink extra water to help keep more fluid in your veins. You may also be asked to drink water 15 to 20 minutes before certain urine tests.

WHAT TYPES OF LAB TESTS REQUIRE SPECIAL PREPARATION?

Some of the most common lab tests that require fasting include:

Blood Glucose Test

Cholesterol Levels Test

Triglycerides Test

Calcitonin Test

Some of the most common lab tests that require other special preparations include:

Creatinine Test, which may require fasting or avoiding cooked meats.

Cortisol Test. For this test, you may need to rest for a bit before your sample is taken. You may also have to avoid eating, drinking, or brushing your teeth for a certain period of time before your test.

Fecal Occult Blood Test. For this test, you may need to avoid certain foods or medicines.

5-HIAA Test. For this test, you may be asked to avoid various foods. These may include avocados, bananas, pineapples, walnuts, and eggplants.

Pap Smear. A woman may be instructed not to douche, use tampons, or have sex for 24 to 48 hours before this test.

IS THERE ANYTHING ELSE I SHOULD KNOW ABOUT PREPARING FOR A LAB TEST?

Talk to your provider if you have questions or concerns about test preparations. They will let you know if there are any special instructions to follow.

Be sure you understand your preparation instructions before the day of your test.

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How to Prepare Your Child for a Lab Test

<https://medlineplus.gov/lab-tests/how-to-prepare-your-child-for-a-lab-test/>

HOW TO PREPARE YOUR CHILD FOR A LAB TEST

WHY WOULD MY CHILD NEED A LAB TEST?

A laboratory (lab) test is a procedure in which a health care provider takes a sample of blood, urine, or other body fluid or tissue. The tests can provide important information about your child's health. They may be used to diagnose, screen for, or monitor diseases and conditions. Lab tests can also be used to check the health of organs and body systems. But lab tests can be scary, especially for children. Fortunately, children don't need to be tested as often as adults. But if your child does need testing, you can take steps to help them feel less scared and anxious. Preparing in advance may also help keep your child calm and less likely to resist the procedure.

HOW DO I PREPARE MY CHILD FOR A LAB TEST?

Here are some simple steps that may make your child feel more at ease before and during a lab test.

Explain what will happen. Tell your child why the test is needed and how the sample will be collected. Use language and terms based on your child's age. Assure your child you will be with them or nearby the whole time. Let them know that adults sometimes get nervous about having lab tests, too.

Be honest but reassuring. Don't tell your child the test won't hurt; it may actually be painful. Instead, say that the test may hurt or pinch a bit, but the pain will go away quickly.

Practice the test at home. Younger children can practice doing the test on a stuffed animal or doll.

Practice deep breathing and other comforting activities with your child. These may include thinking happy thoughts and counting slowly from one to ten.

Schedule the test at the right time. Try to schedule the test for a time when your child is less likely to be tired or hungry. If your child is getting a blood test, eating beforehand will lessen the chance of lightheadedness. But if your child needs a test that requires fasting (not eating or drinking), it's best to schedule the test for first thing in the morning. You should also bring a snack for afterward.

Offer plenty of water. If the test doesn't require limiting or avoiding fluids, encourage your child to drink lots of water the day before and the morning of the test. For a blood test, it can make it easier to draw blood because it puts more fluid in the veins. For a urine test, it can make it easier to urinate when the sample is needed.

Offer a distraction. Bring along a favorite toy, game, or book to help distract your child before and during the test.

Provide physical comfort. If the provider says it's OK, hold your child's hand or provide other physical contact during the test. If your baby needs a test, comfort them with gentle physical contact and use a calm, quiet voice. If you are allowed, hold your baby during the test. If not, stand where your baby can see your face.

Plan a reward for afterward. Offer your child a treat or make a plan to do something fun together after the test. Thinking about a reward may help distract your child and encourage cooperation during the procedure.

Specific preparations and tips will depend on your child's age and personality, as well as the type of test being done.

WHAT HAPPENS TO MY CHILD DURING A LAB TEST?

Common lab tests for children include blood tests, urine tests, swab tests, and throat cultures.

Blood tests are used to test for many different diseases and conditions.

During a blood test, a sample will be taken from a vein in the arm, a fingertip, or a heel.

If done on a vein, a health care professional will take a sample, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial.

A fingertip blood test is done by pricking your child's fingertip.

Heel stick tests are used for newborn screenings, a test given shortly after birth to almost every baby born in the United States. These screenings are used to help diagnose a variety of serious health conditions. During a heel stick test, a

provider will clean your baby's heel with alcohol and poke the heel with a small needle.

Encourage your child to look at you during a blood test rather than at the person drawing the blood. You should also provide physical comfort and distraction.

Urine tests are done to check for different diseases and for infections of the urinary tract. During a urine test, your child will need to provide a urine sample in a special cup. Unless your child has an infection or a rash, a urine test is not painful. But it can be stressful. The following tips may help:

Talk to your child's provider to find out if a "clean catch" method will be needed. For a clean catch urine sample, your child will need to give a urine sample for the test. A health care professional may give them a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect their urine sample. Explain to your child that it is important to follow these instructions so that germs from their skin, toilet paper, or hair don't get into the sample. Depending on your child's age, you may need to help them:

Wash their hands with soap and water and dry them.

Open the container without touching the inside.

Clean their genital area with a cleansing wipe.

For a penis, wipe the entire head (end) of the penis. If they have a foreskin, it will need to be pulled back first.

For a vagina, separate the labia (folds of skin around the vagina) and wipe the inner sides from front to back.

Ask them to urinate into the toilet for a few seconds and then stop the flow. They should start urinating again, this time into the container. Don't let the container touch their body.

Collect at least an ounce or two of urine into the container. The container should have markings to indicate how much urine is needed.

They should finish urinating into the toilet.

Put the cap on the container and return it as instructed.

Wash their hands with soap and water and dry them.

Open the container without touching the inside.

Clean their genital area with a cleansing wipe.

For a penis, wipe the entire head (end) of the penis. If they have a foreskin, it will need to be pulled back first.

For a vagina, separate the labia (folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If they have a foreskin, it will need to be pulled back first.

For a vagina, separate the labia (folds of skin around the vagina) and wipe the inner sides from front to back.

Ask them to urinate into the toilet for a few seconds and then stop the flow. They should start urinating again, this time into the container. Don't let the container touch their body.

Collect at least an ounce or two of urine into the container. The container should have markings to indicate how much urine is needed.

They should finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If a clean catch sample is needed, practice at home. Ask your child to let out a bit of urine in the toilet, stop the flow, and start again.

Encourage your child to drink water before the appointment, but not go to the bathroom. This may make it easier for your child to urinate when it's time to collect the sample.

Turn on the tap. The sound of running water can help your child begin to urinate.

Nasal swab tests help diagnose different types of respiratory infections. During a swab test, a provider will:

Gently insert a cotton-tipped swab inside your child's nostril. For some swab tests, a provider may need to insert the swab deeper until it reaches the uppermost part of the nose and throat (the nasopharynx).

Rotate the swab for 10-15 seconds.

Remove the swab and insert it into your child's other nostril.

Swab the second nostril using the same technique.

Remove the swab.

Swab tests may tickle the throat or cause your child to cough. A swab of the nasopharynx may be uncomfortable and cause a gag reflex when the swab touches the throat. Let your child know beforehand that gagging may happen, but it

will be over quickly. It may also help to tell your child that the swab is similar to the cotton swabs you have at home. Throat cultures are done to check for bacterial infections of the throat, including strep throat. During a throat culture: Your child will be asked to tilt their head back and open their mouth as wide as possible. Your child's provider will use a tongue depressor to hold down your child's tongue. The provider will use a special swab to take a sample from the back of your child's throat and tonsils. A throat swab isn't painful, but like some swab tests, it can cause gagging. Let your child know what to expect and that any discomfort shouldn't last very long. You may need to hold your child if they are younger to provide comfort and keep them from moving.

IS THERE ANYTHING ELSE I SHOULD KNOW ABOUT PREPARING MY CHILD FOR A LAB TEST?

If you have questions or concerns about testing or if your child has special needs, talk to your child's provider. You can work together to discuss the best way to prepare and comfort your child throughout the testing process. Learn more about laboratory tests, reference ranges, and understanding results.

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How to Understand Your Lab Results

<https://medlineplus.gov/lab-tests/how-to-understand-your-lab-results/>

HOW TO UNDERSTAND YOUR LAB RESULTS

WHAT IS A LABORATORY TEST?

A laboratory (lab) test checks a sample of your blood, urine (pee), or other body fluid or tissue to learn about your health. The sample is sent to a lab where health care professionals test it to see if it contains certain substances and, if so, how much.

Some lab tests can clearly show whether you do or don't have a specific health condition. For example, a pregnancy test can show whether or not you are pregnant.

Other lab tests provide more general information about your health and possible problems. A comprehensive metabolic panel (CMP), for example, can give your health care provider a good idea of your overall physical health. Any abnormal results in these kinds of general lab tests can give your provider clues about the type of health problems you may have.

Though lab tests are an important part of your health care, they don't provide a complete picture of your health. To get a full picture, your provider will use your results along with:

Information from your physical exam.

Your health history.

Your family health history.

Any imaging tests you may have had, such as x-rays.

In most cases, combining that information leads to a more accurate diagnosis than the results from any one lab test. So, even if you get your test results in your electronic health record (EHR) and can review them, it's still important to talk with your provider to find out what your results really mean for you.

WHY DO I NEED A LAB TEST?

Your provider may order a lab test to:

Learn more about a specific part of your health.

Get a general picture of your overall health.

If your provider orders a test to learn more about a specific part of your health, they can use the results to:

Diagnose or rule out a certain disease or condition, especially if you have symptoms. For example, if you're sick with a sore throat, swollen tonsils, and a fever, a strep A test can show you whether you're infected with the bacteria that causes strep throat.

Monitor a condition and/or disease. Test results can help your provider see if your condition is getting better, worse, or staying the same. They can also tell your provider if your treatment is working. For example, a blood glucose test is often used to monitor diabetes and diabetes treatment.

Determine the best treatment option for you. Certain tests can give your provider detailed information about your health condition. Your provider can then use the results of these tests to figure out which treatment will be effective. Tumor marker tests, for example, can help your provider decide how best to treat certain types of cancer.

Lab tests that provide a more general picture of your health are often used even if you don't have any signs or symptoms of a health condition. These tests are usually part of a routine checkup and can tell your provider if you:

Are at high risk for a specific disease. For example, a cholesterol test measures the amount of cholesterol in a sample of your blood. The results can help you understand your risk for heart disease.

Need additional testing. These tests look for signs that you may have a health problem so that you can have more testing to find out for sure. For example, one type of test for colon cancer looks for blood in your stool (poop). If blood is found, it could be a sign of cancer. To find out if it is cancer, your provider may order a colonoscopy.

WHAT IS A REFERENCE RANGE?

Lab test results are reported in different ways. Some results are a number. For example, your low-density lipoprotein (LDL) result for a cholesterol test may be 125 or 80. But how do you know what that number means?

To answer that question, your lab report tells you whether your test result is in a normal range called a reference range or "normal values." A reference range is a set of numbers that form the high and low ends of the range of results that's considered to be normal. The ranges are based on the test results from large groups of healthy people. A test may have different reference ranges for different groups. For example, there may be separate ranges for children and adults.

Because these ranges can vary, the term "normal" can be misleading. Reference ranges show the average results in a healthy test group. So:

A test result that is higher or lower than the range that applies to you may or may not be a sign of a health problem. It's common for healthy people to sometimes have results outside the reference range. If your results are outside of the range, your provider will look at other information about your health to understand what may have affected your results.

A test result that is within a reference range isn't always a guarantee of good health. Some people with health problems can have results that are considered "normal." Therefore, even if you have a test result within the normal range, you may need additional testing, especially if you have symptoms of a health condition.

When looking at your lab test results, it's helpful to consider two important facts. First, because labs often use different testing methods, they often use different reference ranges to describe normal results. This means that:

To find out how your test results compare with the normal range, you need to check the reference range listed on your lab report. You can't compare your results to reference ranges that you might find elsewhere. This is why our Medical Test information cannot provide normal reference ranges for most tests.

You can't compare test results from different labs.

If you're looking for trends in your test results over time, it's important to try to use the same lab for testing.

Second, test results are measured with different units of measurement. Just as you might measure ingredients for a recipe in teaspoons, cups, or ounces, different tests use different units of measurement. Examples of measurement you may see include:

Mcg/dL = micrograms per deciliter

Micromole/L = micromoles per liter

Pg/mL = picograms per milliliter

WHAT DO NEGATIVE, POSITIVE, AND INCONCLUSIVE RESULTS MEAN?

Some test results tell you whether a certain substance, germ, type of cell, or gene was or wasn't found in your test sample. On your test report, you may see these terms:

Negative or normal. This means "No, the test didn't find what it was looking for." So, you're unlikely to have the health problem you were tested for, but you may need more tests.

Positive or abnormal. This means, "Yes, the test found what it was looking for." The germ, substance, or gene being tested was in your sample. So, you may have a disease or infection. In certain cases, you may need more tests to confirm a diagnosis.

Inconclusive or uncertain. This means "not sure." Your test wasn't clearly positive or negative. There are many reasons why this may happen. If you get an uncertain result, you will probably be tested again.

Tests for the COVID-19 virus are an example of tests that tell you whether or not a specific germ was found in your sample.

WHAT ARE FALSE POSITIVE AND FALSE NEGATIVE RESULTS?

Tests results are usually accurate, but no test is perfect.

A false positive result means your test shows you have a disease or condition, but you don't really have it.

A false negative result means your test shows you don't have a disease or condition, but you really do.

These incorrect results don't happen often, but they are more likely with certain types of tests, or if testing wasn't done right. If your provider thinks your test result may be inaccurate, you may need to have another test.

WHAT CAN AFFECT MY RESULTS?

The accuracy of certain test results may be affected by various common factors. These can include:

Eating and drinking certain foods and drinks.

Taking certain medicines or supplements.

Exercising hard before your test.

Having a menstrual period at the time of your test.

Whether or not you followed preparation instructions, including fasting.

Your provider will let you know if you need to prepare for your test. Follow your provider's instructions carefully. That will help make sure your test results are as accurate as possible. Before your test, let your provider know about all the prescription and over-the-counter medicines you take as well as vitamins and other supplements.

WHAT IF I DO A HOME TEST?

At-home test kits are available for many types of lab tests. The kits provide everything you need to collect a sample of body fluid or cells to send to a lab. At-home tests should never replace testing that your provider orders. Ask your provider or pharmacist to recommend a test you can trust. And talk with your provider about your results, even if they're normal.

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Human Papillomavirus (HPV) Test

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HUMAN PAPILLOMAVIRUS (HPV) TEST

WHAT IS AN HPV TEST?

Human papillomavirus (HPV) is a group of more than 200 related viruses that can generally be grouped into two types. These are called low-risk HPV and high-risk HPV. Low-risk HPV can cause warts on or around your genitals, anus, mouth, or throat. High-risk HPV can cause:

Cervical cancer.

Anal cancer.

Oropharyngeal cancer, a type of head and neck cancer, also called throat cancer.

Vulvar cancer.

Vaginal cancer.

Penile cancer.

Most HPV types are harmless. However, it's important for your health care provider to check for high-risk HPV infections. They do this by looking for DNA left by the virus in your vagina or cervix (the lower part of the uterus). These infections can stay at the cervix for many years, leading to cell changes called precancer. If these changes are not screened for and treated, they may over time become cancer.

Nearly all cervical cancers are caused by high-risk HPV infections. Because of this, it is important to get routine testing for these viruses. If you do have an infection, your provider can decide how best to manage it. They may recommend that you have other tests and be tested more regularly to check for precancerous cells and cervical cancer.

Other names: genital human papillomavirus, high risk HPV, HPV DNA, HPV RNA

WHAT IS IT USED FOR?

An HPV test screens for the infection that causes cervical precancer or cancer by examining a sample of cervical cells. The test is often done at the same time as a Pap test (sometimes called a Pap smear). A Pap test looks for abnormal changes in cervical cells before they become cancer. When an HPV test and a Pap test are done at the same time, it's called cotesting.

Because HPV causes cancer at other places in the body, an HPV test can be useful for other treatment decisions. This is especially the case in treatment for oropharyngeal (or throat) cancer. This cancer grows in the back of the throat, including the tonsils and the base of the tongue. In the United States, HPV causes most cases of oropharyngeal cancer. If you have an HPV test because you have oropharyngeal cancer, the test will use a sample of throat tissue collected during a biopsy.

WHY DO I NEED AN HPV TEST?

You may need an HPV test to screen for cervical cancer if you:

Had an abnormal result on a Pap test. An HPV test can show whether HPV caused the abnormal changes in your cervical cells.

Are age 30 through 65. If you had an HPV test or an HPV/Pap cotest and had normal results, your provider may say that you can wait five years until your next test.

If you are between the ages of 21 and 29 and your last Pap test result was normal:

You do not need additional HPV testing.

Your provider may recommend that you wait three years until your next Pap test.

Most people have been exposed to HPV, usually through sexual contact. HPV is especially common in the 21-29 age group. However, these infections usually clear up on their own within a year or two without causing any health conditions. Some medical experts recommend starting HPV testing at age 25 and getting tested every 5 years if your

last test was normal.

Cervical cancer is most common in people over age 30. If you have a high-risk HPV infection in your cervix, you're more likely to develop cervical cancer if you:

Had an abnormal HPV test, Pap test, or cervical biopsy in the recent past.

Have a weakened immune system because you:

Have a disease that harms your immune system, such as HIV.

Take medicine to control your immune system, such as certain medicines to treat cancer or autoimmune diseases.

Have a disease that harms your immune system, such as HIV.

Take medicine to control your immune system, such as certain medicines to treat cancer or autoimmune diseases.

Have had cervical cancer or precancer.

Smoke tobacco or breathe secondhand smoke.

Use birth control pills or have given birth to many children.

Were exposed to a drug called DES (Diethylstilbestrol) before you were born. Between 1940-1971, DES was sometimes prescribed to pregnant women to prevent miscarriages. It was later linked to an increased risk of certain cancers in the female children exposed to it before birth.

Ask your provider which cervical cancer screening test is right for you and how often you should be tested.

WHAT HAPPENS DURING AN HPV TEST?

There are two ways to collect the sample needed for an HPV test. Your provider can collect the sample, or you can collect it yourself.

If your provider collects the sample:

You will be asked to undress completely or from the waist down.

You will lie on an exam table with your knees bent and your heels in footrests called stirrups.

Your provider will gently insert a plastic or metal instrument called a speculum into your vagina. This helps to widen your vagina so that your cervix can be seen. This may cause a feeling of cramping or pressure.

Your provider will then use a small, soft brush or swab to collect cells from the cervix. This doesn't usually hurt.

The cell sample is sent to a lab for examination under a microscope.

If you collect the sample yourself:

Your provider will give you a tube with the testing tools and explain the steps you will take.

You will enter a private space in a health care setting.

Open the tube and remove the swab or brush. Be sure not to touch the end that you will insert into your vagina.

Insert the swab or brush into your vagina up to a point indicated on the handle. Once it is inserted, rotate the swab or brush around so that it can pick up cells from the walls of your vagina.

Remove the swab or brush and put it back into the tube.

Return your sample to your provider so that it can be sent to a lab for testing.

HPV testing is often done as part of a routine pelvic exam. During a pelvic exam, your provider examines your uterus, ovaries, vulva, and vagina. But a pelvic exam doesn't always include an HPV test. So, when you have a pelvic exam, ask your provider whether you'll have an HPV test to screen for cervical cancer, too.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you're having an HPV test:

You don't need to do anything to prepare for an HPV test.

If you are having an HPV/Pap cotest:

Avoid the following during the 2-day window before your HPV/Pap cotest

Having vaginal sex

Douching (rinsing the vagina with water or other fluid)

Using vaginal medicines or spermicidal foam, jelly, or cream

If you did have sex before the HPV/Pap cotest, or are having your period, let your provider know. They should still be able to carry out both the HPV and Pap tests.

ARE THERE ANY RISKS TO THE TEST?

You may feel some mild discomfort when your provider collects cells from your cervix. Afterward, you may have some very light bleeding.

WHAT DO THE RESULTS MEAN?

Negative/Normal HPV test results mean that high-risk HPV was not found in your sample. Your provider will let you know when you should be tested again.

Positive/Abnormal HPV test results mean that DNA from a high-risk HPV were found in your sample.

A positive test result does not mean you have precancer or cancer or that you will get cancer. It does mean that you have an HPV infection, which increases your risk for getting cervical precancer or cancer in the future.

There's no treatment for an HPV infection. So, the next steps will depend on your risk for developing cervical precancer or cancer. To estimate your risk, your provider will consider past test results, your age, and other health conditions you have or have had.

Your provider may suggest a colposcopy^Â to check your cervical cells in greater detail. To do this procedure, your provider will use a lighted, magnifying device called a colposcope. They will place the device at the opening of your vagina. The device magnifies the normal view, allowing your provider to see problems that can't be seen by the eyes alone. You may also need more frequent HPV tests or HPV/Pap cotests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN HPV TEST?

There is a vaccine to prevent infection with the most high-risk types of HPV. The vaccine can also protect you against low-risk types that cause genital warts. The vaccine is safe and effective. It prevents cancers caused by HPV everywhere in the body, including in the cervix, throat, anus, vagina, vulva, and penis.

The U.S. Centers for Disease Control and Prevention (CDC) recommends that 11 to 12-year-olds have two shots of HPV vaccine 6 to 12 months apart. But the vaccine can be given as early as age 9. Ask your child's provider about when to vaccinate your child against HPV.

Teens and young adults who get their first HPV shot between ages 15 and 26 need to have 3 doses. If you're over 26 and haven't been vaccinated, talk with your provider about the possible benefits of vaccination.

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Hysteroscopy

<https://medlineplus.gov/lab-tests/hysteroscopy/>

HYSTEROSCOPY

WHAT IS A HYSTEROSCOPY?

A hysteroscopy is a procedure that allows a health care provider to look at the inside of your uterus and cervix. The cervix is the lower part of the uterus (the uterus is the place where a baby grows during pregnancy).

The procedure uses a thin tube called a hysteroscope, which is inserted through your vagina. The hysteroscope is like a telescope with a camera on it. The camera sends images of your uterus onto a video screen.

The procedure may be done to diagnose or treat different uterine diseases, and conditions:

It's called a diagnostic hysteroscopy when it's done to check for the cause of abnormal uterine bleeding.Â

It's called an operative hysteroscopy when it's done to remove polyps, fibroids, or adhesions.Â

Sometimes, you may have both types done at once.

Other names: hysteroscopic surgery, diagnostic hysteroscopy, operative hysteroscopy

WHAT IS IT USED FOR?

A hysteroscopy is most often used to:

Diagnose the cause of abnormal uterine bleeding

Help find the cause of infertility, the inability to get pregnant after at least a year of trying

Find the cause of repeated miscarriages (more than two miscarriages in a row)

Find and remove fibroids and polyps. These are types of abnormal growths in the uterus. They are usually not cancerous.

Remove scar tissue from the uterus.Â These may be called adhesions and may have formed because of an infection or a past surgery.

Remove an intrauterine device (IUD), a small, plastic device placed inside the uterus to prevent pregnancy

Perform a biopsy. A biopsy is a procedure that removes a small sample of tissue for testing.

WHY DO I NEED A HYSTEROSCOPY?

You may need this test if:

You are having heavier than normal menstrual periodsÂ and/or bleeding between periods.

You are bleeding after menopause.

You are having trouble getting or staying pregnant.

You want to remove an IUD.

WHAT HAPPENS DURING A HYSTEROSCOPY?

A hysteroscopy is often done in a hospital or outpatient surgery center. The procedure usually includes the following steps:

You may be asked to empty your bladder to prepare for the procedure.

You will remove your clothing and put on a hospital gown.

You will lie on your back on an exam table with your feet in stirrups.

An intravenous (IV) line may be put in your arm or hand.

You may be given a sedative, a medicine that helps you relax and blocks the pain. In some cases, you may need general anesthesia, which makes you unconscious during the procedure. A specially trained doctor called an

anesthesiologist will give you this medicine.

Your vaginal area will be cleaned with a special soap.

Your provider will insert a tool called a speculum into your vagina. It is used to spread open your vaginal walls.

Your provider may dilate (expand) your cervix.

Your provider will then insert the hysteroscope into the vagina and move it through your cervix and into your uterus.

Your provider may inject a liquid or gas through the hysteroscope and into your uterus. This helps expand the uterus so your provider can view your uterus, uterine lining, and fallopian tubes.

Your provider will be able to see images of the uterus on a video screen.

Your provider may take a sample of tissue for testing (biopsy).

If you are having a uterine growth removed or another uterine treatment, your provider will insert tools through the hysteroscope to perform the treatment.

A hysteroscopy may take 15 minutes to an hour, depending on what is done during the procedure. The medicines you were given may make you drowsy for a while. You should arrange for someone to take you home after the procedure.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know if there are any special instructions. If you will be getting general anesthesia, you may need to fast (not eat or drink) for 6-12 hours before the procedure. Do not use a douche, tampons, or vaginal medicines for 24 hours before the test.

It's best to schedule your hysteroscopy when you are not having your menstrual period. If you get your period unexpectedly, tell your provider. You may need to reschedule.

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

Also, tell your provider if you are pregnant or think you might be. You should not have a hysteroscopy if you are pregnant. The procedure may be harmful to a developing baby. You also shouldn't have a hysteroscopy if you have a pelvic infection.

ARE THERE ANY RISKS TO THE TEST?

A hysteroscopy is a very safe procedure. You may have mild cramping and a little bloody discharge for a few days after the procedure. Serious complications are rare, but they may include heavy bleeding, infection, and tears in the uterus. Call your provider if you have a fever, severe abdominal (belly) pain, or heavy vaginal bleeding.

After your hysteroscopy, your provider may recommend that you not have sex, use tampons, or douche for two weeks after the procedure. They may also tell you to avoid baths, swimming, or hot tubs during this time.

WHAT DO THE RESULTS MEAN?

If your results were not normal, it may mean one of the following conditions:

Fibroids, polyps, or other abnormal growths were found. Your provider may be able to remove these growths during the procedure. They may also take a biopsy, a tissue sample of the growths, for further testing.

Scar tissue was found in the uterus. This tissue may be removed during the procedure.

The size or shape of the uterus did not look normal.

Openings on one or both fallopian tubes are closed.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A HYSTEROSCOPY?

A hysteroscopy is not recommended if you are pregnant or have cervical cancer or pelvic inflammatory disease.

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IGF-1 (Insulin-like Growth Factor 1) Test

<https://medlineplus.gov/lab-tests/igf-1-insulin-like-growth-factor-1-test/>

IGF-1 (INSULIN-LIKE GROWTH FACTOR 1) TEST

WHAT IS AN IGF-1 TEST?

This test measures the amount of IGF-1 (insulin-like growth factor 1) in your blood. IGF-1 is a hormone, a chemical messenger in your bloodstream that controls the actions of certain cells or organs. IGF-1 manages the effects of growth hormone (GH) in your body. Together, IGF-1 and GH promote normal growth of bones and tissues.

GH levels in the blood change throughout the day, depending on your diet and activity levels. But IGF-1 levels are more stable. So, a reliable way to track GH in your body is by measuring the level of IGF-I in the blood.

Levels of IGF-1 and GH change throughout your life. They:

- Are normally low when you are a child

- Increase until they peak during puberty

- Decrease once you are an adult

Besides growth, GH also helps control your metabolism (the process your body uses to make energy from the food you eat). Some conditions, such as pituitary tumors, can cause your body to make too much or too little GH.

Other names: somatomedin C test

WHAT IS IT USED FOR?

An IGF-1 test is used to detect and monitor growth hormone disorders, including:

GH deficiency. In children, GH is essential for normal growth and development. A GH deficiency can cause a child to grow more slowly and be much shorter than children of the same age. In adults, GH deficiency can lead to low bone density and reduced muscle mass.

GH insensitivity, also known as Laron syndrome. This is a rare genetic disorder in which your body is unable to use the growth hormone it produces. It causes a slowed growth rate in children and shorter than normal height.

Gigantism. This rare childhood disorder causes your body to produce too much growth hormone. Children with gigantism are very tall for their age and have large hands and feet.

Acromegaly. This condition affects adults and causes your body to produce too much growth hormone. Adults with acromegaly have thicker than normal bones and enlarged hands, feet, and facial features.

WHY DO I NEED AN IGF-1 TEST?

Your health care provider may order an IGF-1 test if you or your child has symptoms of a GH disorder or to monitor treatment for a GH disorder.

Symptoms of GH deficiency or GH insensitivity in children include:

- Slowed growth rate compared with children of the same age

- Shorter height, arms, and legs, and lower weight than children of the same age

- Small penis in males

- Thin hair

- Poor nail growth

Adults with GH deficiency may have symptoms such as fatigue and decreased bone density and muscle mass. But IGF-1 testing isn't common for adults, as other disorders are much more likely to cause these symptoms.

Symptoms of GH excess (gigantism) in children include:

- Excessive growth compared with children of the same age

- Overly large head

- Larger than normal hands and feet

- Mild to moderate obesity

Symptoms of GH excess (acromegaly) in adults include:

Deep, husky voice

Larger than normal facial features such as lips, nose, and tongue

Excessive sweating and body odor

Thickening of bones

Coarse, oily skin

Irregular menstrual cycles in women

Erectile dysfunction in men

WHAT HAPPENS DURING AN IGF-1 TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You don't need any special preparations for an IGF-1 test.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to you or your child in having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your child's results show lower than normal levels of IGF-1, it probably means they have a GH deficiency or insensitivity to GH. In a child, this may be caused by a genetic disorder or brain disease. Your child may benefit from treatment with GH supplementation. GH supplementation is an injected medicine that contains manufactured human growth hormone. When GH deficiency is diagnosed and treated early, some children can grow several inches in the first year of treatment. Others grow less and more slowly but still benefit from treatment.

If your results show lower than normal IGF-1, it may be due to a normal age-related decrease in the hormone or other conditions. Your provider may order more tests to help make a diagnosis.

Higher than normal levels of IGF-1 may mean gigantism in children or acromegaly in adults. Gigantism and acromegaly are most often caused by a tumor in the pituitary gland, a small organ in the base of the brain that controls many functions, including growth. Treatment for the tumor may include radiation therapy, surgery, and/or medicines. But if the cause is not a tumor, you or your child may need more tests to identify the cause.

Learn more about laboratory tests, reference ranges, understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN IGF-1 TEST?

Your provider may order growth hormone tests to help diagnose a GH disorder. These include:

A GH stimulation test, which helps diagnose a GH deficiency or insensitivity.

A GH suppression test, which helps diagnose a GH excess (too much GH).

IGBP-3 test. IGBP-3 is a protein that is the main carrier of IGF-1. This test can help diagnose a GH deficiency, GH insensitivity, or GH excess.

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Immunoglobulins Blood Test

<https://medlineplus.gov/lab-tests/immunoglobulins-blood-test/>

IMMUNOGLOBULINS BLOOD TEST

WHAT IS AN IMMUNOGLOBULINS BLOOD TEST?

An immunoglobulins blood test measures the amount of immunoglobulins in your blood. Immunoglobulins are also called antibodies. Antibodies are proteins that your immune system makes to fight germs, such as viruses and bacteria. Your body makes different kinds of antibodies to fight germs in different parts of the body and at different stages of an infection.

An immunoglobulins test usually measures the three main types of immunoglobulin (Ig) antibodies that your body makes.

IgM antibodies are the first immunoglobulins your body makes after you're exposed to a germ. They provide short-term protection while your body makes other antibodies. They can be found in your blood and lymph fluid (the substance that carries disease-fighting cells throughout your body).

IgG antibodies are much more common and much more specialized. Your body creates these antibodies as a response to a specific germ and even remembers them after you've cleared the infection. That way, if you're exposed to the same germs again, your immune system can quickly make more antibodies. Though present in all your bodily fluids, they are most commonly found in your blood.

IgA antibodies protect your respiratory tract (the organs you use to breathe) and your digestive system (the organs you use to eat and digest food) from infections. You have IgA antibodies in your blood, saliva, tears, breast milk, and respiratory and gastric fluids.

Other names: quantitative immunoglobulins, total immunoglobulins, IgG, IgM, IgA testing

WHAT IS IT USED FOR?

An immunoglobulins blood test can help your health care provider diagnose different types of health conditions that may affect your immune system. A variety of conditions can cause abnormal levels of IgM, IgG, and/or IgA, such as:

Autoimmune disorders. With these disorders your immune system attacks your own healthy cells by mistake. Examples of autoimmune disorders include rheumatoid arthritis and lupus.

Certain types of cancer that affect your bone marrow, blood, or immune system.

Chronic (long-term) infections.

Genetic diseases you're born with (uncommon).

Infections a baby may be born with, such as syphilis or toxoplasmosis.

WHY DO I NEED AN IMMUNOGLOBULINS BLOOD TEST?

You may need this test if you have symptoms of low immunoglobulin production. One of the main symptoms of low immunoglobulin levels is frequent infections, especially repeated infections from the same germ. Other symptoms often include:

Sinus, throat, and ear infections, especially strep throat.

Respiratory infections, such as pneumonia and bronchitis.

Diarrhea and/or gastrointestinal infections.

Serious infections from germs that don't usually cause problems in healthy people, such as:

Cytomegalovirus (a virus related to chickenpox).

A yeast infection in the mouth, eyes, or digestive tract (thrush).

Cytomegalovirus (a virus related to chickenpox).

A yeast infection in the mouth, eyes, or digestive tract (thrush).

You may also need this test if immunodeficiency runs in your family, or your provider thinks you may have a problem making normal levels of immunoglobulins. Low immunoglobulin levels may be a sign of an immunodeficiency (a condition where your body's immune system can't effectively respond to an infection).

Lastly, you may need this test if your provider thinks your immunoglobulin levels are too high. Some autoimmune diseases and cancers can cause a high level of immunoglobulins. However, those immunoglobulins don't work normally. So, even though your levels are high, you may still get frequent infections.

WHAT HAPPENS DURING AN IMMUNOGLOBULINS BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an immunoglobulins blood test. If you or your child is taking any medicines, be sure to tell your child's or your provider before the test. Some medicines can affect the accuracy of the results. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your provider will compare your levels of IgM, IgG, and IgA. Your results can mean different things, depending on your symptoms, any conditions you may have, and which immunoglobulins are high or low. An immunoglobulins blood test alone cannot diagnose any conditions. So if your results aren't normal, you'll probably need more testing to find out what's affecting your immune system.

Some possible causes of low levels of one or more immunoglobulins are:

Conditions that may reduce the amount of protein in your body, including:

Kidney disease

Serious burns

Certain malabsorption disorders

Malnutrition

Kidney disease

Serious burns

Certain malabsorption disorders

Malnutrition

Conditions that affect your ability to make immunoglobulins, including:

Complications from diabetes

Kidney failure

Complications from diabetes

Kidney failure

A genetic disease that you were born with, such as common variable immunodeficiency disorder (CVID)

Some possible causes of high levels of one or more immunoglobulins are:

An autoimmune disease

Hepatitis

Cirrhosis

A chronic infection

Certain cancers. These cancers often cause a very high level of one type of immunoglobulin and low levels for the other types:

Multiple myeloma

Chronic lymphocytic leukemia (CLL)

Lymphoma

Waldenström macroglobulinemia

Multiple myeloma

Chronic lymphocytic leukemia (CLL)

Lymphoma

Waldenström macroglobulinemia

If your immunoglobulin levels aren't normal, it doesn't always mean you have a condition that needs treatment. Certain medicines can affect your results. If you have questions about your results, talk with your provider. The earlier you start treating abnormal immunoglobulin levels, the more likely you are to treat its underlying cause.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN IMMUNOGLOBULINS BLOOD TEST?

Depending on your results, your provider may order additional tests. These may include:

A lumbar puncture. If your symptoms show that an immune condition may be affecting your spine or brain, your provider may order this test to check the immunoglobulin levels in your cerebrospinal fluid (CSF).

A saliva test. Your provider may order this test to check the IgA levels in your saliva if your symptoms include frequent colds, sinus infections, or diarrhea.

A Serum (blood) and/or urine protein electrophoresis test. This test helps providers check for and monitor conditions associated with abnormal immunoglobulin production.

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Insulin in Blood

<https://medlineplus.gov/lab-tests/insulin-in-blood/>

INSULIN IN BLOOD

WHAT IS AN INSULIN IN BLOOD TEST?

An insulin in blood test measures the amount of insulin in a sample of your blood. Insulin is a hormone that your pancreas makes. It helps move blood glucose (blood sugar) from your bloodstream into your cells where it's used for energy. Glucose comes from many foods you eat. It's your body's main source of energy.

Normally, insulin and blood glucose levels rise and fall together:

Blood glucose levels increase after you eat.

When blood glucose rises, your pancreas releases more insulin into your blood.

The insulin lets glucose get into your cells, which lowers your blood glucose level.

When your blood glucose level returns to a range that's normal for you, your insulin levels decrease, too.

Serious problems can develop if your pancreas doesn't make the right amount of insulin at the right time. For example:

If your pancreas makes too much insulin, you may have too little glucose in your blood. This is called hypoglycemia. If blood glucose levels drop to a very low level (severe hypoglycemia), your brain cells may not get enough glucose to work properly. This is a serious condition that needs medical treatment right away.

If your pancreas makes too little insulin, glucose can't get into your cells from your blood. The glucose builds up in your bloodstream until your blood glucose level is too high. This is called hyperglycemia. If your blood glucose levels stay high over time, you could develop type 2 diabetes, which can cause serious complications in your eyes, heart, and other parts of your body.

If you have signs or symptoms of a health condition related to insulin, an insulin in blood test can help check how much insulin your pancreas is making.

Other names: fasting insulin, insulin serum, total and free insulin

WHAT IS IT USED FOR?

An insulin in blood test may be used with other tests to help:

Find out the cause of hypoglycemia (low blood glucose). This is the main reason for doing insulin testing.

Diagnose insulin resistance. With insulin resistance, your cells don't respond well to insulin and can't easily take in glucose from your blood. Your pancreas may make more insulin to help glucose get into your cells. This can keep your blood glucose in a healthy range for a while. But over time, your pancreas may wear out and stop making enough insulin. This can lead to prediabetes, which means your blood glucose levels are higher-than-normal, but not high enough to be diabetes. If blood glucose levels keep increasing, you can develop type 2 diabetes.

Guide treatment decisions for type 2 diabetes. An insulin test may be used to decide if a person with type 2 diabetes needs to take insulin as part of their diabetes treatment.

To monitor how well pancreatic islet cell transplantation surgery is working. This surgery is used in people who have type 1 diabetes. Type 1 diabetes is an autoimmune disease that destroys the special islet cells in the pancreas that make insulin.

WHY DO I NEED AN INSULIN IN BLOOD TEST?

You may need an insulin in blood test if you:

Had a blood glucose test that showed you have low blood glucose levels.

Have symptoms of hypoglycemia (low blood glucose). Symptoms of mild to moderate hypoglycemia include:

Sweating

Feeling shaky or jittery

Arrhythmia (a problem with the rate or rhythm of your heartbeat)
Confusion
Dizziness
Headache
Hunger

Severe hypoglycemia can cause fainting and seizures. This serious condition needs medical treatment right way.

Sweating
Feeling shaky or jittery
Arrhythmia (a problem with the rate or rhythm of your heartbeat)
Confusion
Dizziness
Headache
Hunger

Severe hypoglycemia can cause fainting and seizures. This serious condition needs medical treatment right way.

Have insulin resistance or have a high risk for developing it. Your risk for insulin resistance is higher if you have:

Prediabetes
Type 2 diabetes
Polycystic ovary syndrome (PCOS)
Heart disease
Acanthosis nigricans (dark, thick, velvety skin around the neck or armpits)
Prediabetes
Type 2 diabetes
Polycystic ovary syndrome (PCOS)
Heart disease
Acanthosis nigricans (dark, thick, velvety skin around the neck or armpits)
Had surgery on your pancreas:

To transplant the part of the pancreas that makes insulin (islet cell transplantation). You may need insulin testing to see if the transplant is working properly.

To remove an insulinoma. This is a tumor in the pancreas that makes too much insulin and causes low blood glucose. After surgery you may need insulin testing to check whether any tumor tissue is left and to see if the tumor has come back. Insulinomas are uncommon and usually aren't cancer.

To transplant the part of the pancreas that makes insulin (islet cell transplantation). You may need insulin testing to see if the transplant is working properly.

To remove an insulinoma. This is a tumor in the pancreas that makes too much insulin and causes low blood glucose. After surgery you may need insulin testing to check whether any tumor tissue is left and to see if the tumor has come back. Insulinomas are uncommon and usually aren't cancer.

WHAT HAPPENS DURING AN INSULIN IN BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your health care provider will tell you how to prepare for an insulin in blood test. You will probably need to fast (not eat or drink) for 8 to 12 hours before the test. If you take biotin supplements (vitamin B7), or supplements that include biotin, you'll need to stop taking them for at least a day before your test. But don't stop taking any medicines without talking with your provider first.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand what your insulin test results mean, your provider will consider your medical history and results of other tests, including a blood glucose test. For example:

If your insulin level is high and your blood glucose is normal or a little above normal for you, you may have insulin resistance.

If your insulin level is high or normal and your blood glucose is low for you, you may have hypoglycemia from too much insulin. Possible causes include:

A pancreatic tumor (insulinoma)

Cushing's syndrome

Taking too much insulin for diabetes

A pancreatic tumor (insulinoma)

Cushing's syndrome

Taking too much insulin for diabetes

If your insulin level is low and your blood glucose is high for you, it may mean that your pancreas can't make enough insulin. Possible causes include type 1 diabetes and pancreatitis.

There are other possible causes of abnormal levels of insulin and glucose. Ask your provider to explain what your test results say about your health.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I SHOULD KNOW ABOUT AN INSULIN IN BLOOD TEST?

An insulin in blood test is often done with a C-peptide test. Your pancreas releases equal amounts of insulin and C-peptide into your bloodstream at the same time. C-peptide doesn't affect your blood glucose levels, but it stays in your blood longer than insulin. So, measuring C-peptide provides a more accurate way to find out how much insulin your pancreas is making.

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Diabetes Type 1
Diabetes Type 2
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Hypoglycemia
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How to Understand Your Lab Results

Ketones in Blood

What You Need to Know About Blood Testing

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Intravenous Pyelogram (IVP)

<https://medlineplus.gov/lab-tests/intravenous-pyelogram-ivp/>

INTRAVENOUS PYELOGRAM (IVP)

WHAT IS AN INTRAVENOUS PYELOGRAM (IVP)?

An intravenous pyelogram (IVP) uses x-rays to take pictures of the organs of your urinary tract. These are the organs that make urine (pee) and remove it from your body. They include your:

Kidneys, two organs located below the rib cage, one on each side of your spine. They filter your blood to remove waste and extra water in urine (pee).

Bladder, a hollow organ in your pelvis (the area below your belly). It stores urine.

Ureters, thin tubes that carry urine from your kidneys to your bladder.

IVP uses contrast dye to make these organs show up well on x-ray images. In males, x-rays from an IVP can also show the prostate. The prostate is a gland in the male reproductive system. It lies just below the bladder and makes the fluid part of semen.

During an IVP, contrast dye is injected into a vein using an intravenous (IV) line. The dye travels through your bloodstream to your kidneys, which filter it out. The dye collects in the organs of your urinary tract and makes them look bright white on the x-rays. This allows your health care provider to see blockages or other problems that affect your urinary organs.

IVP isn't commonly used anymore. Instead, providers usually order CT (computerized tomography) scans to see the urinary tract. CT scans can provide more information than x-rays from an IVP. But IVP may be useful in certain situations.

Your provider can explain the pros and cons of having an IVP or a CT scan. In certain cases, MRI scans (magnetic resonance imaging) or ultrasound may also be options for looking at the urinary tract.

Other names: excretory urography, intravenous urography

WHAT IS IT USED FOR?

An IVP may be used to help find the cause of symptoms that may involve the urinary tract or prostate. It can be used to help diagnose disorders such as:

Kidney stones

Stones in the bladder or ureters

Kidney cysts

Enlarged prostate (BPH)

Tumors in the kidneys, bladder, or ureters (cancerous and not cancerous)

Birth defects that affect the structure of the urinary tract

Scarring from a urinary tract infection (UTI)

IVP usually isn't used during pregnancy or for infants and children.

WHY DO I NEED AN IVP?

You may need an IVP if you have symptoms of a urinary tract disorder. These may include:

Pain in your side or back

Sharp pains in your abdomen (belly), side, groin or back

Blood in your urine

Cloudy or bad-smelling urine

Pain when urinating (peeing)

Needing to urinate frequently, or not being able to urinate

Nausea and vomiting

Swelling in your feet or legs

Fever and chills

You may also need an IVP if you have had many urinary tract infections or have had an injury that may have damaged your urinary organs.

WHAT HAPPENS DURING AN IVP?

An IVP may be done in your provider's office or a hospital. The test is often done by a radiology technician (a health care professional who is trained to do x-ray exams). The procedure takes about an hour and generally includes these steps: You'll usually remove your clothing and put on a gown.

You'll lie down on your back on an x-ray table. The technician will take an x-ray to make sure the machine is set right for you.

The technician will insert an IV into a vein in your arm or hand and inject the contrast dye.

Several x-rays will be taken to see how your kidneys filter the dye. You may be asked to move into different positions.

The technician may wrap a special belt tightly around your belly. This helps keep the contrast dye in your kidneys. After the belt is removed, more x-rays will be taken to show the dye passing from your kidneys through your ureters and into your bladder.

Before the IVP is finished, you'll be asked to use the bathroom or a bedpan to urinate. Then the last x-rays will be taken to see if any urine remains in your bladder. This shows how well your bladder works to empty urine.

After the test, drinking plenty of fluids will help flush the contrast dye out of your body.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will give you instructions for how to prepare for an IVP. You will probably need to fast (not eat or drink) after midnight on the night before your test. You may also be asked to take a mild laxative the evening before the IVP. A laxative is medicine that helps you empty your bowels (poop).

ARE THERE ANY RISKS TO THE TEST?

There's a small chance you could have an allergic reaction to the contrast dye. Reactions can often be managed with medicine. Serious reactions are rare. But certain conditions may increase the chance of problems, such as having other allergies, congestive heart failure, or diabetes. If you have any of these conditions, tell your provider. You may be able to take medicine before the IVP to reduce the chance of reacting to the dye.

When the contrast dye is injected, you may feel itchy, warm, or queasy. Some people have a metallic taste in their mouths. These feelings are harmless and usually go away quickly.

Tell your provider if you are pregnant or think you may be pregnant. An IVP delivers a low dose of radiation. The dose is safe for most people, but it can be harmful to an unborn baby.

WHAT DO THE RESULTS MEAN?

Your IVP images will be studied by a diagnostic radiologist. This is a doctor who specializes in examining x-rays, MRIs, and other medical images to diagnose medical conditions. The radiologist will send a report with your IVP results to your provider. The report will explain exactly what the x-rays showed.

In general:

Normal results mean that:

The organs of your urinary tract looked normal, and no blockages were seen.

Your kidneys, ureters, and bladder worked normally to filter the dye and pass it out of your body.

In males, the prostate was the right size, and no problems were seen.

The organs of your urinary tract looked normal, and no blockages were seen.

Your kidneys, ureters, and bladder worked normally to filter the dye and pass it out of your body.

In males, the prostate was the right size, and no problems were seen.

Abnormal results mean that something irregular was found on your IVP images. Ask your provider to explain your report and diagnosis. Examples of abnormal results include:

One or more of urinary tract organs did not appear to be the right size or shape, or it was not in the usual place.

A tumor, stone, cyst, injury, scar, or some other abnormal area of tissue was found and may be blocking the flow of urine.

Your urinary tract did not filter the dye and/or pass it out of your body as quickly or completely as it should.

In males, the prostate was larger than normal.

One or more of urinary tract organs did not appear to be the right size or shape, or it was not in the usual place.

A tumor, stone, cyst, injury, scar, or some other abnormal area of tissue was found and may be blocking the flow of urine.

Your urinary tract did not filter the dye and/or pass it out of your body as quickly or completely as it should.

In males, the prostate was larger than normal.

Learn more about laboratory tests, reference ranges, and understanding results.

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Iron Tests

<https://medlineplus.gov/lab-tests/iron-tests/>

IRON TESTS

WHAT ARE IRON TESTS?

Iron tests measure different substances in the blood to check iron levels in your body. Iron is a mineral that your body needs for growth and development. Your body uses iron to make hemoglobin, a protein in red blood cells. It carries oxygen from the lungs to all parts of the body. Iron is also important for healthy muscles, bone marrow, and organ function. Your body also needs iron to make some hormones.

You get iron from eating food or taking supplements. There are iron-only supplements, but many multivitamin/mineral supplements also contain iron.

Most people in the United States get enough iron. However, certain people may be more likely to have trouble getting enough iron. For example, those who:

- Have heavy periods

- Are pregnant or breastfeeding

- Are infants (especially if they are premature or low birth weight)

- Are frequent blood donors

- Have cancer, certain digestive diseases, or heart failure

The amount of iron you need each day depends on your age, your sex, and whether you consume a mostly plant-based diet. Iron levels that are too low or too high can cause health concerns. For example, not having enough iron in your body is the most common cause of anemia. Anemia is a condition in which your blood has a lower-than-normal amount of red blood cells.

The amount of iron in your blood varies throughout the day and may be higher in the morning.

There are different types of iron tests. You may just have one type of test. But in some cases, your health care provider might order several different iron tests to understand your results better. The different types of iron tests include:

- Serum iron test, which measures the amount of iron in the blood

- Transferrin test, which measures transferrin, a protein that moves iron throughout the body

- Total iron-binding capacity (TIBC), which measures how well iron attaches to transferrin and other proteins in the blood

- Ferritin blood test, which measures how much iron is stored in the body

Other names: Serum Iron, Serum Fe, Ferritin Test, Serum Ferritin

WHAT ARE THEY USED FOR?

Iron tests are most often used to:

- Check if the iron levels in your blood are too low, a sign of anemia. But not all people who have low iron levels have anemia.

- Diagnose different types of anemia.

- Check if the iron levels in your blood are too high, which could be a sign of hemochromatosis (also called iron overload). This condition causes too much iron to build up in your body. It can be inherited (passed down through families) or be caused by another health condition.

- Measure the amount of iron stored in your liver to check for liver disease.

- Check for restless legs syndrome (tingling or burning feelings in your legs).

- Check for adult Still disease, also called adult-onset Still disease or AOSD. This is not a common condition. Symptoms include a high ferritin level, joint pain, fever, and a rash.

- See if treatments for iron deficiency (low iron levels) or excess iron (high iron levels) are working.

WHY DO I NEED AN IRON TEST?

You may need testing if you have symptoms of iron levels that are too low or too high. However, low iron may not cause symptoms until the level is very low. So, your provider may order a routine iron test to check your iron level.

Symptoms of iron levels that are too low include:

Weakness or fatigue

Dizziness

Shortness of breath

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Symptoms of iron levels that are too high include:

Weakness or fatigue

Joint pain, most often in the knees or hands

Abdominal (belly) pain

Loss of interest in sex or erectile dysfunction (ED)

Changes in skin color, which may look gray, metallic, or bronze

WHAT HAPPENS DURING AN IRON TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider may ask you to fast (not eat or drink) for 12 hours before your test. The test is usually done in the morning. If you have any questions about how to prepare for your test, talk to your health care provider.

ARE THERE ANY RISKS TO IRON TESTS?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand the results of your iron test, your provider will consider your symptoms, age, medical history, and the results of other blood tests.

If one or more iron test results show your iron levels are too low, it may mean you have:

Iron deficiency anemia, a common type of anemia.

Another type of anemia.

Thalassemia, a blood disorder that is inherited (passed down through families). It causes the body to make fewer than normal healthy red blood cells.

If one or more iron test results show your iron levels are too high, it may mean you have:

Hemochromatosis

Lead poisoning

Liver disease

Most conditions that cause too little or too much iron can be treated with iron supplements (for low iron), changes to your diet, medicines, and/or other therapies.

If your iron test results are not normal, it does not necessarily mean you have a medical condition that needs treatment. Some medicines, including birth control pills and estrogen treatments, can affect iron levels. Iron levels may also be lower for women during their menstrual cycles.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT IRON TESTS?

Your provider may order other blood tests to help check your iron levels. These include:

Hemoglobin test

Hematocrit test

Complete blood count

Mean corpuscular volume

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RELATED HEALTH TOPICS

Anemia

Hemochromatosis

Iron

Minerals

Thalassemia

RELATED MEDICAL TESTS

[Complete Blood Count \(CBC\)](#)

[Ferritin Blood Test](#)

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[Hemoglobin Test](#)

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[How to Prepare for a Lab Test](#)

[How to Understand Your Lab Results](#)

[MCV \(Mean Corpuscular Volume\)](#)

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Karyotype Genetic Test

<https://medlineplus.gov/lab-tests/karyotype-genetic-test/>

KARYOTYPE GENETIC TEST

WHAT IS A KARYOTYPE TEST?

A karyotype test is a type of genetic testing. It looks at the size, shape, and number of chromosomes in a sample of cells from your body. Chromosomes are tiny "packages" in your cells that contain your genes. Genes carry information, called DNA, that controls what you look like and how your body works.

Normally, all your cells (except for egg and sperm cells) have a set of the same 46 chromosomes. The chromosomes are grouped into 23 pairs. Each pair has two chromosomes, one coming from each parent. A karyotype test checks the chromosomes in your cells to:

See whether you have a full set of 46 chromosomes. Having too many or too few chromosomes can cause serious problems with health, growth, and normal development, such as Down syndrome (extra chromosome 21), Turner syndrome (missing X chromosome), and various genetic disorders.

Look for changes in the structure of chromosomes, such as broken, missing, or extra parts. These changes may cause a wide variety of problems depending on which chromosome is affected. But some changes in chromosomes don't cause problems.

Certain chromosome problems are present from birth, while others can develop in certain cells later in life. Those that develop later in life can cause specific types of cancer.

Other names: genetic testing, chromosome testing, chromosome studies, cytogenetic analysis

WHAT IS IT USED FOR?

A karyotype test looks for unusual changes in chromosomes in order to:

Check you and/or your family members for specific chromosome problems.

Check a fetus for chromosome problems.

Find out if abnormal chromosomes are causing problems having children, such as male or female infertility, frequent (two or more) miscarriages, or stillbirths.

Help diagnose and/or make treatment decisions for certain types of cancer and blood disorders.

Your health care provider can then use the results of a karyotype test to make important treatment decisions for you and/or your family members.

WHY DO I NEED A KARYOTYPE TEST?

The most common reason for getting a karyotype test is if you're planning to have children or are pregnant. These tests can tell you if you or your partner have abnormal chromosomes that you could pass on to a baby. Your provider may also order this test on your fetus to see if they have a genetic disorder, especially if:

The pregnant parent is age 35 or older. The overall risk of genetic birth defects is small, but the risk increases after age 35.

Either parent has a genetic disorder, a family history of a genetic disorder, or another child with a genetic disorder.

The results of certain prenatal screening tests weren't normal.

You may also need a karyotype test if:

You and/or your child have symptoms of a genetic disorder. Each type of genetic disorder has different symptoms. Karyotype testing may be needed to make a diagnosis.

You've been diagnosed with certain cancers or blood disorders that often involve changes in chromosomes. These changes can affect how your disease will progress and which treatment will work best for you. Karyotype testing may be needed to help diagnose and treat conditions such as leukemia, lymphoma, multiple myeloma, and anemia.

You have a family history of a chromosomal disorder and want to know your risk for developing the disorder.

WHAT HAPPENS DURING A KARYOTYPE TEST?

For a karyotype test, the type of sample you provide depends on the reason for the test.

To test the chromosomes of a fetus, the sample may be gathered using:

Amniocentesis. This procedure collects a sample of the amniotic fluid that surrounds the fetus during pregnancy. The fluid contains cells from the fetus that can be tested. Amniocentesis is usually done between week 15 and 20 of pregnancy.

Chorionic villus sampling (CVS). This procedure collects tissue from the chorionic villi, which are parts of the placenta that usually have the same chromosomes as the fetus. (The placenta is the organ that grows in the uterus to provide nutrients and oxygen to the fetus.) A CVS is usually done between week 10 and 13 of pregnancy.

To test the chromosomes of adults, children, and babies, the sample may be from:

A blood test. This is the most common sample used for karyotype tests. A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

A buccal (cheek) swab. A health care professional will insert a swab into your mouth and rub it across the inside of your cheek to gather a sample of cells.

A bone marrow test. This test may be used if you have a certain type of cancer or blood disorder. Marrow is soft, spongy tissue found inside of bones. Your provider usually takes a sample from the back of the hip bone. You will be given an injection (shot) of medicine to numb the area. You may also be given medicine to help you relax or sleep.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

What you have to do to prepare will depend on the type of test you're getting.

For a CVS or amniocentesis, you may need to drink extra fluid and not urinate before your test so that your bladder is full.

A blood test or cheek swab usually don't require any preparation.

If you're having a bone marrow test, ask your provider for instructions about how to prepare.

ARE THERE ANY RISKS TO THE TEST?

During amniocentesis or CVS, you may feel mild stinging or cramping. Both procedures are safe, but they do have a slight risk of causing miscarriage. Talk with your provider about the risks and benefits of these tests.

A blood test has very little risk. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

A cheek swab has no risk.

After a bone marrow test, you may feel stiff or sore where the sample was taken. This usually goes away in a few days. Serious risks are rare.

WHAT DO THE RESULTS MEAN?

A normal or negative result means that there were 46 chromosomes in the sample without any unusual changes in their structure.

An abnormal or positive result means that unusual changes in the number or structure of chromosomes were found.

Abnormal results can mean many things about your health or your child's health depending on the chromosome changes that were found. Talk with your provider to learn what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A KARYOTYPE TEST?

If you're thinking about having a karyotype test or have received abnormal results, it may help to talk with a genetic counselor. Genetic counselors can give you information about how genetic conditions might affect you or your family. They can help you find support services and make informed decisions about treatment or management options for a genetic disorder.

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RELATED HEALTH TOPICS

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Birth Defects
Down Syndrome
Genetic Testing
Leukemia
Lymphoma
Multiple Myeloma
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Turner Syndrome

RELATED MEDICAL TESTS

Alpha-Fetoprotein (AFP) Test
Amniocentesis (amniotic fluid test)
BCR-ABL1 Genetic Test
Bone Marrow Tests
BRAF Genetic Test
Chorionic Villus Sampling (CVS)
Down Syndrome Tests
How to Cope with Medical Test Anxiety
How to Understand Your Lab Results
Prenatal Cell-Free DNA Screening
PTEN Genetic Test
Sweat Test for Cystic Fibrosis
TP53 Genetic Test
Ultrasound
What You Need to Know About Blood Testing

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Ketones in Blood

<https://medlineplus.gov/lab-tests/ketones-in-blood/>

KETONES IN BLOOD

WHAT IS A KETONES IN BLOOD TEST?

A ketones in blood test measures the level of ketones in your blood. Ketones are acids that your body makes when it breaks down fat for energy.

Normally, the cells in your body use blood glucose, also called blood sugar, for energy. You get glucose from foods you eat, such as carbohydrates. Your body breaks down most of that food into glucose and releases it into your bloodstream. If your cells can't get enough glucose, your body will break down fat for energy instead. This process produces an acid called ketones. If your body breaks down fat too fast, ketones can build up in your blood and urine.

High ketone levels make your blood too acidic. This is a serious condition called ketoacidosis. The most common type of ketoacidosis is a complication of diabetes called diabetic ketoacidosis (DKA). DKA is a medical emergency that often develops quickly and can lead to a coma and be life-threatening.

If you have diabetes, a ketones in blood test can help find high ketone levels early so you can get treatment right away. If your health care provider recommends frequent testing to check for increasing ketone levels, at-home blood testing meters are available that check both glucose levels and ketones. At-home testing for ketones in urine—testing may also be an option. Talk with your provider about which is right for you.

Other names: Ketone bodies (blood), serum ketones, beta-hydroxybutyric acid, acetoacetate, nitroprusside test

WHAT IS IT USED FOR?

With diabetes:

A ketones in blood test is mostly used to check for diabetic ketoacidosis (DKA) in people with diabetes. DKA can affect anyone with diabetes, but it is most common with type 1 diabetes. With type 2 diabetes, DKA is less common.

If you have diabetes, a blood ketones test may be used when you are sick or pregnant. You may also be tested if you develop diabetes during pregnancy.

If you don't have diabetes, you may have your blood ketones checked if you are at risk for developing ketoacidosis and you have symptoms. You are at risk for ketoacidosis if you have a condition that causes your body to use fat for energy instead of glucose, such as:

Eating disorders and starvation

Vomiting or diarrhea that lasts a long time

Alcohol use disorder (AUD)

WHY DO I NEED A KETONES IN BLOOD TEST?

You may need a ketones in blood test if you have diabetes or other conditions that increase your risk for developing high ketone levels, or symptoms of ketoacidosis. You may also need the test if you have symptoms, and you eat a low-carbohydrate diet.

Symptoms of ketoacidosis may include:

Extreme hunger or thirst

Urinating (peeing) a lot more than usual

Dehydration

Headache

Nausea and vomiting

Pain in your abdomen (belly)

Confusion

Trouble breathing

Fatigue

Dry skin and mouth

Breath that has a fruity smell

Muscle stiffness or aches

If you don't get treatment, symptoms can quickly become more severe. If you have severe symptoms of ketoacidosis, seek immediate medical help. Large amounts of ketones can be life-threatening.

WHAT HAPPENS DURING A KETONES IN BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

To check blood ketones at home, you may use a blood glucose meter that also tests for ketones. You will use a device to prick your finger. You'll collect a drop of blood on a ketone test strip that is inserted into the meter. The meter will show your ketone levels. Follow the instructions carefully. You can also test for ketones in urine using an at-home kit. Ask your provider how you should test for ketones, when you should test, and what to do if your levels are high.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a ketones in blood test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A normal or negative test result usually means that your body is not making too many ketones and you don't have ketoacidosis. But a normal test result doesn't always rule out diabetic ketoacidosis (DKA). That's because your body makes three types of ketones. If DKA is in the early stages, only one type of ketone will be high, and lab tests often don't measure this type of ketone.

A high or positive test result means you have ketones in your blood. It's normal to have some blood ketones, but if your levels are very high and you have symptoms, you may have ketoacidosis. If you have diabetes, you likely have DKA. If you don't have diabetes, you may have ketoacidosis from alcohol use disorder, malnutrition, not eating for very long periods of time, or other conditions.

Ketoacidosis from any cause must be treated. Treatment may include going to the hospital.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A KETONES IN BLOOD TEST?

People who eat a low-carbohydrate ketogenic ("keto") diet to lose weight will have blood ketone levels that are higher but are not high enough to be ketoacidosis. This condition is called ketosis.

Ketosis doesn't make your blood too acidic because your body is able to use the ketones, so they don't build up. But it's possible to develop ketoacidosis on a keto diet, so it's important to talk with your provider to see if it's safe for you.

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Ketones in Urine

<https://medlineplus.gov/lab-tests/ketones-in-urine/>

KETONES IN URINE

WHAT IS A KETONES IN URINE TEST?

A ketones in urine test measures ketone levels in your urine. Normally, the cells in your body use blood glucose (also called blood sugar) for energy. It comes from the food you eat. Your body breaks down most of that food into glucose and releases it into your bloodstream. If your cells can't get enough glucose, your body breaks down fat for energy instead. This produces an acid called ketones, which can build up in your blood and urine.

Having some ketones in your urine is normal. But high ketone levels in your urine may be a sign that your blood is too acidic. This condition is called ketoacidosis. The most common type of ketoacidosis is a complication of diabetes called diabetic ketoacidosis (DKA). Diabetic ketoacidosis is a medical emergency that often develops quickly and can be life-threatening.

If you have diabetes, a ketones in urine test can help find high ketone levels early, so you can get treatment right away. Home tests are available if your health care provider recommends frequent testing.

Other names: ketones urine test, ketone test, urine ketones, ketone bodies

WHAT IS IT USED FOR?

A ketones in urine test is often used to help monitor people who are more likely to develop high levels of ketones. You are more at risk if you have a medical condition, diet, or exercise habits that cause your body to use fat for energy instead of glucose, such as:

Diabetes. Diabetic ketoacidosis (DKA) is a life-threatening condition caused by high ketone levels. It's most common in people who have type 1 diabetes. But people who have type 2 diabetes can also develop DKA.

Vomiting or diarrhea that lasts a long time.

Alcohol use disorder (AUD), which can cause alcoholic ketoacidosis.

Eating disorders and starvation.

Certain digestive diseases.

Eating a "keto diet," which is low in carbohydrates (foods that supply glucose for energy).

Exercising too hard for a long time.

Pregnancy.

WHY DO I NEED A KETONES IN URINE TEST?

Your provider may order a ketones in urine test if you have diabetes or other conditions that increase your risk for developing high ketone levels. You may also need this test if you have symptoms of ketoacidosis, which may include:

Early symptoms of ketoacidosis:

Urinating (peeing) a lot

Extreme hunger or thirst

Dehydration

Headache

Severe symptoms of ketoacidosis:

Nausea or vomiting

Pain in your abdomen (belly)

Confusion

Trouble breathing

Feeling very tired or sleepy

Fruity-smelling breath

If you have severe symptoms of ketoacidosis, seek immediate medical help. Large amounts of ketones can be life-threatening.

WHAT HAPPENS DURING A KETONES IN URINE TEST?

A ketones in urine test can be done at home or in a lab. If you go to a lab for your test, you will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

If you do the test at home, follow the instructions in your test kit. Your kit will include a package of test strips. You will either hold a test strip in your urine stream or dip it into a cup of urine that you collect with the clean-catch method. It's important to follow the test instructions exactly. Talk with your provider if you have any questions about how you should test for ketones in urine.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may have to fast (not eat or drink) for a certain period of time before taking a ketones in urine test. Ask your provider if you need to fast or do any other type of preparation before your test.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a ketones in urine test.

WHAT DO THE RESULTS MEAN?

Your test results may be a specific number or listed as a "small," "moderate," or "large" amount of ketones. Normal results can vary depending on your diet, activity level, and other factors.

Because high ketone levels can cause serious health problems, be sure to talk with your provider about what is normal for you and what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A KETONES IN URINE TEST?

Ketone test kits are available at most pharmacies without a prescription. If you are planning to test for ketones at home,

ask your provider which kit would be best for you. At-home urine tests are easy to do and can provide accurate results as long as you carefully follow all the instructions.

Some people use at-home kits to test for ketones if they are on a ketogenic or "keto" diet to try to lose weight. On a keto diet, you eat mainly foods with little or no carbohydrates. This makes your body use fat instead of glucose for energy, which may increase your ketone levels. A keto diet may not be right for people with certain conditions. Talk with your provider before going on a keto diet to see if it is right for you.

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Kidney Stone Analysis

<https://medlineplus.gov/lab-tests/kidney-stone-analysis/>

KIDNEY STONE ANALYSIS

WHAT IS A KIDNEY STONE ANALYSIS?

Kidney stones are small, pebble-like pieces of material that form in one or both of your kidneys. Your kidneys are two bean-shaped organs just below your rib cage. They keep your body's balance of water, minerals, and salts by removing waste to make urine (pee). Kidney stones form when you have high levels of some minerals or salts in your urine (pee). Many things may increase your risk of kidney stones. These can include your diet, some health conditions, and some supplements or medicines.

A kidney stone analysis is a test to find out what a kidney stone is made of. This information helps your health care provider develop a plan to help you lower your risk of forming more stones in the future.

There are four main types of kidney stones:

Calcium is the most common type of stone.

Uric acid is another common type of stone.

Cystine is an uncommon type of stone. It is caused by cystinuria, a kidney disease that you inherit from your parents.

Struvite is the most uncommon type of stone. It usually forms after a urinary tract infection (UTI).

Kidney stones can be as small as a grain of sand or as large as a pea, and sometimes even larger. Very small stones may pass out of your body when you urinate (pee). Larger or odd-shaped stones can get stuck inside your urinary tract and block the flow of urine. This may cause severe pain or bleeding, and you may need treatment to get rid of the stones. But with treatment, kidney stones usually don't cause lasting damage.

Other names: urinary stone analysis, renal calculus analysis

WHAT IS IT USED FOR?

A kidney stone analysis is used to find out what your kidney stone is made of. This helps your provider understand why the stone formed and what will help stop more stones from forming in the future. This is important because once you've had a stone, you're likely to get another one.

WHY DO I NEED A KIDNEY STONE ANALYSIS?

You may need a kidney stone analysis if you have symptoms that could be from a kidney stone. These include:

Sharp pain in your abdomen (belly), side, groin, or back

Blood in your urine (pee)

Frequent need to urinate

Not being able to urinate at all or only urinating a little bit

Pain when urinating

Cloudy or bad-smelling urine

Nausea and vomiting

Fever and chills

WHAT HAPPENS DURING A KIDNEY STONE ANALYSIS?

To collect a kidney stone, you will need a kidney stone strainer to filter your urine and a clean container for your stone. A kidney stone strainer is a device made of fine mesh or gauze. Your provider may give you a strainer, or you may get one from a store that sells medicine and other health care products.

To collect your stone, you will:

Filter all your urine through the strainer.

Check the strainer carefully after each time you urinate to look for a stone. It may look like a grain of sand or a tiny piece of gravel.

If you find a stone, put it in a clean container with a lid.

DO NOT add anything to the container. The stone must be kept dry.

DO NOT put tape or tissue on the stone.

Return the container to your provider or lab as instructed.

A stone may pass at any time of the day or night. So, it's important to filter all your urine (pee) every time you urinate until you find a stone.

Your provider may tell you to drink a lot of water to help pass the stone. If you have pain while the stone is passing, ask your provider about pain medicine.

If your kidney stone is too large to pass, you may need a minor surgical procedure to remove the stone for testing. If you've already passed a kidney stone and you kept it, ask your provider about testing it.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a kidney stone analysis.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a kidney stone analysis. But passing a kidney stone may cause:

Pain depending on the size of the stone

A urinary tract infection

A blocked urinary tract if the stone gets stuck

WHAT DO THE RESULTS MEAN?

Your results will show what your kidney stone is made of. With this information, your provider will be able to recommend steps and/or medicines to help prevent more stones from forming. Your treatment plan will depend on the size, type, and location of the stone. Most plans include drinking plenty of water and changing some of the foods you eat.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A KIDNEY STONE ANALYSIS?

Calcium stones are the most common type, but they are not caused by eating dairy products. In fact, dairy products can help prevent calcium stones. Your provider can explain what diet changes can help reduce your risk for stones. This is important because kidney stones increase your risk for developing chronic kidney disease.

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Lactate Dehydrogenase (LDH) Isoenzymes Test

<https://medlineplus.gov/lab-tests/lactate-dehydrogenase-ldh-isoenzymes-test/>

LACTATE DEHYDROGENASE (LDH) ISOENZYMES TEST

WHAT IS A LACTATE DEHYDROGENASE (LDH) ISOENZYMES TEST?

This test measures lactate dehydrogenase (LDH) isoenzymes in a sample of your blood. It's a general test that can help find out if you have tissue damage from disease or injury. It can also help show which organs and other tissues are likely to be damaged. But an LDH isoenzymes test alone cannot diagnose what's causing the damage.

LDH is also called lactic acid dehydrogenase. It is an enzyme. An enzyme is a protein that speeds up certain chemical reactions in your body. The LDH enzyme helps your cells make energy. LDH is found in almost all the tissues in your body. The largest amounts of LDH are found in your muscles, liver, kidneys, and red blood cells.

There are five forms of the LDH enzyme which are called LDH isoenzymes. Each isoenzyme has a slightly different structure and is found in specific tissues in your body:

LDH-1 is mainly in your heart and red blood cells.

LDH-2 is mainly in your white blood cells. It is also found in your heart and red blood cells, but in smaller amounts than LDH-1.

LDH-3 is mainly in your lungs. Smaller amounts are found in other tissues.

LDH-4 is mainly in your kidneys and pancreas. If you're pregnant, it's also found in the placenta (the organ that brings oxygen and nutrients to the baby).

LDH-5 is mainly in your liver and skeletal muscles (muscles that attach to your bones to help you move).

It's normal to have some LDH in your blood. But if disease or injury damages tissues that contain LDH, the cells release LDH isoenzymes into your bloodstream and your LDH levels will rise above normal.

There are two ways to measure LDH:

A total LDH test measures the total amount of LDH in your blood or other body fluids. It can't tell the difference between the types of LDH isoenzymes. So, this test can show possible tissue damage, but it can't show which tissue may be damaged.

An LDH isoenzyme test measures the amount of each type of isoenzyme in your blood. This information helps your health care provider figure out which tissues may be damaged.

Because each LDH isoenzyme is found in more than one type of tissue, other more specific tests are often used with or instead of an LDH isoenzymes test. These tests can more accurately identify which tissue is damaged and the condition causing the damage.

Other names: LD isoenzymes, lactic dehydrogenase isoenzyme, LDH fractionation

WHAT IS IT USED FOR?

An LDH isoenzymes test is mainly used as a general test to check for tissue damage. It can also help find out how serious the damage may be. An LDH isoenzymes test may be used to show which organs and other tissues are likely to be damaged, but other more specific tests may be used instead.

An LDH isoenzyme test may be used with other tests to help diagnose and monitor many types of acute (sudden) and chronic (long-lasting) conditions. Conditions that may cause high LDH levels include:

Anemia

Kidney disease

Liver disease, including hepatitis and cirrhosis

Pancreatitis

A blood clot in the lung (pulmonary embolism)

Muscular dystrophy

A recent heart attack, although troponin tests have mostly replaced LDH isoenzyme tests for diagnosing damage to the heart muscle from a heart attack

LDH isoenzyme tests may also be used to find out if treatment for many conditions is working.

The test is used to learn how serious certain types of cancer may be and whether the cancer is likely to respond to certain treatments. Regular LDH isoenzyme testing may be done to see whether cancer is getting better during and/or after treatment.

WHY DO I NEED AN LDH ISOENZYMES TEST?

An LDH isoenzyme testing may be used for many conditions, so it's best to ask your provider why you need this test. In general, the test is ordered if you:

Have symptoms that suggest you have some type of tissue damage

Had a total LDH test that showed you have higher than normal levels of LDH

Have a condition that affects LDH levels, and your provider wants to:

Learn more about your condition

Find out if your treatment is helping

Learn more about your condition

Find out if your treatment is helping

WHAT HAPPENS DURING AN LDH ISOENZYMES TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an LDH isoenzymes test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The meaning of your LDH isoenzyme test results depends on the reason you had the test, your symptoms, and the results of other tests you've had. Ask your provider to explain what your results say about your health.

In general, a high level of one or more LDH isoenzymes usually means you have some type of tissue damage. The type of disease or damage depends on which LDH isoenzymes are high and how your isoenzyme levels compare with each other. For example:

LDH-1 that's higher than LDH-2 may be a sign of a certain type of anemia.

LDH-5 that's higher than LDH-4 may be a sign of liver damage or disease.

High levels of two or more isoenzymes may mean you have two different conditions. Or it could be a sign of cancer that has spread to different tissues.

Higher than normal LDH isoenzyme levels don't always mean you have a medical condition that needs treatment. High levels of certain LDH isoenzymes can be caused by intense exercise and certain medicines, including aspirin. It's also possible to have high LDH isoenzymes if many red blood cells broke open when your sample was collected and tested.

Lower than normal LDH isoenzyme levels aren't common and usually aren't considered to be a health problem. Your levels of certain isoenzymes may be affected by taking large amounts of vitamin C or vitamin E. There is also a rare genetic disorder that can cause low LDH levels.

If you have questions about your results, talk with your provider.
Learn more about laboratory tests, reference ranges, and understanding results.

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Lactate Dehydrogenase (LDH) Test

<https://medlineplus.gov/lab-tests/lactate-dehydrogenase-ldh-test/>

LACTATE DEHYDROGENASE (LDH) TEST

WHAT IS A LACTATE DEHYDROGENASE (LDH) TEST?

A lactate dehydrogenase (LDH) test usually measures the level of lactate dehydrogenase (LDH) in a sample of your blood. In certain cases, LDH levels are measured in samples of other body fluids. This includes testing fluid from the spine (cerebrospinal fluid), the belly (peritoneal fluid), and the chest (pleural fluid).

LDH is also called lactic acid dehydrogenase. It is an enzyme. An enzyme is a protein that speeds up certain chemical reactions in your body. LDH helps your cells make energy. It is found in almost all the tissues in your body. The largest amounts of LDH are found in your muscles, liver, kidneys, and red blood cells.

If disease or injury damages tissues that have LDH, their cells release the enzyme into your bloodstream or other body fluids. It's normal to have some LDH in your blood and body fluids. But if your LDH levels are high, it may be a sign of certain diseases or injuries.

An LDH test alone can't show what is damaging your tissues or where the damage is located. So, an LDH test is usually done with other types of tests that can help diagnose the problem.

Other names: LD test, lactic dehydrogenase, lactic acid dehydrogenase

WHAT IS IT USED FOR?

LDH is mainly used to check for tissue damage. Many types of acute (sudden) and chronic (long-lasting) conditions can damage tissues and cause high LDH levels. So LDH testing may be used in many different situations from bone and muscle injuries to monitoring treatment for cancer. For example, an LDH test may be used to:

Help diagnose and/or monitor diseases that affect the:

Blood, including anemia

Liver

Lungs

Kidneys

Heart, including heart attack

Pancreas

Brain and spinal cord, such as encephalitis and bacterial meningitis

Blood, including anemia

Liver

Lungs

Kidneys

Heart, including heart attack

Pancreas

Brain and spinal cord, such as encephalitis and bacterial meningitis

Help monitor conditions that may get worse over time, such as muscular dystrophy and HIV

Help diagnose certain severe infections

Find out how serious certain types of cancer may be, including multiple myeloma, lymphoma, leukemia, testicular cancer, and melanoma.

Check whether treatment for certain types of cancer is working

WHY DO I NEED AN LDH TEST?

Because LDH testing is used for so many conditions, it's best to ask your health care provider why you need an LDH test. In general, the test is ordered if other tests and/or your symptoms suggest you have tissue damage or disease. If you have a disease that affects LDH levels, you may need an LDH test to learn more about your condition and to find out if your treatment is helping.

WHAT HAPPENS DURING AN LDH TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

If your provider thinks you have a condition that may affect LDH levels in other body fluids, you may have a procedure to get a fluid sample. These procedures may include:

A lumbar puncture or "spinal tap" to collect cerebrospinal fluid that flows around your brain and spinal cord

A thoracentesis to remove fluid in the chest

A paracentesis or "abdominal tap" to remove fluid from the abdomen

If you are having one of these tests, your provider will explain the procedure.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an LDH blood test. If your test uses other fluid samples, your provider will tell you how to prepare.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly. If you are having another type of test, your provider will explain the risks.

WHAT DO THE RESULTS MEAN?

LDH testing is used for many reasons. The meaning of your LDH test results depends on the reason you had the test, your symptoms, and the results of other tests you've had. So, ask your provider to explain what your results say about your health.

In general, LDH levels that are higher than normal usually mean you have some type of tissue damage. The damage is usually from disease, infection, or injury. Your provider may order more tests to diagnose your condition.

But higher than normal LDH level doesn't always mean you have a medical condition that needs treatment. High levels can be caused by intense exercise and certain medicines, including aspirin. It's also possible to have a high LDH level if many red blood cells broke open when your sample was collected and tested.

Lower than normal LDH levels aren't common and usually aren't considered to be a health problem. Taking large amounts of vitamin C or vitamin E may cause low levels. Low LDH levels may also be caused by a rare genetic disorder called lactate dehydrogenase deficiency.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN LDH TEST?

If your results showed higher than normal LDH levels, your provider may order a more specific type of LDH test that can help find out where the LDH is coming from. This more specific test is called an LDH isoenzyme test.

An LDH isoenzyme test measures the levels of five different forms of LDH. This information helps your provider find out about the type of tissue that's damaged, where in the body it may be, and how serious the damage may be.

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Lactate Test

<https://medlineplus.gov/lab-tests/lactate-test/>

LACTATE TEST

WHAT IS A LACTATE TEST?

This test measures the level of lactate, also called lactic acid, in a sample of your blood. In certain cases, cerebrospinal fluid (CSF) may be used. Normally, you have low levels of lactate in your blood and CSF. High levels of lactate may be a sign of a wide variety of medical conditions.

Lactate is mostly made in your muscles and red blood cells when they break down food for energy. Most of the time, your cells use oxygen to break down food for energy. But if your cells can't get enough oxygen, they can switch to another process for making energy that doesn't use oxygen. This backup process makes lactate. Your liver and kidneys turn the lactate into glucose (sugar) that your cells use for energy.

It's normal to have brief increases in lactate when you're exercising or doing other hard physical work. That's because your body uses more oxygen when you're physically active, which triggers your cells to make energy without oxygen. Usually, your lactate levels decrease quickly when you stop the activity.

But certain medical conditions can cause a harmful buildup of lactate. If lactate levels get too high, your blood becomes too acidic. This can lead to serious health problems and a life-threatening condition called lactic acidosis.

Many types of conditions can cause lactate buildup. They fall into two main groups:

Conditions that reduce the amount of oxygen that gets to your cells. These are conditions that:

Affect your breathing

Reduce the flow of oxygen-rich blood in your body

Examples include:

Shock

Heart failure

Lung diseases

Anemia

Severe infection and sepsis

Affect your breathing

Reduce the flow of oxygen-rich blood in your body

Shock

Heart failure

Lung diseases

Anemia

Severe infection and sepsis

Conditions that affect how much oxygen your body uses and disorders that affect how your body makes energy or gets rid of extra lactate. Examples include:

Liver and kidney diseases

Diabetes that isn't controlled

Leukemia

Mitochondrial disorders that you inherit

Using certain medicines or being exposed to certain toxins (poisons)

Very strenuous exercise for a long time, such as running a marathon

Liver and kidney diseases
Diabetes that isn't controlled
Leukemia
Mitochondrial disorders that you inherit
Using certain medicines or being exposed to certain toxins (poisons)
Very strenuous exercise for a long time, such as running a marathon
A lactic acid test can help find high levels of lactic acid before they cause serious health problems. But the test cannot diagnose the cause.
Other names: lactic acid test, lactic acid: plasma, L-lactate

WHAT IS IT USED FOR?

A lactate blood test is most often used to diagnose lactic acidosis and find out how severe it is. It can also help evaluate and monitor many types of medical conditions that may make lactate levels rise.
If a person appears to have meningitis, an infection of the brain and spinal cord, the test may be used to help find out if it is caused by bacteria or a virus. Because meningitis is an infection of the brain and spinal cord, both cerebrospinal fluid and blood may be tested.

WHY DO I NEED A LACTATE TEST?

You may need a lactate test:
If you have symptoms of lactic acidosis, such as:

Shortness of breath
Nausea and vomiting
Muscle weakness
Sweating
Abdominal (belly) pain
Shortness of breath
Nausea and vomiting
Muscle weakness
Sweating
Abdominal (belly) pain

To find out if your cells are getting enough oxygen

To find out why you have too much acid in your blood

You may also need this test if your health care provider thinks you have signs and symptoms of any of the many conditions that can cause lactic acidosis.

If you have been diagnosed with high levels of lactate, you may need to have repeated tests to monitor your condition.

If you have symptoms of meningitis, you may need to have your lactate levels measured in your blood and your cerebrospinal fluid. Symptoms of meningitis include:

Severe headache
Fever
Stiff neck
Sensitivity to light

WHAT HAPPENS DURING A LACTATE TEST?

A health care professional will take a blood sample from a vein or an artery. Blood from an artery may provide more accurate results than blood from a vein, so your provider may recommend this type of blood test, which is called an arterial blood gas (ABG) test.

To take a blood sample from a vein, a health care professional will insert a small needle into your arm. After the needle

is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes. Make sure you don't clench your fist during the test, as this can temporarily raise lactate levels.

To take a blood sample from an artery, a provider will usually use an artery inside the wrist. But the sample may be taken from an artery in your arm or groin. For a newborn, the sample may be taken from the baby's heel or the umbilical cord shortly after birth.

If your blood sample is taken from your wrist, the provider will first test your blood circulation. The provider will hold your wrist and apply pressure to the arteries to cut off blood flow to your hand for several seconds. Then the provider will let go of your wrist to check how quickly blood flow returns to your hand. If your blood flow is normal, the provider will collect a blood sample.

A blood sample taken from an artery tends to be more uncomfortable than most blood tests, which use a vein. So, the provider may apply some numbing medicine to your skin first. The provider will insert a needle with a syringe into the artery to remove some blood.

When the syringe is full, the provider will bandage the puncture site. Pressure will be applied to the site for at least 5 minutes to stop the bleeding.

If meningitis is suspected, your provider may also take a sample of your cerebrospinal fluid in a procedure called a spinal tap or lumbar puncture. This is usually done in a hospital.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will tell you how to prepare for your test. Your provider may tell you to:

Avoid exercise for several hours before the test. That's because exercise can cause a temporary increase in lactate levels.

Fast (not eat or drink) for 8 to 10 hours before the test.

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

After a blood test from an artery, you may have some bleeding, bruising, or soreness at the spot where the needle was put in. Very rarely, the needle may damage a nerve or the artery. You may be told to avoid lifting heavy objects for 24 hours after the test.

WHAT DO THE RESULTS MEAN?

The results of a lactate test can only show whether you have abnormal levels of lactate. They can't tell what's causing the problem.

A high level of lactate in a blood sample usually means you have either hyperlactatemia or lactic acidosis:

Hyperlactatemia means your level of lactate is staying mildly or moderately high, but the acidity of your blood (pH) is still within a normal range. If lactate continues to build up, hyperlactatemia can become lactic acidosis.

Lactic acidosis means that your lactate level is high enough to make your blood is too acidic, and it is staying high. Lactic acidosis is more serious than hyperlactatemia and needs to be treated right away.

Treatment for high levels of lactate depends on the condition that's causing the lactate to build up in your blood. Your provider will consider your symptoms, medical history, and the results of other tests to diagnose the cause.

If you had a lactate test using cerebrospinal fluid:

High levels of lactate mean you probably have bacterial meningitis.

Normal or slightly high levels of lactate mean you probably have viral meningitis.

Low levels of lactate are rare and not considered to be a medical concern.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A LACTATE TEST?

Certain medicines may cause high lactate levels, such as:

Metformin, a medicine for type 2 diabetes

Aspirin

Certain medicines to treat HIV called NRTIs

If you're concerned about how these medicines may affect your risk for lactic acidosis, talk with your provider.

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Lactose Tolerance Tests

<https://medlineplus.gov/lab-tests/lactose-tolerance-tests/>

LACTOSE TOLERANCE TESTS

WHAT ARE LACTOSE TOLERANCE TESTS?

Lactose tolerance tests measure your body's ability to break down lactose. Lactose is a type of sugar found in milk, cheese, ice cream, and other dairy products. Normally, an enzyme (a protein that speeds up certain chemical reactions in your body) called lactase breaks down lactose into simpler sugars. These sugars are absorbed by your body and turned into energy. If your body doesn't make enough lactase, you won't be able to properly digest foods that contain lactose.

When you can't digest all of the lactose you eat or drink and it passes into your colon, it's called lactose malabsorption. Lactose malabsorption can cause uncomfortable symptoms such as gas, bloating, and diarrhea within 30 minutes to 2 hours after drinking milk or eating other dairy products.

Getting the symptoms of lactose malabsorption after you eat or drink products with lactose is called lactose intolerance.

There are two types of lactose tolerance tests:

Hydrogen breath test. This test measures the amount of hydrogen gas in your breath before and after you drink a liquid containing lactose. If you can't digest all the lactose, it gets broken down by bacteria in your large intestine. This creates extra hydrogen gas, which you breathe out. This is the most common way to test for lactose intolerance.

Glucose blood test. This test involves a series of blood tests that measure the level of glucose (a type of sugar) in your blood before and after you drink a liquid containing lactose. If you are lactose intolerant, you usually won't have an increase in blood glucose.

Other names: hydrogen breath test, oral lactose tolerance, lactose tolerance serum test, lactose intolerance test, lactose breath test, disaccharide absorption test

WHAT ARE THEY USED FOR?

Lactose tolerance tests are used to:

Diagnose lactose intolerance

Check for other health conditions that may cause similar symptoms, such as:

Irritable bowel syndrome

Celiac disease

Small intestinal bacterial overgrowth (SIBO)

Crohn's disease

Ulcerative colitis

Irritable bowel syndrome

Celiac disease

Small intestinal bacterial overgrowth (SIBO)

Crohn's disease

Ulcerative colitis

WHY DO I NEED A LACTOSE TOLERANCE TEST?

You may need this test if you have symptoms of lactose intolerance. Symptoms usually happen within 30 minutes to 2 hours of eating or drinking dairy products. They can include:

Diarrhea

Abdominal (belly) pain

Bloating

Gas
Nausea

WHAT HAPPENS DURING A LACTOSE TOLERANCE TEST?

Your health care provider may order a hydrogen breath test or a glucose blood test.

During a hydrogen breath test:

You will breathe into a balloon-type container that measures the amount of hydrogen in your breath.

You will drink a flavored liquid that contains lactose.

You will breathe into the container again 30 minutes later and every 30 minutes for a few hours. Your hydrogen levels will be measured each time, and your provider will ask you about your symptoms.

During a glucose blood test:

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial.

You will drink a liquid that contains lactose.

More blood samples will be taken at specific intervals, usually at 30 minutes, one hour, and two hours.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

To prepare for either type of test, you will need to:

Fast (not eat or drink) for eight to 12 hours before the test

Stop taking antibiotics for two to four weeks before the test

Not smoke or exercise strenuously the day before the test

You may also need to stop taking certain medicines before this test, such as laxatives, antacids, and stool softeners, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

If you are having a breath test, you may need to brush your teeth or rinse your mouth shortly before the test.

ARE THERE ANY RISKS TO THIS TEST?

If you are lactose intolerant, you may feel cramps or bloating after drinking the lactose beverage.

There are no other risks to having a hydrogen breath test.

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If you had a hydrogen breath test, your results may show:

An increased level of hydrogen and worsening symptoms after drinking lactose. This probably means you have lactose intolerance.

Little or no increase in hydrogen after drinking lactose. This means your symptoms are probably not caused by lactose intolerance. But sometimes you can have a negative result and still have lactose intolerance. You may need more tests to figure out the cause of your symptoms.

If you had a glucose blood test, your results may show:

Your glucose levels did not increase after drinking lactose. This probably means you have lactose intolerance.

Your glucose levels increased. This means your symptoms are probably not caused by lactose intolerance. But if you have diabetes, your blood glucose may increase even if you have lactose intolerance. You may need more tests to figure out the cause of your symptoms.

If you are diagnosed with lactose intolerance, you should be able to manage your condition by limiting or avoiding foods with lactose, such as milk and cheese. Some people with lactose intolerance can have a limited amount of dairy without symptoms. Other people may need to avoid lactose altogether. Taking a supplement that contains lactase may also help

manage your condition.

Be sure to talk to your provider about planning your diet and/or taking lactase supplements.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT LACTOSE TOLERANCE TESTS?

Lactose intolerance is not the same as an allergy to milk or dairy. An allergy is an immune system response and can cause severe symptoms. While lactose intolerance does not cause serious complications, it can lead to uncomfortable and unpleasant symptoms. Adjusting your diet may help you avoid this discomfort.

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Laparoscopy

<https://medlineplus.gov/lab-tests/laparoscopy/>

LAPAROSCOPY

WHAT IS A LAPAROSCOPY?

A laparoscopy is a type of surgery that lets a surgeon look inside your body without making a large incision (cut). It's used to help diagnose and sometimes treat conditions that develop in your belly or pelvis.

To do a laparoscopy, a surgeon makes a small cut near your belly button that's usually a half-inch long or less. The surgeon inserts a long, thin tube with a camera through the cut and into your body. This tube is called a laparoscope. The camera sends images from inside your body to a video monitor. This allows the surgeon to see inside your body.

The surgeon may make one or two other small cuts in your belly for inserting special surgical tools into your body. Using these tools, the surgeon can remove samples of tissue to check for signs of disease (a biopsy).

Sometimes during a laparoscopy, the surgeon will use these tools to do procedures to treat certain problems. For example, if a laparoscopy finds a tumor, the surgeon may remove it completely during the same surgery. Many common surgical treatments can be done with laparoscopic surgery.

Laparoscopy is sometimes called "minimally invasive surgery" or "keyhole surgery," because it requires smaller cuts than traditional, "open" surgery. The use of smaller cuts has several benefits, including:

- Shorter hospital stays

- Less pain

- Faster recovery and smaller scars

Other names: diagnostic laparoscopy, exploratory laparoscopy

WHAT IS IT USED FOR?

Laparoscopy is used to help diagnose the cause of symptoms in the belly or pelvis. It's usually done if imaging tests, such as x-rays, ultrasounds, and MRI scans, haven't provided enough information to confirm a diagnosis.

The test is often used to help diagnose:

- Digestive diseases

- Urinary disorders

- Disorders in the female reproductive system, which includes the uterus, ovaries, and fallopian tubes

A surgeon uses laparoscopy to:

- Examine organs and glands to look for:

- Abnormal growths, such as cysts and tumors (cancerous and not cancerous)

- Scar tissue and adhesions

- Bleeding

- Infections

- Abnormal growths, such as cysts and tumors (cancerous and not cancerous)

- Scar tissue and adhesions

- Bleeding

- Infections

- Collect tissue samples to check for disease

- Find out if a known cancer has spread in the body

WHY DO I NEED A LAPAROSCOPY?

There are many reasons why you may need a laparoscopy, including if:

- You have pain or other symptoms in your belly or pelvis and imaging tests haven't been able to find the cause.

You've been in a serious accident or have been wounded and may have organ injuries.

You have cancer and your health care provider needs to know if it has spread to other parts of your body. This information helps choose your treatment.

You've had abnormal liver tests and your provider doesn't know why.

If you're female, you may need a laparoscopy to:

Find out what's causing problems getting pregnant

Diagnose symptoms that may involve your reproductive organs, including:

Pelvic inflammatory disease (PID)

Endometriosis

Ectopic pregnancy

Uterine fibroids

Ovarian cysts

Pelvic inflammatory disease (PID)

Endometriosis

Ectopic pregnancy

Uterine fibroids

Ovarian cysts

WHAT HAPPENS DURING A LAPAROSCOPY?

Laparoscopy is usually done in a hospital or an outpatient clinic. In general, it includes these steps:

You'll wear a hospital gown and lay on an operating table.

Usually, you will have general anesthesia, which is medicine to make you sleep during the surgery. The medicine is injected into your bloodstream through an intravenous (IV) line that's inserted into a vein. You may also inhale a gas from a mask to make you sleep.

To do the laparoscopy, the surgeon will:

Make a small cut near your belly button

Put carbon dioxide gas into your belly to open up space between your organs so it's easier to see them

Insert the laparoscope and small camera into your belly

Move the laparoscope to look at your organs and glands on a computer screen

Make a small cut near your belly button

Put carbon dioxide gas into your belly to open up space between your organs so it's easier to see them

Insert the laparoscope and small camera into your belly

Move the laparoscope to look at your organs and glands on a computer screen

If you need to have a biopsy or another procedure, the surgeon may need to make more small cuts to insert surgical tools.

When the surgeon is done, the tools and most of the gas will be removed from your body, and the small incisions will be closed and bandaged.

After surgery, you will be moved to a recovery room. In most cases, you'll be able to go home after a few hours, but that depends on what procedures were done during the laparoscopy. Before you leave, you'll get information about what to expect over the next few days at home.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know how to prepare. Be sure to follow all the instructions. You will need to fast (not eat or drink) for a period of time before a laparoscopy. Ask your provider whether you should take your usual medicines and/or supplements. But don't stop taking any medicine without talking with your provider first.

Plan to wear loose-fitting clothes because after the surgery, your belly may be bloated from the gas and a little sore. You're also likely to feel groggy, so you'll need to plan to have someone take you home.

ARE THERE ANY RISKS TO THE TEST?

You may have mild abdominal pain or discomfort for a few days after a laparoscopy. You may also have neck or shoulder pain. That's because the gas used in the surgery may irritate nerves in your belly that run through your shoulder.

After a laparoscopy, serious problems are very uncommon. But they can include bleeding, infection, blood clots, damage to an organ or blood vessel, and problems from the medicine that made you sleep during the procedure.

WHAT DO THE RESULTS MEAN?

The results of a laparoscopy depend on the reason for doing the procedure. Your provider can explain what was found and what that information means for your health. In general, your provider will be able to make a very accurate diagnosis of your condition based on the information from a laparoscopy.

Learn more about laboratory tests, reference ranges, and understanding results

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Legionella Tests

<https://medlineplus.gov/lab-tests/legionella-tests/>

LEGIONELLA TESTS

WHAT ARE LEGIONELLA TESTS?

Legionella is a type of bacteria. When it affects your lungs, it can cause a severe form of pneumonia known as Legionnaires' disease. Legionella tests look for these bacteria in urine, sputum, or in some cases in lung tissue. A blood test may be done to check if you have been exposed to the bacteria, but it cannot diagnose a Legionella infection.

Legionnaires' disease has similar symptoms to some types of pneumonia, but it can also cause diarrhea and confusion. Legionella bacteria can also cause Pontiac fever, a milder, flu-like illness that doesn't infect your lungs. Together, Legionnaires' disease and Pontiac fever are known as legionellosis.

Legionella bacteria are found naturally in freshwater environments. But the bacteria can make people sick when it grows and spreads in man-made water systems. These types of systems include plumbing systems of large buildings, including hotels, hospitals, nursing homes, and cruise ships. The bacteria may then contaminate water sources, such as hot tubs, fountains, and air-conditioning systems.

Legionellosis infections happen when people breathe in mist or small drops of water that contain the bacteria. You usually can't get the infection through drinking the water, unless you aspirate it (accidentally breathe the water into your lungs). The bacteria do not spread from person to person. But there can be a disease outbreak if many people are exposed to the same contaminated water source.

Not everyone who is exposed to Legionella bacteria will get sick. You are more likely to develop an infection if you:

- Are over the age of 50

- Are a current or former smoker

- Have a chronic disease such as diabetes or kidney failure

- Have a chronic lung disease such as COPD or emphysema

- Have a weakened immune system due to a disease such as HIV or cancer, or are taking medicines that suppress the immune system

- Live in a long-term care facility

- Have recently stayed in a hospital or had surgery requiring anesthesia

Pontiac fever usually clears up on its own, but Legionnaires' disease can be fatal if not treated. Most people will recover if promptly treated with antibiotics.

Other names: Legionnaires' disease testing, Legionellosis testing, Legionella pneumophila

WHAT ARE THEY USED FOR?

Legionella tests are used to find out whether you have Legionnaires' disease. Other lung diseases have symptoms similar to Legionnaires' disease. It's important to get the right diagnosis and treatment to help prevent life-threatening complications.

WHY DO I NEED A LEGIONELLA TEST?

You may need this test if you have symptoms of Legionnaires' disease. Symptoms usually show up two to 14 days after exposure to Legionella bacteria and may include:

- Cough

- High fever

- Chills

- Headache

- Chest pain

- Shortness of breath

Fatigue
Nausea and vomiting
Diarrhea
Confusion

WHAT HAPPENS DURING A LEGIONELLA TEST?

Legionella tests are usually done using urine (pee) and/or sputum. Many providers will test both urine and sputum samples to confirm a diagnosis. A blood test may be done to see if you have been exposed to the bacteria.

During a clean catch urine test:

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

During a sputum test:

Sputum is a thick type of mucus made in your lungs. If you have an infection or chronic illness affecting the lungs or airways, sputum can settle in your lungs, and you can also cough it up. During the sputum test:

You may be asked to rinse your mouth with water before the sample is taken.

Your provider will ask you to breathe deeply and then cough deeply into a special cup.

Your provider may tap you on the chest to help loosen sputum from your lungs.

If you have trouble coughing up enough sputum, your provider may ask you to breathe in a salty mist to help you cough more deeply.

If you can't produce enough sputum even after breathing in a salty mist, your provider may order a bronchoscopy (a procedure that's used to check for the cause of a lung problem).

In some cases, your provider may also order a bronchoalveolar lavage (BAL). This is a procedure that can be done during a bronchoscopy. A bronchoalveolar lavage collects a sample from your lungs and checks it for Legionella bacteria. A provider may do this if:

You have a severe respiratory disease.

You aren't improving from treatment for Legionnaires' disease.

Your provider is unable to determine the type of infection without a sample directly from your lungs.

During a bronchoscopy:

You may need to remove some or all of your clothing. If so, you will be given a hospital gown.

You may also need to remove dentures and any removable dental appliances.

You will lie on a bed or table with your head raised.

You may get medicine (sedative) to help you relax. The medicine will be injected into a vein or given through an IV (intravenous) line placed in your arm or hand.

Your provider will spray a numbing medicine in your mouth and throat so you won't feel any pain during the procedure.

Your provider will insert the bronchoscope (a thin, lighted tube with a camera attached) down your throat and into your

airways. The camera will take photos and connect to a video screen.

As the bronchoscope is moved down, your provider can look into your airways and lungs through the bronchoscope and on the video screen.

For a BAL:

Your provider will put a small amount of saline through the bronchoscope.

After washing the airways, the saline is sucked up into the bronchoscope.

The saline solution will contain cells and other substances, such as Legionella bacteria, which will be taken to a lab for testing.

During a blood test:

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a Legionella urine test.

Because a sputum culture is often scheduled in the morning, your provider may encourage you to drink more water the night prior. This is because drinking more water can increase the amount of sputum. You may also be asked not to eat for one or two hours before the test.

If you get a BAL and/or bronchoscopy, you may be asked to fast (not eat or drink) for up to twelve hours before the test. You may also need to stop taking certain medicines before a sputum culture and a BAL/bronchoscopy. So, tell your provider about everything you take. Antibiotics may affect bacterial growth and your results, especially those of a sputum culture. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to providing a urine or sputum sample. There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is very little risk to having a bronchoscopy or a BAL. Serious complications are rare, but they may include bleeding in the airways, infection, or a pneumothorax (collapse of part of your lung).

WHAT DO THE RESULTS MEAN?

If your results were positive, it probably means you have Legionnaires' disease.

If your results were negative, it may mean you have a different type of infection. It may also mean not enough Legionella bacteria were found in your sample.

If you have questions about your results, talk to your provider. Your provider may consider your symptoms, medical history, and the results of other tests to understand the results of your Legionella test.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT LEGIONELLA TESTS?

Whether your results were positive or negative, your provider may do other tests to confirm or rule out a diagnosis of Legionnaires' disease. These include:

Chest x-rays

Gram Stain

Acid Fast Bacillus (AFB) Tests

Bacteria Culture

Sputum Culture

Respiratory Pathogens Panel

If you have Legionnaires' disease, you may have symptoms such as fatigue and muscle aches for several months after treatment. If there's an outbreak of Legionnaires' disease in a building where you work or live, ask your provider if you need to do anything to reduce your risk of infection.

Legionnaires' disease got its name in 1976 after a group of people attending an American Legion convention became ill with pneumonia.

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Lipase Tests

<https://medlineplus.gov/lab-tests/lipase-tests/>

LIPASE TESTS

WHAT IS A LIPASE TEST?

A lipase test measures the level of lipase in your blood. Lipase helps your body digest fats. It's a type of digestive enzyme. An enzyme is a protein that speeds up certain chemical reactions in your body. Most of your lipase is made in your pancreas, an organ located behind the lower part of your stomach. Lipase is also made in your salivary (spit) glands and in your stomach.

It's normal to have a small amount of lipase in your blood. But if the cells of your pancreas are damaged, they will release larger amounts of lipase. So, high levels of lipase in your blood may mean you have pancreatitis, (an inflamed, swollen pancreas) or another type of pancreatic disease. A lipase tests can help diagnose these conditions and others.

Other names: serum lipase, lipase, LPS

WHAT IS IT USED FOR?

A lipase test may be used to:

Help diagnose diseases of the pancreas, especially acute (sudden) pancreatitis

Check to see if treatment for pancreatic diseases is working

Help diagnose and monitor chronic diseases that affect the pancreas, including:

Cystic fibrosis

Crohn's disease

Celiac disease

Cystic fibrosis

Crohn's disease

Celiac disease

Help diagnose injury or blockage of the intestines

Help diagnose peritonitis (inflammation of the peritoneum, the tissue that lines the abdominal wall)

Help diagnose pancreatic cysts

WHY DO I NEED A LIPASE TEST?

You may need a lipase test if you have symptoms that may be caused by a problem with your pancreas. These include:

Pain in your abdomen (belly) that may spread to your lower back

Nausea and vomiting

Diarrhea

Swollen or tender belly

Fever

Fast heartbeat

Jaundice, or yellowing of the skin and eyes

WHAT HAPPENS DURING A LIPASE TEST?

Lipase is measured with a blood test. During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for 8-12 hours before a lipase blood test.

Certain medicines and supplements can affect your test results. Your provider will let you know if you need to stop taking any medicines before your test. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A very high level of lipase is usually a sign of acute pancreatitis.

Higher than normal levels of lipase may be caused by:

Diseases of the pancreas, including a blocked duct (tube), or pancreatic cancer

Chronic kidney disease

Peptic ulcer

Gallbladder disease

Intestinal problems, such as a blocked intestine

Diabetes

Salivary (spit) gland disorders, such as infection or cancer

Alcohol use disorder (AUD)

Low levels of lipase may be a sign of permanent damage to cells in your pancreas that make lipase. This may happen in certain chronic (long-lasting) diseases, such as cystic fibrosis and chronic pancreatitis.

If your lipase levels are not normal, it doesn't always mean you have a medical condition that needs treatment. Many medicines, including codeine, diuretics ("water pills"), certain cholesterol medicines, and birth control pills can affect your lipase levels.

If you have questions about your lipase test results, talk with your health care provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A LIPASE TEST?

A lipase test is commonly used to diagnose pancreatitis. Pancreatitis can be acute or chronic. Acute pancreatitis is a sudden, short-term condition that can usually be cured with treatment. Chronic pancreatitis is a long-lasting condition that gets worse over time. But it can be managed with medicine and lifestyle changes, such as by not drinking alcohol or smoking and being at a healthy weight. In certain cases, your provider may recommend surgery if other treatments don't help enough.

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Lipoprotein (a) Blood Test

<https://medlineplus.gov/lab-tests/lipoprotein-a-blood-test/>

LIPOPROTEIN (A) BLOOD TEST

WHAT IS A LIPOPROTEIN (A) BLOOD TEST?

A lipoprotein (a) test measures the level of lipoprotein (a) in your blood. A high level of lipoprotein (a) may mean you have a high risk for heart disease and stroke.

Lipoproteins are particles made of protein and fats (lipids). They carry cholesterol through your bloodstream to your cells. The two main groups of lipoproteins are called HDL (high-density lipoprotein) or "good" cholesterol and LDL (low-density lipoprotein) or "bad" cholesterol.

Lipoprotein (a) is a type of LDL. These lipoproteins carry cholesterol to your cells through your arteries (blood vessels that carry oxygen-rich blood from your heart and lungs to your body). If you have high levels of LDL particles, cholesterol can build up in your arteries and form blockages called plaques. This condition is known as atherosclerosis, or "hardening of the arteries," which means that the arteries thicken and become less flexible. It can lead to many serious medical conditions including:

Coronary artery disease, narrow or blocked arteries in your heart

Heart attack

Stroke

Peripheral arterial disease, blocked arteries in your legs or arms

Other blood vessel diseases

Lipoprotein (a) particles are stickier than other types of LDL particles, so they may be more likely to cause blockages and blood clots in your arteries. As a result, high levels of lipoprotein (a) may mean you have a high risk for heart disease, stroke, and other serious conditions related to blockages and blood clots in your arteries.

A lipoprotein (a) blood test can give you a more accurate understanding of your risk than a routine cholesterol test that only measures your total LDL cholesterol level. That's because a routine cholesterol test may show that your LDL cholesterol level is "healthy," but if a large percentage of your LDL cholesterol is carried by lipoprotein (a) particles, your risk for heart disease and stroke could still be high.

Other names: cholesterol Lp(a), Lp(a)

WHAT IS IT USED FOR?

A lipoprotein (a) test is used to get a better understanding of your risk for heart disease, stroke, and other blood vessel diseases. But is not a routine screening test. It is usually ordered if your health care provider thinks you might be at high risk for cardiovascular problems or if having high levels of lipoprotein (a) runs in your family. Researchers are still studying how lipoprotein (a) levels affect health and when the test should be used.

WHY DO I NEED A LIPOPROTEIN (A) TEST?

Your provider may order the test if you have certain signs or health conditions that mean you have a high risk for blockages in your arteries, such as:

A family health history of early heart or blood vessel disease (before age 55 for a father or brother and before age 65 for a mother or sister)

High LDL cholesterol, even though you take medicine to lower it

Heart or blood vessel disease, especially if your cholesterol and triglyceride (another type of fat in your body) levels are normal without taking medicine to lower them

Signs of familial hypercholesterolemia, an inherited (passed down through families) condition that causes high cholesterol levels

Had more than one heart attack or more than one angioplasty, a procedure to open up narrow or blocked arteries in

your heart

The test may also be used to help make decisions about the risks and benefits of taking cholesterol medicine to lower your chance of developing heart and blood vessel disease.

WHAT HAPPENS DURING A LIPOPROTEIN (A) TEST?

A health care professional will take a blood sample from a vein in your arm using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Preparations for a lipoprotein (a) test depend on the lab doing the test. Usually, you need to fast (not eat or drink) for 9 to 12 hours before your blood is drawn. Your provider will let you know if there are any special instructions to follow. Certain substances can affect your test results. Before you get a lipoprotein (a) test, tell your provider if you have been drinking alcohol or taking niacin supplements, aspirin, or oral estrogen hormones.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A high lipoprotein (a) level may mean you have a high risk for heart and blood vessel disease, even if your cholesterol levels are normal and you are healthy.

Lipoprotein (a) levels usually don't change much over time. But test results may be affected by certain health conditions. Talk with your provider about what your test results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A LIPOPROTEIN (A) TEST?

Your genes control how much lipoprotein (a) you make. Genes are parts of DNA in your cells that you inherit from your parents. Genes carry information that controls what you look like and how your body works. By age 5, you usually have your "adult level" of lipoprotein (a), and it tends to remain about the same for the rest of your life. For that reason, diet and exercise may not change your lipoprotein (a) levels.

But if you have a high level of lipoprotein (a), it's very important to take steps to improve your heart health. This will help lower your overall risk for heart and blood vessel problems, even though it may not change your lipoprotein (a) levels.

Your provider may suggest:

Taking medicine to lower your LDL cholesterol if it is high

Eating a healthy diet

Losing weight

Quitting smoking

Getting regular exercise

Reducing stress

Lowering your blood pressure

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Liver Function Tests

<https://medlineplus.gov/lab-tests/liver-function-tests/>

LIVER FUNCTION TESTS

WHAT ARE LIVER FUNCTION TESTS?

Liver function tests (also called a liver panel) use a sample of your blood to measure several substances made by your liver.

The most common liver function tests measure:

Albumin, a protein made in the liver.

Total protein. This test measures the total amount of protein in your blood, which includes albumin and globulins. These proteins are mainly made in your liver.

ALP (alkaline phosphatase), ALT (alanine transaminase), AST (aspartate aminotransferase), and GGT (gamma-glutamyl transferase). These are enzymes that are mainly made in your liver. Enzymes are proteins that speed up certain chemical reactions in your body.

Bilirubin, a waste product your body makes when it breaks down old red blood cells. Your liver removes most of the bilirubin from your body.

Lactate dehydrogenase (LDH), an enzyme found in most of the tissues in your body, but some of the largest amounts are found in your liver.

Prothrombin time (PT), how long it takes your blood to clot. Prothrombin is a protein involved in blood clotting. It's made in your liver.

Some of these tests can show how well your liver is working and others can show whether your liver may be damaged by liver disease or injury. But liver function tests alone usually can't diagnose specific diseases. So, if your results are abnormal, you'll usually need other tests to find the exact cause.

Other names: liver panel, liver function panel, liver profile hepatic function panel, LFT

WHAT ARE THEY USED FOR?

Liver function tests are most often used to help:

Find out if liver disease or damage could be causing certain symptoms

Learn how serious liver disease is after it has been diagnosed

Monitor liver disease over time and/or find out how well treatment is working

Check for side effects of certain medicines that can affect the liver

WHY DO I NEED LIVER FUNCTION TESTING?

Many liver function tests are included in a common blood test called a comprehensive metabolic panel (CMP). Your provider often orders a CMP as part of your routine checkup to screen for liver and other diseases.

You may also need liver function tests if you have symptoms of liver disease or damage. These include:

Nausea and vomiting

Lack of appetite

Fatigue

Weakness

Jaundice, a condition that causes your skin and eyes to turn yellow

Swelling and/or pain in your abdomen (belly)

Swelling in your ankles and legs

Dark-colored urine (pee) and/or light-colored stool (poop)

Frequent itching

Even if you don't have symptoms, you may need to be tested if you have a high risk for liver damage. Your risk may be

high if you have:

A family history of liver disease

Alcohol use disorder (AUD)

Obesity

Diabetes

Been taking certain medicines that may cause liver damage

Hepatitis or have been exposed to hepatitis (swelling of the liver from infection or injury)

If you already have liver disease, you may need to be tested to monitor your condition and/or to see how well your treatment is working.

WHAT HAPPENS DURING A LIVER FUNCTION TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will tell you how to prepare for your test. You will probably need to fast (not eat or drink) for 10-12 hours before the test. Certain medicines can affect your test results, so be sure to tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Understanding abnormal liver function test results is often complicated. So, it's best to ask your provider to explain what your results say about your health.

If any of your liver function test results are abnormal, your provider will compare the results of all the substances that were measured. This is done to look for certain patterns of normal and abnormal results that suggest different types of liver conditions.

In general, the results of your liver function tests can tell you if:

Your liver is inflamed, which means you have hepatitis.

You have hepatitis from drinking alcohol or other causes, such as infection.

Your liver isn't working well and how weak it has become.

You have a problem with your bile ducts. Bile is a digestive "juice" that your liver makes. Bile ducts are the tubes that carry bile through your liver and out of your liver.

Medicines are harming your liver and how serious the damage is.

Abnormal results don't always mean you have a problem with your liver. Other conditions can cause high or low levels of certain substances that these tests measure. For example, high levels of ALP can be a sign of bone disease or liver disease. So, your provider will also consider your symptoms, medical history, risk for liver disease, and any medicines you take.

If you have been ill recently, certain types of abnormal results may be temporary. In this case, your provider may repeat the tests later to see if your results return to normal.

Learn more about laboratory tests, reference ranges, and understanding results.

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Lung Cancer Genetic Tests

<https://medlineplus.gov/lab-tests/lung-cancer-genetic-tests/>

LUNG CANCER GENETIC TESTS

WHAT ARE LUNG CANCER GENETIC TESTS?

If you have lung cancer, your health care provider may order genetic tests. These tests look for changes in the genes of the cancer cells in your body. Gene changes are also called gene variants or mutations. There are many types of gene changes that are found in lung cancer. Knowing what type of gene change you have may help your provider to:

Predict how your lung cancer will develop over time

Match your cancer to a targeted therapy that's most likely to help you

Targeted therapy uses drugs or other substances that mainly attack specific cancer cells and cause less harm to normal cells. It usually has fewer side effects than chemotherapy or radiation therapy.

There are different types of targeted therapies. Each type is designed to work for cancers with specific gene changes. So, lung cancer genetic tests are important to check if your cancer has a gene change that can be treated with targeted therapy.

There are two main types of lung cancer, small cell lung cancer (SCLC) and non-small cell lung cancer (NSCLC). Most people with lung cancer have NSCLC. The most common lung cancer genetic tests are done for NSCLC. That's because this type of lung cancer is more likely to have gene changes that respond to targeted therapies.

Genes are parts of DNA in your cells that you inherit from your parents. Genes carry information that controls what you look like and how your body works. The gene changes found in lung cancer affect genes that control how fast your cells grow and divide to make new cells. Cells with these changes grow out of control and form cancerous tumors that can spread through your body.

The gene changes in lung cancer usually aren't inherited. That means you can't pass them down to your children. You develop gene changes in certain cells in your lungs after you're born. These changes can happen if you're exposed to substances that cause cancer, such as tobacco smoke, radon gas, or air pollution. Gene changes can also happen if your cells make a mistake when they divide to make new cells.

Other names: Lung cancer targeted gene panel

WHAT ARE THEY USED FOR?

Lung cancer genetic testing is most often used to look for gene changes that are often involved in non-small cell lung cancer. The results may help guide treatment choices for targeted therapies.

There are many types of gene changes that can be involved in NSCLC. And researchers are still learning about more.

Lung cancer genetic tests may check a single gene or several genes may be checked in one test.

WHY DO I NEED A LUNG CANCER GENETIC TEST?

You may need a lung cancer genetic testing if you have been diagnosed with lung cancer. The test results may help your provider know which treatments are likely to help you the most and which ones may not help at all.

WHAT HAPPENS DURING A LUNG CANCER GENETIC TEST?

The best sample for lung cancer genetic tests is tissue from a tumor in your lung. The procedure to collect a tissue sample is called a biopsy. If you had a biopsy to diagnose your cancer, the same tissue may be used to test for gene changes.

There are several ways to do a lung biopsy. The type of biopsy you have depends on many things, including where the cancer is growing in your lungs. In certain cases, tissue may be removed with a special hollow needle inserted through

your skin and into your lung.

A tissue sample may also be removed during a bronchoscopy or with a procedure that involves making an incision (cut) in your chest area to insert special tube-like instruments into your lung. Talk with your provider to find out which biopsy method will work for you.

If tumor tissue can't be collected, a blood test may be used for genetic testing in certain cases. Tumor cells shed DNA into your bloodstream. Testing the tumor DNA in your blood is called a "liquid biopsy." This test is not as accurate as tests that use tumor tissue. Researchers are still studying the use of liquid biopsies to test for certain gene changes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Ask your provider if you need to do anything to prepare for your test.

ARE THERE ANY RISKS TO THE TEST?

If you have a biopsy, the risks depend on the type of procedure you have. In general, you may have a little bruising or bleeding at the biopsy site. You may also have some discomfort. Ask your provider about any risks involved with the test you're having.

WHAT DO THE RESULTS MEAN?

If your lung cancer cells have a gene change that may respond well to a targeted therapy, your provider may start you on that treatment.

If your results show you don't have a gene change that matches a targeted therapy, your provider may discuss other treatment options. If you were tested for one gene change, other tests may be done to look for other gene changes that have targeted therapies.

Your test results may also show that your lung cancer has a gene change that:

May prevent a certain therapy from working. This information could spare you from getting treatment that won't help.

Doesn't help make treatment decisions. Researchers don't understand the importance of all the gene changes that are found in cancer.

Genetic testing may take longer than many other types of lab tests. So, you may not get your results for a few days to a week or more. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Lung Function Tests

<https://medlineplus.gov/lab-tests/lung-function-tests/>

LUNG FUNCTION TESTS

WHAT ARE LUNG FUNCTION TESTS?

Lung function tests are a group of tests that check how well your lungs are working. These tests are also called pulmonary function tests, or PFTs. They may be used with other tests and information to help diagnose and monitor certain lung diseases.

There are many types of lung function tests. Most of them involve breathing through a mouthpiece. The mouthpiece is connected to a device that makes different types of measurements, such as:

How much air your lungs can hold

How well you move air in and out of your lungs when you breathe

How well your lungs absorb oxygen from the air you breathe and move it into your blood

Lung function tests also help check the health of your airways. Your airways are the parts of your body that carry air to your lungs, including your nose, throat, trachea (windpipe), and bronchial tubes that connect your windpipe to each lung.

Common lung function tests include:

Spirometry. This lung function test is used the most. A spirometry test measures how much air you breathe out and how fast you can blow air out of your lungs.

Lung volume test, also called as body plethysmography. This test is the most accurate way to measure the amount of air your lungs can hold. It also measures the amount of air that remains in your lungs after you exhale as much as you can.

Lung diffusion capacity test. This test measures how well your lungs deliver oxygen to your blood from the air you breathe.

Exercise tests. There are different types of tests that measure how well your lungs work when you are active, including:

Six minute walk test

Cardiopulmonary (heart and lung) exercise test (CPET)

Six minute walk test

Cardiopulmonary (heart and lung) exercise test (CPET)

Your health care provider may order one or more lung function tests, depending on the reason you need to be tested.

Other names: pulmonary function tests, PFTs

WHAT ARE THEY USED FOR?

Lung function tests are often used to:

Help diagnose and monitor lung diseases, including:

Asthma

COPD (chronic obstructive pulmonary disease)

Emphysema

Pulmonary fibrosis

Pulmonary hypertension

Asthma

COPD (chronic obstructive pulmonary disease)

Emphysema

Pulmonary fibrosis

Pulmonary hypertension

Guide decisions about treating certain lung diseases with medicine

Help diagnose and monitor certain diseases that can affect the lungs, such as scleroderma or sarcoidosis

Check lung function before surgery

Check for lung damage from exposure to chemicals or other substances at home or work

WHY DO I NEED A LUNG FUNCTION TEST?

Lung function tests are done for many different reasons. You may need testing if you:

Have signs and/or symptoms of a breathing problem such as:

Shortness of breath

Wheezing

Coughing

A tight feeling in your chest or trouble breathing deeply

An abnormal chest x-ray

Shortness of breath

Wheezing

Coughing

A tight feeling in your chest or trouble breathing deeply

An abnormal chest x-ray

Have a lung condition or another type of condition that could damage your lungs

Have been exposed to substances that can cause lung damage, such as asbestos, radiation, or certain medicines

Are going to have surgery

Your provider may also order a lung function test as part of your routine exam.

WHAT HAPPENS DURING LUNG FUNCTION TESTING?

There are different ways to do each type of lung function test. But the basic steps are about the same.

For a spirometry test:

You'll sit on a chair with a soft clip on your nose. The clip makes sure you breathe only through your mouth.

You'll place your lips tightly around a mouthpiece that's connected to a device called a spirometer.

You'll be told to take a deep breath in and then blow it out into the mouthpiece as hard as you can. The spirometer will measure how much air you breathe out and how fast you blow it out.

You may be given medicine to inhale (breath in) that opens your airways. Then you'll blow into the tube again to see if the medicine helps you blow out more air faster.

For a lung volume (body plethysmography) test:

You'll sit in a small, airtight room with clear walls. It looks like a telephone booth.

You'll wear a soft clip on your nose to make sure you breathe only through your mouth.

You'll place your lips tightly around a mouthpiece.

You'll be told how to breathe in and out through the mouthpiece.

Air pressure changes inside the room will show how much air your lungs can hold and how much air stays in your lungs after you breath out.

A lung volume test may be done by breathing a special mixture of gases for a few minutes while wearing a clip on your nose. For this test, you do not need to sit in an airtight room.

For a gas diffusion test:

You'll sit on a chair with a soft clip on your nose. The clip makes sure you breathe only through your mouth.

You'll place your lips tightly around a mouthpiece that's connected to a spirometer.

You will be asked to breathe in a very small, safe amount of carbon monoxide gas. Then you'll hold your breath for about 10 seconds.

Next, you'll breathe out through the mouthpiece. The spirometer will measure the amount of carbon dioxide you exhale.

This shows how much of the gas your lungs were able to absorb and move into your blood.

For exercise tests:

Six minute walk test:

Your blood pressure will be measured, and you'll be asked to rate any shortness of breath and fatigue that you feel.

You'll walk as far as possible in six minutes, usually in a hallway.

Your heart rate and the amount of oxygen in your blood will be checked as you walk.

After six minutes, your blood pressure will be measured again, and you'll rate your shortness of breath and fatigue.

If you normally use portable oxygen for a lung condition, you'll be able to use it during this test.

Cardiopulmonary exercise test (CPET):

You'll be attached to monitors and machines that will measure your breathing, heart rate, blood pressure, and blood oxygen levels while you exercise.

You'll breathe through a face mask or mouthpiece connected to a monitor. It measures how much oxygen you breathe in and how much carbon dioxide you breathe out.

You'll exercise on a stationary bicycle or treadmill for a total of 10 to 20 minutes:

You'll begin exercising with the bike or treadmill set to the easiest level.

You'll slowly increase your effort until you feel you can't continue. On a bicycle, the resistance will increase, so it's harder to pedal. On a treadmill, the speed will increase, and the platform may tilt so you feel like you're walking uphill.

As you exercise, you'll be asked to rate how you're feeling. If you have serious symptoms, the test will stop.

The bike or treadmill will return to the easiest level while you keep pedaling or walking to cool down.

You'll begin exercising with the bike or treadmill set to the easiest level.

You'll slowly increase your effort until you feel you can't continue. On a bicycle, the resistance will increase, so it's harder to pedal. On a treadmill, the speed will increase, and the platform may tilt so you feel like you're walking uphill.

As you exercise, you'll be asked to rate how you're feeling. If you have serious symptoms, the test will stop.

The bike or treadmill will return to the easiest level while you keep pedaling or walking to cool down.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TESTS?

Your provider will let you know how to prepare for your lung function test. To make sure your test results are accurate, follow the instructions you're given. In general:

Wear loose, comfortable clothing.

Don't eat a large meal within at least two hours of the test.

You may need to avoid certain activities or foods and beverages before your test. Your provider will tell you for how long. These include:

Heavy exercise

Smoking

Drinking alcohol

Drinking or eating products with caffeine, such as coffee, colas, and chocolate

Heavy exercise

Smoking

Drinking alcohol

Drinking or eating products with caffeine, such as coffee, colas, and chocolate

If you use inhaler medicines (bronchodilators), you may need to stop for a period of time before testing. Your provider will tell you when to stop. But never stop taking any medicine without talking with your provider first.

If you wear dentures, wear them for the test. They'll help you close your lips tightly around the mouthpiece.

ARE THERE ANY RISKS TO THE TESTS?

There is very little risk to having a lung function test. Some people may feel lightheaded, dizzy, or tired from breathing hard. It's also common to cough. Some people may feel nervous being in a small space for a lung volume test. If you have concerns about your test, talk with your provider.

WHAT DO THE RESULTS MEAN?

A normal lung function test result for you depends on the size of your lungs. Your lung size is based on your height, sex,

and age. So, your provider will compare your test results to results from healthy people who are the same height, age, and sex as you.

If your results aren't normal, it may mean you have a lung condition. Lung function tests can't diagnose which specific lung disease you have. But your test results can tell which of the two main types you have:

Obstructive lung conditions cause your airways to become narrow, making it hard for air to flow out of your lungs. Examples of obstructive lung diseases include asthma, bronchitis, and emphysema.

Restrictive lung conditions decrease the amount of air your lungs can hold. This may happen if your lungs or your chest muscles can't expand fully when you breathe in. Examples of restrictive lung conditions include scleroderma, sarcoidosis, pulmonary fibrosis, and interstitial lung diseases.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT LUNG FUNCTION TESTS?

If you need one or more lung function tests, your provider may also order a blood oxygen level test called an arterial blood gas (ABG) test. This blood test measures the amount of oxygen and carbon dioxide in your blood.

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Luteinizing Hormone (LH) Levels Test

<https://medlineplus.gov/lab-tests/luteinizing-hormone-lh-levels-test/>

LUTEINIZING HORMONE (LH) LEVELS TEST

WHAT IS A LUTEINIZING HORMONE (LH) LEVELS TEST?

This test measures the level of luteinizing hormone (LH). A hormone is a chemical messenger in your bloodstream that controls the actions of certain cells or organs. LH is usually measured in a sample of your blood. But in certain cases, it may be measured in urine (pee).

LH plays an important role in sexual development in children and fertility in adults:

In women who menstruate (have periods), LH helps control the menstrual cycle. It also triggers the release of an egg from the ovary. This is called ovulation. LH levels quickly rise just before ovulation. If you're trying to have a baby, this monthly increase in LH tells you when you're most likely to become pregnant.

In men, LH causes the testicles to make the hormone testosterone, which is important for making sperm. Normally, LH levels in men do not change very much.

In children, LH levels are normally low in early childhood. They begin to rise slowly in the years before puberty, usually between ages six and eight. As puberty begins LH continues to rise along with levels of other hormones. The increase in all of these hormones triggers the physical changes of puberty:

In girls, LH helps signal the ovaries to make the hormone estrogen. Estrogen is involved in the growth of breasts, body hair, and the start of menstruation.

In boys, LH helps signal the testicles to make testosterone. Testosterone is involved in the growth of facial and body hair, changes in a boy's voice, and sperm production.

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In boys, LH helps signal the testicles to make testosterone. Testosterone is involved in the growth of facial and body hair, changes in a boy's voice, and sperm production.

LH levels are controlled by a complex system of hormones made in different parts of your body. Abnormal levels of LH may be a sign of a problem with any of these parts. They include your:

Pituitary gland, a small gland at the base of your brain. It makes LH.

Hypothalamus, a part of your brain. It makes hormones that tell your pituitary gland how much LH to make.

Reproductive glands, your ovaries or testicles. The amount of sex hormones they make let your hypothalamus and pituitary gland know when to start and stop making LH.

Other names: lutropin, interstitial cell stimulating hormone, ICSH

WHAT IS IT USED FOR?

LH testing is used to help diagnose conditions that cause too much or too little LH. The specific way the test is used depends on your sex, age, and symptoms.

In women, an LH tests may be used to:

Help find the cause of infertility.

Find out when ovulation is about to happen if a woman is trying to get pregnant. Usually at-home tests are used to track ovulation. These tests measure LH in urine (pee).

Check for medical conditions that affect the ovaries.

Find the reason for irregular or stopped menstrual periods in women of childbearing age.

Help find out if menopause, or perimenopause may have begun:

Menopause happens when the ovaries stop making certain hormones, and menstrual periods have stopped for 12

months in a row. This usually happens around age 50.

Perimenopause is the time leading up to menopause when hormones and periods begin to change. It usually starts around age 45.

Menopause happens when the ovaries stop making certain hormones, and menstrual periods have stopped for 12 months in a row. This usually happens around age 50.

Perimenopause is the time leading up to menopause when hormones and periods begin to change. It usually starts around age 45.

In men, an LH test may be used to:

Help find the cause of:

Infertility

A low sperm count

Low testosterone levels

A low sex drive

Infertility

A low sperm count

Low testosterone levels

A low sex drive

Check for medical conditions that affect the testicles

In women and men, LH testing may be used to help diagnose disorders of the:

Pituitary gland

Hypothalamus

In children and teens, LH testing is most often used to help find out if a medical disorder is causing early or delayed (late) puberty.

An LH test is usually used with a blood test to check the levels of another hormone called follicle-stimulating hormone (FSH). These two hormones work closely together to control sexual development and reproduction. Blood tests to check estrogen, testosterone, and progesterone levels are also commonly used with LH testing.

WHY DO I NEED AN LH TEST?

For women:

You may need this test if:

You've been unable to get pregnant after 12 months of trying.

Your menstrual periods aren't regular or have stopped.

If you're age 45 or older, testing usually isn't needed. That's because high levels of LH and/or FSH are usually a sign of normal perimenopause and menopause. As your ovaries release fewer eggs, your body makes more hormones to try to trigger ovulation. But you may need an LH test along with other hormone tests if there is a medical reason, such as:

Premature menopause (age 40 or younger)

Early menopause (before age 45)

Having symptoms that may be related to menopause, but the cause isn't clear

Premature menopause (age 40 or younger)

Early menopause (before age 45)

Having symptoms that may be related to menopause, but the cause isn't clear

For men:

You may need this test if:

You've been unable to get your partner pregnant after 12 months of trying.

You have less interest in sex than usual.

You have a low sperm count.

You have a loss of muscle and/or body hair.

For men and women:

You may need an LH test if your health care provider thinks you could have a pituitary disorder. Pituitary problems can affect fertility and decrease your interest in sex. They may also cause symptoms, such as:

Fatigue

Weakness

Decreased appetite and/or weight loss

For children and teens:

LH testing may be needed if puberty seems to be starting too early or too late:

Early puberty means showing signs of sexual development before age 8 in girls and age 9 in boys.

Delayed puberty means not showing signs of sexual development by age 13 in girls and age 14 in boys.

WHAT HAPPENS DURING AN LH LEVELS TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

If you're using at-home urine testing to track ovulation, follow the instructions that come with your LH testing kit.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines or supplements before your test. Tell your provider everything you take. But don't stop taking any prescription medicines unless your provider tells you to.

If you have menstrual periods, you may need to have your test at a specific time during your menstrual cycle.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no risk to a urine test.

WHAT DO THE RESULTS MEAN?

To understand the results of your LH test, your provider will consider your sex, age, symptoms, medical history, and the results of other hormone tests, especially your level of follicle stimulating hormone (FSH). Your provider can explain what the results of your LH test mean.

Higher LH in adults:

In women who have menstrual periods, LH levels change through the month. An increase in LH a couple of weeks after your period usually means you're ovulating normally. But high LH levels through the month may mean your ovaries aren't working properly. If you are:

Childbearing age, higher than normal LH levels may mean you may have a disorder that affects your ovaries work, such as:

Ovaries that never developed normally

Primary ovarian insufficiency (POI)

Polycystic ovary syndrome (PCOS)

Thyroid diseases or adrenal gland disorders

A chromosomal disorder, such as Turner syndrome

45 or older, high LH levels may mean you are approaching menopause.

Childbearing age, higher than normal LH levels may mean you may have a disorder that affects your ovaries work, such as:

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45 or older, high LH levels may mean you are approaching menopause.

In men, high levels of LH may be caused by:

A germ cell tumor (a tumor that starts in the cells that become sperm)

Mumps

A physical injury to the testicles

Testicles that never developed normally

A chromosomal disorder, such as Klinefelter syndrome

A germ cell tumor (a tumor that starts in the cells that become sperm)

Mumps

A physical injury to the testicles

Testicles that never developed normally

A chromosomal disorder, such as Klinefelter syndrome

In women and men, high LH levels may be a sign of damage to the ovaries or testicles from:

Autoimmune disease

Radiation therapy and/or chemotherapy

Autoimmune disease

Radiation therapy and/or chemotherapy

Lower-than-normal levels of LH in adults are often a sign of a problem with the pituitary gland or hypothalamus. These problems may make it hard for your body to make enough LH and FSH.

In women, low LH and FSH levels with missed periods may also be linked to extreme exercise, malnutrition, stress, or being very underweight.

Generally, in children with signs of early sexual development (before age 8 in girls or age 9 in boys):

High levels of LH and FSH mean the child has early puberty, also called precocious puberty. In most cases, the cause is unknown. But in certain cases, the cause is a problem in the brain, such as:

A brain tumor

A past brain injury

A past brain infection, such as meningitis or encephalitis

A brain tumor

A past brain injury

A past brain infection, such as meningitis or encephalitis

Normal levels of LH and FSH mean the child has a type of early puberty called peripheral precocious puberty. This often means that a disorder in the testicles, ovaries, or adrenal glands is causing higher than normal levels of estrogen or testosterone. Exposure to medicines that contain sex hormones may also cause this type of early puberty.

Generally, in teens with no signs of sexual development by age 13 for girls or age 14 for boys, levels of LH and FSH may be low, normal, or high depending on the cause of delayed puberty. Many teens with delayed puberty are healthy and go through normal puberty at a later age. But delayed puberty may be caused by certain medical conditions, such as:

Poor nutrition from a long-term illness or eating disorder

Hypogonadism, which is when the ovaries or testicles make little or no hormone. It may be caused by:

Certain genetic disorders, including Kallmann syndrome

Tumors in the brain or pituitary gland

Turner syndrome in girls

Klinefelter syndrome in boys

Certain autoimmune disorders

Radiation therapy or chemotherapy

Certain genetic disorders, including Kallmann syndrome

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Turner syndrome in girls

Klinefelter syndrome in boys

Certain autoimmune disorders

Radiation therapy or chemotherapy

If you have questions about test results, talk with your or your child's provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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- Prolactin Levels
- SHBG Blood Test
- Testosterone Levels Test
- What You Need to Know About Blood Testing

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Lyme Disease Tests

<https://medlineplus.gov/lab-tests/lyme-disease-tests/>

LYME DISEASE TESTS

WHAT ARE LYME DISEASE TESTS?

Lyme disease is an infection caused by bacteria you get from tick bites. Lyme disease tests look for signs of infection in a sample of your blood or cerebrospinal fluid (CSF). CSF is a clear liquid that flows in and around your brain and spinal cord. The test checks your sample for antibodies that your immune system makes to fight the bacteria that cause Lyme disease.

You can get Lyme disease if an infected tick bites you. Ticks can bite you anywhere on your body, but they usually bite in hard-to-see places such as your groin, scalp, backs of the knees, and armpits.

Most Lyme disease is caused by bites from black-legged ticks (also known as deer ticks), which are about the size of a sesame seed. Baby black-legged ticks are about the size of a poppy seed, so you may not know you have been bitten. In general, an infected tick needs to be attached to you for 36 to 48 hours before it can pass Lyme disease bacteria onto you.

Without treatment, Lyme disease can cause serious health problems that may affect your joints, heart, and nervous system. But if you find Lyme disease early, it can usually be cured after a few weeks of treatment with antibiotic medicine. If it's found later, treatment may take up to eight weeks.

Other names: Lyme antibodies detection, Borrelia burgdorferi antibodies test, Borrelia DNA Detection, IgM/IgG by Western Blot, Lyme disease test (CSF), Borrelia antibodies, IgM/IgG

WHAT ARE THEY USED FOR?

Lyme disease tests are used to find out if you have a Lyme disease infection. Your health care provider is most likely to order the test if you have symptoms that could be Lyme disease, and if you were exposed or possibly exposed to infected ticks.

In the United States, infected ticks are most commonly found in states in the:

Northeast

Mid-Atlantic

Upper Midwest

Pacific coast, especially northern California

WHY DO I NEED A LYME DISEASE TEST?

You may need a test if you have symptoms of infection and were exposed or possibly exposed to ticks that carry the bacteria that causes Lyme disease. The first symptoms of Lyme disease usually show up between 3 and 30 days after the tick bite. Symptoms may vary from person to person.

Early signs and symptoms may include:

A reddish, round skin rash that slowly gets bigger over several days. It usually doesn't itch or hurt, but it may be warm. As it gets better, it may look like a bull's-eye (a red ring with a clear center), but not always.

Fever

Chills

Headache

Fatigue

Muscle and joint aches

Swollen lymph nodes ("swollen glands")

Later signs and symptoms of Lyme disease usually involve your joints and nervous system. They include:

Severe headaches and neck stiffness

Rashes on other parts of your body
Facial palsy (drooping of one or both sides of your face)
Arthritis with severe joint pain and swelling, especially in the knees
Pain that comes and goes in your muscles, joints, bones, and tendons
Arrhythmia (a problem with the rate or rhythm of your heartbeat)
Periods of feeling dizzy or short of breath
Shooting pains, numbness, or tingling in your hands or feet

WHAT HAPPENS DURING LYME DISEASE TESTING?

Lyme disease testing is usually done on a sample of blood. In certain cases, a CSF test may be done.

For a Lyme disease blood test:

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a CSF test:

You may need a CSF test if your symptoms could mean that Lyme disease is affecting your nervous system, such as a stiff neck or numb hands or feet. Providers may order a CSF test if the results of your blood test show you are likely to have Lyme disease or if the results are uncertain.

To get a sample of your CSF, a provider will do a procedure called a lumbar puncture, also known as a spinal tap. During the procedure:

You will lie on your side or sit on an exam table.

A provider will clean your back and inject an anesthetic into your skin, so you won't feel pain during the procedure. Your provider may put a numbing cream on your back before this injection.

When the area on your back is completely numb, your provider will insert a thin, hollow needle between two vertebrae in your lower spine. Vertebrae are the small backbones that make up your spine.

Your provider will withdraw a small amount of cerebrospinal fluid for testing. This will take about five minutes.

You'll need to stay very still while the fluid is being withdrawn.

Your provider may ask you to lie on your back for an hour or two after the procedure. This may prevent you from getting a headache afterward.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a Lyme disease blood test.

For a lumbar puncture, you may be asked to empty your bladder (pee) and bowels (poop) before the test.

ARE THERE ANY RISKS TO LYME DISEASE TESTS?

There is very little risk to having a blood test or a lumbar puncture.

With a blood test, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

With a CSF test, you may feel a little pinch or pressure when the needle is inserted. After the test, you may feel some pain or tenderness in your back where the needle was inserted.

You may also have some bleeding at the site or get a headache. The headache may last for several hours or up to a week or more, but your provider may suggest treatment to help relieve the pain.

WHAT DO THE RESULTS MEAN?

Lyme disease is difficult to diagnose. The symptoms are common to many conditions, and test results alone can't diagnose the disease. To make a diagnosis, your provider will consider your test results along with your medical history,

exposure, and symptoms. You may also need other tests.

A negative blood test result means that antibodies to fight the Lyme disease bacteria were not found in your blood. If you had symptoms for longer than 30 days before your test, you probably don't have Lyme disease.

But if you had symptoms for less than 30 days before you gave your blood sample, you may need to have another Lyme disease test. That's because it may take a few weeks for your body to make enough antibodies to show up on a test. If your test was done too soon, you could be infected even though your test was negative. This is called a "false negative."

A positive blood test result means that antibodies to fight the Lyme disease bacteria were found in your blood. In this case, the Centers for Disease Control and Prevention (CDC) recommends a second blood test on the same blood sample. If the second test is positive and you have symptoms of infection, you probably have Lyme disease.

But positive test results don't always mean that Lyme disease is causing your symptoms. You could have antibodies from a past case of Lyme disease that your body successfully fought off months or even years ago.

It's also possible to have a "false positive." That means you don't have antibodies that fight Lyme disease bacteria, even though the test says you do. The test may have mistakenly found antibodies that fight other bacteria or diseases, such as certain autoimmune diseases. And it's possible that these diseases are causing your symptoms, not Lyme disease.

The results from a CSF test can help show whether Lyme disease has spread to your nervous system.

A negative CSF test result means that no Lyme disease antibodies were found in your CSF. But it doesn't rule out Lyme disease in your nervous system. You may need more tests.

A positive CSF test result means that Lyme disease antibodies were found in your CSF. If more antibodies were in your CSF than in your blood, it's possible that Lyme Disease has spread to your nervous system.

Your provider will prescribe antibiotic medicine if you are likely to have Lyme disease. Most people who are treated with antibiotics in the early stage of the disease will recover completely.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT LYME DISEASE TESTS?

At-home test kits for Lyme disease are available. To do these tests, you usually collect a drop of blood from your finger and send it to a lab for testing. If you want to do an at-home test, it's important to talk with your provider first. Some at-home tests may use lab methods that aren't proven to work, so your results may not be accurate.

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RELATED HEALTH TOPICS

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Magnesium Blood Test

<https://medlineplus.gov/lab-tests/magnesium-blood-test/>

MAGNESIUM BLOOD TEST

WHAT IS A MAGNESIUM BLOOD TEST?

A magnesium blood test measures the amount of magnesium in a sample of your blood. Magnesium is a mineral that you get from many kinds of foods you eat. Some examples include nuts, seeds, beans, fortified breakfast cereals, green leafy vegetables, and milk.

Your body needs magnesium to help your muscles, nerves, and heart work properly. Magnesium also helps control blood pressure and blood glucose, also called blood sugar. It's important for building strong bones, and it supports your immune system.

Magnesium is a type of electrolyte. Electrolytes are electrically charged minerals. They help control the amount of fluid and the balance of acids and bases (pH balance) in your body.

Most of your body's magnesium is stored in your bones, organs, and other body tissue. Only a small amount is found in your blood. Your body tightly controls the amount of magnesium in your blood mainly through your:

Kidneys. They filter extra magnesium from your blood and get rid of it through urine (pee). If your magnesium level is low, your kidneys limit the amount lost through urine.

Intestines. They control how much magnesium you absorb from the foods you eat.

Abnormal levels of blood magnesium can be caused by many different conditions. So, a magnesium test may be used to help diagnose a variety of disorders.

Other names: Mg, Mag, Magnesium-Serum

WHAT IS IT USED FOR?

A magnesium blood test is used to check the level of magnesium in your blood. It's done if your health care provider thinks your levels may not be normal:

A low magnesium level is also called magnesium deficiency or hypomagnesemia. This condition is more common than levels that are too high.

A high magnesium level is also called hypermagnesemia. This condition is uncommon. Most cases happen in people who have kidney failure.

A magnesium blood test may also be used to help find the cause of abnormal levels of other minerals, including calcium, potassium, and phosphorus. That's because magnesium plays a role in how your body absorbs these minerals.

WHY DO I NEED A MAGNESIUM BLOOD TEST?

You may need a magnesium blood test if you have symptoms that could be caused by abnormal magnesium levels.

Symptoms of low magnesium include:

Loss of appetite

Nausea and vomiting

Fatigue and weakness

Muscle cramps

Numbness and/or tingling

Irregular heartbeat (arrhythmia)

Seizures (if magnesium levels are extremely low)

Symptoms of high magnesium are much the same as symptoms of low magnesium. They may also include:

Diarrhea

Abdominal (belly) cramping

Cardiac arrest from extremely high levels of magnesium

You may also need to have your magnesium levels checked to monitor your health if you:
Have a condition that's linked to abnormal magnesium levels, such as:

Chronic kidney disease
Diabetes that's not well controlled
Alcohol use disorder
Malabsorption syndrome
Malnutrition
Chronic (long-lasting) diarrhea
Low levels of potassium and/or calcium
Chronic kidney disease
Diabetes that's not well controlled
Alcohol use disorder
Malabsorption syndrome
Malnutrition
Chronic (long-lasting) diarrhea
Low levels of potassium and/or calcium

Take medicine that can decrease magnesium levels, for example:

Diuretics ("water pills")
Certain antibiotics
Proton pump inhibitors to reduce stomach acid
Diuretics ("water pills")
Certain antibiotics
Proton pump inhibitors to reduce stomach acid

Take medicine that can increase magnesium levels, for example:

Aspirin
Medicines that contain magnesium, such as laxatives for constipation and certain antacids for heartburn or gastroesophageal reflux disease (GERD)
Lithium
Aspirin
Medicines that contain magnesium, such as laxatives for constipation and certain antacids for heartburn or gastroesophageal reflux disease (GERD)
Lithium
Are being treated with magnesium and/or calcium.

WHAT HAPPENS DURING A MAGNESIUM BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

There's usually no preparation for a magnesium blood test. If you're having other blood tests at the same time, you may need to fast (not eat or drink) for several hours before your test.

Your provider will let you know if you need to prepare for your test. Be sure to tell your provider about any prescription and over-the-counter medicines and/or supplements that you are taking.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If the results of your magnesium blood test are abnormal, your provider will use your medical history and the results of other tests to diagnose the cause. Many different conditions can affect magnesium levels, so ask your provider to explain what your test results mean.

In general, low magnesium levels may mean that:

You're not getting enough magnesium in your diet. Older adults and people with alcohol use disorder or malnutrition are more likely to have this problem.

Your body isn't able to absorb magnesium, even if you eat the right amount. Causes of this problem include digestive disorders, such as Crohn's disease, or taking certain medicines.

Your body is getting rid of too much magnesium. Causes include kidney problems, long-term diarrhea, and diabetes, which can cause frequent urination (peeing) if blood glucose levels are too high.

Other health conditions linked to low magnesium include underactive parathyroid glands and pregnancy, which increases the need for magnesium.

Higher than normal magnesium levels may mean that:

You're getting too much magnesium from dietary supplements, laxatives, antacids, or certain prescription medicines. It's rare to get too much magnesium from food alone.

Your body has a problem getting rid of magnesium. The most common cause is kidney failure.

Other health conditions linked to high magnesium include dehydration, hypothyroidism, Addison disease, and overactive parathyroid glands.

Treatment for abnormal magnesium levels will depend on your condition.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A MAGNESIUM BLOOD TEST?

The amount of magnesium in your blood may be normal even though the amount of magnesium stored in your body is low. That's because your body will take magnesium from your bones to keep blood levels stable.

To get more information about your magnesium levels, your provider may also order a magnesium in urine test or a magnesium red blood cell (RBC) test. A magnesium RBC test measures the amount of magnesium inside your red blood cells. This test may be better at finding low magnesium levels than a regular magnesium blood test.

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Sodium Blood Test

What You Need to Know About Blood Testing

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Malaria Tests

<https://medlineplus.gov/lab-tests/malaria-tests/>

MALARIA TESTS

WHAT ARE MALARIA TESTS?

Malaria is a serious disease caused by a parasite. Malaria parasites are tiny single-celled life forms. They live and reproduce in the red blood cells of humans. This destroys the red blood cells, which makes you sick.

In most cases, people get malaria when they are bitten by mosquitos that are infected with the parasite. At first, malaria symptoms may be much like the flu. In certain cases, the disease can become life-threatening if it isn't treated quickly. But medicines can usually cure malaria.

Malaria tests look for signs of a malaria infection in a sample of your blood. The tests can diagnose malaria early so the disease can be cured before it causes serious illness. Malaria testing is used if you have symptoms of malaria and you have recently been in parts of the world where malaria is common.

Malaria is rare in the United States. It's mostly found in warm, humid parts of the world. The highest number of cases are in parts of Africa south of the Sahara Desert. In the U.S., most cases are in people who have traveled in tropical and subtropical areas where mosquitos spread a lot of disease.

In rare cases, malaria is spread through contact with the blood of an infected person. This may happen during blood transfusions, organ transplantations, or sharing needles and other items used to inject drugs. If you're pregnant when you have malaria, it's possible to give it to your unborn baby. But malaria isn't catchy like a cold or flu. And it can't be spread through sex or being near someone who has malaria.

Other names: malaria blood smear, malaria rapid diagnostic test, malaria by PCR

WHAT ARE THEY USED FOR?

Malaria tests are used to diagnose malaria. There are different types of malaria parasites that infect humans. The tests can tell which type you have. This information is important for choosing the right treatment to cure the disease.

Malaria testing is also used to help control the spread of the disease. That's because mosquitos become infected with malaria parasites when they bite a person who has malaria. Then they spread the disease by biting others. Testing helps you get treatment to get rid of the parasites before more mosquitos get the parasite by biting you.

WHY DO I NEED A MALARIA TEST?

You may need this test if you have symptoms of malaria and you've been in an area where malaria is common. Depending on the type of malaria parasite, symptoms usually begin between 7 to 30 days after an infected mosquito bites you. But symptoms can take up to a year to appear. So, if you have symptoms, remind your health care provider about any places you've traveled over the past 12 months.

The first symptoms of malaria are often like having the flu, and may include:

- Fever
- Chills
- Fatigue
- Headache
- Muscle pain
- Nausea and vomiting

If you are infected with certain types of malaria parasites, you may have symptoms which come and go every two or three days.

Without treatment, malaria can quickly become a life-threatening disease. As it gets worse, the symptoms will depend on the type of parasite you have. They may include:

- Anemia

Jaundice
Kidney failure
Seizures
Mental confusion
Coma

WHAT HAPPENS DURING A MALARIA TEST?

Your provider will examine you and ask about your symptoms and recent travels. If your provider thinks you could have malaria, you will have a blood test to check for signs of malaria parasites.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

There are two types of blood tests to diagnose malaria. Both tests are done when possible:

Blood smear test. For this test, a drop of your blood is put on a special slide. A laboratory professional will examine the slide under a microscope to look for parasites. This test can identify all types of malaria parasites.

Rapid diagnostic test (RDT). This test looks for certain proteins that come from malaria parasites. It can provide results in under 15 minutes, but an RDT can't accurately show which type of malaria parasite you have. This information affects your treatment. So, if an RDT shows that you have malaria, a blood smear is needed, too.

Your provider may order other blood tests, such as a complete blood count (CBC). If you do have malaria, the results of these tests will show how the disease is affecting you.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't any special preparations for a malaria test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If the results of your malaria test are negative, it means that no signs of malaria parasites were found in your blood sample. But this doesn't rule out malaria. That's because the number of malaria parasites in your blood can vary over time. If you have a negative result, more blood samples are usually taken every 12-24 hours for a total of 3 sets of tests. If these test results are all negative, you probably don't have malaria.

If the results of your malaria test are positive, it means you have malaria. Your provider will prescribe medicine right away. The medicine you take will depend on:

The type of malaria parasite you have

Whether the parasite has become resistant to certain medicines

Your symptoms and medical history

Whether you are pregnant

When treated early, malaria can usually be cured. If your provider strongly believes you have a serious case of malaria, you may start treatment before test results confirm you have the disease.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT MALARIA TESTS?

If you will be traveling to an area where malaria is common, talk with your provider before you go. You may be able to take medicine that can help prevent malaria.

You can also take steps to protect yourself against mosquito bites. This may reduce your risk of getting malaria and other infections transmitted by mosquitos.

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MCV (Mean Corpuscular Volume)

<https://medlineplus.gov/lab-tests/mcv-mean-corpuscular-volume/>

MCV (MEAN CORPUSCULAR VOLUME)

WHAT IS AN MCV BLOOD TEST?

MCV stands for mean corpuscular volume. An MCV blood test measures the average size of your red blood cells. Red blood cells carry oxygen from your lungs to every cell in your body. Your cells need oxygen to grow, reproduce, and make energy. If your red blood cells are too small or too large, it could be a sign of a blood disorder such as anemia, a lack of certain vitamins, or other medical conditions.

Other names: CBC with differential, Mean Cell Volume, Mean Corpuscular Volume

WHAT IS IT USED FOR?

An MCV blood test is often part of a complete blood count (CBC), a group of tests that measures the number and type of cells in your blood. It is used to check your general health.

An MCV test may also be used with other tests to help diagnose or monitor certain blood disorders, including anemia. There are several types of anemia, and each type has a different effect on the size, shape, and/or quality of your red blood cells. An MCV test can help diagnose which type of anemia you have.

WHY DO I NEED AN MCV BLOOD TEST?

Your health care provider may order a complete blood count, which includes an MCV test, as part of your routine checkup. You may also have the test if you have a chronic (long-lasting) condition that could lead to anemia or if you have symptoms of anemia, which could include:

Shortness of breath

Weakness or fatigue

Headache

Dizziness

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

WHAT HAPPENS DURING AN MCV BLOOD TEST?

During the test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an MCV blood test. If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put

in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

An MCV test alone cannot diagnose any disease. Your provider will use the results of your MCV, other test results, and your medical history to make a diagnosis.

If your results show that your red blood cells are smaller than normal, it may be a sign of:

Certain types of anemia, including iron deficiency anemia, the most common form of anemia. It happens when you don't have enough iron in your body.

Thalassemia, which is a group of blood disorders that are inherited (passed down through families). These disorders cause your body to make fewer healthy red blood cells and less hemoglobin. This can cause severe anemia.

If your results show that your red blood cells are larger than normal, it may be a sign of:

Pernicious anemia, which may be caused by:

A lack of vitamin B12

A disease that affects your body's ability to use vitamin B12, such as certain autoimmune diseases, celiac disease, or Crohn's disease

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Anemia caused by a lack of folic acid

Liver disease

It's also possible to have anemia with a normal MCV. This may happen if anemia is caused by conditions, such as:

A sudden loss of blood

Kidney failure

Aplastic anemia (uncommon)

If your MCV levels are not in the normal range, it doesn't always mean you have a medical problem that needs treatment. Your age, diet, activity level, medicines, menstrual period, and other conditions can affect the test results. Talk with your provider to learn what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN MCV BLOOD TEST?

If your provider thinks you may have anemia or another blood disorder, you may have other red blood cell tests with an MCV. These tests may include a red blood cell count and measurements of hemoglobin. All together, these tests are called red blood cell indices.

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Measles and Mumps Tests

<https://medlineplus.gov/lab-tests/measles-and-mumps-tests/>

MEASLES AND MUMPS TESTS

WHAT ARE MEASLES AND MUMPS TESTS?

Measles and mumps are infections caused by similar types of viruses. They are contagious, which makes them easy to spread from person to person but vaccines can help keep you from getting them.

There are different types of measles and mumps tests. These tests can:

Diagnose an infection you (or your child) have now or recently had.

Show whether you are immune (protected against infection) to measles and mumps because you had a vaccination or a past infection.

Measles and mumps are both very contagious:

Measles causes symptoms like a bad cold or the flu. It also causes a flat, red rash. The rash usually starts on your face and spreads all over your body.

Mumps causes symptoms like the flu with painful swelling of the salivary (spit) glands, which makes your cheeks and jaw puffy. In general, mumps is a milder disease than measles.

Most healthy people with measles or mumps infections will get better in about two weeks or less. But sometimes these infections cause serious problems, such as encephalitis (swelling of the brain). Measles may also cause pneumonia and serious problems during pregnancy.

In the United States, most people have been vaccinated against measles and mumps. But, the number of kindergarteners who have been vaccinated against measles and mumps has dropped below 95% (and even more in some communities). This has caused it to spread to more people. Measles can also come from unvaccinated people who have traveled to countries where measles is common and returned home with the disease.

Outbreaks of mumps, however, still happen in the U.S., mainly in group living situations, such as colleges. Some people who have been vaccinated may still get mumps during an outbreak, but they will usually have mild symptoms.

Other names: measles immunity test, mumps immunity test, measles blood test, mumps blood test, measles viral culture, measles viral culture, measles virus by RT-PCR, mumps virus by RT-PCR, measles antibody IgM, IgG, mumps antibody IgM, IgG

WHAT ARE THE TESTS USED FOR?

Measles testing and mumps testing can be used to:

Diagnose an active infection of measles or mumps. An active infection means you have a virus growing in your body now. Two types of tests are used to diagnose measles and mumps:

IgM antibody test. IgM antibodies are proteins your immune system makes when the measles or mumps virus is in your body. The test looks for IgM antibodies in a sample of your blood.

Molecular tests. These tests look for the measles or mumps virus in a sample of your blood, fluid from your nose or throat, or urine. A PCR (polymerase chain reaction) test is a common type of molecular test.

IgM antibody test. IgM antibodies are proteins your immune system makes when the measles or mumps virus is in your body. The test looks for IgM antibodies in a sample of your blood.

Molecular tests. These tests look for the measles or mumps virus in a sample of your blood, fluid from your nose or throat, or urine. A PCR (polymerase chain reaction) test is a common type of molecular test.

Find out whether you are immune to measles or mumps. If you had a vaccination for measles and mumps or were sick with either infection in the past, your body makes IgG antibodies to protect you against future infection, usually for the rest of your life. An IgG antibody test checks for these antibodies in your blood.

Help find and control outbreaks of measles and mumps in the community. Tests help find new and recent cases of measles and mumps so that public health officials can take steps to help stop the disease from spreading.

An MMR (measles, mumps, and rubella) antibody screening is a single blood test that is used to help diagnose measles,

mumps, and another type of viral infection called rubella or to see if you are immune to them. Rubella, also known as German measles, is also contagious, but its symptoms (low-grade fever, sore throat, and rash) are usually mild.

WHY DO I NEED A MEASLES OR MUMPS TEST?

You may need a test to check your immunity to measles and mumps if you:

Are pregnant or planning to become pregnant

Are going to attend a school or start a job that requires proof that you are immune

Are a health care worker who may be exposed to people with measles or mumps

Your health care provider may order a measles or mumps test if you or your child has symptoms of measles or mumps.

Symptoms of measles include:

Rash that starts on the face and spreads to the chest and legs

High fever

Cough

Runny nose

Sore throat

Pink eye (also called conjunctivitis)

Tiny white spots in the mouth

Symptoms of mumps include:

Swollen, painful jaw and puffy cheeks

Headache

Earache

Fever

Muscle aches

Loss of appetite

Painful swallowing

WHAT HAPPENS DURING MEASLES AND MUMPS TESTS?

There are different ways to test for measles and mumps. Antibody tests require a blood sample. Molecular tests often use a sample of blood or fluid from your mouth, throat, or nose:

Blood test. During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Swab test. A health care professional will use a special swab to take a sample from your nose, throat, or cheek.

Nasal aspirate or wash. A health care professional will insert a saline solution (salt water) into your nose and remove the sample with gentle suction.

If your provider thinks you have measles or mumps, you may have more than one test. Also, if your provider thinks measles or mumps has caused encephalitis or meningitis, you may need a spinal tap. For a spinal tap, your provider will insert a thin, hollow needle into your spine and withdraw a small amount of fluid for testing.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THESE TESTS?

You don't need any special preparations for measles testing or mumps testing.

ARE THERE ANY RISKS TO THESE TESTS?

There is very little risk to measles or mumps testing.

A blood test may cause slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

A swab test may cause a gagging feeling or even a tickle when your throat or nose is swabbed.
A nasal aspirate or wash may feel uncomfortable. These effects are temporary.

WHAT DO THE RESULTS MEAN?

Negative test results mean that no signs of measles or mumps were found in your sample. This usually means that you don't have measles or mumps now. If you had a negative result on an antibody test, it also means that you are not immune to measles or mumps.

Positive test results mean different things depending on which test you had:

A positive result on a molecular test (PCR) means that the measles or mumps virus was found in your blood, and you have an infection now.

A positive result on an IgM antibody test means you have a measles or mumps infection now or you have had a recent infection.

A positive result on an IgG antibody test means you are immune to measles or mumps because you had a vaccination or past infection.

Talk with your provider if you have questions about test results.

If you (or your child) have a measles or mumps infection, you should stay home for several days to recover and to make sure you don't spread the disease. Your provider will let you know when it will be okay to return to your regular activities.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT MEASLES AND MUMPS TESTS?

Vaccination is the best protection against measles and mumps and their complications. The Centers for Disease Control and Prevention (CDC) recommends:

For children: A two of total doses of the MMR (measles, mumps, and rubella) vaccine given:

Once between ages 12 and 15 months.

Once between ages four and six years.

Once between ages 12 and 15 months.

Once between ages four and six years.

For adults who aren't immune to measles and mumps: At least one dose of the MMR vaccine. Some people may need two doses.

If you're pregnant, you should not get the MMR vaccine. If you're not sure whether you're immune to measles and mumps, or if you have questions about vaccine safety, talk with your provider.

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Measuring Blood Pressure

<https://medlineplus.gov/lab-tests/measuring-blood-pressure/>

MEASURING BLOOD PRESSURE

WHAT IS A BLOOD PRESSURE MEASUREMENT?

Each time your heart beats, it pumps blood into your arteries (the blood vessels that carry blood from your heart to the tissues and organs in your body). A blood pressure measurement is a test that measures the force (pressure) in your arteries as your heart pumps. Blood pressure is measured as two numbers:

Systolic blood pressure (the first and higher number) measures pressure inside your arteries when your heart beats.

Diastolic blood pressure (the second and lower number) measures the pressure inside the artery when your heart rests between beats.

For example, a blood pressure reading of 120/80 means a systolic of 120 and a diastolic of 80.

Many things can change your blood pressure. It can increase or decrease based on:

Whether you're resting or exercising

If you eat or drink foods with caffeine, alcohol, or tobacco

Your age

If you're taking certain medicines

Your position, such as whether you're sitting, lying down, or standing

If your blood pressure is high on several different measurements on different days, you might have high blood pressure, also known as hypertension. Around half of American adults have high blood pressure. It increases the risk of life-threatening conditions, including heart attack and stroke. But high blood pressure rarely causes symptoms. A blood pressure measurement helps diagnose high blood pressure early, so it may be treated before it leads to serious complications.

Other names: blood pressure reading, blood pressure test, blood pressure screening, sphygmomanometry

WHAT IS IT USED FOR?

A blood pressure measurement is most often used to diagnose high blood pressure.

Blood pressure that's too low, known as hypotension, is much less common. But you may get tested for low blood pressure if you have certain symptoms. Unlike high blood pressure, low blood pressure usually causes symptoms. These include:

Dizziness or lightheadedness

Nausea

Cold, sweaty skin

Pale skin

Fainting

Weakness

WHY DO I NEED A BLOOD PRESSURE TEST?

A blood pressure measurement is often included as part of a regular checkup. Everyone age 3 or older should have their blood pressure checked by their provider at least once a year.

If you have high blood pressure or are at higher risk for it, you may need testing more often. You may be at higher risk if you:

Are 40 years old or older.

Are overweight or have obesity.

Have a family history of heart disease or diabetes.

Have a family history of high blood pressure.

Take birth control pills or other medicines that can increase your blood pressure.

Are Black/African American. Black/African Americans are more likely than other racial and ethnic groups to have high blood pressure.

Your provider can tell you how often you should get your blood pressure checked.

You may also need to have your blood pressure checked:

To see how well your blood pressure medicine is working (if you have high blood pressure)

If you are being treated in the hospital

To check if it's safe for you to have a medical or surgical procedure

To check if it's safe for you to play sports or do other strenuous activities or jobs

If you have symptoms of low blood pressure

WHAT HAPPENS DURING A BLOOD PRESSURE TEST?

A blood pressure test includes the following steps:

You will sit in a chair with your feet flat on the floor.

You will rest your arm on a table or other surface, so your arm is level with your heart. You may be asked to roll up your sleeve.

Your health care provider will wrap a blood pressure cuff around your arm. A blood pressure cuff is a strap-like device. It should fit snugly around your upper arm, with the bottom edge placed just above your elbow.

Your provider will inflate the blood pressure cuff using a small hand pump or by pressing a button on an automated device.

Your provider will measure the pressure manually (by hand) or with an automated device.

If done manually, they will place a stethoscope over the major artery (brachial artery) in your upper arm and elbow joint to listen to the blood flow and pulse as the cuff inflates and deflates.

If using an automated device, the blood pressure cuff automatically inflates, deflates, and measures pressure.

If done manually, they will place a stethoscope over the major artery (brachial artery) in your upper arm and elbow joint to listen to the blood flow and pulse as the cuff inflates and deflates.

If using an automated device, the blood pressure cuff automatically inflates, deflates, and measures pressure.

As the blood pressure cuff inflates, you'll feel it tighten around your arm. This blocks the blood flow in that part of your arm for a short time.

Your provider will then open a valve on the cuff to slowly release air. As the cuff deflates, blood flow will start again in your arm.

As the pressure falls, a measurement is taken when the sound of blood pulsing is first heard. This is the systolic pressure.

As the air continues to be let out, the blood pulsing sound will start to go away. When it completely stops, another measurement is taken. This is the diastolic pressure.

This test only takes about one minute to complete.

Your provider may also ask to take your blood pressure while you stand or lie down. They may also want to repeat the blood pressure measurement on your other arm.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Several things can increase your blood pressure measurement. Tell your provider if you take any medicines, since some medicines may affect your results.

To get a more accurate blood pressure measurement, avoid these things for at least 30 minutes before having your blood pressure taken:

Exercise

Alcohol

Tobacco products

Food or drinks containing caffeine

Also, make sure to go to the bathroom before the test.

ARE THERE ANY RISKS TO THE TEST?

You may have a little discomfort when the blood pressure cuff inflates and squeezes your arm. But this feeling only lasts for a few seconds.

WHAT DO THE RESULTS MEAN?

Your results, also known as a blood pressure reading, will contain two numbers. The top or first number is the systolic pressure. The bottom or second number is the diastolic pressure. Blood pressure readings are also labeled by categories, ranging from normal to crisis. Blood pressure is measured in millimeters of mercury (mm Hg).

Your reading may show your blood pressure is:

Your blood pressure measurement can go up and down. Your provider will only usually diagnose you with high blood pressure if you have two or more high blood pressure readings on different days. They may also consider your blood pressure readings over a period of time, your medical and family history, and if you have other symptoms before diagnosing you with high blood pressure.

If your child is younger than 13, your provider will compare their blood pressure readings with what is common for children the same age, sex, and height.

If you've been diagnosed with high blood pressure, your provider may recommend:

Lifestyle changes and/or medicines to control your blood pressure.

You to regularly check your blood pressure at home with an automated blood pressure monitor. An at-home blood pressure monitor usually includes a blood pressure cuff and a digital device to record and display blood pressure readings.

At-home blood pressure monitoring is not a replacement for regular visits with your provider. But it can provide important information, such as whether your treatment is working or if your condition may have gotten worse.

Also, home monitoring may make the test less stressful. Many people get nervous about getting their blood pressure taken at a provider's office. This is called "white coat syndrome." It can cause a temporary rise in blood pressure, making the results less accurate.

To get the most accurate results when taking your blood pressure at home:

Make sure the blood pressure cuff fits snugly but isn't too small.

Use a blood pressure cuff that goes around your upper arm. It is more accurate than the ones that measure on your wrist or finger.

Take your blood pressure at the same time of day each time.

Rest in the chair for about five minutes before taking your blood pressure.

Sit up straight with your legs uncrossed and your feet flat on the floor.

Seek immediate medical attention or call 911 if your blood pressure is over 180/120 mmHg, you have chest pain, shortness of breath, back pain, numbness or weakness, trouble speaking, or vision changes.

For more information about home monitoring of blood pressure, talk to your provider.

If you were tested for low blood pressure, a blood pressure reading of 90 systolic, 60 diastolic (90/60 mmHg) or lower is considered abnormal. Treatments for low blood pressure may include medicines and certain dietary changes.

Seek immediate medical attention or call 911 if you have symptoms of extremely low blood pressure, such as confusion, a weak and rapid pulse, or cold skin.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A BLOOD PRESSURE MEASUREMENT?

If you were diagnosed with high blood pressure, your provider may recommend some lifestyle changes such as:

Getting regular physical activity. Staying active can help lower your blood pressure and also help manage your weight. Check with your provider before beginning an exercise program.

Being at a healthy weight. Losing as little as 5 pounds can lower your blood pressure if you are overweight.

Eating a healthy diet that includes fruits, vegetables, and whole grains. Limit foods high in saturated fat and total fat. The DASH eating plan is a heart-healthy style of eating.

Reducing salt in your diet. Most adults and children over age 14 should have less than 2,300 mg of salt per day.

Limiting alcohol use. Talk to your provider about how much alcohol you drink. They may recommend that you reduce the amount of alcohol you drink or that you stop drinking alcohol.

Quitting smoking (if you smoke).

Managing stress.

Getting enough good-quality sleep.

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Meningococcal Disease Tests

<https://medlineplus.gov/lab-tests/meningococcal-disease-tests/>

MENINGOCOCCAL DISEASE TESTS

WHAT IS A MENINGOCOCCAL DISEASE TEST?

Meningococcal disease is the name for any infection caused by meningococcal bacteria (called *Neisseria meningitidis*). Meningococcal disease tests look for these bacteria in your blood or cerebrospinal fluid (CSF), the fluid in spaces around your brain and spinal cord.

Meningococcal disease can be spread from person to person by coming into contact with someone's respiratory secretions (saliva) or throat secretions (spit). The bacteria aren't as contagious as germs that cause the common cold or the flu. For the bacteria to spread, it takes close contact (kissing) or lengthy contact (living together).

The most common signs and symptoms of meningococcal disease are a sudden high fever, neck stiffness, confusion, nausea, vomiting, fatigue, and rash. Diagnosing meningococcal disease can be difficult because the signs and symptoms are often similar to other illnesses. If not treated early and with the right type of treatment, meningococcal infection can get worse quickly and cause death.

If your health care provider thinks you might have meningococcal disease, they will order one or more of these meningococcal disease tests:

Meningococcal culture test, a bacterial culture test that helps find meningococcal bacteria in a sample of your blood or CSF. Your sample will be sent to a lab where the bacteria cells will be put in a dish with a special substance that lets the bacteria grow.

Meningococcal polymerase chain reaction (PCR) test, a test that checks for small amounts of the genetic material of meningococcal bacteria in a sample of your blood or CSF.

Meningococcal antigen test, a test that looks for antigens (proteins from a foreign substance that trigger your immune system to fight it off) specific to the meningococcal bacteria.

The two most common types of meningococcal infections are meningitis and bloodstream infections.

In meningococcal meningitis, the lining of the brain and spinal cord become infected by the bacteria, which causes inflammation (swelling).

In meningococcal bloodstream infections, the bacteria get into the blood and cause damage to the walls of the blood vessels, which causes bleeding in the skin and organs.

Anyone can get meningococcal disease, but some people have a higher risk. This includes those who:

Travel outside of the U.S. to areas where meningococcal disease is common

Are younger than one year old

Are teenagers and young adults ages 16-23 years old

Are an adult aged 65 years and older

Have a medical condition that weakens the immune system such as HIV, having no spleen or an injured spleen, or a disorder of the complement system, which helps the body fight infections

Live in close quarters with many other people like on a college campus or military base

Take a type of medicine called a complement inhibitor, which is used to treat certain rare diseases

Other names: *Neisseria meningitidis* Tests, *N. meningitidis* Tests

WHAT IS IT USED FOR?

Meningococcal disease tests are used to find out whether you have meningococcal disease.

WHY DO I NEED A MENINGOCOCCAL DISEASE TEST?

Meningococcal meningitis and bloodstream infections are serious and can become deadly in a matter of hours. It's important to seek medical attention immediately and get the right diagnosis and treatment to help prevent serious health

problems.

Symptoms are different for meningococcal meningitis and bloodstream infections.

The most common symptoms of meningococcal meningitis for adults include:

Fever

Headache

Stiff neck

Additional symptoms of meningitis for adults include nausea, vomiting, light sensitivity, and confusion.

The most common symptoms of meningococcal meningitis for babies include:

Appearing slow or inactive

Irritability

Feeding poorly

Having a soft spot on the skull

Having abnormal reflexes

Vomiting

The most common symptoms of a meningococcal bloodstream infection in adults include:

Cold hands and feet

Diarrhea or nausea without vomiting

Fatigue

Fever and chills

Rapid breathing

Severe aches or pain in the muscles, joints, chest, or abdomen (belly)

Dark, purple rash (in the later stages)

Call your provider right away if you think you or your child might have meningococcal disease.

WHAT HAPPENS DURING A MENINGOCOCCAL DISEASE TEST?

Your provider will take samples from your blood and may also take a sample from your CSF to do a spinal tap (lumbar puncture) to check for the *Neisseria meningitidis* bacteria in your body. These samples will then be sent to a laboratory to test for meningococcal disease and figure out which antibiotic will work best for treatment.

If the meningococcal bacteria are in the samples, the laboratory will be able to culture (grow) them using the meningococcal culture test. If it's not clear from this test whether your samples have the meningococcal bacteria in them, your provider may order additional tests such as the meningococcal antigen test or PCR test.

During a blood test:

If your provider thinks you might have a meningococcal infection, they may collect a blood sample for a meningococcal culture test, PCR test, or antigen test.

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

During a spinal tap:

If your provider thinks you might have a meningococcal infection, they may collect a CSF sample for a meningococcal culture test, PCR test, or antigen test.

To get a sample of CSF, a provider will do a procedure called a spinal tap, also known as a lumbar puncture. A spinal tap is usually done in a hospital. During the procedure:

You will lie on your side or sit on an exam table.

Your provider will clean your back and inject an anesthetic into your skin, so you won't feel pain during the procedure. They may put a numbing cream on your back before this injection.

When the area on your back is completely numb, your provider will insert a thin, hollow needle between two vertebrae in your lower spine. Vertebrae are the small backbones that make up your spine.

Your provider will withdraw a small amount of cerebrospinal fluid for testing. This will take about five minutes.

You'll need to stay very still while the fluid is being withdrawn.

Your provider may ask you to lie on your back for an hour or two after the procedure. This may prevent you from getting a headache afterward.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

Talk to your provider and ask about any special preparations. You may need to fast¹ (not eat or drink) before the test. You may also need to stop taking certain medicines, so be sure to tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

With a spinal tap, there is very little risk. You may feel a little pinch or pressure when the needle is inserted. After the test, you may feel some pain or tenderness in your back or have some bleeding where the needle was inserted. You may also get a headache. The headache may last for several hours or up to a week or more, but your provider may suggest treatment to relieve the pain.

WHAT DO THE RESULTS MEAN?

The results will show whether you have meningococcal disease.

A positive result means that meningococcal bacteria (*Neisseria meningitidis*) were found in your blood or CSF sample. You most likely have meningococcal disease.

A negative result means that no meningococcal bacteria were found in your blood or CSF sample. Most likely, your symptoms are not caused by meningococcal disease.

If your provider thinks you have meningococcal disease, you will get treatment with antibiotics. Ideally treatment will begin less than 24 hours after your diagnosis is confirmed.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A MENINGOCOCCAL DISEASE TEST?

There are three types of meningococcal vaccines² available in the U.S. While meningococcal disease is uncommon, it can cause serious illness and death in people of all ages. Meningococcal vaccines help protect against meningococcal disease.

The CDC recommends the meningococcal vaccine for all preteens, teens, and other children and adults at increased risk.

Ask your provider if they recommend that you or your child get a vaccine.

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Mental Health Screening

<https://medlineplus.gov/lab-tests/mental-health-screening/>

MENTAL HEALTH SCREENING

WHAT IS A MENTAL HEALTH SCREENING?

A mental health screening is a standard set of questions that a person answers to help a health care provider check for signs of a mental disorder. The questions help the provider learn about a person's mood, thinking, behavior, and memory.

A mental health screening is a way to catch mental health conditions early. If the screening shows signs of a disorder, more testing is usually needed to diagnose a specific mental disorder. Mental disorders are also called mental illnesses, and a mental health screening may be called a "mental illness test" or a "psychology test."

More than half of all Americans will have a mental disorder at some point in their lives. Their symptoms may range from mild to severe. Common mental disorders include:

Mood disorders, which include depression, bipolar disorder, seasonal affective disorder (SAD), and self-harm.

Anxiety disorders, which include panic disorder, phobias, and obsessive-compulsive disorder (OCD). Anxiety is a common disorder in children.

Eating disorders, which include anorexia and bulimia.

Attention deficit hyperactivity disorder (ADHD). ADHD is one of the most common mental health disorders in children. It can also continue into adulthood.

Post-traumatic stress disorder (PTSD).

Personality disorders.

Substance use disorders, which include alcohol use disorder and drug use and addiction.

Psychotic disorders, which include schizophrenia.

These and other mental disorders affect people of all ages, including children. So, there are special mental health screening tests designed for children, teenagers, and older adults. Some screening tests look for general signs of the most common mental disorders. Other screening tests look for signs related to specific types of disorders.

Mental health screening can be an important part of your total health at every stage of life. Mental health symptoms can be a sign of certain physical conditions. And certain mental disorders can increase the risk of developing physical health problems. With proper mental health screening, diagnosis, and treatment, people with mental health disorders can get better and many recover completely.

Other names: mental health assessment, mental illness test, psychological evaluation, psychology test, psychiatric evaluation

WHAT IS IT USED FOR?

A mental health screening is usually used as the first step to find out if a person has signs of a mental disorder. It may be part of a routine checkup. A screening test is used to see whether a person:

Has a risk for developing a mental disorder

Needs more testing to diagnose or rule out a mental health disorder

Needs mental health care right away, before diagnostic testing can be finished

Mental health screening may also be used to see if treatment for a mental disorder is working.

A primary care provider may use mental health screening to help decide if a person needs to see a mental health provider for testing and/or treatment. A mental health provider is a health care professional who specializes in diagnosing and treating mental health problems. A mental health provider may use the results of a screening test to choose which other tests are needed to diagnose or rule out a specific mental disorder.

WHY DO I NEED A MENTAL HEALTH SCREENING?

You or your child may need a mental health screening if you have symptoms of a mental disorder. Different disorders have specific symptoms.

General symptoms of mental disorders may include:

Eating or sleeping too much or too little

Pulling away from people and usual activities

Fatigue and lack of energy

Feeling helpless, hopeless, or numb like nothing matters

Smoking, drinking, or using drugs more than usual

Feeling unusually confused, forgetful, nervous, angry, worried, or scared

Severe mood swings that cause problems in relationships

Thoughts and memories that you can't get out of your head

Hearing voices or believing things that aren't true

Thinking about death, suicide or harming yourself or others

Signs of mental disorders in children may also include:

Frequent tantrums and other behavior problems

Frequent stomachaches or headaches without a known medical cause

Difficulty in school

Repeating actions or constantly checking things out of fear that something bad will happen

Talking a lot about fears, worries, death, or suicide

If you or someone you know needs immediate help for a mental health disorder:

Call 911 or go to your local emergency room

Contact a crisis hotline. In the United States, you can reach the National Suicide and Crisis Lifeline at any time:

Call or text 988

Chat online with Lifeline Chat

TTY users: Use your preferred relay service or dial 711 then 988

Call or text 988

Chat online with Lifeline Chat

TTY users: Use your preferred relay service or dial 711 then 988

Veterans can contact the Veterans Crisis Line:

Call 988 then press 1

Text 838255

Chat online

Call 988 then press 1

Text 838255

Chat online

Call your mental health provider or other provider

Reach out to a loved one or close friend

WHAT HAPPENS DURING A MENTAL HEALTH SCREENING?

If you are having a mental health screening, you'll answer a set of questions about your symptoms. The questions will be about your feelings, mood, sleep, appetite, and other parts of your life. A provider may ask you the questions or you may fill out a questionnaire and discuss your answers afterwards. It's important that your answers are honest and complete.

If your primary care provider is doing the screening, you may also have a physical exam and blood tests. There are no medical tests that can diagnose mental health disorders. But certain blood tests can show if a physical condition, such as thyroid disease or an electrolyte imbalance, is causing mental health symptoms.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

If your provider thinks your symptoms could be caused by physical problems with your brain or nerves, you may have a neurological exam or imaging tests of your brain.

If your child is having a mental health screening, the test will be geared to your child's age and abilities. You may be asked to fill out a questionnaire about your child's behavior.

The screening may be done by your child's primary care provider or a mental health provider who works with children and teenagers. Your child's provider may also order medical tests to look for physical causes of mental disorders. Some schools provide mental health screening services.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A MENTAL HEALTH SCREENING?

You usually don't need any special preparations for a mental health screening. If your child is having a screening, you may be asked to keep notes on your child's behavior for a few days before the test.

ARE THERE ANY RISKS TO SCREENING?

There is no risk to having a physical exam or taking a questionnaire.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Usually, the provider who did the screening will explain the results. If the results show signs of a mental health disorder, the next steps depend on the type of disorder and how serious it may be. If a primary care provider did the screening, the provider may:

- Talk with you about treatments

- Order tests to check for other health problems that may be causing symptoms

- Refer you or your child to a mental health provider

A mental health provider may do more tests to confirm a diagnosis and help develop a treatment plan. Starting treatment as soon as possible may improve the chance of recovery.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A MENTAL HEALTH SCREENING?

There are many types of mental health providers who treat mental disorders. Your or your child's primary health care provider or school can help you find the right support.

These are some of the providers who have training to treat mental health disorders:

Psychiatrists are medical doctors who specialize in mental health. Psychiatrists can prescribe medicine.

Psychologists generally have doctoral degrees, but they do not have medical degrees. They can't prescribe medicine unless they have a special license. Some psychologists work with providers who can prescribe medicine. Psychologists may use one-on-one counseling and/or group therapy sessions.

Psychiatric-mental health nurses are nurses with special training in mental health problems. Nurses who may have a master's or doctoral degree in psychiatric-mental health nursing include, advanced practice registered nurses (APRNs), certified nurse practitioners (CNPs), and clinical nurse specialists (CNSs). In some states, certain nurses can prescribe medicines.

Licensed clinical social workers have at least a master's degree in social work with special training in mental health. They can't prescribe medicine, but may work with providers who can prescribe medicine. Providers who are licensed clinical social workers usually have LCSW or LICSW after their names.

Licensed professional counselors (LPC) may also be called clinicians or therapists. States have different names of these licenses, such as LMFT (licensed marriage and family therapist). These professionals usually have a master's degree in a field related to mental health. They can't prescribe medicine but may work with providers who can prescribe.

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RELATED HEALTH TOPICS

Alcohol Use Disorder (AUD)

Antidepressants

Anxiety

Attention Deficit Hyperactivity Disorder

Bipolar Disorder

Child Mental Health

Depression

Drug Use and Addiction

Eating Disorders

How to Improve Mental Health

Mental Disorders

Mental Health

Opioids and Opioid Use Disorder (OUD)

Post-Traumatic Stress Disorder

Postpartum Depression

Psychotic Disorders

Suicide

RELATED MEDICAL TESTS

ADHD Screening

Alcohol Use Screening Tests

Depression Screening

Differential Diagnosis

Drug Use Screening Tests

Neurological Exam

Obsessive Compulsive Disorder (OCD) Test

Panic Disorder Test

Postpartum Depression Screening

Suicide Risk Screening

Tricyclic Antidepressant (TCA) Screen

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Methylmalonic Acid (MMA) Test

<https://medlineplus.gov/lab-tests/methylmalonic-acid-mma-test/>

METHYLMALONIC ACID (MMA) TEST

WHAT IS A METHYLMALONIC ACID (MMA) TEST?

This test measures the amount of methylmalonic acid (MMA) in your blood or urine. MMA is a substance made in small amounts when you digest protein. MMA is needed for metabolism, which is the process that your body uses to change food into energy.

Vitamin B12 plays an important role in metabolism. Part of this role is breaking down MMA into substances that help with metabolism. If your body doesn't have enough vitamin B12, it will continue to make more MMA than is broken down. High MMA levels can be a sign of a vitamin B12 deficiency.

Vitamin B12 deficiency can happen if you are not eating enough B12 in your diet or if your body has difficulty absorbing it. Your body cannot make vitamin B12 on its own, so you have to get it from the foods you eat. This vitamin is found in foods such as:

Seafood such as fish, clams, and oysters

Dairy products

Chicken

Eggs

Meat

Vitamin B12 is also found in some fortified foods, meaning the food has vitamins added to it. It's also available as a supplement.

Vitamin B12 also helps to make red blood cells. A deficiency can lead to health issues, including anemia. This is a condition in which your blood has a lower-than-normal amount of red blood cells.

Other names: MMA

WHAT IS IT USED FOR?

An MMA test is most often used to diagnose a vitamin B12 deficiency.

The test is usually also included as part of a series of tests called a newborn screening. A newborn screening checks for a variety of health conditions. An MMA test checks for methylmalonic acidemia, a rare genetic disorder. Symptoms of this disorder can range from mild to severe and may include:

Vomiting

Dehydration

Developmental delays

Intellectual disability, which means that a person has trouble with intellectual functioning (such as learning and communicating) and everyday social and life skills.

WHY DO I NEED AN MMA TEST?

You may need this test if you have symptoms of a vitamin B12 deficiency. These may include:

Fatigue

Diarrhea

Dizziness

Shortness of breath

Loss of appetite

Tingling in hands and/or feet

Mood changes

Confusion

Irritability

Pale skin

You might be at a higher risk for B12 deficiency if you:

Are a strict vegetarian or vegan

Have a digestive disease, such as celiac disease or Crohn's disease

Are a person with an alcohol use disorder (AUD)

Are older than 75 years

Have Sjogren's syndrome, a disease that causes dry eyes and mouth

Have had certain types of stomach or intestinal surgery

If you have a new baby, they will probably be tested as part of a newborn screening.

WHAT HAPPENS DURING AN MMA TEST?

MMA levels may be checked in blood or urine.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

During a newborn screening, a health care provider will clean your baby's heel with alcohol and poke the heel with a small needle. The provider will collect a few drops of blood and put a bandage on the site.

MMA urine testing may be ordered as a 24-hour urine sample test or a random urine test.

For a 24-hour urine sample test, you'll need to collect all your urine during a 24-hour period. This is called a 24-hour urine sample test. You'll be given a special container to collect your urine and instructions on how to collect and store your samples. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

For a random urine test, your sample of urine may be collected any time of the day.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for several hours before your test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to you or your baby during an MMA blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

Your baby may feel a little pinch when the heel is poked, and a small bruise may form at the site. This should go away quickly.

There is no known risk to having a urine test.

WHAT DO THE RESULTS MEAN?

If your results show higher-than-normal levels of MMA, it may mean you have a vitamin B12 deficiency. The test can't show how much of a deficiency. To help make a diagnosis, your provider may compare your results with other tests including a homocysteine blood test and/or vitamin B tests.

Lower-than-normal levels of MMA are not common and are not considered a health problem.

If your baby has moderate or high levels of MMA, it may mean that they have methylmalonic acidemia. If it's not treated,

it can cause life-threatening complications. If your baby is diagnosed with this disorder, talk to your child's provider about treatment options.

If you have questions about your results, talk to your provider. Your provider will consider your symptoms, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

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Microalbumin Creatinine Ratio

<https://medlineplus.gov/lab-tests/microalbumin-creatinine-ratio/>

MICROALBUMIN CREATININE RATIO

WHAT IS A MICROALBUMIN CREATININE RATIO TEST?

A microalbumin creatinine ratio urine test checks the levels of the protein albumin in your urine (pee). Abnormal amounts of this protein may be an early sign of kidney disease.

Your kidneys are two organs on either side of your spine near your waist. Filters in your kidneys remove waste and extra water from your blood and gets rid of them through urine. Albumin, however, is not a waste product. It is a common protein found in blood. Therefore, very little, if any, of it should pass into your urine.

In healthy kidneys, only a trace of albumin, at most, will get through the filters. This trace is often called microalbumin. But if your kidneys are damaged, large amounts of albumin may pass into your urine.

Creatinine, however, often appears in your urine. It is a waste product that comes from daily wear and tear on your muscles. Your kidneys usually get rid of this protein at a steady rate. The microalbumin creatinine ratio test compares the levels of albumin in your urine to your levels of creatinine. This ratio can help your health care provider tell if your body is getting rid of albumin at an increased rate.

Other names: albumin-creatinine ratio; urine albumin; microalbumin, urine; ACR; UACR

WHAT IS IT USED FOR?

A microalbumin creatinine ratio test is most often used to look for signs of kidney disease in people who have a high risk of developing it but don't have symptoms. You're more likely to develop kidney disease if you:

Have diabetes

Have high blood pressure

Have a family health history of kidney disease, diabetes, or high blood pressure

Have heart disease

Are over 50 years old

Smoke

Have obesity

A microalbumin creatinine ratio test may be used to monitor your kidneys if you have been diagnosed with kidney disease. It can also help your provider see how well any treatment for kidney disease is working.

WHY DO I NEED A MICROALBUMIN CREATININE RATIO TEST?

Early kidney disease usually doesn't have any symptoms. So, if you have a high risk for kidney disease, you need a microalbumin creatinine ratio test so you can determine if you have early signs of kidney disease. If you do, treating it early can keep it from getting worse.

Because diabetes is the most common causes of kidney disease, it is important for anyone with this condition to get tested for kidney disease every year.

If you have other conditions that increase your risk for getting kidney disease, such as high blood pressure or heart disease, ask your provider how often you need to get tested.

WHAT HAPPENS DURING A MICROALBUMIN CREATININE RATIO TEST?

There are several ways to collect a urine sample for a microalbumin creatinine ratio test:

A "random" or "spot" urine sample means that you collect a single urine sample at any time of the day. You will usually collect this sample at your provider's office or at a medical lab.

An early morning or timed urine sample is collected first thing in the morning or after not going to the bathroom for four hours. You'll likely be given a kit and instructions on how to collect your urine at home.

A 24-hour urine sample requires you to collect all your urine over a 24-hour period. This test is usually used to follow up on abnormal results from a test on a single sample of urine. A 24-hour urine test is the most accurate way to measure albumin in urine.

For a random urine sample, a health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

For home collection— for an early morning or timed urine sample, the instructions will be about the same as for a random urine sample. Be sure to follow the instructions that come with your collection kit.

For a 24-hour urine sample, you'll be given a special container to collect your urine over a full day and instructions on how to collect and store your sample. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or in a cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

If you have hemorrhoids— that bleed or are having your menstrual period, tell your provider before your test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Before providing your urine sample, you may need to avoid:

Intense exercise. Hard exercise may increase the amount of albumin in your urine for a short time.

Eating meat. Meat can affect your creatine levels, so you may be asked not to eat any for a day before your test.

Check with your provider about how to prepare for your test. Be sure to ask if any medicines or supplements you take could affect your results.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to providing a urine sample for a microalbumin creatinine ratio test.

WHAT DO THE RESULTS MEAN?

If your microalbumin creatinine ratio shows an abnormal amount of albumin in your urine, you will likely need more tests to confirm your results. That's because temporary increases in albumin levels may be caused by exercise, certain medicines, fever, and inflammation in the body.

Usually, you will have two more tests during the three to six months after your first test. If two out of three tests show abnormal levels of albumin in your urine, you may have early-stage kidney disease.

Higher or increasing amounts of albumin in your urine usually mean you have:

More serious kidney disease that's likely to get worse faster.

A higher risk for developing problems with your heart and blood vessels (vascular diseases).

If you have higher-than-normal amounts of albumin in your urine, you may need to be tested more often.

Smaller amounts of albumin in your urine don't always mean you have kidney disease. Other conditions that cause inflammation may increase albumin in urine, including:

Urinary tract infections.

Gum infections (periodontitis).

Hepatitis.

Your test results may also be affected by your muscle density and your diet, age, and race.

If you have an abnormal result, your provider will usually order other kidney tests, including a blood test called an estimated glomerular filtration rate (eGFR) test. Talk with your provider about what your test results mean. If you're diagnosed with early kidney disease, there are steps you can take to reduce the amount of albumin in your urine and protect your health.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A MICROALBUMIN CREATININE RATIO?

It's easy to confuse prealbumin with albumin. Although they sound similar, prealbumin is a different type of protein. A prealbumin test¹ is not part of testing your kidney health.

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Mononucleosis (Mono) Tests

<https://medlineplus.gov/lab-tests/mononucleosis-mono-tests/>

MONONUCLEOSIS (MONO) TESTS

WHAT ARE MONONUCLEOSIS (MONO) TESTS?

Mononucleosis (mono) is an infectious disease caused by viruses, including:

Epstein-Barr virus (EBV), which is the most common cause

Cytomegalovirus

Toxoplasmosis

HIV

Rubella

Hepatitis A, B, C

Adenovirus

These viruses are commonly spread through body fluids. Mono is sometimes called the kissing disease because it is spread through saliva. So, besides kissing, you can get it if you share a fork, spoon, drinking glass, straw, food, toothbrush, or lip balm with a person who has mono. These viruses can also spread through blood and semen during sexual contact, blood transfusions, and organ transplants.

EBV is a type of herpes virus that is very common. Most people get infected with EBV at some point in their lives, but only some of them will get symptoms of mono.

Many people become infected with EBV in childhood. EBV infections in children usually do not cause symptoms. But when they do, it's difficult to tell the difference between the symptoms of mono and the symptoms of other childhood illnesses such as the flu.

Teens and young adults, though, are more likely to get mono. In fact, at least one out of four teens and young adults who get EBV will develop mono. They are also more likely to have symptoms that are typical of mono, such as:

Extreme fatigue

Fever

Sore throat

Swollen lymph nodes in the neck

Enlarged spleen

Swollen liver

Rash

Mono is rarely serious, but symptoms can linger for weeks or months.

Mono is typically diagnosed based on your symptoms. However, in some cases, specific lab tests may be needed to identify the cause of illness. An example would be if someone is having severe or life-threatening illness.

Types of mono tests include:

Monospot test. This test looks for specific antibodies in the blood. These antibodies show up during or after certain infections, including mono.

EBV antibody test. This test looks for EBV antibodies, the main cause of mono. There are different types of EBV antibodies. If certain types of antibodies are found, it may mean you were infected recently. Other types of EBV antibodies may mean you were infected in the past.

Other names: monospot test, mononuclear heterophile test, heterophile antibody test, EBV antibody test, Epstein-Barr virus antibody test, mononucleosis spot test

WHAT ARE THEY USED FOR?

Mono tests are used to help diagnose a mono infection. Your provider may use a monospot to get fast results. Results are usually ready within an hour. But this test has a high rate of false negatives. This means your test shows you don't have mono, but you actually do have it. So, your provider may also order an EBV antibody test and other tests that look for infections. These include:

Complete blood count and/or blood smear, which checks for high levels of white blood cells, a sign of infection.
Throat culture, to check for strep throat, which has similar symptoms to mono. Strep throat is a bacterial infection treated with antibiotics. Antibiotics don't work on viral infections like mono.

WHY DO I NEED A MONO TEST?

Your provider may order one or more mono tests if you or your child has symptoms of mono. Symptoms include:

Fever
Sore throat
Swollen glands, especially in the neck and/or armpits
Fatigue
Headache
Rash
Enlarged liver, spleen, or both
Swollen tonsils

An enlarged spleen and a swollen liver are less common symptoms. For some people, their liver or spleen, or both, may still be enlarged even after their fatigue ends.

Most people get better in two to four weeks; however, some people may feel fatigued for several more weeks. Occasionally, the symptoms of mono can last for six months or longer.

WHAT HAPPENS DURING A MONO TEST?

You will need to provide a sample of blood from your fingertip or a vein.

For a fingertip blood test, a health care professional will prick your middle or ring finger with a small needle. After wiping away the first drop of blood, he or she will place a little tube on your finger and collect a small amount of blood. You may feel a pinch when the needle pricks your finger.

For a blood test from a vein, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

Both types of tests are quick, usually taking less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a fingertip blood test or blood test from a vein.

ARE THERE ANY RISKS TO MONO TESTS?

There is very little risk to having a fingertip blood test or blood test from a vein. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your monospot test result is positive, it may mean you or your child has mono. If it is negative, but you or your child still have symptoms, your provider will probably order an EBV antibody test.

If the monospot test is negative, but you or your child still have symptoms, your provider will probably order an EBV antibody test.

If your EBV test is negative, it means you don't currently have an EBV infection and were never infected with the virus. A negative result means your symptoms are probably caused by another disorder.

If your EBV test is positive, it means EBV antibodies were found in your blood. The test will also show which types of

antibodies were found. This allows your provider to find out whether you were infected recently or in the past.

While there is no cure for mono, you can take steps to relieve symptoms. These include:

Get plenty of rest

Drink lots of fluids

Suck on lozenges to soothe a sore throat

Take over-the-counter medicines for pain and fever. But don't give aspirin to children or teens because it may cause Reye syndrome, a serious, sometimes fatal, disease that affects the brain and liver.

Antibiotics are not used to treat mono. If you have a bacterial infection along with mono, you should not take penicillin antibiotics like ampicillin or amoxicillin. Otherwise, you may experience a reaction, such as a rash. If you have severe symptoms, your provider may suggest additional treatment based on which organs in your body are affected by the mono. Mono usually goes away on its own within a few weeks. Fatigue may last a bit longer.

Mono can cause an enlarged spleen, which could rupture and cause a medical emergency. To try to protect the spleen, providers recommend avoiding intense exercise and contact sports until you fully recover (about a month). If you have questions about your results or treatment for mono, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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MPV Blood Test

<https://medlineplus.gov/lab-tests/mpv-blood-test/>

MPV BLOOD TEST

WHAT IS AN MPV BLOOD TEST?

MPV stands for mean platelet volume. Platelets (also called thrombocytes) are small blood cells that stick together to make blood clots that stop or slow bleeding when you have a cut or injury. Platelets are made in your bone marrow. An MPV blood test measures the mean (average) size of your platelets. Looking at the size of platelets provides information about how well they're working. This test can help diagnose bleeding disorders and diseases of the bone marrow.

Other names: Mean Platelet Volume

WHAT IS IT USED FOR?

An MPV blood test is used to help diagnose or monitor many blood-related conditions. It may also help diagnose other health conditions or find out how serious they may be. A test called a platelet count, which measures the total number of platelets in your blood, is often done with an MVP test.

WHY DO I NEED AN MPV BLOOD TEST?

Your health care provider may order an MPV blood test as part of a complete blood count (CBC). This is a test that measures many different parts of your blood, including platelets. It is often part of a routine checkup.

You may also need an MPV test if you have symptoms of a blood disorder that may involve having too many or too few platelets. These symptoms can include:

Bleeding that takes a long time to stop, even from minor cuts

Nosebleeds

Bleeding from brushing your teeth

Small red or purplish spots on the skin

Bruising easily

Abnormal vaginal bleeding (especially heavy menstrual periods)

Feeling weak or dizzy

Lasting headache and dizziness

Pain and burning in the feet and hands

WHAT HAPPENS DURING AN MPV BLOOD TEST?

During the test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an MPV blood test. If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

An MPV test alone can't diagnose a health condition. Your provider will use your MPV results, platelet counts, and other test results to get a more complete picture of your health.

Newly made platelets are larger than older platelets. So looking at the average size of your platelets can provide information about whether your bone marrow is making a lot of new platelets.

With a high MPV test result, your platelets are larger than usual. This may mean that your older (smaller) platelets are being destroyed at an increased rate and your bone marrow is making many new platelets. That can be a sign of:

Thrombocytopenia, not having enough platelets

Myeloproliferative diseases, blood cancers in which the bone marrow makes too many platelets or other blood cells

Preeclampsia, a complication that usually starts after the 20th week of pregnancy and causes high blood pressure

Heart disease

Diabetes

Hemolytic anemia

With a low MPV test result, your platelets are smaller than usual. This may mean your bone marrow isn't making enough new platelets. That can be a sign of:

Certain cancers

Side effects from certain medicines

Aplastic anemia

Autoimmune diseases

Bacterial or viral infections

Genetic conditions

Alcohol use disorder (AUD)

To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN MPV BLOOD TEST?

Many things can affect the results of your MPV blood test. Your age, sex, race and ethnicity, lifestyle (diet, smoking, drinking alcohol, physical activity), and genes can affect both your MPV and platelet counts.

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MRSA Tests

<https://medlineplus.gov/lab-tests/mrsa-tests/>

MRSA TESTS

WHAT ARE MRSA TESTS?

A MRSA test looks for MRSA bacteria in a sample of fluid or tissue from your body. The sample is often taken from your nose or a wound. MRSA stands for methicillin-resistant *Staphylococcus aureus*. It is a type of staph bacteria.

Many people have MRSA or other staph bacteria living on their skin or in their noses. These bacteria usually don't cause any harm. Having the bacteria without an infection is sometimes called "carrying" or being "colonized" with the bacteria. But even when you don't have an infection, you could still spread the bacteria to others.

But staph bacteria that enters your body through a cut, scrape, or other open wound, often causes a skin infection. Most staph skin infections are minor. They usually heal on their own or after treatment with antibiotics. But if you are carrying the MRSA bacteria on your skin and it enters through an open wound, the usual treatments may not work well.

MRSA bacteria are different than other staph bacteria. In a normal staph infection, antibiotics will kill the bacteria and stop them from growing. In a MRSA infection, the antibiotics usually used to treat staph infections don't work. The bacteria are not killed, and they continue to grow. When common antibiotics don't work on bacterial infections, it's known as antibiotic resistance. Antibiotic resistance makes it difficult to treat certain bacterial infections.

In the past, MRSA infections mostly happened in hospitals or nursing homes, putting people with weaker immune systems at risk. This is called hospital-associated MRSA (HA-MRSA). But now, MRSA is becoming more common in healthy people. This is known as community-associated MRSA (CA-MRSA). In both cases, it's the same MRSA bacteria, but how you treat it and reduce its spread may be different.

Anyone can get MRSA. You could have MRSA and not have any signs or symptoms. MRSA is not spread through the air like a cold or flu virus, but it can live on surfaces for a long time. There are several different ways that MRSA can spread:

Through contact with objects that are contaminated with the bacteria, such as sports or medical equipment.

By sharing personal items that are contaminated with the bacteria, such as a towel, bedding, or a razor.

Through close, personal contact with someone who has an infected wound. This is more likely to happen when large groups of people are close together, such as in a college dorm, locker room, prison, health care setting, or military barracks.

By sharing contaminated needles, which can happen when people are using injectable drugs.

If it's not treated, a MRSA infection may become more serious. If the bacteria spreads to your bloodstream, it can be dangerous, and you need to get quick medical treatment. Untreated MRSA can also spread to other people.

Other names: MRSA screening, methicillin-resistant *Staphylococcus aureus* screening, methicillin-resistant *Staphylococcus aureus* culture

WHAT ARE THEY USED FOR?

This test is most often used to find out if you have a MRSA infection. The test may also be used to see if treatment for a MRSA infection is working.

WHY DO I NEED A MRSA TEST?

You may need this test if you have symptoms of a MRSA infection. Your symptoms will depend on where the infection is located.

A MRSA infection on the skin may look like a type of rash. A MRSA rash looks like red, swollen bumps on the skin. The bumps may be filled with fluid or pus. Some people may mistake a MRSA rash for a spider bite. However, unless you actually see the spider, the bumps are probably not from a spider bite. With MRSA, the infected area may also be:

Warm to the touch

Painful

Symptoms of a MRSA infection in the bloodstream or other parts of the body include:

Fever

Chills

Headache

MRSA rash

WHAT HAPPENS DURING A MRSA TEST?

A health care provider will take a fluid sample from your wound, nose, blood, or urine. Steps may include the following:

Wound sample:

A provider will use a special swab to collect a sample from the site of your wound.

Nasal swab:

A health care professional will use a special swab to take a sample from your nose.

Blood test:

A provider will take a sample of blood from a vein in your arm.

Urine test:

You will provide a sterile sample of urine in a cup, as instructed by your provider.

After your test, your sample will be sent to a lab for a bacteria culture test. Most tests take 24-48 hours to get results.

That's because it takes time to grow enough bacteria to be detected. There is also a molecular test, which can deliver results much faster. The test, which is done on nasal swabs, can find MRSA bacteria in as little as five hours.

Talk to your provider to see which test would be a good choice for you.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a MRSA test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a wound sample, swab, or urine test.

You may feel a little pain when a sample is taken from a wound. A nasal swab may be slightly uncomfortable. These effects are usually mild and temporary.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results are positive, it means you have a MRSA infection. Treatment will depend on how serious the infection is. For mild skin infections, your provider may clean, drain, and cover the wound. You may also get an antibiotic to put on the wound or take by mouth. Certain antibiotics still work for some MRSA infections.

For more serious cases, you may need to go to the hospital and get treated with powerful antibiotics through an IV (intravenous line).

To understand the results of a MRSA test, your provider will consider your symptoms, medical history, and the results of other tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT MRSA TESTS?

The following steps can lower your risk of getting a MRSA infection:

Wash your hands often and thoroughly, using soap and water.

Use alcohol-based hand sanitizer if you can't wash your hands.

Keep cuts and scrapes clean and covered until they are fully healed.

Don't share personal items such as towels and razors.

Use disinfecting sprays or wipes on light switches, remote controls, or other areas that are touched frequently.

You can also take steps to reduce antibiotic-resistant infections. Antibiotic resistance happens when people don't use antibiotics in the right way. To prevent antibiotic resistance:

Take antibiotics as prescribed, making sure to finish the medicine even after you feel better.

Don't use antibiotics if you don't have a bacterial infection. Antibiotics don't work on viral infections.

Don't use antibiotics prescribed for someone else.

Don't use old or leftover antibiotics.

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MTHFR Gene Test

<https://medlineplus.gov/lab-tests/mthfr-gene-test/>

MTHFR GENE TEST

WHAT IS AN MTHFR GENE TEST?

An MTHFR gene test checks a sample of your blood for two very common changes in a gene called MTHFR. Changes in your genes are also called gene variants or mutations.

Genes are parts of DNA in your cells that you inherit from your parents. You inherit two copies of each gene, one coming from each parent. Genes carry information that controls what you look like and how your body works.

The MTHFR gene provides instructions for making the MTHFR protein. This protein helps your body use folate, a type of B vitamin that you get from various foods. Normally, your body uses folate and other B vitamins to break down an amino acid called homocysteine. Because of this, only small amounts of homocysteine stay in your bloodstream. If you have a change in the MTHFR gene, however, the MTHFR protein may not process folate as well as it should. As a result, you may have increased levels of homocysteine in your blood.

In general, medical researchers aren't certain that increased homocysteine levels directly cause health problems. However, some studies have shown that they can damage the inside of your blood vessels. This may increase the risk for blood clots, heart disease, and stroke.

Higher homocysteine levels during pregnancy can also increase the risk of having:

A baby with neural tube defects.

Early pregnancy loss (a miscarriage).

Preeclampsia.

To lower homocysteine levels, your health care provider will usually recommend taking vitamin B supplements that include folic acid, which is a form of folate.

However, if you have the gene variant MTHFR, your provider may recommend methylfolate (5-MTHF) instead since your body may be able to use this form more easily. The extra vitamins can make up for an MTHFR gene that's not fully working. Your provider may recommend eating more foods with folate, too. Before making changes in your health habits, talk with your provider.

Studies show that taking folic acid greatly reduces the risk of neural tube defects in babies. So, even if you have the MTHFR gene variant, the U.S. Centers for Disease Control and Prevention (CDC) recommends folic acid supplements for everyone who may become pregnant.

Other names: methylenetetrahydrofolate reductase mutation, C677T and A1298C, MTHFR DNA assay, MTHFR DNA testing, MTHFR gene variants, MTHFR gene polymorphisms 677T>C & 1298A>C mutations

WHAT IS IT USED FOR?

An MTHFR gene test looks for the two most common MTHFR gene variants. These are called C677T and A1298C. It can be done alone or as part of a group of genetic tests.

Having either of these variants does not necessarily increase your risk for any health condition. Nor does it guarantee that you will have high homocysteine levels. So, your health care provider may only check for these variants if a homocysteine blood test has already shown that your homocysteine levels are high.

If a homocysteine blood test does show high homocysteine levels, an MTHFR test can:

Determine whether the C667T and/or A1298C variants are causing these high levels.

Estimate your risk of developing blood clots or early heart or blood vessel disease, especially if you have a family history of these conditions.

However, even if a homocysteine blood test shows that your homocysteine levels are higher than normal, an MTHFR gene test is generally not needed. That's because:

The treatment for high homocysteine levels is the same whether or not it's caused by one of the two most common variants of the MTHFR gene.

Researchers aren't certain whether homocysteine levels affect your risk for developing blood clots or early heart or blood

vessel disease. Because of this, some medical experts recommend against using MTHFR testing for this purpose. To lower homocysteine levels, your provider will usually recommend taking vitamin B supplements that include folic acid, which is a form of folate. However, if you have the gene variant MTHFR, your provider may recommend methylfolate (5-MTHF) instead since your body may be able to use this form more easily. The extra vitamins can make up for an MTHFR gene that's not fully working. Your provider may also order MTHFR testing if a newborn has a positive result in a routine homocystinuria screening test. Most newborns in the U.S. receive this screening shortly after birth.

WHY DO I NEED AN MTHFR GENE TEST?

Medical experts do not recommend testing for common MTHFR gene changes in most cases. However, your provider may order this test if you had a blood test that showed higher than normal levels of homocysteine and:

A close relative was diagnosed with an MTHFR gene change.

You and/or close family members have early heart or blood vessel disease.

You may also need an MTHFR gene test if your provider prescribes a medicine called methotrexate. Methotrexate is used to treat certain types of cancer, psoriasis, and rheumatoid arthritis. The MTHFR protein helps your body use this medicine properly. The results of a MTHFR gene test may help your provider choose the dose that will be safe for you.

WHAT HAPPENS DURING AN MTHFR GENE TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an MTHFR gene test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The results of an MTHFR gene test tell you whether you have one or both of the two most common changes in the MTHFR gene. Test results are usually reported as positive or negative:

A positive test result means that you have an MTHFR gene change in one or both of your MTHFR genes. Having a change in only one of your MTHFR genes is unlikely to cause any health problems. If your homocysteine levels are high and your MTHFR gene test shows that you have:

Two copies of the C677T gene change or one copy each of C677T and A1298C, then these gene changes are probably causing your increased homocysteine levels

Two copies of the A1298C gene change, then your increased homocysteine levels are probably not caused by these gene changes

A negative test result means that neither of the common MTHFR gene changes were found in your blood sample. If you have an increased homocysteine level, it's probably from another condition, such as a lack of B vitamins, taking certain medicines, older age, hypothyroidism, kidney disease, or another disease.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN MTHFR GENE TEST?

Different types of gene tests are used to look for rare types of MTHFR gene changes that can cause a genetic disease called homocystinuria. This disease prevents your body from making an enzyme (a protein that speeds up chemical reactions in your body) that helps break down homocysteine. It can cause very high homocysteine levels. Symptoms usually show up in the first year of life. In mild cases, symptoms may not appear until childhood or later.

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Mucus in Urine

<https://medlineplus.gov/lab-tests/mucus-in-urine/>

MUCUS IN URINE

WHAT IS A MUCUS IN URINE TEST?

Mucus is a thick, slippery fluid that coats and moistens certain parts of the body, including the nose, mouth, throat, and urinary tract. A small amount of mucus in your urine (pee) is normal. Too much mucus may be a sign of a urinary tract infection (UTI) or other medical condition. A urinalysis¹ test checks a sample of your urine. It can find out if there is too much mucus in your urine.

Other names: microscopic urine analysis, microscopic examination of urine, urine test, urine analysis, UA

WHAT IS IT USED FOR?

A mucus in urine test may be part of a urinalysis. You may have a urinalysis during a routine checkup² to check your general health or to monitor a medical condition. Your health care provider may also order the test if you have signs of a urinary tract infection (UTI) or other condition that can cause mucus in your urine.

A urinalysis may include a visual check of your urine sample, tests for certain chemicals, and an examination under a microscope to look for certain types of cells. A mucus in urine test is part of a microscopic exam of urine.

WHY DO I NEED A MUCUS IN URINE TEST?

A urinalysis is often part of a routine checkup. Your provider may include a mucus in urine test in your urinalysis if you have symptoms of a UTI. These may include:

Frequent urge to urinate, even when you have little urine in your bladder

Pain or burning while urinating

Dark, cloudy, or reddish-colored urine

Bad smelling urine

Pain or discomfort in your lower abdomen (belly)

Weakness

Fatigue

WHAT HAPPENS DURING A MUCUS IN URINE TEST?

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine

is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for this test. If your provider has ordered other urine or blood tests, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a urinalysis or a test for mucus in urine.

WHAT DO THE RESULTS MEAN?

A normal test result usually shows a small or moderate amount of mucus in your urine.

A large amount of mucus may be a sign of a medical condition, including:

A urinary tract infection (UTI)

A sexually transmitted infection (STI)

Kidney stones

Irritable bowel syndrome (IBS)

Bladder cancer

If your test shows a large amount of mucus in your urine, you may need more tests to find out why. To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A MUCUS IN URINE TEST?

If a urinalysis is part of your regular checkup, your urine will be tested for various substances along with mucus. These include red and white blood cells, proteins, acid and glucose levels, and the concentration of particles in your urine.

If you get frequent UTIs, your provider may recommend more testing and steps to help prevent future infections.

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Myasthenia Gravis Tests

<https://medlineplus.gov/lab-tests/myasthenia-gravis-tests/>

MYASTHENIA GRAVIS TESTS

WHAT ARE MYASTHENIA GRAVIS (MG) TESTS?

Myasthenia gravis (MG) tests are used to diagnose MG, a chronic (long-lasting) autoimmune disease that causes weakness in muscles throughout your body. In autoimmune diseases, your immune system mistakenly attacks the healthy cells of your organs and tissues. If you have MG, your immune system makes antibodies that block or change the connection between your nerves and muscles.

MG causes weakness in the voluntary muscles in your body. These are muscles that you can control. The symptoms of MG may begin suddenly. There are different types:

Ocular MG affects the muscles controlling the eyelids. These are often the first and sometimes the only muscles that MG affects.

In generalized MG, the weakness may also be in your eyelids. But it affects other muscles as well, such as those that control eye movement and those in your neck, arms, and/or legs. If the muscles that control your breathing are affected, this is called a myasthenic crisis. It is a life-threatening emergency.

Researchers aren't sure why people get MG. It is not inherited (passed down in families). But occasionally, the disease affects more than one member of the same family.

What is known is that you may have an increased chance of developing MG if you:

Have another autoimmune disorder, such as rheumatoid arthritis or systemic lupus erythematosus.

Have a problem with your thymus gland, a small organ in your upper chest. It is part of your immune system. If you have MG, your thymus gland might be larger than usual, or you may have a thymoma (thymus tumor). These tumors are usually benign (not cancer), but they can become cancerous.

Are a woman under age 40 or a man over age 60. This is the age range MG most commonly affects (although you can develop it at any age).

MG testing can help you get diagnosed and treated for the disease. While there is no cure for MG, there are treatments that can relieve symptoms and may even allow you to have periods without any symptoms (remission).

Other names: Ice pack test, edrophonium test, tensilon test, nerve conduction study

WHAT ARE THEY USED FOR?

MG tests are used to diagnose MG.

WHY DO I NEED MG TESTING?

You may need to be tested if you have symptoms of MG. Your symptoms will depend on which muscles are affected. The symptoms often include:

Drooping of one or both eyelids

Weakness of the eye muscles

Blurred or double vision

Changes in facial expression

Weakness in arms, hands, fingers, legs, and/or neck

Trouble swallowing and/or chewing

Trouble talking

Fatigue

Shortness of breath

MG symptoms usually get worse after periods of activity, such as exercise, and improve after rest.

WHAT HAPPENS DURING MG TESTING?

MG can be hard to diagnose because muscle weakness can be a symptom of many disorders. So, your health care provider may order more than one type of MG test to help make a diagnosis. Types of MG tests include:

Neurological exam. During this test, your provider will take your medical history and do a physical exam, which will include checking your:

Reflexes

Muscle strength

Muscle tone

Coordination

Balance

Sense of touch and sight

Ice pack test. This test may be done if you have a droopy eyelid. During the test:

Your provider will place an ice pack or bag filled with ice on your eyelid.

The ice will be removed after two minutes.

If drooping improves, it may mean you have MG. This is because cooling temporarily improves muscle weakness that's caused by MG.

Antibody test. This is a blood test that looks for certain types of antibodies that are found in people with MG. During the test:

A health care professional will take a blood sample from a vein in your arm, using a small needle.

After the needle is inserted, a small amount of blood will be collected into a test tube or vial.

Electromyography (EMG). This test looks at the electrical signals your muscles make when they are at rest and when used. A healthy muscle should not give off electrical signals when you aren't moving. If your muscle is damaged, it may show electrical activity while at rest or activity that is not normal while using it. During the test:

Your provider will place a small needle with an electrode into your muscle. This is connected to a machine used to record your muscle's electrical activity. You may have slight pain or discomfort when the electrode is inserted.

A machine will record the muscle activity while your muscle is at rest.

Then, you will be asked to tighten (contract) the muscle slowly and steadily.

The machine will record the muscle activity while your muscle is contracted.

The electrode may be moved to record activity in different muscles.

The electrical activity is recorded and shown on a video screen. The activity may also be sent to an audio speaker. The audio may make popping sounds when you're contracting your muscles.

Nerve conduction study. This test measures how fast and well your body's electrical signals move along your nerves. A damaged nerve has a slower and weaker signal. This test can help check for nerve damage. During the test:

Your provider will stick electrodes to your skin above a nerve they're checking using tape or a paste. The electrodes, called stimulating electrodes, deliver a mild electrical pulse.

Your provider will attach recording electrodes to the muscles controlled by those nerves. These electrodes will record the muscle's response to the electrical stimulation from the nerve.

Your provider will send a small pulse of electricity through the stimulating electrodes to trigger the nerve to signal the muscle. This may cause a mild tingling feeling.

Your provider will record the time it takes for your muscle to respond to the nerve signal.

Imaging tests. Your provider may order a CT scan (computed tomography) or an MRI (magnetic resonance imaging) to see if you have an enlarged thymus or a thymus tumor, which can be a sign of MG.

During a CT scan:

You will lie on a narrow table that slides into a scanning machine.

Once you are in the scanner, the machine will take pictures as it rotates around you.

During an MRI:

You will lie on a narrow table that will slide into a large tunnel-shaped scanner.

The scanner will take pictures using powerful magnets and radio waves. It does not use radiation.

For both imaging tests, you will need to hold as still as possible. You may also be asked to hold your breath at times while images are being taken.

Edrophonium test. Edrophonium is a drug that briefly relieves muscle weakness in people with MG. It's mostly used to test weakness in the eye muscles. During the test:

A provider will inject the drug through an IV (intravenous) line.

If muscle weakness briefly improves, it may mean you have MG.

Lung function tests. These tests measure breathing and how well your lungs are working. There are several types of lung function tests. For most tests, you will breathe into a mouthpiece or machine as your breath is measured. For some tests, you may need to inhale a type of gas or medicine.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THESE TESTS?

You don't need any special preparations for a neurological exam, ice pack test, or antibody test.

For the other tests, you may need to fast (not eat or drink) or avoid certain foods and beverages before the test. Tell your provider if you are pregnant, have pieces of metal in your body, or if you have a pacemaker or cardiac defibrillator. Special steps may need to be taken before the test. Also, tell your provider if you take any medicines. If you're taking blood thinners, you may need to watch for more bleeding from the needle.

Your provider will let you know if you need to make any additional preparations.

ARE THERE ANY RISKS TO THESE TESTS?

There is no risk to having a neurological exam or an ice pack test.

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

You may feel a little pain or cramping during an EMG test. If your pain becomes very uncomfortable, let your provider know, as this may change your test results. You may feel tingly during a nerve conduction study, like mild static electricity. Your tested muscles may be sore for a few days, and you may have bruises from where the needles were placed.

CT scans and MRIs are painless, but can be a little uncomfortable. Some people feel claustrophobic in an MRI scanning machine.

An edrophonium test may cause an allergic reaction. It is not common, but contact your provider if you have breathing problems, vision changes, and/or feel faint after taking the medicine.

There is very little risk to having lung function tests. Some people may feel lightheaded or dizzy during the procedures.

WHAT DO THE RESULTS MEAN?

If one or more of the tests show that you have MG, your provider will work with you on a treatment plan. Treatments can help reduce your symptoms and improve muscle weakness. The treatment options may include:

Medicines. Different medicines are used to help improve muscle strength and/or decrease your body's immune system responses.

Plasmapheresis (plasma exchange) and intravenous (IV) immunoglobulin. These procedures remove abnormal antibodies from your blood. They are usually only used in severe cases. They can help with your symptoms for a few weeks or months.

Surgery. Your provider may recommend surgical removal of your thymus gland, which is often abnormal in people with MG. The procedure can reduce symptoms, even in people without thymus tumors.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT MG TESTING?

If you've been diagnosed with MG, the following lifestyle changes may also help boost muscle strength:

Eating a healthy diet that includes plenty of protein and carbohydrates.

Doing regular gentle exercise to strengthen your muscles. But check with your provider before starting any exercise program.

Avoiding heat and stress, which can make symptoms worse.
Wearing an eye patch if you have double vision.
Taking naps or rest breaks throughout the day.

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Myelography

<https://medlineplus.gov/lab-tests/myelography/>

MYELOGRAPHY

WHAT IS MYELOGRAPHY?

Myelography, also called a myelogram, is an imaging test that checks for problems in your spinal canal. The spinal canal contains your spinal cord, nerve roots, and the subarachnoid space. The subarachnoid space is a fluid-filled space between the spinal cord and the membrane that covers it.

A regular x-ray only shows your bones. So during a myelography, contrast dye is injected into the spinal canal. Contrast dye is a substance that makes specific organs, blood vessels, and tissue show up more clearly on an x-ray. This clear view of the bones, disks, and other tissues surrounding your spinal canal may help your health care provider find out if something is pressing on your nerves or spinal cord. This helps them choose the best treatment.

Myelography involves using one of these two imaging procedures:

Fluoroscopy, a type of x-ray that shows internal tissues, structures, and organs moving in real time as a live video.

CT scan (computerized tomography), a procedure that combines a series of x-ray images taken from different angles around your body.

Other names: myelogram

WHAT IS IT USED FOR?

Myelography is used to look for conditions and diseases that affect the nerves, blood vessels, and structures in the spinal canal. These can include:

Herniated disk. Spinal disks are rubbery cushions (disks) that sit between the bones of your spine. A herniated disk is a condition in which the disk bulges out and presses on spinal nerves or the spinal cord.

Tumors

Spinal stenosis, which happens when the spaces in the spine narrow and create pressure on the spinal cord and nerve roots.

Infections of the membranes and tissues of the spinal cord. This includes meningitis, which causes inflammation of the membranes.

Arachnoiditis, a condition that causes inflammation of a membrane that covers the spinal cord

Your provider may also recommend myelography if you have new pain or weakness in your back after a recent surgery.

WHY DO I NEED MYELOGRAPHY?

You may need this test if you have symptoms of a spinal disorder, such as:

Pain in the back, neck, and/or leg

Tingling sensations

Weakness

Trouble walking

Trouble with tasks that involve small muscle groups, such as buttoning a shirt

WHAT HAPPENS DURING MYELOGRAPHY?

A myelography may be done at a radiology center or in the radiology department of a hospital. The procedure usually includes the following steps:

You will need to remove any jewelry, removable dental appliances, eyeglasses, and any metal objects that may interfere with the x-ray.

You may need to remove your clothing. If so, you will be given a hospital gown.

You will lie on your side or stomach on a pillow or a padded x-ray table.

Your provider will clean your back with an antiseptic solution (to kill the germs on your skin).

You will be injected with numbing medicine so that you won't feel any pain during the procedure. Once the area is numb, your provider will use a thin needle to inject contrast dye into your spinal canal.

You may feel some pressure when the needle goes in, but it should not hurt. The contrast dye may cause a warm or flushed feeling, nausea, or a metallic taste in your mouth.

Your provider may remove a sample of spinal fluid (cerebrospinal fluid) for testing.

You'll need to stay very still while the fluid is being withdrawn.

Your x-ray table will be tilted in different directions to allow the contrast dye to move to different areas of your spinal cord. You will need to remain still so the images won't blur.

Your provider will remove the needle.

Your provider will capture and record images using fluoroscopy or a CT scan.

The procedure usually takes about an hour. Your provider may not receive your test results for a few days.

After the test, you may be monitored for one to two hours and will need someone to take you home. Your provider may ask you to:

Rest at home with your head elevated (slightly raised) for a few hours.

Avoid strenuous activity for one to two days after the test.

Drink plenty of fluids.

Avoid alcohol.

Use an over-the-counter pain reliever if you have a headache.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know if there are any special instructions to follow. They may ask you to drink extra fluids the day before the test. On the day of the test, you will probably be asked not to eat or drink anything except clear fluids. These include water, clear broth, tea, and black coffee.

Tell your provider about any medicines you are taking. Certain medicines should not be taken before your test. These include aspirin and blood thinners. Your provider will let you know how long you need to avoid these medicines. It may be as long as 72 hours before the test. Don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

You should not have this test if you are pregnant or think you may be pregnant. Radiation can be harmful to an developing baby.

For others, there is little risk to having this test. The radiation dose is very low and is not considered harmful for most people. But talk to your provider about all the x-rays you've had in the past. The risks from radiation exposure may be linked to the number of x-ray treatments you've had over time.

There is a small risk of an allergic reaction to the contrast dye. Tell your provider if you have any allergies, especially to shellfish or iodine, or if you've ever had a reaction to contrast material.

Other risks include headache and nausea and vomiting. The headache may last for up to 24 hours. Serious reactions are rare but may include seizures, infection, and a blockage in the spinal canal.

Contact your provider right away if you have any of the following:

A headache that doesn't go away in 24 hours

Fever

A stiff neck

Tingling or numbness in your legs

Trouble going to the bathroom

WHAT DO THE RESULTS MEAN?

If your results were not normal, it may mean you have one of the following conditions:

Herniated disk

Spinal stenosis

Tumor

Nerve injury

Bone spurs

Arachnoiditis (inflammation of the membrane surrounding the spinal cord)

A normal result means your spinal canal and structures were normal in size, position, and shape. Your provider may want to do more tests to find out what is causing your symptoms.

If you have questions about your results, talk to your provider. To understand the results of myelography, your provider may consider your symptoms, medical history, and the results of other tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT MYELOGRAPHY?

You may hear different names for the test, such as cervical or lumbar myelography. This is the same type of test. It only refers to the part of your spine that needs imaging.

MRI (magnetic resonance imaging) has often replaced the need for myelography. MRIs use a magnetic field and radio waves to create images of organs and structures inside the body. But myelography can be more useful in diagnosing some conditions, such as certain spinal tumors and spinal disk problems. It's also used for people who cannot have an MRI because they have metal or electronic devices in their bodies. These include a pacemaker, surgical screws, and cochlear implants.

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Nasal Swab

<https://medlineplus.gov/lab-tests/nasal-swab/>

NASAL SWAB

WHAT IS A NASAL SWAB?

A nasal swab is a test that checks for viruses and bacteria that cause respiratory infections.

There are many types of respiratory infections. A nasal swab test can help your health care provider diagnose the type of infection you have and determine which treatment would be best for you. The test may be done by taking a sample of cells from inside your nostrils or from the nasopharynx, the uppermost part of your nose and throat.

Other names: anterior nares test, nasal mid-turbinate swab, NMT swab nasopharyngeal culture, nasopharyngeal (NP) swab

WHAT IS IT USED FOR?

A nasal swab is used to diagnose certain infections of the respiratory system. These can include:

The flu.

COVID-19.

Respiratory syncytial virus (RSV), a common and usually mild respiratory infection. But it can cause serious lung infections in infants and older adults.

Whooping cough, a respiratory infection that can cause coughing fits. Coughing can become violent and rapid.

Meningitis, a disease caused by inflammation of the membranes that surround the brain and spinal cord.

MRSA (methicillin-resistant *Staphylococcus aureus*), a serious type of bacterial infection that can be very difficult to treat.

WHY DO I NEED A NASAL SWAB?

You may need this test if you have symptoms of a respiratory infection. They can include:

Cough

Fever

Stuffy or runny nose

Sore throat

Headache

Fatigue

Muscle aches

WHAT HAPPENS DURING A NASAL SWAB?

A nasal swab may be taken from:

The front part of your nostrils (anterior nares).

You or your provider might collect the sample.

It's less uncomfortable than other types of swab tests.

You or your provider might collect the sample.

It's less uncomfortable than other types of swab tests.

The back of your nostrils, in a procedure known as nasal mid-turbinate (NMT) swab.

You or your provider might collect the sample.

You or your provider might collect the sample.

Your nasopharynx (the uppermost part of your nose and throat), which is deeper into your respiratory system.

Your provider must collect the sample.

It's less comfortable than other types of nasal swab tests.

It's more effective for detecting viruses or bacteria in the upper respiratory tract.

Your provider must collect the sample.

It's less comfortable than other types of nasal swab tests.

It's more effective for detecting viruses or bacteria in the upper respiratory tract.

During an anterior nares test, you will start by tilting your head back. Then you or your provider will:

Gently insert a swab inside your nostril.

Rotate the swab in four or five sweeping circles for 10-15 seconds.

Remove the swab and insert it into your second nostril.

Swab the second nostril using the same technique.

Remove the swab.

If you only swab one nostril, don't rotate the swab in your nostril, or only keep the swab in one area, you may not get a good sample.

If you are doing the test yourself, your provider will let you know how to prepare and package your sample. This often involves placing your swab in a sterile tube, closing it, and sealing it in a special bag. You would then give that bag to your provider or follow the instructions on how to return your sample for testing.

During an NMT swab, you will start by tilting your head back. Then you or your provider will:

Gently insert a swab onto the bottom of your nostril, pushing it until you feel it stopping.

Rotate the swab for 10 to 15 seconds.

Remove the swab and insert it into your second nostril.

Swab the second nostril using the same technique.

Remove the swab.

If you only swab one nostril, don't rotate the swab in your nostril, or only keep the swab in one area, you may not get a good sample.

If you are doing the test yourself, the provider will let you know how to prepare and package your sample. This often involves placing your swab in a sterile tube, closing it, and sealing it in a special bag. You would then give that bag to your provider or follow the instructions on how to return your sample for testing.

During a nasopharyngeal (NP) swab:

You will tilt your head back.

Your provider will insert a long swab into your nostril until it reaches your nasopharynx (the upper part of your throat).

Your provider will rotate the swab for 10 to 15 seconds and remove it.

If your provider gets enough of a sample from one of your nostrils, they may only need to do one. But they may need to get a sample from your other nostril as well if they had trouble getting the sample from the first side.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a nasal swab.

ARE THERE ANY RISKS TO THE TEST?

The test may tickle your throat or cause you to cough. A nasopharyngeal swab may be uncomfortable and cause coughing or gagging. All these effects are temporary.

WHAT DO THE RESULTS MEAN?

A negative result means no harmful viruses or bacteria were found in your sample.

A positive result means a specific type of harmful virus or bacteria was found in your sample. If you are diagnosed with

an infection, follow your provider's recommendations for treating your illness. This may include medicines and steps to prevent spreading the infection.

To understand the results of a nasal swab test, your provider may consider your symptoms, medical history, and the results of other tests.

Learn more about laboratory tests, reference ranges, and understanding results.

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Natriuretic Peptide Tests (BNP, NT-proBNP)

<https://medlineplus.gov/lab-tests/natriuretic-peptide-tests-bnp-nt-probnp/>

NATRIURETIC PEPTIDE TESTS (BNP, NT-PROBNP)

WHAT ARE NATRIURETIC PEPTIDE TESTS (BNP, NT-PROBNP)?

Natriuretic peptides are proteins that your heart and blood vessels make. Natriuretic peptide tests measure the amount of these proteins in a sample of your blood. They are mainly used to help confirm or rule out heart failure in people who have symptoms.

Heart failure is also called congestive heart failure. If you have this condition, your heart has trouble pumping enough oxygen-rich blood to meet your body's needs.

Two types of natriuretic peptides can be measured to check for possible heart failure. Your health care provider will decide which of these peptides to test:

Brain natriuretic peptide (BNP) is a protein that's a type of hormone. A hormone is a chemical messenger in your bloodstream that controls the actions of certain cells or organs. BNP has "brain" in its name because that's where researchers first discovered it. Your heart makes and releases BNP into your bloodstream when it's working harder than normal to pump blood.

The BNP tells your blood vessels to open wider and your kidneys to get rid of water and salt through urine (pee). This helps reduce the workload on your heart by lowering blood pressure and reducing the amount of blood your heart has to pump.

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N-terminal pro b-type natriuretic peptide (NT-proBNP) is a protein that's an "ingredient" for making the BNP hormone. Like BNP, your heart makes larger amounts of NT-proBNP when it has to work harder to pump blood.

It's normal to have some BNP and NT-proBNP in your bloodstream. But higher than normal levels for your age and sex may be a sign of heart failure.

Other names: brain natriuretic peptide, NT-pro B-type natriuretic peptide test, B-type natriuretic peptide, N terminal proBNP

WHAT ARE THEY USED FOR?

A BNP test or an NT-proBNP test is mainly used to help diagnose or rule out heart failure in a person who is having symptoms. Shortness of breath is the most common symptom of heart failure, but other conditions can cause the same type of breathing problems.

If a person with shortness of breath has normal BNP or NT-proBNP levels, heart failure can usually be ruled out. If high BNP or NT-proBNP levels are found, other tests will usually be done to confirm that heart failure is causing the symptoms.

For people who have already been diagnosed with heart failure, the test may be used to:

- Find out how serious their heart condition is

- Predict the chances that their condition will get worse

- Check if an increase in symptoms means that heart failure has gotten worse

In certain cases, the test may also be used to monitor the heart health of people who:

- Have had a heart attack

- Have a heart condition that suddenly reduces the flow of blood to the heart, such as unstable angina

WHY DO I NEED A NATRIURETIC PEPTIDE TEST?

You may need a BNP test or an NT-proBNP test if you have symptoms that could mean you have heart failure. These include:

Feeling short of breath

Swelling in your abdomen (belly), feet, legs, and/or veins in your neck

Fatigue and/or general weakness

Coughing

Inability to sleep lying flat

Needing to urinate (pee) a lot, especially at night

Loss of appetite and nausea

If you have already been diagnosed with heart failure, your provider may order one of these tests to learn more about your condition.

WHAT HAPPENS DURING A NATRIURETIC PEPTIDE TEST?

For a BNP test or an NT-proBNP test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

If you're being tested in a hospital, you may have a fingerstick test. Your finger will be pricked with a small needle and a few drops of blood will be gathered for testing. This method provides quicker results than using blood from a vein.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a BNP test or an NT-proBNP test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The meaning of your test results will depend on your age, sex, medical history, family history, and the results of other tests. Ask your provider to explain what the results of your BNP or NT-proBNP test say about your health and how this information may affect your treatment.

A BNP or NT-proBNP test alone can't diagnose the cause of your symptoms. But in general, if you have symptoms that could be caused by heart failure:

Normal results on a BNP or NT-proBNP usually mean that you're unlikely to have heart failure. Your provider may order other tests to find out what's causing your symptoms.

If you have obesity, your weight may affect the accuracy of your test results, making your levels lower than they really are. If your provider suspects heart failure, you may have other heart health tests even if your BNP or NT-proBNP levels are normal.

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Higher than normal levels of BNP or NT-proBNP mean that you could have heart failure. Your provider will probably

order other heart health tests to help make a final diagnosis.

In most cases, the higher the level of your test results, the more serious your heart failure may be. But other health conditions can increase BNP and NT-proBNP levels. That's why your medical history is important for understanding whether high test results mean you have heart failure.

Other conditions that can cause high levels of BNP and NT-proBNP include:

Kidney failure

Other types of heart disease, including conditions that affect your heart valves or heart muscle

Certain lung disorders, including a blood clot in your lung (pulmonary embolism) and high blood pressure in your lungs (pulmonary hypertension)

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results

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Neurological Exam

<https://medlineplus.gov/lab-tests/neurological-exam/>

NEUROLOGICAL EXAM

WHAT IS A NEUROLOGICAL EXAM?

A neurological exam is a group of questions and tests to check for disorders of your nervous system. It's sometimes called a "neuro" exam.

Your nervous system includes your:

Brain and spinal cord, which are also called your central nervous system

Peripheral nerves that carry signals back and forth between your central nervous system and all parts of your body, including your skin, muscles, and organs

Your nervous system plays a role in almost every part of your health and well-being. Because it controls so many body processes, a neurological exam has many different types of tests. The tests you have will depend on why you're having the exam. There are nerve tests to check for problems with your:

Muscle movement, balance, and coordination. These activities are controlled by nerves called motor nerves.

Breathing, heartbeat, digestion, and other processes that happen without thinking. Autonomic nerves control these activities.

Sense of touch, smell, hearing, and vision. Sensory nerves carry this information from your senses to your brain.

Thinking and memory. Certain parts of your brain control these and other types of complex mental activity, including your emotions.

Other names: neuro exam

WHAT IS IT USED FOR?

A neurological exam may be used to:

Check the health of your nervous system during a routine checkup.

Help find out whether a nervous system disorder is causing certain symptoms. There are hundreds of disorders that affect the brain and nerves. Some examples include:

Degenerative nerve diseases, such as certain types of dementia and Parkinson's disease

Diabetic nerve problems

Epilepsy

Headache disorders, such as migraines and cluster headaches

Meningitis

Multiple sclerosis

Degenerative nerve diseases, such as certain types of dementia and Parkinson's disease

Diabetic nerve problems

Epilepsy

Headache disorders, such as migraines and cluster headaches

Meningitis

Multiple sclerosis

Help find out if an injury has damaged part of the nervous system and how serious the damage may be.

Monitor a known nerve condition and/or see if treatment is helping.

WHY DO I NEED A NEUROLOGICAL EXAM?

You may need a neurological exam if you have symptoms that may be from a nervous system disorder. Because your nerves affect every part of your body, nervous system disorders can cause many different types of symptoms, including:

Pain in your back, neck, head, or along a nerve, such as sciatica
Tremors
Weak or stiff muscles
Problems with balance and/or coordination
Numb or tingling skin
Changes in any of your senses (hearing, vision, taste, smell, and touch)
Slurred speech
Confusion or other changes in mental ability
Seizures

You may also need a neurological exam if you have had an injury that may have damaged your peripheral nerves, spinal cord, or caused a traumatic brain injury (TBI).

WHAT HAPPENS DURING A NEUROLOGICAL EXAM?

A neurological exam is often done by a neurologist. A neurologist is a doctor who specializes in diagnosing and treating disorders of the nervous system. Certain other providers may also do a neurological exam.

The exam is usually done in a provider's office. If you've had a serious injury, the exam may be done in the emergency room or in the hospital.

First, the provider will usually:

Ask questions about any symptoms you may have

Ask about your medical history

Do a physical exam, including checking your heart and lungs

Next, the provider will do specific tests to check how different parts of your nervous system are working. The tests you have will depend on your symptoms. The tests may check your:

Mental status. This includes your memory, problem-solving ability, alertness, and mood. During a mental status exam, you may answer questions about the date, time, and where you are. You may also be asked to remember a list of items, name objects, repeat words, and/or draw specific shapes.

Cranial nerves. These 12 nerves connect your brain with your eyes, ears, nose, face, tongue, throat, shoulders, and certain organs. The provider will test the nerves that may be involved with your symptoms. For example, to test your sense of smell, you may be asked to sniff certain smells and identify what they are. If you're having speech problems, you may be asked to try to talk while you stick out your tongue.

Coordination, balance, and walking. These tests check how well your nervous system controls your muscle movements. You may be asked to walk in a straight line, placing one foot directly in front of the other. Other tests include checking your handwriting and having you touch your finger to your nose with your eyes closed.

Reflexes. A reflex is your body's automatic movement in response to certain triggers. For example, if your knee is tapped with a rubber hammer, your lower leg will jerk on its own. There are many types of reflexes that are tested in different ways. Reflex tests show how well nerves between your spinal cord and muscles are working.

Sensory nerves. The provider may test how well you can feel touch, hot and cold temperatures, vibrations, and pain. These tests involve gently touching part of your skin with different objects, such as a dull needle or a cotton swab. You will be asked to describe what you can feel.

Autonomic nervous system. A neurological exam tests the part of your nervous system that controls your breathing, heart rate, digestion, and other processes that happen without thinking. Examples of these tests include checking your blood pressure and heartbeat. Another test checks how your eyes respond to light.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A NEUROLOGICAL EXAM?

You don't need any special preparations for a neurological exam.

ARE THERE ANY RISKS TO THE EXAM?

There is no risk to having a neurological exam.

WHAT DO THE RESULTS MEAN?

If the results of any part of your neurological exam are not normal, your provider will probably order more tests to help make a diagnosis. The tests will depend on what type of condition your provider thinks you could have. They may include:

Blood and/or urine tests

Imaging tests, such as an MRI

Cerebrospinal fluid (CSF) testing, also called a lumbar puncture

Biopsy

Electroencephalography (EEG) or electromyography (EMG), which use small electric sensors to measure brain activity and nerve function

If you have questions about your results, talk with your neurologist or other provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A NEUROLOGICAL EXAM?

Nervous system disorders and mental health problems can have similar symptoms. For example, problems paying attention could be a symptom of either type of disorder. So in certain cases, your provider may do a mental health screening before or after a neurological exam.

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Nitrites in Urine

<https://medlineplus.gov/lab-tests/nitrites-in-urine/>

NITRITES IN URINE

WHAT IS A NITRITES IN URINE TEST?

A urinalysis test checks for different cells, chemicals, and other substances in a sample of your urine (pee). This includes checking for nitrites. Nitrites in your urine may be a sign of a urinary tract infection (UTI).

Different types of bacteria can cause UTIs. Many of these types of bacteria change a normal chemical in your urine, called nitrates, into another chemical, called nitrites. So, if you have nitrites in your urine, it usually means that you have a urinary tract infection (UTI).

UTIs are one of the most common types of infections, especially in women. Most UTIs affect the bladder and urethra (the tube that carries urine out of your body). Fortunately, most UTIs are not serious. They are usually treated with antibiotics. UTIs that aren't treated could spread to your kidneys, which can be serious. So, it's important to diagnose and treat UTIs early.

Other names: urine test, urine analysis, microscopic urine analysis, microscopic examination of urine, UA

WHAT IS IT USED FOR?

A urinalysis, which includes a test for nitrites in urine, is used to check your general health, including the health of your urinary tract and kidneys. It may also be used to check for a UTI.

WHY DO I NEED A NITRITES IN URINE TEST?

Your health care provider may order a urinalysis as part of a routine checkup or if you have symptoms of a UTI. Symptoms of a UTI may include:

Frequent urges to urinate (pee), even when you have little urine in your bladder

Pain or burning when you pee

Dark, cloudy, or reddish-colored urine

Bad smelling urine

Weakness, fatigue, or confusion, mainly in older people

Fever, which may be a sign of a kidney infection

If you have symptoms of a UTI, it's important to see your provider so you can start treatment right away, if needed.

WHAT HAPPENS DURING A NITRITES IN URINE TEST?

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

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Urinate into the toilet for a few seconds, then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

If your provider has ordered other urine or blood tests, you may need to fast^Â (not eat or drink) for several hours before the test. They will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a urinalysis or a nitrites in urine test.

WHAT DO THE RESULTS MEAN?

If there are nitrites in your urine, it may mean that you have a UTI. But, even if no nitrites are found, you may still have an infection. That's because some types of bacteria don't change nitrates into nitrites.

Your provider will also look at other results of your urinalysis, especially tests that look for signs of white blood cells in your urine. White blood cells in urine may also mean you have a UTI. To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A NITRITES IN URINE TEST?

At-home test kits for UTIs are available without a prescription. These test kits, also called urine dipstick tests or UTI test strips, check for nitrites and white blood cells in your urine.

At-home UTI tests are quick and easy to use, but they don't always provide accurate results. Your results may say you don't have a UTI when you really do. So, you may still need to see your provider for a nitrites in urine test. If you think you have a UTI, ask your provider which test is right for you.

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If you have signs of a UTI, your provider may order a urine culture test to see how much and what type of bacteria may be in your urine. This information helps your provider decide which type of antibiotic medicine to prescribe. In a urine culture test, bacteria from your urine are grown in a lab. Results usually take a couple of days.

If a urinalysis is part of your regular checkup, your urine will be tested for several other substances. These include blood, proteins, acid and sugar levels, cell fragments, and crystals in your urine.

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Nonstress Test

<https://medlineplus.gov/lab-tests/nonstress-test/>

NONSTRESS TEST

WHAT IS A NONSTRESS TEST?

A nonstress test (NST) is a safe and painless test given during pregnancy. The test measures the heart rate of your developing baby as the baby moves or when you have contractions in your uterus. Your uterus is the place where your baby grows during pregnancy.

In most healthy babies, the heart rate, also known as the fetal heart rate, increases during movement. If your nonstress test results showed that the fetal heart rate was not normal, it may mean that your baby is not getting enough oxygen. If this happens, you may need more testing or treatment. In some cases, your health care provider may want to induce labor so you can deliver your baby. Your provider will give you medicine or use other methods to start labor before it begins naturally.

Other names: fetal nonstress test, NST

WHAT IS IT USED FOR?

A nonstress test is used to check your baby's health before birth. The test is usually done in the third trimester of pregnancy, most often after 28 weeks. A baby born between 39 and 41 weeks is considered full-term.

WHY DO I NEED A NONSTRESS TEST?

You may need this test if:

Your baby doesn't seem to be moving as much as usual.

Your pregnancy is overdue (at 42 weeks or beyond).

You have a chronic medical condition that makes your pregnancy high risk, such as high blood pressure, diabetes, heart disease, or a clotting disorder.

You had complications in a previous pregnancy.

Your blood is Rh negative, and your baby's is Rh-positive, which is called Rh incompatibility. This condition can cause your immune system to attack your baby's red blood cells.

You are having more than one baby (twins, triplets, or more).

WHAT HAPPENS DURING A NONSTRESS TEST?

The test may be done in your provider's office or a hospital's prenatal area. It generally includes the following steps:

You will lie on a reclining chair or exam table.

Your provider will spread a special gel on the skin over your abdomen (belly).

Your provider will attach two belt-like devices around your abdomen. One will measure your baby's heartbeat. The other will record your contractions.

Your provider will move the device over your abdomen until the baby's heartbeat is found.

The baby's heart rate will be recorded on a monitor, while your contractions are recorded on paper.

You may be asked to press a button on the device each time you feel your baby move. This allows your provider to record the heart rate during movement.

The test usually lasts about 20 minutes to 30 minutes.

If your baby isn't active or moving during that period, they may be asleep. To wake up the baby, your provider may place a small buzzer or other noisemaker over your abdomen. This won't harm the baby but it may help a sleepy baby become more active. Your baby may also wake up if you have a snack or sugary drink.

Your provider will remove the belts. They will likely review the results with you soon after the test. The procedure is very safe. It's called a "nonstress" test because no stress, or risk, is placed on the baby during the test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need to do any special preparations for a nonstress test. However, your provider may ask you to go to the bathroom to empty your bladder before the test.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to you or your baby from having a nonstress test.

WHAT DO THE RESULTS MEAN?

Nonstress test results are given as one of the following:

Reactive or Reassuring. This means your baby's heart rate increased two or more times during the testing period.

Nonreactive. This means your baby's heartbeat didn't increase when moving or the baby wasn't moving much.

A nonreactive result doesn't always mean your baby has a health problem. Your baby may have been asleep and not easily awoken. Nonreactive results may also be caused by certain medicines taken during pregnancy. But if the result was nonreactive, your provider will probably take more tests to find out if there is cause for concern. If your baby is found to be at risk, you may need treatment or monitoring, or to have labor induced if it is late enough in your pregnancy. If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A NONSTRESS TEST?

Additional noninvasive tests to check your developing baby's heart rate and health can include:

Biophysical profile. This test combines a nonstress test with an ultrasound. An ultrasound imaging test uses sound waves to create a picture. The ultrasound checks your baby's breathing, muscle tone, and amniotic fluid level.

Contraction stress test. This test checks how your baby's heart reacts when your uterus contracts. To make your uterus contract, you may be asked to rub your nipples through your clothing or may be given a medicine called oxytocin, which can cause contractions.

These tests pose no known risks to you or your baby.

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Obesity Screening

<https://medlineplus.gov/lab-tests/obesity-screening/>

OBESITY SCREENING

WHAT IS OBESITY SCREENING?

Obesity screenings check for obesity and overweight in adults and children, starting as early as age 2. In general, having obesity and being overweight both mean that your weight is higher than what is considered healthy for your height:

Obesity means that you have too much body fat.

Overweight usually means that you have too much body fat. But the extra weight may come from muscle, bone, and/or holding too much water in your body.

Extra fat from obesity and overweight increases your risk of developing serious, long-lasting health problems, such as:

Heart and blood vessel diseases, including heart attack and stroke

High blood cholesterol

Type 2 diabetes

Nonalcoholic fatty liver disease (NAFLD)

Arthritis

Certain types of cancer

The more fat you have, the higher your risk of developing these problems. Children who have obesity may develop many of the same health problems as adults with obesity. They are also more likely to have obesity in adulthood and develop serious health problems later in life.

Obesity screening helps find out if too much body fat could be a health concern for you. Measuring body fat is difficult and expensive. So, obesity screening estimates how much body fat you have. The estimate is made with a calculation called a BMI (body mass index).

HOW IS A BMI CALCULATED?

BMI for adults ages 20 and older is calculated from height and weight information. The result is a number called a BMI score. Higher BMI scores are linked to larger amounts of body fat and a higher risk for certain health problems. For most adults:

A healthy weight is a BMI between 18.5 and 24.9

Overweight is a BMI between 25 and 29.9

Obesity is a BMI of 30 and above

BMI for children over age 2 and teens is also based on weight and height. But age and sex are part of figuring out if a child or teen has too much body fat. That's because the normal amount of body fat is different at each stage of growth. And the normal amount of fat is different for boys and girls.

To adjust for these differences, your child's BMI is compared to standard growth charts for children of their age and sex. The result is called a BMI percentile. For example, if your child's BMI is in the 25th percentile, it means that 25% percent of children the same age and sex have a lower BMI, and 75% percent have a higher BMI.

Because height and weight changes with growth, your child's BMI will be tracked over time to understand if obesity or overweight may be a health problem. For most children and teens:

A healthy weight is a BMI between the 5th and 85th percentile.

Overweight is a BMI between the 85th and 94th percentile.

Obesity is a BMI in the 95th percentile or higher.

A BMI is a fairly accurate way to screen most people for too much body fat, but it's not perfect. Your BMI can't tell if your weight is from extra fat or muscle. This means a very muscular person could have a BMI in the overweight range even though they don't have too much body fat.

Also, some people who have a healthy BMI may still have high amounts of body fat. This may be more common in groups that tend to have lower BMI scores, such as Asians and older adults who have lost muscle. Talk with your health

care provider if you have questions about how accurate a BMI is for your body type.

WHAT IS AN OBESITY SCREENING USED FOR?

An obesity screening with a BMI is used to find out if an adult or a child has an unhealthy body weight for their height. This helps a provider understand a person's risk for diseases that are linked to extra body fat. But obesity screening doesn't show the amount of fat a person has, and it can't diagnose any health conditions, so other tests may be needed. A BMI may also be used to find out if weight-loss efforts are working.

WHY DO I NEED OBESITY SCREENING?

Most adults and children ages 2 and older should be screened with a BMI at least once a year.

Yearly obesity screening tracks your BMI over time. If it gets higher, it usually means you're gaining extra fat. Even if your higher BMI stays in the healthy range, weight gain from fat can still increase your risk for weight-related diseases. Obesity screening can help catch weight gain early so you can take steps to control your weight before it harms your health.

If you already have obesity or are overweight, screening helps your provider monitor your risk for health problems related to body fat. Information about your risk level helps you and your provider consider the pros and cons of different weight-loss treatments.

WHAT HAPPENS DURING AN OBESITY SCREENING?

An obesity screening is usually part of a routine checkup that includes a physical exam. Your provider will measure your height and weight. That information is usually entered into an online BMI calculator. If you know your height and weight, you can find your BMI online:

Use this BMI calculator for adults.

Use this BMI percentile calculator for children and teens.

Your provider may also use other tests to estimate how much body fat you have. These include:

A waist measurement. Adults who have too much fat around their abdomen (belly) have an increased risk of developing conditions related to obesity. Even if you have a healthy BMI, your risk may be high if your waist size is:

More than 40 inches for men

More than 35 inches for women who aren't pregnant

More than 40 inches for men

More than 35 inches for women who aren't pregnant

Skinfold measurements. A skinfold test measures the thickness of pinch of skin and fat at several places on your body. A special tool gently measures skinfolds on your belly, back, thigh, the back of your upper arm, and/or other places on your body. The results are used to estimate how much of your body is fat.

If your obesity screening shows that you may have too much body fat, your provider may ask questions to help find out why. This often includes discussing your:

Medical history and medicines you take. Some health conditions cause people to gain weight, such as metabolic syndrome, hypothyroidism, and polycystic ovary syndrome (in females only). Your provider may order blood tests to confirm or rule out these and other conditions.

Many medicines can cause weight gain, too. Examples include:

Certain antidepressants

Beta-blockers, which are used to treat high blood pressure

Steroids, which are often used to treat autoimmune disease

Certain diabetes medicines

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Eating habits. Obesity and overweight develop over time when you take in more calories than you use. So, your provider will want to know how much you eat and drink. What you eat and drink matters, too. Too much saturated fat and sugar can cause weight gain.

Physical activity. A lack of physical activity is linked with weight gain. Your provider will want to know how much exercise you get and how much time you spend sitting.

Sleep habits. Not getting enough sleep or poor-quality sleep can make you more likely to overeat. That's because sleep affects hormones that control hunger.

Stress levels. Stress can affect your brain and trigger changes in your hormones that make you eat more and store more fat.

Family history. You're more likely to become obese if close family members have obesity. You may have inherited genes that affect weight. The eating and exercise habits of the people around you can affect your habits and weight, too.

WILL I NEED TO DO ANYTHING TO PREPARE FOR AN OBESITY SCREENING?

You don't need to prepare for an obesity screening. If you're also having certain blood tests to learn more about your health, you may need to fast (not eat or drink) for a period of time. Your provider will let you know how to prepare.

ARE THERE ANY RISKS TO THE SCREENING?

There is no risk to having a BMI or a measurement of your waist or skinfolds.

WHAT DO THE RESULTS MEAN?

If an obesity screening shows that you have obesity or are overweight, your provider will work with you to develop a treatment plan. The goal of the plan is to reduce your risk of health problems linked to extra body fat.

Your plan will depend on how much weight you need to lose and the cause of your weight gain. Most plans are likely to include lifestyle changes, such as:

Eating a healthier, lower calorie diet

Getting more exercise

Getting enough good-quality sleep

Depending on your BMI and health, your provider may also suggest:

Support from a dietary or nutritional counselor

A behavioral weight-loss treatment program

Prescription weight-loss medicines combined with lifestyle changes

Weight loss surgery or devices, such as balloons, that are put in the stomach to help you lose weight

If you have questions about obesity screening or treatments to lower your BMI, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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RELATED HEALTH TOPICS

Diabetes Type 2

Diets

Heart Diseases

How Much Exercise Do I Need?

Obesity

Obesity in Children

Weight Control

Weight Loss Surgery

RELATED MEDICAL TESTS

Blood Glucose Test

Cortisol Test

How to Cope with Medical Test Anxiety

TSH (Thyroid-stimulating hormone) Test

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Obsessive Compulsive Disorder (OCD) Test

<https://medlineplus.gov/lab-tests/obsessive-compulsive-disorder-ocd-test/>

OBSESSIVE COMPULSIVE DISORDER (OCD) TEST

WHAT IS AN OBSESSIVE-COMPULSIVE DISORDER (OCD) TEST?

An obsessive-compulsive disorder (OCD) test can help you find out if certain symptoms are caused by OCD or a physical condition. This helps to make sure you get the right treatment to reduce your symptoms and improve your quality of life.

OCD is a type of anxiety disorder. People with OCD may have obsessions, compulsions, or both:

Obsessions are repeated unwanted thoughts, urges, or images that can cause distress or anxiety.

Compulsions are behaviors that you feel like you need to do over and over to try to reduce your anxiety or stop the obsessive thoughts.

OCD is different than regular habits and routines. It's not unusual to brush your teeth at the same time every morning or sit in the same chair for dinner every night. But with OCD, the compulsive behaviors usually take up more than an hour a day. You may be frustrated or embarrassed by your compulsive behaviors but feel unable to stop them. And if you're unable to perform the compulsive behavior, you may become more distressed. Stress may worsen these behaviors. The compulsions can take up a lot of your time, reduce your quality of life, and get in the way of your daily responsibilities.

OCD usually starts in teens or young adults but can start in childhood. The cause of OCD is unknown. However, many researchers believe that genetics, brain biology and chemistry, and your environment may play a role. It often runs in families.

An OCD test can help find out if you have obsessive-compulsive disorder so you can get treated. Treatment can reduce your symptoms and improve your quality of life.

Other names: OCD screening

WHAT IS IT USED FOR?

This test is used to find out if certain symptoms are being caused by OCD.

WHY DO I NEED AN OCD TEST?

This test may be done if you or your child is having obsessive thoughts and/or showing compulsive behaviors.

Common obsessions include:

Fear of germs or contamination

Fear that harm will come to yourself or your loved ones

An overwhelming need for neatness and order

Constant worries that you've left something undone, like left the stove on or the door unlocked

Fear of losing or misplacing something

Fear of losing control over your behavior

Forbidden thoughts or images involving sex or religion

Aggressive thoughts toward yourself or others

Common compulsions include:

Repeated hand washing, often until chapped or sore

Checking and rechecking that appliances and lights are turned off or that the door is locked

Ordering or arranging items in a particular way

Repeating certain actions, such as sitting down and getting up from a chair

Constantly cleaning

Repeatedly checking buttons and zippers on clothing

Counting in a certain pattern over and over

WHAT HAPPENS DURING AN OCD TEST?

Your health care provider may give you a physical exam and order blood tests to find out if your symptoms are being caused by certain medicines, another mental illness, or other physical disorders.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

You may be tested by a mental health provider in addition to or instead of your primary care provider. A mental health provider is a health care professional who specializes in diagnosing and treating mental health conditions.

If you are being tested by a mental health provider, they may ask you detailed questions about your thoughts and behaviors. You may also be asked to fill out a series of questions about these issues.

WILL I NEED TO DO ANYTHING TO PREPARE FOR AN OCD TEST?

You don't need any special preparations for an OCD test.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to having a physical exam or filling out a series of questions for a mental health provider.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand the results of an obsessive-compulsive disorder test, your provider will consider your symptoms, medical history, and the results of other tests.

Your symptoms would usually be considered obsessive-compulsive disorder when:

The thoughts or behaviors take up an hour a day or more of your day

You can't control your excessive thoughts and behaviors

The behaviors may give you some relief from anxiety, but you don't enjoy performing the behaviors

The thoughts and behaviors interfere with personal relationships, work, and other important parts of daily life

They are not caused by a substance use disorder

Treatment for OCD usually includes one or more of the following:

Talk therapy, also called psychotherapy or counseling.

Cognitive-behavioral therapy (CBT), a type of talk therapy that helps you change negative thoughts or how you react to things that cause you to feel anxiety.

Cognitive-behavioral therapy (CBT), a type of talk therapy that helps you change negative thoughts or how you react to things that cause you to feel anxiety.

Other mental health counseling.

Medicines such as certain types of antidepressants.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN OCD TEST?

If you are diagnosed with OCD, your provider may refer you to a mental health provider for treatment. There are many types of providers who treat mental health disorders. Some specialize in OCD. The most common types of mental health providers include:

Psychiatrists are medical doctors who specialize in mental health disorders. Psychiatrists can prescribe medicine. Psychologists are professionals trained in psychology. Psychologists may use one-on-one counseling and/or group therapy sessions. They often have doctoral degrees, but they do not have medical degrees. They can't prescribe medicine unless they have a special license. Some psychologists work with providers who can prescribe medicine. Psychiatric or mental health nurses are nurses with special training in mental health disorders. Nurses who may have a master's or doctoral degree in psychiatric-mental health nursing include advanced practice registered nurses (APRNs), certified nurse practitioners (CNP), and clinical nurse specialists (CNSs). In some states, certain nurses can prescribe medicines.

Licensed clinical social workers have at least a master's degree in social work with special training in mental health. They can't prescribe medicine, but they may work with providers who can prescribe medicine. Providers who are licensed clinical social workers usually have LCSW or LICSW after their names.

Licensed professional counselors (LPC) may also be called clinicians or therapists. States have different names for these licenses, such as LMFT (licensed marriage and family therapist). These professionals usually have a master's degree in a field related to mental health. They can't prescribe medicine but may work with providers who can prescribe it.

If you don't know which type of mental health provider you should see or how to find a mental health provider who can best treat your OCD, talk to your regular health care provider.

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RELATED HEALTH TOPICS

Antidepressants

Anxiety

Depression

Mental Disorders

Mental Health

Obsessive-Compulsive Disorder

RELATED MEDICAL TESTS

Depression Screening

Mental Health Screening

Panic Disorder Test

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Opioid Testing

<https://medlineplus.gov/lab-tests/opioid-testing/>

OPIOID TESTING

WHAT IS OPIOID TESTING?

Opioid testing looks for opioids in a sample of your urine (pee), blood, saliva (spit), hair, or sweat.

Opioids include powerful medicines that health care providers may prescribe to relieve pain from serious injuries, illnesses, or surgery. Opioids also include illegal drugs, such as heroin.

The terms "opioids" and "opiates" are often used to mean the same thing, but they're slightly different. Opioids include: Opiates, which are natural opioids made from the opium poppy plant. Opiates include the medicines codeine and morphine, and the illegal drug heroin.

Synthetic and semi-synthetic opioids, which are medicines made in labs, including:

Oxycodone.

Hydrocodone.

Hydromorphone.

Oxymorphone.

Methadone.

Fentanyl.

Oxycodone.

Hydrocodone.

Hydromorphone.

Oxymorphone.

Methadone.

Fentanyl.

Prescription opioids are relatively safe for reducing pain if you take them for a short time according to your provider's instructions. But opioids can increase feelings of pleasure and well-being in some people. For these reasons, misuse of prescription opioids is common.

Prescription opioid misuse means taking the medicine in a different way or for a different purpose than your provider prescribed. It includes:

Taking more than your prescribed dose or taking it more often.

Getting and using prescription pills from friends or family members, even if you're using the pills for a medical condition.

Taking prescription drugs to get high.

Mixing prescription opioids with alcohol or other drugs.

Opioid misuse is dangerous because opioids are highly addictive. Misusing them can lead to opioid use disorder (OUD), which is sometimes called opioid addiction. OUD can cause difficulties with work, school, and homelife. It can even lead to overdose and death.

Opioid testing can't diagnose opioid misuse or OUD. But it can be used with questionnaires called drug use screening tests to help find out who may need help with a drug problem.

Other names: opioid screening, opiate screening, opiate testing

WHAT IS IT USED FOR?

Opioid testing may be used for different purposes, including:

Therapeutic drug monitoring. Some opioids can be used during certain medical treatments. During these treatments, your provider may monitor opioid levels to make sure the medicine is working and that you don't have a dangerous amount of opioids in your blood.

Monitoring misuse of prescription opioids. If your provider prescribed opioids for pain, opioid tests may be used to make

sure you're taking the right amount of medicine. Your provider may also order this test if they suspect you are taking opioid medicines that they didn't prescribe for you.

As part of an overall drug testing program to check for a variety of drugs, including marijuana, cocaine, and opioids. Drug testing may be used for:

Employment. Employers may test you for drugs before hiring you and/or after hiring to check for on-the-job drug use.

Legal evidence. Testing may be part of a criminal or motor vehicle accident investigation. Drug testing may also be ordered as part of a court case.

Drug treatment. Drug testing may be used to monitor treatment in programs for drug or alcohol use disorder.

Sports. Professional and other athletes are often tested for drugs.

Employment. Employers may test you for drugs before hiring you and/or after hiring to check for on-the-job drug use.

Legal evidence. Testing may be part of a criminal or motor vehicle accident investigation. Drug testing may also be ordered as part of a court case.

Drug treatment. Drug testing may be used to monitor treatment in programs for drug or alcohol use disorder.

Sports. Professional and other athletes are often tested for drugs.

Drug testing may be scheduled ahead of time or done randomly without notice.

WHY DO I NEED OPIOID TESTING?

Your provider may order opioid testing if you're taking prescription opioids to treat:

Chronic pain.

Post-surgical pain.

A persistent cough.

Severe pain due to trauma.

Cancer-related pain.

Certain tests can tell if you are taking more medicine than prescribed, which can be a sign of misuse or addiction.

Your provider may also order opioid testing if you have other signs or symptoms of opioid misuse or addiction, which may include:

Calling early for refills or asking for higher doses.

Complaining of increasing pain even though your condition isn't getting worse.

Showing signs of opioid use when you don't have a prescription for opioid medicines. Some common signs of opioid use include:

Constricted (abnormally small) pupils.

Trouble staying awake or falling asleep at inappropriate times.

Flushed, itchy skin.

Personality changes.

Social withdrawal.

Neglecting responsibilities.

Constricted (abnormally small) pupils.

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Flushed, itchy skin.

Personality changes.

Social withdrawal.

Neglecting responsibilities.

Having withdrawal symptoms when not taking opioids, which may include:

Frequent yawning.

Wide open pupils (the openings that let light into your eyes).

Uncontrollable leg movements.

Increased heart rate and blood pressure.

Nausea and vomiting.

Frequent yawning.

Wide open pupils (the openings that let light into your eyes).

Uncontrollable leg movements.

Increased heart rate and blood pressure.

Nausea and vomiting.

In a hospital emergency room, a provider may order opioid testing if you have signs of a possible drug overdose, such as slow shallow breathing and very low blood pressure.

You may also be asked to take a drug test, which includes testing for opioids, for a job, to participate in certain organized sports, or as part of a police investigation or court case.

WHAT HAPPENS DURING AN OPIOID TEST?

Opioid testing may be done in a variety of locations, including labs, hospitals, drug treatment centers, and workplaces. Most opioid tests require a urine sample. Other opioid tests may use samples of your blood, saliva, sweat, or hair.

For a urine test, you will be given instructions for how to collect your urine sample. Follow them carefully. In certain cases, a health care professional or other person may need to be present while you provide your sample. This is to make sure the urine is yours and isn't contaminated with anything that might affect the test results.

For a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a saliva test, a sample may be collected in one of two ways:

A swab or pad will be placed inside of your cheek for a few minutes until it is soaked with saliva. OR

You'll spit into a tube.

For a sweat test, you'll wear a patch on your skin for five to seven days. The patch will absorb your sweat.

For a hair test, scissors will be used to cut a small sample of hair from near your scalp. Hair may be cut from other parts of your body instead. Hair samples can show long-term opioid use.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Be sure to tell the testing professional or your provider if you are taking any prescription or over-the-counter medicines. Some of these medicines may show up as opioids on your test results. Poppy seeds can also show up as opioids. So, you should avoid foods with poppy seeds for at least three days before your test.

ARE THERE ANY RISKS TO THE TEST?

There are no known risks to having a urine, saliva, sweat or hair test. There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your test results depend on the type of opioid testing you had.

Tests to screen for drugs in your system only show whether opioids are in your sample. They can't show what kind or how much:

Positive results mean that opioids were found. But positive test results may not be accurate (false positives), so follow-up testing is usually required to confirm the results.

Negative results mean that either:

No opioids were found in your sample. That may be because you haven't taken any, or you took them but they are no longer in your body.

A very small amount of opioids was found, but too little to be called a positive test result. In certain cases, follow-up tests

may be done to get more information.

No opioids were found in your sample. That may be because you haven't taken any, or you took them but they are no longer in your body.

A very small amount of opioids was found, but too little to be called a positive test result. In certain cases, follow-up tests may be done to get more information.

Confirmation tests may be done if there is a question about the results of a screening test. These tests can show more detail about the amount and/or type of opioids found in your sample.

Opioid test results can be complicated to understand. They may be affected by how fast your body uses opioids, other medicines you take, and the different ways labs do the tests. So, it's best to ask your provider to explain what your results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT OPIOID TESTING?

You can buy at-home tests for many opioids and other prescription and illegal drugs. These tests only check whether opioids show up in your sample. There are two types:

At-home tests let you do the entire test at home and get rapid results. But if the test result finds opioids, you'll need to have a follow-up lab test to check the accuracy of the result. Some at-home tests include a kit for collecting a sample, usually urine or saliva, to send to a lab in case the home test result is positive.

A self-collection test has a kit for gathering urine, saliva, or another type of sample to send to a lab for testing.

It's best to talk with your provider about using home drug testing so you can decide if it's right for your situation.

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RELATED HEALTH TOPICS

Drug Use and Addiction

Heroin

Opioid Overdose

Opioid Use Disorder (OUD) Treatment

Opioids and Opioid Use Disorder (OUD)

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At-Home Medical Tests

Drug Testing

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How to Cope with Medical Test Anxiety

How to Understand Your Lab Results

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Osmolality Tests

<https://medlineplus.gov/lab-tests/osmolality-tests/>

OSMOLALITY TESTS

WHAT ARE OSMOLALITY TESTS?

Osmolality tests measure the amount of certain substances in your blood, urine, or stool (poop). The tests can show how your kidneys are working and whether you have an unhealthy fluid balance.

The substances that the tests measure include:

Glucose (sugar).

Urea (a waste product made in the liver).

Several electrolytes, such as sodium, potassium, and chloride. Electrolytes are minerals that have an electrical charge when they are dissolved in water or body fluids. They help control the balance of fluids in the body. They also help balance the acids and bases (pH balance) in your body.

Osmolality is the concentration of particles dissolved in your body fluid. The concentration is based on the amount of particles that are in a specific amount of that fluid. Different factors can make it go up or down:

Your blood osmolality increases when the amount of water in your blood decreases or the number of substances such as sodium, chloride, and glucose increases. When your blood osmolality goes up it causes your kidney to hold onto water. This makes your urine more concentrated.

Your blood osmolality decreases when the amount of water in your blood increases. This makes your urine more diluted. This process helps your body maintain a healthy fluid balance. But sometimes this balance can be thrown off. Many different conditions and factors can cause an unhealthy fluid balance, such as too much salt intake, kidney disease, heart disease, and some types of poisoning.

Other names: serum osmolality, plasma osmolality, urine osmolality, stool osmolality, osmotic gap

WHAT ARE THEY USED FOR?

Osmolality tests may be used for a variety of reasons. A blood osmolality test, also known as a serum osmolality test, is most often used to:

Check the balance between water and certain substances in your blood.

Find out if you have swallowed a poison such as antifreeze or rubbing alcohol.

Check for an overdose of drugs such as salicylates, commonly found in aspirin.

Help diagnose dehydration, a condition in which your body loses too much fluid.

Help diagnose overhydration, a condition in which your body retains (holds in) too much fluid.

Help diagnose diabetes insipidus, a condition that affects the kidneys. It causes frequent urination, which can lead to dehydration.

Sometimes, your blood plasma is also tested for osmolality by measuring the amount of sodium dissolved in it.

A urine osmolality test is often used along with a serum osmolality test to check your body's fluid balance. The urine test may also be used to find out the reason for increased or decreased urination.

A stool osmolality test is most often used to find out the reason for chronic diarrhea that is not caused by a bacterial or parasitic infection.

WHY DO I NEED AN OSMOLALITY TEST?

You may need a serum osmolality or urine osmolality test if you have symptoms of a fluid imbalance, diabetes insipidus, or certain types of poisoning.

Symptoms of a fluid imbalance and diabetes insipidus are similar and may include:

Excessive thirst (if dehydrated)

Nausea and vomiting

Headache

Confusion

Fatigue

Seizures

Symptoms of poisoning will be different depending on the type of substance that was swallowed but may include:

Nausea and vomiting

Convulsions, a condition that causes uncontrollable shaking of your muscles

Trouble breathing

Slurred speech

You may also need a urine osmolality test if you have trouble urinating or are urinating too much. This test measures how well your kidneys are working.

You may need a stool osmolality test if you have chronic diarrhea that can't be explained by a bacterial or parasitic infection or another cause such as intestinal damage.

WHAT HAPPENS DURING AN OSMOLALITY TEST?

During a blood test (serum osmolality or plasma osmolality):

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

During a urine osmolality test:

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

During a stool osmolality test:

You will need to provide a fresh sample of loose or liquid stool for your test. Your provider will probably give you a container or kit with instructions on how to collect the sample. There are different ways to collect stool samples so follow the instructions carefully. In general, you'll need to:

Label the container with your name, the collection date, and time.

Collect a stool sample as instructed. This usually involves using a clean, dry container, or special paper or plastic wrap placed over the toilet to catch the stool.

Make sure the stool doesn't mix with any urine, toilet paper, or water from the toilet.

Close the container tightly.

Wash your hands well with soap and water.

Return the container according to the instructions. This is important to get an accurate test result.

If you're collecting a sample from diapers, you'll get special instructions for using plastic wrap inside of a clean diaper. In certain cases, a provider may use a swab to take a stool sample from the rectum.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast^Â (not eat or drink) for 6 hours before the test or limit fluids 12 to 14 hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO OSMOLALITY TESTS?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no risk to having a urine or stool test.

WHAT DO THE RESULTS MEAN?

If your serum osmolality results were not normal, it may be a sign of:

Dehydration or overhydration

Too much or too little salt in the blood

Antifreeze or other types of poisoning

Diabetes insipidus

Stroke

If your urine osmolality results were not normal, it may be a sign of:

Dehydration or overhydration

Heart failure

Liver disease

Kidney disease

If your stool osmolality results were not normal, it may be a sign of:

Factitious diarrhea, a condition caused by the overuse of laxatives

Malabsorption, a condition that affects your ability to digest and take in nutrients from food

If you have questions about your results, talk to your provider. To understand the results of an osmolality test, your provider may consider your symptoms, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT OSMOLALITY TESTS?

Your provider may order more tests along with or after your osmolality test. These may include:

Complete blood count (CBC)

Blood urea nitrogen (BUN) test

Blood glucose test

Electrolyte panel

Albumin blood test

Fecal occult blood test (FOBT)

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RELATED HEALTH TOPICS

Dehydration
Diabetes Insipidus
Diarrhea
Fluid and Electrolyte Balance
Minerals
Poisoning
Sodium

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Albumin Blood Test
Blood Glucose Test
BUN (Blood Urea Nitrogen)
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Complete Blood Count (CBC)
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How to Prepare for a Lab Test
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Sodium Blood Test
What You Need to Know About Blood Testing

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Ova and Parasite Test

<https://medlineplus.gov/lab-tests/ova-and-parasite-test/>

OVA AND PARASITE TEST

WHAT IS AN OVA AND PARASITE TEST?

An ova and parasite test looks for intestinal parasites and their eggs (ova) by checking a sample of your stool (poop) under a microscope. The test is also called an O&P.

Intestinal parasites are tiny life forms that can live and reproduce in your digestive system. They most commonly infect your intestines (the long, tube-shaped organ in your belly that helps digest food). A parasitic infection can cause diarrhea, gas, stomach cramps, and other abdominal (belly) symptoms.

When people and animals have intestinal parasites, eggs and parasites are found in their stool. Tiny amounts of contaminated stool that you can't see, smell, or taste can get into water, soil, food, and beverages. If a person with an infection doesn't wash their hands well after using the bathroom, parasites can also get on any surfaces that they touch. If you swallow parasites or their eggs in contaminated water or food, you can develop an infection. You can also become infected if you put anything in your mouth that has touched contaminated surfaces.

There are many types of intestinal parasites. The most common types in the U.S. include:

Giardia

Cryptosporidium

Entamoeba histolytica, which causes a disease called "amebiasis." In the US, amebiasis happens mostly in people who have been in countries where it is common. This includes tropical areas with poor sanitary conditions.

These parasites are often found in water, including rivers, lakes, ponds, and streams. They are also found in swimming pools, hot tubs, and drinking water. That's because certain types of intestinal parasites are not easily killed by chlorine in water.

Other intestinal parasites that aren't common in the U.S. include many types of worms, such as pinworms. These parasites also spread through contaminated soil, water, food, and drink. But some of them, such as hookworms, can infect you by getting into your skin, especially if you walk barefoot on contaminated ground or other surfaces.

Healthy people often recover from intestinal parasite infections without treatment. When treatment is needed, it usually works to get rid of the parasites.

But intestinal parasites can cause severe, long-lasting diarrhea and other serious conditions in people with weakened immune systems. This includes people with HIV, cancer, certain genetic disorders, and people who take certain medicines that affect their immune systems. Infants and older adults also have weaker immune systems.

Other names: parasitic examination (stool), stool sample exam, stool O&P, fecal smear, egg cyst

WHAT IS IT USED FOR?

An ova and parasite test is used to see if intestinal parasites are causing symptoms that appear to be from an intestinal infection. This information helps your health care provider know how to treat your illness. The test may also be used to see if treatment for intestinal parasites is working.

WHY DO I NEED AN OVA AND PARASITE TEST?

Your provider may order an O&P test if you have symptoms that could be caused by an intestinal parasite, and the symptoms have lasted for a few days. They include:

Diarrhea

Abdominal pain

Blood and/or mucus in the stool

Unusually foul-smelly, greasy stools that float

Nausea and vomiting

Gas
Fever
Weight loss

Sometimes these symptoms go away without treatment, so testing is not always needed. Your provider will be more likely to order a test for parasites if:

You have developed serious symptoms, including an electrolyte imbalance or dehydration.

You have a weakened immune system because of disease, medicines you take, or your age. Older adults and very young children often have weaker immune systems.

You could have been exposed to parasites. This includes:

Drinking or accidentally swallowing water from a stream and/or lake.

Traveling to another country where intestinal parasites are more common.

Being around another person who has an intestinal parasite infection.

Drinking or accidentally swallowing water from a stream and/or lake.

Traveling to another country where intestinal parasites are more common.

Being around another person who has an intestinal parasite infection.

WHAT HAPPENS DURING AN OVA AND PARASITE TEST?

You will need to provide a stool sample for your test. Your provider will probably give you a container or kit with instructions on how to collect the sample. There are different ways to collect stool samples so follow the instructions carefully. In general, you'll need to:

Label the container with your name, the collection date, and time.

Collect a stool sample as instructed. This usually involves using a clean, dry container, or special paper or plastic wrap placed over the toilet to catch the stool.

Make sure the stool doesn't mix with any urine, toilet paper, or water from the toilet.

Close the container tightly.

Wash your hands well with soap and water.

Return the container according to the instructions.

If you're collecting a sample from diapers, you'll get special instructions for using plastic wrap inside of a clean diaper. In certain cases, a provider may use a swab to take a stool sample from the rectum.

To get accurate test results, you may need to collect several stool samples over a few days. That's because parasites may not show up in your stool every day. Testing several samples will increase the chance that the parasites will be found.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an ova and parasite test.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having an ova and parasite test.

WHAT DO THE RESULTS MEAN?

A negative or normal test result means that no parasites or eggs were found in your stool sample. This may mean:

You don't have parasites.

There were not enough parasites in your sample to show up in a test.

Your provider may order another test and/or different tests to help find out what's causing your symptoms.

A positive or abnormal test result means that parasites and/or eggs were found in your stool sample. You have intestinal

parasites. The results will also show the type of parasites you have and how many were in your sample. Treatment for an intestinal parasite infection will depend on the type of parasite you have and your general health. Treatment may include medicines that get rid of the parasites and/or relieve symptoms. You will also need to drink plenty of fluids to prevent dehydration (the loss of too much fluid from your body) from the diarrhea and vomiting. You may be told not to swim until all your symptoms have been gone for at least two weeks. That can help make sure you don't spread parasites to others. If you have questions about your results, talk with your provider. Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN OVA AND PARASITE TEST?

There are many steps you can take to protect yourself from getting infected with intestinal parasites. For example: Wash your hands with warm water and soap, which work better than hand sanitizers. Always wash:

Before, during, and after preparing food

Before and after eating and caring for someone who is or may be infected with parasites

After going to the bathroom, changing a diaper, handling animals, and gardening or working with soil

Before, during, and after preparing food

Before and after eating and caring for someone who is or may be infected with parasites

After going to the bathroom, changing a diaper, handling animals, and gardening or working with soil

Don't drink untreated water from lakes, streams, or rivers. And be careful not to swallow water when you swim.

When camping or traveling to countries where the water may not be safe, avoid tap water, ice, and uncooked foods washed with tap water. Bottled water is safe.

If you are unsure if water is safe, boil it before drinking. Boiling water for one to three minutes will kill the parasites. Wait until the water cools before drinking.

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RELATED HEALTH TOPICS

- Abdominal Pain
- Bowel Movement
- Cryptosporidiosis
- Dehydration
- Diarrhea
- Germes and Hygiene
- Giardia Infections
- Infectious Diseases
- Nausea and Vomiting
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RELATED MEDICAL TESTS

- Bacteria Culture Test
- C. diff Testing
- How to Cope with Medical Test Anxiety
- How to Understand Your Lab Results
- White Blood Cell (WBC) in Stool
- Xylose Testing

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Panic Disorder Test

<https://medlineplus.gov/lab-tests/panic-disorder-test/>

PANIC DISORDER TEST

WHAT IS A PANIC DISORDER TEST?

A panic disorder test can help you find out if certain symptoms are caused by a panic disorder or a physical condition. This helps to make sure you get the right treatment.

Panic disorder is a type of anxiety disorder in which you have repeated panic attacks. A panic attack is a sudden feeling of intense fear or anxiety. Besides emotional distress, a panic attack can cause physical symptoms. They include chest pain, rapid heartbeat, dizziness, and shortness of breath. A panic attack can last anywhere from a few minutes to over an hour.

During a panic attack, some people think they are having a heart attack.

Panic attacks may occur as part of an anxiety disorder or if you have another mental health condition such as depression. Usually, panic attacks cannot be predicted. Some panic attacks happen in response to a stressful or scary situation. Other attacks happen without a clear reason. Panic attacks can occur as often as a few times a day or only a few times a year.

If you have panic attacks, it doesn't mean you will develop a panic disorder. Many people only have one or two panic attacks in their lifetime and get better without treatment. But having a panic disorder is not common. It only affects a small percentage of adults, but it's more common in women than in men.

While panic disorder is not life-threatening, it can be upsetting and affect your quality of life. If left untreated, it can lead to other health conditions, including depression and substance use disorder.

Other names: panic disorder screening

WHAT IS IT USED FOR?

A panic disorder test is used to find out if certain symptoms are caused by a panic disorder or a physical condition, such as a heart attack.

WHY DO I NEED A PANIC DISORDER TEST?

You may need a panic disorder test if you've had two or more recent panic attacks for no clear reason and are afraid of having more panic attacks. Panic attacks have at least four of these symptoms at the same time:

Palpitations (racing or pounding heart)

Chest pain or discomfort

Shortness of breath or feeling as if you're suffocating

Sweating, chills, or heat sensations

Feeling dizzy, light-headed, or unsteady

Trembling or shaking

Nausea or upset stomach

Fear of losing control

Fear of dying

Feeling like you are choking

Depersonalization (feeling as if you are watching yourself from outside of your body)

Derealization, (feelings of unreality, as if you are dreaming)

Numbness or tingling

WHAT HAPPENS DURING A PANIC DISORDER TEST?

Your health care provider may give you a physical exam and ask you about your feelings, mood, and other symptoms. Your provider may also order blood tests and/or tests on your heart to rule out a heart attack or other physical conditions such as thyroid problems. Conditions related to substance use will also be considered since some of its symptoms can be similar to panic attacks.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

You may also be tested by a mental health provider. A mental health provider is a health care professional who specializes in diagnosing and treating mental health conditions.

If you are being tested by a mental health provider, they may ask you more detailed questions about your feelings and actions. You may also be asked to fill out a series of questions about these issues.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A PANIC DISORDER TEST?

You don't need any special preparations for a panic disorder test.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to having a physical exam or filling out a series of questions.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand the results of a panic disorder test, your provider will consider your symptoms, medical history, and the results of other tests. Symptoms of panic disorder often begin before age 25 but may occur in the mid-30s and include:

Frequent, unexpected panic attacks

Ongoing worry about having another panic attack

Avoiding places and situations because you're worried that they may cause a panic attack

Fear of losing control

No other cause of a panic attack, such as substance use, a mental health condition or physical condition

Treatment for panic disorder usually includes one or more of the following:

Talk therapy, also called psychotherapy or counseling, to help you understand your feelings.

Cognitive-behavioral therapy (CBT), a type of talk therapy that helps you change negative thoughts or how you react to things that cause you to feel anxiety.

Mental health counseling.

Anti-anxiety, antidepressant or other medicines.

Your provider may also urge you to practice other healthy habits to reduce panic attacks. These may include avoiding alcohol, reducing caffeine, eating regular meals, and getting enough sleep and exercise. They may also recommend that you join a support group.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PANIC DISORDER TEST?

If you are diagnosed with panic disorder, your provider may refer you to a mental health provider for treatment. There are many types of providers who treat mental disorders. The most common types of mental health providers include:

Psychiatrists are medical doctors who specialize in mental health disorders. Psychiatrists can prescribe medicine.

Psychologists are professionals trained in psychology. Psychologists may use one-on-one counseling and/or group therapy sessions. They often have doctoral degrees, but they do not have medical degrees. They can't prescribe

medicine unless they have a special license. Some psychologists work with providers who can prescribe medicine. Psychiatric or mental health nurses are nurses with special training in mental health disorders. Nurses who may have a master's or doctoral degree in psychiatric-mental health nursing include advanced practice registered nurses (APRNs), certified nurse practitioners (CNPs), and clinical nurse specialists (CNSs). In some states, certain nurses can prescribe medicines.

Licensed clinical social workers have at least a master's degree in social work with special training in mental health. They can't prescribe medicine, but they may work with providers who can prescribe medicine. Providers who are licensed clinical social workers usually have LCSW or LICSW after their names.

Licensed professional counselors (LPC) may also be called clinicians or therapists. States have different names for these licenses, such as LMFT (licensed marriage and family therapist). These professionals usually have a master's degree in a field related to mental health. They can't prescribe medicine but may work with providers who can prescribe it.

If you don't know which type of mental health provider you should see, talk to your regular health care provider.

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RELATED HEALTH TOPICS

Antidepressants

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Mental Disorders

Mental Health

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RELATED MEDICAL TESTS

Depression Screening

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Pap Smear

<https://medlineplus.gov/lab-tests/pap-smear/>

PAP SMEAR

WHAT IS A PAP SMEAR?

A Pap smear is a test to screen for cervical cancer. If cervical cancer is found and treated early, there is a better chance it can be cured. This type of cancer starts in the cells of the cervix. Your cervix is the lower part of your uterus, the place where a developing baby grows during pregnancy. The cervix opens into your vagina.

Cervical cancer develops slowly over time. Before cancer appears in the cervix, the cells of the cervix go through changes in which abnormal cells begin to appear in the cervical tissue. This is called dysplasia. Over time, if not destroyed or removed, the abnormal cells may become cancer cells. They can grow and spread into other parts of the cervix and the areas around it.

During a Pap smear, your health care provider collects cells from your cervix and sends them to a lab. There, the cells are checked under a microscope for cancer or signs that they are abnormal and may become cancer. Cells that may become cancer are called precancerous cells. Finding and treating precancerous cells can help prevent cervical cancer. The Pap smear is also a reliable way to find cancer early, when it's easier to treat. A pap smear may also find certain infections or inflammation.

Other names for a Pap smear: Pap test, cervical cytology, Papanicolaou test, Pap smear test, vaginal smear technique

WHAT IS IT USED FOR?

A Pap smear looks for abnormal changes in cervical cells before they become cancer. Sometimes, the cells collected during a Pap smear are also checked for HPV (human papillomaviruses). HPV is the name of a group of related viruses, some of which can spread through sexual contact. HPV can cause cell changes that may lead to cancer, including in the cervix. Long-lasting infections with high-risk types of HPV cause almost all cervical cancers. So if you have a cervix and are infected with these types of HPV, you have a higher risk of getting cervical cancer.

Pap smears and HPV tests are cervical cancer screening tests that look for cancer before you have any symptoms. Cervical cancer screening can reduce your risk of getting cervical cancer or dying from the disease. Ask your provider which test is right for you or if you should have both a Pap smear and an HPV test.

HOW OFTEN DO I NEED A PAP SMEAR?

Your age and health history will help determine how often you should have a Pap smear. If you have a cervix, it's recommended that you have your first Pap smear at age 21.

If you're between ages 21 and 29 and your last Pap test result was normal:

Your provider may say you can wait three years until your next Pap test.

At age 25, you should begin screening with an HPV test and have them every five years through age 65.

If you're between ages 30 and 65 and your last Pap smear result was normal:

Your provider may say you can wait three years until your next Pap smear.

If you also had a normal HPV test result, your provider may say that you can wait five years until your next test.

If you had an HPV test and a Pap smear, called co-testing, and had normal results for both, your provider may say that you can wait five years until your next test.

If you're over age 65, talk with your provider about whether you still need Pap smears. If you've had regular screenings with normal results, they may tell you that you don't need Pap smears anymore.

You don't need cervical cancer screening if you had a total hysterectomy (surgery to remove your uterus and cervix) because of a condition that was not cancer. But if your hysterectomy was related to cervical cancer or precancer, ask your provider whether you need screening.

If you are at higher risk of developing cervical cancer, your provider may recommend screening more often. You may be

at higher risk if you:

Had an abnormal HPV test, Pap smear, or cervical biopsy in the recent past.

Have had a diagnosis of cervical cancer.

Have HIV. You may also need screenings after age 65.

Have a weakened immune system.

Were exposed to a drug called DES (Diethylstilbestrol) before you were born. Between the years 1940-1971, DES was sometimes prescribed to prevent miscarriages. It was later linked to an increased risk of certain cancers in the female children exposed to it before birth.

If you're under age 21, cervical cancer screening is not recommended. Your risk of cervical cancer is very low. Also, any changes in cervical cells will likely go away on their own.

If you are unsure whether you need a Pap smear, talk with your provider.

WHAT HAPPENS DURING A PAP SMEAR?

During a Pap smear:

You will be asked to undress completely or from the waist down.

You will lie on an exam table with your knees bent and your heels in footrests called stirrups.

Your provider will gently insert a plastic or metal instrument called a speculum into your vagina. This helps to widen your vagina so that your cervix can be seen. This may cause a feeling of cramping or pressure.

Your provider will then use a small, soft brush or swab to collect cells from the cervix. This doesn't usually hurt.

The cell sample is sent to a lab for examination under a microscope.

A Pap smear is often done as part of a pelvic exam. During a pelvic exam, your provider examines your uterus, ovaries, and genital area. But a pelvic exam doesn't always include a Pap smear. So, when you have a pelvic exam, ask your provider whether you'll have a Pap smear, too.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You should not have a Pap smear while you are having your period because this may affect your results. A good time to have the test is about five days after the last day of your period. For two days before your test, you should not:

Use tampons

Use birth control foam, jelly, or cream

Use other creams or medicines in your vagina

Douche (rinse the vagina with water or other fluid)

Have vaginal sex

ARE THERE ANY RISKS TO THE TEST?

You may feel some mild discomfort during a Pap smear and have some very light bleeding afterward, but there are no known risks to having a Pap smear.

WHAT DO THE RESULTS MEAN?

A Pap smear test could have the following results:

Normal or "negative" result, which usually means that no abnormal changes were found in your cervix cells. Your provider may tell you that you can wait three years for your next test. If you also had a normal HPV test result, you may be able to wait five years for your next test, depending on your age and medical history.

Unsatisfactory results, which may mean that the lab sample did not have enough cells, or the cells may have been clumped together or hidden by mucus. Your provider may ask you to come in for another test in a few months.

Unclear results, which may mean that your cells look abnormal, but it's unclear if it's due to HPV. Unclear results could also be related to pregnancy, menopause, or an infection. An HPV test may help to find out if unclear results are due to

HPV. Your provider will let you know if other tests need to be done.

Abnormal or "positive" result, which usually means that abnormal changes were found in your cervical cells. Most of the time, abnormal results do not mean you have cervical cancer. Minor changes in the cells usually go back to normal on their own. But your provider may recommend a follow-up test to check. More serious cell changes may turn into cancer if they are not removed. Finding and treating these cells early can help prevent cancer from developing.

A Pap smear doesn't diagnose cancer. It's also possible to receive a false-positive result, in which your results appear abnormal, but there are no precancerous or cancerous cells. Or you could have a false-negative, which happens when your results appear normal but there are precancer or cancerous cells.

Talk with your provider to learn what your Pap smear results mean. If they are concerned about your results, they may perform a colposcopy. This is an exam that uses a special magnifying instrument to look closely at your cervix to help diagnose or rule out cervical cancer.

Learn more about laboratory tests, reference ranges, and understanding results.

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Cervical Cancer Screening
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Parainfluenza Tests

<https://medlineplus.gov/lab-tests/parainfluenza-tests/>

PARAINFLUENZA TESTS

WHAT IS A PARAINFLUENZA TEST?

Parainfluenza viruses are a group of four types of viruses. They are a common cause of respiratory (breathing) illnesses. These illnesses can affect your nose, throat, lungs, and bronchial tubes (the airways that carry air to your lungs). Parainfluenza tests check whether a parainfluenza virus is causing an illness. The tests usually use a sample of fluid from your nose or mucus from your lungs.

Parainfluenza viruses are also called human parainfluenza viruses (HPIVs). They aren't the same as flu (influenza) viruses. The flu is more common in winter. In the U.S., parainfluenza viruses are more common in the spring, summer, and fall.

Like the flu, parainfluenza spreads easily from person to person. It can spread through the air when a person with the infection coughs or sneezes. You can also get the flu by touching a surface that has the flu virus on it, and then touching your mouth, nose, or eyes.

Parainfluenza viruses commonly cause respiratory illness in babies and younger children, but anyone can get sick from these viruses. You can also have parainfluenza illnesses many times in your life. If you're healthy, you'll probably have mild symptoms that are much like a common cold, such as a runny nose, fever, or cough. And you'll usually get better on your own.

Older adults and people of all ages who have weakened immune systems are more likely to develop more serious illness from parainfluenza infections. These illnesses include croup, bronchitis, and pneumonia.

The most common test for parainfluenza viruses is a molecular test, such as a PCR (polymerase chain reaction) tests. These tests can find very small amounts of genetic material from parainfluenza viruses in the fluid sample from your nose or lungs.

Other tests for parainfluenza virus are used less often than molecular tests. Rapid antigen tests are less accurate and must be done within a week after symptoms begin. Viral culture tests take much longer than other tests. And blood tests are mainly used for research.

Other names: human parainfluenza virus test; HPIV test; HPIV-1 test; HPIV-2 test, HPIV-3 test; paraflu 1-4 PCR; parainfluenza 1-4 by PCR; parainfluenza PCR panel; PIV

WHAT IS IT USED FOR?

Most healthy people who have symptoms of a respiratory illness do not need parainfluenza testing. That's because they usually get better on their own. But you may need this test if you have a high risk of getting seriously ill from a parainfluenza infection because you:

Have a condition that affects your immune system, such as:

Blood stem cell or bone marrow transplants

Organ transplantation

Blood cancers

HIV

Blood stem cell or bone marrow transplants

Organ transplantation

Blood cancers

HIV

Have a heart or lung disorder

Are in the hospital with community-acquired pneumonia (pneumonia that you catch during your usual daily life, not from spending time in a hospital or other health facility)

Are an older adult

For these people, treatment decisions depend on knowing whether they have a viral or bacterial infection. So, parainfluenza testing may be done as part of a respiratory pathogens panel. This is a group of tests that checks for several viruses and bacteria that cause respiratory illnesses. There is no cure for parainfluenza viruses, but bacterial infections may be treated with antibiotics.

WHY DO I NEED A PARAINFLUENZA TEST?

You or your child may need a parainfluenza test if either of you have:

Symptoms of a respiratory illness.

And another condition that increases your risk of getting seriously sick from a parainfluenza illness.

Symptoms of parainfluenza and other respiratory illnesses may include:

Fever

Sore throat

Wheezing

A long-lasting cough that may sound like a barking in children (croup)

Shortness of breath

Too much mucus in the lungs or airways (congestion)

Rapid breathing

WHAT HAPPENS DURING A PARAINFLUENZA TEST?

Tests to diagnose parainfluenza viruses are usually done on a sample of fluid from your nose or chest.

A sample from your nose may be used if you've had symptoms between 3 and 7 days. Most tests are done with this type of sample. The sample may be collected using one of these methods:

Nasal aspirate or wash. To do a nasal aspirate, a health care professional will insert a saline solution (salt water) into your nose and remove it with gentle suction.

Swab test. A health care professional will use a special swab to take a sample from either:

Deep in the back of your nose

An inch deep inside both of your nostrils and from your throat using a separate swab inserted through your mouth

Deep in the back of your nose

An inch deep inside both of your nostrils and from your throat using a separate swab inserted through your mouth

A sample from your chest may be used at any time during your illness. The fluid may be collected with one of these methods:

Sputum test. Sputum is a thick mucus from the lungs. It is different from saliva (spit). To get a sample, you will be asked to cough up sputum into a special container.

Bronchoalveolar lavage (BAL). A BAL collects a sample of fluid from your lung. This type of sample is usually used only if you are very ill. A doctor will insert a narrow, flexible tube (a bronchoscope) down your throat and into your lung. A small amount of salt water will be injected through the tube into your lung and then suctioned out. Numbing medicine will be used in your mouth and throat. You may also have medicine to help you relax.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you have a BAL, you may need to fast (not eat or drink) for several hours before the test. You should plan to have someone take you home because you may feel sleepy if you are given medicine to relax.

You don't need any special preparations for other parainfluenza tests.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a parainfluenza test:

A nasal aspirate or wash may feel uncomfortable. These effects are temporary.

Nasal and throat swabs may cause a brief gagging feeling.

A BAL may cause a sore throat for a few days. Serious problems are rare. They include bleeding in the airways, infection, or a partly collapsed lung.

WHAT DO THE RESULTS MEAN?

A negative result means that no signs of parainfluenza viruses were found in your sample. Another illness is probably causing your symptoms. But a negative test result doesn't rule out a parainfluenza illness. It's possible that there was too little virus in your sample for the test to find.

A positive result means that you likely have a parainfluenza infection. But it's possible for a PCR test to show signs of a virus even though you don't have enough virus in your body to make you sick.

Talk with your provider about what your test results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

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- Breathing Problems
- Common Cold
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- At-Home Medical Tests
- How to Cope with Medical Test Anxiety
- How to Understand Your Lab Results
- Nasal Swab
- PCR Tests
- Respiratory Pathogens Panel

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Parathyroid Hormone (PTH) Test

<https://medlineplus.gov/lab-tests/parathyroid-hormone-ptb-test/>

PARATHYROID HORMONE (PTH) TEST

WHAT IS A PARATHYROID HORMONE (PTH) TEST?

This test measures the level of parathyroid hormone (PTH) in your blood. PTH, is made by your parathyroid glands. They are four pea-sized glands in your neck.

Hormones are chemical messengers in your bloodstream. They control the actions of certain cells or organs. PTH controls the level of calcium in your blood. Calcium is one of the most important minerals in your body. Most of it is stored in your bones and teeth. Having the right amount of calcium in your blood is important for your nerves, muscles, and heart to work properly.

If your calcium blood levels are too low, your parathyroid glands will release just enough PTH into your blood to return your calcium levels to normal. PTH raises your blood calcium by:

Telling your bones to release calcium into your blood

Helping your intestines to absorb calcium from the food you eat

Helping your kidneys to keep calcium in your blood instead of getting rid of it through your urine (pee)

When your calcium blood levels return to normal, your parathyroid glands stop making PTH and your blood PTH levels decrease.

Because the amount of PTH in your blood changes with the amount of calcium in your blood, your health care provider will usually order a calcium blood test along with a PTH test. Comparing the results of both tests helps your provider understand if your parathyroid glands are working properly to control your calcium levels.

Other names: parathormone, intact PTH

WHAT IS IT USED FOR?

A PTH test may be used to:

Help find out whether parathyroid disorders are the cause of abnormal blood calcium levels. This is the most common use.

Help find the cause of abnormally low results on a phosphate blood test. That's because PTH also helps control the amount of phosphate (phosphorus) in blood.

Monitor people with chronic kidney disease.

Help find the cause of severe osteoporosis, a condition that causes your bones to become weak and break easily.

Check whether treatment for a known parathyroid disorder is working.

Check whether all overactive parathyroid tissue has been removed during surgery to treat parathyroid disorders that cause too much PTH.

WHY DO I NEED A PTH TEST?

You may need a PTH test if you:

Had a calcium test that showed your blood calcium levels aren't normal.

Have symptoms of too much calcium in your blood. These symptoms may include:

Digestive symptoms, such as:

Constipation

Nausea and vomiting

Abdominal (belly) pain
Loss of appetite

Increased thirst
Urinating (peeing) more than usual
Kidney stones
Bone, joint, and muscle symptoms, such as weakness, aches, and pains
Fatigue
A change in mental health, such as depression or confusion
Digestive symptoms, such as:

Constipation
Nausea and vomiting
Abdominal (belly) pain
Loss of appetite

Constipation
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Kidney stones
Bone, joint, and muscle symptoms, such as weakness, aches, and pains
Fatigue
A change in mental health, such as depression or confusion

Have symptoms of too little calcium in your blood. These symptoms can range from mild to severe. They may include:

Dry skin, coarse hair, and nails that easily break (after a long period of low levels)
Muscle cramps, spasms, or stiffness
Tingling in the lips, tongue, fingers, and feet
Irregular heartbeat (arrhythmia)
Seizures, if calcium levels are extremely low
Dry skin, coarse hair, and nails that easily break (after a long period of low levels)
Muscle cramps, spasms, or stiffness
Tingling in the lips, tongue, fingers, and feet
Irregular heartbeat (arrhythmia)
Seizures, if calcium levels are extremely low
Have a parathyroid disorder or chronic kidney disease.

WHAT HAPPENS DURING A PTH TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You probably won't need any special preparations for a PTH test but check with your provider. Some providers may ask you to fast (not eat or drink) before your test. You may also need to take the test at a certain time of day.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Many health conditions can cause abnormal PTH levels. To understand what your PTH test results mean, your provider will consider the results of other tests, especially a blood calcium test. Your symptoms, medical history, and family history will also be considered.

If your PTH level is higher than normal, it's called hyperparathyroidism. There are two types. The type you have generally depends on your blood calcium level:

Primary hyperparathyroidism is high PTH levels with high blood calcium levels. This often means that a problem with your parathyroid glands is causing them to make too much PTH. The extra PTH raises your blood calcium levels. Possible causes of primary hyperparathyroidism include:

A benign tumor (not cancer) in one of your parathyroid glands, called a parathyroid adenoma. This is the most common cause of high PTH levels.

Hyperplasia of your parathyroid glands, which means that they have become larger than normal.

A rare inherited condition, such as multiple endocrine neoplasia type 1 or familial hypocalciuric hypercalcemia.

Parathyroid cancer, which is rare.

A benign tumor (not cancer) in one of your parathyroid glands, called a parathyroid adenoma. This is the most common cause of high PTH levels.

Hyperplasia of your parathyroid glands, which means that they have become larger than normal.

A rare inherited condition, such as multiple endocrine neoplasia type 1 or familial hypocalciuric hypercalcemia.

Parathyroid cancer, which is rare.

Secondary hyperparathyroidism is high PTH levels with low blood calcium levels. Your parathyroid glands are working correctly by making extra PTH to try to increase your blood calcium. It's likely that a problem outside of your parathyroid glands is preventing your blood calcium levels from increasing.

Your provider will probably order more tests to find out what's causing your low blood calcium. Possible causes include:

Having low levels of vitamin D (vitamin D deficiency), phosphate, and/or magnesium

Not getting enough calcium

Digestive disorders, such as malabsorption syndromes, that make it hard for your body to absorb calcium from food

Kidney failure, which causes your body to lose too much calcium through urine

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Kidney failure, which causes your body to lose too much calcium through urine

If your PTH levels are lower than normal and your blood calcium level is also low, it may mean your parathyroid glands can't make enough PTH. This is called hypoparathyroidism. This condition is less common than hyperparathyroidism.

Possible causes include:

Damage to your parathyroid glands from surgery on your neck or radiation therapy for cancer

Certain autoimmune disorders that destroy parathyroid tissue

Certain genetic disorders, such as DiGeorge syndrome, also called 22q11.2 deletion syndrome

Getting too much vitamin D and/or calcium from supplements or from antacids that contain calcium

Serious illnesses

If your PTH levels aren't normal when compared to your blood calcium level and other test results, your provider will usually order other blood tests to learn more. If a parathyroid tumor is suspected, you may also have imaging tests. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Partial Thromboplastin Time (PTT) Test

<https://medlineplus.gov/lab-tests/partial-thromboplastin-time-ptt-test/>

PARTIAL THROMBOPLASTIN TIME (PTT) TEST

WHAT IS A PTT (PARTIAL THROMBOPLASTIN TIME) TEST?

A partial thromboplastin time (PTT) test uses a blood sample to measure how long it takes for your blood to make a clot. Normally, when you get a cut or injury that causes bleeding, many different types of proteins in your blood work together to make a clot to stop the bleeding. These proteins are called coagulation factors or clotting factors.

If any of your clotting factors are missing, at a low level, or not working properly, your blood may:

Clot too slowly after an injury or surgery. If this happens, you have a bleeding disorder. Bleeding disorders can cause serious blood loss. Hemophilia is one type of bleeding disorder.

Clot too much and/or too quickly, even without an injury. This condition may lead to clots that block your blood flow and cause serious conditions, such as heart attack, stroke, or clots in the lungs.

A PTT test helps check a specific group of clotting factors. It helps show how much of these clotting factors you have and how well they're working. A PTT test is often done with other tests that check clotting factors and how well they all work together.

Other names: activated partial thromboplastin time, aPTT, intrinsic pathway coagulation factor profile

WHAT IS IT USED FOR?

A PTT test is used to check for problems with a specific group of blood clotting factors. The test is done to:

Find the cause of too much bruising or bleeding.

Find the cause of clotting problems. Causes can include certain autoimmune diseases such as lupus and antiphospholipid syndrome (APS).

Monitor people taking heparin, a type of medicine that's used to prevent and treat blood clots. PTT testing can help make sure the dose is safe and effective.

Check the risk for possible bleeding problems before surgery or medical procedures. (A PTT test is not always used as a routine test before surgery. It may be used for certain people who may have a risk for bleeding problems).

WHY DO I NEED A PTT TEST?

You may need a PTT test if you:

Have problems with bleeding or bruising and the cause is not known

Have a blood clot in a vein or artery

Have liver disease (your liver makes most of your clotting factors).

Have had several miscarriages

Have been diagnosed with a bleeding or clotting disorder and don't know which clotting factors are involved

Are taking heparin (to check how this medicine is affecting you, your health care provider may use a PTT test or another test instead).

WHAT HAPPENS DURING A PTT TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a PTT test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your PTT test results will show how much time it took for your blood to clot. Results are usually given as a number of seconds. A PTT test is often ordered along with another blood test called a prothrombin time (PT) test. A PT test measures other clotting factors that a PTT test doesn't check. Your provider will usually compare the results of both tests to understand how your blood is clotting. Ask your provider to explain what your test results mean for your health.

In general, if your blood took longer than normal to clot on a PTT test, it may be a sign of:

Liver disease.

A lack of vitamin K.

Certain genetic disorders that you inherit from your parents. These disorders affect certain clotting factors and increase your risk of bleeding. They include:

Von Willebrand disease.

Hemophilia.

Von Willebrand disease.

Hemophilia.

Too much heparin.

Certain types of leukemia.

Autoimmune diseases, such as antiphospholipid antibody syndrome or lupus anticoagulant syndrome. These diseases cause your body to make proteins called antibodies. The antibodies related to these diseases cause too much clotting. But the results of a PTT test may show a slow clotting time. That's because the chemicals in the PTT test react with the antibodies in your blood sample. This chemical reaction makes the blood sample clot more slowly than the blood in your body. If your provider thinks that an autoimmune disease is causing a clotting problem, you will usually have other tests to make a diagnosis.

If your blood clotted faster than normal on a PTT test, it may be a sign of:

The early stage of disseminated intravascular coagulation (DIC). This rare but serious condition may develop if you have an infection or damage to organs or tissues that affects blood clotting. In the early stage, you have too much blood clotting. Later on, DIC starts to use up clotting factors in your blood, which leads to bleeding problems.

Cancer of the ovaries, colon, or pancreas that is advanced, which means the cancer has spread to other parts of the body and is unlikely to be controlled with treatment.

Talk with your provider to learn what your test results mean.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PTT TEST?

If your provider thinks you may have a clotting disorder linked to lupus, you may have a test called an LA-PTT. This is a type of PTT test that is designed to look for a protein that's linked to increased clotting and having many miscarriages.

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PCR Tests

<https://medlineplus.gov/lab-tests/pcr-tests/>

PCR TESTS

WHAT ARE PCR TESTS?

PCR (polymerase chain reaction) tests are a reliable and accurate way to diagnose certain infectious diseases, some types of cancer, and certain genetic changes. The tests check for small amounts of genetic material of a pathogen (disease-causing organism) or abnormal cells in a sample of your blood, saliva, mucus, or tissue. The genetic material could be:

DNA, which contains the genetic information needed for a person and most other living things to develop and grow. DNA is copied from one generation to the next.

RNA, which contains information copied from DNA. Many types of RNA help your cells make proteins. Some viruses use RNA instead of DNA to carry their genetic information.

Most viruses and other pathogens contain DNA or RNA.

Unlike many other tests, PCR tests can find signs of disease in the earliest stages of infection. They can also find pieces of a virus you had, after you are no longer infected. Other tests may miss early signs of disease because there aren't enough viruses, bacteria, or other pathogens in your sample, or your body hasn't had enough time to develop an antibody response. Antibodies are proteins your immune system makes to attack foreign substances, such as viruses and bacteria. PCR tests can detect disease when there is only a very small amount of pathogens in your body.

As part of a PCR test, a small amount of genetic material in your sample is copied multiple times. The copying process is known as amplification. If your sample contains pathogens, amplification can make them much easier to detect.

Other names: polymerase chain reaction, rtPCR, reverse transcription PCR, qPCR, quantitative PCR, real-time PCR

HOW ARE THEY USED?

PCR tests are used to:

Diagnose certain infectious diseases

Identify certain genetic changes that can cause disease

Find small amounts of certain cancer cells that might be missed in other types of tests

HOW DO THEY WORK?

To do a PCR test:

A health care professional will take a sample of blood, saliva, mucus, or tissue.

The sample will contain your own DNA and possibly the DNA of a pathogen or cancer cell.

The sample, an enzyme called polymerase, and certain chemicals are put in a thermal cycler. This machine makes copies of a special segment of DNA through heating and cooling cycles.

The copying process is repeated multiple times. After about an hour, billions of copies are made. If a virus or other pathogen is present, it will be shown on the machine.

Certain viruses, including COVID-19, are made up of RNA rather than DNA. For these viruses, the RNA must be changed into DNA before copying. This process is called reverse transcription PCR (rtPCR).

WHAT HAPPENS DURING A PCR TEST?

There are different ways to get a sample for a PCR test. Common methods include blood tests and nasal swabs.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting

when the needle goes in or out. This usually takes less than five minutes.

A nasal swab may be taken from the front part of your nostrils (anterior nares). It also may be taken from the back of your nostrils, in a procedure known as a nasal mid-turbinate (NMT) swab, or from the nasopharynx, the uppermost part of your nose and throat. In some cases, your provider will ask you to do an anterior nares test or an NMT swab yourself. During an anterior nares test, you will start by tilting your head back. Then you or your provider will:

Gently insert a swab inside your nostril.

Rotate the swab in four or five sweeping circles for 10 to 15 seconds.

Remove the swab and insert it into your second nostril.

Swab the second nostril using the same technique.

Remove the swab.

If you only swab one nostril, don't rotate the swab in your nostril, or keep the swab in one area, you may not get a good sample.

If you are doing the test yourself, your provider will let you know how to prepare and package your sample. This often involves placing your swab in a sterile tube, closing it, and sealing it in a special bag. You would then give the specimen bag to your provider or follow the instructions on how to return your specimen for testing.

During an NMT swab, you will start by tilting your head back. Then you or your provider will:

Gently insert a swab onto the bottom of your nostril, pushing it until you feel it stopping.

Rotate the swab for 10 to 15 seconds.

Remove the swab and insert it into your second nostril.

Swab the second nostril using the same technique.

Remove the swab.

If you only swab one nostril, don't rotate the swab in your nostril, or only keep the swab in one area, you may not get a good sample.

If you are doing the test yourself, your provider will let you know how to prepare and package your sample. This often involves placing your swab in a sterile tube, closing it, and sealing it in a special bag. You would then give the specimen bag to your provider or follow the instructions on how to return your specimen for testing.

During a nasopharyngeal (NP) swab:

You will tilt your head back.

Your provider will insert a long swab into your nostril until it reaches your nasopharynx (the upper part of your throat).

Your provider will rotate the swab for 10 to 15 seconds and remove it.

If your provider gets enough of a sample from one of your nostrils, they may only need to do one. But they may need to get a sample from your other nostril as well if they had trouble getting the sample from the first side.

DO I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You don't need any special preparations for a PCR test.

ARE THERE ARE ANY RISKS TO A PCR TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

A nasal swab may tickle your throat or cause you to cough. A nasopharyngeal swab may be uncomfortable and cause coughing or gagging. All these effects are temporary.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT PCR TESTS?

PCR tests are considered the best and most effective method for identifying many infectious diseases, including COVID-19 and HPV. Because they often diagnose infections before symptoms start, PCR tests play a very important role in preventing the spread of diseases.

Learn more about laboratory tests, reference ranges, and understanding results.

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PDL1 (Immunotherapy) Tests

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PDL1 (IMMUNOTHERAPY) TESTS

WHAT IS A PD-L1 TEST?

A PD-L1 test uses a sample of cancerous tumor tissue to measure how much of a protein called programmed-death ligand 1 (PD-L1) is found on the cancer cells. Normally, PD-L1 is found on certain healthy cells. It helps keep your immune system from attacking healthy tissue by acting as a kind of "brake" on your immune system cells (called T-cells). However, in some cancers, PD-L1 keeps your immune system from attacking cancerous cells.

If you have one of these kinds of cancers and high levels of PD-L1, your provider can treat your cancer with immunotherapy (a treatment option that helps your immune system fight cancer). In cancers with high PD-L1 levels, health care providers use a type of immunotherapy called "immune checkpoint inhibitors." These medicines stop PD-L1 from binding to your T-cells. In other words, they "release the brakes," allowing your immune system to target and kill cancer cells.

Other names: programmed death-ligand 1, PDLI, PDL-1 by immunohistochemistry (IHC)

WHAT IS IT USED FOR?

PD-L1 testing is mainly used to help your provider decide on treatment options for certain cancers. However, there are different types of PD-L1 tests. The test your provider orders will depend on:

The type of cancer you have.

The specific immunotherapy medicine that your provider is considering for you.

The testing method used by your laboratory.

With the results from these tests, your provider can then not only decide whether immunotherapy will work, but how best to use immunotherapy. They may, for example, decide between:

Monotherapy. This means using only immunotherapy to treat your cancer.

Combination therapy. This means using both immunotherapy and chemotherapy. Combination therapy is most often used for advanced cancer.

Though immunotherapy can help stop or slow the growth of many types of cancer that have PD-L1, it isn't always the best option. This is because it can cause serious side effects in some people, and not everyone benefits from it.

WHY DO I NEED A PD-L1 TEST?

You may need a PD-L1 test if you've been diagnosed with certain types of cancer that commonly cause high PD-L1 levels. These cancers include:

Non-small cell lung cancer

Melanoma

Hodgkin lymphoma

Bladder cancer

Kidney cancer

Breast cancer

Head and neck cancer (squamous cell carcinoma)

Cancer of the esophagus (squamous cell carcinoma)

Stomach cancer (adenocarcinoma)

Cervical cancer

Your provider may also order a PD-L1 test to monitor your treatment progress.

WHAT HAPPENS DURING A PD-L1 TEST?

PD-L1 tests are done on a tissue sample from a tumor. If you're having surgery to remove a tumor, a sample will be taken for testing.

If you aren't having the tumor removed, you may have a biopsy, which is a procedure to remove a small amount of tissue for testing. If you already had a biopsy to diagnose your cancer, the same tissue sample may be used to test for PD-L1.

There are many ways to do a biopsy. The type of biopsy you have depends on the type of cancer you have and where the tumor is located. In general, a biopsy may be done using:

A hollow needle inserted through your skin. Imaging tests, such as an ultrasound, may be used to guide the needle: Â Â

A fine needle aspiration biopsy uses a very thin needle to remove a sample of cells and/or fluid.

A core needle biopsy uses a larger needle to remove a sample.

A fine needle aspiration biopsy uses a very thin needle to remove a sample of cells and/or fluid.

A core needle biopsy uses a larger needle to remove a sample.

Surgery. Surgery may be done to remove a sample of tissue (an incisional biopsy). In certain cases, the entire tumor will be removed (an excisional biopsy).

Endoscopic biopsy. A flexible tube with a camera (scope) is inserted through an opening in your body or through a small cut to view and collect tissue samples. Examples of scope procedures include bronchoscopy, laparoscopy, colposcopy, and cystoscopy (to examine the inside of the bladder). Â

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Preparations for your test depend on how your tissue sample will be taken. You may need to stop taking certain medicines, such as blood thinnersÂ like aspirin, before this test. So, tell your provider about all the medicines and supplements you take. Don't stop or start taking anything unless your provider tells you to.

Before you get any procedure, you should tell your provider about any recent surgeries or known allergies. You should also tell them if you're pregnant.

If you're having general anesthesia, you will probably need to fastÂ (not eat or drink) for several hours before surgery. If you have general anesthesia or medicine to relax, you may be groggy after the procedure, so plan to have someone take you home.Â

Ask your provider how to prepare for your test.

ARE THERE ANY RISKS TO THE TEST?

The risks depend on how your tissue sample is collected.

With a needle biopsy, you may have slight bruising, soreness, or bleeding at the site.

With a surgical biopsy, you may have mild pain, a small scar, or a risk of infection at the incision site.

These side effects are usually mild and temporary. Your health care team will explain any risks and how to care for the biopsy area.

WHAT DO THE RESULTS MEAN?

Your test results will show whether you have enough PD-L1 protein in your tumor for you to benefit from a specific immunotherapy medicine. Some immunotherapy medicines will not be helpful if a small percentage of your cancer cells have PD-L1.

If your test results show that:

Your tumor cells have enough PD-L1 for you to use immunotherapy medicine, you may be able to start that medicine.

Common examples include pembrolizumab, nivolumab, and atezolizumab. This may be called a "positive" test result. Your tumor cells don't have enough PD-L1, then immunotherapy likely won't help you. This may be called a "negative" test result. Your provider will consider other types of cancer treatment that will be more effective. Your test results may also have different scoring systems and values. For example, you may see: Combined Positive Score (CPS), which measures PD-L1 on both tumor cells and T-cells. Tumor Propositional Score (TPS), which measures the percentage of tumor cells that have high amounts of PD-L1. These scoring systems and values can vary depending on what type of PD-L1 test your provider used. Because of these differences, you should talk to your provider about what your results mean. Your provider may also use other tests to help decide on the best treatment for you. Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PD-L1 TEST?

Immunotherapy is a quickly changing area of medicine. For some immunotherapy medicines, PD-L1 testing is required to see if you are a good match for that treatment. For other medicines, however, testing is recommended but not required before treatment. New research may change how tests are used to choose treatments. So, check with your provider to find out how PD-L1 testing may affect your treatment plan.

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PET Scan

<https://medlineplus.gov/lab-tests/pet-scan/>

PET SCAN

WHAT IS A PET SCAN?

A PET (positron emission tomography) scan is a nuclear imaging test that looks at how well your tissues and organs are working. It also checks for signs of cancer. For the scan, you will get an injection of a small amount of a tracer, which is a safe radioactive substance.

The tracer settles in areas of your body that have high levels of chemical activity. This activity can be a sign of cancer or other diseases. The tracer can also help measure blood flow, oxygen use, and changes in the metabolism of a particular tissue or organ. Metabolism is the process your body uses to make energy from the food you eat.

Chemical changes in your body can show up before you even have symptoms of a disease. Since a PET scan can see how your organs are working in real-time, it can find signs of disease at an early stage, often before the disease can be seen on other imaging tests.

Other names: positron emission tomography

WHAT IS IT USED FOR?

A PET scan is most often used to:

Diagnose or monitor certain cancers, including breast, thyroid, and lung cancers

Find out how well your heart muscle is working

Check blood flow to your heart

Check for signs of certain brain disorders, such as Parkinson's disease, Huntington's disease, Alzheimer's disease, and other types of dementia

A PET scan is often done along with a CT scan, a type of x-ray that takes a series of pictures as it rotates around you.

WHY DO I NEED A PET SCAN?

You may need a PET scan to find out if you have cancer. If you've already been diagnosed with cancer, you may need this test to see if your cancer treatment is working.

A PET scan also helps diagnose and monitor heart and brain diseases. So you may need this test if you:

Have a family history of heart disease.

Are being treated for heart disease. This test can show if the treatment is working.

Have symptoms of a brain disease, such as tremors, memory problems, and/or seizures.

WHAT HAPPENS DURING A PET SCAN?

Before the scan, you will change into a hospital gown. You may be asked to empty your bladder. During the scan:

A health care provider will inject the radioactive tracer into your vein through an intravenous (IV) line. The tracer sends out a form of energy called gamma rays. The scanner picks up the rays to create images of the inside of your body.

You will need to wait for your body to absorb the tracer, which takes 45 to 60 minutes.

You will then lie on a narrow, padded table, which will slide into a large, donut-shaped scanner.

The scanner will move slowly across your body to capture images. As this happens, you will need to be very still for about 30 minutes.

You will hear clicking or buzzing noises as the images are taken.

The scanner will send images to a computer monitor for your provider to review.

Your provider will then remove the IV line.

The test takes about two hours to complete, including the time you wait for the tracer to be absorbed by your body.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Talk to your provider about how to prepare for the test.

You may be asked to fast (not eat or drink) for four to six hours before the test.

You may be asked to avoid tobacco products and food or drinks with caffeine or alcohol for 24 hours before the test.

If you have diabetes and use insulin, you may need to change the timing of your regular dose. Your provider will give you specific instructions about adjusting your insulin since people with diabetes may not absorb the sugar in the tracer. This may affect the results of your scan.

You may need to stop taking other medicines or supplements before the test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

Also, tell your provider if you have claustrophobia (fear of enclosed spaces). Your provider may decide to give you a medicine to help you relax during the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little exposure to radiation in a PET scan. Only a small amount of radioactive substance is used, and it will leave your body through your urine and stool (poop) within 2 to 10 hours. Drink more water after the test to help remove the remaining tracer from your body.

While radiation exposure in a PET scan is safe for most adults, it can be harmful to a developing baby. So be sure to tell your provider if you are pregnant or think you may be pregnant. Also, tell your provider if you are breastfeeding, since the tracer may contaminate your breast milk.

Allergic reactions to the tracer are rare and usually mild.

WHAT DO THE RESULTS MEAN?

Depending on which part of the body was being scanned, your results may show signs of:

Cancer. Cancer cells show up as bright spots on a PET scan.

Heart disease. The scan can show decreased blood flow to the heart.

A brain disorder. The scan may show changes in certain brain chemicals that can be a sign of a disease.

Some cancers don't show up on a PET scan, so your provider may order additional tests based on your symptoms and medical history. If you also had a CT scan, your provider will review the results of the two scans to help make a diagnosis.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PET SCAN?

Your PET scan results will be looked at by a radiologist, a doctor who specializes in diagnosing and treating medical conditions using imaging technologies. They will share the results with your provider.

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Pharmacogenetic Tests

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PHARMACOGENETIC TESTS

WHAT IS PHARMACOGENETIC TESTING?

Pharmacogenetics (also called pharmacogenomics) is the study of how your genes affect the way your body responds to certain medicines. Genes are parts of DNA in your cells that you inherit from your parents. You inherit two copies of each gene, one from each parent. Genes carry information that controls what you look like and how your body works.

Some genes also affect how your body uses and breaks down medicines. For example, certain changes (or variants) in these genes may make it harder for your body to absorb a certain medicine or dosage. Because your body can't absorb it, the medicine won't work as expected.

These genetic differences can be the reason why some people may benefit from a certain medicine while others may not benefit at all. They may also be the reason why only some people have serious side effects.

Therefore, before treating you, your health care provider may want to see if you have any gene variants that may affect your response to certain medicines. To do this, they will use pharmacogenetic testing, which looks for these gene changes in a sample of your blood, saliva (spit), or cells swabbed from your cheek.

Other names: pharmacogenomics, pharmacogenomic testing

WHAT IS IT USED FOR?

Because not everyone responds to medicine the same way, providers consider a variety of factors before deciding which medicine to prescribe. These factors include:

Your personal medical history.

Your family medical history.

Your lifestyle and environment.

What other medicines and supplements you're taking.

This approach is called precision medicine, and it also includes pharmacogenetic testing. Providers use pharmacogenetic testing to look for genes related to specific health conditions and medicines. So, if you are being treated for high cholesterol, your provider may test for a specific gene variant that can affect your response to some cholesterol-lowering medicines.

Once your provider is aware of these various factors, they can:

Find out whether a certain medicine could be effective for you.

Find out how much of the medicine you need.

Predict whether you will have a serious side effect from a medicine.

WHY DO I NEED PHARMACOGENETIC TESTING?

Pharmacogenetic tests are not available for all medicines or used for all health conditions. Providers most often rely on pharmacogenetic testing when considering the following health conditions:

High cholesterol. If you have certain variants of the SLCO1B1 gene, you're more likely to have muscle pain and weakness when taking some statins such as atorvastatin and fluvastatin.

Depression. If you have certain variants of the CYP2D6 or CYP2C19 gene, you're more likely to have trouble breaking down some antidepressants such as sertraline and venlafaxine.

Blood clot prevention. If you have certain genetic variants, you're more likely to need lower doses of the anticoagulant medicine warfarin.

HIV. If you have a certain variation of the HLA-B gene, you're more likely to have a severe skin reaction to Abacavir. A certain variant of the CYP2B6 gene can affect your nervous system if you take efavirenz.

Multiple sclerosis. If you have certain variants of the TPMT or NUDT15 gene, you're more likely to have suppressed

immune system responses if you take azathioprine.

When treating breast cancer, colon cancer, or acute lymphoblastic leukemia (ALL), your provider may use pharmacogenetic testing to make specific treatment decisions. For example:

Trastuzumab only works for people with a kind of breast cancer that makes too much of the HER2 protein.

Typical doses of mercaptopurine used to treat ALL can cause severe side effects if you have a certain variant of the TPMT gene.

Irinotecan treatment for colon cancer can cause severe diarrhea and increased infection risk if you have a certain variant of the UGT1A1 gene.

You may also need a pharmacogenetic test if you are recovering from an organ transplant or taking a medicine that's not working and/or causing serious side effects.

Pharmacogenetics is a growing field of medicine. This means that, in the future, genetic testing may be used more frequently to make treatment decisions.

WHAT HAPPENS DURING A PHARMACOGENETIC TEST?

Testing is usually done on blood, saliva (spit), or a sample of cells swabbed from the inside of your cheek.

For a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a saliva test, you will be instructed how to provide your sample. Usually, you will either spit into a small tube, or your saliva will be collected with a special swab or pad.

For a cheek swab, a health care professional will gently rub the inside of your cheek with a special swab.

At-home test kits are available for certain pharmacogenetic tests. With a home test, you collect a saliva sample to send to a lab for testing. In general, these tests should not be used for making treatment decisions because they may not be accurate. Talk with your provider if you're considering an at-home test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You usually don't need any special preparations for a blood test. If you are getting a saliva test, you should not eat, drink, smoke, or chew gum for 30 minutes before the test. If you have a cheek swab, you may be asked to rinse your mouth first.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no risk to having a saliva test or a cheek swab.

WHAT DO THE RESULTS MEAN?

Your test results will describe any changes in the gene that was tested. Changes in these genes can show how certain medicines are likely to affect you. But they cannot tell you anything about your health condition.

Your provider will use the results of your pharmacogenetic test along with other information about your condition to recommend treatment or to make changes in your treatment. If your test results show that a certain medicine isn't right for you, your provider will consider other treatments that may help you. Talk with your provider to find out how your test results may change your treatment options.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT PHARMACOGENETIC TESTING?

Pharmacogenetic testing is not the same thing as genetic testing. Genetic tests that are done for health reasons can help diagnose diseases. They may also provide information about your risk for certain diseases. Pharmacogenetic testing cannot diagnose any conditions or tell you about your risk for developing them.

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Phenylketonuria (PKU) Screening

<https://medlineplus.gov/lab-tests/phenylketonuria-pku-screening/>

PHENYLKETONURIA (PKU) SCREENING

WHAT IS A PKU SCREENING TEST?

A PKU screening test measures the amount of a protein called phenylalanine (Phe) in a sample of a newborn's blood. Phe is in all foods that contain protein such as milk, meats, and nuts. It's also in an artificial sweetener called aspartame. If a newborn has high levels of Phe in their blood, it could mean that they have phenylketonuria (PKU).

PKU is a rare genetic disorder, meaning it's caused by changes (also called variants or mutations) in a gene. Genes are parts of DNA in your cells that you inherit from your parents. You inherit two copies of each gene, one from each parent. Genes carry information that controls what you look like and how your body works.

The gene involved in PKU helps make a substance that breaks down part of Phe in food. So, a change in that gene can make it harder for your body to break Phe down. This means that if you have PKU and eat foods with Phe, the Phe will build up in your blood. High levels of Phe can permanently damage your nervous system and brain, leading to various health problems such as seizures and possibly learning disabilities.

PKU can vary from mild to severe, depending on whether the changed gene still works somewhat or has stopped working completely.

Other names: PKU newborn screening, PKU test

WHAT IS IT USED FOR?

A PKU test is used to see if a newborn has high blood levels of Phe in their blood. Though PKU is rare, all newborns in the United States are tested for it. That's because:

The test is easy and has almost no health risks. And doing the test may save a baby from lifelong brain damage and/or other serious health problems.

If PKU is found early, a special diet that's low in protein and Phe may help prevent health problems.

If a newborn does have high Phe levels, that may mean that they have PKU. However, more tests may be needed to confirm or rule out a diagnosis.

WHY DOES MY BABY NEED A PKU SCREENING TEST?

Newborns in the United States are required to get a PKU test. A PKU test is usually part of a series of tests called a newborn screening that is performed between 24 and 72 hours after birth. Some older infants and children may need testing if they were born in another country or if the results of their newborn screening were negative for PKU, especially if they have any symptoms of PKU. Symptoms include:

Seizures.

Skin Rashes, such as eczema.

Delayed development.

Intellectual and/or learning difficulties.

A musty odor in the breath, skin, and/or urine (pee).

Unusually small head (microcephaly).

Psychiatric problems such as mental health disorders.

Nausea and vomiting.

WHAT HAPPENS DURING A PKU SCREENING TEST?

A health care provider will clean your baby's heel with alcohol. Then the provider will stick the baby's heel with a small

needle to collect a few drops of blood. The heel will then be bandaged.

To get accurate test results, the baby needs to have some protein from breast milk or formula before the blood sample is taken. That's why the test is usually done a day or two after birth. If your baby wasn't born in the hospital or if you left the hospital early, be sure to talk with your child's provider to schedule a PKU test as soon as possible.

WILL I NEED TO DO ANYTHING TO PREPARE MY BABY FOR THE TEST?

There are no special preparations needed for a PKU test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to your baby from a needle stick blood test. Your baby may feel a little pinch when the heel is poked. A small bruise may form on the heel, but it should go away quickly.

WHAT DO THE RESULTS MEAN?

If the results are normal, it means that the amount of Phe in your baby's blood was normal. So, your baby is unlikely to have PKU. But if the test was done sooner than 24 hours after birth, your baby may need to be tested again at 1 to 2 weeks of age.

If both parents know they carry a gene for PKU and your baby's test is normal, the baby should have genetic testing. This blood test will check the baby's genes for changes. The results will diagnose PKU or rule it out for certain. If your baby does have PKU, genetic testing can also help your provider understand how serious the condition is and how best to treat it.

If your baby's results are not normal, it means high levels of Phe were found in your baby's blood. Your baby most likely has PKU. Your provider may order more tests to confirm or rule out PKU. These tests may include more blood tests and/or urine tests.

If your baby is diagnosed with PKU, you can safely give your baby formula that doesn't contain Phe. Breastmilk does contain Phe, so if you would like to breastfeed, talk with your baby's provider to find out if your baby can safely have any amount of breastmilk.

After infancy, people with PKU are generally treated with a special-low protein diet, including some food supplements, to make sure they have proper nutrition.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PKU SCREENING TEST?

If you're pregnant, your baby could have PKU even if you and the baby's other parent don't have it. That's because you have two copies of all your genes, one from each parent. To have PKU, you must have changes in both copies of the gene that causes PKU.

If each parent has one changed gene for PKU, a fetus could inherit two changed genes and be born with the disease.

A prenatal screening test^Â can show whether your baby may have PKU. The test uses either a sample of:

Amniotic fluid that surrounds the fetus in the uterus. Your provider will remove a small amount of fluid by inserting a needle into your belly during a procedure called amniocentesis.

Tissue from part of the placenta, the organ that grows in the uterus to provide nutrients and oxygen to the fetus. This sample is taken using a needle inserted into your abdomen (belly) or a tube inserted through your vagina in a procedure called chorionic villus sampling.^Â

Your provider or a genetic counselor can help you decide if prenatal testing for PKU is right for you.

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Phosphate in Blood

<https://medlineplus.gov/lab-tests/phosphate-in-blood/>

PHOSPHATE IN BLOOD

WHAT IS A PHOSPHATE IN BLOOD TEST?

A phosphate in blood test measures the amount of phosphate in a sample of your blood. Phosphate contains the mineral phosphorus. So, a phosphate test is sometimes called a phosphorus test.

Phosphate is a type of electrolyte. Electrolytes are electrically charged minerals. They help control the amount of fluid and the balance of acids and bases (pH balance) in your body. Your body also needs phosphates for many other important processes, such as:

Building strong bones and teeth. Most of the phosphate in your body is found in your bones where it is combined with the mineral calcium.

Making energy.

Helping your nerves and muscles work properly.

The phosphorus in phosphate comes from the foods you eat, including nuts and seeds, dairy products, dried beans, meats, poultry, and eggs. Your body tightly controls the amount of phosphate in your blood mainly through your:

Kidneys. They filter extra phosphate from your blood and get rid of it through urine (pee). If your phosphate level is low, your kidneys limit the amount lost through urine.

Intestines. They control how much phosphorous you absorb from the foods you eat.

The amount of phosphate in your blood is also linked to your levels of:

Calcium. When blood calcium levels increase, phosphate levels decrease. And when calcium levels decrease, phosphate levels increase.

Vitamin D. It helps your body use phosphate.

Parathyroid hormone (PTH). This hormone is made by parathyroid glands in your neck. It helps balance phosphate and calcium levels in your blood.

Abnormal phosphate levels may be a sign of problems with the systems in your body that control phosphate levels. To help understand the cause of abnormal levels, the test is often done with blood tests that measure calcium, vitamin D, and PTH.

Other names: phosphorus test, P, PO₄, phosphorus-serum, phosphate, inorganic phosphorus

WHAT IS IT USED FOR?

A phosphate in blood test is often used with other tests to help diagnose and/or monitor:

Kidney disease, especially chronic kidney disease. High phosphate levels are a common sign that the kidneys aren't working well to get rid of extra phosphate in the blood.

Bone disorders. High phosphate levels can pull calcium out of bones and weaken them over time.

Parathyroid disorders. Parathyroid hormones control the balance of phosphate and calcium in the blood. A phosphate test can help show how well the parathyroid glands are working.

A phosphate test may also be used to monitor people who:

Have diabetes that isn't well controlled

Have signs of an acid-base imbalance (having too much or too little acid in your blood)

WHY DO I NEED A PHOSPHATE IN BLOOD TEST?

Phosphate levels that are either high or mildly low usually don't cause any symptoms on their own. But you may need a phosphate test with other tests if you:

Have had an abnormal result on a calcium blood test. A calcium test may be part of a routine exam. Abnormal levels are often linked to abnormal phosphate levels.

Have symptoms of low calcium levels. High levels of phosphate can lower calcium levels in your blood, which can cause symptoms, such as:

Nails that break easily, and/or hair that becomes coarse
Rash and/or itchy skin
Muscle cramps
Bone pain
Seizures
Irregular heartbeat (arrhythmia)

Nails that break easily, and/or hair that becomes coarse
Rash and/or itchy skin
Muscle cramps
Bone pain
Seizures
Irregular heartbeat (arrhythmia)

Have symptoms of severely low phosphate levels. This condition can become life-threatening and needs immediate treatment. Symptoms may include:

Muscle pain and/or weakness
Bone pain
Changes in your mental condition, such as feeling confused or very irritable
Seizures
Coma
Muscle pain and/or weakness
Bone pain
Changes in your mental condition, such as feeling confused or very irritable
Seizures
Coma

You may also need to have your phosphate levels checked if you have or may have a condition that can cause abnormal phosphate levels, such as:

Kidney disease
Malnutrition
Malabsorption syndrome
A parathyroid gland disorder
Using certain medicines

WHAT HAPPENS DURING A PHOSPHATE IN BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your health care provider will tell you how to prepare for your test. You may need to fast (not eat or drink) for several hours before you give your blood sample.

Certain medicines, vitamins and supplements can affect the accuracy of your test results. So tell your provider about everything you take, including over-the-counter medicines. But don't stop taking any prescription medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The results of your phosphate test may say "phosphorus levels" or "phosphate levels." These terms mean the same thing. The meaning of your test results depends on your symptoms, medical history, and the results of other tests.

In general:

A higher than normal phosphate level is called hyperphosphatemia. High levels are linked to many conditions, including: Late stages of chronic kidney disease and kidney failure. If damaged kidneys can't filter out extra phosphate, it can build up in your blood. If you have a condition that affects your kidneys, eating too much phosphorus-rich food may also lead to high phosphate levels.

Hypoparathyroidism. With this condition, your parathyroid glands don't make enough parathyroid hormone (PTH). When PTH is low, calcium levels are low and phosphate levels are high.

Acidosis. This means having too much acid in your blood from a pH imbalance. It may be caused by a lung disorder or other health conditions, including kidney disease.

Long-term use of certain medicines. These include steroids, laxatives, and enemas that contain phosphate.

A lower than normal phosphate level is called hypophosphatemia. Mildly low levels often aren't a health problem. But many conditions that cause low levels may need treatment, including:

Hyperparathyroidism. With this condition, your parathyroid glands make too much parathyroid hormone, which leads to high levels of blood calcium and low levels of phosphate.

Vitamin D deficiency. A lack of this vitamin can lead to osteomalacia, a condition that causes soft bones. In children, this condition is called rickets.

Getting too little phosphorus from food. This is uncommon in the U.S. But some people are more likely to have trouble getting enough phosphorus, including:

Premature babies

People who have certain rare genetic disorders, severe malnutrition, alcohol use disorder, or problems absorbing nutrients

Premature babies

People who have certain rare genetic disorders, severe malnutrition, alcohol use disorder, or problems absorbing nutrients

Long-term use of certain medicines. These include antacids that contain aluminum hydroxide or calcium carbonate and certain prescription diuretics ("water pills").

If your phosphate levels are not normal, it doesn't always mean you have a medical condition that needs treatment. Also, children often have higher phosphate levels because their bones are still growing. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PHOSPHATE IN BLOOD TEST?

Your provider may order a phosphate in urine test instead of or with a phosphate in blood test.

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Calcium in Urine Test

Glomerular Filtration Rate (GFR) Test

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Phosphate in Urine

<https://medlineplus.gov/lab-tests/phosphate-in-urine/>

PHOSPHATE IN URINE

WHAT IS A PHOSPHATE IN URINE TEST?

A phosphate in urine test measures the amount of phosphate in a urine (pee) sample that you collect over a 24-hour period. Phosphate contains the mineral phosphorus. So, a phosphate test is sometimes called a phosphorus test.

Phosphate is an electrolyte. Electrolytes are electrically charged minerals. They help control the amount of fluid and the balance of acids and bases (pH balance) in your body. Your body also needs phosphates for many other important processes, such as:

Building strong bones and teeth. Most of the phosphate in your body is found in your bones where it is combined with the mineral calcium.

Making energy.

Helping your nerves and muscles work properly.

The phosphorus in phosphate comes from the foods you eat, including nuts and seeds, dairy products, dried beans, meats, poultry, and eggs. Your body tightly controls the amount of phosphate in your blood mainly through your:

Kidneys. They filter extra phosphate from your blood and get rid of it through urine (pee). If your phosphate level is low, your kidneys limit the amount lost through urine.

Intestines. They control how much phosphorus you absorb from the foods you eat.

The amount of phosphate in your body is also linked to your levels of:

Calcium. When calcium levels increase, phosphate levels decrease. And when calcium levels decrease, phosphate levels increase.

Vitamin D. It helps your body use phosphate.

Parathyroid hormone (PTH). This hormone is made by parathyroid glands in your neck. It helps balance phosphate and calcium levels.

Abnormal phosphate levels may be a sign of problems with the systems in your body that control phosphate levels. These include various kidney conditions and disorders of the parathyroid glands.

Other names: phosphorous test, P, PO₄, inorganic phosphate, PO₄ urine concentration

WHAT IS IT USED FOR?

Your health care provider may use a phosphate in urine test, along with other tests, to help:

Understand the cause of high or low phosphate in your blood.

Find out if too much phosphate in urine is causing kidney stones, small, pebble-like material that form in the kidneys.

Diagnose kidney problems.

Diagnose disorders of the parathyroid glands.

WHY DO I NEED A PHOSPHATE IN URINE TEST?

You may need a phosphate in urine test if you have:

Had an abnormal result on a phosphate in blood test.

Had a kidney stone.

Had an abnormal result on a blood or urine calcium test. Calcium and phosphate work together, abnormal calcium levels may mean you have abnormal phosphate levels, too.

Symptoms of a kidney problem.

Problems with your electrolyte balance, which affects your acid/base (pH) level.

WHAT HAPPENS DURING A PHOSPHATE IN URINE TEST?

You'll need to collect all your urine during a 24-hour period. This is called a 24-hour urine sample test. You will be given a special container to collect your urine and instructions on how to collect and store your sample. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or in a cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a phosphate in urine test. Be sure to carefully follow all the instructions for providing a 24-hour urine sample.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a phosphate in urine test.

WHAT DO THE RESULTS MEAN?

The results of your phosphate test may say "phosphorus levels" or "phosphate levels." These terms mean the same thing. Your results may include guidelines for normal, low, and high levels, but your provider will consider your health, age, and other factors when deciding what is normal for you.

To understand what your test results mean, your provider will consider your symptoms, medical history, and the results of other tests. That's why it's best to ask your provider what your test results say about your health. In general:

High phosphate/phosphorous levels in your urine may mean you have:

Kidney disease that affects your kidneys' ability to hold on to phosphate.

Hyperparathyroidism, a condition in which your parathyroid gland produces too much parathyroid hormone.

Low phosphate/phosphorous levels in your urine may mean you have:

Kidney disease that affects your kidneys' ability to get rid of extra phosphate.

Hypoparathyroidism, a condition in which your parathyroid gland produces too little parathyroid hormone.

If your phosphate/phosphorous levels are not normal, it doesn't always mean you have a medical condition that needs treatment. Eating a meal high in phosphorous or using laxatives with sodium phosphate can affect your results. A laxative is a medicine that can help you have a bowel movement (poop). Also, children often have higher phosphate levels because their bones are still growing. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PHOSPHATE IN URINE TEST?

Phosphate is often tested in blood. Your provider may order both a blood and urine test to see how much phosphate your kidneys are removing.

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RELATED HEALTH TOPICS

- Calcium
- Fluid and Electrolyte Balance
- Kidney Diseases
- Kidney Stones
- Kidney Tests
- Minerals
- Parathyroid Disorders
- Vitamin D

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- Calcium Blood Test

[Calcium in Urine Test](#)

[Crystals in Urine](#)

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[Kidney Stone Analysis](#)

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Platelet Tests

<https://medlineplus.gov/lab-tests/platelet-tests/>

PLATELET TESTS

WHAT ARE PLATELET TESTS?

Platelets, also known as thrombocytes, are small blood cells. They form in your bone marrow, the sponge-like tissue in your bones. Platelets are essential for blood clotting. A blood clot is a mass of blood that forms when platelets, proteins, and cells in the blood stick together. When you get hurt, your body forms a blood clot to stop the bleeding.

There are two types of platelet tests: a platelet count test and platelet function tests.

A platelet count test measures the number of platelets in your blood:

A lower-than-normal platelet count is called thrombocytopenia. This condition can cause you to bleed too much after a cut or other injury that causes bleeding.

A higher-than-normal platelet count is called thrombocytosis. This can make your blood clot more than you need it to. Blood clots can be dangerous because they can block blood flow.

Platelet function tests check your platelets' ability to form clots. Platelet function tests may include one or more of the following:

Closure time. This test measures the time it takes for platelets in a blood sample to plug a small hole in a tiny tube. It helps screen for different platelet disorders.

Viscoelastometry. This test measures the strength of a blood clot as it forms. A blood clot has to be strong to stop your bleeding.

Platelet aggregometry. This group of tests is used to measure how well platelets clump together (aggregate).

Lumiaggregometry. This test measures the amount of light produced when certain substances are added to a blood sample. It can help show if there are defects in the platelets.

Flow cytometry. This test uses lasers to look for proteins on the surface of platelets. It can help diagnose inherited platelet disorders. This specialized test is only available at certain hospitals and laboratories.

Bleeding time. This test measures how long it takes for bleeding to stop after small cuts are made in your forearm. It was once commonly used to screen for various platelet disorders but is not used much anymore.

Other names: platelet count, PLT, PLT count thrombocyte count, platelet function tests, PFT, platelet function assay, PFA, platelet aggregation studies

WHAT ARE THEY USED FOR?

A platelet count is most often used to monitor or diagnose conditions that cause too much bleeding or clotting. A platelet count may be included in a complete blood count, a test that is often done as part of a regular checkup.

Platelet function tests may be used to:

Help diagnose certain platelet diseases.

Check platelet function during complex surgical procedures, such as coronary artery bypass surgery and trauma surgery. These types of procedures have an increased risk of bleeding.

Check patients before surgery if they have a personal or family history of bleeding disorders.

Monitor people who are taking blood thinners. These medicines may be given to reduce clotting in people at risk for heart attack or stroke.

WHY DO I NEED A PLATELET TEST?

You may need platelet count and/or platelet function testing if you have symptoms of having lower-than-normal or higher-than-normal platelets.

Symptoms of lower-than-normal platelets include:

Bleeding for a long time after a minor cut or injury.

Nosebleeds.

Unexplained bruising.

Pinpoint sized red spots on the skin, known as petechiae.

Purplish, red, or brown spots on your skin, known as purpura. These may be caused by bleeding under the skin.

Menstrual periods that have a heavy flow or last a long time.

Blood in your urine, stool (poop), or from your rectum (where stool passes out of your body).

Symptoms of higher-than-normal platelets include:

Numbness of hands and feet

Headache

Dizziness

Weakness

Pain, swelling, and warmth in your lower legs

You may also need platelet function testing if you are:

Having a complicated surgery

Taking medicines to lower blood clotting

WHAT HAPPENS DURING A PLATELET TEST?

Most platelet tests are done on a blood sample.

During the test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a platelet count test.

If you are getting a platelet function test, you may need to stop taking certain medicines, such as aspirin and ibuprofen, before your test. Tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. They will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results show a lower-than-normal platelet count (thrombocytopenia), it may be a sign of:

A cancer that affects the blood, such as leukemia or lymphoma.

A viral infection, such as mononucleosis, hepatitis, or measles.

An autoimmune disease. This is a type of disease that causes your body to attack its own healthy tissues, which can include platelets.

Infection or damage to your bone marrow.

Cirrhosis.

Vitamin B12 deficiency (not enough).

Gestational thrombocytopenia. A common, but mild, low-platelet condition that can occur during pregnancy. It is not known to cause any harm to the mother or fetus. It usually gets better on its own during pregnancy or after birth.

If your results show a higher-than-normal platelet count (thrombocytosis), it may be a sign of:

Certain types of cancer, such as lung cancer or breast cancer

Anemia

Inflammatory bowel disease, such as Crohn's disease^Â and ulcerative colitis

Rheumatoid arthritis

A viral or bacterial infection

Severe blood loss

If your platelet function test results were not normal, it may be a sign of an inherited or acquired platelet disorder:

Inherited disorders are passed down from your family. These conditions are present at birth, but you may not have symptoms until you are older.

Acquired disorders are not present at birth. They may be caused by other diseases, medicines, or environmental exposure. Sometimes, the cause is unknown.

Inherited platelet disorders include:

Von Willebrand disease, a genetic disorder^Â that reduces the production of platelets or causes the platelets to work less effectively. It can cause excess bleeding.

Glanzmann thrombasthenia, a disorder that affects platelets' ability to clump together.

Bernard-Soulier syndrome, another disorder that affects platelets' ability to clump together.

Storage pool disease, a condition that affects platelets' ability to release substances that help platelets clump together.

Acquired platelet disorders may be due to chronic diseases such as:

Kidney failure

Certain types of leukemia

Myelodysplastic syndrome (MDS), a disease of the bone marrow

Your provider may consider your symptoms, medical history, and the results of other blood tests to understand the results of your platelet tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT PLATELET FUNCTION TESTS?

Platelet tests are sometimes done along with one or more of the following blood tests:

MPV blood test, which measures the size of your platelets

Partial thromboplastin time (PTT) test, which measures the time it takes for your blood to clot

Prothrombin time and INR test, which measures how many seconds it takes for a clot to form in your blood sample

If you are taking anti-platelet medicine, like aspirin, your results show how well your medicine is working.

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RELATED HEALTH TOPICS

Bleeding Disorders
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[Prothrombin Time Test and INR \(PT/INR\)](#)

[What You Need to Know About Blood Testing](#)

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Pleural Fluid Analysis

<https://medlineplus.gov/lab-tests/pleural-fluid-analysis/>

PLEURAL FLUID ANALYSIS

WHAT IS A PLEURAL FLUID ANALYSIS?

A pleural fluid analysis is a group of tests that examine a sample of abnormal fluid that builds up in the space between your lungs and chest wall. This space is called the pleural space.

The pleura is a thin tissue that covers the outside of your lungs and lines the inside wall of your chest. Normally, a small amount of fluid, called pleural fluid, fills the space between the two layers of pleural tissue. This fluid keeps the pleura moist and helps the layers of tissue glide smoothly over each other as you breathe.

If too much fluid builds up in your pleural space, it's called a pleural effusion. Many different types of conditions can cause pleural effusion. The fluid buildup doesn't always cause symptoms, but if it presses on your lungs, you may have trouble breathing, even if your lungs are healthy.

A pleural fluid analysis can help find the cause of a pleural effusion. Treating the cause may help keep the fluid from building up in the future. The fluid buildup you have now usually needs treatment, too. Without treatment, it may lead to a collapsed lung and other serious health problems.

Other names: pleural fluid aspiration, diagnostic thoracentesis, pleural fluid tap, pleural fluid culture, pleural fluid cytology, pleural effusion testing, LDH body fluid

WHAT IS IT USED FOR?

A pleural fluid analysis helps find the cause of a pleural effusion. There are different types of fluid that can build up in the pleural space. The type of fluid you have will depend on the cause. A pleural fluid analysis is used to figure out which type of fluid you have. This information helps narrow down the possible causes.

A pleural fluid analysis alone may not find the specific condition causing your pleural effusion. But the results help your health care provider decide which other tests to order to help make a final diagnosis.

WHY DO I NEED A PLEURAL FLUID ANALYSIS?

You may need this test if you have symptoms of pleural effusion and an imaging test, such as an MRI scan or chest x-ray, shows that you have fluid buildup around your lungs.

The symptoms of pleural effusion depend on the cause and the amount of fluid. They may include:

Chest pain that doesn't have another explanation

Dry cough that doesn't bring up mucus

Trouble breathing

Fatigue

If the cause of your pleural effusion is known, you may not need a pleural fluid analysis even if you have symptoms.

Pleural effusions don't always cause symptoms, so sometimes a pleural effusion is found on a chest imaging test that's done for another reason. If this happens, a pleural fluid test is often done, even though you don't have symptoms.

WHAT HAPPENS DURING A PLEURAL FLUID ANALYSIS?

Your provider will remove some pleural fluid using a hollow needle. This is done with a procedure called thoracentesis. The procedure may be done in a doctor's office or hospital. It may include these steps:

You may be asked to remove your clothes and put on a paper or cloth gown.

You may breathe oxygen through a face mask or short tubes that go into your nostrils.

You will sit up straight with your arms resting on a table. If you can't sit, you may lie on your side.

Your provider will clean an area on your back where the needle will be inserted.
You'll be given an injection (a shot) to numb the area so you won't feel any pain.
When the area is numb, your provider will insert a needle into your back between your ribs. The needle will go into the pleural space. Your provider may use ultrasound imaging to help see where to insert the needle.
You may feel some pressure when the needle goes in and when your provider removes the fluid.
At times, you may be asked to hold your breath during the procedure.
When enough fluid has been removed, the needle will be taken out and the fluid will be sent to a lab. A small bandage will be placed where the needle went into your back.
After the test, you may have a chest x-ray to check for any problems.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

In general, a thoracentesis doesn't require special preparations. Your provider will let you know if there's anything you need to do or avoid doing. For example, you may need to stop taking certain medicines before your test. But never stop taking any prescription medicines without talking with your provider first. or a blood test.

ARE THERE ANY RISKS TO THE TEST?

Thoracentesis is a generally safe procedure. Serious complications are uncommon. They include:

A collapsed lung if the needle goes into your lung

Bleeding that causes blood to collect near the lung where the needle was put in

Pulmonary edema, a condition in which fluid builds up inside your lungs if too much pleural fluid is removed or if the fluid is removed too quickly

Infection if bacteria get into your body during the procedure

Injury to your liver or spleen if the needle goes into either organ

WHAT DO THE RESULTS MEAN?

A pleural fluid analysis usually includes results from several tests done on your fluid sample. Your provider will consider all your test results, your symptoms, and your medical history to understand what your results mean. Ask your provider to explain what your test results say about your health.

In general, the results of your pleural fluid analysis will show which type of fluid is in your pleural space. There are two main types of fluid, and each type has different possible causes. To find out which type of fluid you have, your test usually uses a set of guidelines called Light's criteria.

Light's criteria tell providers how to compare the amount of certain proteins in your pleural fluid to the amount of those proteins measured in your blood. The results of the comparison will show whether your pleural fluid is transudate or exudate:

Transudate is a watery fluid that has low amounts of protein. It's caused by pressure in your blood vessels. The pressure makes fluid leak from your blood vessels into the pleural space. Transudate fluid is most often caused by:

Heart failure

Kidney disease

Liver disease, including cirrhosis

Heart failure

Kidney disease

Liver disease, including cirrhosis

Exudate is a cloudy fluid that has high amounts of protein. It can be caused by conditions that involve inflammation, infections, and other conditions that affect blood vessels or lymphatic vessels, which are part of your immune system. These conditions include:

Pancreatitis

Certain autoimmune diseases, including rheumatoid arthritis and lupus
Cancer, most often leukemia, lymphoma, breast or lung cancer
Lung diseases that aren't cancer, including infections, such as pneumonia or tuberculosis
Lung or chest injuries
Pancreatitis

Certain autoimmune diseases, including rheumatoid arthritis and lupus
Cancer, most often leukemia, lymphoma, breast or lung cancer
Lung diseases that aren't cancer, including infections, such as pneumonia or tuberculosis
Lung or chest injuries

Exudate contains different substances depending on the condition that caused it. For example, exudate from an infection may have signs of pus, bacteria, or fungus. Exudate from cancer usually has cancer cells. A chest injury may cause exudate that includes blood.

To help find the specific cause of your pleural effusion, your provider will order other tests on your fluid sample to check for certain substances. The tests you have will depend on the most likely causes of your pleural effusion, based on all the information about your condition.

In certain cases, you may also need other types of tests, including blood and/or urine tests, imaging tests, a bronchoscopy, or a biopsy.

If you have questions about your test results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PLEURAL FLUID ANALYSIS?

If your pleural effusion causes symptoms, your provider will remove enough fluid during a thoracentesis to relieve your discomfort. This helps treat the pleural effusion, but not the cause.

If your provider thinks the fluid may quickly build up again, a special tube, called a catheter, may be inserted when your pleural fluid sample is taken. The tube will be left in place to let the fluid drain out of your body. If pleural fluid stops building up, the tube may be removed.

Other treatments for pleural effusion will depend on how severe it is and what's causing it.

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Pneumococcal Disease Tests

<https://medlineplus.gov/lab-tests/pneumococcal-disease-tests/>

PNEUMOCOCCAL DISEASE TESTS

WHAT ARE PNEUMOCOCCAL DISEASE TESTS?

Pneumococcal disease is the name for any infection caused by pneumococcus bacteria (*Streptococcus pneumoniae*). Pneumococcal disease tests look for these bacteria in your:

Blood

Urine (pee)

Sputum, the thick mucus coughed up from your lungs

Cerebrospinal fluid (CSF), the fluid in spaces around your brain and spinal cord

Your health care provider will decide which type of test you need, based on your symptoms.

You may have one or more of these pneumococcal disease tests, depending on where in your body you have an infection:

A pneumococcal PCR test, which is a test that checks for small amounts of the genetic material of pneumococcus bacteria in a sample of your blood.

A pneumococcal antigen test, which is used to identify pneumococcus bacteria in your urine or cerebrospinal fluid.

A pneumococcal culture, which is a bacteria culture test that helps find pneumococcus bacteria in a sample of your blood, urine, sputum, or other tissue.

Pneumococcal disease often causes a fever. Which other symptoms you have and how serious they are can vary, depending on where your infection is.

If your infection is mild, the infection may be considered "noninvasive" (an infection that doesn't spread to your major organs or blood). If you have a noninvasive infection, the bacteria may cause:

An ear infection

A sinus infection

You may have more serious symptoms if the infection is considered "invasive" (an infection that can or has spread to other parts of your body). If you have an invasive infection, the bacteria may get into your:

Lungs, which could lead to pneumonia

Bloodstream, which is called bacteremia and can lead to sepsis

Tissue surrounding your brain and spinal cord, which could lead to meningitis

Pneumococcal bacteria can spread from person to person through droplets from infected saliva (spit) or mucus. For example, this can happen when:

Someone who has the bacteria coughs or sneezes near you

You touch an item that has infected droplets on it

You kiss someone who has the bacteria

Many people, especially children, have this bacteria in their nose or throat at one time or another. Most of the time, they don't get sick, and this is called being a "carrier" of the bacteria. But they can still spread it to others.

Pneumococcal disease often spreads through crowded settings and is more common during winter and early spring.

Anyone can get pneumococcal disease, but some people have a higher risk. They include people who:

Are age 65 or older. Older adults tend to have more severe symptoms and complications.

Are younger than five years old.

Live in a long-term care facility.

Are current or former smokers.

Have a chronic (long-lasting) lung disease such as COPD or emphysema.

Have a weakened immune system due to a disease such as HIV or because you are taking certain medicines such as corticosteroids or chemotherapy.

Have a chronic health condition such as diabetes, coronary artery disease, or kidney failure.

Have alcohol use disorder (AUD).

Have no spleen.

Have a cochlear implant (an implanted device that helps you hear).

Are from certain racial and ethnic groups, such as Alaska Native people, African American people, and certain American Indian people.

Other names: Streptococcus pneumoniae test, S. pneumoniae test, Pneumococcal Culture, Pneumococcal Antigen Test, Pneumococcal PCR Test

WHAT IS IT USED FOR?

Pneumococcal disease tests are used to find out whether you have pneumococcal disease.

WHY DO I NEED A PNEUMOCOCCAL DISEASE TEST?

Symptoms of pneumococcal infections can vary, depending on which part of your body is affected. They most often affect your lungs, middle ear, or sinuses. Your symptoms could begin suddenly. They can range from mild to severe. It's important to get the right diagnosis and treatment to help prevent health complications.

Symptoms of an ear infection can include:

A red, swollen eardrum

Ear pain

Fever

Fluid draining from the ear

Symptoms of a sinus infection can include:

Cough

Facial pain or pressure

Headache

Post-nasal drip (mucus dripping down the throat)

Runny or stuffy nose

Sore throat

Symptoms of pneumococcal pneumonia, a lung infection, can include:

Sharp chest pain, often on one side

Cough

Fever and chills

Rapid or difficulty breathing

In older adults, confusion or not being alert

Symptoms of pneumococcal bacteremia, a bloodstream infection, can include:

Chills

Fever

Confusion

Symptoms of pneumococcal meningitis, an infection of the lining of your brain and spinal cord, can include:

Confusion

Fever

Headache

Your eyes being more sensitive to light (photophobia)

A stiff neck, which may make lowering your chin to your chest difficult

In babies, not eating or drinking as well, being less alert, and sometimes vomiting

Call your provider if you think that you or your child could have a serious pneumococcal infection. Severe infections can lead to sepsis or long-term health problems.

WHAT HAPPENS DURING A PNEUMOCOCCAL DISEASE TEST?

The type of pneumococcal disease test you have is usually based on your symptoms. If you have a mild infection, such as an ear or sinus infection, your provider may diagnose and treat you based on your symptoms, medical history, and a physical exam.

Depending on your symptoms, your provider may want a sample of certain body fluids. They may want one or more samples, which could be from your:

Blood

Urine

Sputum

Nostrils (nasopharyngeal swab)

Cerebrospinal fluid (CSF)

Your provider may order a pneumococcal culture, pneumococcal antigen test, and/or a pneumococcal PCR test to help identify the bacteria.

Your sample may be tested using a Gram Stain and/or bacterial culture which involves growing bacteria in your sample to help identify what type is causing your infection. Knowing the cause may help your provider figure out which antibiotic will work best, since some pneumococcal bacteria may have become resistant to certain antibiotics.

During a blood test:

If your provider thinks you might have meningitis or a bloodstream infection, they may collect a blood sample for a pneumococcal PCR test.

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

During a clean catch urine test:

If your provider suspects that you might have pneumonia, they may want a urine sample for a pneumococcal antigen test, which is often used to test for the pneumococcal bacteria.

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

During a sputum test:

Your provider may request a sputum sample for a pneumococcal culture. Sputum is a thick type of mucus made in your lungs. If you have an infection or chronic illness affecting your lungs or airways, sputum can settle in your lungs, and you can also cough it up. During the sputum test:

You may be asked to rinse your mouth with water before the sample is taken.

Your provider will ask you to breathe deeply and then cough deeply into a special cup.

Your provider may tap you on the chest to help loosen sputum from your lungs.

If you have trouble coughing up enough sputum, your provider may ask you to breathe in a salty mist to help you cough more deeply.

In certain situations, if you can't produce enough sputum even after breathing in a salty mist, your provider may order a bronchoscopy (a procedure that's used to check for the cause of a lung problem). They may also order a bronchoalveolar lavage (BAL). This is a procedure that can be done during a bronchoscopy. A bronchoalveolar lavage collects a sample from your lungs and checks it for *Streptococcus pneumoniae*. A provider may do this if:

You have a severe respiratory disease.

You aren't improving from treatment for pneumococcal disease.

Your provider is unable to determine the type of infection without a sample directly from your lungs.

During a bronchoscopy:

You may need to remove some or all of your clothing. If so, you will be given a hospital gown.

You may also need to remove dentures and any removable dental appliances.

You will lie on a bed or table with your head raised.

You may get medicine (sedative) to help you relax. The medicine will be injected into a vein or given through an IV (intravenous) line placed in your arm or hand.

Your provider will spray a numbing medicine in your mouth and throat so you won't feel any pain during the procedure.

Your provider will insert the bronchoscope (a thin, lighted tube with a camera attached) down your throat and into your airways. The camera will take photos and connect to a video screen.

As the bronchoscope is moved down, your provider can look into your airways and lungs through the bronchoscope and on the video screen.

For a BAL:

Your provider will put a small amount of saline through the bronchoscope.

After washing the airways, the saline is sucked up into the bronchoscope.

The saline solution will contain cells and other substances, such as *Streptococcus pneumoniae*, which will be taken to a lab for testing.

During a nasal swab test:

Your provider may want to do a pneumococcal culture with a sample of cells from inside your nostrils or from the nasopharynx, the uppermost part of your nose and throat.

During a nasopharyngeal (NP) swab:

You will tilt your head back.

Your provider will insert a long swab into your nostril until it reaches your nasopharynx (the upper part of your throat).

Your provider will rotate the swab for 10 to 15 seconds and remove it.

If your provider gets enough of a sample from one of your nostrils, they may only need to do one. But they may need to get a sample from your other nostril as well if they had trouble getting the sample from the first side.

During a spinal tap:

If your provider thinks you might have meningitis or a bloodstream infection, they may collect a sample of your cerebrospinal fluid for a pneumococcal antigen test.

To get a sample of cerebrospinal fluid, a provider will do a procedure called a spinal tap, also known as a lumbar puncture. A spinal tap is usually done in a hospital. During the procedure:

You will lie on your side or sit on an exam table.

A provider will clean your back and inject an anesthetic into your skin, so you won't feel pain during the procedure. Your provider may put a numbing cream on your back before this injection.

When the area on your back is completely numb, your provider will insert a thin, hollow needle between two vertebrae in your lower spine. Vertebrae are the small backbones that make up your spine.

Your provider will withdraw a small amount of cerebrospinal fluid for testing. This will take about five minutes.

You'll need to stay very still while the fluid is being withdrawn.

Your provider may ask you to lie on your back for an hour or two after the procedure. This may prevent you from getting a headache afterward.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

For most pneumococcal disease tests, you don't need any special preparation. Talk to your provider and ask about any special preparations. You may need to fast (not eat or drink) before the test. You may also need to stop taking certain medicines, so be sure to tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put

in, but most symptoms go away quickly.

There is no risk to providing a urine or sputum sample.

A nasopharyngeal swab may be uncomfortable and cause coughing or gagging. All these effects are temporary.

With a spinal tap, there is very little risk. You may feel a little pinch or pressure when the needle is inserted. After the test, you may feel some pain or tenderness in your back or have some bleeding where the needle was inserted. You may also get a headache. The headache may last for several hours or up to a week or more, but your provider may suggest treatment to relieve the pain.

There is very little risk to having a bronchoscopy or a BAL. Serious complications are rare, but they may include bleeding in the airways, infection, or a pneumothorax (collapse of part of your lung).

WHAT DO THE RESULTS MEAN?

The results will show whether you have pneumococcal disease.

A positive result means that pneumococcus bacteria were found in your specimen. You most likely have pneumococcal disease.

A negative result means that no pneumococcus bacteria were found in your specimen. Most likely, your symptoms are not caused by pneumococcal disease.

If you have questions about your results, talk to your provider. They will consider your symptoms and medical history to understand the results of your pneumococcal disease test. They may also want to use a physical exam and/or other tests to help them make a diagnosis and decide on the best treatment for you. These other tests could include:

Chest x-ray

Complete blood count (CBC) and differential

Erythrocyte sedimentation rate (ESR)

C-reactive protein (CRP) test

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PNEUMOCOCCAL DISEASE TEST?

Pneumococcal disease may be treated with antibiotics, fluids, pain relievers, and rest. But don't give aspirin to children, because it may cause Reye syndrome. This is a rare, serious illness that can affect the brain and liver.

There are two types of vaccines to help prevent pneumococcal disease. Vaccines can reduce the risk of infection, especially for children younger than five years old, adults 50 years or older, and people with health conditions that increase their risk for the disease. Ask your provider if they recommend that you or your child get a vaccine.

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Porphyrin Tests

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PORPHYRIN TESTS

WHAT ARE PORPHYRIN TESTS?

Porphyrin tests measure the level of porphyrins in your blood, urine (pee), or stool (poop). Your body uses porphyrins to make heme. Heme is part of hemoglobin, which is a protein in your red blood cells that carries oxygen from your lungs to the rest of your body.

It's normal to have a small amount of porphyrins in your blood and other body fluids. But too much porphyrin may mean you have a type of porphyria. Porphyrias are a group of diseases that happen if you lack one of the proteins necessary to change porphyrins into heme. If your body doesn't use porphyrins correctly, they can build up and cause illness.

Porphyrias are very uncommon disorders that are usually inherited. That means these diseases tend to run in families because of a change in a gene that's passed down from parents to children. There are two main groups of porphyrias:

Acute porphyrias happen suddenly and usually last days or weeks. They mainly affect the nervous system. Sometimes they affect the skin, too.

Cutaneous porphyrias are long-lasting and affect only the skin, causing blisters or pain when you are exposed to sunlight

There are eight types of porphyria that fall into either of these groups. Your symptoms and where excess porphyrins are in your body will help your health care provider determine a diagnosis of acute or cutaneous porphyria.

Other names: protoporphyrin; protoporphyrin, blood; protoporphyrin, stool; porphyrins, feces; uroporphyrin; porphyrins, urine; Mauzerall-Granick test; acid; ALA; porphobilinogen; PBG; free erythrocyte protoporphyrin; fractionated erythrocyte porphyrins; FEP

WHAT ARE THEY USED FOR?

Porphyrin tests are used to help diagnose or to monitor all types of porphyria.

WHY DO I NEED A PORPHYRIN TEST?

You may need a porphyrin test if you have symptoms of porphyria. Each porphyria group has different symptoms:

Symptoms of acute porphyria may be mild or severe. Without early treatment, they may get worse and even become life-threatening. Symptoms may include:

Pain in the abdomen (belly), back, or arms and legs

Constipation

Nausea and vomiting

Mental changes, such as anxiety, confusion, and seizures

Problems urinating (peeing), leaking urine, or red or brown urine

Problems with nerves that control movement, which may cause muscle weakness, paralysis, and breathing problems

Skin blisters when skin is in sunlight (for certain types of acute porphyria)

Symptoms of cutaneous porphyria happen when skin is exposed to sunlight.

Symptoms of some types of cutaneous porphyria include:

Blisters

Fragile skin that is easily wounded and slow to heal

Infections in blisters or wounds

Changes in skin color or scarring

Other types of cutaneous porphyria usually don't cause blisters. Instead, sunlight may cause these skin symptoms:

Pain, burning, stinging, or tingling

Redness

Swelling

If someone in your family has porphyria, you may need a porphyrin test to see if you have inherited the condition. But the most common type of porphyria isn't inherited. It usually develops after age 30 and may be caused by conditions, such as:

Hemochromatosis, a disease where too much iron builds up in the body

Alcohol use disorder (AUD)

Smoking

Viral infections, such as hepatitis C and HIV

Taking estrogen in medicines such as birth control pills or hormone replacement therapy

WHAT HAPPENS DURING PORPHYRIN TESTING?

Porphyrins can be tested in blood, urine, or stool. The most common types of porphyrin tests are listed below. The type of test your provider will use to diagnose you will depend on the type of porphyria they think you might have.

Blood Test

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

24-Hour Urine Sample

You will collect all your urine during a 24-hour period. For this test, your provider or laboratory will give you a container and specific instructions on how to collect your samples at home. Be sure to follow all instructions carefully. This 24-hour urine sample test is used because the amounts of substances in urine, including porphyrin, can vary throughout the day. So, collecting several samples in a day may give a more accurate picture of your urine content.

You will collect all your urine during a 24-hour period. For this test, your provider or laboratory will give you a container and specific instructions on how to collect your samples at home. Be sure to follow all instructions carefully. This 24-hour urine sample test is used because the amounts of substances in urine, including porphyrin, can vary throughout the day. So, collecting several samples in a day may give a more accurate picture of your urine content.

Random Urine Test

You can provide your sample at any time of day, with no special preparations or handling needed. This test is often done in your provider's office or a lab.

You can provide your sample at any time of day, with no special preparations or handling needed. This test is often done in your provider's office or a lab.

Stool Test (also called protoporphyrin in stool)

You will collect a sample of your stool and place it in a special container. Your provider will give you instructions on how to prepare your sample and send it to a lab.

You will collect a sample of your stool and place it in a special container. Your provider will give you instructions on how to prepare your sample and send it to a lab.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for blood or urine tests.

For a stool test, you may be instructed to not eat meat or take any aspirin-containing medicines for three days prior to your test.

ARE THERE ANY RISKS TO PORPHYRIN TESTS?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There are no known risks to urine or stool tests.

WHAT DO THE RESULTS MEAN?

Porphyria is a rare disease and is hard to diagnose. Many of the symptoms of porphyria are similar to those of other, more common diseases.

If high levels of porphyrin are found in your blood, urine, or stool, your provider will probably order more tests to confirm a diagnosis and to find out what kind of porphyria you have.

While there is no cure for porphyria, the condition can be managed. Certain lifestyle changes and/or medicines can help prevent the symptoms and complications of the disease. Specific treatment depends on the type of porphyria you have. If you have questions about your results or about porphyria, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT PORPHYRIN TESTS?

While most types of porphyria are inherited, other types of porphyria can also be acquired. Acquired porphyria can be caused by a variety of factors, including overexposure to lead, HIV, hepatitis C, excess iron intake, and/or heavy alcohol use.

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Postpartum Depression Screening

<https://medlineplus.gov/lab-tests/postpartum-depression-screening/>

POSTPARTUM DEPRESSION SCREENING

WHAT IS A POSTPARTUM DEPRESSION SCREENING?

A postpartum depression screening is a standard set of questions you answer to help your health care provider find out whether you have depression related to having a baby. Postpartum means "after birth."

Postpartum depression is a common but serious mood disorder. Though there is no clear cause for this condition, there are several factors that may increase your risk of getting it. These include:

Sudden changes in hormone levels that happen after pregnancy.

Lack of sleep.

Stress from new routines.

It differs from what are commonly called the "baby blues." With the baby blues, you may feel sad, weepy, or anxious starting about three days after childbirth. These feelings tend to come and go and usually get better on their own within a week or two. Postpartum depression, however, may:

Last for months.

Cause more intense symptoms than the baby blues.

Affect how you think and behave.

Lead to feelings of extreme hopelessness and/or a lack of interest in the baby.

Not get better without treatment.

Symptoms of postpartum depression usually start between one to three weeks after birth. However, they may begin at any time during the first year of parenthood. Because of this, your and/or your baby's provider may repeatedly screen you for postpartum depression throughout the first year of your baby's life. Doing so can help them detect signs of postpartum depression early, and so better treat it with medicine and/or talk therapy.

Other names: postpartum depression assessment, EPDS test

WHAT IS IT USED FOR?

Your provider will use a postpartum depression screening to help find out if you have depression after giving birth. This screening is usually part of a routine checkup a few weeks after your baby is born and may include questions about:

Your mood since having the baby.

Your sleep habits.

Sources of stress in your life, such as new routines.

Personal or family history of depression.

WHY DO I NEED POSTPARTUM DEPRESSION SCREENING?

Because anyone who has had a baby can develop postpartum depression, you'll likely be tested for it shortly after childbirth. However, if you have symptoms of postpartum depression before or after your routine checkup, contact your provider as soon as possible.

Symptoms of postpartum depression include:

Feeling sad or empty most of the time.

Eating too much or too little.

Sleeping too much or too little.

Crying more than usual or for no clear reason.

Feeling more angry than usual.

Pulling away from family and friends.

Worrying or feeling anxious often.

Not having any interest in your baby.

Constant doubts about your ability to care for your baby.

Thinking about hurting yourself or your baby.

If you have thoughts about suicide, or hurting yourself or your baby get help right away:

Call 911 or go to your local emergency room.

Contact a crisis hotline. In the United States, you can reach the National Suicide and Crisis Lifeline at any time:

Call or text 988

Chat online with Lifeline Chat

TTY users: Use your preferred relay service or dial 711 then 988

Call or text 988

Chat online with Lifeline Chat

TTY users: Use your preferred relay service or dial 711 then 988

Veterans can contact the Veterans Crisis Line:

Call 988, then press 1

Text 838255

Chat online

Call 988, then press 1

Text 838255

Chat online

Call your mental health provider or other provider.

Reach out to a loved one or close friend.

Postpartum depression can happen after any childbirth. However, you may be more likely to develop this condition if you:

Have had depression or bipolar disorder or a family health history that includes these conditions.

Don't have support from family or friends.

Had a multiple birth (twins, triplets, or more).

Gave birth while in your teens.

Had health problems during pregnancy.

Went into preterm labor or had any other birth problems.

Have a baby with special needs.

Have ever experienced domestic violence.

Experience financial difficulties.

Had an unplanned pregnancy.

You may also be screened if you have depressive symptoms during pregnancy. This is called perinatal depression, and it has many of the same symptoms as depression after birth. If left untreated, it can:

Increase the risk of your depressive symptoms lasting into the first weeks or months of parenthood.

Make it harder to bond with your baby.

Contribute to sleeping problems.

Cause difficulties with breastfeeding.

Because of this, medical experts recommend that providers screen for depression during routine pregnancy checkups.

WHAT HAPPENS DURING A POSTPARTUM DEPRESSION SCREENING?

During a postpartum depression screening you'll answer a set of questions. Your provider may ask the questions, or you may fill out a questionnaire form to discuss with your provider later. One of the most common questionnaires is called the Edinburgh Postnatal Depression Scale (EPDS). The EPDS includes 10 questions about your mood and thoughts.

Your provider may also order a blood test to find out if a physical condition, such as hyperthyroidism or hypothyroidism, may be causing your depression.

During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A POSTPARTUM DEPRESSION SCREENING?

You usually don't need any special preparations for a postpartum depression screening.

ARE THERE ANY RISKS TO SCREENING?

There is no risk to having a physical exam or taking a questionnaire.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your screening shows you may have postpartum depression, it's important to get treatment as soon as possible. You may need treatment from a mental health provider. A mental health provider is a health care professional who specializes in diagnosing and treating mental health problems. They may treat you using:

Counseling (also called talk therapy).

Medicine, such as antidepressants. If you are breastfeeding your baby, let your provider know. This may affect what medicines your provider can give you.

A mental health provider may also follow-up with you throughout your treatment to monitor your progress.

Along with these treatment options, there are things you can do at home that may help you feel better, such as:

Asking for help caring for the baby and doing other household chores.

Spending time with other adults.

Taking time for yourself.

Getting some physical activity, such as walking.

Resting when the baby rests.

Joining a support group for new moms.

In the most serious cases, when symptoms don't respond to medicine or any other treatment, your provider may use brain stimulation therapies, such as electroconvulsive therapy (ECT), also called "shock therapy."

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A POSTPARTUM DEPRESSION SCREENING?

A rare but more serious form of postpartum depression is called postpartum psychosis. Postpartum psychosis may cause hallucinations (seeing or hearing things that aren't real) and confusion. It is a medical emergency and it's important to get help immediately by calling 911 or going to the nearest emergency room.

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Twins, Triplets, Multiple Births

RELATED MEDICAL TESTS

Depression Screening

Mental Health Screening

Suicide Risk Screening

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Potassium Blood Test

<https://medlineplus.gov/lab-tests/potassium-blood-test/>

POTASSIUM BLOOD TEST

WHAT IS A POTASSIUM BLOOD TEST?

A potassium blood test measures the amount of potassium in your blood. Potassium is a type of electrolyte. Electrolytes are electrically charged minerals that help control the amount of fluids and the balance of acids and bases (pH balance) in your body. They also help control muscle and nerve activity and perform other important functions.

Your cells, nerves, heart, and muscles need potassium to work properly. You can usually get most of the potassium that you need from the foods that you eat. Almost all of your body's potassium is inside your cells, but a small amount is in your blood. Your kidneys remove potassium in your urine (pee) to help keep the right amount of potassium in your blood. Potassium levels that are too high or too low may be a sign of a medical condition and may cause serious health issues.

Other names: potassium serum, serum potassium, serum electrolytes, K

WHAT IS IT USED FOR?

A potassium blood test measures the amount of potassium in your blood. The test is often part of a group of routine blood tests called an electrolyte panel. It may be used to monitor or diagnose conditions related to abnormal potassium levels. These conditions include kidney disease, high blood pressure, and heart disease.

WHY DO I NEED A POTASSIUM BLOOD TEST?

Your health care provider may order a potassium blood test as part of your routine checkup or to monitor an existing condition, such as diabetes, kidney disease, or adrenal gland disorders. You may also need this test if you take medicines that could affect your potassium levels or if you have symptoms of having too much or too little potassium.

If your potassium levels are too high (hyperkalemia), your symptoms may include:

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Fatigue

Muscle weakness

Nausea

Numbness or tingling

If your potassium levels are too low (hypokalemia), your symptoms may include:

Irregular heartbeat

Muscle cramps

Weak or twitching muscles

Fatigue

Nausea

Constipation

WHAT HAPPENS DURING A POTASSIUM BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a potassium blood test or an electrolyte panel. If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Too much potassium in the blood (hyperkalemia) is often the result of two or more causes, which could include: Kidney disease. Too much potassium in your blood may mean your kidneys aren't working well.

Addison disease, a disorder of the adrenal glands.

Injuries, burns, or surgery, which can cause your cells to release extra potassium into your blood.

Type 1 diabetes that is not well controlled.

The side effects of certain medicines, such as diuretics ("water pills") or antibiotics.

A diet too high in potassium (not common). Bananas, apricots, green leafy vegetables, avocados, and many other foods are good sources of potassium that are part of a healthy diet. But eating very large amounts of potassium-rich foods or taking potassium supplements can lead to health problems.

Too little potassium in the blood (hypokalemia) may be a sign of:

The side effects of prescription diuretics

Fluid loss from diarrhea, vomiting, or heavy sweating

Using too many laxatives

Adrenal gland disorders, including Cushing's syndrome and aldosteronism

Kidney disease

Alcohol use disorder (AUD)

A diet too low in potassium (not common)

If your test results are not in the normal range, it doesn't always mean that you have a medical condition that needs treatment.

Certain prescription and over-the-counter medicines and supplements may raise your potassium levels.

You should not eat natural black licorice for at least two weeks before your test because it may lower your levels. Most licorice products in the U.S. don't contain natural licorice. But if you eat licorice, check the ingredients label to be sure.

To learn what your results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A POTASSIUM BLOOD TEST?

Repeated clenching and relaxing of your fist just before or during your blood test may temporarily increase the potassium levels in your blood. This may lead to an incorrect result.

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Prealbumin Blood Test

<https://medlineplus.gov/lab-tests/prealbumin-blood-test/>

PREALBUMIN BLOOD TEST

WHAT IS A PREALBUMIN BLOOD TEST?

A prealbumin blood test measures prealbumin levels in your blood. Prealbumin is a protein that's made in your liver and is found in your blood. Prealbumin is used to make other proteins and helps carry thyroid hormones and vitamin A through your bloodstream. It also helps control how your body uses energy.

If your prealbumin levels are lower than normal, it may be a sign of malnutrition. Malnutrition is a condition where your body doesn't get enough of the nutrients that you need for good health, such as protein, vitamins, and minerals. Children with malnutrition may not grow and develop normally.

Malnutrition can happen if you:

- Don't eat enough food

- Eat enough, but don't eat healthy, nutritious foods

- Have problems absorbing nutrients

- Have an increased need for nutrients because of infection, a serious injury, cancer, recovery after surgery, or other conditions

Other names: thyroxine binding prealbumin, PA, transthyretin test, transthyretin, tryptophan-rich prealbumin

WHAT IS IT USED FOR?

Until recently, a prealbumin test was used to help:

- Find out if you are getting enough nutrients, especially protein, in your diet

- Check for nutrition problems if you have a risk of malnutrition because you:

 - Are in the hospital with certain diseases

 - Have had surgery

 - Have a serious or chronic (long-lasting) illness

 - Get dialysis treatment for kidney disease and need to avoid eating many types of food

 - Are fed by parenteral nutrition (artificial feeding), which means that you get a nutritional liquid through an intravenous (IV) tube

 - Are in the hospital with certain diseases

 - Have had surgery

 - Have a serious or chronic (long-lasting) illness

 - Get dialysis treatment for kidney disease and need to avoid eating many types of food

 - Are fed by parenteral nutrition (artificial feeding), which means that you get a nutritional liquid through an intravenous (IV) tube

While some health care professionals still use the prealbumin test for these reasons, many now believe that changes in prealbumin can actually be a sign of inflammation (swelling), infection, or injury. So, they think that this test should not be used to measure your nutrition levels or test for malnutrition anymore. That being said, there are health care professionals who think the test can still be helpful to check for malnutrition in people who are very sick or hospitalized.

WHY DO I NEED A PREALBUMIN BLOOD TEST?

Your health care provider may order a prealbumin test to help diagnose and monitor problems with your nutritional health if they think the test is still helpful for this. You may also need this test before surgery or if you have symptoms of malnutrition, such as:

- Weight loss

Weakness
Dizziness
Dry skin
Dry, brittle hair and/or hair loss
Feeling cold
Aching joints

WHAT HAPPENS DURING A PREALBUMIN BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a prealbumin test. Certain medicines may affect the results of your test. Tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Prealbumin levels that are lower than normal may happen with:

Malnutrition
Inflammation
Trauma, such as a burn injury
Serious or long-term illness, such as cancer
Liver disease
Serious infection
Hyperthyroidism
Certain digestive diseases

If you have low prealbumin levels and you are ill, injured, or recovering from surgery, it may be difficult to know exactly what is causing the low levels. Your provider may need to do other tests to find out more about your nutritional health.

Prealbumin levels that are higher than normal may happen with several conditions, including:

Hodgkin disease
Kidney failure
Alcohol use disorder (AUD)
High-dose corticosteroid therapy

Prealbumin testing is not used to diagnose or monitor these conditions. If you have high prealbumin levels, your provider may order other tests to diagnose your condition.

If your prealbumin levels are not normal, it doesn't always mean you have a condition that needs treatment. Certain medicines and pregnancy can affect your prealbumin levels. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PREALBUMIN BLOOD TEST?

Remember, some medical experts don't think a prealbumin test is the best way to diagnose or monitor malnutrition. That's because other medical conditions can cause low prealbumin levels. More research is needed to understand how prealbumin works in the body and what abnormal test results may mean. But many providers still find prealbumin blood tests useful for monitoring nutrition, especially in people who are seriously ill or in the hospital.

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Pregnancy Test

<https://medlineplus.gov/lab-tests/pregnancy-test/>

PREGNANCY TEST

WHAT IS A PREGNANCY TEST?

A pregnancy test can tell whether you're pregnant by checking a sample of your urine (pee) or blood for a hormone called human chorionic gonadotropin (hCG).

This hormone is made by the placenta, the organ that grows in your uterus to provide oxygen and nutrients to the fetus. The placenta starts making hCG after a fertilized egg attaches to the inside wall of your uterus. Levels then continue to rise for ten weeks, almost doubling every three days.

hCG helps your body prepare for pregnancy by:

Telling your body to produce progesterone, which thickens the lining of your uterus.

Telling your body to produce estrogen, which prepares your body for labor and helps your fetus develop.

Stopping your menstrual periods.

There are two types of pregnancy tests:

Urine tests, which can be done at your health care provider's office or with an at-home test kit. These tests are 97-99% accurate when done a week or two after you've missed your period and can give you the results quickly. If you do an at-home test, follow the instructions carefully.

Blood tests, which are always done in your provider's office or a lab. These tests can find very small amounts of hCG and show whether you're pregnant before you've missed your period. However, they aren't commonly used to check for pregnancy. That's because urine tests are less expensive, very accurate, and provide quicker results than blood tests, which may take hours to more than a day.

Other names: human chorionic gonadotropin test, HCG test, qualitative hCG blood test, quantitative hCG blood test, Beta-hCG urine test, total chorionic gonadotropin, hCG total OB, chorionic gonadotropin

WHAT IS IT USED FOR?

Both urine and blood tests check for hCG to find out whether you're pregnant.

There are two types of blood tests. A quantitative blood test measures the exact amount of hCG in your blood, and a qualitative hCG blood test simply tells you whether you are pregnant or not.

The amount of hCG in your body can give your provider important information about your pregnancy and the health of your fetus. Quantitative tests are sometimes used to help:

Find out the age of the fetus if you're very early in your pregnancy.

Monitor your pregnancy if you have a high risk of miscarriage.

Check for an ectopic pregnancy, which is when a fertilized egg tries to grow outside of the uterus. The egg cannot grow into a fetus when it's in the wrong place. It must be removed to avoid damage to your organs. This can be a medical emergency.

Check for a molar pregnancy (hydatidiform mole), which is an abnormal growth of tissue in the uterus. It's caused by a fertilized egg with such severe genetic problems that it cannot become a fetus. The growth can turn into cancer and must be removed.

Detect any problems in the fetus, including Down syndrome, other chromosome problems, and certain birth defects. hCG testing is usually part of a group of prenatal screening tests called a "triple" or "quadruple" screen test.

In some cases, your provider may also order a quantitative hCG blood test to help diagnose or monitor health conditions that aren't related to pregnancy. These include ovarian and testicular cancer along with other conditions that can increase hCG levels.

WHY DO I NEED A PREGNANCY TEST?

You may need this test if you think you're pregnant. Symptoms of pregnancy vary, but the most common sign of early pregnancy is a missed period. Other common signs of early pregnancy may include:

Swollen, tender breasts

Fatigue

Frequent need to urinate (pee)

Nausea and vomiting (also called morning sickness)

Feeling bloated or swollen in your abdomen (belly) or body

You may also need a pregnancy test to make sure you are not pregnant if you need to have medical treatment that could harm a fetus.

Lastly, you may need a quantitative blood test if you:

Are at high risk for a miscarriage. The most common risk factors include active infections, thyroid disease, uncontrolled diabetes, obesity, and uterus or cervix problems.

Have symptoms of an ectopic pregnancy. These can include abnormal vaginal bleeding, pain in your lower back and/or abdomen, and cramping on one side of the pelvis.

Have symptoms of a molar pregnancy. These can include abnormal vaginal bleeding, pain in your abdominal or pelvic area, high blood pressure, and severe nausea.

WHAT HAPPENS DURING A PREGNANCY TEST?

Home pregnancy tests are quick and easy to use. You can buy a home pregnancy test kit without a prescription. The kits include test sticks or strips that react to hCG in your urine. The steps for doing a test depend on the brand, so it's very important to follow the instructions that come with your test. For most test kits, you'll either:

Hold the test stick or strip in your urine stream

Collect your urine in a cup and dip the test stick or strip into the cup

After waiting a certain number of minutes, you'll check your results on the test stick or strip. The instructions will tell you what to look for. In general, to get the most accurate results with any home pregnancy test, you'll need to:

Check the expiration date before you use the test.

Read the instructions for your test to see if it has a control symbol. This symbol shows that your test worked correctly and should appear on your stick/strip regardless of the result. It does not tell you whether you're pregnant or not. If your test uses a control symbol and you don't see it on the stick/strip, the test didn't work, and you'll need to take another one.

Test your first morning urine. Morning urine usually has more hCG than urine later in the day.

Use a timer. If you guess the timing, your results may not be accurate.

Blood tests are done at your provider's office or a lab. A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This process usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you're doing a urine test, don't drink large amounts of fluid before collecting your sample. That could dilute the hCG in your urine, and it may not show up on the test.

Taking medicine to help you get pregnant (fertility drugs) could affect your pregnancy test results. If you are taking fertility drugs, talk to your provider if you get a positive result. Your provider may need to verify it.

Otherwise, you don't need any special preparations for a pregnancy test that uses a sample of your urine or blood.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a urine test.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will show whether you're pregnant.

A negative result means hCG wasn't found in your sample, so you may not be pregnant. However, this isn't always the case. If you did a home urine test too soon after getting pregnant, your body may not have made enough hCG to show up in the results. This is called a false-negative. A false negative result means your test shows that you aren't pregnant, but you really are.

Because hCG levels increase every day during early pregnancy, it's a good idea to repeat the test a week after you took it the first time. If you get negative (not pregnant) results on two home tests, but you still think you're pregnant, call your provider.

A positive result means that hCG was found in your sample. That usually means that you're pregnant. It's important to see your provider as soon as possible to make sure you get the right care. If you did a home test, your provider may do another test to confirm your pregnancy.

In some cases, a home test may say that you're pregnant when you aren't. This is called a false-positive. A false-positive result means your test shows that you are pregnant, but you really aren't. Taking a pregnancy test while also taking fertility medicines could lead to a false-positive test result. If your results are positive but you think you might not be pregnant, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Prenatal Cell-Free DNA Screening

<https://medlineplus.gov/lab-tests/prenatal-cell-free-dna-screening/>

PRENATAL CELL-FREE DNA SCREENING

WHAT IS A PRENATAL CELL-FREE DNA (CFDNA) SCREENING?

Prenatal cell-free DNA (cfDNA) screening is a blood test given during pregnancy. During pregnancy, some of the fetus's DNA circulates in the mother's bloodstream. A cfDNA screening checks this DNA to find out if the baby is more likely to have certain conditions caused by an abnormal number of chromosomes, such as Down syndrome.

Chromosomes are tiny "packages" in your cells that contain your genes. Genes carry information, called DNA, that controls what you look like and how your body works.

Normally, you're born with a set of 46 chromosomes in each cell. The chromosomes are grouped into 23 pairs. Each pair has two chromosomes, one coming from each parent.

If one of these pairs has an extra copy of a chromosome, it's called a trisomy. A trisomy causes changes in the way the body and brain develop.

If there is an extra copy of chromosome 21, it's called Down syndrome. This is also known as trisomy 21. Down syndrome is the most common chromosome disorder in the United States.

Other trisomy disorders include Edwards syndrome (trisomy 18), where there is an extra copy of chromosome 18, and Patau syndrome (trisomy 13), where there is an extra copy of chromosome 13. These disorders are rare but more serious than Down syndrome. Most babies with trisomy 18 or trisomy 13 die within the first year of life.

There is very little risk to you and your baby with a cfDNA screening. But a cfDNA screening can't tell you for sure whether your baby has a chromosome disorder. Your health care provider will need to order other tests to confirm or rule out a diagnosis.

Other names: cell-free fetal DNA, cffDNA, non-invasive prenatal tet, NIPT, non-invasive prenatal screening, NIPS

WHAT IS USED FOR?

A cfDNA screening is most often used to show if your unborn baby has an increased risk for one of the following chromosome disorders:

Down syndrome (trisomy 21)

Edwards syndrome (trisomy 18)

Patau syndrome (trisomy 13)

The screening may also be used to:

Determine a baby's sex. This may be done if an ultrasound shows that a baby's genitals are not clearly male or female. This may be caused by a disorder of the sex chromosomes.

Check Rh blood type. Rh is a protein found in red blood cells. If you have the protein, you are considered Rh positive. If you don't, you are Rh negative. If you are Rh negative and your unborn baby is Rh positive, your body's immune system may attack your baby's blood cells. If you find out you are Rh negative early in pregnancy, you can take medicines to protect your baby from dangerous complications.

A cfDNA screening can be done as early as the 10th week of pregnancy.

WHY DO I NEED A PRENATAL CFDNA SCREENING?

Many providers recommend this screening if you are pregnant and are at a higher risk of having a baby with a chromosome disorder. You may be at higher risk if:

You are age 35 or older. The mother's age is the main risk factor for having a baby with Down syndrome or other trisomy disorders. The risk increases as you get older.

You've had another baby with a chromosome disorder.

Your fetal ultrasound didn't look normal.

Other prenatal test results were not normal.

Many providers recommend screening if you are pregnant even if you aren't at higher risk of having a baby with a chromosome disorder. This is because the screening has almost no risk and has a high rate of accuracy.

You and your provider should discuss if a cfDNA screening is right for you.

WHAT HAPPENS DURING A PRENATAL CFDNA SCREENING?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You may want to speak to a genetic counselor before you get tested. A genetic counselor is a specially trained professional in genetics and genetic testing. He or she can explain the possible results and what they might mean to you and your baby.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to your unborn baby and very little risk to you. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results were negative, it's unlikely that your baby has Down syndrome or another trisomy disorder.

If your results were positive, it means there is an increased risk that your baby has one of these disorders. But it can't tell you for sure if your baby is affected. To confirm a diagnosis, you'll need other tests, such as amniocentesis and chorionic villus sampling (CVS):

Amniocentesis. This test takes a sample of the amniotic fluid that surrounds the unborn baby in the uterus. The fluid contains cells from the baby that can be tested. The test is usually done between the weeks 15 and 20 of pregnancy.

Chorionic villus sampling (CVS). A CVS collects tissue from parts of the placenta called the chorionic villi. The placenta is the organ that grows in the uterus to provide nutrients and oxygen to the baby. Cells from the chorionic villi usually have the same chromosomes as the unborn baby. A CVS may be done between weeks 10 and 13 of pregnancy.

These tests are usually very safe procedures, but they do have a slight risk of causing a miscarriage.

If you have questions about your results, talk to your provider and/or a genetic counselor.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PRENATAL CFDNA SCREENING?

cfDNA screenings are not as accurate if you are pregnant with more than one baby (twins, triplets, or more).

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Prenatal Panel

<https://medlineplus.gov/lab-tests/prenatal-panel/>

PRENATAL PANEL

WHAT IS A PRENATAL PANEL?

A prenatal panel is a group of blood tests done in early pregnancy. The tests check for diseases and infections that can affect the health of you and your developing baby. The results of the tests can guide treatments, which may help prevent serious complications. A prenatal panel usually includes the following tests:

Complete blood count (CBC). This test measures many different parts of your blood, including red blood cells, white blood cells, and platelets. It also measures the physical features of some of the cells. A CBC can help diagnose various health conditions, such as anemia, clotting disorders, and infections.

Blood type and Rh factor. This test finds out your blood type (A, B, AB, or O) and checks your Rh factor. Rh factor is a type of protein found on the surface of red blood cells. If your blood cells have this protein, you are Rh-positive. If they don't, you are Rh-negative. If you are Rh-negative and your developing baby is Rh-positive, it's called Rh incompatibility. It may cause your body to make antibodies against your baby's blood, which can damage your baby's red blood cells.

Rubella, also known as German measles, is a viral infection. This test shows if you are immune to (protected against) rubella. You would be immune if you have been vaccinated against rubella or infected with it in the past. If you are not immune and you get the infection during pregnancy, it could put your baby at risk for congenital anomalies (health conditions that can develop before birth).

Hepatitis B and C are viruses that infect your liver. If you're pregnant, the hepatitis virus can be passed to your baby. Hepatitis B is a test usually given during pregnancy. Hepatitis C is not routinely tested because it is not common. But you may be tested if you have certain risk factors.

Sexually transmitted infections (STIs). Usually, you will be tested for chlamydia and syphilis early in your pregnancy. You may also be tested for gonorrhea if you have certain risk factors. An STI can lead to miscarriage or infect your baby during delivery. An STI can be dangerous to a newborn. It may cause blindness, breathing problems, or other health issues.

HIV is a virus that attacks your immune system. If you have HIV, you may pass along the virus to your baby. If you are pregnant, you may be tested for HIV, and some states require testing.

Other names: obstetric panel, OB panel

WHAT IS IT USED FOR?

A prenatal panel is used to find health problems early in your pregnancy. Many conditions can be treated during pregnancy to avoid complications. The test may also guide treatments for your baby immediately after birth.

WHY DO I NEED A PRENATAL PANEL?

Your health care provider will probably order these tests as part of a routine prenatal visit. The tests are done in the first trimester of pregnancy, often at your first prenatal visit.

WHAT HAPPENS DURING A PRENATAL PANEL?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THESE TESTS?

You don't need any special preparations for a prenatal panel.

ARE THERE ANY RISKS TO THE TESTS?

There is no risk to your developing baby and very little risk to you. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results were not normal, you and/or your baby might get treatment to prevent serious health conditions. Examples of some abnormal results and treatments can include:

Rh incompatibility. You will receive medicine that prevents your body from making the antibodies that may damage your baby's red blood cells.

Infections (hepatitis, STIs, HIV). You will receive medicine to treat the infection. If you have a hepatitis B infection, your baby will get a vaccine within a few hours of birth.

No immunity to rubella. You'll need to avoid anyone who has rubella while you are pregnant. After your baby is born, you should get vaccinated.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PRENATAL PANEL?

In addition to a prenatal panel, you may also get a urine test during your first trimester. Pregnancy urine tests are used to:

Diagnose a urinary tract infection.

Check glucose levels. High levels of glucose in your urine may be a sign of gestational diabetes, a type of diabetes that can develop during pregnancy. If your urine glucose was high, your provider may order a blood glucose test to confirm the diagnosis.

Check protein levels. High levels of protein in your urine can be a sign of several health conditions, including kidney disease, infection, or stress. If your urine protein level was high, your provider will probably order more tests.

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Procalcitonin Test

<https://medlineplus.gov/lab-tests/procalcitonin-test/>

PROCALCITONIN TEST

WHAT IS A PROCALCITONIN TEST?

A procalcitonin test measures the level of procalcitonin in your blood. Normally, you have very low levels of procalcitonin in your blood. But if you have a serious bacterial infection, the cells in many parts of your body will release procalcitonin into your bloodstream. A high level of procalcitonin in your blood may be a sign of a serious infection or sepsis (an overactive and extreme response to infection).

Sepsis (also called septicemia) is your immune system's extreme response to an infection, usually from bacteria. Sepsis happens when an infection you already have spreads into your bloodstream and triggers a chain reaction throughout your body. It causes inflammation (swelling) and blood clots. Without quick treatment, sepsis can rapidly lead to tissue damage, organ failure, or even death.

Infections that lead to sepsis most often start in your lungs, urinary tract, skin, or digestive system. A procalcitonin test can help your health care provider diagnose if you have sepsis from a bacterial infection or if you have a high risk of developing sepsis. This may help you get the right treatment quickly before your condition worsens.

Other names: PCT test

WHAT IS IT USED FOR?

A procalcitonin test is mostly used if you are seriously ill and your provider thinks you may have a systemic infection (an infection that affects your entire body). The test helps find out whether bacteria or a virus is causing your infection. For example, a procalcitonin test can help tell the difference between bacterial and viral pneumonia. This matters because antibiotics help most bacterial infections, but not viral infections.

A procalcitonin test may be used to help:

- Diagnose or rule out a bacterial infection and/or sepsis

- Find out how serious a sepsis infection may be

- Make treatment decisions (such as the need for antibiotics)

- Monitor how well treatment is working

- Diagnose kidney infections in children with urinary tract infections

This test is usually used for very sick people admitted to the hospital.

WHY DO I NEED A PROCALCITONIN TEST?

You may need this test if you have symptoms of sepsis or a serious bacterial infection that could become sepsis.

Symptoms of sepsis include:

- Fever or chills

- Clammy or sweaty skin

- Confusion

- Extreme pain

- Rapid heartbeat

- Shortness of breath, feeling like you can't get enough air

- Low blood pressure

- Reduced amounts of urine (pee)

Sepsis is a medical emergency. If you or your loved one has an infection that's not getting better or is getting worse, get medical help right away.

WHAT HAPPENS DURING A PROCALCITONIN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparation for a procalcitonin test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

High procalcitonin levels mean that you:

- Most likely have sepsis

- May have a high risk of developing severe sepsis and septic shock, a life-threatening condition when your organs don't get enough blood to work properly

- May have a serious systemic bacterial infection that increases your risk for sepsis

- The higher your procalcitonin levels, the higher your risk for sepsis and septic shock.

Moderate to mildly high levels of procalcitonin may be sign of:

- The earlier stages of a systemic bacterial infection

- Kidney infection, in children only

- Conditions other than infections, such as tissue damage from trauma, serious burns, recent surgery, severe heart attack

Slightly high levels of procalcitonin mean that you are unlikely to develop sepsis, but they may be a sign of:

- A local bacterial infection, such as a urinary tract infection

- An infection from another cause, such as a virus

- A systemic bacterial infection that is just beginning

If you are being treated for a bacterial infection, decreasing or low procalcitonin levels mean that your treatment is working.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PROCALCITONIN TEST?

A procalcitonin test won't show what type of bacteria is causing an infection. To make a full diagnosis, your provider will likely order other tests. But a procalcitonin test does give you important information about your risk for sepsis, so you can start treatment sooner, if needed, and avoid more serious illness.

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Progesterone Test

<https://medlineplus.gov/lab-tests/progesterone-test/>

PROGESTERONE TEST

WHAT IS A PROGESTERONE TEST?

A progesterone test measures the level of progesterone in a sample of your blood. Progesterone is a hormone that's made mainly by the ovaries, which are two glands in the female reproductive system that contain eggs.

Each month, progesterone prepares your uterus for pregnancy. During the first half of a normal menstrual cycle, estrogen makes the lining of your uterus grow thicker. In the second half, an ovary releases an egg and your progesterone levels begin to rise. Progesterone gets the lining of your uterus ready for a fertilized egg to attach (implant) and grow into a fetus.

If you don't become pregnant, your progesterone levels will fall. The lining of your uterus will thin, and your uterus will get rid of the extra blood and tissue when your menstrual period begins.

If you become pregnant, progesterone levels will continue to rise to support the pregnancy. High levels of progesterone:

Cause cells in the lining of your uterus to produce proteins that are necessary for the attachment of a fertilized egg.

Cause the creation of white blood cells that will make sure your immune system doesn't attack the fertilized egg.

Prevent the uterus from contracting (squeezing) and causing pre-term labor.

Play a role in the development of the placenta, an organ that grows in the uterus to provide nutrients and oxygen to the fetus.

Once the placenta is fully formed, it will make much of the progesterone you need for a healthy pregnancy.

Other names: serum progesterone, progesterone blood test, PGSN

WHAT IS IT USED FOR?

A progesterone test can help show whether low progesterone levels are causing female infertility (problems getting pregnant after a year of trying) or problems during pregnancy.

A progesterone test may also be used to help:

Check to see if fertility treatments are working.

Find out your risk of a miscarriage (loss of a pregnancy before 20 weeks).

Diagnose an ectopic pregnancy, which is a fertilized egg that tries to grow outside of the uterus. The egg cannot grow into a fetus when it's in the wrong place. It must be removed to avoid damage to your organs. This can be a medical emergency.

Check the health of you and your fetus during a high-risk pregnancy.

Diagnose ovarian cancer or problems with your adrenal glands. Normally, your adrenal glands make small amounts of progesterone. High levels of progesterone may be a sign of an adrenal gland disorder in both females and males.

Diagnose the cause of abnormal uterine bleeding.

WHY DO I NEED A PROGESTERONE TEST?

You may need a progesterone test if:

You are having trouble getting pregnant. A progesterone test can help your health care provider see if your ovaries are releasing eggs (ovulating) normally.

You are having fertility treatments.

You are having abnormal bleeding when you're not pregnant.

You have signs of low progesterone such as irregular menstrual periods, hot flashes, and trouble sleeping.

You have signs of high progesterone such as vaginal dryness and bloating.

It's common to have your progesterone levels tested as part of a panel of tests you may get if you're pregnant. You may also need a progesterone test while pregnant if:

You're having progesterone therapy to prevent a miscarriage.

Your provider wants to check the health of your pregnancy because you have a higher risk for miscarriage or other pregnancy complications.

You have signs of a miscarriage or ectopic pregnancy, such as cramps or bleeding.

WHAT HAPPENS DURING A PROGESTERONE TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You usually don't need any other preparations for a progesterone test. In certain cases, you may need to know the first day of your last menstrual period. Your provider will tell you if you need to stop taking any medicines before the test. But never stop taking medicine unless your provider tells you.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The meaning of your test results depends on:

Why you were tested.

Where you were in your menstrual cycle when your blood sample was taken.

Whether or not you're pregnant.

High progesterone levels:

When you're not pregnant may be linked to:

A cyst on your ovaries.

Ovarian cancer.

Adrenal gland problems, including adrenal gland cancer, or an inherited disorder called congenital adrenal hyperplasia (CAH).

A cyst on your ovaries.

Ovarian cancer.

Adrenal gland problems, including adrenal gland cancer, or an inherited disorder called congenital adrenal hyperplasia (CAH).

Higher than expected progesterone levels when you are pregnant, may mean you're having two or more babies.

High progesterone levels may also be a sign of a molar pregnancy, which is an abnormal growth of tissue in the uterus. It's caused by a fertilized egg with such severe genetic problems that it cannot become a fetus. The growth can turn into cancer and must be removed.

Low progesterone levels:

When you're not pregnant, may be linked to:

Not ovulating normally.

Not having a menstrual period (amenorrhea).

Polycystic ovary syndrome (PCOS), a condition that causes missed periods and unpredictable ovulation.

High stress. Too much of the stress hormone cortisol can affect your body's ability to make progesterone.

Perimenopause. This is the time just before menopause.

Low cholesterol.

Not ovulating normally.

Not having a menstrual period (amenorrhea).

Polycystic ovary syndrome (PCOS), a condition that causes missed periods and unpredictable ovulation.

High stress. Too much of the stress hormone cortisol can affect your body's ability to make progesterone.

Perimenopause. This is the time just before menopause.

Low cholesterol.

When you're pregnant, may be linked to:

Ectopic pregnancy.

Miscarriage or a high risk of miscarriage.

Pre-eclampsia.

Ectopic pregnancy.

Miscarriage or a high risk of miscarriage.

Pre-eclampsia.

Your provider can explain what your test results say about your health.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PROGESTERONE TEST?

Because progesterone levels change throughout your pregnancy and menstrual cycle, you may need to be tested several times.

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Prolactin Levels

<https://medlineplus.gov/lab-tests/prolactin-levels/>

PROLACTIN LEVELS

WHAT IS A PROLACTIN TEST?

A prolactin (PRL) test measures the level of prolactin in a sample of your blood. Prolactin is a hormone produced by the pituitary gland, a small gland at the base of the brain. Hormones are chemical messengers in your bloodstream that control the actions of certain cells or organs.

Prolactin affects many body processes. But its main role is to signal breast tissue to grow during pregnancy and make milk for breastfeeding after birth. So, prolactin levels in pregnant women and new mothers are normally high.

In nonpregnant women and men, prolactin levels are normally low.

Having higher-than-normal prolactin levels (hyperprolactinemia) can affect the body in different ways:

In women, it can affect how the ovaries work and can cause menstrual problems and infertility.

In men, it can affect how the testicles work and can cause a lower sex drive and erectile dysfunction (ED). Also known as impotence, ED is the inability to get or maintain an erection.

In men and women who are not pregnant or breastfeeding, it can signal your body to start making breastmilk.

Higher-than-normal prolactin levels can have different causes:

Prolactinoma, a benign (noncancerous) tumor growing on the pituitary gland. This is the most common cause. This type of tumor makes prolactin, which leads to higher-than-normal levels of prolactin in the blood.

Certain medicines.

Certain health conditions, such as hypothyroidism (underactive thyroid), kidney disease, and chest injuries

Other pituitary gland tumors.

A prolactin test can't show the cause of higher-than-normal prolactin levels. But finding out what your levels are can help your health care provider decide which other tests you need to diagnose the cause of your symptoms.

Other names: PRL test, prolactin blood test

WHAT IS IT USED FOR?

A prolactin test is most often used with other tests to help:

Diagnose a prolactinoma

Find the cause of irregular menstrual periods

Find the cause of infertility in men and women

Find the cause of a man's low sex drive and/or erectile dysfunction

Monitor a known prolactinoma to see if:

Treatment is working.

The tumor has come back after treatment.

Treatment is working.

The tumor has come back after treatment.

WHY DO I NEED A PROLACTIN TEST?

You may need a prolactin levels test if you are having symptoms that could be caused by higher-than-normal prolactin levels. The symptoms are different for women and men.

In women, symptoms may include:

Changes in menstruation (irregular periods or absence of periods)

Production of breast milk when not pregnant or breastfeeding

Nipple discharge

Infertility

Symptoms also depend on whether you have gone through menopause, which is when your periods stop, normally around age 50.

In men, symptoms may include:

Nipple discharge

Erectile dysfunction/impotence

Enlarged breasts (gynecomastia)

Decreased facial and body hair

Low sex drive

If you have higher-than-normal levels of prolactin because of a prolactinoma, you may also have other symptoms. A prolactinoma that is not treated can grow large enough that it presses against nearby parts of the pituitary gland and the brain. This can cause:

Headaches

Changes in vision, if the tumor is pressing on the optic nerve (the nerve at the back of the eye)

Low levels of other pituitary hormones, such as thyroid hormones and cortisol

If you've already been diagnosed with high prolactin levels from a prolactinoma, you may need prolactin testing to check whether your treatment is working. After treatment, you may be retested to see if the tumor has come back.

WHAT HAPPENS DURING A PROLACTIN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know if you need to do anything to prepare for your test. You may be asked to fast (not eat or drink) for several hours before your test.

Prolactin levels change throughout the day. For accuracy, your blood sample will usually be scheduled for three to four hours after you wake up.

Certain medicines and vitamins can affect prolactin levels. So, be sure to tell your provider about anything you are taking. But never stop taking any medicine without talking with your provider first.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand what the results of your prolactin levels test mean, your provider will consider your symptoms, sex, and medical history, and the results of any other tests you had.

In general, for men and women who aren't pregnant or breastfeeding, higher-than-normal prolactin levels may be a sign of many types of conditions, such as:

Prolactinoma or another pituitary disorder.

A disease of the hypothalamus. The hypothalamus is an area of the brain that controls the pituitary gland and other body functions.

Hypothyroidism.

Cirrhosis (scarring of the liver).

Kidney disease.

Polycystic ovary syndrome (PCOS) in women.

If your results show high prolactin levels, your provider will usually order other blood tests to help diagnose the cause. If your provider thinks you may have a prolactinoma, you may also have an MRI (magnetic resonance imaging) scan to check your pituitary gland for tumors.

Treatment for high prolactin levels depends on the cause. Treatment is usually medicine. Surgery may be needed for certain pituitary tumors.

Higher-than-normal prolactin levels don't always mean you have a medical condition that needs treatment. Diets high in protein and certain medicines can cause high levels. Small increases in prolactin may be caused by stress, exercise, and even having sex. So, depending on your symptoms and medical history, your provider may suggest testing you again to see if your prolactin levels remain high.

Lower-than-normal prolactin levels are rare but may be a sign of a pituitary disorder. Without enough prolactin, a woman may not make enough milk to breastfeed after birth. This is usually the only symptom of abnormally low prolactin levels.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Prostate-Specific Antigen (PSA) Test

<https://medlineplus.gov/lab-tests/prostate-specific-antigen-psa-test/>

PROSTATE-SPECIFIC ANTIGEN (PSA) TEST

WHAT IS A PROSTATE-SPECIFIC ANTIGEN (PSA) TEST?

A prostate-specific antigen (PSA) test is a blood test that measures the level of PSA in a sample of your blood. PSA is a protein made by your prostate. The prostate is a gland in the male reproductive system. It lies just below the bladder. It makes the fluid part of semen.

It's normal to have a low level of PSA in your blood. A high PSA level may be caused by:

Prostate cancer

An enlarged prostate (also called benign prostatic hyperplasia or BPH)

Prostatitis, inflammation (swelling and pain) in the prostate gland

Other common prostate problems

Urinary tract infections (UTIs)

Taking certain medicines

A PSA test can't show what is causing abnormal PSA levels. So, if your level is high, you may need other tests.

Other names: total PSA, free PSA

WHAT IS IT USED FOR?

A PSA test is used to screen for prostate cancer. Cancer screening means looking for signs of cancer before it causes symptoms. But screening tests can't diagnose cancer. If a screening test finds signs of cancer, you'll need other tests to find out if you have cancer and how serious it may be.

Most types of prostate cancer grow very slowly. They don't spread beyond the prostate and may never cause health problems. In fact, you can live a long life with prostate cancer and never know you have it. The goal of prostate cancer screening is to help find cancers that may be more likely to spread so they can be treated early. But there are challenges and possible harms from using a PSA test to screen for prostate cancer:

A PSA test can't tell the difference between abnormal PSA levels from prostate cancer and noncancerous conditions. If your PSA level is high, a prostate biopsy is the only way to find out if the cause is cancer. And prostate biopsies have possible harms, such as bleeding at the biopsy site, blood in your semen or urine, infection, or difficulty urinating.

A PSA test may lead to finding and treating prostate cancer that would never have affected your health. If prostate cancer is found:

It can be difficult to tell the difference between slow-growing cancers and those that are likely to grow faster and spread in your body.

You could have prostate cancer treatment that you never really needed. And cancer treatment may cause serious harms, such as:

Erectile dysfunction, when a man has trouble getting or keeping an erection

Urinary incontinence, uncontrolled urination causing you to leak pee

Bowel incontinence, problems controlling your bowels (poop)

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Erectile dysfunction, when a man has trouble getting or keeping an erection

Urinary incontinence, uncontrolled urination causing you to leak pee

Bowel incontinence, problems controlling your bowels (poop)

To decide whether a PSA test to screen for prostate cancer is right for you, talk with your health care provider about:

Your risk for developing a serious type of prostate cancer. If your risk is high, the possible benefits of finding cancer early may outweigh the possible harms.

Your general health. Are you well enough to have treatment for prostate cancer if it's found?

Your preferences. How do you feel about the possible benefits and harms of screening, diagnosis, and treatment?

PSA testing may also be used to:

Help diagnose the cause of prostate conditions that aren't cancer

Monitor treatment for a prostate condition, including cancer

WHY DO I NEED A PSA TEST?

It's your choice whether to have a PSA test to screen for cancer. You and your provider may consider your risk for developing a serious cancer that could spread if you don't catch it early. Your risk for serious prostate cancer may be higher depending on your:

Age. The risk of prostate cancer increases after age 50.

Your family health history. If members of your family have had prostate cancer, your risk may be higher.

Your race. Prostate cancer is more common in Black men, who also have a higher risk of developing prostate cancer at a younger age and having more serious disease.

You may also have a PSA test if:

You have symptoms of a prostate condition, such as:

Painful, slow, and/or frequent urination (peeing)

Blood in urine or semen

Pelvic and/or back pain

Urinary incontinence

Painful, slow, and/or frequent urination (peeing)

Blood in urine or semen

Pelvic and/or back pain

Urinary incontinence

You have prostate cancer. Your provider may use PSA testing to monitor your condition or to see how well treatment is working.

WHAT HAPPENS DURING A PSA TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You will need to avoid having sex or masturbating for 24 hours before your PSA test. That's because releasing semen can increase your PSA levels, which may make your results less accurate. Also, certain medicines may affect your test results, so tell your provider about any medicines you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

There is no specific normal or abnormal level for PSA in blood. In general, the higher your PSA level, the more likely it is that you have cancer. But it's possible to have a high PSA without prostate cancer, or a low PSA with prostate cancer.

If you had a PSA test for a prostate cancer screening or because you have prostate symptoms:

High PSA levels can mean you have prostate cancer or a prostate condition that's not cancer, such as an infection (causing prostatitis) or an enlarged prostate. If your PSA levels are higher than normal, your provider may talk with you about having more tests to diagnose the cause. These tests may include:

Another PSA test, more commonly if you don't have any symptoms. PSA levels can go up and down, so it's helpful to see if your PSA levels change over time.

A digital rectal exam (DRE). For this test, your provider inserts a gloved, lubricated finger into your rectum to feel your prostate for lumps or anything unusual.

A urine test. A sample of your urine is tested for infection.

A prostate biopsy. A biopsy is minor surgery. A provider removes samples of tissue from your prostate so it can be studied under a microscope to look for cancer cells. A biopsy is the only way to diagnose cancer. It may be recommended if your provider thinks you may have prostate cancer.

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A prostate biopsy. A biopsy is minor surgery. A provider removes samples of tissue from your prostate so it can be studied under a microscope to look for cancer cells. A biopsy is the only way to diagnose cancer. It may be recommended if your provider thinks you may have prostate cancer.

If you had a PSA test to monitor prostate cancer or treatment, ask your provider what a high PSA level means. Your provider will usually look at several tests results over time to get a better understanding of your condition.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Protein C and Protein S Tests

<https://medlineplus.gov/lab-tests/protein-c-and-protein-s-tests/>

PROTEIN C AND PROTEIN S TESTS

WHAT ARE PROTEIN C AND PROTEIN S TESTS?

Protein C and protein S tests use a sample of your blood to measure how many of these proteins you have and how well they're working. Though they are often done together, these are separate tests. A protein C test measures protein C in your blood; a protein S test measures protein S.

Normally, proteins C and S work together to prevent your blood from clotting too much. Tests of protein C and S use a sample of your blood to measure how much of these proteins you have and how well they're working.

However, if you have a protein C or S deficiency (when your body doesn't make enough of these proteins), your blood may clot too much. It may also clot even when you don't have an injury. These deficiencies can cause:

Deep vein thrombosis (DVT), a condition where blood clots form in the veins in your legs or pelvis.

Pulmonary embolism, a life-threatening condition where blood clots break loose and travel to your lungs.

Most protein C and protein S deficiencies are acquired. That means that the deficiency is caused by another health condition or certain medicines. Acquired protein C and S deficiencies may develop with:

Infections.

Liver or kidney disease.

A lack of vitamin K.

Cancer and chemotherapy.

Certain medicines, including birth control pills and warfarin, a blood-thinner.

Pregnancy.

Disseminated intravascular coagulopathy (DIC), a rare condition that causes blood clotting throughout the body, followed by bleeding.

In rare cases, protein C or protein S deficiencies are inherited. That means that your condition is caused by a change in a gene that one or both of your parents passed down to you.

There are separate tests for protein C and S. But they are often done at the same time.

Other names: protein C antigen, protein S antigen, PC, PS, free PS, autoproteins IIA

WHAT ARE THEY USED FOR?

A health care provider may use protein C and protein S tests to:

Find the cause of a blood clot that can't be explained and check your risk for developing more blood clots in the future.

Check for a protein C or S deficiency.

Determine the type of protein C or S deficiency you may have.

Check your protein C or S levels to see if they've gotten better or worse. This is done if you have had a change in the health condition that is causing a protein C or S deficiency.

To determine whether you have a protein C or S deficiency, your provider will measure these proteins' activity. These are called protein C and protein S functional tests, and they can show whether one or both of these proteins are the cause of any clotting problems.

To determine the type of protein C or protein S deficiency, your provider will measure the quantity of these two proteins. If the functional tests show that these proteins aren't working as they should, your provider can see how many of them you have in your blood. If your body isn't making enough protein C and/or S, they might not be able to stop your blood from clotting.

There are several types of protein C and S deficiency, but generally they can be broken down into two categories:

Those that are caused by a low quantity of protein C or S. In other words, your body doesn't make enough of these proteins.

Those that are caused by dysfunctional proteins C or S. In other words, your body makes enough of these two proteins, but they don't work as they should.

Knowing the deficiency type can help your provider make important treatment decisions.

WHY DO I NEED PROTEIN C AND PROTEIN S TESTS?

Your provider may need to test protein C and S if you've had:

A blood clot before the age of 50 without a known reason.

A blood clot in an unusual location, such as in the veins of your kidneys, liver, or brain.

Several blood clots.

Treatment for a clotting disorder, such as a protein C and/or S deficiency. Your provider may want to use the tests to see if your treatment is working.

In certain cases, you may need testing if you have a family member who has a serious inherited protein C or S deficiency or who had a clot at a young age. Testing may also be needed for a newborn with a clotting disorder.

You may also need protein C and protein S tests if you have a condition or are taking certain medicines that can cause a protein C or S deficiency. For example, you may need these tests if you have liver or kidney disease or if you have had severe infections.

WHAT HAPPENS DURING PROTEIN C AND PROTEIN S TESTING?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you are having protein C and S tests because you had a blood clot, you must wait until you have fully recovered to have this test. If you have the test too soon, the results will not be accurate. Your provider will let you know when you're ready to be tested.

Your provider may tell you not to take blood thinners (medicine that helps prevent blood clots) for several days or longer before your test. But never stop any prescription medicine without talking with your provider first.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Higher than normal levels of protein C or S are not known to cause any health problems.

Lower than normal levels of protein C or S, or proteins that aren't working well, mean that you have a higher risk of developing a blood clot. Your level of risk depends on how abnormal your test results are.

If your provider thinks your protein C or S deficiency is inherited, you may need a genetic test to find out for sure. Having an inherited deficiency doesn't mean that you'll develop a blood clot. But it does mean that your risk for developing a clot will be increased for the rest of your life.

If your protein C or S deficiency is acquired, the deficiency may be temporary or long-lasting. In certain cases, acquired deficiencies may get worse over time.

With both inherited and acquired deficiencies, treatment may help reduce your risk of developing blood clots. Your provider may recommend treatment depending on your test results and health history. You may be able to lower your risk of blood clots by:

Taking medicines (blood thinners, also called "anticoagulants") to help prevent too much clotting.

Treating the condition that caused an acquired deficiency.

Avoiding other things that can increase your risk of blood clots, including smoking and birth control pills.

If you have questions about your results or how to manage a protein C or S deficiency, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT PROTEIN C AND PROTEIN S TESTS?

If you have been diagnosed with a protein C or S deficiency, it's important to talk with your provider whenever you face situations that can trigger blood clots, such as:

Surgery

Pregnancy

Broken bones or other trauma injuries

Not moving for long periods of time due to bedrest or long car rides and/or airplane trips.

Your provider can help you reduce your risk of getting a clot during those higher-risk times. They can:

Make suggestions for traveling such as walking around on the plane or stopping often to stretch your legs if you're driving.

Prescribe medicines that may reduce your risk.

Teach you how to recognize certain warning signs such as sudden swelling or pain in your leg, chest pain, or shortness of breath.

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RELATED HEALTH TOPICS

Blood Thinners

Deep Vein Thrombosis

Genetic Disorders

Pulmonary Embolism

Vascular Diseases

RELATED MEDICAL TESTS

Coagulation Factor Tests

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Partial Thromboplastin Time (PTT) Test

Platelet Tests

Prothrombin Time Test and INR (PT/INR)

V/Q Scan

What You Need to Know About Blood Testing

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Protein Electrophoresis by Immunofixation Blood Test

<https://medlineplus.gov/lab-tests/protein-electrophoresis-by-immunofixation-blood-test/>

PROTEIN ELECTROPHORESIS BY IMMUNOFIXATION BLOOD TEST

WHAT IS AN IMMUNOFIXATION BLOOD TEST?

An immunofixation (IFE) blood test measures the number and type of proteins in your blood. Proteins have many important functions, including giving your body energy, rebuilding muscles, and helping your immune system. There are different types of proteins, and each one has a different job for your body. This test checks to see if your body is making or losing too much of certain proteins. Your body might make too much or too little protein if you have certain health conditions.

There are two main types of proteins in the blood: albumin and globulin. Albumin makes up most of the protein in the blood, while the rest are called globulin.

The immunofixation blood test uses a process called electrophoresis to separate the proteins into subgroups. The subgroups are based on their size, shape, and electrical charge. Measuring the number of proteins in each subgroup can help check for different health conditions.

The protein subgroups are:

Albumin

Alpha-1 globulin

Alpha-2 globulin

Beta globulin

Gamma globulin

Other names: serum protein electrophoresis (SPEP), protein electrophoresis, SPE, immunofixation electrophoresis, IFE, serum immunofixation

WHAT IS IT USED FOR?

This test is most often used to help check for or monitor many different conditions that affect the amount of protein in the body. Some of these health conditions might be hard to find without having an immunofixation blood test.

These conditions include:

Multiple myeloma, a cancer of plasma cells, which are a type of white blood cell. This is the most common use for an immunofixation test.

Other forms of cancer, such as lymphoma (cancer of the lymph system, which is part of your immune system) or leukemia (cancers of the blood cells).

Kidney disease.

Liver disease.

Certain autoimmune and inflammatory diseases, which cause your immune system to attack healthy tissues. They include rheumatoid arthritis and Crohn's disease.

Certain neurologic diseases, including multiple sclerosis (MS).

Malnutrition or malabsorption, conditions in which your body is not getting enough nutrients from the foods you eat.

Monoclonal gammopathy of unknown significance (MGUS), a condition that causes high levels of certain proteins for no reason.

WHY DO I NEED AN IMMUNOFIXATION BLOOD TEST?

You may need testing if you have symptoms of health conditions that affect the amount of protein in the body, such as multiple myeloma, multiple sclerosis, malnutrition, or malabsorption. Your health care provider may also order an immunofixation blood test if you have other blood tests that are not normal, such as your albumin level, total protein, high calcium levels, or low white or red blood cell counts.

Symptoms of multiple myeloma may include:
Weakened bones, which can lead to

Bone pain, especially in the back or ribs

Bone lesions

Easily broken bones (fractures)

Bone pain, especially in the back or ribs

Bone lesions

Easily broken bones (fractures)

Fatigue.

Anemia (low level of red blood cells).

Fever for no known reason or frequent infections.

Easy bruising or bleeding.

Trouble breathing.

High levels of calcium in the blood (hypercalcemia). It can affect many organs in the body. The symptoms it causes may depend on which organs are affected.

Symptoms of multiple sclerosis may include:

Numbness, tingling, or pain in the face, arms, and/or legs

Vision problems

Muscle weakness, stiffness, and spasms

Trouble walking

Fatigue

Dizziness and vertigo

Bladder control problems

Symptoms of malnutrition or malabsorption may include:

Weight loss, especially the loss of fat and muscle

Weakness

Fatigue

Diarrhea

Frequent infections

WHAT HAPPENS DURING AN IMMUNOFIXATION BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an immunofixation blood test.

ARE THERE ANY RISKS TO AN IMMUNOFIXATION BLOOD TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will show that your protein levels are in the normal range, or if certain proteins were too high or too low.

Many conditions can cause certain protein levels to be high or low. Common causes may include:

An immunofixation test can also be used to monitor a known condition that affects your protein levels to see if the condition is getting worse or if treatment is helping.

To understand the results of an immunofixation blood test, your provider will consider your symptoms, age, medical history, and the results of other tests.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN IMMUNOFIXATION BLOOD TEST?

Other things can affect your results, such as some vaccines (if you got them in the past six months) and certain medicines.

Immunofixation tests can also be done in urine. Urine immunofixation tests are often done if immunofixation blood test results were not normal.

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Alpha-1 Antitrypsin Deficiency

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How to Understand Your Lab Results

What You Need to Know About Blood Testing

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Protein in Urine

<https://medlineplus.gov/lab-tests/protein-in-urine/>

PROTEIN IN URINE

WHAT IS A PROTEIN IN URINE TEST?

A protein in urine test measures how much protein is in your urine (pee). Normally, you have very little protein in your urine. A large amount of protein in urine (proteinuria) may mean you have a kidney problem. Your kidneys are organs that filter extra water and waste out of your blood to make urine.

Proteins are large molecules that are essential for your body to work properly. They have many important functions, including giving your body energy, rebuilding muscles, and helping your immune system. Protein is found in all parts of your body, including your blood. When your kidneys clean waste from your blood, tiny filters prevent the large protein molecules from leaving your body through your urine.

If there is a problem with your kidneys, protein can leak into your urine. High protein levels in your urine over a period of time may be the first sign that kidney disease or another condition has damaged the filters in your kidneys. A protein in urine test can help you find kidney damage early. This may help your health care provider determine the best treatment. You can also start making changes to protect your kidneys.

Other names: urine protein, 24-hour urine protein; urine total protein; ratio; reagent strip urinalysis

WHAT IS IT USED FOR?

A protein in urine test is often part of a urinalysis, which may include a visual check of your urine (pee) sample, tests for certain chemicals, and an examination under a microscope to look for certain types of cells. A urinalysis is used to check your general health, including the health of your urinary tract (which includes your kidneys, ureters, bladder, and urethra). This test may also be used to diagnose or monitor kidney disease and other health problems.

WHY DO I NEED A PROTEIN IN URINE TEST?

Your provider may order a protein in urine test as part of your routine checkup, to monitor a condition affecting your kidneys, or if you have symptoms of kidney disease. With some types of kidney disease, such as chronic kidney disease, you may not have symptoms until the later stages of the disease. These symptoms may include:

Swelling in the hands and feet or puffy eyelids

Dry skin, itching, or numbness

Fatigue

Increased or decreased urination

Urine that is bloody or foamy

Loss of appetite and weight loss

Muscle cramps

Nausea and vomiting

Shortness of breath

Sleep problems

Trouble thinking clearly

Protein in urine may be an early sign of kidney disease before you have symptoms. So your provider may order a protein in urine test if you have a high risk for kidney disease. You're more likely to develop kidney disease if you have:

A family history of kidney disease

Diabetes

High blood pressure

Heart disease

Taking certain prescription and over-the-counter medicinesÂ can damage your kidneys, too. For example, nonsteroidal

anti-inflammatory drugs (NSAIDs, such as ibuprofen and naproxen sodium) can harm your kidneys if you take them for a long period or if you have certain medical conditions. Talk with your provider about your risk for kidney disease.

WHAT HAPPENS DURING A PROTEIN IN URINE TEST?

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

In certain cases, your provider may ask you to collect all of your urine during a 24-hour period. A "24-hour urine sample test" may provide more complete results because the amount of protein and other substances in your urine can vary throughout the day.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations to test for protein in urine. If your provider has ordered a 24-hour urine sample, you will be given a special container to collect your urine and instructions on how to collect and store your samples.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a urinalysis or a protein in urine test.

WHAT DO THE RESULTS MEAN?

If your result is negative, your urine did not contain too much protein when you were tested.

If a large amount of protein is found in your urine sample, it doesn't always mean you have a medical problem that needs treatment. Your urine protein levels can also be high for a short time because of:

Strenuous exercise

Dehydration

Your diet

Stress

Pregnancy

Your provider may recommend other urinalysis tests to see whether you continue to have a high protein level in your urine over time. If your urine protein remains high, it is likely a sign of kidney disease or kidney damage from other conditions.

The amount of protein in your urine is linked to the amount of kidney damage you may have. But you will need more

tests to diagnose what is causing the damage. To learn what your results mean, talk with your provider. Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PROTEIN IN URINE TEST?

At-home test kits that check for protein in urine tests are available without a prescription. The kits usually include dipsticks and instructions for how to collect and test your urine sample. At-home urine tests are easy to do and provide accurate results as long as you carefully follow all instructions. Ask your provider which protein in urine test is best for you.

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Prothrombin Time Test and INR (PT/INR)

<https://medlineplus.gov/lab-tests/prothrombin-time-test-and-inr-ptinr/>

PROTHROMBIN TIME TEST AND INR (PT/INR)

WHAT IS A PROTHROMBIN TIME TEST WITH AN INR (PT/INR)?

A prothrombin time (PT) test measures how many seconds it takes for a clot to form in a blood sample. Since the results of this test can vary with different labs or test methods, your results may be used to create an INR (international normalized ratio). Using this ratio makes it easier for health care providers to compare and better understand results from different labs or test methods.

Prothrombin is a protein made by the liver. It is one of several substances known as clotting (coagulation) factors. When you get a cut or other injury that causes bleeding, your clotting factors work together to form a blood clot. Blood clotting is an important process that prevents you from losing too much blood when you are injured. Clotting factor levels that are too low can cause you to bleed too much after an injury. Levels that are too high can cause dangerous clots to form in your arteries or veins.

A PT/INR test helps find out if your blood is clotting normally. It also checks to see if a medicine that prevents blood clots is working the way it should.

Other names: prothrombin time/international normalized ratio, PT, pro time

WHAT IS IT USED FOR?

A PT/INR test is most often used to:

See how well warfarin, an anti-coagulant or blood-thinning medicine, is working to prevent blood clots. (Coumadin is a common brand name for warfarin.) Your provider may change your dose based on these results.

Find out the reason for abnormal blood clots.

Find out the reason for unusual bleeding.

Check clotting ability before a procedure or surgery that may cause bleeding.

Check for liver problems.

Check for vitamin K deficiency. Vitamin K helps with blood clotting.

A PT/INR test is often done along with a partial thromboplastin time (PTT) test. A PTT test also checks for clotting problems but looks at different coagulation factors than the PT test.

WHY DO I NEED A PT/INR TEST?

If you are regularly taking warfarin, you may need this test to make sure you are taking the right dose. Your provider will let you know how often you need to get the test to monitor your dose.

If you are not taking warfarin, you may need this test if you have symptoms of a bleeding or clotting disorder.

Symptoms of a bleeding disorder include:

Unexplained heavy bleeding

Bruising easily

Unusually heavy nose bleeds

Unusually heavy menstrual periods in women

Symptoms of a clotting disorder include:

Leg pain or tenderness

Leg swelling

Redness or darkening of the skin on the leg

Skin that feels warm

Trouble breathing

Cough

Chest pain

Rapid heartbeat and breathing

In addition, you may need a PT/INR test if you are scheduled for surgery. It helps make sure your blood is clotting normally, so you won't lose too much blood during the procedure.

WHAT HAPPENS DURING A PT/INR TEST?

The test may be done on a blood sample from a vein or a fingertip.

For a blood sample from a vein:

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a blood sample from a fingertip:

A fingertip test may be done in a provider's office or in your home. If you have been taking warfarin for a long time, you may be trained to take your blood sample with an at-home testing kit. Your provider may recommend you test your blood regularly using an at-home PT/INR test kit. During this test, you or your provider will:

Use a small needle to puncture your fingertip.

Collect a drop of blood and place it onto a test strip or other special instrument.

Place the instrument or test strip into a device that calculates the results. At-home devices are small and lightweight.

If you are using an at-home test kit, you will need to review your results with your provider. Your provider will let you know how they would like to receive the results.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you are taking warfarin, you may need to delay your daily dose until after testing. Your provider will let you know if there are any other special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If you were tested because you are taking warfarin or have certain health conditions, your results will probably be in the form of INR levels:

INR levels that are too low may mean you are at risk for dangerous blood clots because your blood clots too easily.

INR levels that are too high may mean you are at risk for dangerous bleeding because your blood is taking longer than normal to form a clot.

If your INR is not normal, your provider will probably change your dose of warfarin.

If you are not taking warfarin, your results may be in the form of a PT time, which is the number of seconds it takes for your blood sample to clot (prothrombin time).

If you are not taking warfarin and your INR or prothrombin time results are not normal, it may be a sign of that your blood is not clotting normally.

If your PT/INR levels are too low, your blood is clotting too quickly. This may be due to:

A clotting disorder, a condition in which your body forms too many clots in arteries or veins. Clots can form when you don't have an injury, or they don't dissolve when they should.

Supplements with vitamin K, or a diet high in vitamin K.

Certain medicines.

A clotting disorder, a condition in which your body forms too many clots in arteries or veins. Clots can form when you don't have an injury, or they don't dissolve when they should.

Supplements with vitamin K, or a diet high in vitamin K.

Certain medicines.

If your PT/INR levels are too high, your blood is clotting too slowly. This may be due to:

A bleeding disorder, a condition in which your body can't clot blood well. When you have an injury or a procedure that may cause bleeding, it may cause too much bleeding.

Liver disease.

Vitamin K deficiency. Vitamin K plays an important role in blood clotting.

A bleeding disorder, a condition in which your body can't clot blood well. When you have an injury or a procedure that may cause bleeding, it may cause too much bleeding.

Liver disease.

Vitamin K deficiency. Vitamin K plays an important role in blood clotting.

Be sure to tell your provider if you are taking any medicines or supplements. Some may affect your results. Other things that may affect your PT/INR results include:

Alcohol

Foods high in vitamin K, such as green leafy vegetables

Soybean and canola oils

High-fat diets

Ask your provider what you should avoid before your PT/INR test.

To understand the results of a PT/INR test, your provider will consider your symptoms, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PT/INR TEST?

Sometimes certain liver tests are ordered along with a PT/INR test. These include:

Aspartate Aminotransferase (AST)

Alanine Aminotransferase (ALT)

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RELATED HEALTH TOPICS

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PTEN Genetic Test

<https://medlineplus.gov/lab-tests/pten-genetic-test/>

PTEN GENETIC TEST

WHAT IS A PTEN GENETIC TEST?

A PTEN genetic test uses a sample of your blood or other body tissue to look for changes in a gene called PTEN. Changes in your genes are also called gene variants or mutations.

Genes are parts of DNA in your cells that you inherit from your parents. You inherit two copies of each gene, one from each parent. Genes carry information that controls what you look like and how your body works.

PTEN is a type of gene called a "tumor suppressor." Tumor suppressor genes provide instructions for making proteins that help prevent or slow the growth of cells that may become tumors. The proteins work by:

Preventing cells from dividing too quickly to make new cells.

Repairing damaged DNA in cells.

Starting the normal process of cell death in cells that can't be repaired.

If you have certain changes in a PTEN gene, the gene or the protein it makes may not work well or may stop working completely. As a result, your cells may grow out of control, which can lead to tumors and cancer. Changes in your PTEN gene can be acquired or inherited.

Acquired changes in your genes are not passed down from your parents. They develop at some time after you're born. These kinds of changes are commonly found in many different types of cancer, including prostate cancer, uterine cancer, and some brain tumors.

Where you develop these gene changes, and how likely you are to develop them, depends on a variety of factors. These include:

Errors in DNA repair or when cells divide to make new cells. These errors can happen at any time. However, they become more common with age.

Being exposed to certain carcinogens (substances that increase your likelihood of getting cancer). For example, exposure to UV radiation from the sun and some artificial sources like tanning beds can lead to PTEN gene changes in your skin.

Having certain chronic viral infections. These viruses can damage your DNA and lead to PTEN gene changes that can cause cancer.

Inherited changes in your genes are changes that you're born with. They show up in all the cells in your body, including your egg or sperm cells. That means they can be passed from parent to child.

Inherited changes in the PTEN gene aren't common. But when they happen, they can cause several disorders that trigger the growth of benign tumors (not cancer). These tumors are called hamartomas. Cowden syndrome and Bannayan-Riley-Ruvalcaba syndrome are the two main disorders associated with inherited PTEN gene changes. They are often grouped together and called PTEN hamartoma tumor syndrome (PHTS).

Hamartomas and other symptoms of PHTS can show up in infancy or later. PHTS increases your risk for certain types of cancer, such as cancer of the breast, thyroid, and kidney. However, these changes usually don't develop until adulthood.

Other names: PTEN gene, full gene analysis; PTEN sequencing and deletion/duplication, PTEN FISH, chromosome 10 deletion, CHR 10

WHAT IS IT USED FOR?

The PTEN genetic test is used to look for changes in your PTEN gene. It's not a routine test. It may be used to:

Find out if a change in your PTEN gene is involved in cancer. If you do have a PTEN gene change, your health care provider may prioritize some treatments over others that wouldn't work on cancers without this gene change.

Check for inherited changes in your PTEN gene if you have symptoms of PHTS or family members who have PTEN gene changes.

WHY DO I NEED A PTEN GENETIC TEST?

If you have cancer, you may need a PTEN genetic test to check whether your cancer cells have a PTEN gene change. Genetic changes in cancer can affect how the disease responds to treatment and how it may behave over time. So, your test results can help guide your treatment plan.

If you have a family history of PTEN gene changes and/or symptoms of PHTS, a PTEN genetic test can help you understand and manage your cancer risk. The test can also show whether you have a genetic change that you could pass on to your children.

You or your child may need a test if you have symptoms of PHTS, which include:

Having many hamartomas, especially in the mouth and intestines, or on the breasts or skin.

A larger than normal head size (macrocephaly).

Developmental delays, learning disabilities, and/or autism.

Dark freckles on the penis.

Thyroid, kidney, or colorectal cancer.

Uterine or breast cancer (in females only).

If you're pregnant and have a family history of PTEN gene changes, you may consider having the test as part of your prenatal testing. The test can show whether your baby could be born with PHTS.

WHAT HAPPENS DURING A PTEN GENETIC TEST?

If you're having a PTEN genetic test because you have cancer, the type of sample used will depend on your cancer:

For solid tumors, the test often uses a sample of tissue from the tumor. The tissue sample may be taken during surgery to remove the tumor. If you're not having surgery, you may have a biopsy of the tumor to remove a small amount of tissue. Health care professionals can do this by performing:

A fine needle aspiration biopsy. This procedure uses a very thin needle to remove a sample of cells from the tumor.

A core needle biopsy. This procedure uses a larger needle to remove a sample.

A fine needle aspiration biopsy. This procedure uses a very thin needle to remove a sample of cells from the tumor.

A core needle biopsy. This procedure uses a larger needle to remove a sample.

For blood or bone marrow cancer, a bone marrow test is often used. Marrow is soft, spongy tissue found inside of bones. Most bone marrow samples are taken from the back of the hip bone. You will be given an injection (shot) of medicine to numb the area. You may also be given medicine to help you relax or sleep. A bone marrow test has two parts that are usually done one after the other:

For a bone marrow aspiration, the provider pushes a hollow thin needle into the bone. A sample of bone marrow fluid is removed using a syringe attached to the needle.

For a bone marrow biopsy, the provider inserts a large hollow needle into the bone and twists the needle to take out a small piece of bone marrow tissue.

For a bone marrow aspiration, the provider pushes a hollow thin needle into the bone. A sample of bone marrow fluid is removed using a syringe attached to the needle.

For a bone marrow biopsy, the provider inserts a large hollow needle into the bone and twists the needle to take out a small piece of bone marrow tissue.

If you're having the test to look for inherited gene changes, you may have:

A blood test. During a blood test, a health care professional will use a small needle to remove a blood sample from a vein in your arm.

A buccal (cheek) swab. A health care professional will insert a swab into your mouth and rub it across the inside of your cheek to gather a sample of cells.

If you're testing a fetus, a sample may be gathered using:

Amniocentesis. This procedure collects a sample of the amniotic fluid that surrounds the fetus during pregnancy. The fluid contains cells from the fetus that can be tested. Amniocentesis is usually done between week 15 and 20 of pregnancy.

Chorionic villus sampling (CVS). This procedure collects tissue from the chorionic villi, which are parts of the placenta. The placenta is the organ that grows in the uterus to provide nutrients and oxygen to the fetus. The chorionic villi usually have the same genes as the fetus. A CVS is usually done between week 10 and 13 of pregnancy.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you're having a tissue biopsy or bone marrow test, ask your provider for instructions about how to prepare.

A blood test or cheek swab usually doesn't require any preparation.

For a CVS or amniocentesis, you may need to drink extra fluid and not urinate (pee) before your test so that your bladder is full.

ARE THERE ANY RISKS TO THE TEST?

The risks of a biopsy depend on the type of biopsy you have. In general, you may have soreness, pain, or bleeding where the tissue was removed.

After a bone marrow test, you may feel stiff or sore where the sample was taken. This usually goes away in a few days.

Serious risks are rare, but may include swelling, bleeding at the site, or fever.

A blood test has very little risk. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

A cheek swab has no risks.

During amniocentesis or CVS, you may feel mild stinging or cramping. Both procedures are safe, but they do have a slight risk of causing miscarriage. Talk with your provider about the risks and benefits of these tests.

WHAT DO THE RESULTS MEAN?

If you have cancer and your test results show you have an acquired change in your PTEN gene, your provider can explain what it means for your treatment plan and future health.

If your test results show that you have certain inherited changes in your PTEN gene, it means you have PHTS. It doesn't mean you have cancer. However, your risk of cancer may be higher than most people. Your provider may discuss steps you can take to reduce your cancer risk, including:

- Having a yearly exam

- Having frequent cancer screening tests to help find cancer early when it may be easier to treat

- Making lifestyle changes, such as getting more exercise, eating healthy foods, and quitting smoking

- Taking medicine or having surgery (mastectomy) to reduce the risk of breast cancer

- Having surgery (hysterectomy) to reduce the risk of uterine cancer

For children, your provider may recommend a yearly exam that includes a skin check. Yearly thyroid ultrasound tests may begin at age seven. Your child may also need a checkup to see how their brain and nervous system are developing.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A PTEN GENETIC TEST?

If you're thinking about getting tested for an inherited change in your PTEN gene, it may help to talk with a genetic counselor. Genetic counselors are specially trained to help you understand the risks and benefits of testing for you and your family. This can help you make informed decisions.

If you've been diagnosed with an inherited change in your PTEN gene, a counselor can explain your risk for cancer. The counselor can also help you decide about testing your children and how to manage your risk over time.

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Pulse Oximetry

<https://medlineplus.gov/lab-tests/pulse-oximetry/>

PULSE OXIMETRY

WHAT IS PULSE OXIMETRY?

Pulse oximetry is a test that uses a small, clip-like device called a pulse oximeter (Pulse Ox) to measure oxygen levels in your blood. When you breathe, your lungs take in oxygen and send it into your bloodstream. This oxygen-rich blood is carried by your red blood cells into your heart, which pumps it to the rest of your body.

But if a health condition causes your blood oxygen level (oxygen saturation) to get too low (a condition called hypoxemia), it can make it hard for your body to work properly. The longer your blood oxygen level stays low, the higher the risk is of damage to your tissues and organs.

Pulse oximetry is a painless and quick way to find out if your oxygen level is too low. Quick treatment of low blood oxygen may help you avoid serious complications.

Other names: pulse ox, oxygen saturation monitor, finger pulse oximeter

WHAT IS IT USED FOR?

Pulse oximetry is used to check your blood oxygen level. The test may be done in your health care provider's office, a clinic, a hospital, or even your home. An at-home pulse oximeter may be helpful if you have a serious and/or chronic lung disease.

WHY DO I NEED PULSE OXIMETRY?

Your provider may give you a pulse-oximetry test as part of a routine checkup. You may also get the test if you:

Have a condition that affects your lung function. These can include chronic obstructive pulmonary disease (COPD), lung cancer, heart failure, asthma, and COVID-19.

Are taking medicine to treat lung disease. The test can show how well the medicine is working.

Are having surgery. Your blood oxygen level may be checked before, during, and after your procedure.

Have sleep apnea. This is a disorder that causes you to repeatedly stop breathing during sleep.

Use extra oxygen (oxygen therapy). This may help your provider determine if you are getting the right oxygen flow rate from your oxygen device.

Have shortness of breath (feeling like you can't get enough air).

WHAT HAPPENS DURING PULSE OXIMETRY?

During pulse oximetry:

A small electronic device called a pulse oximeter will be clipped to a part of your body, usually your fingertip.

The oximeter uses a special type of light that passes through your skin and into your blood. The light may make your fingertip look red, but you won't feel any pain.

A sensor on the oximeter measures the amount of light. That measurement is used to figure out your blood oxygen level.

After a few seconds, the oximeter will provide a readout of your heart rate (pulse) and oxygen level.

If you have a condition that affects your lung function, your provider may recommend you use an at-home finger pulse oximeter or other oximeter to monitor your condition. Your provider will guide you on how to purchase and use your at-home device.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Dark nail polish on the finger being tested can block the light emitted from the oximeter. Other things that may cause inaccurate results can include:

Artificial nails

A darker skin pigmentation (color)

Poor circulation

Cold skin

Tobacco use

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having pulse oximetry.

WHAT DO THE RESULTS MEAN?

Results are often given as oxygen saturation levels. A normal oxygen saturation level ranges between 95% and 100%. Saturation levels may be somewhat lower and considered acceptable if you have a lung disease such as COPD or pneumonia. Your levels may also be lower if you live in an area with a higher elevation, where there is less oxygen in the air.

If you are using an at-home oximeter, watch for signs or symptoms that you have low oxygen levels, such as:

Lips, nails, or face that look bluish (cyanosis)

Trouble breathing or shortness of breath

Chest pain or tightness

Fast heart rate

Contact your provider if your oxygen saturation level is 92% or lower. If it falls to 88% or lower, seek immediate medical attention.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT PULSE OXIMETRY?

Pulse oximetry results may be 2% to 4% higher or lower than your actual blood oxygen level. For a more accurate result, your provider may do a blood test called an arterial blood gas. You may need both types of tests to manage your condition.

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Rapid Tests

<https://medlineplus.gov/lab-tests/rapid-tests/>

RAPID TESTS

WHAT ARE RAPID TESTS?

Rapid tests, also known as rapid diagnostic tests or RDTs, are easy-to-use tests that provide quick results, usually in 20 minutes or less. Most standard tests have to be sent to a lab, and you have to wait for the results. With a rapid test, you can have the test done and get the results at your provider's office or a clinic. For some of these tests, you can also do them and get the results at home.

Rapids tests usually include the following steps:

You provide a sample of your body fluid, such as blood, urine, saliva, or nasal secretions.

The sample is mixed with a special substance that will cause a chemical reaction if a disease or other condition is present.

The results show on a dipstick or rapid test cassette:

A dipstick is a thin, plastic strip coated with chemicals. If the test is positive, the dipstick will change color when placed in the sample.

A rapid test cassette is a small plastic case with a well that holds the sample. The well shows a letter or a symbol to indicate the result.

A dipstick is a thin, plastic strip coated with chemicals. If the test is positive, the dipstick will change color when placed in the sample.

A rapid test cassette is a small plastic case with a well that holds the sample. The well shows a letter or a symbol to indicate the result.

Most rapid tests can provide quick, reliable results, but the results are not always as accurate as standard lab tests. The accuracy of your test results may be affected by:

The type of test used

How the test is stored and transported

The type of specimen taken

Your symptoms

When your symptoms started

What you are testing for

Using a test that is past its expiration date

Other names: rapid diagnostic tests, RDT's point-of-care tests

HOW ARE THEY USED?

Rapid tests are most often used to diagnose infectious diseases, including:

Flu.

Strep throat.

Malaria.

HIV.

COVID 19. Rapid tests for COVID-19 usually provide results in about 30 minutes.

Rapid tests are also used in certain home-based tests, such as pregnancy tests.

WHAT ARE THE ADVANTAGES AND DISADVANTAGES OF RAPID TESTS?

The advantages of rapid tests include:

Fast results. This may allow you to get treatment faster. Faster treatment can reduce the severity of your symptoms and

may help prevent the spread of disease.

Easy to use. They can be done by non-medical workers and volunteers who have been trained to do the test. You may even be able to perform a rapid test on yourself.

Little or no special equipment is required. This is especially useful in areas with limited resources or access to specialized labs.

The disadvantages of rapid tests include:

Lower sensitivity than lab tests. Sensitivity means how well a test can identify a disease or condition. Rapid tests are not as good as lab tests at finding diseases in the earliest stages of infection.

False negatives are more common. A false negative result means your test shows you don't have a disease or condition, but you actually do have it.

Because of these possible disadvantages, the instructions for some tests (such as over-the-counter COVID-19 tests) recommend doing the test again after a certain period of time if you got a negative result. A negative result means that the test showed that you don't have the disease.

Your health care provider may take a second sample and send it to a lab for testing to confirm or rule out the results of a rapid test. But you may still be able to get treatment and help prevent the spread of disease while you wait for lab results.

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How to Understand Your Lab Results

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Pregnancy Test

Strep A Test

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Rash Evaluation

<https://medlineplus.gov/lab-tests/rash-evaluation/>

RASH EVALUATION

WHAT IS A RASH EVALUATION?

A rash evaluation is a test to find out what is causing a rash. As part of the evaluation, your health care provider will ask about your medical history and your rash. They may ask you if you have started using any new soaps, lotions, cosmetics, or household products. Also, tell your provider if your skin comes in contact with any irritating substances, for example while you are at work or doing a hobby.

Your provider will also review your symptoms and examine your skin. They may also order additional tests, such as a patch test, skin culture, or skin biopsy.

A rash, also known as dermatitis, is an area of red, irritated, and usually itchy skin. A skin rash may also be dry, scaly, and/or painful. Most rashes happen when your skin touches a substance that irritates it. This is known as contact dermatitis. There are two main types of contact dermatitis: allergic contact dermatitis and irritant contact dermatitis.

Allergic contact dermatitis happens when your body's immune system treats a normally harmless substance as a threat. These substances are called allergens. When exposed to an allergen, your immune system sends out chemicals in response. These chemicals affect your skin, causing you to have an allergic reaction to the substance and develop a rash. Common causes of allergic contact dermatitis include:

Poison ivy and related plants, like poison sumac and poison oak. A poison ivy rash is one of the most common types of contact dermatitis.

Cosmetics.

Fragrances.

Jewelry metals, such as nickel.

Rubber, including latex.

Allergic contact dermatitis usually causes more itching than pain. Sometimes the rash may not develop until several days after you were exposed to the allergen. A rash may happen after the first time you are exposed to the allergen, or after you have been exposed to it repeatedly.

Irritant contact dermatitis happens when an area of your skin responds to an irritating substance. This causes a skin rash to form. Common causes of irritant contact dermatitis include:

Household products such as detergents and drain cleaners.

Strong soaps.

Pesticides.

Nail polish remover.

Body fluids, such as urine and saliva. These rashes, which include diaper rash, most commonly affect babies.

Irritant contact dermatitis is usually more painful than itchy. The rash usually develops quickly after exposure to the irritating substance. Irritant contact dermatitis is more common than allergic contact dermatitis.

You might be at a higher risk of developing contact dermatitis if you work around irritating chemicals or allergens, for example if you are a hairstylist, mechanic, or construction worker.

In addition to contact dermatitis, a rash may be caused by:

Skin disorders such as eczema and psoriasis.

Infections such as chickenpox, shingles, and measles.

Insect bites.

Heat. If you get overheated, your sweat glands can get blocked. This can cause a heat rash. Heat rashes often happen in hot, humid weather. While it can affect people of any age, heat rashes are most common in babies and young children.

Other names: patch test, skin biopsy

WHAT IS IT USED FOR?

A rash evaluation is used to diagnose the cause of a rash. Rashes can occur anywhere on your skin where you come into contact with an allergen or irritant. Most rashes can be treated at home with over-the-counter anti-itch creams or antihistamines. Your rash may go away quickly if you know what caused your reaction and avoid that allergen. But sometimes, a rash is a sign of a more serious condition and should be checked by a provider.

WHY DO I NEED A RASH EVALUATION?

You may need a rash evaluation if you have symptoms that don't respond to at-home treatment. Symptoms of a contact dermatitis rash include:

Redness or darker than natural skin tone

Swollen or bumps that look like hives or blisters

Itching

Pain (more common with an irritant rash)

Dry, cracked skin

Other types of rashes may have similar symptoms. Additional symptoms can vary, depending on the cause of the rash.

While most rashes are not serious, in some cases, a rash can be a sign of a serious health condition. Call your provider if you or your child has a skin rash with any of the following symptoms:

Severe pain.

Blisters, especially if they affect the skin around the eyes, mouth, or genitals.

Yellow or green fluid, warmth, and/or red streaks in the rash area. These are signs of infection.

Fever. This could be a sign of a viral or bacterial infection. These include scarlet fever, shingles, and measles.

Itching that is constant.

A rash that doesn't go away after a week of treatment.

Sometimes, a rash can be the first sign of a severe and dangerous allergic reaction called anaphylaxis. Call 911 or seek immediate medical attention if:

The rash is sudden and spreads quickly

You have trouble breathing

Your face is swollen

WHAT HAPPENS DURING A RASH EVALUATION?

There are different ways to do a rash evaluation. The type of test you get will depend on your symptoms and medical history.

To test for allergic contact dermatitis, your provider may give you a patch test:

During a patch test:

A provider will place small patches on your skin. The patches look like adhesive bandages. They contain small amounts of specific allergens.

You'll wear the patches for 48 to 96 hours and then return to your provider's office. You'll need to keep the patches dry.

Your provider will remove the patches and check for rashes or other reactions.

There is no test for irritant contact dermatitis. But your provider may make a diagnosis based on a physical exam, your symptoms, and your exposure to certain substances.

A rash evaluation may also include a blood test and/or a skin biopsy.

During a blood test:

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

During a biopsy:

A provider will use a special tool or a blade to remove a small piece of skin for testing.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. Some medicines may affect the results of your test. But don't stop any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a patch test. If you feel intense itching or pain under the patches once you are home, remove the patches and call your provider.

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

After a biopsy, you may have a little bruising, bleeding, or soreness at the biopsy site. If these symptoms last longer than a few days or they get worse, talk to your provider.

WHAT DO THE RESULTS MEAN?

If you had a patch test and have itchy, red bumps or swelling at any of the testing sites, it means you are probably allergic to the substance tested.

If you had a blood test, abnormal results may mean you:

- Are allergic to a certain substance

- Have a viral, bacterial, or fungal infection

If you had a skin biopsy, abnormal results may mean you:

- Have a skin disorder such as psoriasis or eczema

- Have a bacterial or fungal infection

If you have questions about your results, talk to your provider. To understand the results of a rash evaluation, your provider will consider your symptoms, medical history, and the results of other tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A RASH EVALUATION?

To relieve symptoms of a skin rash, your provider may suggest over-the-counter medicines and/or at-home treatments, such as cool compresses and baths. Other treatments will depend on your specific diagnosis.

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RELATED HEALTH TOPICS

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- Viral Infections

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- Allergy Blood Test
- Allergy Skin Test
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- Chickenpox and Shingles Tests
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- How to Prepare Your Child for a Lab Test
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RDW (Red Cell Distribution Width)

<https://medlineplus.gov/lab-tests/rdw-red-cell-distribution-width/>

RDW (RED CELL DISTRIBUTION WIDTH)

WHAT IS A RED CELL DISTRIBUTION WIDTH TEST?

A red cell distribution width (RDW) test measures how much the volume and size of your red blood cells (erythrocytes) varies.

Red blood cells are made in your bone marrow (the spongy tissue inside your large bones). They contain hemoglobin, an iron-rich protein that carries oxygen from your lungs to every cell in your body. Your cells need oxygen to grow, reproduce, and make energy for you to function. Normally, your red blood cells are about the same size. Differences in the size of your red blood cells may affect how well they can deliver oxygen through your body.

An RDW test uses a special graph called a histogram. If your red blood cells are about the same size, they will be close together on the histogram, and the RDW will be low. A high RDW means that the size of your red blood cells varies more than it should. The sizes will be more spread out on the histogram. This may be a sign of a medical condition.

Other names: RDW-SD (standard deviation) test, Erythrocyte Distribution Width

WHAT IS IT USED FOR?

The RDW blood test is often part of a complete blood count (CBC), a group of tests that measures the number and type of cells in your blood, including red cells. The RDW test is commonly used to help diagnose or determine what's causing anemia, a condition in which your body does not make enough red blood cells.

The RDW test may also be used with other tests to help diagnose other conditions, including thalassemia, an inherited disease that can cause severe anemia.

WHY DO I NEED AN RDW TEST?

Your health care provider may have ordered a complete blood count (CBC), which includes an RDW test, as part of a routine exam, or if you have:

Symptoms of anemia, which may include:

Shortness of breath

Weakness or fatigue

Headache

Dizziness

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Your provider may also order an RDW test for other reasons, for example if you have:

A family history of thalassemia, sickle cell anemia, or other inherited blood disorder

A chronic illness such as Crohn's disease, diabetes, or HIV

A diet low in iron and other minerals

A long-term infection

Excessive blood loss from an injury or surgical procedure

If you have anemia, your RDW test may help them determine what's causing your anemia.

WHAT HAPPENS DURING AN RDW TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This process generally takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

No special preparation is necessary.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to a blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

RDW results help your provider understand how much your red blood cells vary in size and volume. Results are usually shown as a percentage and compared to a reference range, which may vary depending on the lab used.

A normal RDW result means your red blood cells are within the reference range. Even if your RDW results are normal, your red blood cells may still be larger or smaller than usual. You may still have anemia or another medical condition that may need treatment.

A high RDW result means your red blood cells are higher than the reference range. This means that your red blood cells vary in size more than what is considered normal. This may be a sign of certain conditions, such as:

Chronic liver disease

Heart disease

Diabetes

Kidney disease

Cancer, especially colorectal cancer

A low RDW result means your red blood cells are below the reference range. This means that they are similar in size and close together. This is not a sign of anemia and isn't usually something to worry about.

If you have questions about your results, talk to your provider. They will usually look at your RDW results along with the results of other blood tests. The combined test results can show a more complete picture of your red blood cells. This can help diagnose a variety of conditions, including:

Iron deficiency

Different types of anemia

Thalassemia

Sickle cell anemia

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A RED CELL DISTRIBUTION WIDTH TEST?

If your test results indicate you have a chronic blood disorder, such as anemia, you may be put on a treatment plan to increase the amount of oxygen your red blood cells can carry. Depending on your specific condition, your provider may recommend iron supplements, medicines, and/or changes in your diet.

Be sure to talk to your provider before taking any supplements or changing your eating plan.

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RELATED HEALTH TOPICS

Anemia

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Red Blood Cell Antibody Screen

<https://medlineplus.gov/lab-tests/red-blood-cell-antibody-screen/>

RED BLOOD CELL ANTIBODY SCREEN

WHAT IS AN RBC ANTIBODY SCREEN?

An RBC (red blood cell) antibody screen is a blood test that looks for RBC antibodies in your blood. These antibodies destroy red blood cells that are foreign (different from your own).

Antibodies are proteins that your immune system makes to attack germs and other foreign substances in your body, including red blood cells that aren't like yours. Your immune system recognizes foreign red blood cells by chemical markers on the cells, called antigens.

All red blood cells have antigens. Your blood type is based on the kind of antigens you have on your cells. The main antigens found on red blood cells are A, B, and Rh Factor (also called D antigen). If you're exposed to red blood cells with antigens that aren't like yours, the foreign antigens will trigger your immune system to make RBC antibodies to destroy the foreign red blood cells.

You may have RBC antibodies in your blood if you have been exposed to foreign red blood cells in a blood transfusion, or if you have been pregnant with a fetus whose blood type was different from your own. RBC antibodies that attack foreign blood cells won't harm your health, but these antibodies could cause problems if you are exposed to foreign red blood cells again, for example:

If you have a blood transfusion, RBC antibodies could cause serious illness.

If you are pregnant with a fetus who has a blood type that's different from yours, certain RBC antibodies could harm your fetus.

Once your body makes RBC antibodies, you will always have them.

Other names: antibody screen, indirect antiglobulin test, indirect anti-human globulin test, IAT, indirect coombs test, erythrocyte Ab, RBC antibody identification

WHAT IS IT USED FOR?

An RBC antibody screen is used to check your blood for RBC antibodies before you have a blood transfusion or when you're pregnant:

Before a blood transfusion, the test can help your health care provider select donor blood (the blood you will get) that's well matched with your blood. This prevents you from getting blood that triggers an immune reaction that could make you very sick.

During pregnancy, the test can show whether you have RBC antibodies that could attack the red blood cells of your fetus, causing a very serious type of fetal anemia.

RBC antibodies that may cause problems in pregnancy include Rh antibodies and Kell antibodies.

Rh antibodies attack an antigen on red blood cells called Rh factor. You inherit the Rh factor from your father or your mother. Most people have this antigen, but some don't. Early in your pregnancy, or even before you're pregnant, you'll have a prenatal blood test to find out whether you have the Rh antigen:

If you have the Rh antigen, you're Rh positive. That means your immune system won't make antibodies to attack blood cells with Rh antigens because they aren't foreign. But you may still have an RBC antibody screen to look for other less common red blood cell antibodies that could affect your fetus.

If you don't have the Rh antigen, you're Rh negative. If your fetus could be Rh positive, you have Rh incompatibility, which may harm them. If blood from your fetus gets into your bloodstream, the Rh antigens may trigger your immune system to make Rh antibodies. These antibodies could enter the fetal bloodstream and attack their red blood cells. If you

have Rh incompatibility, you may have several RBC antibody screens during your pregnancy to find out whether you've made Rh antibodies.

Rh incompatibility isn't common, but it causes most cases of anemia in babies. It's unlikely to harm your baby the first time it happens. That's because exposure to your baby's blood mostly happens during childbirth. They will likely be born before your body has had time to make antibodies. But once you have Rh antibodies, they could cause problems in future pregnancies, or if you need a blood transfusion.

Kell antigens are a group of red blood cell antigens that can trigger a strong immune response if you don't have Kell antigens in your own blood. During pregnancy, if your fetus has Kell antigens and you don't, you may make Kell RBC antibodies. They can cause a very severe anemia that may be life-threatening for the fetus.

Most people with Kell RBC antibodies develop them over time after having many transfusions with blood that has Kell antigens.

If you have the Rh antigen, you're Rh positive. That means your immune system won't make antibodies to attack blood cells with Rh antigens because they aren't foreign. But you may still have an RBC antibody screen to look for other less common red blood cell antibodies that could affect your fetus.

If you don't have the Rh antigen, you're Rh negative. If your fetus could be Rh positive, you have Rh incompatibility, which may harm them. If blood from your fetus gets into your bloodstream, the Rh antigens may trigger your immune system to make Rh antibodies. These antibodies could enter the fetal bloodstream and attack their red blood cells. If you have Rh incompatibility, you may have several RBC antibody screens during your pregnancy to find out whether you've made Rh antibodies.

WHY DO I NEED AN RBC ANTIBODY SCREEN?

Your provider may order an RBC antibody screen if:

You need a blood transfusion. The test helps make sure you get donor blood that's safe for you.

You're pregnant. The test is usually done in early pregnancy, as part of routine prenatal testing. The test helps your provider know what care you will need during your pregnancy.

WHAT HAPPENS DURING AN RBC ANTIBODY SCREEN?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an RBC screen.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If you are getting a blood transfusion:

A positive result means you have one or more RBC antibodies in your blood that could destroy red blood cells in certain types of donor blood. You will likely need more tests to find out exactly what type of RBC antibodies you have so your provider can select donor blood that's a good match for you.

A negative result means no RBC antibodies were found.

If you are pregnant:

A positive test result means that you have RBC antibodies in your blood that could harm your fetus. You will likely need more tests to find out what type of RBC antibodies you have.

If you have Rh antibodies, you may have more frequent tests during your pregnancy. There is no treatment to stop your body from making more Rh antibodies.

If you have other RBC antibodies, but no Rh antibodies, your pregnancy will be closely watched. If you are Rh-negative, your provider may give you an injection (shot) to prevent your body from making Rh antibodies.

If you have Rh antibodies, you may have more frequent tests during your pregnancy. There is no treatment to stop your body from making more Rh antibodies.

If you have other RBC antibodies, but no Rh antibodies, your pregnancy will be closely watched. If you are Rh-negative, your provider may give you an injection (shot) to prevent your body from making Rh antibodies.

A negative result means that no RBC antibodies were found. If you are Rh-negative, your provider may give you an injection to prevent your body from making Rh antibodies.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN RBC ANTIBODY SCREEN?

Once you have RBC antibodies, they never go away. But you may have too few to show up on a test. If you have had an RBC antibody screen that showed you have RBC antibodies, make sure to tell your provider, especially if you need a blood transfusion or are pregnant.

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What You Need to Know About Blood Testing

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Red Blood Cell (RBC) Count

<https://medlineplus.gov/lab-tests/red-blood-cell-rbc-count/>

RED BLOOD CELL (RBC) COUNT

WHAT IS A RED BLOOD CELL (RBC) COUNT?

A red blood cell (RBC) count measures the number of red blood cells, also known as erythrocytes, in your blood. Red blood cells are made in your bone marrow, the spongy tissue inside your large bones. They contain hemoglobin, an iron-rich protein that carries oxygen from your lungs to every cell in your body. Your cells need oxygen to grow, reproduce, and make energy for you to function. An RBC count that is higher or lower than normal is often the first sign of an illness. So, the test may allow you to get treatment even before you have symptoms.

Other names: erythrocyte count, red count, RBC Count

WHAT IS IT USED FOR?

A red blood cell (RBC) count is almost always part of a complete blood count (CBC), a group of tests that measures the number and type of cells in your blood. The RBC measurement is used to help diagnose red blood cell disorders, such as anemia, a condition in which your body does not make enough healthy red blood cells.

WHY DO I NEED A RED BLOOD CELL COUNT?

You may get this test as part of a CBC, which is often included in a routine checkup. You may also need this test if you are getting treatment that might affect your blood cell counts, such as chemotherapy, to monitor health conditions, such as chronic kidney disease, or if you have symptoms of a low or high red blood cell count.

Symptoms of a low red blood cell count may include:

Shortness of breath

Weakness or fatigue

Headache

Dizziness

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Pale skin

Symptoms of a high red blood cell count may include:

Headache

Dizziness

Vision problems

WHAT HAPPENS DURING A RED BLOOD CELL COUNT?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You don't need any special preparations for a red blood cell (RBC) count.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will show whether you have a normal red blood cell count or a count that is too low or too high.

Conditions that may cause a low red blood cell count can include:

Anemia

Leukemia, a type of blood cancer

Malnutrition, a condition in which your body does not get the calories, vitamins, and/or minerals needed for good health

Multiple myeloma, a cancer of the bone marrow

Kidney failure

Blood loss from injury or other problems

Alcohol use disorder (AUD)

Pregnancy

Conditions that may cause a high red blood cell count can include:

Dehydration

Heart disease

Polycythemia vera, a disease which causes your bone marrow to make too many red blood cells

Scarring of the lungs, often due to cigarette smoking

Lung disease

Kidney cancer

Sleep apnea, a disorder that causes you to repeatedly stop breathing during sleep

Misuse of certain drugs for athletic performance (such as anabolic steroids) and living at a high altitude can also cause a high red blood cell count.

Talk to your health care provider if you have questions about your results. Your provider will consider your symptoms, medical history, and the results of other blood tests to understand your red blood cell count results.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN RBC COUNT?

If results showed you had a low or a high red blood cell count, you may need more tests to help your provider make a diagnosis. These can include:

Reticulocyte count, which measures the number of reticulocytes in your blood. Reticulocytes are immature (still developing) red blood cells. This test helps check if your bone marrow is making the right amount of red blood cells.

Iron tests, which measure iron levels in your blood. Iron is essential for making red blood cells.

Vitamin B test, which measures the amount of one or more B vitamins in your blood. B vitamins are important for making red blood cells and keeping your body healthy.

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Red Blood Cell (RBC) Indices

<https://medlineplus.gov/lab-tests/red-blood-cell-rbc-indices/>

RED BLOOD CELL (RBC) INDICES

WHAT ARE RED BLOOD CELL (RBC) INDICES?

Red blood cell (RBC) indices measure your red blood cells' size, shape, and quality. Red blood cells are also known as erythrocytes. They are made in your bone marrow (the spongy tissue inside your large bones). They contain hemoglobin, an iron-rich protein in your red blood cells that carries oxygen from your lungs to every cell in your body. Your cells need oxygen to grow, reproduce, and make energy.

Knowing the size and shape of your red blood cells can help your provider determine if you have a certain type of anemia, a condition in which your body does not make enough healthy red blood cells. There are four types of red blood cell indices:

Mean corpuscular volume (MCV), which measures the average size of your red blood cells.

Mean corpuscular hemoglobin (MCH), which measures the average amount of hemoglobin in a single red blood cell.

Mean corpuscular hemoglobin concentration (MCHC), which measures how concentrated (close together) the hemoglobin is in your red blood cells. It also includes a calculation of the size and volume of your red blood cells.

Red cell distribution width (RDW), which measures differences in the volume and size of your red blood cells. Healthy red blood cells are usually about the same size.

If one or more of these indices are not normal, it may mean you have some type of anemia.

Other names: erythrocyte indices

WHAT ARE THEY USED FOR?

Red blood cell (RBC) indices are part of a complete blood count, a group of tests that measures the number and type of cells in your blood. The results of RBC indices are used to diagnose different types of anemia. There are several types of anemia, and each type has a different effect on the size, shape, and/or quality of red blood cells.

WHY DO I NEED RED BLOOD CELL INDICES TESTING?

You may get this test as part of a complete blood count, which is often included in a routine checkup. You may also need this test if you have symptoms of anemia, which may include:

Shortness of breath

Weakness or fatigue

Headache

Dizziness

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Pale skin

Cold hands and feet

WHAT HAPPENS DURING A RED BLOOD CELL INDICES TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This test usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THESE TESTS?

You don't need any special preparations for a red blood cell (RBC) indices test.

ARE THERE ANY RISKS TO THESE TESTS?

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

You will get results for each of the indices. Abnormal results may include one or more of the following:

Mean corpuscular volume (MCV)

If your red blood cells are smaller than normal, it may mean you have:

Iron deficiency anemia, the most common form of anemia. It happens when you don't have enough iron in your body.

Thalassemia, which is a group of blood disorders that are inherited (passed down through families). These disorders cause your body to make fewer healthy red blood cells and less hemoglobin. This can cause severe anemia.

If your red blood cells are larger than normal, it may mean you have:

Anemia caused by a vitamin B₁₂ deficiency

Liver disease

Mean corpuscular hemoglobin (MCH)

If the amount of hemoglobin is lower than normal, it may mean you have:

Iron deficiency anemia

If the amount of hemoglobin is higher than normal, it may mean you have:

Anemia caused by a vitamin B deficiency

Mean corpuscular hemoglobin concentration (MCHC)

If the average amount of hemoglobin is lower than normal, it may mean you have:

Iron deficiency anemia

Thalassemia

If the average amount of hemoglobin is higher than normal, it may mean you have:

Hemolytic anemia, a type of anemia that happens when red blood cells are destroyed faster than they can be replaced.

Hereditary spherocytosis, a rare genetic disorder that causes anemia and gallstones

Red cell distribution width (RDW)

If your results were normal, it means your red blood cells are normal in size and are all about the same size.

If your results were not normal, it means there are differences in the size of your red blood cells.

This measurement is not enough for your provider to make a diagnosis, so RDW results are usually combined with the results of other indices and other blood tests. This combination of results can help confirm a diagnosis.

Talk to your health care provider if you have questions about your results. Your provider will consider your symptoms, medical history, and the results of other blood tests to understand your results.

Learn more about laboratory tests, reference ranges, and understanding results.

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Renin Test

<https://medlineplus.gov/lab-tests/renin-test/>

RENIN TEST

WHAT IS A RENIN TEST?

This test measures the level of renin in your blood. Renin is an enzyme, a protein that speeds up certain chemical reactions in your body. It's made by your kidneys. When your blood pressure or sodium decreases or your potassium increases, your kidneys release renin. That's because renin also acts as a hormone (a chemical messenger in your bloodstream that controls the actions of certain cells or organs) to help control your blood pressure.

Renin controls the production of another hormone called aldosterone, which is made in your adrenal glands. These are two small glands located above the kidneys. Aldosterone helps keep your blood pressure stable. It does this by helping to balance the levels of two electrolytes in your blood: potassium and sodium.

Usually, if renin increases or decreases, aldosterone does the same. These levels are highest in the morning and change throughout your day. Your position (sitting, standing, etc.), stress level, and certain medicines can affect your renin and aldosterone levels.

If your renin and/or aldosterone levels are not normal, it can be a sign of an adrenal gland disorder. Because the two hormones work together, an aldosterone test is often done at the same time as a renin test.

Other names: renin blood test, plasma renin activity (PRA) aldosterone-renin ratio (ARR)

WHAT IS IT USED FOR?

A renin test (or renin and aldosterone test) is used to find out if your adrenal glands are making too much or too little aldosterone.

The test is especially useful in diagnosing primary aldosteronism (PA), a disease caused by having too much aldosterone. PA, also known as primary hyperaldosteronism or Conn syndrome, causes high blood pressure. If not treated, high blood pressure may increase your risk of stroke, heart attack, and kidney failure.

WHY DO I NEED A RENIN TEST?

You may need a renin test if you've been diagnosed with high blood pressure, especially if it doesn't respond well to standard blood pressure medicines. The test can show whether PA is causing the condition.

Sometimes, PA causes low potassium levels, so you may also need this test if you have symptoms of low potassium. Your symptoms can vary, depending on the cause and how low your potassium level is. The symptoms can include:

- Muscle cramps
- Muscle weakness
- Excessive thirst
- Fatigue
- Headache
- Constipation
- Racing heartbeat

WHAT HAPPENS DURING A RENIN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You may be asked to stop taking certain medicines before your test, so tell your health care provider about everything you take. But don't stop taking any medicines unless your provider tells you to. Some medicines that may affect the results of your test can include:

Birth control pills

Blood pressure medicines

Diuretics (water pills)

You may also be asked to limit the amount of salt you eat each day before your test. Your provider will explain how to do this and for how long.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results show a higher-than-normal amount of renin, it may be a sign of:

Addison disease (also called adrenal insufficiency), a condition in which your adrenal glands don't make enough of certain hormones

Cirrhosis, scarring of the liver

Dehydration

If your results show a lower-than-normal amount of renin, it may be a sign of:

Primary aldosteronism (PA)

Kidney disease

Most often, your renin test results will be compared with your aldosterone test results. These results may show one of the following:

Lower than normal renin/Lower than normal aldosterone. This may mean you have Cushing's syndrome, a disorder in which the adrenal glands make too much of a hormone called cortisol.

Lower than normal renin/Higher than normal aldosterone. This may mean you have PA.

Higher than normal renin/Lower than normal aldosterone. This may mean you have Addison disease.

Higher than normal renin/Higher than normal aldosterone. This may mean you have secondary aldosteronism. With this disorder, your adrenal glands are healthy, but another medical condition is causing them to make too much aldosterone. These conditions can include diseases of the heart, liver, and kidneys.

Depending on your results, your provider may recommend medicines and/or diet and lifestyle changes to treat your condition.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A RENIN TEST?

You should not eat natural black licorice for at least two weeks before your test, because it can affect your results. Most licorice products in the United States don't contain natural licorice. Before you eat any licorice, check the ingredients label to be sure.

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Respiratory Pathogens Panel

<https://medlineplus.gov/lab-tests/respiratory-pathogens-panel/>

RESPIRATORY PATHOGENS PANEL

WHAT IS A RESPIRATORY PATHOGENS (RP) PANEL?

A respiratory pathogens (RP) panel checks for pathogens in your respiratory tract. A pathogen is a virus, bacteria, or other organism that causes an illness. Your respiratory tract, or respiratory system, includes the parts of your body that are involved in breathing. This includes your lungs, nose, and throat.

Many types of viruses and bacteria can cause respiratory tract infections. The symptoms are often similar, but their treatments can be different. Many respiratory infections clear up on their own with no specific treatment other than resting and drinking fluids. But for other types of infections, you may need to take medicine.

For most respiratory infections, your health care provider will make a diagnosis and determine treatment based on:

Your symptoms

Your medical history

The season of the year; some infections are more common during certain times of the year

Your recent travel (if any)

Which respiratory infections are most common at the time

But if you have a serious illness, are hospitalized, or are at risk for complications, your provider may order a respiratory pathogens panel. Identifying what is causing your respiratory infection can help determine the best treatment. It can also help you avoid taking unnecessary antibiotics, which are not used to treat viral infections.

An RP panel only needs a single sample to test for various viruses and bacteria. The results usually come in a few hours. This is faster than other respiratory tests, where the results can take a few days. Getting results earlier can allow you to start the right treatment more quickly.

Other names: RP panel, respiratory virus profile, syndromic multiplex panel, multiplex respiratory panel, multi-pathogen molecular assay

WHAT IS IT USED FOR?

Which pathogens an RP panel tests for can vary from one panel to another. But many of the panels often test for:

Viral infections such as:

Flu.

Common cold.

Respiratory syncytial virus (RSV). This is a common and usually mild respiratory infection. But it can be dangerous to babies and the elderly.

Adenovirus infection. Adenoviruses cause many different types of infections. These include pneumonia and croup, an infection that causes hoarse, barking coughs.

COVID-19.

Bacterial infections such as:

Whooping cough.

Bacterial pneumonia.

WHY DO I NEED A RESPIRATORY PATHOGENS PANEL?

You may need this test if you have symptoms of a respiratory infection and are seriously ill, hospitalized, or at risk for complications. Most respiratory infections cause mild to moderate symptoms. But they can be serious or even life-threatening to young children, the elderly, and people with weakened immune systems.

Symptoms of a respiratory infection include:

Coughing

Trouble breathing
Sore throat
Stuffy or runny nose
Fatigue
Loss of appetite
Fever

WHAT HAPPENS DURING A RESPIRATORY PATHOGENS PANEL?

There are two ways your provider may take a sample for testing:

Nasopharyngeal swab:

You will tip your head back.

Your provider will gently insert a swab into your nostril, pushing it until you feel it stopping. This is about 1 to 2 inches into your nostril, until it reaches the top part of your throat.

Your provider will rotate the swab and remove it.

Nasal aspirate:

Your provider will squirt a saline solution (salt water) into your nose and remove the sample with gentle suction.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a respiratory pathogens panel.

ARE THERE ANY RISKS TO THE TEST?

The swab test may tickle your throat or cause you to cough. The nasal aspirate may be uncomfortable or cause your eyes to tear. These effects are temporary.

WHAT DO THE RESULTS MEAN?

A negative result may mean your symptoms were caused by a pathogen not included in the panel of tests. It could also mean that your sample was not collected correctly, the sample wasn't large enough, or you have a condition not caused by a virus or bacteria.

A positive result means a specific pathogen was found. It tells you which type of infection you have. If more than one part of the panel was positive, it means you may be infected with more than one pathogen. This is known as a co-infection.

Based on your results, your provider will recommend treatment and/or order more tests. These may include a bacteria culture, viral blood tests, and a Gram stain. The tests may help confirm your diagnosis and guide treatment.

If you have questions about your results, talk to your provider. To understand the results of a respiratory pathogens panel, your provider may consider your symptoms, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A RESPIRATORY PATHOGENS PANEL?

Respiratory infections are common and can be contagious. They may be spread through coughing, sneezing, and contact with contaminated surfaces even before you start showing symptoms. You may help prevent getting or spreading a respiratory infection by:

Washing your hands thoroughly and often

Using hand sanitizer with at least 60% alcohol

Avoiding close contact with people who have a respiratory infection

Cleaning and disinfecting surfaces
Getting a yearly flu shot
Getting COVID-19 vaccines (as recommended by your provider)

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Respiratory Syncytial Virus (RSV) Tests

<https://medlineplus.gov/lab-tests/respiratory-syncytial-virus-rsv-tests/>

RESPIRATORY SYNCYTIAL VIRUS (RSV) TESTS

WHAT IS AN RSV TEST?

RSV stands for respiratory syncytial virus. It is a viral infection that affects the respiratory tract. Your respiratory tract includes your lungs, nose, and throat. RSV testing checks a fluid sample from your nose to see if the virus is in your body and causing symptoms of a respiratory infection.

RSV is very contagious, which means it spreads easily from person to person. It's also very common and goes away after 1-2 weeks without the need for testing or specific treatment. Most children get RSV by the age of two.

RSV usually causes mild symptoms that are like a cold (such as a cough, runny nose, and congestion). But the virus may lead to serious breathing problems, especially in certain groups of people, including:

Infants, especially those 6 months and younger

Older adults, especially those 65 years and older

People with heart or lung diseases

People with weakened immune systems

Two types of tests are commonly used to diagnose RSV infections:

Rapid RSV antigen tests are the most common test for RSV. They check a fluid sample from your nose for certain proteins from the virus called antigens. RSV antigens trigger your immune system to attack the virus. Rapid antigen tests can provide results in an hour or less.

Molecular tests called RT-PCR (polymerase chain reaction) tests look for genetic material from the virus in your sample. These tests can find smaller amounts of the virus than antigen tests (which means that they are more sensitive and are better at being able to tell if the virus is in your body). So, RT-PCR tests may be used for older children and adults who tend to have less of the virus in their noses than infants and younger children. Samples are usually sent to a lab for testing. In certain cases, your health care provider may order a molecular test called a respiratory pathogens panel. This test checks for RSV and other respiratory viruses like those that cause the flu and COVID-19.

Other names: respiratory syncytial antibody test, RSV rapid detection

WHAT IS IT USED FOR?

An RSV test is used to see if RSV is the cause of moderate to severe cold symptoms in infants, older adults, and people with weakened immune systems. People with mild symptoms may be tested as part of public health efforts to track the spread of the virus.

The test is usually done during the "RSV season," the time of year when RSV outbreaks are more common. In the United States, RSV season usually starts in mid-fall and ends in early spring, but the season may vary in different parts of the country.

WHY DO I NEED AN RSV TEST?

Healthy adults and older children usually don't need RSV testing. Most RSV infections only cause mild symptoms such as a runny nose, sneezing, and headaches. But an infant, younger child, or an adult over age 65 may need an RSV test if they have symptoms of moderate or serious infection. These include:

Fever

Wheezing

Serious cough

Breathing faster than normal or trouble breathing

Blue skin color

In infants less than six months old, the only symptoms of RSV may be:

Acting irritable
Moving less than usual
Lacking appetite
Pausing while breathing

WHAT HAPPENS DURING AN RSV TEST?

Tests to diagnose RSV are done on a sample of fluid from your nose. There are different ways to collect the sample: Nasal aspirate or wash. This is the most common way to collect a sample for RSV testing. To do a nasal aspirate, a health care professional will insert a saline solution (salt water) into your nose and remove it with gentle suction. Nasal swab test. This method is not preferred because it's not good at collecting enough of the virus to be able to test for it. A health care professional will use a special swab to take a sample from your nose. RSV test samples should be taken during the first few days after symptoms begin. That's because the amount of the virus in your nose decreases over time, which may make test results less accurate. At-home test kits are available to buy without a prescription. The test checks for RSV, flu, and COVID-19 using one sample. The kit includes a nasal swab to collect the sample to send to a lab for testing. Talk with your provider about using an at-home test. Blood tests aren't usually used to diagnose RSV, but they can show if a recent illness was an RSV infection. So public health officials may use blood tests to measure the size of an RSV outbreak in the community.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an RSV test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to RSV testing.
A nasal aspirate or wash may feel uncomfortable. These effects are temporary.
A swab test may cause a gagging feeling or brief discomfort when your nose is swabbed.

WHAT DO THE RESULTS MEAN?

A negative result means that no signs of the virus were found in your sample. This may mean that another illness is causing your symptoms.
But a negative test result does not rule out RSV. It's possible that there was not enough of the virus in the sample for the test to find it.
A positive result means that you likely have an RSV infection.
Infants, young children and older adults may need treatment in the hospital if they are having trouble breathing or have lost too much fluid and become dehydrated. Treatment may include oxygen and intravenous (IV) fluids. In the most serious cases, a breathing machine called a ventilator may be needed, but this is uncommon.
Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN RSV TEST?

If you have RSV symptoms, but are otherwise in good health, your provider probably won't order RSV testing. Most healthy adults and children with RSV will get better in 1-2 weeks. Your provider may recommend over-the-counter medicines and drinking plenty of fluids to relieve your symptoms. Talk with your child's provider before giving your child any medicines.
RSV can infect anyone, but a few ways to prevent getting or spreading the infection include to:

Practice good hygiene by washing your hands thoroughly and often, keeping kitchen and bathroom countertops clean, and washing toys regularly.

Cover your mouth and nose when you sneeze and don't share drinking glasses.

Talk to your provider about whether your infant or young child should take medicine to protect against RSV. Nirsevimab and palivizumab are two medicines that have antibodies (proteins specifically made to attack bacteria or a virus) against RSV.

Consider an RSV vaccination (as recommended by your provider) if you are pregnant, an older adult (age 75 years and older), or age 60-74 with a weaker immune system.

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Reticulocyte Count

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RETICULOCYTE COUNT

WHAT IS A RETICULOCYTE COUNT?

A reticulocyte count (retic count) measures the number of reticulocytes in your blood. Reticulocytes are immature (still developing) red blood cells (RBCs). Your body makes reticulocytes in your bone marrow. Then it sends them into your blood, where they mature into red blood cells within one to two days. The main job of the red blood cells is to move oxygen from your lungs to every cell in your body.

Your bone marrow constantly creates new red blood cells as your body needs them. A reticulocyte count checks to see if your bone marrow is making the right amount of red blood cells when old ones die off. If you have too few, your tissues may not get enough oxygen. If you have too many, you could be at risk for blood clots or other health concerns. This test can check for certain health conditions or show how well a treatment is working.

If the reticulocyte count is too high or too low, it can be a sign of a serious health condition, including anemia and conditions of the bone marrow, liver, and kidneys.

Other names: retic count, reticulocyte percent, reticulocyte index, corrected reticulocyte, reticulocyte production index, RPI

WHAT IS IT USED FOR?

A reticulocyte count is most often used to:

Diagnose specific types of anemia (a condition in which your blood has a lower-than-normal amount of red blood cells)

See if treatment for anemia is working

See if bone marrow is producing the right amount of red blood cells

Check bone marrow function after chemotherapy or a bone marrow transplant

WHY DO I NEED A RETICULOCYTE COUNT?

You may need this test if:

Other blood tests show your red blood cell levels are not normal. These tests may include a complete blood count, hemoglobin test, and/or hematocrit test.

You are being treated with radiation or chemotherapy.

You recently received a bone marrow transplant.

You may also need this test if you have symptoms of anemia. They include:

Shortness of breath

Weakness or fatigue

Headache

Dizziness

Arrhythmia (a problem with the rate or rhythm of your heartbeat)

Sometimes newborn babies may have a reticulocyte count test to check for a blood disorder called hemolytic disease of the newborn (HDN). This condition may happen if your blood does not match the blood type of your unborn baby. This is known as Rh incompatibility. It causes your immune system to attack your baby's red blood cells.

You most likely will be tested for Rh incompatibility during a routine prenatal screening. This screening helps to prevent HDN by providing early treatment.

WHAT HAPPENS DURING A RETICULOCYTE COUNT?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

To test a newborn, a health care provider will clean your baby's heel with alcohol and poke the heel with a small needle. The provider will collect a few drops of blood and put a bandage on the site.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a reticulocyte count test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is very little risk to your baby with a needle stick test. Your baby may feel a little pinch when the heel is poked, and a small bruise may form at the site. This should go away quickly.

WHAT DO THE RESULTS MEAN?

To understand the results of a reticulocyte count test, your provider will consider your symptoms, medical history, and the results of other blood tests.

If your results show a higher-than-normal reticulocyte count (reticulocytosis), it may mean:

You have hemolytic anemia, a type of anemia in which red blood cells are destroyed faster than the bone marrow can replace them.

You have had new or ongoing bleeding (hemorrhage), and your body is making many more red blood cells to replace what you've lost.

Your baby has hemolytic disease of the newborn (HDN), a condition that limits the ability of a baby's blood to carry oxygen to organs and tissues.

If your results show a lower-than-normal reticulocyte count, it may mean you have:

Iron deficiency anemia, the most common type of anemia. It happens when you don't have enough iron in your body.

Pernicious anemia, a type of megaloblastic anemia which can happen when your body is not getting enough of certain B vitamins (B12 and folate) in your diet or when your body can't absorb enough B vitamins.

Aplastic anemia, an uncommon type of anemia, happens when the bone marrow isn't able to make enough blood cells.

Bone marrow failure, which may be caused by an infection or cancer.

Kidney disease.

Cirrhosis, or scarring of the liver.

If your reticulocyte test results were not normal, it doesn't always mean you have anemia or another health condition.

For example, your reticulocyte count may not be normal because:

Your reticulocyte counts may be higher during pregnancy.

You may have an increase in your count for a short time if you move to a place with a high altitude. The count should return to normal once your body gets used to the lower oxygen levels that happen in higher altitudes.

Some medicines may increase or decrease your reticulocyte count.

If you have questions about your results or your child's results, talk to your health care provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Rheumatoid Factor (RF) Test

<https://medlineplus.gov/lab-tests/rheumatoid-factor-rf-test/>

RHEUMATOID FACTOR (RF) TEST

WHAT IS A RHEUMATOID FACTOR (RF) TEST?

A rheumatoid factor (RF) test checks a sample of your blood for rheumatoid factor, a protein made by the immune system.

Normally, your immune system makes proteins called antibodies to attack germs that could make you sick. Rheumatoid factor, however, does not attack germs. It attacks healthy cells, usually immunoglobulin G (IgG), one of the most common antibodies made to fight germs. Your immune system makes rheumatoid factor by mistake or when it's overactive fighting an infection.

Many people have small amounts of RF in their blood and are healthy. Even high amounts aren't always a sign of a health condition. But if you have certain symptoms and higher levels of rheumatoid factors, you may have an autoimmune disorder or another health condition related to high RF levels.

Other names: RF Blood Test

WHAT IS IT USED FOR?

RF testing is mostly used with other tests to help diagnose rheumatoid arthritis. Rheumatoid arthritis (RA) is a type of autoimmune disorder that damages your joints and causes pain, swelling, and stiffness. It is a chronic (long-lasting) condition that can also affect your organs and cause other symptoms.

An RF test can also be used to check for other autoimmune disorders, certain infections, and certain types of cancer.

WHY DO I NEED AN RF TEST?

You may need an RF test if you have symptoms of rheumatoid arthritis. These may include:

Joint pain or stiffness when moving. This is usually worse in the morning or after inactivity. RA usually affects both joints. If one of your hands or knees is affected, so is the other.

Joint tenderness, redness, and warmth.

Joint swelling that may make activities hard. These could include things such as combing your hair, buttoning your clothes, or bending your knees.

Fatigue, fever, and a loss of appetite.

Hard bumps (rheumatoid nodules) under your skin near the joints.

You may also need an RF test if your health care provider wants to check for other conditions that can cause high RF levels. These may include:

Other autoimmune disorders, including:

Sjögren syndrome.

Lupus.

Scleroderma.

Juvenile idiopathic arthritis in children and teens.

Sjögren syndrome.

Lupus.

Scleroderma.

Juvenile idiopathic arthritis in children and teens.

Chronic infections, including:

Hepatitis C (liver).

Tuberculosis (mostly affects the lungs).

Endocarditis (heart).

Hepatitis C (liver).

Tuberculosis (mostly affects the lungs).

Endocarditis (heart).

Certain types of cancer, including leukemia.

People with rheumatoid arthritis often develop anemia, so if you have anemia, your provider may want to give you an RF test. It is rare to get tested for high RF levels if you don't have symptoms of any of the above conditions.

WHAT HAPPENS DURING AN RF TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

At-home tests for rheumatoid factor are available. The test kit provides everything you need to collect a sample of blood by pricking your finger. You'll mail your sample to a lab for testing. If you do a home test, remember to share your results with your provider.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an RF test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

The results of an RF test alone cannot diagnose a condition. However, your provider can use these results, along with your medical history and the results of other tests, to determine if you have rheumatoid arthritis or any other condition that causes high RF levels. Generally, your results will be either positive or negative.

A negative (normal) result means that you have little or no rheumatoid factor in your blood. But that doesn't rule out rheumatoid arthritis or another health problem. If you have symptoms of rheumatoid arthritis, but your RF test results are normal, your provider may order more tests to make a diagnosis.

A positive (abnormal) result means that a higher level of rheumatoid factor was found in your blood. Your provider will also consider other test results along with your symptoms and medical history to find out if you have:

Rheumatoid arthritis or another autoimmune disease.

A chronic infection.

Certain cancers.

Sometimes, rather than a negative or positive result, your results may tell you how much RF was in the sample. Your results may also give a normal range for RF for that laboratory.

Generally, the higher your RF levels, the more likely it is that you have a condition linked to rheumatoid factors. However, people with rheumatoid arthritis can have little or no rheumatoid factor. Similarly, people without rheumatoid arthritis can have a lot of RF in their blood. Your provider may refer you to a rheumatologist (a provider who specializes in autoimmune disorders, including RF) to get more testing if your results are inconclusive.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN RF TEST?

An RF test is not used to diagnose osteoarthritis. Although rheumatoid arthritis and osteoarthritis both affect the joints, they are very different diseases. Osteoarthritis is not an autoimmune disease and does not involve rheumatoid factor.

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Salicylates Level

<https://medlineplus.gov/lab-tests/salicylates-level/>

SALICYLATES LEVEL

WHAT IS A SALICYLATES LEVEL TEST?

This test measures the amount of salicylates in the blood. Salicylates are a type of drug found in many over-the-counter and prescription medicines. These can include:

Aspirin, which is a common pain reliever

Bismuth subsalicylate, which is used to treat diarrhea, heartburn, and upset stomach

Magnesium salicylate, which is used to treat pain

Methyl salicylate (oil of wintergreen), which is which is most often found in creams to treat muscle pain

Aspirin is the most common type of salicylate. It is used to treat pain, fever, and inflammation. Popular brand name aspirins include Bayer and Ecotrin. Some other medicines may also contain aspirin, including some antacids, pain relievers, and cough and cold medicines.

Aspirin and other salicylates affect blood clotting. A blood clot is a mass of blood that forms when platelets, proteins, and cells in the blood stick together. Blood clotting is an important process that prevents you from losing too much blood when you get injured. But too much blood clotting can cause a heart attack or stroke. People at risk for blood clots may be advised to take baby aspirin or other low-dose aspirin every day to help prevent dangerous blood clots.

Even though it's called baby aspirin, it should not be given to children or teens unless your health care provider tells you to. For these age groups, taking aspirin when they have a viral infection can cause a life-threatening disorder called Reye syndrome.

Aspirin and other salicylates are usually safe and effective for adults when taken at the proper dose. However, if you take too much, it can cause a serious and sometimes fatal condition called aspirin or salicylate poisoning. Check with your provider before taking medicines containing salicylates for more than ten days.

Other names: acetylsalicylic acid level test, salicylate serum test, aspirin level test

WHAT IS IT USED FOR?

A salicylates level test is most often used to:

Help diagnose acute or gradual aspirin or other salicylate poisoning:

Acute poisoning happens when you take too much aspirin or other salicylate at once.

Gradual, or chronic, poisoning happens when you take lower doses over a long period of time.

Acute poisoning happens when you take too much aspirin or other salicylate at once.

Gradual, or chronic, poisoning happens when you take lower doses over a long period of time.

Monitor people taking prescription-strength aspirin for arthritis or other inflammatory conditions. The test can show whether you are taking enough to treat your disorder or are taking a harmful amount.

WHY DO I NEED A SALICYLATES LEVEL TEST?

You may need this test if it's possible that you have acute poisoning from aspirin or other salicylates. The symptoms of acute poisoning usually happen three to eight hours after an overdose. The symptoms can vary, based on the amount of salicylates in your blood and whether you are an adult, teenager, or child. But the symptoms often include:

Nausea and vomiting

Rapid breathing (hyperventilation)

Ringing in the ears (tinnitus)

Sweating

You may also need this test if you have symptoms of gradual poisoning from aspirin or other salicylates. The symptoms

may take days or weeks to show up and can include:

Rapid heartbeat

Fatigue

Headache

Confusion

Hallucinations

If you think you or your child has taken too much aspirin or other salicylates, call your provider right away. If you notice symptoms of aspirin poisoning, call 911 or the poison control center, or go to the nearest emergency room. Early treatment can prevent damage to your organs or death.

WHAT HAPPENS DURING A SALICYLATES LEVEL TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you regularly take aspirin or other salicylate, you may need to stop taking it for at least four hours before your test. Your provider will let you know if there are any other special instructions to follow.

ARE THERE ANY RISKS TO A SALICYLATES LEVEL TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If your results show a high level of salicylates, you may need immediate treatment. If levels get too high, it can be fatal. The treatment will depend on the amount of overdose.

If you have questions about your results, talk to your provider. To understand the results of a salicylate test, your provider may consider your symptoms, medical history, and the results of other blood tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A SALICYLATES LEVEL TEST?

A daily dose of low-dose or baby aspirin used to be recommended as a way to reduce the risk of a heart attack or stroke for many older adults. But daily aspirin use may cause bleeding in the stomach or brain. That is why it is no longer recommended for adults without heart disease risk factors.

Because heart disease is usually more dangerous than the complications from bleeding, it may still be recommended for those at high risk. Risk factors for heart disease include family history and a previous heart attack or stroke.

Before you stop or start taking aspirin, be sure to talk with your provider.

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Semen Analysis

<https://medlineplus.gov/lab-tests/semen-analysis/>

SEMEN ANALYSIS

WHAT IS A SEMEN ANALYSIS?

A semen analysis, also called a sperm count, measures the quantity and quality of semen and sperm. Semen is the thick, white fluid released from the penis during sexual climax (orgasm). This release is called ejaculation. Semen has sperm, the cells made in the male reproductive system that carry half of the genetic material needed to make a baby.

Problems with the quantity or quality of your semen and sperm, such as low sperm count or sperm that don't move properly, can affect your sperm's ability to reach and fertilize an egg. An egg is the cell made in the female reproductive system that carries the other half of the genetic material needed to make a baby. Problems with your sperm's ability to fertilize an egg can cause infertility, which is the inability to conceive a baby after a year of trying.

Other names: sperm count, sperm analysis, semen testing, male fertility test

WHAT IS IT USED FOR?

A semen analysis helps your health care provider find out if there is a problem with your semen or sperm. This test checks your semen and sperm using a variety of measurements:

Volume is the amount of semen in your sample.

Sperm count is the number of sperm in the entire sample.

Sperm concentration is the number of sperm present in a milliliter of semen.

Sperm movement (motility) is the percentage of sperm that are moving efficiently (in a straight line or in large circles).

Sperm shape (morphology) is the number of sperm that have a normal shape.

pH describes the acidity of your semen.

Time to liquefaction is how quickly your semen changes into a fluid.

Abnormal results in any of these measurements may be causing infertility.

The test may also be used to see if a vasectomy has been successful. A vasectomy is a surgery on the male reproductive system that's done for birth control. To prevent pregnancy, a vasectomy cuts the tubes that carry sperm out of your testicles (testes) so that the sperm can't get into your semen. A semen analysis is usually done 8 to 16 weeks after a vasectomy to make sure your semen has no sperm.

WHY DO I NEED A SEMEN ANALYSIS?

You may need a semen analysis if:

You and your partner have been trying to have a baby for at least 12 months without success.

You've recently had a vasectomy and need to make sure that you don't have any sperm in your semen.

WHAT HAPPENS DURING A SEMEN ANALYSIS?

You will need to provide a semen sample. For accurate results, the semen must be examined quickly before the sperm begin to die. You will usually provide a sample in a private room at a lab by masturbating and collecting your semen in a sterile container.

To get accurate results:

Urinate (pee) and wash your hands and penis before collecting your sample. This will keep bacteria from your skin out of the sample.

Don't touch the inside of the container.

Don't use lubricants or saliva. They could harm your sperm.

Make sure to collect all the semen you produce. If you miss collecting any of it, let your provider know. Don't try to collect any semen that didn't make it into the container. This could contaminate the sample.

You may also collect a semen sample at home. The two main options are:

At-home self-test kits. To use these, you collect a semen sample and follow the kit's instructions on how to do a sperm count. Some kits may also check motility and concentration. These tests are useful, but they can't provide as much information as a full semen analysis.

Self-collection kits. To use these, collect a semen sample and send it to a lab. It is important that you keep the sample at room temperature and get it to a lab within one hour.

If you prefer, you may be able to collect your sample at home during sex by using a special type of condom that your provider gives you. But as with self-collection kits, you will have to keep the sample at body temperature and take it to the lab within 30 to 60 minutes after collecting. Be sure to follow any additional instructions your provider gives you. For example, you may need to transfer your sample from the condom to a sterile container.

Because your sperm count and semen quality can vary from day to day, you may need to provide two or more semen samples within a week or two. Talk with your provider if you have questions or concerns about providing a semen sample. You can also talk to your provider about which type of test you prefer.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You'll need to avoid all sexual activity that results in ejaculating sperm for 2-7 days before the sample is collected. This will help make sure your sperm count is at its highest level. Your provider will give you instructions.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to a semen analysis.

WHAT DO THE RESULTS MEAN?

Results to check fertility:

Your semen analysis will include the measurements listed above. It will also often include the normal ranges for these measurements:

Volume: 1.5 to 7.6 milliliters. Anything lower than 1.5 milliliters may cause infertility.

Sperm count: More than 39 million sperm in the entire sample. A count lower than this may make it difficult to get your partner pregnant.

Sperm concentration: 15 to 259 million sperm per milliliter of semen. Anything lower than 15 million may make it difficult to get your partner pregnant.

Sperm movement (motility): Over 40%. If less than 40% of your sperm are able to move efficiently, the odds of one reaching and fertilizing the egg may be lower.

Sperm shape (morphology): Over 4% of your sperm should have a normal shape. Sperm with abnormal shapes may not be able to fertilize an egg.

pH: 7.2 to 8.0 pH. A pH outside of this range can kill sperm or affect their ability to move.

Time to liquefaction: 15 to 30 minutes after collection. Your results may give a timeframe, or they may say normal.

White blood cells (leukocytes): Less than 1 million white blood cells per 1 milliliter of semen. More than this may be a sign that an infection is affecting your fertility.

If your semen analysis results aren't all normal, it doesn't mean you're permanently infertile. But it does show that your sperm may be part of the reason you and your partner are having difficulties getting pregnant.

Your provider may order more tests, such as ultrasounds and blood tests, to help find the cause of the problem so it can be treated. There are many possible causes, including:

Certain habits you may be able to change, such as smoking.

A physical problem in your testicles, such as an enlarged vein (varicocele).

Certain genetic disorders or other health conditions, including hormone disorders, diabetes, and cystic fibrosis.

Treatments for infertility depend on the cause. To help improve your chances of getting your partner pregnant, your

provider may refer you to a provider who specializes in infertility.

Results for semen analysis to check a vasectomy:

The results of your semen analysis will tell you if there's any sperm in your semen. It's not unusual to find sperm in several ejaculations after a successful vasectomy.

If sperm is found, you'll need repeat testing until your sample has no sperm. In the meantime, you and your partner will have to use other birth control to prevent pregnancy.

If no sperm is found, you and your partner should be able to stop using other forms of birth control.

Learn more about laboratory tests, reference ranges, and understanding results.

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Sentinel Lymph Node Biopsy

<https://medlineplus.gov/lab-tests/sentinel-lymph-node-biopsy/>

SENTINEL LYMPH NODE BIOPSY

WHAT IS A SENTINEL LYMPH NODE BIOPSY?

A sentinel lymph node biopsy (SLNB) is a surgical procedure that checks your lymph nodes for cancer cells. An SLNB is only done if you have already been diagnosed with cancer. This procedure finds, removes, and checks the sentinel lymph node to determine the stage of your cancer. Cancer staging is usually based on the size of your tumor and whether the cancer has spread (metastasized) to nearby areas, lymph nodes, or other parts of your body. Knowing the stage of your cancer can help your health care provider plan the best treatment for you.

Your lymph nodes are located throughout your body, including your underarms, neck, chest, abdomen (belly), and groin. They are part of the lymphatic system, which helps to protect you from illness and maintain healthy fluid levels in your body.

Your lymphatic system is part of your body's immune system. Your lymph fluid carries fluid and cells throughout your body to help fight infections. It may also filter cancer cells (if there are any). But sometimes, with certain cancers such as breast cancer and melanoma (the most serious type of skin cancer), some cancer cells may spread to your lymph nodes.

Cancers spread when cells break off from the original (primary) tumor and are carried by the lymphatic system to other parts of your body. The sentinel lymph node is the first lymph node where these types of cancers are most likely to spread. It's usually located near the original tumor site and may receive drainage from the tumor. Your provider tests your sentinel lymph node to see if your cancer has spread. Sometimes, you may have more than one sentinel lymph node.

Other names: lymph node biopsy, sentinel node biopsy, sentinel lymph node mapping and biopsy

WHAT IS IT USED FOR?

A sentinel lymph node biopsy is used to find out whether an early-stage cancer has spread through your lymphatic system to your lymph nodes or other parts of your body. It's most often used for people who have breast cancer or melanoma and is usually done when the tumor is being removed.

WHY DO I NEED A SENTINEL LYMPH NODE BIOPSY?

You may need this test if you've been diagnosed with breast cancer, melanoma, or certain other types of cancer. It can show whether your cancer is spreading and help your provider create the best treatment plan for you.

WHAT HAPPENS DURING A SENTINEL LYMPH NODE BIOPSY?

A sentinel lymph node biopsy may be done in a hospital or an outpatient surgical center. The procedure usually includes a special type of imaging test called lymphoscintigraphy. Lymphoscintigraphy may also be called lymph node mapping. It is the process of taking pictures of your lymphatic system to locate the sentinel lymph node. The procedure includes the following steps:

Your provider will apply medicine to numb the skin over the procedure area.

You will be injected with a small amount of a radioactive substance called a tracer near the tumor. The tracer will travel and collect in the sentinel lymph node, or nodes. You may also be injected with a blue dye that stains the lymph node, making it easier to see.

You will be given general anesthesia. General anesthesia is a medicine that makes you unconscious. It makes sure you won't feel any pain during the surgery. A specially trained doctor called an anesthesiologist will give you this medicine.

Once you're unconscious, a camera will take images of the area. The camera finds and records the location of the tracer, which will have settled in the sentinel node.

The node will be removed and sent to a lab, where it will be checked for cancer cells. This part of the test is known as a biopsy.

The original tumor is also usually removed during the procedure.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You will probably need to fast (not eat or drink) for several hours before the test. You may also need to stop taking blood thinners, such as aspirin, for a certain time before your test. Your provider will let you know when you need to stop taking your medicine and anything else to do to prepare for the test.

In addition, be sure to arrange for someone to take you home. You may be groggy and confused after you wake up from the procedure.

ARE THERE ANY RISKS TO THE TEST?

You may have a little bleeding, pain, or bruising at the biopsy site.

There is also a small risk of infection, which can be treated with antibiotics.

Allergic reactions to the tracer are rare and usually mild.

If only the sentinel lymph node is removed, there is a low risk of lymphedema (swelling, usually of the arms or legs) from the lymph fluid. The risk increases with the number of lymph nodes removed.

There is very little exposure to radiation in a sentinel lymph node biopsy. While radiation exposure from the tracer in a sentinel lymph node biopsy is safe for most adults, it can be harmful to a developing baby. So be sure to tell your provider if you are pregnant or think you may be pregnant.

WHAT DO THE RESULTS MEAN?

The results will be given as positive or negative.

If your results are positive, it means cancer was found in the sentinel lymph node and may have spread to nearby lymph nodes and/or other organs. Your provider may recommend removing and testing more lymph nodes to check for cancer cells.

If your results are negative, it means no cancer cells were found in the sentinel lymph node, and it's unlikely that the cancer has spread. You will not need to have any more lymph nodes removed.

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A SENTINEL LYMPH NODE BIOPSY?

While sentinel lymph node biopsies are mostly used to stage breast cancer or melanoma, it is sometimes used for penile and uterine cancer. It is also being studied for use with other types of cancer, including:

Colon cancer

Stomach cancer

Thyroid cancer

Non-small cell lung cancer

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Sexually Transmitted Infection (STI) Tests

<https://medlineplus.gov/lab-tests/sexually-transmitted-infection-sti-tests/>

SEXUALLY TRANSMITTED INFECTION (STI) TESTS

WHAT ARE STI TESTS?

STIs (sexually transmitted infections) or sexually transmitted diseases (STDs) are infections that are passed from one person to another through sexual contact. STIs are usually spread during vaginal, oral, or anal sex. But sometimes, they can spread through other sexual contact involving the penis, vagina, mouth, or anus. This is because some STIs, like herpes and HPV (human papillomavirus), are spread by skin-to-skin contact.

Other ways that some STIs may be spread include:

Through close personal contact, such as kissing

During pregnancy to the fetus or during childbirth

During breastfeeding

Through blood transfusions

By sharing needles

Certain bacteria, viruses, and parasites cause STIs. Millions of Americans get an STI every year, and many of those infections are in teens and young adults. Common STIs include chlamydia, gonorrhea, genital herpes, HIV, and HPV infection.

Left untreated, certain STIs can cause long-term health problems such as blindness, brain damage, infertility, birth defects, and even death. Fortunately, all STIs can be treated, and some can be cured. However, that doesn't prevent you from getting infected again if you don't take steps to prevent infection. STI testing can help you get diagnosed and treated so you may avoid serious complications.

Other names: STD screening, STI (sexually transmitted infection) tests, STI screening, STD testing

WHAT ARE THEY USED FOR?

STI tests are used to find out if you have been infected with an STI. Testing can help you get the treatment you need and may help prevent the spread of infection.

WHY DO I NEED AN STI TEST?

STIs don't always cause symptoms, so catching or spreading an infection is possible even if you or your partner seem healthy. That's why STI screening tests are important. They look for diseases before you have symptoms.

Anyone who is sexually active could get an STI. So talk to your health care provider about your risk for HIV and other STIs and whether you need to be tested. If you are sexually active, you might consider getting a yearly test for HIV. Other STI testing is mostly recommended for people who are at higher risk for certain STIs:

Sexually active women under the age of 25. Experts recommend that women in this age group get a chlamydia test and a gonorrhea test every year. Sexually active women 25 years and older should be tested if they have new or multiple sex partners or have a partner who has been diagnosed with an STI.

Women 21 to 30 years old. Women in this age group should get a Pap smear to check for cell changes in the cervix that could become cancer if not treated. These can be caused by HPV (human papillomavirus).

Women 30 years and older. Women in this age group should talk to their provider about whether to get a Pap smear, HPV test, or both.

If you are pregnant, you will usually be tested for HIV, hepatitis B, chlamydia, and syphilis. If you are pregnant and have certain risk factors, you may also be tested for gonorrhea and hepatitis C.

Sexually active men and women who have unprotected sex (sex without a condom) and have more than one sexual partner. They should be tested regularly for HIV and other STIs as recommended by their provider.

Men who have sex with men. Compared with other groups, men who have sex with men and bisexual men have a

higher risk of STIs. Men in this group should be tested at least once a year for syphilis, chlamydia, gonorrhea, and HIV. Men with multiple partners may benefit from more frequent testing.

People with HIV. If you have HIV, you are at higher risk of getting another STI. People in this group should be tested for syphilis, gonorrhea, and herpes. A chlamydia test is also usually recommended.

People who use injectable drugs. Sharing needles puts you at risk for HIV. Regular HIV testing is recommended for this group.

Some STIs do not cause symptoms or may only cause mild symptoms. So it is possible to have an infection and not know it. Symptoms vary depending on the type of STI, but you may need testing if you have one or more of the following symptoms:

Painful or frequent urination (peeing)

Pain during sex

Unusual odor in vaginal discharge

Vaginal itching

Discharge and/or itching from the penis

Sores or bumps in the genital or rectal area

Blisters or sores in or around the mouth

Anal itching, soreness, or bleeding

Even without symptoms, STIs can still be harmful and may be passed on to your partner during sex.

WHAT HAPPENS DURING AN STI TEST?

Some STIs may be diagnosed during a physical exam, blood tests, or a microscopic examination of a sore or fluid swabbed from the vagina, penis, or anus. Depending on the type of suspected infection, you may get one of the following types of tests:

Blood tests

Used to diagnose syphilis, HIV, hepatitis B, and sometimes herpes.

During the test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial.

Urine tests

Used to diagnose trichomoniasis, chlamydia, and sometimes gonorrhea.

During the test, you will provide a sterile sample of urine in a cup as instructed by your provider.

Swab tests

Used to diagnose HPV, chlamydia, gonorrhea, and herpes.

During the test, a provider will use a special swab to take a sample from the site of the infection. In women, samples may be taken from the vagina or cervix. In men, samples may be taken from the penis or urethra, the tube that carries urine out of the body.

Lumbar puncture, also known as a spinal tap

This is not a frequently used STI test, but it may be ordered if your provider thinks you have an advanced stage of syphilis or if a herpes infection has affected your brain or spinal cord.

For this test, a provider will inject an anesthetic into your back so you won't feel any pain during the procedure.

Once the area is numb, the provider will insert a thin, hollow needle between two vertebrae in your lower spine. Vertebrae are the small bones that make up your spine. Your provider will then withdraw a small amount of fluid for testing.

You may also be able to check for STIs with an at-home test. Home test kits are available for several STIs, including chlamydia, gonorrhea, syphilis, trichomoniasis, and hepatitis C.

These tests usually involve using a device (lancet) to prick your finger for a drop of blood, collecting a urine sample, or taking an oral swab. You should discuss your at-home STI test results with your provider to see if you need further testing and/or treatment.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a blood test. Women may need to avoid douches and vaginal creams for 24

hours before a urine or swab test.
You may need to empty your bladder before a lumbar puncture.
Your provider will let you know if you need other preparations.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There are no known risks to having urine or swab tests.

If you had a lumbar puncture, you may have pain or tenderness in your back where the needle was inserted. You may also get a headache after the procedure. The headache can last for several hours or up to a week or more, but your provider may suggest treatment to relieve the pain.

WHAT DO THE RESULTS MEAN?

The test result will show whether you have an STI.

If you have an STI, be sure to take the medicine as prescribed by your provider. It's also very important that you tell your partner so they can get tested and treated if necessary. Some types of STIs can cause serious health problems if they are not treated.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT STI TESTS?

You can take steps to help protect yourself from STIs. The only way to completely prevent them is by not having sex. But if you do decide to have sex, you can lower your risk of getting an STI by:

Having you and your partner tested for STIs before having sex.

Reducing your number of sex partners.

Being in a long-term relationship with "mutual monogamy" with just one partner who has tested negative for STIs. This means that you agree to be sexually active only with them, and they agree to be sexually active only with you.

Using latex condoms correctly every time you have sex. Correct use of condoms greatly reduces but does not eliminate, the risk of catching or spreading STIs. If you or your partner is allergic to latex, you can use polyurethane condoms.

Getting vaccinated against HPV and hepatitis B.

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SHBG Blood Test

<https://medlineplus.gov/lab-tests/shbg-blood-test/>

SHBG BLOOD TEST

WHAT IS AN SHBG BLOOD TEST?

An SHBG blood test measures the level of a protein called sex hormone binding globulin (SHBG) in a sample of your blood. SHBG is made mostly in your liver. It binds (attaches) to sex hormones—such as testosterone and estrogen and helps control how much of these hormones are active in your body.

All of the sex hormones in your blood are either bound to proteins or unbound (free). Free sex hormones work with your tissues, helping with a variety of bodily functions including bone and muscle growth and reproductive health. Sex hormones that are bound to proteins, such as SHBG, however, cannot work with your tissues.

An SHBG test is most often done if you have signs and/or symptoms of having too much or too little testosterone. Though often considered a male sex hormone, testosterone is present in females in smaller amounts. In both sexes, it helps control your sex drive and contributes to muscle and bone growth. In males, it is also involved in:

Body hair growth.

A deepening voice during puberty.

The production of sperm.

The creation of new red blood cells.

An increase in height.

Other names: testosterone-estrogen binding globulin, TeBG

WHAT IS IT USED FOR?

Measuring SHBG levels can provide information about how much of your testosterone is active in the tissues of your body. The results of this test can help your provider understand how testosterone is working in the body for:

Adult males who have symptoms that may be caused by too little testosterone.

Adult females who have symptoms that may be caused by too much testosterone.

Usually, your provider will order a total testosterone blood test—first. This test measures both free testosterone and testosterone attached to proteins such as SHBG. However, this test can't tell the difference between the two types. It can't tell your provider how much of your testosterone is free for your body to use. This means that you could have symptoms from too little or too much free testosterone even when your total testosterone level is normal.

By measuring the amount of free testosterone you have, an SHBG test can help your provider better understand if abnormal testosterone levels are causing your symptoms. SHBG and total testosterone tests may be done at the same time.

Your provider may also order an SHBG test to:

Help diagnose some hormonal disorders such as polycystic ovary syndrome—(PCOS), a condition that affects the female reproductive system. An SHBG test alone cannot diagnose this condition, but it can help providers make a diagnosis when paired with other test results.

Monitor treatments for hormonal disorders such as PCOS.

Help diagnose early puberty in children. Normally, SHBG levels are high in all children. The high levels prevent sex hormones from affecting body tissues. If SHBG levels are low, children may start puberty too early.

WHY DO I NEED AN SHBG BLOOD TEST?

You may need this test if you have signs and/or symptoms of abnormal testosterone levels, especially if results from a total testosterone test didn't explain your symptoms.

If you're male, you may need this test if you have symptoms of low testosterone levels, including:

Low sex drive.

Difficulty getting an erection.

Fatigue or loss of muscle mass.

Fertility problems.

If you're female, you may need this test if you have symptoms of high testosterone, including:

Too much hair growth on the face and body.

Deepening of voice.

Having menstrual periods that aren't regular or no periods at all.

Acne.

Weight gain.

Fertility problems.

Your provider may also order an SHBG test if:

You are undergoing hormone replacement therapy or treatment for hormone disorders.

You have signs and/or symptoms of PCOS, such as irregular menstrual periods.

A child shows signs of early puberty.

WHAT HAPPENS DURING AN SHBG BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an SHBG test. Tell your provider about all medicines, vitamins, and supplements you take. Certain medicines that contain hormones - such as birth control pills, estrogen therapy, or anabolic steroids - can affect SHBG levels. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your provider will usually look at your SHBG blood test results along with the results of a total testosterone test. This helps your provider estimate how much free testosterone is in your blood.

If your SHBG levels are too low, it means that more of your total testosterone is likely free and available for your body to use.

Low levels of SHBG may be related to:

Hypothyroidism (underactive thyroid).

Type 2 diabetes or insulin resistance.

Obesity or being very overweight.

Overuse of steroid medicines and/or anabolic steroids.

Cushing's syndrome.

In females, polycystic ovary syndrome (PCOS).

It's also normal for SHBG levels to decrease with older age as the production of sex hormones also decreases.

If your SHBG levels are too high, it likely means that less of your total testosterone is available for your tissues to use.

So, your tissues may not be getting enough testosterone.

High levels of SHBG may be related to:

Liver disease.

Hyperthyroidism (overactive thyroid).

Certain eating disorders.

Use of estrogen-containing medicines, such as hormone replacement therapy or birth control pills, in females.

Decreased production of sex hormones in males.

Your provider may order other tests to check if your hormones are in balance with each other. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Skin Biopsy

<https://medlineplus.gov/lab-tests/skin-biopsy/>

SKIN BIOPSY

WHAT IS A SKIN BIOPSY?

A skin biopsy is a procedure that removes a small sample of skin for testing. The skin sample is looked at under a microscope to check for a variety of conditions, including skin cancer.

There are three main ways to do a skin biopsy. The type of biopsy you get depends on the location and size of the abnormal area of skin, known as a skin lesion. These types include:

A punch biopsy, which uses a special tool with a round blade to remove a small, round piece of skin. This may be used to get a sample that goes deeper into the skin.

A shave biopsy, which removes a sample from the topmost layers of your skin with a razor blade. This is the most used of the three types.

An excisional biopsy, which removes the skin sample with a small knife called a scalpel (a small cutting blade).

Most skin biopsies can be done in a health care provider's office or other outpatient facility.

Other names: punch biopsy, shave biopsy, excisional biopsy, skin cancer biopsy, basal cell biopsy, squamous cell biopsy, melanoma biopsy

WHAT IS IT USED FOR?

A skin biopsy is used to help diagnose a variety of skin conditions including:

Skin disorders such as psoriasis and eczema

Bacterial or fungal infections of the skin

A provider can also use a skin biopsy to see if a suspicious mole or other growth is cancerous. Because skin cancer is the most common type of cancer in the United States, it is important for providers to check for skin cancer often. Its most common types are:

Squamous cell carcinoma. This type forms in the upper parts of the epidermis (the topmost layer of the skin) and rarely spreads. It's most often found in sun-exposed parts of the body such as your face, arms, and hands.

Basal cell carcinoma. This type forms in the basal cell layer (the lower parts) of the epidermis and also rarely spreads. This is the most common type of skin cancer and usually forms on the face, head, neck, and arms.

Melanoma. This type forms in the cells that make your skin's pigment (or color). These cells also protect deeper layers of the skin from some of the sun's harmful effects. It is less common than the other two, but also more dangerous. This is because it is more likely to spread.

If your provider thinks you may have basal or squamous cell carcinoma, they may perform a punch or shave biopsy. If they think you may have melanoma, they will likely perform an excisional biopsy to make sure they remove all of the suspicious tissue.

WHY DO I NEED A SKIN BIOPSY?

You may need a skin biopsy if you have certain skin symptoms such as:

A persistent rash.

Scaly or rough skin.

Open sores that won't heal.

Blistering skin, which may be a sign of pemphigus, an autoimmune disease.

A mole or new growth that has the "A-B-C-D-Es" of melanoma:

Asymmetrical -- the shape is not regular.

Border -- the edge is jagged.

Color -- the color is uneven.

Diameter -- the size is larger than a pea.

Evolving -- the mole or growth has changed in the past few weeks or months.

Asymmetrical -- the shape is not regular.

Border -- the edge is jagged.

Color -- the color is uneven.

Diameter -- the size is larger than a pea.

Evolving -- the mole or growth has changed in the past few weeks or months.

WHAT HAPPENS DURING A SKIN BIOPSY?

A provider will clean the site and inject an anesthetic (medicine) so you won't feel any pain during the procedure. The rest of the procedure steps depend on which type of skin biopsy you are getting:

Punch biopsy

A provider will place a special circular tool over the abnormal skin area (lesion) and rotate it to remove a small piece of skin (about the size of a pencil eraser).

The sample will be lifted out with a special tool.

If a larger skin sample was taken, you may need one or two stitches to cover the biopsy site.

Pressure will be applied to the site until the bleeding stops.

The site will be covered with a bandage or sterile dressing.

Shave biopsy

A provider will use a razor or a scalpel to remove a sample from the top layer of your skin.

Pressure will be applied to the biopsy site to stop the bleeding. You may also get a medicine that goes on top of the skin (also called a topical medicine) to help stop the bleeding.

Excisional biopsy

A surgeon will use a scalpel to remove the entire skin lesion (the abnormal area of skin).

The surgeon will close the biopsy site with stitches.

Pressure will be applied to the site until the bleeding stops.

The site will be covered with a bandage or sterile dressing.

After the biopsy, keep the area covered with a bandage until you've healed, or until your stitches come out. If you had stitches, they will be taken out 3-14 days after your procedure.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a skin biopsy.

ARE THERE ANY RISKS TO THE TEST?

You may have a little bruising, bleeding, or soreness at the biopsy site. If these symptoms last longer than a few days or they get worse, talk to your provider.

WHAT DO THE RESULTS MEAN?

If your results were normal, it means no cancer or skin disease was found. If your results were not normal, you may be diagnosed with one of the following conditions:

A bacterial or fungal infection.

A skin disorder such as psoriasis.

Skin cancer. Your results may mean you have one of three types of skin cancers: basal cell, squamous cell, or melanoma.

If you are diagnosed with basal cell or squamous cell cancer, the entire cancerous lesion may be removed at the time of

the skin biopsy or soon after. Often, no other treatment is needed. If you are diagnosed with melanoma, you will need more tests to see if the cancer has spread. Then you and your provider can develop a treatment plan that's right for you. Learn more about laboratory tests, reference ranges, and understanding results.

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Skin Cancer Screening

<https://medlineplus.gov/lab-tests/skin-cancer-screening/>

SKIN CANCER SCREENING

WHAT IS A SKIN CANCER SCREENING?

A skin cancer screening checks your skin for signs of skin cancer before you have any symptoms. To do this, you or your health care provider will examine your skin for moles, birthmarks, or other areas that have an unusual color, size, shape, or texture. If an area of skin doesn't look normal, you may need tests to find out if it's cancer.

Skin cancer is the most common cancer in the United States. The most common types of this cancer are:

Squamous cell carcinoma. This type forms in the upper parts of the epidermis (the topmost layer of the skin) and rarely spreads. It's most often found in sun-exposed parts of the body such as your face, arms, and hands.

Basal cell carcinoma. This type forms in the basal cell layer (the lower parts) of the epidermis and also rarely spreads. It is the most common type of skin cancer and usually forms on the face, head, neck, and arms.

Melanoma. This type forms in the cells that make your skin's pigment (or color). These cells also protect deeper layers of the skin from some of the sun's harmful effects. Melanoma is less common than the other two, but it's also much more serious. That's because it's more likely to spread.

The earlier your provider can detect these cancers, the more effective the treatments can be. This is especially true for melanoma.

Other names: skin exam

WHAT IS IT USED FOR?

Skin cancer screening, like other types of cancer screening, is used to find signs of cancer before you have symptoms. It's not used to diagnose cancer. If, during a screening test, your provider finds an unusual mole or other mark on your skin, you may need to have a test called a skin biopsy to find out whether you have cancer.

Your provider may also screen you for skin cancer more often if you have certain types of birthmarks or moles, or if you have a lot of moles on your body. If you notice any changes to these areas of skin, or if you develop any new moles, you should contact your provider.

WHY DO I NEED A SKIN CANCER SCREENING?

Some medical experts recommend checking your own skin regularly starting at age 18. That's because skin cancer is very common, and people of all skin colors can get it.

You may need a skin cancer screening with your provider or with a dermatologist (a doctor who specializes in skin disorders) if you:

Have had skin cancer in the past. In this case, it's usually recommended to have a regular yearly skin cancer screening with your provider or a dermatologist.

Have a higher-than-normal risk for getting skin cancer.

Find a suspicious area of skin during a self-exam.

A suspicious area of skin can mean:

Moles or spots of skin that have changed in size, shape, or color.

Moles or other skin spots that ooze, bleed, or become scaly or crusty.

Moles that are painful to the touch.

Sores that haven't healed within two weeks.

Shiny pink, red, pearly white, or clear-looking bumps.

You may have a higher risk of basal cell or squamous cell carcinoma if you have had:

Actinic keratosis, a pre-cancerous condition caused by too much exposure to the sun. This condition causes patches of thick, scaly skin.

Long-term inflammation or injuries such as severe burns.

Radiation therapy for cancer.

A weakened immune system.

Exposure to arsenic.

Your risk for melanoma is higher if you have:

Had many blistering sunburns, especially as a child or teenager.

A personal and/or family health history of melanoma.

A family health history of unusual moles, such as Gorlin syndrome or xeroderma pigmentosum.

Several large or many small moles.

Your risk for all types of skin cancer is higher if you have:

Had frequent exposure to natural sunlight or artificial sunlight, such as tanning beds.

Pale skin that burns and freckles easily.

Skin that tans a little or not at all.

Blond or red hair.

Light colored eyes, including blue or green.

Ask your provider how often to do a self-exam and whether you need to have regular skin cancer screenings from a provider.

WHAT HAPPENS DURING A SKIN CANCER SCREENING?

For a self-exam to screen for skin cancer, you'll check your skin for any suspicious areas of skin. This can mean unusual moles or sores and any marks on your skin that have changed. As a good rule of thumb, follow the "ABCDEs" of melanoma. This stands for:

Asymmetry: Does the mole or spot have an irregular shape with two parts that look very different?

Border: Is the border of the mole ragged or irregular?

Color: Is the color uneven?

Diameter: Is the mole or spot bigger than the size of a pea or a pencil eraser?

Evolving: Has the mole or spot changed during the past few weeks or months?

To do a head-to-toe self-exam:

Choose a well-lit room with a full-length mirror. You'll also need a hand-held mirror.

Check your scalp. Part your hair and look with a hand-held mirror for any unusual marks on your skin. It may also help to use a blow dryer to move your hair as you look. Check the back of your neck, too.

Look at your face, ears, and front of your neck.

Look at the front of your chest and belly. Lift your breasts to check the skin underneath.

Raise your arms and check the skin on your left and right sides, including your underarms.

Look at the front and back of your arms.

Check your hands, including between your fingers and your fingernails (without nail polish).

Check your back and buttocks with a hand-held mirror.

Sit down to check the front and sides of your legs and use the hand-held mirror to check the backs of your legs and your genitals.

Check your feet, including the bottoms, the spaces between your toes, and the nail of each toe (without nail polish).

For a skin cancer screening by a provider, you'll remove your clothing and put on a gown. Your provider will do a full exam that includes your scalp, behind your ears, fingers, buttocks, and feet. Your provider may use a special magnifying glass with a light to look more closely at certain moles or spots. The exam should take 10-15 minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You should not wear makeup or nail polish. You will need to have your hair loose so your scalp can be checked.

ARE THERE ANY RISKS TO THE TEST?

Skin cancer screening isn't always helpful and can result in unnecessary treatments. For example, your provider could find a cancer that would never cause health problems and treat it with chemotherapy, radiation, or surgery. These procedures may not always be necessary and can cause side effects such as scarring from a skin biopsy and anxiety about having cancer

WHAT DO THE RESULTS MEAN?

If you find a mole or other spot on your skin that concerns you, contact your provider. If you or your provider finds a sign of skin cancer, you'll probably have a skin biopsy to find out whether you have cancer.

A skin biopsy is a procedure that removes a small sample of skin for testing. The skin sample is checked under a microscope to look for cancer cells. Not all suspicious spots turn out to be skin cancer. If the biopsy shows that you do have skin cancer, your provider will talk with you about your treatment options.

In some cases, your screening test results may be misleading. For example, they could show that:

You have skin cancer, but you really don't. This is called a "false positive." If you have a false positive, you may have other tests.

You don't have skin cancer when you really do. This is called a "false negative." If your results are negative but you're still concerned about an unusual area of skin, talk to your provider about getting additional tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A SKIN CANCER SCREENING?

There are mobile phone apps that use the camera of your phone to check skin moles and spots to help find skin cancer. These apps need to be studied to see if they are accurate and useful for skin cancer screening. If you do use them to check skin moles and spots, always check in with your provider about the results the app gives you.

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Sleep Study

<https://medlineplus.gov/lab-tests/sleep-study/>

SLEEP STUDY

WHAT IS A SLEEP STUDY?

A sleep study, also known as polysomnography, is a test that measures and records different body functions while you sleep. These include your:

- Breathing rate
- Blood oxygen level
- Heart rate
- Brain waves
- Leg movements
- Eye movements

The test checks how well you are sleeping and can diagnose sleep disorders. Sleep disorders are conditions that cause problems with sleeping. They include trouble falling asleep, getting too much sleep, and irregular breathing during sleep. Sleep disorders can affect your overall health, safety, and quality of life. Lack of sleep may increase your risk of serious conditions such as depression, diabetes, and heart disease. Daytime sleepiness may increase your risk of an accident while driving. Early diagnosis and treatment of a sleep disorder may help you avoid health complications.

Other names: polysomnography, sleep apnea study, home sleep study

WHAT IS IT USED FOR?

A sleep study is used to diagnose sleep disorders. Common sleep disorders include:

Sleep apnea, a disorder that causes you to repeatedly stop breathing during sleep.

Insomnia, a disorder that can make it hard for you to fall asleep and/or stay asleep throughout the night.

Restless leg syndrome, a condition that causes uncomfortable feelings in your legs and a strong urge to move them while you try to fall asleep.

Narcolepsy, a neurologic condition. It causes extreme sleepiness in the daytime. You may suddenly fall asleep during the day.

WHY DO I NEED A SLEEP STUDY?

You may need this test if you have symptoms of a sleep disorder, such as:

- Loud snoring during sleep
- Waking up from sleep and gasping for breath
- Trouble falling and/or staying asleep
- Daytime sleepiness

WHAT HAPPENS DURING A SLEEP STUDY?

A sleep study is done during your normal sleeping hours. The study may be done at a hospital, a sleep study clinic, or in your own home. A home sleep study, also known as a sleep apnea study, is used to help diagnose sleep apnea. It does not diagnose other types of sleep disorders.

During a study at a hospital or sleep clinic:

You will arrive at the hospital or clinic in the evening and stay overnight.

You may bring your bedclothes to wear.

You will stay in a private, comfortable bedroom with a private bathroom, similar to a hotel room.

Many sleep study rooms will have a low-light video camera to record your sleep movements during the night. A health care provider will attach electrodes (small metal disks) to your scalp, eyelids, chin, chest, and legs. After the sensors are connected, your provider may ask you to move these body parts to make sure the sensors are working well. The electrodes will be connected to a computer with wires. The wires will be long enough to allow you to move during sleep. If you have to get up at night to use the bathroom, you may need the staff to assist you with the wires. The electrodes record different body functions, such as breathing, heart rate, muscle movement, brain waves, and leg and eye movements.

A small clip, known as a pulse oximeter, will also be placed on your finger or ear to monitor the oxygen level in your blood.

The lights will be turned off.

While you sleep, specially trained providers known as polysomnography technologists will monitor you through the night with video and audio equipment. They'll check for when you fall asleep and monitor your breathing, heart rate, and other body functions.

If you need assistance during the night, you can talk to a polysomnography technologist through the monitoring equipment.

You will be woken up in the morning, and your electrodes will be removed.

You may be asked to fill out a questionnaire about your night's sleep.

During a home sleep study:

You will pick up sleep study equipment at your provider's office, or it may be delivered to you.

Your provider will provide thorough instructions on how to use the equipment. Be sure to ask if you have any questions.

On the night of your study, you will attach the electrodes to your body and connect them to a sleep monitoring device. The electrodes will record different body functions, including your breathing rate, heart rate, blood oxygen levels, and snoring.

In the morning, you'll remove the electrodes and return the sleep monitoring device as directed by your provider.

A home sleep study uses fewer sensors than a sleep study at a hospital or sleep clinic. It is used to diagnose obstructive sleep apnea. But this test is not recommended if you have other sleep disorders or health conditions that could make your breathing more difficult.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

The preparations are the same whether you are being tested at a hospital, clinic, or your home. To prepare for your sleep study:

Avoid alcohol and caffeine during the afternoon and evening before your test. They can disrupt your normal sleep patterns.

Don't nap during the day. It may make it harder for you to fall asleep at night.

Don't use lotions, hair gels, or makeup before the test. They can affect the way the electrodes work.

Avoid using nail polish or artificial nails. The pulse oximeter is attached to your finger, and this can affect the readings.

If you normally take sleep medicines, talk to your provider about whether or not you should take the medicine before the test.

ARE THERE ANY RISKS TO THE TEST?

You may have some mild skin irritation from the electrodes. There are no other known risks to having a sleep study.

WHAT DO THE RESULTS MEAN?

Your results may include measurements of the following:

Apnea Hypopnea Index (AHI). This looks at how many episodes of sleep apnea happen in a night. If you have more than five episodes, it probably means you have sleep apnea.

Sleep efficiency. This is a calculation based on the total number of minutes you sleep during the night compared with the amount of time you spend in bed. So if you spend a long time in bed without being able to sleep, you would have low

sleep efficiency.

Oxygen Desaturation Index (ODI). This measures the number of times your oxygen level drops during sleep. Lower than 90 percent oxygen is considered abnormal.

Heart rate. A normal heart rate is between 60 and 100 beats per minute. This result will show if your heart is beating faster or slower than normal.

Your provider will review all the results and, if appropriate, provide a treatment plan. If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A SLEEP STUDY?

You may not sleep as well during the sleep study, but a full night's sleep isn't necessary to get results.

If you've been diagnosed with sleep apnea, your provider may recommend:

Lifestyle changes, such as losing weight, avoiding alcohol, and quitting smoking. Your provider may also recommend avoiding sleeping on your back, which may worsen your sleep apnea.

A continuous positive airway pressure (CPAP) machine. A CPAP provides a gentle and steady pressure of air into your airway to keep it open. You wear a mask over your nose and mouth while sleeping. The mask is connected by a hose to a small machine by your bed that pumps air through the hose and mask.

Dental devices, such as a:

Mouth guard, which helps prevent you from grinding your teeth during sleep (sleep bruxism). Many people with sleep apnea also have sleep bruxism.

Mandibular advancement device, a small plastic device that pulls the lower jaw and tongue forward. This helps keep the airways open during sleep.

Tongue-retaining device, which uses suction to keep the tongue forward. If the tongue moves back, it can block your airway.

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If other treatments have failed, your provider may recommend a surgical procedure to correct problems in your throat, nose, or jaw.

Treatments for other sleep disorders will depend on the specific condition and your specific health needs.

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Smooth Muscle Antibody (SMA) Test

<https://medlineplus.gov/lab-tests/smooth-muscle-antibody-sma-test/>

SMOOTH MUSCLE ANTIBODY (SMA) TEST

WHAT IS A SMOOTH MUSCLE ANTIBODY (SMA) TEST?

This test looks for smooth muscle antibodies (SMAs) in a sample of your blood. High levels of SMAs are usually a sign of a chronic (long-lasting) liver condition called autoimmune hepatitis.

Autoimmune hepatitis is a type of autoimmune disease. Normally, your immune system makes antibodies to attack foreign substances that could make you sick, such as viruses and bacteria. But in autoimmune diseases, your immune system makes certain types of antibodies - called autoantibodies - that attack your body's own healthy cells and organs by mistake.

SMAs are autoantibodies that mainly attack the liver. They are sometimes called anti-smooth muscle antibodies (ASMAs). They get their name because they attack certain proteins that are found in smooth muscle (the muscle that lines the inside of hollow organs, such as your intestines, stomach, and blood vessels). But these proteins are also found in large amounts in liver cells.

When SMAs attack your liver, they cause liver swelling and damage. Without treatment, autoimmune hepatitis can lead to serious liver problems, including cirrhosis and liver failure.

Treatment often includes medicines that decrease the activity of your immune system. Early diagnosis and treatment can lower the chances of developing cirrhosis and other liver conditions linked to autoimmune hepatitis.

Other names: anti-smooth muscle antibody, ASMA, actin antibody, ACTA, F-actin antibody

WHAT IS IT USED FOR?

An SMA test is mainly used with other tests to help find out if autoimmune hepatitis could be the cause of liver damage.

There are two types of autoimmune hepatitis. Each type involves different autoantibodies. So, an SMA can also help find out which type of autoimmune hepatitis a person may have:

Type 1 involves SMAs. It's the most common type of autoimmune hepatitis in both adults and children. Like most autoimmune diseases, it's more common in women.

Type 2 involves anti-liver kidney microsomal type 1 antibodies (anti-LKM-1) and/or anti-liver cytosol type 1 antibodies (anti-LC1). Type 2 autoimmune hepatitis isn't common. It mostly develops during childhood and mainly affects girls.

SMA tests may be done with tests that check for other autoantibodies, including:

ANAs (antinuclear antibodies). ANAs attack the nucleus (center) of healthy cells. A high level of ANAs in your blood means you're more likely to have an autoimmune disease. But the test can't diagnose any specific type of autoimmune diseases.

LKM-1 antibodies. These are autoantibodies that attack a protein that's mainly found in liver cells. LKM-1 antibodies are a sign of Type 2 autoimmune hepatitis.

WHY DO I NEED AN SMA TEST?

You may need an SMA test if you have symptoms of autoimmune hepatitis. The test is usually ordered after the results of liver function tests show that your liver may be damaged.

Symptoms of autoimmune hepatitis may include:

Fatigue

Jaundice (a condition that causes your skin and eyes to turn yellow)

Abdominal pain in the upper part of your belly

Joint pain

Nausea

Skin conditions, including rashes, psoriasis, acne, or vitiligo (white patches on your skin)

Loss of appetite

Dark-colored urine (pee)

Light colored stool (poop)

In the early stages, autoimmune hepatitis may not cause symptoms. But you may need an SMA test if you had abnormal results on a routine blood test, such as a comprehensive metabolic panel (CMP). A CMP measures several substances made by your liver. Abnormal amounts of these substances are usually a sign of liver inflammation and/or damage.

WHAT HAPPENS DURING AN SMA TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for an SMA test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Diagnosing autoimmune hepatitis can be difficult. The process may involve tests to rule out other diseases that can affect your liver. To understand what your SMA test results mean, your provider will consider the results of other tests you've had. Your provider can explain what these results say about your health.

In general:

A positive SMA test result means SMAs were found in your blood sample:

High levels of SMAs often mean you have type 1 autoimmune hepatitis. But this test can't diagnose the disease. If you have high levels of both SMAs and ANAs, your provider may suggest a biopsy to remove a small sample of liver tissue for testing. The results of a biopsy can confirm whether you have autoimmune hepatitis.

SMAs that are somewhat high may be a sign of other conditions, including:

Primary biliary cholangitis, an autoimmune disease that damages the bile ducts (tubes) in your liver

Infectious mononucleosis

Hepatitis C

Certain types of cancer

Primary biliary cholangitis, an autoimmune disease that damages the bile ducts (tubes) in your liver

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A negative SMA test result means SMAs were not found in your blood sample. You're unlikely to have type 1 autoimmune hepatitis. But if you had a test that showed you have anti-LKM-1 antibodies, you may have type 2 autoimmune hepatitis. You may need a biopsy to confirm the diagnosis.

In general, if SMA, ANA, and anti-LKM-1 testing doesn't find any autoantibodies in your blood, it usually means you don't have autoimmune hepatitis. But it doesn't rule it out completely. It's possible for autoimmune hepatitis to be caused by other autoantibodies that are not commonly tested.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT AN SMA TEST?

In certain cases, a provider may order a test that checks for a specific type of SMA. This type of SMA attacks F-actin. F-actin is a protein found in the liver and other body tissues. Tests that check for F-actin antibodies may help find more cases of autoimmune hepatitis than general SMA tests, but F-actin antibody tests are not available in all labs.

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RELATED HEALTH TOPICS

Autoimmune Diseases

Cirrhosis

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Immune System and Disorders

Liver Diseases

RELATED MEDICAL TESTS

ALT Blood Test

ANA (Antinuclear Antibody) Test

AST Test

Comprehensive Metabolic Panel (CMP)

How to Cope with Medical Test Anxiety

How to Understand Your Lab Results

What You Need to Know About Blood Testing

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Sodium Blood Test

<https://medlineplus.gov/lab-tests/sodium-blood-test/>

SODIUM BLOOD TEST

WHAT IS A SODIUM BLOOD TEST?

A sodium blood test measures the amount of sodium in your blood. Sodium is a type of electrolyte. Electrolytes are minerals that have an electrical charge when they are dissolved in water or body fluids. You have electrolytes in your blood, urine (pee), tissues, and other body fluids. They help control the amount of fluid and the balance of acids and bases (pH balance) in your body. Sodium also helps your nerves and muscles work properly.

You get most of the sodium you need in your diet. If you take in too much, your kidneys get rid of the extra sodium in your urine (pee). Normally, your body keeps your sodium levels in a very narrow range. If your sodium blood levels are too high or too low, it may mean that you have a kidney problem, dehydration, or another type of medical condition.

Other names: Na test

WHAT IS IT USED FOR?

A sodium blood test is a routine test that may be used to check your general health. It may be used to help find and monitor conditions that affect the balance of fluids, electrolytes, and acidity in your body.

The test is often done as part of a group of tests called an electrolyte panel. It may also be part of two other groups of tests called a basic metabolic panel (BMP) and a comprehensive metabolic panel (CMP).

WHY DO I NEED A SODIUM BLOOD TEST?

Your health care provider may order a sodium blood test as part of your routine checkup or if you take medicines that can affect your fluid balance. You may also have this test if you have symptoms of too much or too little sodium in your blood.

Symptoms of high sodium levels (hypernatremia) include:

- Thirst
- Urinating (peeing) very little
- Vomiting
- Diarrhea
- Confusion
- Muscle twitching
- Seizures

Without treatment, extremely high levels of sodium may lead to a coma and become life-threatening.

Symptoms of low sodium levels (hyponatremia) include:

- Weakness
- Fatigue
- Confusion
- Muscle twitching
- Seizures

Without treatment, extremely low levels of sodium may also lead to a coma and become life-threatening.

WHAT HAPPENS DURING A SODIUM BLOOD TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle

goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Your provider will let you know how to prepare for a sodium blood test. You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop any medicines unless your provider tells you to.

You may also need to fast (not eat or drink) for several hours before the test. Ask your provider if you have any questions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Blood sodium levels that are too high or too low can be a sign of many different conditions. To make a diagnosis, your provider will usually look at the results of your sodium blood test with the results of other tests, such as measurements of other electrolytes.

Sodium blood test results that are higher than normal may be a sign of a condition such as:

Dehydration, which may be caused by not drinking enough, diarrhea, or taking diuretic medicines (water pills)

A disorder of the adrenal glands

A kidney disease

Diabetes insipidus (uncommon)

Sodium blood test results that are lower than normal may be a sign of a condition, such as:

A loss of sodium from diarrhea or vomiting

A condition that may cause the body to hold onto extra fluid (which dilutes sodium), including:

Kidney diseases

Cirrhosis of the liver

Heart failure

Certain brain and lung diseases

Certain types of cancer

Certain medicines, including many over-the-counter pain relievers and some antidepressants

Kidney diseases

Cirrhosis of the liver

Heart failure

Certain brain and lung diseases

Certain types of cancer

Certain medicines, including many over-the-counter pain relievers and some antidepressants

Addison disease

Malnutrition

If your results are not in the normal range, it doesn't always mean that you have a medical condition that needs treatment. Certain medicines, your age, sex, and certain medical conditions can increase or decrease your sodium levels. If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I SHOULD KNOW ABOUT A SODIUM BLOOD TEST?

Sodium levels are often measured with other electrolytes in another test called the anion gap. An anion gap test looks at the difference between negatively charged and positively charged electrolytes. The test checks for acid imbalances and other conditions.

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RELATED HEALTH TOPICS

Dehydration
Fluid and Electrolyte Balance
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Ultrasound

<https://medlineplus.gov/lab-tests/sonogram/>

ULTRASOUND

WHAT IS AN ULTRASOUND?

An ultrasound is an imaging test that uses sound waves to make pictures of organs, tissues, and other structures inside your body. It allows your health care provider to see into your body without surgery. Ultrasound is also called ultrasonography or sonography. Ultrasound images may be called sonograms.

Ultrasound can be used to treat certain medical conditions. But it's mostly used to help:

Monitor the health and development of an unborn baby during pregnancy. Pregnancy ultrasound can help check if your baby is growing normally. It can screen for certain conditions, such as birth defects that can be seen in images. It can also check for pregnancy problems. For example, an ultrasound can show if your placenta (the organ that brings oxygen and nutrients to the baby) is in the right position. Pregnancy ultrasound may also be called "prenatal ultrasound," "fetal ultrasound," or "obstetrical ultrasound."

Diagnose the cause of a wide variety of medical conditions. Ultrasound is best used to learn about conditions that involve soft tissues, such as organs, glands, and blood vessels. Diagnostic ultrasound may be used if you have signs or symptoms of a problem, and an ultrasound may help diagnose or rule out possible causes.

Guide certain biopsy procedures. Some biopsies use a needle to remove a sample of fluid or tissue from the body for testing. An ultrasound can find the abnormal area and guide the needle to the right place to collect the sample.

There are different types of ultrasounds. One type, called Doppler ultrasound, can show movement in your body. For example, it can show your heart beating and the speed and direction of blood flowing through your blood vessels. It can also show the beating heart and movement of an unborn baby. Another type of ultrasound can create 3-D (three-dimensional) images.

Other names: sonogram, ultrasonography, pregnancy sonography, fetal ultrasound, obstetric ultrasound, diagnostic medical sonography, diagnostic medical ultrasound

WHAT IS IT USED FOR?

A pregnancy ultrasound may be used to:

Check the size, position, heart rate, and age of the unborn baby

See if there is more than one baby

Screen for:

Genetic disorders, such as Down syndrome

Birth defects in the heart, brain and spinal cord, or other parts of the body

Genetic disorders, such as Down syndrome

Birth defects in the heart, brain and spinal cord, or other parts of the body

Check the amount of amniotic fluid (the liquid in the sac surrounding an unborn baby) and the location of the placenta

Guide the collection of test samples taking during amniocentesis and chorionic villus sampling (CVS)

Diagnostic ultrasound has many uses. For example, ultrasound can help:

Find the cause of pain, swelling, and other symptoms

Look for blockages, growths, and structural problems in organs, glands, and blood vessels

Tell the difference between cysts (fluid-filled sacs) and solid tumors

Ultrasound may help diagnose medical conditions that involve many parts of the body, such as the:

Heart and heart valves

Blood vessels

Organs in the abdomen (belly), including the liver, gallbladder, pancreas, or spleen

Organs in the pelvis, including the urinary tract, and male and female reproductive organs

Thyroid and parathyroid glands

Kidneys
Breasts
Brain, spine, and hips in infants

WHY DO I NEED AN ULTRASOUND?

If you're pregnant, it's common to have a routine ultrasound between weeks 18 and 22 of pregnancy to check on the health of your baby. If your provider suspects a problem, you may need an ultrasound at other times during your pregnancy.

You may need a diagnostic ultrasound if you have signs or symptoms of certain types of medical conditions and your provider needs to look inside your body to help find the cause. If you're having a needle biopsy to remove fluid or tissue for a test, you may have an ultrasound as part of the procedure.

WHAT HAPPENS DURING AN ULTRASOUND?

An ultrasound is often done by a sonographer, a health care professional who has special training to do ultrasound exams. Ultrasounds may be done in different ways depending on the part of your body that's being examined. Most ultrasound exams include these general steps:

You will remove clothing from the area that will be examined and lie on a table.

The sonographer will spread a special gel on your skin over the area that will be examined.

The sonographer will hold a wand-like device, called a transducer, and move it across your skin. The device sends sound waves into your body. The gel prevents air from getting between the device and your skin, which would block the soundwaves from entering your body. The sound waves are a very high pitch, so you can't hear or feel them.

The soundwaves "bounce" off the structures inside your body. The ultrasound device picks up the echoes and turns them into images on a computer screen. You may be able to see the images on the screen during your exam.

After the exam is over, the sonographer will wipe the gel off your skin.

For certain ultrasound exams, the ultrasound device is placed inside a body opening to get a clearer image. Depending on the organs being checked, the device may be placed in the:

Vagina. This is called a transvaginal ultrasound. It helps view the uterus and ovaries.

Rectum. This is called a transrectal ultrasound. It's usually done to view the prostate gland.

Esophagus (the tube that connects your mouth and stomach). This is called a transesophageal echocardiogram (TEE). It's done to get clear images of the heart.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Preparations for an ultrasound exam depend on what part of your body is being checked. Some ultrasound exams require no preparation at all. Your provider will tell you if you need to prepare for your ultrasound and what to do.

For example, if you're having an ultrasound to view your urinary tract, you may need to drink water so that you have a full bladder. You will have to hold your urine (pee) until the test is over. For other ultrasounds, you may need to fast (not eat or drink) for several hours before your test.

ARE THERE ANY RISKS TO THE TEST?

Ultrasound has not been linked to any health harms. It's generally considered safe when trained sonographers use it correctly. Ultrasound doesn't use ionizing radiation like x-rays use, which makes it safer than x-ray. That's why ultrasound is the most widely used medical imaging method for viewing an unborn baby during pregnancy.

But in certain cases, ultrasound can affect fluids and tissues in the body. That's why most medical experts recommend using ultrasound only when it's needed to provide important medical information.

WHAT DO THE RESULTS MEAN?

The results of your ultrasound depend on the type of ultrasound you had. Your provider can explain what your results mean for your health.

If you had a pregnancy ultrasound, normal results mean that your baby appears to be developing normally. But an ultrasound can't guarantee you'll have a healthy baby. If your results weren't normal, you may need more tests, including another ultrasound exam.

Learn more about laboratory tests, reference ranges, and understanding results.

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Down Syndrome Tests
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Karyotype Genetic Test
Nonstress Test
Prenatal Cell-Free DNA Screening

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Sputum Culture

<https://medlineplus.gov/lab-tests/sputum-culture/>

SPUTUM CULTURE

WHAT IS A SPUTUM CULTURE?

A sputum culture is a test that checks for bacteria or another type of germ that may be causing a respiratory tract infection (an infection in your lungs or airways). Sputum, also known as phlegm, is a thick type of mucus made in your lungs. If you have an infection or chronic illness affecting the lungs or airways, sputum can settle in your lungs, and you may also cough it up.

Sputum is not the same as spit or saliva. Sputum contains cells from your immune system that help fight bacteria, fungi, or other foreign substances in your lungs or airways. The thickness of sputum helps trap the foreign material. This allows cilia (tiny hairs) in the airways to push it through the mouth and be coughed out.

Sputum can be one of several different colors. The colors can help identify the type of infection you may have or check if a chronic illness has become worse. This helps your health care provider determine the best treatment. Sputum colors can include:

Clear. This usually means you don't have an infection, but large amounts of clear sputum may be a sign of lung disease.

White or gray. This may also be normal, but increased amounts may mean lung disease.

Dark yellow or green. This often means a bacterial infection, such as pneumonia. Yellowish-green sputum is also common in people with cystic fibrosis. Cystic fibrosis is an inherited disease that causes mucus to build up in the lungs and other organs.

Black. This often shows up in people who smoke. It is also a common sign of black lung disease. Black lung disease is a serious condition that can happen if you have long-term exposure to coal dust.

Brown or brown spots. This may be a sign of old blood. This can happen if you have cystic fibrosis, bacterial pneumonia, bacterial bronchitis, or some other lung diseases. It can also be caused by breathing in dust or other brown materials or toxins.

Pink. This may be a sign of pulmonary edema, a condition in which excess fluid builds up in the lungs. Pulmonary edema is common in people with heart failure.

Red. This may be an early sign of lung cancer. It may also be a sign of a pulmonary embolism (PE), a life-threatening condition in which a blood clot from a leg or other part of the body breaks loose and travels to the lungs. If you are coughing up red or bloody sputum, call 911 or seek immediate medical attention.

Other names: respiratory culture, bacterial sputum culture, routine sputum culture, bacterial culture, respiratory

WHAT IS IT USED FOR?

A sputum culture is most often used to:

Find and diagnose bacteria or fungi that may be causing an infection in the lungs or airways

See if a chronic illness of the lungs has worsened

See if treatment for a lung or airway infection is working

Help to diagnose respiratory conditions such as pneumonia, tuberculosis, and bronchiectasis

A sputum culture is often done with another test called a Gram stain. A Gram stain is a test that checks for bacteria at the site of a suspected infection or in body fluids such as blood or urine. It can help identify the specific type of infection you may have.

A sputum test is not usually used if your provider thinks you may have a viral infection.

WHY DO I NEED A SPUTUM CULTURE?

You may need this test if you have symptoms of pneumonia or another serious infection of the lungs or airways. These include:

Cough which produces a lot of sputum

Fever

Chills

Shortness of breath

Chest pain that gets worse when you breathe deeply or cough

Fatigue

Confusion, especially in older people

Your provider may also order a sputum culture after an x-ray or other imaging tests show that you may have a lung infection. The sputum culture can help confirm this and determine the cause of the infection.

WHAT HAPPENS DURING A SPUTUM CULTURE?

Your provider will need to get a sample of your sputum. During the test:

You may be asked to rinse your mouth out with water before the sample is taken.

Your provider will ask you to breathe deeply and then cough deeply into a special cup.

A health care provider will ask you to breathe deeply and then cough deeply into a special cup.

Your provider may tap you on the chest to help loosen sputum from your lungs.

If you have trouble coughing up enough sputum, your provider may ask you to breathe in a salty mist to help you cough more deeply.

If you still can't cough up enough sputum, your provider may perform a procedure called a bronchoscopy. In this procedure, you'll first get a medicine to help you relax, and then a numbing medicine so you won't feel any pain.

Then a camera attached to a thin, lighted tube will be put through your mouth or nose and into your airway. This helps your provider see the inside of your lungs and airway.

Your provider will collect a sample from your airway using a small brush or suction.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

The test is often scheduled in the morning. To increase the amount of sputum, you may be encouraged to drink more water the night prior. You may also be asked not to eat for one or two hours before the test.

If you will be getting a bronchoscopy, you may be asked to fast (not eat or drink) for up to twelve hours before the test.

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. Antibiotics may affect bacterial growth and the results of your sputum culture test. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to providing a sputum sample into a container. If you had a bronchoscopy, your throat may feel sore after the procedure.

If you had a bronchoscopy:

Your throat may feel sore after the procedure

It may take a few hours to be able to cough normally.

You will not be permitted to eat or drink until you can cough normally again.

WHAT DO THE RESULTS MEAN?

Your sputum sample will be put in a dish with a special substance that allows bacteria to grow. It usually takes a few days to get the results of your sputum culture, but it may take weeks. How long it takes to get your results will depend on the type of bacteria that you are being tested for, since some types of bacteria take longer to grow in the lab

If your results were normal, it means that no harmful bacteria or fungi were found.

If your results were not normal, it may mean you have some kind of bacterial or fungal infection. Your provider may need

to do more tests to find the specific type of infection you have. The most common types of harmful bacteria found in a sputum culture include those that cause:

Pneumonia

Bronchitis

Tuberculosis

An abnormal sputum culture result may also mean a flare-up of a chronic condition, such as cystic fibrosis or chronic obstructive pulmonary disease (COPD). COPD is a lung disease that makes it hard to breathe.

If you have questions about your results, talk to your provider. To understand the results of your sputum culture, your provider will consider your symptoms, medical history, and the results of other tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A SPUTUM CULTURE?

Sputum may be referred to as phlegm or mucus. All terms are correct, but sputum and phlegm only refer to the mucus made in the respiratory tract (lungs and airways). Sputum (phlegm) is a type of mucus. Mucus can also be made elsewhere in the body, such as in the urinary or genital tract.

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- Antibiotics
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- Bacteria Culture Test
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- Gram Stain
- How to Cope with Medical Test Anxiety
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- Legionella Tests
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Stool Elastase

<https://medlineplus.gov/lab-tests/stool-elastase/>

STOOL ELASTASE

WHAT IS A STOOL ELASTASE TEST?

A stool elastase test measures the amount of elastase in your stool (poop). Elastase is one of a few digestive enzymes ("digestive juices") that your pancreas makes to help digest food. Your pancreas is a gland that sits behind your stomach.

Elastase helps break down fats, carbohydrates, and proteins so your body can use them for energy, growth, and repairing cells. Your pancreas releases elastase into your small intestine through a duct (a small tube). If your pancreas is working well, you'll have elastase in your stool.

If little or no elastase is found in your stool, it can mean that your pancreas can't make and/or release enough elastase and other digestive juices. This is called exocrine pancreatic insufficiency. It's often called pancreatic insufficiency or EPI for short.

If you have EPI, your body can't digest fats very well. This can cause symptoms, such as abdominal (belly) pain, and diarrhea. EPI also makes it hard for your body to absorb vitamins, minerals, and calories from food. This is called malabsorption and it can lead to serious health problems from malnutrition.

EPI is caused by conditions that damage the pancreas or block the ducts that let elastase flow into the intestine. The two main causes of EPI are:

Chronic (long-lasting) pancreatitis in adults. Pancreatitis is inflammation of the pancreas.

Cystic fibrosis in children. This inherited disease causes mucus to build up in pancreas, lungs, and other organs.

Other causes of EPI include:

Acute (short-term) pancreatitis

Pancreatic cancer

Crohn's disease

Ulcerative colitis

Diabetes

Weight-loss surgery and other surgery on the digestive system

Certain genetic disorders, such as Shwachman-Diamond syndrome in children

Other names: pancreatic elastase, fecal pancreatic elastase, fecal elastase, FE-1, EL-1, PE-1, PE stool

WHAT IS IT USED FOR?

A stool elastase test is used to check for exocrine pancreatic insufficiency (EPI) when a person has belly pain and other digestive symptoms that don't have a known cause. This test is better at finding severe EPI than mild or moderate cases.

A stool elastase test is also used to monitor how well the pancreas is working in people who have cystic fibrosis, diabetes, or chronic pancreatitis.

WHY DO I NEED A STOOL ELASTASE TEST?

You or your child may need a stool elastase test if you have certain digestive problems and a health care provider thinks exocrine pancreatic insufficiency (EPI) could be the cause. Signs and symptoms of EPI include:

Greasy, unusually foul-smelling stools (poops)

Abdominal (belly) pain

Bloating and gas

Weight loss without trying

Malabsorption

Malnutrition

Delayed growth and trouble gaining weight in children

If you or your child have a condition that can cause EPI, such as chronic pancreatitis or cystic fibrosis, you may need to have stool elastase testing from time to time. The tests are done to check whether your pancreas is making enough digestive juices.

WHAT HAPPENS DURING A STOOL ELASTASE TEST?

You will need to provide a stool sample for your test. Your provider will probably give you a container or kit with instructions on how to collect the sample. The stool must be formed, not watery.

There are different ways to collect stool samples, so follow the instructions carefully. In general, you'll need to:

Label the container with your name, the collection date, and time.

Make sure the stool doesn't mix with any urine, toilet paper, or water from the toilet.

Close the container tightly.

Wash your hands well with soap and water.

Return the container promptly according to the instructions. You may be told to refrigerate or freeze the sample first.

If you're collecting a sample from diapers, you'll get special instructions for using plastic wrap inside of a clean diaper.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

If you are taking pancreatic enzyme supplements, you may need to stop taking them for five days before the test. Don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a stool elastase test.

WHAT DO THE RESULTS MEAN?

Your stool elastase test results will tell you how much elastase was in your stool sample:

A normal amount of elastase means that you may not have exocrine pancreatic insufficiency (EPI). But a normal test result doesn't rule out EPI. If you were tested because you have symptoms, you may need other tests to find out what's causing them.

A less than normal amount of elastase may mean that you have EPI. Test results that aren't normal may be described as either "moderate" or "severe pancreatic insufficiency." Your provider will probably order other tests to confirm EPI and the cause.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A STOOL ELASTASE TEST?

If you are diagnosed with exocrine pancreatic insufficiency (EPI), treatment may include:

Replacing the pancreatic enzymes that you're missing. This is called pancreatic enzyme replacement therapy (PERT). Your provider will usually prescribe enzymes to take with meals.

Taking vitamins that your provider prescribes.

Eating frequent small meals with plenty of healthy fats. Your provider may refer you to a dietician for support. A dietician is a professional who has special training to help you learn the best way to eat for your condition.

Not drinking alcohol or smoking.

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Strep A Test

<https://medlineplus.gov/lab-tests/strep-a-test/>

STREP A TEST

WHAT IS A STREP A TEST?

Group A Streptococcus, also known as group A strep, is a type of bacteria that often lives in the nose and throat. It causes strep throat and other infections. Strep throat is an infection that affects the throat and tonsils. You can get strep throat at any age, but it's most common in children ages 5 to 15 years old.

The infection can spread from person to person through the air, for example from being close to someone who is coughing or sneezing. It can also spread when you touch something that has the bacteria on it and then touch your eyes, mouth, or nose. Another way you can get it is if you share a glass, straw, or utensils with someone who has the infection.

The infection can spread more easily among people living together or in crowded places. Some people infected with group A strep do not have symptoms or seem sick. People who are sick with strep throat are more contagious than those who do not have symptoms.

Once you have been exposed to the group A strep bacteria, it usually takes two to five days before you start to feel sick. Strep throat can be easily treated with antibiotics. But if it's not treated, strep throat can lead to serious health problems (complications). These include rheumatic fever, a disease that can damage the heart and joints, and glomerulonephritis, a type of kidney disease.

Strep A tests check for strep A infections. There are two types of strep A tests:

Rapid strep test. This test looks for antigens to strep A. These antigens are substances in the Strep A bacteria that trigger your immune system to fight the bacteria. A rapid strep test can provide results in 10-20 minutes. If a rapid test is negative, but your health care provider thinks you or your child has strep throat, they may order a throat culture.

Throat culture. This test looks for strep A bacteria. It provides a more accurate diagnosis than a rapid test, but it can take 24-48 hours to get results.

Other names: strep throat test, throat culture, group A Streptococcus (GAS) throat culture, rapid strep test, Streptococcus pyogenes

WHAT IS IT USED FOR?

A strep A test is most often used to find out if a sore throat and other symptoms are being caused by strep throat or by a viral infection. Strep throat needs to be treated with antibiotics to prevent complications. But most sore throats are caused by viruses, and antibiotics don't work on viral infections. Viral sore throats will usually go away on their own.

WHY DO I NEED A STREP A TEST?

Your provider may order a strep A test if you or your child has symptoms of strep throat. These include:

- A sudden and severe sore throat
- Pain or trouble swallowing
- Fever of 101 °F (38 °C) or more
- Swollen lymph nodes
- Red and swollen tonsils

White patches or streaks of pus on the tonsils
Tiny, red spots on the roof of the mouth, called petechiae

Some people, especially children, may have other symptoms too. Other symptoms may include:

- Headache
- Stomach pain
- Nausea or vomiting

Your provider may also order a strep A test if you or your child has a rough, red rash that starts on the face and spreads to another part of the body. This type of rash is a sign of scarlet fever, an illness that can happen a few days after you've been infected with strep A. Like strep throat, scarlet fever is easily treated with antibiotics.

If you have symptoms such as a cough or runny nose along with your sore throat, it's more likely that you have a viral infection rather than strep throat.

WHAT HAPPENS DURING A STREP A TEST?

To find out what type of illness you have, your provider will first ask about your symptoms and do a physical exam. If your provider thinks you might have strep throat, they will swab your throat to test for strep throat.

There are two types of tests for strep throat: a rapid strep test and a throat culture. Both tests are done in the same way. During the procedure:

You will be asked to tilt your head back and open your mouth as wide as possible

Your provider will use a tongue depressor to hold down your tongue

They will use a special swab to take a sample from the back of your throat and tonsils

The sample may be used to do a rapid strep test in the provider's office. Sometimes the sample is sent to a lab

Your provider may take a second sample and send it to a lab for a throat culture if necessary.

A rapid strep test quickly shows if group A strep bacteria are causing the illness. But a throat culture takes time to see if group A strep bacteria grow from the swab. Although it takes more time, a throat culture sometimes finds infections that the rapid strep test misses. It's important to do a throat culture in children and teens since they can get rheumatic fever from an untreated strep throat infection.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a rapid strep test or a throat culture.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to having swab tests, but they may cause slight discomfort and/or gagging.

WHAT DO THE RESULTS MEAN?

If the rapid test is positive, it means you or your child has strep throat or another strep A infection. No further testing will be needed.

If the rapid test is negative but the provider thinks you or your child might have strep throat, they may order a throat culture. If you or your child has not already provided a sample, you will get another swab test.

If the throat culture is positive, it means you or your child has strep throat or other strep infection.

If the throat culture is negative, it means your symptoms are not being caused by strep A bacteria. Your provider will probably order more tests to help make a diagnosis.

If you or your child is diagnosed with strep throat, you will need to take antibiotics. Most often you will need to take them for 10 days. After a day or two of taking the medicine, you or your child should start to feel better. Most people are no longer contagious after taking antibiotics for 24 hours. But it's important to take all the medicine as prescribed. Stopping early can lead to serious complications, such as:

Rheumatic fever

An abscess (pocket of pus) around the tonsils or in the back of the throat

Swollen lymph nodes in the neck

Sinus infections

Ear infections

People with strep throat should stay home from work, school, or daycare until they no longer have a fever and have taken antibiotics for at least 12-24 hours.

It's important to take the antibiotics as prescribed and to finish the medicine even if you start feeling better. If you stop taking them too soon, some bacteria may survive and re-infect you.

Someone who tests positive for strep throat but has no symptoms (called a "carrier") usually does not need antibiotics. They are less likely to spread the bacteria to others and very unlikely to have complications.

People can get strep throat more than once. Having strep throat does not protect someone from getting it again in the future. To prevent group A strep infections, you should:

Cover your mouth and nose with a tissue when you cough or sneeze. Put your used tissue in the waste basket.

If you don't have a tissue, cough or sneeze into your upper sleeve or elbow.

Wash your hands often with soap and water for at least 20 seconds.

If soap and water are not available, you can use an alcohol-based hand sanitizer that contains at least 60% alcohol.

Wash glasses, utensils, and plates after they are used, especially if the person who used them is sick.

Regularly clean and disinfect frequently touched surfaces and objects.

If you have questions about your results or your child's results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A STREP A TEST?

Strep A can cause other infections besides strep throat. These infections are less common than strep throat but are often more serious. They include toxic shock syndrome and necrotizing fasciitis, also known as flesh-eating bacteria.

There are also other kinds of strep bacteria. These include strep B, which can cause a dangerous infection in newborns, and *Streptococcus pneumoniae*, which causes the most common type of pneumonia. *Streptococcus pneumoniae* bacteria can also cause infections of the ear, sinuses, and bloodstream.

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Strep B Test

<https://medlineplus.gov/lab-tests/strep-b-test/>

STREP B TEST

WHAT IS A GROUP B STREP TEST?

Strep B, also known as group B strep (GBS), is a type of bacteria commonly found in the digestive tract, urinary tract, and genital area. Many healthy people have GBS bacteria in their bodies.

Most of the time, GBS bacteria are not harmful and do not make people feel sick or have any symptoms. But in rare cases, these bacteria cause certain infections, known as GBS disease. This is more likely in people who are age 65 and older, have certain chronic health problems, or have a condition that weakens their immune system. And sometimes GBS infections can be very serious, or even life-threatening, in newborns.

In most cases, it is not known how people spread GBS bacteria to others. If you are pregnant, you can pass the bacteria to your baby during a vaginal delivery. GBS is usually found in the vagina and rectum, so the baby can be exposed to the GBS bacteria during labor. It is rare, but it can cause serious illness in the baby.

There are two types of GBS infections in babies. The types are based on the age of the infant when signs and symptoms begin to show:

In early-onset GBS disease, the signs and symptoms start during the first six days of life

In late-onset GBS disease, the signs and symptoms start during between 7 to 89 days of life. The source of late-onset GBS infection is unknown.

In newborns, GBS can cause:

Bacteremia (infection in the bloodstream)

Pneumonia (inflammation of the lung)

Meningitis (inflammation of the thin tissue that surrounds the brain and spinal cord)

Other serious illnesses

GBS bacteria are a leading cause of meningitis and bloodstream infections in a newborn's first three months of life.

If you are pregnant, GBS can cause:

Bacteremia

Chorioamnionitis (infection of the placenta and amniotic fluid)

Endometritis (infection of the membrane lining the uterus)

Urinary tract infections

In older adults and people with chronic health conditions, GBS bacteria can cause:

Bacteremia

Urinary tract infection

Pneumonia

Bone and joint infections

Endocarditis (infection of the heart valves)

Meningitis

Skin or soft-tissue infections

A group B strep test checks for GBS bacteria. If you are pregnant and the test shows that you have GBS, taking antibiotics during labor can protect your baby from infection.

Other names: group B streptococcus, group B beta-hemolytic streptococcus, *Streptococcus agalactiae*, beta-hemolytic strep culture

WHAT IS IT USED FOR?

If you are pregnant, a group B strep test is most often used to look for GBS bacteria during your routine prenatal screening. It may also be used to test infants who show signs of infection.

WHY DO I NEED A GROUP B STREP TEST?

You may need a strep B test if you are pregnant. The American College of Obstetricians and Gynecologists recommends GBS testing if you are pregnant. Testing is usually done between 36 and 38 weeks of pregnancy. If you go into labor earlier than 36 weeks, you may be tested at that time.

Symptoms depend on the part of the body that is infected. Symptoms of GBS disease are different in newborns compared to people of other ages who get GBS disease. Most newborns who get sick in the first week of life (early onset) have symptoms on the day of birth. In contrast, babies who develop disease later (late onset) can appear healthy at birth and during their first week of life.

A baby may need a group B strep test if he or she has symptoms of infection. These include:

High fever

Trouble with feeding

Trouble breathing

Lack of energy (hard to wake up)

Irritability

Blue-ish skin color

Unstable blood pressure (a pattern where the blood pressure suddenly spikes and then falls back to normal)

Kidney problems

If you are pregnant, you usually do not feel sick or have any symptoms of GBS.

Symptoms of bacteremia include:

Fever

Chills

Low alertness

Symptoms of pneumonia include:

Fever

Chills

Cough

Rapid breathing or trouble breathing

Chest pain

Skin and soft tissue infections often appear as a bump or infected area on the skin that may be:

Red

Swollen or painful

Warm to the touch

Full of pus or other drainage

Fever, in some cases

Bone and joint infections often appear as pain in the infected area and might also include:

Fever

Chills

Swelling

Stiffness or being unable to move the affected joint(s)

WHAT HAPPENS DURING A GROUP B STREP TEST?

If you are pregnant, your health care provider may order a swab test or a urine test.

For a swab test, you will lie on your back on an exam table. Your provider will use a small cotton swab to take a sample of cells and fluids from your vagina and rectum.

For a urine test, a health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back. Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If your baby needs testing, a provider may do a blood test or a lumbar puncture (spinal tap).

For a blood test, a health care professional will use a small needle to take a blood sample from your baby's heel. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. Your baby may feel a little sting when the needle goes in or out.

A lumbar puncture, also known as a spinal tap, is a test that collects and looks at spinal fluid, the clear liquid that surrounds the brain and spinal cord. During the procedure:

A nurse or other provider will hold your baby in a curled-up position.

A provider will clean your baby's back and inject an anesthetic into the skin, so your baby won't feel pain during the procedure. The provider may put a numbing cream on your baby's back before this injection.

The provider may also give your baby a sedative and/or pain reliever to help him or her better tolerate the procedure.

Once the area on the back is completely numb, your provider will insert a thin, hollow needle between two vertebrae in the lower spine. Vertebrae are the small backbones that make up the spine.

The provider will withdraw a small amount of cerebrospinal fluid for testing. This will take about five minutes.

The provider may also order a chest X-ray to help determine if someone has GBS disease.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't any special preparations for group B strep tests.

ARE THERE ANY RISKS TO THE TEST?

There is no risk to you from a swab or urine test. Your baby may have slight pain or bruising after a blood test, but that should go away quickly. Your baby will likely feel some pain after a spinal tap, but that shouldn't last too long. There is also a small risk of infection or bleeding after a lumbar puncture.

WHAT DO THE RESULTS MEAN?

If you are pregnant and results show you have GBS bacteria, you will be given antibiotics intravenously (by IV) during labor, at least four hours before delivery. This will prevent you from passing the bacteria to your baby. Taking antibiotics earlier in your pregnancy is not effective, because the bacteria can grow back very quickly. It's also more effective to take antibiotics through your vein (by IV), rather than by mouth.

You may not need antibiotics if you are having a planned cesarean delivery (C-section). During a cesarean delivery, a baby is delivered through the mother's abdomen rather than vaginally. But you still should be tested during pregnancy because you may go into labor before your scheduled cesarean delivery.

If your baby's results show a GBS infection, he or she will be treated with antibiotics. If your provider suspects a GBS infection, he or she may treat your baby before test results are available. This is because GBS can cause serious illness or death.

If you have questions about your results or your baby's results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A GROUP B STREP TEST?

Strep B is one type of strep bacteria. Other forms of strep cause different types of infections. These include strep A,

which causes strep throat, and *Streptococcus pneumoniae*, which causes the most common type of pneumonia. *Streptococcus pneumoniae* bacteria can also cause infections of the ear, sinuses, and bloodstream.

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Stress Tests

<https://medlineplus.gov/lab-tests/stress-tests/>

STRESS TESTS

WHAT ARE STRESS TESTS?

Stress tests show how well your heart works when it's pumping hard. Some heart diseases are easier to find when your heart is working its hardest to pump blood through your body. So stress tests check your heart while you exercise on a treadmill or stationary bicycle. If you're not able to exercise, medicine can be used to make your heart work harder, as if you were exercising.

There are different types of stress tests. They all check:

- Blood flow in your heart

- Your blood pressure

- The rate and rhythm of your heartbeat

- The strength of the electrical signals that control your heartbeat

Some stress tests also take pictures of your heart at rest and when it's working hard. The pictures provide more detail about how your heart is working.

Other names: exercise stress test, treadmill test, stress EKG, stress ECG, exercise tolerance test, exercise EKG, stress echocardiogram, nuclear stress test, thallium stress test, sestamibi stress test, cardiac MRI stress test, myocardial perfusion imaging test, cardiac single photon emission computed tomography (SPECT), stress perfusion test

WHAT ARE THEY USED FOR?

Stress tests are most often used to find the cause of symptoms that may be from a heart problem.

The tests can help diagnose certain heart conditions, including:

- Coronary artery disease (CAD)

- Angina

- Arrhythmia

- Heart failure

- Heart valve diseases

- Cardiomyopathy

Stress testing is also used:

- To find out how serious a known heart condition is, including the chance that you'll have a heart attack in the future.

- To help make treatment decisions for a heart condition.

WHY DO I NEED A STRESS TEST?

You may need a stress test if you have symptoms that could be from a heart condition. Symptoms may include:

- Chest pain or discomfort without a known cause

- Shortness of breath

- Irregular or rapid heartbeat that may feel like a fluttering in your chest

- Feeling dizzy or lightheaded

You may also need a stress test to check your heart health if you:

- Have a heart condition with new or worsening symptoms.

- Are going to have surgery. Your health care provider may want to see if your heart is strong enough for the operation.

- Are being treated for heart disease. The test can show how much treatment is helping, including heart surgery.

- Have a high risk for heart disease. Your risk may be higher than normal if you have a family history of heart disease and/or certain conditions, such as diabetes, that are linked to heart disease.

- Plan to start an exercise program. If you have a heart condition or a high risk for a heart condition, a stress test can

show what level of exercise is safe for you. (Always talk with your provider before starting any strenuous, new exercise.)

WHAT HAPPENS DURING A STRESS TEST?

Stress tests may be done with or without imaging (pictures of your heart). The most common type of test is an exercise stress test. It doesn't include pictures. But the main steps of an exercise stress test are part of all stress tests:

You'll have a blood pressure cuff on your arm to check your blood pressure.

You'll have a blood pressure cuff on your arm to check your blood pressure.

Electrodes will be placed on your body for an electrocardiogram test (also called EKG or ECG). Electrodes are small sensors that stick to your skin. Wires connect the electrodes to a computer or an EKG machine that records the electrical activity in your heart during the stress test.

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If you're unable to exercise: An intravenous (IV) line will be inserted into a vein in your arm. You'll get medicine through the IV. The medicine will make your heart work harder for 10 to 20 minutes while an EKG records your heart's electrical activity. Medicine can be used instead of exercise for all types of stress tests.

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If you can exercise: You'll walk on a treadmill or ride a stationary bicycle. On a treadmill, the speed will slowly increase. The treadmill may also tilt so you feel like you're walking uphill. On a bicycle, the resistance will slowly increase, so it's harder to pedal.

You'll exercise for about 10 to 15 minutes until you reach a target heart rate based on your age and fitness level. But you may stop the test sooner if:

You develop chest pain, shortness of breath, dizziness, fatigue or other symptoms

The EKG shows a problem with your heart

If you can exercise: You'll walk on a treadmill or ride a stationary bicycle. On a treadmill, the speed will slowly increase. The treadmill may also tilt so you feel like you're walking uphill. On a bicycle, the resistance will slowly increase, so it's harder to pedal.

You develop chest pain, shortness of breath, dizziness, fatigue or other symptoms

The EKG shows a problem with your heart

The EKG shows a problem with your heart

After all types of stress tests, you'll be monitored for 10-15 minutes or until your heart rate returns to normal.

After all types of stress tests, you'll be monitored for 10-15 minutes or until your heart rate returns to normal.

Stress tests with pictures include stress echocardiograms, nuclear stress tests, and cardiac (heart) MRI stress tests. These tests have extra steps:

A stress echocardiogram or "echo" takes moving pictures of your heart using doppler ultrasound. The pictures show the size and shape of your heart and blood flow through your heart. Pictures will be taken before and after your heart has worked its hardest. You'll lie on your left side on a table as a provider moves an ultrasound device on your chest. If you exercise on a bicycle, the second set of pictures may be taken while you're pedaling.

A nuclear stress test takes pictures of your heart using a small amount of a radioactive substance called a "tracer" and a special camera that scans your heart. A provider injects the tracer into your bloodstream through an IV line that's placed in a vein. Your heart and blood vessels absorb the tracer, which makes them show up more clearly in the pictures.

Pictures are taken while you lie on a table before and after your heart has worked its hardest. After the test, the tracer naturally leaves your body in your urine (pee). Drinking lots of water will help remove it faster.

A cardiac MRI stress test takes the most detailed pictures of your heart using radio waves, magnets, and a computer. It is a newer test that's mostly used for very serious heart problems. For an MRI stress test:

A provider may inject dye into your bloodstream through an IV line placed in your arm. Some cardiac MRIs, but not all, use dye to help show very small details on the pictures.

A provider may inject dye into your bloodstream through an IV line placed in your arm. Some cardiac MRIs, but not all, use dye to help show very small details on the pictures.

Pictures of your heart may be taken before and after your heart has worked hard. To take the pictures, you'll lie on a table that slides into a large, tunnel-like MRI machine. You'll hear loud sounds as the machine takes pictures. If you're exercising for the test, you may use a treadmill near the MRI, or you may exercise while lying in the MRI machine using special devices to move your arms or legs.

Pictures of your heart may be taken before and after your heart has worked hard. To take the pictures, you'll lie on a table that slides into a large, tunnel-like MRI machine. You'll hear loud sounds as the machine takes pictures. If you're exercising for the test, you may use a treadmill near the MRI, or you may exercise while lying in the MRI machine using special devices to move your arms or legs.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Wear comfortable shoes and clothing for exercise. Your provider will tell you what else to do before a stress test. You'll usually need to fast (not eat or drink) and stop smoking for at least a few hours before the test. You may need to avoid drinks with caffeine, such as coffee, tea, and colas, for a longer period of time.

If you take medicines or supplements, ask your provider if you should take them as usual. Don't stop taking any medicine unless your provider tells you to. If your test uses MRI, tell your provider about any metal you have in your body. An MRI can heat certain metal and cause burns.

ARE THERE ANY RISKS TO THE TEST?

Stress tests are usually safe. You will be closely watched during the test. If a problem develops, you'll be treated quickly. Medicines that make your heart work harder sometimes cause symptoms such as chest pain, dizziness, or nausea. An IV may bruise your arm.

The tracers and dyes used in nuclear and MRI stress tests may cause an allergic reaction, but this is rare. If you are or could become pregnant or are breastfeeding, talk with your provider before you have these tests.

WHAT DO THE RESULTS MEAN?

A normal test result means no blood flow problems were found. Your heart is working well.

If your test result was not normal, it may mean you have heart disease. Ask your provider to explain what condition you may have. You may need more tests to find out how serious your condition is.

Learn more about laboratory tests, reference ranges, and understanding results.

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Suicide Risk Screening

<https://medlineplus.gov/lab-tests/suicide-risk-screening/>

SUICIDE RISK SCREENING

WHAT IS A SUICIDE RISK SCREENING?

Suicide is death that happens when someone harms themselves because they want to end their life. It is a major public health concern and is a leading cause of death in the United States and in the world.

Suicide and suicide attempts can have lasting harmful effects:

People who survive a suicide attempt may have serious injuries that can affect their long-term health. They may also have depression or other mental health concerns.

When people die by suicide, it affects their family, friends, and community. They may feel grief, shock, anger, and guilt. Some may develop post-traumatic stress disorder (PTSD), depression, or anxiety. They may also be at risk for suicidal thoughts.

Suicide is often preventable. A suicide risk screening can help determine how likely it is that someone will attempt suicide. During most screenings, a health care provider will ask some questions about behavior and feelings. There are specific questions and guidelines that providers can use. These are known as suicide risk assessment tools. If you or a loved one is found to be at risk for suicide, medical, psychological, and emotional support are available and can help avoid a tragic outcome.

Other names: suicide risk assessment

WHAT IS IT USED FOR?

A suicide risk screening is used to find out if someone is at risk for attempting suicide.

WHY DO I NEED A SUICIDE RISK SCREENING?

You or a loved one may need a suicide risk screening if you notice any of the following warning signs:

Feeling hopeless and/or trapped

Talking about being a burden to others

Increased use of alcohol or drugs

Having extreme mood swings

Having increased anxiety

Withdrawing from social situations or wanting to be alone

A change in eating and/or sleeping habits

A suicide risk screening can be very helpful for people with these warning signs and risk factors. Other warning signs may indicate that someone is at immediate risk for suicide. They include:

Talking about suicide or wanting to die

Searching online for ways to end your life, getting a gun, or stockpiling medicines such as sleeping pills or pain medicines

Talking about having no reason to live

If you or a loved one have any of these warning signs, seek help right away. If there is an immediate life-threatening situation, call 911. Otherwise contact a mental health provider or the National Suicide Prevention Lifeline at 988.

You may also need a screening if you have certain risk factors. You may be more likely to try to harm yourself if you:

Have attempted suicide before

Have a mental health disorder such as depression, another mood disorder, or post-traumatic stress disorder (PTSD)

Have a chronic illness and/or chronic pain

Have legal problems

Have lost your job

Have financial problems
Have trouble in or recently lost a relationship
Are being bullied
Have a history of suicide in your family
Have a history of trauma or abuse
Have a history of substance or alcohol use disorder

WHAT HAPPENS DURING A SUICIDE RISK SCREENING?

A suicide risk screening may be done by your health care provider or a mental health provider. A mental health provider is a health care professional who specializes in diagnosing and treating mental health problems.

Your health care provider may give you a physical exam and ask you about your use of drugs and alcohol, changes in eating and sleeping habits, and mood swings. These could have many different causes. He or she may ask you about any prescription medicines you are taking. This is important because in some cases, antidepressants can increase suicidal thoughts, especially in children, teenagers, and young adults (under the age of 25). You may also get blood, other lab, or imaging tests) to see if a physical disorder could be causing your suicidal symptoms.

Your provider may also use one or more suicide risk assessment tools. A suicide risk assessment tool is a type of questionnaire or guideline for providers. These tools help providers evaluate your behavior, feelings, and suicidal thoughts. The most commonly used assessment tools include:

Patient Health Questionnaire-9 (PHQ9). This tool is made up of nine questions to screen for, diagnose, monitor, and measure the severity of depression.

Ask Suicide-Screening Questions (ASQ). This tool includes four questions and is geared toward youth and adults.

If the results of the ASQ show that someone is at risk for suicide, then a brief suicide safety assessment (BSSA) should be done. The BSSA is done by a trained provider (such as a social worker, nurse practitioner, physician assistant, physician, or other mental health provider). It is done to determine if a more comprehensive mental health evaluation is needed.

If the results of the ASQ show that someone is at risk for suicide, then a brief suicide safety assessment (BSSA) should be done. The BSSA is done by a trained provider (such as a social worker, nurse practitioner, physician assistant, physician, or other mental health provider). It is done to determine if a more comprehensive mental health evaluation is needed.

Suicide Assessment Five-Step Evaluation and Triage (SAFE-T). This assessment uses a five-step evaluation and triage plan. It is used to determine a person's level of suicide risk. It also includes making a treatment plan that lists what kind of interventions are needed, based on the person's suicide risk level. Examples of possible interventions could include transferring the person to a local psychiatric hospital or referring the person for further evaluation

The Columbia-Suicide Severity Rating Scale (C-SSRS). This is a suicide risk assessment scale based on a series of simple, easy-to-understand questions that measure suicide risk.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A SUICIDE RISK SCREENING?

You don't need any special preparations for this screening.

ARE THERE ANY RISKS TO SCREENING?

There is no risk to having a physical exam or a questionnaire.

WHAT DO THE RESULTS MEAN?

If the results of your physical exam or blood, lab, or imaging test show a physical disorder or a problem with a medicine,

your provider may provide treatment and change or adjust your medicines as needed.

The results of a suicide risk assessment tool or suicide risk assessment scale can show how likely it is you will attempt suicide. Your treatment will depend on your risk level. If you are at very high risk, you may be admitted to a hospital. If your risk is more moderate, your provider may recommend one or more of the following:

Brief interventions such as safety planning and follow-up phone calls. A safety plan includes lists of coping skills and people you can call when you need help, as well as a plan on how to keep you away from objects you could use to harm yourself.

Psychological counseling (talk therapy) from a mental health professional. There are different types of talk therapy which may be used, including family therapy.

Medicines, such as antidepressants. However, younger people on antidepressants should be closely monitored. These medicines sometimes increase suicide risk in children and young adults.

Treatment for substance use disorder (if needed).

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A SUICIDE RISK SCREENING?

If you feel you are at risk for attempting suicide, get help right away. There are many ways to get help. You can:

Call 911 or go to your local emergency room.

Call or text 988 to reach the Suicide & Crisis Hotline. Veterans can call and then press 1 to reach the Veterans Crisis Line.

Text the Veterans Crisis Line at 838255 or chat with them online

Chat with someone online at 988lifeline.org.

Call your health care or mental health provider.

Reach out to a loved one or close friend.

If you are worried that a loved one is at risk for suicide, don't leave them alone. You should also:

Encourage them to seek help. Assist them in finding help if needed.

Let them know you care. Listen without judgement and provide encouragement and support.

Keep them away from weapons, pills, and other items that could cause harm.

You may also want to call the Suicide & Crisis Lifeline at 988 for advice and support.

Learn more about laboratory tests, reference ranges, and understanding results.

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Sweat Test for Cystic Fibrosis

<https://medlineplus.gov/lab-tests/sweat-test-for-cystic-fibrosis/>

SWEAT TEST FOR CYSTIC FIBROSIS

WHAT IS A SWEAT TEST?

A sweat test measures the amount of chloride in your sweat. Chloride is a type of electrolyte. Electrolytes are electrically charged minerals that help control the amount of fluids and the balance of acids and bases (pH balance) in your body. Chloride and sodium form the salt found in your sweat.

Normally, chloride moves in and out of your cells through a protein called the cystic fibrosis membrane conductance regulator (CFTR). This protein can be found in the organs that make sweat and mucus, such as your lungs, sweat glands, and intestines. Mucus is the slippery substance that protects the linings of your airways, digestive tract, and other organs and tissues.

If the CFTR protein isn't working as it should, however, chloride can't as easily move in and out of the cells in your mucus/sweat-producing organs. This condition is called cystic fibrosis (CF). If you have CF, you will also have much more chloride in your sweat.

With CF, chloride builds up in your cells, causing your mucus to become thick and sticky. When this happens, mucus can build up in your organs. This can not only make it hard to breathe but can lead to frequent lung infections and organ damage.

Other names: sweat chloride test, cystic fibrosis sweat test, sweat electrolytes

WHAT IS IT USED FOR?

A sweat test is used to diagnose cystic fibrosis (CF).

CF is an inherited disease that can affect people of all ages. To have cystic fibrosis, you must inherit a changed CFTR gene from both your parents. (Changes in genes are also called variants or mutations.) If you inherit one changed gene, you won't have the disease, but you will be a carrier. Being a carrier means that if you have children with someone who also has changes in one or both of their CFTR genes, your children could inherit two changed genes and have CF.

CF most commonly affects the lungs. However, it can also cause malnutrition because the disease may reduce your body's ability to absorb nutrients from food.

WHY DO I NEED A SWEAT TEST?

A sweat test may be ordered if your health care provider suspects you or your child has cystic fibrosis (CF). It's most commonly ordered for newborns. In the United States, babies have newborn screening tests to check for a variety of conditions, including CF. If a screening test shows that your baby may have CF, a sweat test will be needed to make a diagnosis.

It's best if the test is done when your baby is between 10 days and 4 weeks old. Starting treatment as soon as possible can delay or prevent health problems from CF.

A provider may also order a cystic fibrosis sweat test for a child or adult if they have symptoms of CF or if someone in the family has CF. Most people with CF have symptoms, but some people don't. In milder cases, symptoms may not show up until adulthood. The symptoms of CF depend on which organs are affected and may include:

A wheeze or cough that may produce mucus or blood.

Clubbing of fingers and toes, which means that your fingernails and toenails are wide and rounded like the back of a spoon. This is usually due to a low level of oxygen in your blood that happens if your lungs aren't working well. The lack of oxygen may also cause changes in your nails.

Fever, which may include night sweats (heavy sweating during sleep).

Digestive symptoms, such as severe abdominal (belly) pain, chronic (long-lasting) diarrhea, or constipation. Children may also have oily or loose stools.

Being underweight.
Muscle and joint pain.
Delayed growth.
Delayed puberty.
Salty skin and saltier than normal sweat.
Sinus infections.
Infertility, especially in men.
If your provider suspects CF, it's important to get tested. There is no cure for CF, but treatment can help you and/or your child live a longer, healthier life.

WHAT HAPPENS DURING A SWEAT TEST?

To do a sweat test, a health care professional will collect a sample of sweat from you or your child. This takes about an hour and usually includes the following steps:
Your provider will wash and pat dry a small area of the forearm or leg.
To cause sweating, an odorless chemical (pilocarpine) will be applied to the small area of the forearm or leg. An electrode will be taped over the area.
A weak electrical current will be sent through the electrode to make the chemical seep into the skin. This may cause a little tingling or warmth.
After about 5 minutes, the electrode will be removed. The health care professional will collect the sweat by taping a piece of filter paper or gauze on the area or by using a small plastic coil.
Sweat will be collected for 30 minutes.
The sweat sample will be sent to a lab for testing.
Some babies may not produce enough sweat for the test on the first try. If this happens, the test will usually be repeated on another day.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a sweat test, but you should avoid applying any creams or lotions to the skin for 24 hours before the test.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to a sweat test. The electrode may cause a tingling or tickling sensation from the electric current, but this is not painful.

WHAT DO THE RESULTS MEAN?

Chloride in sweat is measured in mmol/L (millimoles of chloride per liter of sweat):
A chloride level of 60 or greater means that you or your child most likely has cystic fibrosis. To confirm the diagnosis, your provider may have you repeat the test or order a CF genetic test.
A chloride level of 30 to 59 means that more testing is needed because the test result is unclear, and CF is possible. Your provider may order another sweat test or a different type of test, such as genetic testing.
A chloride level less than 29 is normal. This means that you or your child is unlikely to have CF. But a normal test result doesn't rule out CF. Certain people with CF have normal sweat test results. If you or your child have symptoms of CF or if your baby had a newborn screening test that showed possible CF, your provider may order more tests.
Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A SWEAT TEST?

In rare cases, conditions other than CF may result in high chloride levels on a sweat test. These conditions include hypothyroidism, nephrogenic diabetes insipidus, and Addison disease.

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Synovial Fluid Analysis

<https://medlineplus.gov/lab-tests/synovial-fluid-analysis/>

SYNOVIAL FLUID ANALYSIS

WHAT IS A SYNOVIAL FLUID ANALYSIS?

A synovial fluid analysis is a group of tests that looks for changes in your synovial fluid. These tests check for the cause of pain and swelling in your joints. Joints are places in the body where two bones meet, such as your knees, shoulders, hips, hands, and feet. Synovial fluid, also known as joint fluid, is a thick liquid located between your joints. The fluid cushions and protects the ends of bones and reduces friction during movement.

Normally, there is only a small amount of synovial fluid between the joints. If you have a joint problem, extra fluid can build up, causing pain, stiffness, and inflammation. Sometimes, removing the excess fluid is enough to reduce the pain you may have in the affected joint.

A synovial fluid analysis is a group of tests that checks for A synovial fluid analysis can help find out if you may have a disorder that affects the joints. The analysis usually includes:

A check of the appearance of the synovial fluid, such as its color and thickness

Chemical tests to check for changes in the synovial fluid's chemicals, such as glucose, protein acid, and uric acid

Microscopic analysis to look for crystals, bacteria, and other substances

Other names: joint fluid analysis

WHAT IS IT USED FOR?

A synovial fluid analysis is used to help diagnose the cause of joint pain and inflammation. Inflammation is your body's response to injury or infection. It can cause pain, swelling, redness, and loss of function in the affected area. Causes of joint problems include:

Osteoarthritis, the most common form of arthritis. This type of arthritis only affects the joints, usually in the hands, knees, hips, neck, and lower back. It's a chronic (long-lasting) disease that causes the joint cartilage to break down and become rough. It can be painful and lead to loss of mobility and function.

Gout or calcium pyrophosphate deposition (pseudogout), types of inflammatory arthritis. They can cause pain, swelling, and redness in one or more joints. Having crystals in the synovial fluid may be a sign of these types of arthritis.

Rheumatoid arthritis, a condition in which the body's immune system attacks healthy cells in your joints, causing pain, swelling, stiffness, and loss of function in your joints. It most commonly occurs in the wrist and fingers.

Joint effusion, a condition that happens when too much fluid builds up around a joint. It often affects the knee. When it affects the knee, it may be referred to as knee effusion or fluid on the knee.

Infection in a joint.

A bleeding disorder, such as hemophilia. Hemophilia is a rare disorder in which the blood does not clot well. This can lead to sudden bleeding inside your body, such as in your joints, muscles, and organs. Sometimes the excess blood ends up in the synovial fluid.

WHY DO I NEED A SYNOVIAL FLUID ANALYSIS?

You may need this test if you have symptoms of a joint disorder. These include:

Joint pain with a fever

Joint swelling

Redness over the joint

A joint that feels warm to the touch

Difficulty bending a joint

This test may also be used to monitor a known joint condition.

WHAT HAPPENS DURING A SYNOVIAL FLUID ANALYSIS?

Your synovial fluid will be collected in a procedure called arthrocentesis, also known as joint aspiration. Your provider can then check the appearance, chemical properties, and what makes up your synovial fluid to check for any signs of infection or disease.

During the procedure:

You will be asked to sit or lie down so your provider can access the affected joint.

The provider will clean the skin on and around the affected joint.

The provider will inject an anesthetic and/or apply a numbing cream to the skin. You shouldn't feel any pain during the procedure, but you may have some discomfort. If your child is getting the procedure, they may also be given a sedative. Sedatives are medicines that have a calming effect and help reduce anxiety.

Once the needle is in place, your provider will withdraw a sample of synovial fluid and collect it in the syringe of the needle.

Your provider will put a small bandage on the spot where the needle was inserted.

The procedure usually takes less than two minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if you need to fast and if there are any special instructions to follow.

Tell your provider if you are taking any blood thinner medicines.

ARE THERE ANY RISKS TO THE TEST?

Your joint may be sore for a couple of days after the procedure. Your provider may ask that you limit your activity or recommend that you use ice packs or elevate the joint.

Serious complications, such as infection and bleeding may happen, but are uncommon. Let your provider know if you have increased redness, swelling, or pain at the joint.

WHAT DO THE RESULTS MEAN?

Synovial fluid is usually light yellow, clear, and a little sticky or stringy. Abnormal synovial fluid may be cloudy and thicker or thinner than normal.

If your results show your synovial fluid was not normal, it may be a sign of one of the following conditions:

A type of arthritis, such as osteoarthritis, rheumatoid arthritis, or gout

A bleeding disorder

A bacterial infection

Your specific results will depend on what abnormalities were found. If you have questions about your results, talk to your provider.

To understand the results of a synovial fluid analysis, your provider will consider your symptoms, medical history, and the results of other tests.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A SYNOVIAL FLUID ANALYSIS?

Arthrocentesis, the procedure used to do a synovial fluid analysis, may also be done to remove excess fluid from a joint. This procedure can help relieve pain and other symptoms.

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Syphilis Tests

<https://medlineplus.gov/lab-tests/syphilis-tests/>

SYPHILIS TESTS

WHAT ARE SYPHILIS TESTS?

Syphilis tests generally look for certain antibodies in your blood. Antibodies are proteins that your immune system makes when it finds harmful substances in your body. In this case, the harmful substance is the bacterium that causes syphilis, one of the most common sexually transmitted infections (STIs.)

Testing usually involves two steps. The first is used to screen for antibodies that your immune system makes in response to syphilis, some autoimmune diseases, and other infections. The two most common types of this test are:

Rapid plasma reagin (RPR), which is a blood test that looks for an antibody called reagin. Your body often makes reagin in response to a syphilis infection.

Venereal Disease Research Laboratory (VDRL) Test, which can be done on blood or spinal fluid. This test looks for a broader range of antibodies that your immune system might make in response to a syphilis infection.

The second step in syphilis testing is to confirm the findings of the first step. If the result of your screening test shows you have antibodies linked to syphilis infections, you'll need a second test to confirm whether you have syphilis.

Usually, the second test looks for antibodies that your immune system makes only to fight off syphilis. If you have these antibodies, it means you either currently have a syphilis infection or had a syphilis infection that's been treated.

In some cases, a health care provider will use a test that looks for actual syphilis bacteria, instead of antibodies. These tests are less common because they can only be done in specialized labs.

Other names: rapid plasma reagin (RPR), Venereal Disease Research Laboratory (VDRL), fluorescent treponemal antibody absorption (FTA-ABS) test, agglutination assay (TP-PA), darkfield microscopy, Treponema pallidum particle agglutination assay (TP-PA), Fluorescent treponemal antibody absorption (FTA-ABS) test, Microhemagglutination assay for antibodies to *T. pallidum* (MHA-TP), *T. pallidum* hemagglutination assay (TPHA), *T. pallidum* enzyme immunoassay (TP-EIA), Chemiluminescence immunoassays (CLIA)

WHAT ARE THEY USED FOR?

All syphilis tests are used to check your blood for signs of a syphilis infection. Like other STIs, syphilis spreads through vaginal, oral, or anal sexual contact with someone who has the infection. It can also pass during pregnancy to the fetus.

Syphilis usually develops in the following four stages.

First stage. This usually starts 2-12 weeks after exposure and can go unnoticed. The main symptom is a small, painless sore (chancre) on your genitals or in your mouth, anus, or rectum.

Second stage. This usually starts 1-6 months after the chancre goes away. There are more symptoms in this stage, including fever and swollen glands.

Third (or latent) stage. If you don't receive treatment, then your infection will enter this stage, which can last for years. Though you likely won't have symptoms during this phase, that does not mean you no longer have syphilis. The infection can still spread and cause damage.

Fourth stage. This can start years after exposure if it isn't treated. Its symptoms are far more severe, including dementia (a mental disorder that causes memory loss) and blindness.

People are more likely to pass a syphilis infection to someone else during the infection's first phase, when it's more likely they won't have any obvious symptoms. It's therefore important to find and test for the infection early. Doing so may help to prevent the spread of syphilis to others.

Syphilis is also easiest to cure in the early stages of infection. If it's not treated and develops to a late stage, it can cause permanent damage to your health. Treatment can still help, but it will not reverse the damage. In rare cases, untreated syphilis can even cause death.

WHY DO I NEED SYPHILIS TESTS?

You should get tested for syphilis if you have symptoms of the infection or if your sexual partner was recently diagnosed with it. Most symptoms usually appear in the second stage. Along with fever and swollen glands, these can include:
Rough, red rash, usually on the palms of the hands or the bottom of the feet.

Headache or muscle aches.

Sore throat.

Fatigue.

Weight loss.

Hair loss in patches.

Your provider may also test you for syphilis if you show any of the more serious signs of the infection's fourth stage. Along with dementia and blindness, these can include:

Deep bone pain.

Numbness.

Dizziness.

Heart damage.

Hearing loss.

Even if you don't have symptoms, you should get tested regularly if you have a high risk of getting syphilis. You're more likely to get syphilis if you have:

Multiple sex partners.

A partner with multiple sex partners.

Unprotected sex (sex without using a condom).

An HIV infection and are sexually active.

Another sexually transmitted disease, such as gonorrhea.

Sex with men who have sex with men (MSM).

Ask your provider how often you should get tested.

You will also need a syphilis test if you are pregnant. If you pass it on to your fetus, it could cause serious, and sometimes deadly, health problems. If you are pregnant, the Centers for Disease Control and Prevention recommends that you have a syphilis test at your first prenatal visit. If you are more likely to become infected with syphilis, you should be tested again at 28 weeks of pregnancy and at delivery.

WHAT HAPPENS DURING A SYPHILIS TEST?

A syphilis test is usually a blood test. During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

At any stage, syphilis can affect your brain, spinal cord, and nerves. If your symptoms show that syphilis may be affecting your brain and nervous system, your provider may order a syphilis test of your cerebrospinal fluid (CSF).

CSF is a clear fluid found in your brain and spinal cord. To get a sample of the fluid, a provider will do a procedure called a spinal tap, also known as a lumbar puncture. A spinal tap is usually done in a hospital. During the procedure:

You will lie on your side or sit on an exam table.

A health care provider will clean your back and inject an anesthetic into your skin, so you won't feel pain during the procedure. Your provider may put a numbing cream on your back before this injection.

When the area on your back is completely numb, your provider will insert a thin, hollow needle between two vertebrae in your lower spine. Vertebrae are the small backbones that make up your spine.

Your provider will withdraw a small amount of cerebrospinal fluid for testing. This will take about five minutes.

You'll need to stay very still while the fluid is being withdrawn.

Your provider may ask you to lie on your back for an hour or two after the procedure. This may prevent you from getting a headache afterward.

At-home syphilis test kits are also available. These kits let you collect a sample of your blood at home which you send to a lab for testing. However, these are only screening tests. In other words, they only test for antibodies to the syphilis bacteria. So, you'll still need another test to confirm your diagnosis. Talk with your provider to find out if a self-collection test is right for you.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a syphilis blood test. For a lumbar puncture, you may be asked to empty your bladder (pee) and bowels (poop) before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

With a spinal tap, there is also very little risk. You may feel a little pinch or pressure when the needle is inserted. After the test, you may feel some pain or tenderness in your back or have some bleeding where the needle was inserted. You may also get a headache. The headache may last for several hours or up to a week or more, but your provider may suggest treatment to relieve the pain.

WHAT DO THE RESULTS MEAN?

If your screening test results are negative (normal), it means you probably don't have a syphilis infection. However, it doesn't completely rule out the possibility. Antibodies can take a couple of weeks to develop. So, you may need another screening test if you think you were exposed to syphilis. Your provider can tell you whether you need another test.

If your screening test results are positive, it means you have antibodies that may be from a syphilis infection. Your provider will likely order a second test to confirm whether or not you have syphilis.

If your follow-up test confirms you have syphilis, you will probably be treated with penicillin, a type of antibiotic. Antibiotic treatment cures most early-stage syphilis infections. Later-stage syphilis is also treated with antibiotics, but this treatment can't undo any damage that the syphilis caused.

If you have questions about your test results, or about syphilis, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT SYPHILIS TESTS?

If you are diagnosed with syphilis, you need to tell anyone you've had sexual contact with so that they can get tested and treated if necessary. You will likely need repeat syphilis screening tests to see how well your treatment is working. Avoid sexual contact with others until testing shows you are cured.

In certain parts of the world, there are bacteria that are closely related to the bacteria that cause syphilis. These other bacteria usually spread through non-sexual skin contact with sores on a person who has an infection. They can cause diseases called yaws, bejel, and pinta. Blood tests cannot tell the difference between syphilis and their related bacteria. But a provider can make a diagnosis by looking at skin sores and asking about your medical and travel history.

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Testosterone Levels Test

<https://medlineplus.gov/lab-tests/testosterone-levels-test/>

TESTOSTERONE LEVELS TEST

WHAT IS A TESTOSTERONE LEVELS TEST?

A testosterone levels test measures the amount of the hormone testosterone in a sample of your blood. Hormones are chemical messengers in your bloodstream that control the actions of certain cells or organs.

Though often considered a male sex hormone, testosterone is present in females in smaller amounts. In both sexes, testosterone helps control your sex drive and helps with muscle and bone growth. In males, it is also involved in

Body hair growth.

A deepening voice during puberty.

The production of sperm.

The making of new red blood cells.

An increase in height.

All the testosterone in your blood is either bound to proteins or unbound (free). Free testosterone is more available for your body to use when building bones and muscles. However, most of your testosterone is bound to proteins such as sex-hormone-binding globulin (SHBG) and albumin. When testosterone is bound, your body can't use it as easily. This is necessary for your health. Having most of your testosterone bound to proteins makes sure your body doesn't use too much.

Your health care provider can measure your testosterone levels by ordering one of three tests. These are:

A total testosterone test, which measures both bound and free testosterone. This is the most common type of test.

A free testosterone test, which measures only the unattached form of testosterone. This test is less common, but it may help your provider diagnose certain medical conditions.

A bioavailable testosterone test, which measures free testosterone and testosterone that's attached to proteins other than SHBG. This test isn't commonly done, but like a free testosterone test, it may help diagnose certain medical conditions.

Other names: serum testosterone, total testosterone, free testosterone, bioavailable testosterone

WHAT IS IT USED FOR?

A testosterone levels test may be used with other tests to help your provider find the cause of abnormal testosterone levels. It can also be used to monitor your health if you:

Have had an abnormal testosterone test result in your past.

Are taking medicine that may affect your testosterone levels.

Are taking hormone therapy.

In children and teens, your provider may use a testosterone levels test to figure out what's causing early or delayed puberty.

WHY DO I NEED A TESTOSTERONE LEVELS TEST?

You may need this test if you have symptoms of abnormal testosterone levels.

For males who have gone through puberty, the test is mostly used for symptoms of low T levels (testosterone levels which are too low).

The symptoms of low T may include:

Low sex drive.

Erectile dysfunction or "ED" (difficulty getting or keeping an erection).

Infertility (difficulty getting someone pregnant).

Enlarged breasts.

Lack of face or body hair.

Thinning bones or anemia without a known cause.

Loss of muscle mass.

For females, the test is mostly used for symptoms of high T levels.

Symptoms of high T may include:

Abnormal hair growth on the body or face.

Hair loss on the head that may be like male-pattern baldness (hair loss at the temples and balding on the top of the head).

Voice changes.

Irregular or no menstrual periods.

Infertility (difficulty getting pregnant).

Acne.

WHAT HAPPENS DURING A TESTOSTERONE LEVELS TEST?

Most testosterone levels tests involve a blood draw in a medical setting. Your provider will take a blood sample in the morning, between 7 and 10. During these hours, your testosterone levels are typically the highest.

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

You may be able to test your testosterone levels with an at-home kit. These have everything you need to collect a blood or saliva (spit) sample to send to a lab for testing. Talk to your provider if you think an at-home test is right for you. If you do a home test, be sure to follow your provider's instructions for collecting the sample and for returning the kit for testing.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

In certain cases, you may need to fast (not eat or drink) for several hours before the test. You may also need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. Ask your provider how you should prepare for a testosterone levels test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

A testosterone level test alone can't diagnose any health conditions, but abnormal results with symptoms may be a sign of a health problem.

In males, low T levels may be caused by different conditions, including:

Problems with the testicles (the part of the male reproductive system that makes sperm), which are the main producers of testosterone in males.

Chronic (long-lasting) diseases such as liver, or kidney disease.

A pituitary gland disorder.

Benign tumors (not cancer).

Injury to the testicles.

Alcohol use disorder (AUD).

Certain genetic diseases such as Klinefelter syndrome.

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Benign tumors (not cancer).
Injury to the testicles.
Alcohol use disorder (AUD).
Certain genetic diseases such as Klinefelter syndrome.
In females low T levels are normal, but extremely low levels may be a sign of:

An adrenal gland disorder.
A pituitary gland disorder.
A disorder of the ovaries (the part of the female reproductive system that makes eggs).
An adrenal gland disorder.
A pituitary gland disorder.
A disorder of the ovaries (the part of the female reproductive system that makes eggs).
In males, high T levels may be caused by:

A tumor in the testicles.
Steroid use.
An adrenal gland disorder, such as congenital adrenal hyperplasia (CAH).
A tumor in the testicles.
Steroid use.
An adrenal gland disorder, such as congenital adrenal hyperplasia (CAH).
In females, high T levels may be caused by:

Polycystic ovary syndrome (PCOS).
Ovarian cancer or other problems with the ovaries.
An adrenal gland disorder, such as congenital adrenal hyperplasia (CAH).
A disorder of the pituitary gland.
Polycystic ovary syndrome (PCOS).
Ovarian cancer or other problems with the ovaries.
An adrenal gland disorder, such as congenital adrenal hyperplasia (CAH).
A disorder of the pituitary gland.

Your test results may mean different things depending on which test was done, your age, your symptoms, the results of other tests, and other conditions you may have. If your results are not normal, it doesn't always mean you have a medical condition that needs treatment. It's normal for testosterone levels to decrease with age. If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Therapeutic Drug Monitoring

<https://medlineplus.gov/lab-tests/therapeutic-drug-monitoring/>

THERAPEUTIC DRUG MONITORING

WHAT IS THERAPEUTIC DRUG MONITORING (TDM)?

Therapeutic drug monitoring (TDM) is testing that measures the amount of certain medicines in your blood. It checks whether the amount of medicine you take is safe and effective.

All medicines don't require therapeutic monitoring. Usually, your health care provider can determine the right dose of medicine for you without special testing. But some medicines must maintain a certain level in your blood to work well. This is called the "therapeutic dose level."

As soon as you take medicine, your body begins to process and remove the medicine from your body. This means that the level of the medicine in your blood will vary over time. Many things can affect how quickly this process happens, including:

Aging.

Pregnancy.

Illness.

Infection.

If you're taking a medicine that needs to be kept at a certain level, it can be hard to figure out which dose provides enough medicine to treat your condition without causing dangerous side effects. TDM helps your provider figure out if you are taking the right dose of your medicine.

Other names: medicine levels blood test, therapeutic drug levels

WHAT IS IT USED FOR?

Therapeutic drug monitoring (TDM) helps determine the best dosages if you're taking certain hard-to-dose medicines. Many medicines that require TDM are ones that you might take for a long time. While you are taking the medicines, you may have changes that could affect the medicine levels in your blood. These may include:

Changes in your health.

Any other medicines you take.

Aging.

For each of these changes, you may need to change your medicine dose. Otherwise, you might have too little medicine, which might not be effective, or too much (toxicity), which might not be safe.

TDM may also be used to make sure you are taking your medicine as prescribed.

There are many different medicines that are monitored; some of the most common ones include:

You may also need therapeutic drug monitoring if you are taking certain biologic medicines. This type of medicine is made from living cells instead of chemicals. These cells can be used to make proteins (like insulin), antibiotics, or altered versions of your cells. By working with specific proteins or cells in your immune system, biologic medicines can often treat conditions that traditional medicine cannot (such as cancer and autoimmune diseases).

Like with other medicines that need therapeutic monitoring, however, your provider may closely monitor the levels of biologic medicine in your blood to make sure you don't have:

Low levels of the medicine in your blood, which could make the drug less effective.

High levels of the medicine in your blood, which could make the drug unsafe.

An immune response to the medicine, which is when your immune system rejects the drug.

Monoclonal antibodies (mAb) are some of the most common biologic medicines. Laboratories make monoclonal antibodies to only work with specific cells, such as those that are cancerous. Because their targets are so specific, these artificial antibodies can make treatment much more effective. Your provider may monitor mAb levels during treatment for:

Cancer.

Organ transplant rejection.

Inflammation and autoimmune diseases.
Osteoporosis.
Severe viral infections.
Nervous system disorders.
Other examples of biologic medicines include:
Adalimumab.
Apixaban.
Nivolumab.
Pembrolizumab.
Rituximab.

WHY DO I NEED TDM?

You may need testing when you first start taking a medicine. This helps your provider figure out the most effective dose for you. Once that dose is determined, you may be tested regularly to make sure the medicine is still effective without being harmful. You may also need testing if you have symptoms of a serious side effect. Side effects can vary, depending on the medicine. Your provider will tell you which symptoms to watch out for.

WHAT HAPPENS DURING TDM?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Depending on the type of medicine you are taking, you may need to schedule your test before or after taking your regular dose. Be sure to tell your provider if you are taking other medicines or supplements, in case they affect the results.

ARE THERE ANY RISKS TO TDM?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will show if medicine levels in your blood are in a range that is medically helpful but not dangerous. This is called the "therapeutic range." This range varies, depending on the type of medicine you are taking and your own health needs. If your results are not in this range, your provider may need to adjust your dosages. If your dosages change, you may get repeated tests until your medicine levels fall into the therapeutic range.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Thyroglobulin

<https://medlineplus.gov/lab-tests/thyroglobulin/>

THYROGLOBULIN

WHAT IS A THYROGLOBULIN TEST?

A thyroglobulin test measures the amount of a protein called thyroglobulin (Tg) in a sample of your blood. Your thyroid (a small, butterfly-shaped gland in your neck) makes thyroglobulin along with many other important hormones. Together, these hormones control bodily activities such as your heart rate and how fast you burn calories from food.

Normally, your thyroid releases small amounts of thyroglobulin into your bloodstream. However, various conditions that affect the thyroid can lead to higher levels of this protein. The most common causes of high thyroglobulin levels are two types of thyroid cancer called papillary carcinoma and follicular thyroid cancer. In both conditions, cancerous cells produce thyroglobulin. Because of this, thyroglobulin is usually measured as a tumor marker. Tumor markers are substances made by cancer cells and/or by normal cells in response to cancer in your body.

Other conditions can, however, affect your thyroglobulin levels. These include various thyroid diseases such as:

Hyperthyroidism, a condition where your thyroid becomes overactive and makes too much thyroglobulin.

Hypothyroidism, a condition where your thyroid becomes underactive and makes too little thyroglobulin.

Thyroiditis, a condition caused by inflammation of the thyroid, which can lead to a temporary rise in thyroglobulin levels.

Because such a wide range of conditions can affect thyroglobulin levels, health care providers don't use the results from a thyroglobulin test to diagnose cancer.

Other names: Tg, TGB, thyroglobulin tumor marker

WHAT IS IT USED FOR?

Though a thyroglobulin test alone cannot be used to diagnose cancer, it can give you and your provider valuable information about any thyroid cancer you already have. For example, a thyroglobulin test can help your provider:

See if thyroid cancer treatment was successful. Usually, the goal of treatment for thyroid cancer is to get rid of all thyroid tissue. This means both healthy cells and cancer cells. So, if treatment is successful, there should be little or no thyroglobulin in your blood. If you still have thyroglobulin in your blood after treatment, you may need to go through treatment again.

Predict how your cancer will behave over time. This prediction is called a prognosis, and it can tell you how likely it is that your cancer will return and respond to treatment.

See if cancer has returned after successful treatment. Because successful treatment involves the removal of all thyroid tissue, if you have any thyroglobulin in your blood after treatment, it may mean your cancer has returned.

Your provider may also order a thyroglobulin test to diagnose and monitor other, noncancerous conditions such as hyperthyroidism and hypothyroidism. In rarer cases, a thyroglobulin test can see if there is any inflammation in your thyroid that's been caused by a viral infection, such as COVID-19.

WHY DO I NEED A THYROGLOBULIN TEST?

You may need a thyroglobulin test before starting treatment for thyroid cancer. Your provider will compare these test results to your test results after treatment.

You may also need this test:

Four to six weeks after treatment for thyroid cancer to check whether all thyroid tissue has been removed.

Regularly after successful treatment to make sure cancer hasn't come back.

If your provider is monitoring a thyroid condition that isn't cancer, a thyroglobulin test can help evaluate how active your thyroid cells are and how your treatment is working.

WHAT HAPPENS DURING A THYROGLOBULIN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You usually don't need any special preparations for a thyroglobulin test. But you need to avoid taking certain vitamins or supplements so tell your provider about everything you take.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Understanding the results of a thyroglobulin test after treatment can be complicated. The meaning of your results depends on your health history, the type of treatment you had, and the results of other tests. So, it's best to ask your provider to explain what your test results say about your health.

In general, if you were tested after treatment for thyroid cancer:

Very low levels or no thyroglobulin may mean that your cancer treatment successfully removed all thyroid tissue, including cancerous tissue. However, you will still need more testing over time to make sure.

Thyroglobulin levels that stay high or increase may mean that your treatment did not successfully remove all thyroid tissue or that you still have thyroid cancer. It may also mean that your cancer has grown and/or spread.

Thyroglobulin levels that were low after treatment but later increased may mean that your thyroid cancer has come back after treatment.

Because other thyroid conditions and certain illnesses can also affect thyroglobulin, your provider will likely check your levels over time instead of relying on a single test result.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A THYROGLOBULIN TEST?

Labs use different methods to measure the amount of thyroglobulin in your blood sample. The test method can affect your results. So, it's important to have your tests done the same way, and usually in the same lab. This allows your provider to compare your results over time. If you have questions about this, talk with your provider.

A thyroglobulin test may not be useful for monitoring your thyroid cancer treatment if you have thyroglobulin antibodies in your blood. These antibodies are proteins that your immune system may make. They attach to thyroglobulin and can make your thyroglobulin levels appear lower than they really are.

Your provider will usually order a thyroid antibody test to find out whether thyroglobulin testing will be useful. If you have thyroglobulin antibodies, other tests will be used to see if your cancer treatment was successful.

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Thyroid Antibodies

<https://medlineplus.gov/lab-tests/thyroid-antibodies/>

THYROID ANTIBODIES

WHAT IS A THYROID ANTIBODY TEST?

A thyroid antibody test measures the level of thyroid antibodies in a sample of your blood. Your thyroid is a small, butterfly-shaped gland in your neck. It makes hormones that control many activities in your body, including your heart rate and how fast you burn calories from food.

Antibodies are proteins that your immune system makes to fight foreign substances, such as viruses and bacteria. But sometimes antibodies attack the healthy cells of your own tissues and organs by mistake. This is called autoimmune disease.

If antibodies attack your thyroid, they can cause serious thyroid autoimmune diseases:

Hashimoto's disease, also known as Hashimoto's thyroiditis, is the most common cause of hypothyroidism (underactive thyroid). Hypothyroidism happens when your thyroid doesn't make enough thyroid hormones to meet your body's needs. Thyroid hormones affect the way your body uses energy. So, without enough thyroid hormones, many of your body's functions slow down.

Graves' disease is the most common cause of hyperthyroidism (overactive thyroid). With hyperthyroidism, your thyroid makes more thyroid hormones than your body needs. This causes many of your body's functions to speed up.

If your health care provider has diagnosed you with hypothyroidism or hyperthyroidism, thyroid antibody tests can help find out if a thyroid autoimmune disease is causing the problem. There are different tests that check for different types of thyroid antibodies. Your provider will choose tests for you based on your symptoms, the results of other tests, and information about your medical history and family health history.

Other names: thyroid autoantibodies, thyroid peroxidase antibody, TPO, anti-TPO, antithyroid microsomal antibody, thyroid-stimulating immunoglobulin, TSI, TSH receptor antibody, Long-acting thyroid stimulator, LATS, TSH receptor-binding inhibitory immunoglobulin, Human thyroid stimulator

WHAT IS IT USED FOR?

If tests of your thyroid hormone levels show that your thyroid isn't working properly, your provider may order thyroid antibody tests to see if an autoimmune disease is causing the problem.

Thyroid antibody tests are used to check for different thyroid antibodies that affect your thyroid in different ways. These antibodies include:

Thyroid peroxidase antibodies (TPOAb). High levels of these antibodies are a sign that Hashimoto's disease is causing hypothyroidism.

Thyroglobulin antibodies (TgAb). High levels of these antibodies are a sign of Hashimoto's disease.

Thyrotropin receptor antibodies (TRAb). These antibodies can be a sign of Graves' disease.

WHY DO I NEED A THYROID ANTIBODY TEST?

You may need a thyroid antibody test if you have symptoms of Hashimoto's disease or Graves' disease. But first, your provider will usually order blood tests to check your thyroid hormone levels to see if you have hyperthyroidism or hypothyroidism. Thyroid hormone tests include T3, T4, and TSH (thyroid-stimulating hormone).

Sometimes these test results, along with your health history, your family, an exam, and your symptoms, are enough to make a diagnosis with thyroid antibody testing.

Symptoms of Hashimoto's disease include:

Weight gain

Fatigue

Hair loss

Low tolerance for cold temperatures
Irregular menstrual periods
Constipation
Depression
Joint pain
Symptoms of Graves' disease include:
Weight loss
Bulging of the eyes
Tremors in the hand
Low tolerance for heat
Trouble sleeping
Anxiety
Increased heart rate
Swollen thyroid, known as goiter

WHAT HAPPENS DURING A THYROID ANTIBODY TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

There are no special preparations necessary for a thyroid antibodies blood test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results may show one of the following:

Negative: no thyroid antibodies were found. This means your thyroid symptoms are probably not caused by an autoimmune disease.

Positive: thyroid antibodies, TPOAb and/or TgAb, were found. This may mean you have Hashimoto's disease. Most people with Hashimoto's disease have high levels of one or both of these types of antibodies.

Positive: thyroid antibodies, TPOAb and/or TRAb, were found. This may mean you have Graves' disease.

The more thyroid antibodies you have, the more likely it is that you have an autoimmune disorder of the thyroid. If you are diagnosed with Hashimoto's disease or Graves' disease, there are medicines you can take to manage your condition.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A THYROID ANTIBODY TEST?

Thyroid disease can get worse during pregnancy. This can harm both the mother and her unborn baby. If you have ever had thyroid disease and are pregnant, you may be tested for thyroid antibodies along with tests that measure thyroid hormones. Medicines to treat thyroid disease are safe to take during pregnancy.

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Thyroxine (T4) Test

<https://medlineplus.gov/lab-tests/thyroxine-t4-test/>

THYROXINE (T4) TEST

WHAT IS A THYROXINE (T4) TEST?

Thyroxine is also called T4. T4 is a hormone that your thyroid gland makes. A T4 test measures the level of T4 in a sample of your blood. Too much or too little T4 may be a sign of a thyroid problem.

Your thyroid is a small, butterfly-shaped gland in the front of your neck. It makes hormones that control the way your body uses energy. These hormones affect nearly every organ in your body and control many of your body's most important functions. For example, they affect your breathing, heart rate, weight, digestion, and mood. In children, thyroid hormones affect growth, too.

T4 is the main hormone that your thyroid makes. There are two forms of it in your blood:

Free T4 is the active form of thyroxine hormone that enters your body tissues where you need it.

Bound T4 is thyroxine that attaches or binds to certain proteins which prevent it from entering your body tissues. It stays in your bloodstream as a "backup supply" until your tissues need it.

T4 levels can be measured with either a free T4 test or a total T4 test:

A free T4 test measures the amount of free T4 in your blood. Medical experts believe this test is more accurate than a total T4 test, so it's used more often.

A total T4 test measures free and bound T4 together.

A T4 test alone can't provide enough information to diagnose thyroid problems. So, it's usually done with a TSH blood test. TSH stands for thyroid stimulating hormone. It's a hormone made by your pituitary gland, a small gland at the base of your brain. TSH tells your thyroid how much hormone to make. Normally, if your T4 levels are too low, your pituitary makes more TSH to make your thyroid work harder. If your T4 levels are too high, your pituitary stops making TSH.

Other names: free thyroxine, free T4, total T4 concentration, Free T4 index (FTI), thyroxine screen, free T4 concentration, thyroxine test by equilibrium dialysis

WHAT IS IT USED FOR?

A T4 test is usually used with other thyroid tests to help diagnose and monitor thyroid disease and to gather more information about other conditions that may affect your thyroid. It may be used to:

Diagnose:

Hypothyroidism, or underactive thyroid, which is when your thyroid gland doesn't make enough thyroid hormone to meet your body's needs.

Hyperthyroidism, or overactive thyroid, which is when your thyroid gland makes more thyroid hormone than your body needs.

Hypothyroidism, or underactive thyroid, which is when your thyroid gland doesn't make enough thyroid hormone to meet your body's needs.

Hyperthyroidism, or overactive thyroid, which is when your thyroid gland makes more thyroid hormone than your body needs.

Help learn more about:

Other thyroid conditions, such as if you have thyroid nodules (growths on your thyroid that aren't cancer) or a goiter (an enlarged thyroid that may make your neck look swollen). Sometimes these conditions can cause high T4 levels.

Disorders of your pituitary gland, abnormal T4 levels are usually caused by thyroid problems. But sometimes they are a sign of a pituitary problem that causes too much or too little TSH.

Disorders of your hypothalamus, an area of your brain that controls your pituitary gland and other body functions.

Other thyroid conditions, such as if you have thyroid nodules (growths on your thyroid that aren't cancer) or a goiter (an enlarged thyroid that may make your neck look swollen). Sometimes these conditions can cause high T4 levels.

Disorders of your pituitary gland, abnormal T4 levels are usually caused by thyroid problems. But sometimes they are a sign of a pituitary problem that causes too much or too little TSH.

Disorders of your hypothalamus, an area of your brain that controls your pituitary gland and other body functions.

Check a newborn for congenital hypothyroidism, which is hypothyroidism that is present at birth.

Check your T4 levels, if you are taking thyroid hormone medicine to treat hypothyroidism.

In certain cases, a T4 test may be done as part of a group of thyroid tests called a thyroid panel.

WHY DO I NEED A THYROXINE (T4) TEST?

You may need a T4 test if:

You had abnormal results on a TSH test.

You have symptoms of hypothyroidism. Not having enough thyroid hormone slows down your body functions. The symptoms can vary from person to person and may include:

Fatigue
Weight gain
Being very sensitive to cold
Joint and muscle pain
Dry skin
Dry, thinning hair
Heavy or irregular menstrual periods
Fertility problems in women
Slow heart rate
Depression
Constipation
Fatigue
Weight gain
Being very sensitive to cold
Joint and muscle pain
Dry skin
Dry, thinning hair
Heavy or irregular menstrual periods
Fertility problems in women
Slow heart rate
Depression
Constipation

You have symptoms of hyperthyroidism. Having too much thyroid hormone speeds up your body functions. The symptoms can vary from person to person and may include:

Weight loss, even though you may be eating more than usual
Rapid or irregular heartbeat
Feeling nervous or irritable
Trouble sleeping
Fatigue
Shaky hands, muscle weakness
Sweating or being very sensitive to heat
Frequent bowel movements (pooping a lot) or diarrhea
Goiter
Weight loss, even though you may be eating more than usual
Rapid or irregular heartbeat
Feeling nervous or irritable

Trouble sleeping

Fatigue

Shaky hands, muscle weakness

Sweating or being very sensitive to heat

Frequent bowel movements (pooping a lot) or diarrhea

Goiter

A member of your family has had thyroid disease. Thyroid disease tends to run in families.

You're taking thyroid hormone medicine for hypothyroidism. A T4 test may be used to check how well your treatment is working.

You have symptoms that could be caused by another thyroid condition or a problem with your pituitary gland.

WHAT HAPPENS DURING A THYROXINE (T4) TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

In most cases, you don't need any special preparations for a T4 blood test. Certain medicines and supplements may affect your test results, so let your provider know what you're taking, but don't stop any medicines without talking with your provider.

If your provider has ordered more tests on your blood sample, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

If you had a free T4 test, your test results may be reported as "free T4." If you had a total T4 test, your results may be reported as "free T4 index (FTI)." FTI is the amount of free T4 in your blood based on a calculation using your total T4 test results.

To fully understand your results, your provider will usually need to compare them with the results of other thyroid tests. So, it's best to talk with your provider about what your results mean.

In general, T4 results that are lower than normal may be a sign of:

Hypothyroidism

Hashimoto's disease, an autoimmune disorder that causes your immune system to attack your thyroid

Certain stages of thyroiditis (inflammation of the thyroid)

A pituitary problem (uncommon) or getting too much or too little iodine in your diet

Congenital hypothyroidism

Low T4 levels may also be caused by treatment for hyperthyroidism or thyroid cancer, including thyroid radiation therapy, radioactive iodine, and surgery to remove all or part of your thyroid.

In general, T4 results that are higher than normal may be a sign of:

Hyperthyroidism

Graves' disease, an autoimmune disorder that causes your immune system to attack your thyroid

Certain stages of thyroiditis

Toxic goiter, an enlarged thyroid that makes too much thyroid hormone

Toxic thyroid nodule, a lump growing in the thyroid that makes large amounts of thyroid hormone on its own

Too much iodine, which is found in certain medicines and seaweed

A noncancerous (not cancer) tumor in your pituitary gland

Taking too much thyroid hormone medicine to treat hypothyroidism

A pituitary problem (uncommon)

If your T4 results are abnormal, you may need more testing to find the cause. But abnormal T4 levels don't always mean you have a medical condition that needs treatment. Many things can affect your T4 levels, including severe illness and certain medicines, such as birth control pills and steroids.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A THYROXINE (T4) TEST?

Thyroid disease can develop during pregnancy, but it's not common. If this happens, your provider will treat you, if necessary. After you give birth, you'll usually have your thyroid checked again.

If you have had thyroid disease, be sure to tell your provider if you are pregnant or are thinking of becoming pregnant.

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The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

Total Protein and Albumin/Globulin (A/G) Ratio

<https://medlineplus.gov/lab-tests/total-protein-and-albumin-globulin-a-g-ratio/>

TOTAL PROTEIN AND ALBUMIN/GLOBULIN (A/G) RATIO

WHAT IS A TOTAL PROTEIN AND ALBUMIN/GLOBULIN (A/G) RATIO TEST?

A total protein and albumin/globulin (A/G) ratio test measures the total amount of protein in your blood. Proteins have many important functions, including giving your body energy, rebuilding muscles, and helping your immune system. Your body might make too much or too little protein if you have certain health conditions.

There are two major types of protein in the blood: albumin and globulin. Albumin makes up most of the protein in the blood, while the rest are called globulins.

Albumin, which helps keep fluid from leaking out of blood vessels. It also helps move hormones (chemical messengers in your bloodstream that control the actions of certain cells or organs), medicines, vitamins, and other important substances throughout your body. Albumin is made in the liver.

Globulins, which help fight infection and move nutrients throughout your body. Some globulins are made by the liver. Others are made by the immune system.

The test also compares the amount of albumin in your blood to the amount of globulin. This comparison is called the albumin/globulin (A/G) ratio.

This test can check how well your liver and kidneys are working and if you are lacking some nutrients needed for good health. If your total protein levels or A/G ratio results are not normal, it can be a sign of a serious health problem.

Other names: total serum protein, TP albumin/globulin ratio, Total protein with A/G ratio

WHAT IS IT USED FOR?

A total protein and A/G ratio test is often included as part of a comprehensive metabolic panel (CMP), a test that measures proteins and other substances in your blood. A total protein and A/G ratio test may also be used to help diagnose kidney disease, liver disease, or nutritional problems.

WHY DO I NEED A TOTAL PROTEIN AND A/G RATIO TEST?

You may get this test as part of a comprehensive metabolic panel, which is often included in a routine checkup. You may also need this test to check for or monitor a health condition that affects the amount of protein in your body or if you have symptoms that indicate abnormal protein levels. These can include:

Swelling in the feet, ankles, legs, and/or abdomen (belly), which is caused by extra fluid in your tissues

Fatigue

Unexplained weight loss

Loss of appetite

Nausea and vomiting

Jaundice (yellowing of the skin or eyes). This is a common symptom of liver disease.

Blood in the urine, a common symptom of kidney disease

WHAT HAPPENS DURING A TOTAL PROTEIN AND A/G RATIO TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THIS TEST?

You may need to stop taking certain medicines before this test, so tell your health care provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

If your total protein and A/G ratio test is part of another blood test, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THIS TEST?

There is very little risk to having a blood test. There may be slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your results will show whether you have normal, low, or high total protein levels. They will also show if you have a normal, low, or high albumin to globulin (A/G) ratio.

If your total protein levels were low, it may be a sign of one of the following conditions:

Liver disease

Kidney disease

Malnutrition, a condition in which your body does not get the calories, vitamins, and/or minerals needed for good health

A malabsorption syndrome, a type of disorder in which your small intestine can't absorb enough nutrients from food.

Types of malabsorption syndromes include celiac disease and Crohn's disease.

If your total protein levels were high, it may be a sign of one of the following conditions:

An infection such as HIV or viral hepatitis

Multiple myeloma, a type of blood cancer

If your A/G ratio was low, it may be caused by:

An autoimmune disease, such as lupus

Liver disease, including cirrhosis

Kidney disease

If your A/G ratio was high, it may be caused by:

Certain types of genetic disorders

Leukemia (cancer of the blood cells)

Your provider will consider your symptoms, medical history, and the results of other blood tests to understand the results of this test. If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A TOTAL PROTEIN AND A/G RATIO TEST?

In addition to total protein levels, your provider may order a separate blood test for albumin and/or a test for globulins. The globulins test is called serum electrophoresis. It is a blood test that measures the levels of four different types of globulins.

These results may show the following:

Low albumin levels may be a sign of:

Liver disease, including cirrhosis

Kidney disease

Malnutrition

Thyroid disease

Severe burns

if you are pregnant, a low albumin level is normal.

High albumin levels may be a sign of:

Severe dehydration

Diarrhea

Low globulin levels may be a sign of:

Liver disease

Kidney disease

High globulin levels may be a sign of:

Certain types of blood cancers, such as multiple myeloma, Hodgkin disease, or leukemia

Hemolytic anemia

An autoimmune disease, such as lupus or rheumatoid arthritis

Tuberculosis

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TP53 Genetic Test

<https://medlineplus.gov/lab-tests/tp53-genetic-test/>

TP53 GENETIC TEST

WHAT IS A TP53 GENETIC TEST?

A TP53 genetic test uses a sample of your blood, bone marrow, or tissue from a tumor to look for changes in the tumor protein 53 gene, or TP53 for short. Changes in your genes are also called gene variants or mutations.

Genes are parts of DNA in your cells that you inherit from your parents. You inherit two copies of each gene, one from each parent. Genes carry information that controls what you look like and how your body works.

TP53 is a type of gene called a "tumor suppressor." Tumor suppressor genes provide instructions for making proteins that help prevent or slow the growth of cells that may become tumors. The proteins work by:

Preventing cells from dividing too quickly to make new cells.

Repairing damaged DNA in cells.

Starting the normal process of cell death in cells that can't be repaired.

If you have certain changes in a TP53 gene, the gene or the protein it makes may not work well or may stop working completely. As a result, your cells may grow out of control, which can lead to tumors and cancer. Changes in your TP53 gene can be acquired or inherited.

Acquired changes in your genes are not passed down from your parents. They develop at some time after you're born. These kinds of changes are extremely common, accounting for more than half of all adult cancers. Where you develop these gene changes, and how likely you are to develop them, depends on a variety of factors. These include:

Errors in DNA repair or when cells divide to make new cells. These errors can happen at any time. However, they become more common with age. They are also more common in some cells than others. For example, TP53 gene changes are especially common in breast tissue cells.

Being exposed to certain carcinogens (substances that increase your likelihood of getting cancer). For example, exposure to cigarette smoke or diesel exhaust can lead to TP53 gene changes in your lungs. Similarly, exposure to UV radiation from the sun and some artificial sources like tanning beds can lead to these changes in your skin.

Having certain chronic viral infections, such as HPV. These viruses can damage your DNA and lead to TP53 gene changes in your cervix.

Inherited changes in your genes are changes that you're born with. They show up in all the cells in your body, including your egg or sperm cells. That means they can be passed from parent to child. Inherited changes in the TP53 gene cause a rare genetic condition called Li-Fraumeni syndrome (LFS). It increases your risk of developing certain types of cancer, including cancer of the breast, bone, and muscles.

Other names: TP53 mutation analysis, TP53 full gene analysis, TP53 somatic mutation

WHAT IS IT USED FOR?

A TP53 genetic test is used to look for changes in your TP53 genes. It's not a routine test. It is often used to:

Help choose treatment for people who have cancer. If you do have a TP53 gene change, your health care provider may prioritize some treatments over others that wouldn't work on cancers without this gene change.

Check for inherited changes in the TP53 gene in people who have a personal or family history of cancers that may be linked to Li-Fraumeni syndrome.

WHY DO I NEED A TP53 GENETIC TEST?

You may need a TP53 genetic test if you have:

Been diagnosed with cancer.

A personal and/or family health history suggesting that you may have inherited Li-Fraumeni Syndrome.

If you have been diagnosed with cancer, this test can check whether an acquired change in your TP53 gene is causing

it. This information can help your provider select the right treatment and predict how your cancer may behave over time. If you have a personal and/or family health history that suggests Li-Fraumeni Syndrome, your provider may not only test you, but also some of your family members. Talk with your provider or a genetic counselor about testing if you or your family members have had cancers that are closely linked to this syndrome. These cancers include:

Osteosarcoma, a form of bone cancer.

Soft tissue sarcoma, cancer of the muscle and connective tissues.

Breast cancer.

Brain tumors.

Acute (sudden) leukemia.

Adrenal cortical carcinoma, a type of adrenal gland cancer.

WHAT HAPPENS DURING A TP53 GENETIC TEST?

A TP53 genetic test is often done on a sample of blood. Tissue from a tumor or bone marrow may sometimes be used instead. Bone marrow is a spongy tissue found inside most bones.

For a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For tumor tissue testing, tissue is removed during a procedure called a biopsy. There are many ways to do a biopsy. The type of biopsy you have depends on the type of cancer you have and where the tumor is located. In general, a biopsy may be done using:

A hollow needle inserted through your skin. Imaging tests, such as ultrasound, may be used to guide the needle.

A fine needle aspiration biopsy uses a very thin needle to remove a sample of cells and/or fluid.

A core needle biopsy uses a larger needle to remove a sample.

A fine needle aspiration biopsy uses a very thin needle to remove a sample of cells and/or fluid.

A core needle biopsy uses a larger needle to remove a sample.

Surgery. Surgery may be done to remove a sample of tissue (an incisional biopsy). In certain cases, the entire tumor will be removed (an excisional biopsy).

Endoscopy. This procedure uses a scope to look inside your body. Special tools may be used with a scope to remove a sample of tumor tissue. Different types of endoscopy may be used depending on where the tumor is located.

For a bone marrow test, a provider usually takes a sample from the back of your hip bone. You will be given an injection (shot) of medicine to numb the area. You may also be given medicine to help you relax or sleep.

A bone marrow test has two parts that are usually done one after the other:

For a bone marrow aspiration, the provider pushes a hollow thin needle into the bone. A sample of bone marrow fluid is removed using a syringe attached to the needle.

For a bone marrow biopsy, the provider inserts a larger hollow needle into the bone and twists the needle to take out a small piece of bone marrow tissue.

The procedure takes only a few minutes. You may feel pressure or a brief, sharp pain when the samples are being taken. Afterwards, the area will be bandaged. You'll stay lying down for about 15 minutes to make sure the bleeding stops. You may need to stay longer if you had medicine to relax or sleep.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

A blood test usually doesn't require any preparation. If you're having a tissue biopsy or a bone marrow test, ask your provider how you should prepare. If you will be given medicine to relax or sleep, plan to have someone take you home after the test in case you are drowsy.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put

in, but most symptoms go away quickly.

After a tissue biopsy or a bone marrow test, you may have some soreness where the sample was taken. This usually goes away in a few days. If you need pain relief, talk with your provider.

WHAT DO THE RESULTS MEAN?

If you have cancer and your results show you have an acquired change in your TP53 gene, ask your provider how this information affects your treatment plan and future health.

If your test results show you have an inherited change in one copy of your TP53 gene, you have Li-Fraumeni syndrome. This doesn't mean you have cancer. But your risk of getting cancer is higher than most people. Talk with your provider about the steps you can take to reduce your risk of cancer. Your provider may discuss:

Having frequent cancer screening tests to help find cancer early when it may be easier to treat.

Having a yearly physical exam.

Making lifestyle changes, such as getting more exercise, eating a healthier diet, and quitting smoking.

Chemoprevention, which is taking certain medicines or other substances to reduce the risk of developing cancer or to keep it from coming back.

Mastectomy to help prevent breast cancer.

The right steps for you will depend on your age, health history, and family health history.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A TP53 GENETIC TEST?

If you think you may have Li-Fraumeni syndrome, it may help to talk with a genetic counselor before getting tested. The counselor can help you understand the risks and benefits of testing for you and your family. This can help you make informed decisions.

If you are diagnosed with Li-Fraumeni syndrome, a counselor can explain your risk for cancer and discuss whether family members should be tested. A genetic counselor can also help you make plans to manage your risk over time.

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Trichomoniasis Test

<https://medlineplus.gov/lab-tests/trichomoniasis-test/>

TRICHOMONIASIS TEST

WHAT ARE TRICHOMONIASIS TESTS?

Trichomoniasis, often called trich, is one of the most common sexually transmitted infections (STIs). It's caused by infection with a parasite. The parasite that causes this STI is a tiny single-celled life form. A trichomoniasis infection mostly involves the genital area.

Trichomoniasis testing usually looks for the parasite in a sample of body fluid that's swabbed from the infected area. In women, the infection is usually in the vagina. In men, the infection is usually in the urethra, a tube in the penis that carries urine (pee) out of the body.

You can get trichomoniasis by having sex without a condom with a partner who has the infection. Women can get the infection from men or women who are infected. Men usually get it only from women who are infected. Trichomoniasis is more common in women.

Most people who have trichomoniasis infections don't have any symptoms. That means you can have trichomoniasis and pass it to a sex partner without knowing it. The infections are rarely serious, and they can be cured with antibiotics. But if trichomoniasis isn't treated, it can:

Increase your risk of getting or spreading other STIs, including HIV

Cause infections in the prostate gland (a gland in the male reproduction system)

Increase the chance of problems in pregnancy, including:

Giving birth too early (premature birth)

Having a baby with a low birth weight (less than five and a half pounds)

Giving birth too early (premature birth)

Having a baby with a low birth weight (less than five and a half pounds)

If you think you could have an infection, it's important to get tested so you can get treatment if you need it.

Other names: T. vaginalis, trichomonas vaginalis testing, wet prep

WHAT IS IT USED FOR?

Trichomoniasis testing is used to:

Diagnose trichomoniasis infections in people who have symptoms.

Screen for infection in women who have a high risk of infection.

Check for infection after treatment for trichomoniasis is finished. This is done because people are often reinfected if their partners don't get tested and treated. Also, some parasites may be resistant to certain medicine. This means that they have changed so that the medicine no longer works well to get rid of them.

A trichomoniasis infection can increase your risk for different STIs. So, the test is often done with other STI tests.

WHY DO I NEED A TRICHOMONIASIS TEST?

You may need to get tested for trichomoniasis:

If you have symptoms that could be an STI. Usually, trichomoniasis doesn't cause symptoms, But when they happen, they usually show up within 5 to 28 days after getting the infection. Some people may develop symptoms much later. The symptoms can come and go. They may range from mild to severe and can include:

In women:

Itching, burning, redness or soreness around the genitals
Discomfort when urinating (peeing) or having sex
A clear, white, yellowish, or greenish vaginal discharge with a fishy smell

In men:

Itching or irritation inside the penis
Burning after urinating or ejaculating (releasing semen)
Abnormal discharge from the penis

In women:

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In men:

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Burning after urinating or ejaculating (releasing semen)
Abnormal discharge from the penis
Itching or irritation inside the penis
Burning after urinating or ejaculating (releasing semen)
Abnormal discharge from the penis

If you're a woman with a high risk for getting an STI. Even if you don't have symptoms, your health care provider may recommend testing if you have:

Sex without using a condom
Multiple sex partners
Had trichomoniasis in the past or other STIs
Sex without using a condom
Multiple sex partners
Had trichomoniasis in the past or other STIs

If you've completed treatment for trichomoniasis. You may need to be checked to see if you've been infected again.
Talk with your provider about your need for testing.

WHAT HAPPENS DURING A TRICHOMONIASIS TEST?

Before your trichomoniasis test, your provider will usually ask about your symptoms and examine your genitals for signs of infection.

Next, your provider will usually use a small brush or swab to take a sample of body fluid. The preferred method of testing is with swab samples taken from either the urethra in the penis or the vagina and/or cervix, which is the lower part of the uterus. You may feel some brief discomfort when the sample is taken.

You can also buy at-home collection kits to test for trichomoniasis and other STIs. With these kits, you collect a sample of urine or vaginal fluid to send to a lab for testing. It's important to follow all the instructions carefully. And be sure to

discuss the results with your provider because you may need more testing.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a trichomoniasis test.

ARE THERE ANY RISKS TO THE TEST?

There are no known risks to having a trichomoniasis test.

WHAT DO THE RESULTS MEAN?

In most cases, your first test for trichomoniasis will check a sample of body fluid under a microscope to look for parasites. This test is called "wet prep microscopy." It's often done first because it's quick and inexpensive.

The results of a wet prep microscopy test will be either negative or positive:

A negative result means that no parasites were seen in your sample. But wet prep microscopy results aren't always accurate. So, if your provider thinks you have trichomoniasis, more accurate tests may be done on your sample. They include:

Nucleic acid amplification tests (NAATs). These tests look for genetic material from the parasites that cause trichomoniasis.

Rapid antigen tests. These tests are only done on samples of vaginal fluid. They look for certain proteins, called antigens, from the parasite. Rapid antigen tests can be done at your provider's office. The results are usually ready in under 15 minutes.

Culture tests. For these tests, your sample is sent to a lab, where the cells will grow for up to a week. Then the sample will be checked under a microscope. If your sample had parasites, there will now be more of them, so they'll be easier to find.

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A positive result means that parasites were found in your sample. You have a trichomoniasis infection. Your provider will prescribe medicine to cure the infection. Be sure to:

Finish all your medicine according to your provider's instructions.

Tell your recent sexual partners about your infection so they can get tested, too.

Let your provider know if you still have symptoms after treatment.

Get tested for other STIs.

Finish all your medicine according to your provider's instructions.

Tell your recent sexual partners about your infection so they can get tested, too.

Let your provider know if you still have symptoms after treatment.

Get tested for other STIs.

To prevent any future trichomoniasis infections, take appropriate precautions whenever having sex. These include:

Having sex with only one partner who has tested negative for STIs and has sex only with you.

Using condoms correctly every time you have sex.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A TRICHOMONIASIS TEST?

Your provider may be able to diagnose trichomoniasis when testing a sample of your urine (pee) for conditions with similar symptoms. For example, your provider can tell if you have trichomoniasis if they find trichomonas during general STI screening or when testing for a urinary tract infection. If they find trichomonas during this test, they may perform an NAAT or other tests to confirm the diagnosis.

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Tricyclic Antidepressant (TCA) Screen

<https://medlineplus.gov/lab-tests/tricyclic-antidepressant-tca-screen/>

TRICYCLIC ANTIDEPRESSANT (TCA) SCREEN

WHAT IS A TRICYCLIC ANTIDEPRESSANT (TCA) SCREEN?

This test checks whether tricyclic antidepressants (TCA) can be found in your blood or urine. It can also measure how much TCA is in your blood or urine. TCAs are a type of antidepressant. Antidepressants are prescription medicines that treat depression, a serious mood disorder. Depression can affect how you think, feel, and behave.

There are several types of antidepressants. They each work in slightly different ways. TCAs were one of the first types of antidepressants developed. They can be effective, but they tend to cause more side effects than newer types of antidepressants. Most of the newer antidepressants belong to one of these three groups:

Selective serotonin reuptake inhibitors (SSRIs)

Serotonin and norepinephrine reuptake inhibitors (SNRIs)

Atypical antidepressants, which are newer antidepressants that don't fit into the other groups.

Health care providers often use newer antidepressants rather than TCAs since they are safer and have fewer side effects. But TCAs can be a good option to treat depression when other antidepressants are not working.

Health care providers sometimes use TCAs to treat other health conditions, such as anxiety, chronic pain, migraines, and insomnia.

Some commonly prescribed TCAs include:

Doxepin

Nortriptyline

Amitriptyline

Imipramine

Desipramine

TCAs have to be carefully dosed. It can be hard to find a dose that is effective without being dangerous. In fact, TCAs are the cause of many prescription drug overdose deaths in the United States.

Other names: TCA test, TCA screen, Tricyclic Screen

WHAT IS IT USED FOR?

A TCA screen is used to:

Help your provider prescribe the right dose of a TCA.

Find out if you are misusing TCA (taking it without a prescription or taking more than the prescribed dose). The test may be included as part of a drug (toxicology) screen, which checks for the presence of several different prescription and illegal drugs.

Diagnose a TCA overdose. Prompt treatment of an overdose can be lifesaving.

WHY DO I NEED A TCA SCREEN?

You may need this test if you have been prescribed a TCA to make sure you are getting a dose that is safe and effective. You may also need this test if you have symptoms of a TCA overdose. These include:

Dry mouth

Blurred vision

Constipation

Trouble urinating

Drowsiness

Lightheadedness

Seizures

Confusion

Tremors

If you think you have taken too much of a TCA, or you have symptoms of a TCA overdose, call 911 or go to the nearest emergency center.

WHAT HAPPENS DURING A TCA SCREEN?

A TCA screen can be done as a blood test or a urine test.

During a TCA blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a TCA urine test, you will need to provide a sample of urine into a sterile cup. Your provider will give more information on how to collect your sample.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a TCA screen.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no risk to having a urine test.

WHAT DO THE RESULTS MEAN?

Your blood test results will show the levels of TCA in your bloodstream. Depending on which TCA you are taking and what the levels are, your provider may adjust your dosage.

There are two types of urine test results. One type will show whether TCA was found but not how much. The other type shows the levels of TCA.

If blood or urine test results show you have dangerously high levels of TCA, you will be treated immediately.

If you have questions about your results, talk to your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A TCA SCREEN?

TCAs are most often prescribed by a mental health professional, a specialist in diagnosing and treating mental health problems, such as depression. There are many types of mental health professionals. Some types, usually medical doctors, are allowed to prescribe medicines, including TCAs. Other mental health professionals work with providers who can write prescriptions. The most common types of mental health professionals include:

Psychiatrists are medical doctors who specialize in mental health disorders. Psychiatrists can prescribe medicine.

Psychologists are professionals trained in psychology. Psychologists may use one-on-one counseling and/or group therapy sessions. They often have doctoral degrees, but they do not have medical degrees. They can't prescribe medicine unless they have a special license. Some psychologists work with providers who can prescribe medicine.

Psychiatric or mental health nurses are nurses with special training in mental health disorders. Nurses who may have a master's or doctoral degree in psychiatric-mental health nursing include advanced practice registered nurses (APRNs), certified nurse practitioners (CNPs), and clinical nurse specialists (CNSs). In some states, certain nurses can prescribe medicines.

Licensed clinical social workers have at least a master's degree in social work with special training in mental health.

They can't prescribe medicine, but they may work with providers who can prescribe medicine. Providers who are licensed clinical social workers usually have LCSW or LICSW after their names.

Licensed professional counselors (LPC) may also be called clinicians or therapists. States have different names for these licenses, such as LMFT (licensed marriage and family therapist). These professionals usually have a master's degree in a field related to mental health. They can't prescribe medicine but may work with providers who can prescribe it.

If you have depression, it's important to get treatment. Talk to your primary care provider if you need help finding a mental health provider who can treat your depression.

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Triglycerides Test

<https://medlineplus.gov/lab-tests/triglycerides-test/>

TRIGLYCERIDES TEST

WHAT IS A TRIGLYCERIDES TEST?

A triglycerides test is a blood test that measures the amount of a fat in your blood called triglycerides. High blood triglyceride levels may increase your risk for heart disease and stroke. A triglycerides test can help you decide if you need to take action to lower your risk.

Some triglycerides are important for good health. Your body uses them for energy. If you eat more calories than you need, your body turns the extra calories into triglycerides and stores them in your fat cells to use later. When your body needs energy, your cells release triglycerides into your bloodstream to provide fuel for your muscles to work.

But high triglyceride levels in your blood may affect your arteries and increase your risk of coronary artery disease, heart attack, and blood vessel disorders. Extremely high levels of triglycerides also increase the risk of acute pancreatitis in adults and children.

High blood triglycerides are very common and have many possible causes, including:

Eating a lot of foods that are high in fat and sugar, and other simple carbohydrates or "carbs," such as foods made with white flour

Not getting enough exercise

Having certain medical conditions, such as:

Kidney or liver disease

Thyroid disease

Diabetes that's not well-controlled

Being overweight or having obesity

Alcohol use disorder (AUD)

Inherited conditions (passed down through families)

Kidney or liver disease

Thyroid disease

Diabetes that's not well-controlled

Being overweight or having obesity

Alcohol use disorder (AUD)

Inherited conditions (passed down through families)

Taking certain medicines, such as diuretics (water pills) and some that treat breast cancer, and HIV

High blood triglyceride levels usually don't cause any symptoms, so routine testing is important to help you find and treat high triglyceride levels before they harm your health.

Other names for a triglycerides test: TG, TRIG, lipid panel, fasting lipoprotein panel

WHAT IS IT USED FOR?

A triglycerides test is used to understand your risk for heart disease, stroke, and other conditions that involve your arteries, such as peripheral arterial disease. The test is also used to help monitor heart conditions and treatments to lower your risk of heart disease.

A triglycerides test is usually done as part of a group of tests called a lipid profile. Lipid is another word for fat. A lipid profile measures the level of fats in your blood, including triglycerides and cholesterol.

Information from a lipid profile helps your health care provider calculate your risk for a heart attack or stroke. High blood levels of triglycerides with high LDL (bad) cholesterol and/or low HDL (good) cholesterol can increase your risk.

WHY DO I NEED A TRIGLYCERIDES TEST?

Your provider may order a lipid profile, including a blood triglycerides test, as part of a routine checkup. If you are being treated for high cholesterol and/or triglycerides, you may also need this test to see how well your treatment is working. How often you need a lipid profile test depends on your age, sex, family history, and your risk of developing heart disease. Ask your provider how often you need to have your blood lipid levels tested.

These are the general recommendations for testing blood lipid levels, including triglycerides, in healthy people:

For people who are age 19 or younger:

The first test should be between ages 9 and 11.

Children should have the test again every 5 years.

Some children may have this test starting at age 2 if there is a family history of high cholesterol, heart attack, or stroke.

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For people who are ages 20 to 65:

Younger adults should have the test every 5 years.

Men ages 45 to 65 and women ages 55 to 65 should have it every 1 to 2 years.

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Men ages 45 to 65 and women ages 55 to 65 should have it every 1 to 2 years.

For people older than 65:

They should be tested every year.

They should be tested every year.

You may need to be tested more often if you already have heart disease or have a higher risk for heart disease because you:

Have a family health history of early heart disease. That means your father or brother had heart disease before age 55 and/or your mother or sister had heart disease before age 65.

Smoke.

Have excess weight or obesity.

Have unhealthy eating habits and/or don't get enough exercise.

Have prediabetes or diabetes.

Have high blood pressure.

WHAT HAPPENS DURING A TRIGLYCERIDES TEST?

During the test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to fast (not eat or drink) for 9 to 12 hours before your blood is drawn. Your provider will let you know if you need to fast and if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Triglycerides are usually measured in milligrams (mg) of triglycerides per deciliter (dL) of blood. The commonly used guidelines for normal and higher levels of triglycerides in adults are:

Guidelines for children and teens are different than adults. Ask your child's provider to explain what your child's test results mean.

Triglyceride levels that are higher than normal may mean you have a higher risk of heart disease, stroke, and other conditions that affect your arteries. Having high blood triglycerides also increases your risk of metabolic syndrome. Very high triglyceride levels increase your risk of sudden pancreatitis.

To lower your blood triglycerides, your provider may recommend lifestyle changes, such as:

Eating heart-healthy foods and limiting added sugar and foods high in saturated fat

Getting regular physical activity

Aiming for a healthy weight

Managing stress

Quitting smoking

Getting enough sleep

Limiting alcohol

In certain cases, you may also need medicine to help lower your triglyceride levels. Before making any major changes to your diet or exercise routine, talk with your provider about what treatment is best for you.

Triglyceride levels that are too low are very uncommon.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT TRIGLYCERIDE TESTS?

If your blood triglyceride levels are borderline, your provider may order another blood test called apolipoprotein B or "apo B." The results of this test can help your provider understand how high your risk of heart and blood vessel problems may be. This can help you decide about the best treatment to lower your risk.

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Triiodothyronine (T3) Tests

<https://medlineplus.gov/lab-tests/triiodothyronine-t3-tests/>

TRIIODOTHYRONINE (T3) TESTS

WHAT IS A TRIIODOTHYRONINE (T3) TEST?

A triiodothyronine (T3) test measures the level of T3 in a sample of your blood. It's used to help diagnose thyroid disease.

Your thyroid is a small, butterfly-shaped gland at the base of your throat. It makes two hormones, T3 and T4 (thyroxine). Hormones are chemical messengers in your bloodstream that control the actions of certain cells or organs.

Thyroid hormones work together to control the way your body uses energy. They affect your weight, heart, body temperature, muscle strength, and even your mood. In children, thyroid hormones affect growth, too.

There are two forms of T3 in your blood:

Free T3 is the active form that enters your tissues where it's needed.

Bound T3 is attached to certain proteins which prevent it from entering your tissues. Most of your T3 is bound.

There are different tests for measuring T3:

A total T3 test measures both bound and free T3 together. Medical experts think that this test is the more accurate way to measure T3.

A free T3 test only measures free T3.

Other names: thyroid function test; total triiodothyronine, free triiodothyronine, FT3, TT3, radioimmunoassay FT3

WHAT IS IT USED FOR?

A T3 test is used to help:

Diagnose hyperthyroidism, a condition in which your thyroid gland makes too much thyroid hormone. The test can also help find out what's causing the condition.

Diagnose pituitary disorders. The pituitary is a pea-sized gland at the base of your brain. It makes a hormone called TSH (thyroid stimulating hormone). TSH tells the thyroid how much T3 and T4 to make. So, abnormal T3 levels may be a sign of a pituitary problem.

Monitor treatment for thyroid disease, including hyperthyroidism and hypothyroidism, which is when the thyroid gland makes too little thyroid hormone.

T3 tests are usually ordered with blood tests that measure T4 and TSH (thyroid stimulating hormone) tests.

WHY DO I NEED A T3 TEST?

You may need a T3 test if you have symptoms of hyperthyroidism (too much thyroid hormone). Symptoms may include:

Anxiety, nervousness, irritability

Weight loss, even though your appetite may increase

Shaky hands, muscle weakness

Sweating or being very sensitive to heat

Irregular heartbeat (arrhythmia)

Frequent bowel movements (poops)

Goiter (an enlarged thyroid gland)

Trouble sleeping and fatigue

You may also need a T3 test if:

You take thyroid hormone medicine to treat a thyroid problem. A T3 test can check whether you're taking the right dose to keep your thyroid hormone levels in a healthy range.

You had abnormal results on a TSH and/or T4 test. Providers usually order these tests first to look into a possible thyroid disorder. If your test results are abnormal or unclear, your provider may use a T3 test to confirm or rule out

hyperthyroidism.

T3 testing isn't commonly used to help check for hypothyroidism (too little thyroid hormone). That's because other thyroid tests can diagnose hypothyroidism earlier than a T3 test.

WHAT HAPPENS DURING A T3 TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a T3 blood test. Certain medicines and supplements can affect your test results. Your provider will let you know if you need to stop taking any medicines before your test. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand the results of your T3 test, your provider will usually compare them with the results of your TSH and T4 tests. Like T3, T4 tests may measure either free or total T4, but providers usually measure free T4.

Your provider will look for different combinations of high, low, and normal results. Each combination suggests a different thyroid condition. Ask your provider to explain what your test results say about your health.

In general:

Normal T3 levels (free or total) alone do not rule out thyroid disease. But if your TSH and free T4 levels are also normal, you probably don't have a thyroid condition.

High T3 levels (free or total) may mean you have hyperthyroidism (too much thyroid hormone).

High T3 levels with:

Low TSH and high free T4 levels may mean that a problem with your thyroid gland is causing too much thyroid hormone.

Normal or high TSH and high free T4 levels may mean that a pituitary gland condition is causing too much thyroid hormone.

Low TSH and high free T4 levels may mean that a problem with your thyroid gland is causing too much thyroid hormone.

Normal or high TSH and high free T4 levels may mean that a pituitary gland condition is causing too much thyroid hormone.

Low T3 levels (free or total) may mean hypothyroidism (too little thyroid hormone).

Low or normal T3 levels with:

High TSH levels and low free T4 levels may mean that a problem with your thyroid gland is causing too little thyroid hormone.

Low TSH and low free T4 levels may mean that a pituitary gland condition is causing too little thyroid hormone.

High TSH levels and low free T4 levels may mean that a problem with your thyroid gland is causing too little thyroid

hormone.

Low TSH and low free T4 levels may mean that a pituitary gland condition is causing too little thyroid hormone.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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What You Need to Know About Blood Testing

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Troponin Test

<https://medlineplus.gov/lab-tests/troponin-test/>

TROPONIN TEST

WHAT IS A TROPONIN TEST?

A troponin test measures the level of troponin in a sample of your blood. Troponin is a protein that's found in the cells of your heart muscle.

Normally, troponin levels in blood are so low that only the most sensitive types of tests can measure them. But if your heart muscle is damaged, troponin leaks into your bloodstream, and your troponin blood levels will rise.

Troponin testing is mainly used to help diagnose heart attacks. Heart attacks damage the heart by suddenly blocking the blood flow that brings oxygen to part of the heart muscle. Without oxygen from blood, heart muscle cells die and release troponin.

Troponin test results can confirm damage to the heart muscle from a heart attack. The more damage there is to the heart, the more troponin is released into the blood. So, measuring the amount of troponin in the blood can also help estimate how much of the heart has been damaged. Higher than normal troponin levels may also be found in other conditions that can damage heart muscle.

Two types of troponin maybe be measured to diagnose heart damage. They are called troponin I and troponin T. Tests can measure either type.

Other names: cardiac troponin I (cTnI), cardiac troponin T (cTnT), cardiac troponin (cTn), cardiac-specific troponin I and troponin T

WHAT IS IT USED FOR?

A troponin test is mainly used to:

Confirm if a person is having a heart attack or recently had a heart attack. This is the most common reason for troponin testing. The test is usually done in the emergency room of a hospital along with other heart tests, such as an EKG (also called an ECG or electrocardiogram).

Diagnose and monitor unstable angina. Angina is chest pain that happens if part of the heart muscle doesn't get as much blood as it needs. Unstable angina is a type of angina that happens at any time, even during rest. It's a medical emergency because it can lead to a heart attack.

Check heart health after a surgery that could damage the heart. In this case, a troponin test may be done before and after surgery so that the results can be compared.

WHY DO I NEED A TROPONIN TEST?

You may need a troponin test if you have symptoms of a heart attack. The symptoms can start slowly or suddenly. They may be mild or intense, and they can come and go over several hours. They include:

Chest pain, heaviness, or discomfort in the center or left side of the chest

Pain or discomfort in one or both arms, your back, shoulders, neck, jaw, or above your belly button

Trouble breathing when resting or doing light physical activity

Nausea and vomiting

Feeling unusually tired for no reason, sometimes for days

Dizziness and light-headedness

Sweating a lot for no reason

Rapid or irregular heartbeat (arrhythmia)

If you have symptoms of a heart attack, call 911 immediately. Quick medical attention could save your life.

You may also need troponin testing if you're having surgery that could affect your heart.

WHAT HAPPENS DURING A TROPONIN TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a troponin test. But, if you take vitamin B7, also called biotin, make sure you tell the health care provider who orders your test. Biotin and supplements that contain biotin can make your troponin levels appear lower than they really are.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

To understand the results of a troponin test for chest pain and/or other symptoms of a heart attack, your provider will consider your symptoms, medical history, and the results of other tests.

In general:

If your first troponin test results are normal, you'll probably be retested during the next 12 hours. That's because increased troponin levels usually don't show up for about 2 to 3 hours after a heart attack begins. So, results from a later test may show higher than normal troponin levels. In the meantime, the results of other tests and exams will help your provider know whether to start treatment for a heart attack right away.

If your troponin test results show normal troponin levels 12 hours after your symptoms began, you probably did not have a heart attack.

If your test results show higher than normal amounts of troponin in your blood, it usually means you have some damage in your heart muscle. Your provider will consider all the information about your condition to diagnose whether a heart attack caused the damage. Very high troponin levels most often mean you've had a heart attack.

Higher than normal troponin levels don't always mean you've had a heart attack. They can be caused by unstable angina many other conditions that can damage the heart, including:

Irregular heartbeat (arrhythmia), especially a heartbeat that's too fast

Heart failure

Heart valve disease

Infection or inflammation of the heart muscle (myocarditis)

Heart surgery

Intense exercise over a long time, such as running a marathon

Chest injuries

Chronic kidney disease

A blood clot in your lungs (pulmonary embolism)

Sepsis

A severe COVID-19 infection

If you have high troponin levels, but your provider doesn't think you've had a heart attack, you may have other tests to find the cause.

If you have questions about your test results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A TROPONIN TEST?

Troponin levels can continue to rise for about 24 hours after a heart attack begins. So, if your test results show high troponin levels, you'll probably be tested two or more times over a 24-hour period.

The results of these tests show how fast your troponin level increased and the highest level it reached. That information helps estimate how much of your heart muscle is damaged and how well you might recover.

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TSH (Thyroid-stimulating hormone) Test

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TSH (THYROID-STIMULATING HORMONE) TEST

WHAT IS A TSH TEST?

TSH stands for thyroid stimulating hormone. Another name for it is thyrotropin. A TSH test is a blood test that measures this hormone. A TSH level that is too high or too low may be a sign of a thyroid problem.

The thyroid is a small, butterfly-shaped gland in the front of your neck. Your thyroid makes hormones that control how your body uses energy. Thyroid hormones affect nearly every organ and control many of your body's most important functions. For example, they affect your breathing, heart rate, weight, digestion, and even your mood. If you don't have enough thyroid hormones in your blood, many of your body functions will slow down. But if you have too much, many body functions will speed up.

Your thyroid is controlled by a gland in your brain called the pituitary gland. The pituitary gland makes thyroid-stimulating hormone (TSH). TSH tells your thyroid how much thyroid hormone it needs to make.Â

If the thyroid hormone level in your blood is too low, your pituitary gland makes larger amounts of TSH to tell your thyroid to work harder. If your thyroid hormone level is too high, the pituitary gland makes little or no TSH. By measuring the TSH level in your blood, you can find out if your thyroid is making the right level of hormones.

Other names: thyrotropin test, thyrotropic hormone

WHAT IS IT USED FOR?

A TSH test is used to find out how well your thyroid is working. It can tell if you have hyperthyroidism (too much thyroid hormone) or hypothyroidism (too little thyroid hormone) in your blood. But a TSH test can't show what is causing a thyroid problem.

If you take prescription thyroid hormone medicine because of hypothyroidism or because you had your thyroid removed, you'll have regular TSH tests to check your thyroid hormone levels. This testing is done to check that you are taking the right dose of the medicine. TSH tests are also used to monitor your thyroid hormone levels after treatment for hyperthyroidism. Â

WHY DO I NEED A TSH TEST?

You may need a TSH test if you have symptoms of too much or too little thyroid hormone in your blood.

Hyperthyroidism is also called overactive thyroid. Having more thyroid hormones than you need speeds up your body functions and causes symptoms that can vary from person to person. The symptoms may include::

Weight loss, even though you may be eating more than usual

Rapid or irregular heartbeat

Feeling nervous or irritable

Trouble sleeping

Fatigue

Shaky hands, muscle weakness

Sweating or being very sensitive to heat

Frequent bowel movements (pooping a lot) or diarrhea

Goiter (enlarged thyroid that may cause your neck to look swollen)

Adults over age 60 may have different symptoms than younger adults. For example, they may lose their appetite or withdraw from other people. Sometimes, this can be mistaken for depression or dementia.Â

Hypothyroidism (too little thyroid hormone) is also called underactive thyroid. Not having enough thyroid hormone slows down your body functions and causes symptoms that can vary from person to person. The symptoms may include:

Fatigue

Weight gain
Being very sensitive to cold
Joint and muscle pain
Dry skin
Dry, thinning hair
Heavy or irregular menstrual periods
Fertility problems in women
Depression
Constipation

Because hypothyroidism develops slowly, many people don't notice symptoms of the disease for months or even years. You may also have a TSH test and other tests to help diagnose unusual bumps or lumps on your thyroid. The tests will help your health care provider check for thyroid cancer or thyroid nodules (growths on your thyroid that aren't cancer).

WHAT HAPPENS DURING A TSH TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. If your health care provider has ordered other blood tests, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your TSH test results can tell you if your thyroid is making too much or too little hormones. But the test can't explain why your TSH levels may be too high or too low.

If your test results aren't normal, your provider will probably order other thyroid blood tests to find out what's causing your thyroid problem. These blood tests may include:

T4 thyroid hormone test

T3 thyroid hormone test

Thyroid antibodies test to help diagnose an autoimmune thyroid disorder, such as:

Graves' disease, the most common cause of hyperthyroidism

Hashimoto's disease, that the most common cause of hypothyroidism

Graves' disease, the most common cause of hyperthyroidism

Hashimoto's disease, that the most common cause of hypothyroidism

In certain cases, an abnormal TSH result may be a sign of a pituitary gland problem, but this doesn't happen often.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A TSH TEST?

Your TSH level may be high or low even when your thyroid gland is healthy. For example, the following can affect your TSH level:

Some serious illnesses (that are not related to your thyroid) may cause a lower TSH level for a short time.

The TSH level may be higher in people over age 80, even though they don't have any thyroid problems.

Pregnancy also affects your TSH level. They are often a little low during the first three months. But sometimes, thyroid disease develops during pregnancy. If you develop thyroid disease during pregnancy, your provider will monitor your condition throughout your pregnancy and after your baby is born. That's because hyperthyroidism, and less often, hypothyroidism, can continue after you give birth. If you have a history of thyroid disease, be sure to talk with your provider if you are pregnant or are thinking of becoming pregnant.

Newborn babies are routinely screened with a TSH test to see if they were born with hypothyroidism (congenital hypothyroidism).

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TUBERCULOSIS SCREENING

WHAT IS A TUBERCULOSIS (TB) SCREENING TEST?

A tuberculosis screening test checks to see if you have the bacteria (germs) that cause tuberculosis (TB) in your body. TB is a bacterial infection that mainly attacks the lungs. But it can also affect other parts of the body, including the brain, spine, and kidneys.

TB germs cause two types of conditions:

Inactive (latent) TB infection happens when you have TB germs in your body, but you're not sick because the germs aren't active. This means you don't have symptoms, and you can't spread the disease to others. However, the germs could start to multiply and make you sick in the future. So, inactive TB infection is usually treated to prevent you from getting sick with active TB.

Active TB disease means that TB germs are actively growing in your body and making you sick. If TB germs grow in your lungs or throat, you can spread the disease to other people by coughing or sneezing. Antibiotics can almost always cure active TB, but without treatment, it can cause serious illness or even death.

There are two types of TB screening tests that are used to detect TB: a TB skin test (also called a Mantoux test) and a TB blood test (also called an IGRA test). Your health care provider or your local health department can let you know which test is best for you.

Other names: TB test, TB skin test, purified protein derivative (PPD) test, latent TB infection test, Mantoux tuberculin skin test; IGRA test, TB blood test

WHAT IS IT USED FOR?

Your provider can use a TB screening test to see if the bacteria that cause tuberculosis are in your body. But to actually diagnose TB, your provider will have to review your medical history, your symptoms, and possibly the results from other tests.

If either type of TB screening test finds signs of TB, it may mean you have a TB infection. However, it can't tell whether the infection is active or inactive.

Providers and local health departments can also use TB screening tests to help control TB transmission in communities, workplaces, or health care settings.

WHY DO I NEED A TB SCREENING TEST?

You may need a TB screening test if you:

- Were exposed to someone with active TB disease.

- Are starting a new job or entering a new school.

- Have a high risk of exposure.

Your risk of exposure is higher if you:

- Work in health care.

- Work or live in a place where TB is more common, such as jails, nursing homes, and shelters for people without homes.

- Were born in or often travel to countries where TB disease is common.

- Are having an organ transplant.

- Use illegal drugs that are injected with needles.

- Are receiving treatments for rheumatoid arthritis or Crohn's disease.

- Have a weakened immune system or a health condition that increases your risk of getting active TB. These conditions include diabetes or diseases that weaken your immune system such as HIV.

HIV can especially put you at higher risk for TB. This is because HIV weakens the immune system and so makes it

harder for your body to fight off the bacteria that cause TB. Having HIV can also increase the risk that an inactive TB infection becomes active TB disease. Getting treatment for inactive TB can, however, prevent this risk.

You may need a TB test if you have symptoms of an active TB disease, including:

A cough that lasts longer than three weeks.

Coughing up blood or sputum (a thick mucus from the lungs).

Chest pain.

Night sweats (heavy sweating during sleep).

Losing weight without trying.

Loss of appetite.

Fever.

Weakness or fatigue.

WHAT HAPPENS DURING A TB SCREENING TEST?

You can have either a TB skin test or a TB blood test, depending on:

Your age.

Your overall health.

The test's cost and availability.

Generally, TB skin tests are used more often, but blood tests for TB are becoming more common. Both types are usually done in a clinic or provider's office.

For a TB skin test (also called a Mantoux tuberculin skin test), you will need two visits to complete the test. The first visit is for doing the test and the second is for reading the results.

On the first visit:

A health care professional will use a tiny needle to place a small amount of fluid under the skin in the lower part of your arm. The fluid is called tuberculin. It contains a protein that comes from the bacteria that cause TB. It cannot make you sick.

The fluid will make a small bump on your arm.

The test spot will be left uncovered.

You will need to avoid touching the test spot or putting lotions on it. It's okay to get it wet. If it itches, you can put an ice cube or cold cloth on it.

After two to three days, you will return for the second visit. A health care professional will look at the test spot on your arm to see if your skin reacted. If there is a bump of hard skin, the professional will measure the bump. The meaning of the bump depends on how large it is, your risk of exposure, and your risk for developing active TB disease.

For a TB blood test (also called an IGRA test), a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a TB skin or blood test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a TB skin test or blood test. For a TB skin test, you may feel a pinch when the fluid is placed under your skin.

For a blood test, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your TB skin or blood test results will usually be positive, negative, or borderline (indeterminate).

A positive result means that you have been infected with TB bacteria. You will need more tests to find out if you have an inactive TB infection or TB disease. These tests may include a chest x-ray or a sputum culture. If you had a positive result on a TB skin test, you may have a TB blood test to confirm the result.

Occasionally, a TB blood or skin test can result in a false positive. This means that the test says you have a TB infection when you don't. A TB test can result in a false positive if you've been vaccinated with the bacille Calmette-Guerin (BCG) TB vaccine. This type of vaccine is rarely used in the United States, but if you have had this vaccination, let your provider know. You may also have a false positive if you have an infection from another type of bacteria from the same family as the TB bacteria.

A negative result means that your skin or blood did not react to the test. You are unlikely to have an inactive TB infection or TB disease. But you may still need more testing if you:

Have symptoms of TB.

Were tested sooner than six to eight weeks after exposure to TB.

Have HIV.

Had a TB skin test after being around someone with TB disease.

Occasionally, a TB test can also result in a false negative. This means that the test says you don't have a TB infection when you do. A TB test can result in a false negative if you've contracted TB within the past 8 to 10 weeks or if you've had certain measles or smallpox vaccines.

Sometimes a TB blood test result will be "borderline," which means the test could not show for sure whether you have a TB infection. If this happens, you will likely be tested again.

Overall, TB tests tend to be accurate. However, TB blood tests are more accurate than TB skin tests. Your provider will consider whether anything about your health history might affect the accuracy of your test results.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A TB SCREENING?

Both active TB disease and inactive TB infections should be treated. The treatment for both conditions is antibiotics. To make sure you get rid of all the TB germs in your body, you'll need to follow the directions for taking your medicine. Treatment may last a few months to a year. Stopping treatment too soon can cause the infection to come back and make it harder to treat.

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Tumor Marker Tests

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TUMOR MARKER TESTS

WHAT ARE TUMOR MARKER TESTS?

These tests look for tumor markers, which are sometimes called cancer markers. Tumor markers are substances that are made by cancer cells or by normal cells in response to cancer. For example, some cancer cells make certain proteins in larger amounts than normal cells do. These proteins can act as tumor markers because their presence may mean cancerous cells are present.

Certain tumor markers may be found in samples of body fluids, such as blood or urine (pee). Other tumor markers are found in samples of cells that are removed from a tumor during a biopsy.

Not all cancers have known tumor markers, and the tumor markers that are known don't always provide accurate information. That's because:

Some conditions that aren't cancer may also cause high levels of certain tumor markers. Tumor marker tests can't tell whether tumor markers come from cancer or from another condition.

Some people don't make high levels of the tumor markers that are commonly found in their type of cancer.

WHAT ARE THEY USED FOR?

Tumor marker tests are most often used after you have a cancer diagnosis. When used with other tests, tumor markers may help:

Find out whether cancer has spread to other parts of your body (cancer stage).

Predict how fast your cancer may grow, the chance of recovery, and whether cancer is likely to return.

Select the right treatment for your type of cancer. Some treatments work only with cancers that have certain tumor markers. Tumor markers that help plan treatment are also called biomarkers.

Monitor how well your treatment is working. If tumor marker levels go down, it usually means your treatment is helping.

Find any cancer that remains or comes back after treatment.

Some tumor marker tests that use body fluids (mainly blood or urine) have a limited role in screening[^] for certain types of cancer. The tests are mostly used to screen people who:

Have a high risk for the type of cancer that's linked to the tumor marker being measured. This could mean a family history of that type of cancer or that you have certain environmental or behavioral risk factors.

Have symptoms that could be from that type of cancer.

Tumor marker tests that are used to screen for cancer can't diagnose cancer. If you have a high level of tumor markers, it only means that you're more likely to have cancer. A biopsy is usually needed to diagnose or rule out cancer.

Tumor marker tests that use cells from a tumor may help diagnose cancer. These "tumor cell markers" are usually removed during a biopsy. They may be used with other tests to confirm a cancer diagnosis and decide on the best treatment.

WHY DO I NEED A TUMOR MARKER TEST?

You may need a tumor marker test if you:

Are currently being treated for cancer.

Have finished cancer treatment.

Have a high risk of getting a certain type of cancer because it runs in your family, or you have other conditions that increase your risk.

The type of test you have will depend on your health and health history, cancer diagnosis, and the symptoms you may have.

WHAT HAPPENS DURING A TUMOR MARKER TEST?

Blood tests are the most common type of tumor marker tests. However, urine tests or biopsies are also frequently used to check for tumor markers. A biopsy is a minor procedure that involves removing a small piece of tissue for testing.

For a blood test: A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a urine test: A provider may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

For a biopsy: A provider will remove a small piece of tissue. There are many ways to do a biopsy. The type of biopsy you have depends on the type of cancer you have and where the tumor is located. In general, a biopsy may be done using:

A hollow needle inserted through your skin. Imaging tests, such as ultrasound, may be used to guide the needle.

A fine needle aspiration biopsy uses a very thin needle to remove a sample of cells and/or fluid.

A core needle biopsy uses a larger needle to remove a sample.

A fine needle aspiration biopsy uses a very thin needle to remove a sample of cells and/or fluid.

A core needle biopsy uses a larger needle to remove a sample.

Surgery. Surgery may be done to remove a sample of tissue (an incisional biopsy). In certain cases, the entire tumor will be removed (an excisional biopsy).

Endoscopy. This procedure uses a scope to look inside your body. A special tool may be used with a scope to remove a sample of tumor tissue. Different types of endoscopies may be used depending on where the tumor is located.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

For a blood or urine test: You usually don't need any special preparations.

For a biopsy: You may need to fast (not eat or drink) for several hours before the procedure.

Talk with your provider if you have any questions about preparing for your test.

ARE THERE ANY RISKS TO THE TEST?

For a blood test: There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

For a urine test: There is no risk to a urine test.

For a biopsy: You may have a little bruising or bleeding at the biopsy site. You may also have a little discomfort for a day.

or two.

WHAT DO THE RESULTS MEAN?

Your results will vary depending on the kind of tumor marker your provider tested for. Some tumor markers are linked with only one type of cancer. Others are linked to many different types. Some of the most common tumor markers are:

See if cancer treatment is working

See if cancer has come back after you've finished treatment

Screen for prostate cancer

Help diagnose prostate cancer

Monitor treatment

Check to see if cancer has come back after you've finished treatment

See if cancer treatment is working

See if cancer has come back after you've finished treatment

Help diagnose liver cancer

Find out if cancer has spread (the stage of cancer)

See if cancer treatment is working

Predict chances for recovery

See if cancer treatment is working

Predict chances for recovery

Your provider will review your tumor marker test results along with other information about your condition. Together, you can discuss how your results affect your diagnosis, treatment, and schedule for future testing.

Learn more about laboratory tests, reference ranges, and understanding results.

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RELATED HEALTH TOPICS

Biopsy
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Prostate Cancer
Prostate Cancer Screening
Stomach Cancer
Testicular Cancer
Thyroid Cancer

RELATED MEDICAL TESTS

Alpha Fetoprotein (AFP) Tumor Marker Test
Beta 2 Microglobulin (B2M) Tumor Marker Test
BRAF Genetic Test
BRCA Genetic Test
Breast Biopsy
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Uric Acid Test

<https://medlineplus.gov/lab-tests/uric-acid-test/>

URIC ACID TEST

WHAT IS A URIC ACID TEST?

This test measures the amount of uric acid in a sample of your blood or urine (pee). Uric acid is a normal waste product that your body makes when it breaks down chemicals called purines. Purines come from your cells when they die. Purines are also found in many foods and beverages.

Most uric acid dissolves in your blood. Your kidneys filter the uric acid out of your blood, and it leaves your body in your urine. If uric acid builds up in your blood, it can form needle-shaped crystals in and around your joints. This condition is called gout.

Gout is a type of arthritis that causes painful swelling in your joints. High uric acid levels can also cause kidney stones, or kidney failure. But not everyone with high levels of uric acid will have these problems.

Other names: serum urate, UA, uric acid: serum and urine

WHAT IS IT USED FOR?

A uric acid blood test may be used to:

Help diagnose gout, usually when done with a synovial fluid analysis.

Monitor uric acid levels in people who are having cancer chemotherapy or certain other cancer treatments. When treatment kills cancer cells quickly, they release large amounts of purines into the blood. This can lead to serious problems from high uric acid levels. Testing helps catch an increase in uric acid so it can be treated early.

A uric acid urine test may be used to:

Help find out whether high levels of uric acid are causing kidney stones

Monitor the risk of developing kidney stones in people who have gout

WHY DO I NEED A URIC ACID TEST?

You may need a uric acid blood test if you:

Have symptoms of gout. Symptoms usually happen in one joint at a time. The big toe is most commonly affected, but your other toes, ankle, or knee may have symptoms, which include:

Intense pain

Swelling

Redness

Feeling warm

Intense pain

Swelling

Redness

Feeling warm

Had or are having certain types of cancer treatments.

You may also need a uric acid urine test if you:

Have symptoms of a kidney stone, including:

Sharp pain in your lower abdomen (belly), side, groin or back

Blood in your urine

Frequent urge to urinate (pee)

Not being able to urinate at all or only urinating a little bit

Pain when urinating
Cloudy or bad-smelling urine
Nausea and vomiting
Fever and chills
Sharp pain in your lower abdomen (belly), side, groin or back
Blood in your urine
Frequent urge to urinate (pee)
Not being able to urinate at all or only urinating a little bit
Pain when urinating
Cloudy or bad-smelling urine
Nausea and vomiting
Fever and chills
Have gout and your health care provider wants to monitor your risk for developing kidney stones

WHAT HAPPENS DURING A URIC ACID TEST?

A uric acid test may be done as a blood test or a urine test. Your provider will decide which test you need.
For a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.
For a uric acid urine test, you'll need to collect all your urine during a 24-hour period. This is called a 24-hour urine sample test. You will be given a special container to collect your urine and instructions on how to collect and store your samples. Your provider will tell you what time to start. The test generally includes the following steps:
To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.
For the next 24 hours, collect all your urine in the container.
During the collection period, store the urine container in a refrigerator or in a cooler with ice.
24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.
Return the container with your urine to your provider's office or the laboratory as instructed.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Usually, you don't need any special preparations for a uric acid blood test. Many medicines, including aspirin, can affect your uric acid levels. Taking niacin (vitamin B-3) can also affect your results. So, be sure to tell your provider about all the medicines you take. But don't stop taking any medicines without talking with your provider first.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a uric acid blood or urine test.

WHAT DO THE RESULTS MEAN?

High uric acid levels in a blood test may happen if:
Your body makes too much uric acid
Your kidneys can't remove uric acid from your blood
You are eating too many foods that cause high levels of purines such as:

Red meat and organ meats, including liver and kidney
Certain kinds of seafood, including shellfish, anchovies, sardines, trout, and tuna
Alcohol (all types)
High fructose corn syrup in soda and sweets

Red meat and organ meats, including liver and kidney

Certain kinds of seafood, including shellfish, anchovies, sardines, trout, and tuna

Alcohol (all types)

High fructose corn syrup in soda and sweets

Your provider may order more tests to help diagnose what's causing the problem. High levels of uric acid may be related to many conditions, including:

Gout

Kidney disease

Leukemia, multiple myeloma, or cancer that has spread in your body (metastatic cancer)

Side effects from certain cancer treatments

Alcohol use disorder

Preeclampsia, a condition that can cause dangerously high blood pressure in pregnant women

Low levels of uric acid in blood are uncommon and usually don't cause health problems. Health conditions that are related to low uric acid levels are usually diagnosed using other tests.

High uric levels in urine, may be related to:

Gout

Kidney stones or having a high risk for kidney stones

Leukemia, multiple myeloma, or cancer that has spread in your body

Obesity

An inherited genetic condition, such as Lesch-Nyhan syndrome, that makes it hard for your body to get rid of uric acid

Low levels of uric acid in urine may be related to kidney disease, lead poisoning, or alcohol use disorder.

If your results show a high level of uric acid in your blood or urine, it doesn't always mean you have a condition that needs treatment. Many people have high levels of uric acid without having health problems.

If you have questions about your results, talk with your provider. Medicines and/or changes in your diet may help you keep your uric acid levels in a healthy range.

Learn more about laboratory tests, reference ranges, and understanding results.

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RELATED HEALTH TOPICS

- Chronic Kidney Disease
- Gout
- Kidney Diseases
- Kidney Failure
- Kidney Stones

RELATED MEDICAL TESTS

- Calcium Blood Test
- Calcium in Urine Test
- Crystals in Urine
- How to Cope with Medical Test Anxiety
- How to Understand Your Lab Results
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- Synovial Fluid Analysis

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Urobilinogen in Urine

<https://medlineplus.gov/lab-tests/urobilinogen-in-urine/>

UROBILINOGEN IN URINE

WHAT IS A UROBILINOGEN IN URINE TEST?

A urobilinogen in urine test measures the amount of urobilinogen in your urine (pee). Urobilinogen comes from bilirubin. Your body makes bilirubin during the normal process of breaking down old red blood cells. Your liver uses the bilirubin to make bile, a fluid that helps you digest food in your intestines. Some bile flows through your bile ducts (small tubes) from your liver directly into your intestines. The rest is stored in your gallbladder for when you need it.

Good bacteria in your intestines breaks down the bilirubin in your bile and makes urobilinogen. Some of the urobilinogen leaves your body in your stool (poop). Some of it enters your bloodstream and returns to your liver, where it's "recycled" into bile. Normal urine contains some urobilinogen.

High levels of urobilinogen in urine may be a sign that:

Your liver can't recycle urobilinogen into bile because of a liver disease such as hepatitis or cirrhosis.

Your liver is making too much bilirubin because your body breaks down red blood cells faster than it can make them.

This condition is called hemolytic anemia.

Little or no urobilinogen in urine may be a sign of:

Something blocking bile from flowing into your intestines.

Other problems with your liver, gallbladder, or bile ducts.

Other names: urine test; urine analysis; UA, chemical urinalysis

WHAT IS IT USED FOR?

A urobilinogen in urine test may be part of a urinalysis, a test which may include a visual check of your urine sample, tests for certain chemicals, and an examination under a microscope to look for certain types of cells. A urinalysis is often used to check your general health, including the health of your urinary tract and kidneys. It can also help diagnose diseases that affect the liver and many other medical conditions.

WHY DO I NEED A UROBILINOGEN IN URINE TEST?

Your health care provider may order this test as part of your routine checkup or to monitor an existing liver condition. You may also have this test if you have symptoms of liver disease or hemolytic anemia.

Liver disease/damage symptoms may include:

Nausea and vomiting

Lack of appetite

Fatigue

Weakness

Jaundice, a condition that causes your skin and eyes to turn yellow

Swelling and/or pain in your abdomen (belly)

Swelling in your ankles and legs

Dark-colored urine (pee) and/or light-colored stool (poop)

Frequent itching

Hemolytic anemia symptoms may include:

Fatigue

Dizziness

Heart palpitations (feeling that your heart is pounding or racing)

Headache

Problems concentrating or thinking

Jaundice

Abdominal (belly) pain due to an enlarged liver or spleen

WHAT HAPPENS DURING A UROBILINOGEN IN URINE TEST?

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations. If your provider has ordered other urine or blood tests, you may need to fast (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having this test.

WHAT DO THE RESULTS MEAN?

If your test results show too little or no urobilinogen in your urine, it may be a sign of:

A blockage in the ducts that carry bile from your liver or gallbladder to your intestines

A blockage in the blood flow through the liver

A problem with liver function

If your test results show a higher-than-normal level of urobilinogen, it may be a sign of:

Hepatitis

Cirrhosis

Liver damage due to medicines

Hemolytic anemia

If your test results aren't normal, it doesn't always mean you have a medical condition that needs treatment. Certain medicines and supplements can affect your results, so be sure to let your provider know what you are taking. But don't stop taking any medicines unless your provider tells you to.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A UROBILINOGEN IN URINE TEST?

A urobilinogen in urine test alone cannot diagnose a specific condition. If your test results aren't normal, your provider may order other tests to help diagnose liver disease or hemolytic anemia. To learn what your results mean, talk with your provider.

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Anemia
Liver Diseases
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How to Cope with Medical Test Anxiety
How to Understand Your Lab Results

Liver Function Tests

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Videonystagmography (VNG)

<https://medlineplus.gov/lab-tests/videonystagmography-vng/>

VIDEONYSTAGMOGRAPHY (VNG)

WHAT IS VIDEONYSTAGMOGRAPHY (VNG)?

Videonystagmography (VNG) is a test that measures a type of eye movement that you can't control. These eye movements are called nystagmus. With nystagmus, your eyes move from side to side, up and down, or in a circle. The movements can be slow or fast, steady or jerky.

Normally, nystagmus happens briefly as your eyes adjust when you move your head into certain positions. If it happens at other times or doesn't happen when it should, you may have a disorder of the balance system in your inner ear. This balance system is called the vestibular system.

Good balance partly depends on the organs and nerves in your vestibular system working together with your vision and the muscles in your neck and eyes. When everything is working properly, your brain gets the information it needs to keep your eyes focused on an object when your body changes position. This helps you keep your balance when you are moving.

Problems with certain parts of your balance system can cause nystagmus, which may make you feel dizzy or unsteady. If you're having symptoms of a balance problem, VNG can help find out if a vestibular disorder is causing your symptoms.

Other names: VNG

WHAT IS IT USED FOR?

VNG is used to find out if you have a disorder of the vestibular system (the balance system in your inner ear). It can also help find problems with the nerves or parts of your brain that are part of your sense of balance.

WHY DO I NEED A VNG?

You may need a VNG if you have symptoms of a vestibular disorder. The main symptom is dizziness. Dizziness means different things to different people. It can include:

Vertigo, a feeling that you or everything around you is spinning

Feeling as if you're going to fall

Feeling lightheaded or as if you are going to faint

Other symptoms of a vestibular disorder may include:

Ringing in your ears (tinnitus)

Feeling of fullness or pressure in your ear

WHAT HAPPENS DURING A VNG?

A VNG is usually done by a specialist, such as:

An audiologist, a health care provider who is trained to diagnose hearing loss and balance disorders. Audiologists can also provide certain treatments to improve these conditions.

An otolaryngologist (ENT), a doctor who specializes in treating diseases of the ears, nose, throat, and head and neck.

A neurologist, a doctor who specializes in diagnosing and treating disorders of the brain and nervous system

During a VNG test, you'll sit in a dark room wearing special goggles. A camera in the goggles will record your eye movements during these three main parts of the test:

Ocular testing. Without moving your head, you will follow moving lights with your eyes and stare at steady lights.

Positional testing. You'll be asked to move your head and body into different positions. The provider will also move your

head into certain positions. Your eyes will be checked to see if certain movements cause nystagmus.

Caloric testing. For this part of the VNG, your ears will be tested one at a time. This can show if the vestibular system in one ear is working better than the other. Cool water or air will be put into your ear. The cool temperature should make your eyes move in specific ways. You will be tested again with warm water or air in the same ear. If your eyes don't move as expected, it may mean you have damage in your inner ear or the part of your brain that controls balance.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A VNG?

Your provider will let you know how to prepare for your test. You may need to avoid certain foods or medicines before your test. But don't stop taking any medicine without talking with your provider first.

ARE THERE ANY RISKS TO A VNG?

The test may make you feel dizzy for a few minutes. You may want to plan to have someone bring you home in case the dizziness lasts longer. Some people have nausea for a short time during caloric testing, but vomiting is uncommon.

WHAT DO THE RESULTS MEAN?

If your VNG results were not normal, it may mean you have a disorder of the vestibular system in your inner ear. The disorder may be in one or both ears. Your provider can explain what type of disorder you may have. You may need more tests, including other balance tests, to confirm your diagnosis.

A VNG can help diagnose several vestibular disorders, including:

Benign paroxysmal positional vertigo (BPPV). BPPV is the most common cause of vertigo in adults. With this condition, you feel a spinning sensation when you move your head in certain ways. Normally, part of your inner ear has grains of calcium that help tell your brain about your head's position and movement. BPPV happens if these grains slip out of place and get into a part of your inner ear where they don't belong. When the grains move around in the wrong place, they trigger confusing signals about how your head is moving.

Meniere's disease. This disorder causes vertigo, hearing loss, tinnitus (ringing in the ears), and a full feeling in your ears. It happens when too much fluid builds up in the vestibular system. The cause is unknown.

Labyrinthitis. This balance disorder can cause dizziness and temporary hearing loss. It happens if parts of your inner ear become irritated and inflamed (swollen), usually from a cold or flu. It sometimes goes away on its own.

Vestibular neuritis. This is irritation and inflammation (swelling) of the vestibular nerve. This nerve carries signals from the vestibular system in your inner ear to your brain. It's usually caused by a viral infection. The main symptoms are nausea and vertigo.

Acoustic neuroma. This is a tumor which develops on the nerves involved with hearing and balance. It's usually benign (not cancer), but it can cause hearing loss, tinnitus, and/or dizziness.

An abnormal VNG result may also mean you have a condition that affects the parts of your brain that help control your balance. Treatment options depend on the type of balance disorder you have.

If you have questions about your results or treatment, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A VNG?

Another test, called electronystagmography (ENG), measures the same type of eye movements as a VNG. Instead of using a camera to record eye movements, an ENG measures eye movements with electrodes placed near your eyes. The electrodes are small sensors that either stick to your skin or are put in a headband that you wear.

ENG testing is still useful in certain cases. For example, it may be a good test for people with eyelids that partly cover their pupils or for people who feel anxious about wearing goggles. But VNG testing is more common and more accurate. It is also quicker to do, but more costly.

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Vision Screening

<https://medlineplus.gov/lab-tests/vision-screening/>

VISION SCREENING

WHAT IS VISION SCREENING?

A vision screening is a brief test that mainly checks how well you can see things up close and far away. It's also called an eye test. The test usually involves reading letters on an eye chart. A vision screening is a quick way to find out if you need a comprehensive (complete) eye exam. A complete exam checks both your vision and eye health. It looks for signs of serious eye disorders that may not have symptoms, such as glaucoma.

Children usually have vision screening tests as part of routine health checkups. Older children may also have vision screening tests at school. For infants and toddlers, health care providers use different screening tests to check the health of the child's eyes and whether the eye muscles are working correctly.

Adults may have their vision screened during routine health checkups. But usually, their vision is screened as part of a complete eye exam from an eye care specialist. Eye care specialists who do complete exams include ophthalmologists and optometrists. Ophthalmologists are medical doctors who diagnose and treat all types of vision and eye disorders. Optometrists have advanced training to diagnose and treat certain types of vision and eye disorders.

Vision screening may miss certain types of eye disorders, so it's important for children and adults to have regular eye exams. Your or your child's provider or eye care specialist can tell you how often to have eye exams.

Other names: eye test, vision test

WHAT IS IT USED FOR?

For infants and children: Your child's provider will use special vision screening tests to look for signs of common eye conditions that need early treatment to prevent long-term loss of vision. These eye conditions include:

Amblyopia. Children with amblyopia have poor vision that usually happens in just one eye. It's caused by a problem with how the brain and eye work together. It's sometimes called "lazy eye." Amblyopia is the most common cause of vision loss in children.

Strabismus. This condition causes each eye to look in a different direction. One or both eyes may turn in ("cross-eyes") or turn out ("walleyes"). If strabismus isn't corrected, it can cause amblyopia and permanent eye damage.

For adults and children starting at about age 3, screening for problems with near and far vision is used to help find common vision problems that can be corrected with eyeglasses or contact lenses. In certain cases, eye surgery may also be an option. These conditions include:

Nearsightedness (myopia), a condition that makes far away things look blurry.

Farsightedness (hyperopia), a condition that makes close-up things look blurry.

Astigmatism, which causes generally blurry vision and makes it hard to see at night.

Presbyopia (only in middle-aged adults and older), this condition makes it hard to see things up close. It's a normal part of aging that makes the lens of the eye less flexible. Presbyopia often begins around age 45.

WHY DO I NEED VISION SCREENING?

For adults, vision screening helps find out whether you need a complete eye exam from an eye care specialist. If you have vision screening as part of a complete eye exam, the test will show how well you're seeing at different distances, including with corrective lenses (eyeglasses or contact lenses). If you have problems with your eyes or vision, contact your provider or eye care specialist.

Babies and children need regular vision screening. This helps find and correct vision problems before they affect learning or cause permanent vision loss. Ask your child's provider how often your child should have vision screening tests. In general:

Newborns should be checked for eye infections and signs of other eye disorders.

At 6 months to a year, eyes and vision should be checked during a regular well-baby visit. These tests are needed to check for early signs of strabismus.

At 3 years, screening for near and far vision may begin for children who are able to describe what they see on an eye chart. This screening can be used with other tests to check for signs of amblyopia. All children should have at least one vision screening between age 3 and 5.

If your child has symptoms of an eye disorder, a vision screening may help decide if an exam from an eye care specialist is needed.

For babies under a year old, symptoms include:

After 3 months: Not being able to follow a moving toy or other object with their eyes.

After 4 months: Eyes that don't look properly lined up. Before 4 months, it's normal if your baby's eyes briefly look in different directions from each other.

For older children, other symptoms of vision problems include:

Squinting or frowning

Closing or covering one eye

Trouble reading and/or doing close-up work

Complaining that things are blurry

Blinking more than usual

Crankiness when using close-up vision, such as looking at books

One or both eyes that are watery, red, swollen, or crusted

WHAT HAPPENS DURING VISION SCREENING?

There are several types of vision screening tests.

Vision screening for children and adults includes:

Distance vision test. This test is also called a "visual acuity" test. It checks how well you can see things far away. Usually, you'll read rows of letters off a wall chart or a video monitor. Each row of letters is smaller than the one before it. To take the test, you usually stand 20 feet from the chart and cover one eye. You read the letters one row at a time until you get to a row with letters that are too small for you to see. Each eye is tested separately. Some eye charts use a capital E that points in different directions. Special charts for young children may use pictures or symbols.

Close-up vision test. For this test, you hold a small card about 14 inches away from your face. The card has several lines of printed text that get smaller and smaller. You read the text out loud using both eyes at the same time. This test checks for farsightedness in children and adults, and for presbyopia in adults.

Color blindness test. Color blindness means you see colors differently than most people. You may also have trouble telling certain colors apart. Children are usually tested once for color vision. They're shown a picture of colored numbers or symbols in a background of multicolored dots. If they can't see the numbers or symbols, they are color blind.

Vision screening for infants includes checking:

How well your baby's eyes can follow a moving object, such as a toy

How your baby's pupils (the black center part of the eye) respond to a bright light

Whether your baby blinks when a light shines in the baby's eye

Whether both eyes focus together

WILL I NEED TO DO ANYTHING TO PREPARE FOR VISION SCREENING?

If you or your child wears glasses or contact lenses, bring them with you to the screening. Your provider may want to check the prescription.

ARE THERE ANY RISKS TO SCREENING?

There is no risk to a vision screening.

WHAT DO THE RESULTS MEAN?

If vision screening shows a possible vision problem or eye disorder, you or your child will likely be referred to an eye care specialist for a more complete eye exam and treatment.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT VISION SCREENING?

There are different types of eye care specialists. These are some of the most common types:

Ophthalmologists are medical doctors who specialize in preventing, diagnosing, and treating eye disease. They provide complete eye exams, prescribe eyeglasses and contact lenses, and perform eye surgery.

Optometrists are health care professionals who specialize in vision problems and eye disorders. They provide many of the same services as ophthalmologists, including eye exams, prescribing eyeglasses and contact lenses, and treating some eye disorders. In some states, optometrists can perform certain eye surgeries. For complex eye disorders or surgeries, you will need to see an ophthalmologist.

Opticians are health professionals who are trained to fill prescriptions for eyeglasses. They fit you for frames and prepare your eyeglasses. Many opticians also provide contact lenses. An optician can't give you a complete eye exam, but may screen your vision with an eye chart.

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Vitamin B Test

<https://medlineplus.gov/lab-tests/vitamin-b-test/>

VITAMIN B TEST

WHAT IS A VITAMIN B TEST?

B vitamins are nutrients that your body needs to work well and stay healthy. There are eight different types. A vitamin B test checks the level of one or more B vitamins in a sample of your blood or urine (pee).

All the B vitamins together are called the "B vitamin complex." The B vitamins are:

B1, thiamine

B2, riboflavin

B3, niacin (also called nicotinic acid and nicotinamide)

B5, pantothenic acid

B6, pyridoxal phosphate

B7, biotin

B9, folate (folic acid)

B12, cobalamin

Each type of vitamin B plays different important roles in your body, such as helping to:

Maintain normal metabolism (the process your body uses to make energy from food you eat)

Make healthy blood cells

Keep your nervous system working properly

You get B vitamins from many foods, including leafy green vegetables, meats and fish, eggs, whole grains, milk, and other dairy products. A lack of vitamin B is rare in the United States, because many foods have added B vitamins. These foods include cereals, breads, and pasta.

But certain diets and health conditions can affect how well your body absorbs vitamin B. And if you lack B vitamins, it can cause serious health problems. For example, a lack of B12 can cause anemia and nerve disorders.

A lack of B vitamins is called a vitamin B deficiency. It's usually treated with supplements. But it's important to talk with your health care provider before taking vitamin B supplements. That's because large amounts of some B vitamins can cause health problems, too. And some B vitamins may affect the way certain medicines work in your body.

Other names: vitamin B testing, vitamin B complex, thiamine (B1), riboflavin (B2), niacin (B3), pantothenic acid (B5), pyridoxal phosphate (B6), biotin (B7), vitamin B12, folic acid, folate, and cobalamin test

WHAT IS IT USED FOR?

Vitamin B testing is used if you have symptoms that suggest you may not be getting enough of one or more B vitamins. For example, a vitamin B12 and folate test is often used for symptoms of certain types of anemia.

If you have a condition that increases your risk for a vitamin B deficiency, your provider may check your vitamin B levels even if you don't have symptoms.

WHY DO I NEED A VITAMIN B TEST?

You may need a vitamin B test if you have symptoms that could mean you're lacking certain B vitamins. The symptoms depend on which type of a vitamin B deficiency you have.

For example, common symptoms from a lack of vitamin B6 may include:

Itchy rashes

Cracks at the corners of the mouth

Scaly skin on the lips

Swollen tongue

Common symptoms from a lack of vitamin B12 may include:

Fatigue

Arrhythmia (problems with the rate or rhythm of your heartbeat)

Weight loss when you're not trying to lose weight

Tingly or numb hands and feet (a sign of nerve problems)

You may also need vitamin B testing if you have a condition or habits that increase your risk of having a vitamin B deficiency, such as:

Celiac or Crohn's disease

Alcohol use disorder (AUD)

Malabsorption syndrome

Not eating foods that are rich in B vitamins, such as animal products (meat, eggs, and/or dairy foods)

Having had surgery on your digestive system, including gastric bypass surgery for weight loss

Taking certain medicines that affect how your body uses certain B vitamins

WHAT HAPPENS DURING A VITAMIN B TEST?

Vitamin B levels may be checked in blood or urine.

For a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

For a urine test, you may have a random urine test or a 24-hour urine sample test.

For a random urine test, you will provide a single sample of urine at any time of the day. You'll usually collect your sample at your provider's office or a lab.

For a 24-hour urine sample test, you'll need to collect all your urine during a 24-hour period. You will be given a special container to collect your urine and instructions on how to collect and store your sample. Your provider will tell you what time to start. The test generally includes the following steps:

To begin, urinate in the toilet as usual. Do not collect this urine. Write down the time you urinated.

For the next 24 hours, collect all your urine in the container.

During the collection period, store the urine container in a refrigerator or cooler with ice.

24 hours after starting the test, try to urinate if you can. This is the last urine collection for the test.

Return the container with your urine to your provider's office or the laboratory as instructed.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Before a vitamin B blood test, you may need to fast (not eat or drink) for several hours before the test. A urine test doesn't require any special preparations.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may experience slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is no known risk to having a urine test.

WHAT DO THE RESULTS MEAN?

Your results will show the amount of each B vitamin that was tested. If your vitamin B levels are normal, but you have symptoms, you may need other tests to diagnose your condition.

If you're lacking in one or more B vitamins, you may need other tests to find out why. Your provider may recommend taking vitamin B supplements. In certain cases, vitamins may be given by intravenous (IV) treatment.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT VITAMIN B TESTING?

The meaning of vitamin B test results isn't always clear. That's because several things can affect the results, including inflammation in your body and certain medicines. And medical experts aren't sure what a normal level is for certain B vitamins. Different labs have different ranges for normal levels, too. Your provider can explain what your test results say about your health.

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Vitamin D Test

<https://medlineplus.gov/lab-tests/vitamin-d-test/>

VITAMIN D TEST

WHAT IS A VITAMIN D TEST?

A vitamin D test measures the levels of vitamin D in your blood. Vitamin D helps your body absorb calcium to build healthy bones and teeth. It also helps keep your muscles, nerves, and immune system working normally. Having vitamin D deficiency (very low levels of vitamin D) can lead to bone disorders and other medical conditions.

You can get vitamin D from three sources:

Sunlight. Your body makes vitamin D when your bare skin, without sunscreen, is exposed to sunlight when you are outdoors.

Certain foods. Only a few foods, such as egg yolks, liver, and fatty fish, naturally contain vitamin D. That's why vitamin D is added to many foods, including breakfast cereals, milk, and other dairy items.

Supplements. You can take vitamin D supplements in pill form or liquid drops.

Before your body can use vitamin D, your liver must change it into another form called 25 hydroxyvitamin D, or 25(OH)D. Most vitamin D blood tests measure the level of 25(OH)D in your blood because that's the most accurate way to see if you have enough vitamin D.

Another type of vitamin D test measures a different form of vitamin D in your blood. It's called "active vitamin D." (It may also be called calcitriol or 1,25-dihydroxyvitamin D.) This test isn't usually used to check if you have enough vitamin D. But it may be used to monitor kidney problems or to help find the cause of abnormal calcium levels in your blood.

Other names: 25-hydroxyvitamin D, 25(OH)D, cholecalciferol test, ergocalciferol test, calcidiol test, vitamin D2 test, vitamin D3 test

WHAT IS IT USED FOR?

A vitamin D test is used to check the levels of vitamin D in your blood. The test is usually done if your health care provider thinks that a bone or other health condition you have could be caused by very low vitamin D levels.

Routine vitamin D testing is not recommended for everyone. Your provider can let you know whether a vitamin D test is right for you.

WHY DO I NEED A VITAMIN D TEST?

Your provider may order a vitamin D test if you:

Have been diagnosed with a medical condition that may be related to vitamin D deficiency, such as:

Osteomalacia, soft bones, often with muscle weakness

Low bone density, osteopenia, or osteoporosis

Rickets, a problem with bone growth in children

Osteomalacia, soft bones, often with muscle weakness

Low bone density, osteopenia, or osteoporosis

Rickets, a problem with bone growth in children

Have signs or symptoms of a condition that may be related to vitamin D deficiency such as:

Bone pain

Muscle weakness or aches

Soft or deformed bones

Weak bones and fractures (broken bones)

Bone pain

Muscle weakness or aches

Soft or deformed bones

Weak bones and fractures (broken bones)

Have a high risk for developing vitamin D deficiency. Your risk may be high if you:

Rarely expose your skin to sunlight because you stay indoors, cover up outside, use sunscreen, or live where there is little sunlight

Don't eat enough food or have malnutrition

Have had weight loss surgery

Have a condition that makes it difficult to absorb nutrients in food, such as Crohn's disease, ulcerative colitis, or celiac disease

Have kidney or liver disease that affects your ability to change vitamin D into a form your body can use

Take medicines that affect vitamin D levels, including certain statins to lower cholesterol, steroids, and weight-loss medicines

Rarely expose your skin to sunlight because you stay indoors, cover up outside, use sunscreen, or live where there is little sunlight

Don't eat enough food or have malnutrition

Have had weight loss surgery

Have a condition that makes it difficult to absorb nutrients in food, such as Crohn's disease, ulcerative colitis, or celiac disease

Have kidney or liver disease that affects your ability to change vitamin D into a form your body can use

Take medicines that affect vitamin D levels, including certain statins to lower cholesterol, steroids, and weight-loss medicines

Babies and children can develop serious health problems from a lack of vitamin D. A provider may order a vitamin D test for:

Babies that are mainly fed breastmilk. Breastmilk is low in vitamin D. All babies need vitamin D supplements shortly after birth, unless they are fed only formula, which contains vitamin D.

Children with diets low in vitamin D.

If you are taking vitamin D supplements for vitamin D deficiency, your provider may order a test to see if your vitamin D levels are improving.

Getting too much vitamin D from supplements can harm your health. This rarely happens, but if you take supplements and have symptoms of high levels of vitamin D, your provider may order a test. The symptoms of too much vitamin D include:

Nausea and vomiting

Poor appetite

Constipation

Muscle weakness

Weight loss

WHAT HAPPENS DURING A VITAMIN D TEST?

A vitamin D test is a blood test. During a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a vitamin D test. But be sure to tell your provider about all the medicines, vitamins, and supplements you take, because they may affect your test results. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Your test results may be reported as "total vitamin D," or as the amounts of vitamin D2 and vitamin D3. Vitamin D2 and D3 work about the same in your body. If you add vitamin D2 and D3 together, you get your total vitamin D. Total vitamin D is the important number to check.

Your test results will usually describe total vitamin D levels as:

Deficient, which means very low vitamin D levels that are likely to affect your bones and general health

Insufficient, which means low vitamin D levels that may weaken your bones and affect your health, even if you don't have symptoms

Sufficient or optimal, which means vitamin D levels that are enough for good bone and general health for most people

Toxic or possible toxicity, which means very high vitamin D levels that might cause health problems

If your total vitamin D levels are deficient or insufficient, it may mean you:

Don't get enough vitamin D from your diet and/or exposure to sunlight

Have trouble absorbing vitamin D from your food, which may be a sign of a malabsorption disorder

Have trouble changing vitamin D into a form your body can use, which may be a sign of kidney or liver disease

Take a medicine that affects vitamin D levels

To increase your vitamin D levels, your provider may recommend taking vitamin D supplements and/or eating more foods rich in vitamin D. This is usually safer than getting more sun, which may cause skin cancer.

If your total vitamin D levels are high or toxic, it usually means you're getting too much from supplements. You'll need to stop taking the supplements to reduce your vitamin D levels. That's because too much vitamin D can cause serious damage to your organs and blood vessels.

To learn what your vitamin D test results mean, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

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Vitamin E (Tocopherol) Test

<https://medlineplus.gov/lab-tests/vitamin-e-tocopherol-test/>

VITAMIN E (TOCOPHEROL) TEST

WHAT IS A VITAMIN E (TOCOPHEROL) TEST?

A vitamin E test measures the amount of vitamin E in your blood. Vitamin E (also known as tocopherol or alpha-tocopherol) is a nutrient that is found in every cell of your body. It helps your nerves and muscles work well, prevents blood clots, and boosts your immune system so it can fight off infections from germs.

Vitamin E is a type of antioxidant, which means that it protects cells from damage. But if you have too little or too much vitamin E in your body, it can cause serious health problems.

Most people get the right amount of vitamin E from foods, including vegetable oils, nuts, seeds, avocados, and green, leafy vegetables. Vitamin E is also added to foods such as certain cereals, fruit juices, and margarine. The amount of vitamin E you get from foods doesn't cause harmfully high levels of vitamin E in your body.

High levels usually happen from taking too many vitamin E supplements. Low levels are often caused by digestive diseases, including malabsorption disorders that make it difficult for your body to digest fat. Your body needs fat to absorb vitamin E.

Other names: tocopherol test, alpha-tocopherol test, vitamin E serum test

WHAT IS IT USED FOR?

A vitamin E test may be used to:

Find out if you are absorbing enough vitamin E, especially if you have a disorder that affects how your body uses vitamin E.

Check vitamin E levels in premature babies. Babies born before 37 weeks of pregnancy often lack vitamin E and may develop serious health problems if they aren't treated.

Find out if you are getting too much vitamin E, especially if you have symptoms and have been taking vitamin E supplements.

WHY DO I NEED A VITAMIN E TEST?

You may need a vitamin E test if you have symptoms of vitamin E deficiency (low levels of vitamin E) or vitamin E excess (high levels of vitamin E).

Symptoms of a vitamin E deficiency often involve damage to the nerves and muscles. They include:

Muscle weakness

Loss of feeling in your arms and legs

Problems walking and controlling your body movements

Vision problems

Vitamin E deficiency is very uncommon in healthy people. Most of the time, it's caused by conditions that prevent your body from properly absorbing vitamin E, including:

Crohn's disease, a chronic (long-lasting) disease that causes inflammation in your digestive tract

Liver disease

Cystic fibrosis, a disease of the mucus and sweat glands that's inherited (passed down through families)

Celiac disease, a chronic digestive and immune disorder that damages your small intestine

Pancreatitis, inflammation of the pancreas

Certain genetic disorders, including ataxia with vitamin E deficiency and abetalipoproteinemia (a very uncommon inherited disorder)

Eating an extremely low-fat diet can also lead to a vitamin E deficiency.

Symptoms of too much vitamin E include:

Diarrhea
Nausea
Fatigue
Muscle weakness

Vitamin E excess is uncommon. It usually happens from taking too many vitamin E supplements. Without treatment, too much vitamin E can increase your risk of bleeding, including bleeding in the brain, called a hemorrhagic stroke.

WHAT HAPPENS DURING A VITAMIN E TEST?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You will probably need to fast (not eat or drink) for 12-14 hours before the test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

Normal vitamin E levels vary with age. Your test results should say if your results are normal for your age.

Low vitamin E levels mean you are not getting or absorbing enough vitamin E. Your health care provider will use your medical history and the results from other tests to diagnose the cause. You can usually treat vitamin E deficiency by taking vitamin supplements and treating any conditions that may be blocking absorption.

High vitamin E levels mean you are getting too much vitamin E. If you have been taking vitamin E supplements, you will need to stop. If vitamin E has caused problems with bleeding, your provider may also prescribe medicines to treat you.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A VITAMIN E TEST?

Some people believe vitamin E supplements can help prevent certain disorders. But there is no solid evidence that vitamin E has any effect on heart disease, cancer, eye disease, or mental function.

Taking vitamin E supplements may affect how certain medicines work, including certain blood thinners and cancer medicines. It may also affect your body's ability to use other vitamins. So, before taking any dietary supplements, always talk with your provider to find out if they are safe for you.

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V/Q Scan

<https://medlineplus.gov/lab-tests/vq-scan/>

V/Q SCAN

WHAT IS A V/Q SCAN?

A V/Q scan consists of two imaging tests that look for certain lung problems.

The tests are:

A ventilation scan measures how air moves in and out of your lungs

A perfusion scan measures circulation (how blood flows in the lungs)

The two scans may be done separately or together.

A V/Q scan uses a small amount of a radioactive substance called a tracer. The tracer can help your health care provider see how well air and blood flow through your lungs. The scans help diagnose different lung conditions, but they are most often used to check for a pulmonary embolism (PE). A PE is a blood clot in your lungs.

A PE causes a sudden blockage in a lung artery (a blood vessel that carries oxygen-rich blood from your heart and lungs to your body). This results in your lungs not getting enough air or blood flow.

A PE usually happens due to deep vein thrombosis (DVT). DVT is a blood clot that forms in a vein deep in your body. Most DVT clots occur in your lower leg or thigh. When a DVT blood clot breaks loose and travels through the bloodstream to the lungs it can cause a PE. This is a serious condition and can be life-threatening.

Other names: ventilation/perfusion scan, pulmonary ventilation/perfusion scan, lung scan, lung V/Q scan, lung scintigraphy

WHAT IS IT USED FOR?

A V/Q scan is most often used to check for a pulmonary embolism (PE). It may also be used to:

Find problems with blood flow in the lungs.

Check your lung function before lung surgery.

Test lung function in people with certain lung diseases, such as COPD (chronic obstructive pulmonary disease), a disease that causes coughing, wheezing, and shortness of breath (feeling like you can't get enough air).

WHY DO I NEED A V/Q SCAN?

You may need a V/Q scan if you have symptoms of a pulmonary embolism (PE). These include:

Shortness of breath

Rapid breathing

Chest pain or discomfort, which usually gets worse when you cough or take a deep breath

Increased heart rate

Coughing up blood

Very low blood pressure, lightheadedness, or fainting

Many people with a PE don't have symptoms. But your provider may order a V/Q scan based on your physical exam and/or if you have certain risk factors. These could include:

Family history of blood clots or PE

Long periods of inactivity, for example sitting for too long on long car trips or flights or bed rest

Recent surgery

Having obesity

Older age

Smoking

You may also be at higher risk if you have:

A clotting disorder

Cancer
Heart disease
Other lung disease, such as COPD

WHAT HAPPENS DURING A V/Q SCAN?

V/Q scans are usually performed in a radiology clinic or a hospital. You may have a ventilation scan or a perfusion scan, or you may get both scans. If you get both, one scan will be done right after the other.

For both types of scans:

You will lie very still on a special table while the scanner takes pictures of your lungs or sit in a chair in front of a special camera.

Before the scan, you will be given a substance called a radioactive tracer. The tracer sends out a form of energy called gamma rays. The rays are picked up by the scanner to create images of your lungs.

During a ventilation scan:

You will wear a face mask and breathe in a harmless gas that contains the radioactive tracer.

The tracer will attach to the air in your lungs.

Your provider will use the scanner to take pictures of your lungs while you hold your breath.

The tracer will show up on the pictures so your provider can check the airflow in your lungs.

Your provider may reposition your body as they continue to take pictures while you breathe in the tracer gas for a few more minutes.

After the tracer gas has collected in your lungs, your provider will remove your face mask. As you breathe normally, the tracer will leave your lungs.

During a perfusion scan:

Your provider will inject the radioactive tracer into your vein through an intravenous (IV) line.

The tracer will collect in the blood vessels of your lungs to show how the blood flows in your lungs.

Your provider will use the scanner to take pictures of your lungs.

You will be moved into several positions during the test so the scanner can capture images of your lungs from different angles.

Each test takes about 15 minutes, so if you have both tests, it will take about 30 to 60 minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

A chest x-ray is usually done before a V/Q scan. This can help your provider rule out other lung issues while giving them more information about your lungs so they can better understand your V/Q results.

ARE THERE ANY RISKS TO THE TEST?

There is very little exposure to radiation in a V/Q scan. Only a small amount of radioactive substance is used, and all the radiation leaves your body within a few days.

While radiation exposure in a V/Q scan is safe for most adults, it can be harmful to a developing baby. So be sure to tell your provider if you are pregnant or think you may be pregnant. They may recommend a CT scan or a different test to check for lung issues. Also, tell your provider if you are breastfeeding, since the tracer may contaminate your breast milk.

You may have a little discomfort when the tracer is injected during a perfusion scan.

Allergic reactions to the tracer are rare and usually mild.

WHAT DO THE RESULTS MEAN?

If your ventilation and/or perfusion scan results were not normal, it may mean something is keeping your lungs from working properly. If the pictures from the ventilation and perfusion scan don't match, it could mean you have a

pulmonary embolism (PE) and will need medical treatment right away.

The scans may also show that you have a different condition affecting your lungs. These include:

COPD

Pneumonia

Heart failure

If you have questions about your results, talk to your provider.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A V/Q SCAN?

If you are diagnosed with a pulmonary embolism (PE), your treatment may include medicines such as blood thinners, which help prevent clots from forming, or clot busters, which help dissolve clots quickly.

If the medicines don't work or you have a very large clot, you may need a surgical procedure to remove the clot. A procedure may also be done to prevent more clots from getting into your lungs.

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RELATED HEALTH TOPICS

- Blood Clots
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- Breathing Problems
- COPD
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- How to Cope with Medical Test Anxiety
- How to Understand Your Lab Results
- Partial Thromboplastin Time (PTT) Test
- Protein C and Protein S Tests

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What You Need to Know About Blood Testing

<https://medlineplus.gov/lab-tests/what-you-need-to-know-about-blood-testing/>

WHAT YOU NEED TO KNOW ABOUT BLOOD TESTING

WHAT ARE BLOOD TESTS?

Blood tests are used to measure or look at cells, chemicals, proteins, or other substances in your blood. Blood testing, also known as blood work, is one of the most common types of lab tests. Blood work is often included as part of a regular checkup. Blood tests are also used to:

Help diagnose certain diseases and conditions.

Monitor a chronic (long-lasting) disease or condition, such as diabetes or high cholesterol.

Find out if treatment for a disease is working.

Check how well your organs are working. Your organs include your liver, kidneys, heart, and thyroid.

Help diagnose bleeding or clotting disorders.

Find out if your immune system is having trouble fighting infections.

WHAT ARE THE DIFFERENT TYPES OF BLOOD TESTS?

There are many different types of blood tests. Common ones include:

Complete blood count (CBC). This test measures different parts of your blood, including red and white blood cells, platelets, and hemoglobin. A CBC is often included as part of a regular checkup.

Basic metabolic panel (BMP). This group of tests measures certain chemicals in your blood, including glucose, calcium, and electrolytes.

Blood enzyme tests. An enzyme is a protein that speeds up certain chemical reactions in your body. There are many types of blood enzyme tests. Some of the most common types are troponin and creatine kinase tests. These tests are used to find out if you've had a heart attack and/or if your heart muscle is damaged.

Blood tests to check for heart disease. These include cholesterol tests and a triglyceride test.

Blood clotting tests, also known as a coagulation panel. These tests can show if you have a disorder that causes too much bleeding or too much clotting.

WHAT HAPPENS DURING A BLOOD TEST?

A health care provider will need to take a sample of your blood. This is also called a blood draw. When a blood draw is taken from a vein, it's known as venipuncture.

During venipuncture, a lab professional, known as a phlebotomist, will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Venipuncture is the most common way to do a blood test.

Other ways to do a blood test are:

A finger prick test. This test is done by pricking your fingertip to obtain a small amount of blood. Finger prick testing is often used for at-home test kits and rapid tests. Rapid tests are easy to use, provide fast results, and require little or no special equipment.

A heel stick test. This is most often done on newborns. During a heel stick test, a provider will clean your baby's heel with alcohol and poke the heel with a small needle. The provider will collect a few drops of blood and put a bandage on the site.

Arterial blood test. This test measures your oxygen levels. Blood from arteries has higher levels of oxygen than blood from a vein. So, for this test, blood is taken from an artery instead of a vein. Blood may be taken from your wrist, arm, or groin area. It is often drawn by a specially trained provider. You may feel a sharp pain when the provider inserts the needle into the artery to get the blood sample.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for most blood tests. For some tests, you may need to fast (not eat or drink) for several hours before your test.

You may need to stop taking certain medicines before your test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

Your provider will let you know if there are any special instructions to follow.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a finger prick test or venipuncture. During venipuncture, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is very little risk to your baby with a heel stick test. Your baby may feel a little pinch when the heel is poked, and a small bruise may form at the site.

Collecting blood from an artery is more painful than collecting from a vein, but complications are rare. You may have some bleeding, bruising, or soreness at the spot where the needle was put in. Also, you should avoid lifting heavy objects for 24 hours after the test.

IS THERE ANYTHING ELSE I SHOULD KNOW ABOUT BLOOD TESTING?

Blood testing can provide important information about your health. But it doesn't always give enough information about your condition. If you've had blood work, you may also need other types of tests before your provider can make a diagnosis.

To understand the results of your blood test, your provider may consider your symptoms, medical history, and the results of other tests.

Learn more about laboratory tests, reference ranges, and understanding results.

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RELATED HEALTH TOPICS

Blood

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White Blood Cell (WBC) in Stool

<https://medlineplus.gov/lab-tests/white-blood-cell-wbc-in-stool/>

WHITE BLOOD CELL (WBC) IN STOOL

WHAT IS A WHITE BLOOD CELL (WBC) IN STOOL TEST?

This test looks for white blood cells in a sample of your stool (poop). White blood cells are also called leukocytes, so this test is sometimes called a leukocytes in stool test.

White blood cells are part of your immune system. They help your body fight off infections and other diseases. White blood cells in your stool are usually a sign that you have an infection or other condition in your digestive system that's causing inflammation.

A white blood cell in stool test is used if you have diarrhea that could be caused by inflammation. Diarrhea from inflammation is severe and lasts for more than a couple days. It can include blood in stool, abdominal (belly) pain, and fever. It may be caused by:

Certain bacterial infections, such as:

Clostridium difficile (C. diff). This infection mostly happens after taking antibiotics that kill the "good germs" that keep C. diff bacteria from growing in your body.

Some types of bacteria that cause "food poisoning," including:

Salmonella. Most infections come from contaminated food, but you can get salmonella from infected animals, especially reptiles and birds.

Shigella. Infection with shigella is called shigellosis. You can get it from food and touching contaminated surfaces and then touching your mouth.

Campylobacter. Infection can happen from food and contact with dog and cat feces (poop).

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Inflammatory bowel disease (IBD), which includes:

Ulcerative colitis

Crohn's disease

Ulcerative colitis

Crohn's disease

A parasite (*Entamoeba histolytica*) which causes a disease called "amebiasis." Amebiasis is most common in tropical places with poor sanitary conditions.

A white blood cell in stool test can't diagnose the exact condition that's causing diarrhea. But it can help narrow down the possible causes to conditions that either do or don't involve inflammation.

Other names: leukocytes in stool, stool WBC, fecal WBC, fecal leukocyte test, FLT, FGST, fecal leukocyte gram stain screen

WHAT IS IT USED FOR?

A white blood cell in stool test is used to find out whether diarrhea is caused by conditions that involve inflammation in the digestive system.

It's mainly used for people who have diarrhea symptoms that suggest inflammation. These symptoms include:

Watery diarrhea three or more times a day that doesn't get better after several days

Abdominal (belly) cramps and/or pain

Blood and/or mucus in stool

Fever

Fatigue

Weight loss

A white blood cell in stool test can't diagnose the condition that's causing diarrhea, so it's often used with other tests.

WHY DO I NEED A WHITE BLOOD CELL IN STOOL TEST?

Most diarrhea goes away on its own after a day or two, and your health care provider doesn't need to find the cause. But you may need a white blood cell in stool test and other tests if you have symptoms of diarrhea from inflammation that don't improve after a few days.

Knowing the cause of your diarrhea helps your provider choose the right treatment. That's important because ongoing diarrhea can lead to serious health problems, including dehydration and electrolyte imbalances.

See your provider right away if you have:

Signs of dehydration, including dry mouth, dark urine (pee), fatigue, or dizziness

Diarrhea for more than 2 days, or for more than 24 hours in a young child or infant

A fever of 102 degrees or higher

Severe pain in the abdomen (belly) or rectum

Stool that is black and tarry or contains blood or pus

In adults: frequent vomiting or diarrhea 6 or more times in 24 hours

WHAT HAPPENS DURING A WHITE BLOOD CELL IN STOOL TEST?

You will need to provide a stool sample for your test. Your provider will probably give you a container or kit with instructions on how to collect the sample. There are different ways to collect stool samples, so follow the instructions carefully. In general, you'll need to:

Label the container with your name, the collection date, and time.

Collect a stool sample as instructed. This usually involves using a clean, dry container, or special paper or plastic wrap placed over the toilet to catch the stool.

Make sure the stool doesn't mix with any urine, toilet paper, or water from the toilet.

Close the container tightly.

Wash your hands well with soap and water.

Return the container according to the instructions.

If you're collecting a sample from diapers, you'll get special instructions for using plastic wrap inside of a clean diaper. In certain cases, a provider may use a swab to take a stool sample from the rectum.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

Certain medicines and foods may affect the test results. Ask your provider if there are any specific things you need to avoid before the test.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a white blood cell in stool test.

WHAT DO THE RESULTS MEAN?

A positive or abnormal test result means that white blood cells were found in your stool sample. That means you have inflammation in your digestive tract. This information helps rule out conditions that don't cause inflammation, including viral infections, certain bacteria, and most parasites. Other tests can help find out what is causing your illness.

A negative or normal test result means that no white blood cells were found in your stool sample. That may mean that inflammation isn't causing your illness. But a normal test result can't rule out conditions that cause inflammation. That's because white blood cells don't last long in a stool sample.

If your stool sample contained a small number of cells that broke apart during testing, they may not show up. So, you could have inflammation in your digestive system even though your test didn't find any white blood cells in your stool.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A WHITE BLOOD CELL IN STOOL TEST?

Instead of a white blood cell in stool test, your provider may order a more accurate test to check for inflammation in your digestive system. It's called a calprotectin fecal test. Calprotectin is a protein that certain white blood cells release in the intestine when there is inflammation.

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White Blood Count (WBC)

<https://medlineplus.gov/lab-tests/white-blood-count-wbc/>

WHITE BLOOD COUNT (WBC)

WHAT IS A WHITE BLOOD COUNT (WBC)?

A white blood count measures the number of white blood cells (WBCs) in your blood. White blood cells, also called leukocytes, are part of your immune system. They are a type of blood cell made in your bone marrow and found in your blood and lymph tissue (part of your immune system). If you are injured or get sick, the white blood cells will travel through your bloodstream and tissues to where they are needed. There, they will help your body fight off infections and other diseases.

When you get sick, your body makes more white blood cells to fight the bacteria, viruses, or other foreign substances causing your illness. This increases your white blood count.

Other diseases can cause your body to make fewer white blood cells than you need. This lowers your white blood count. Diseases that can lower your white blood count include some types of cancer and HIV, a viral disease that attacks white blood cells. Certain medicines, including chemotherapy, may also lower the number of your white blood cells.

There are five major types of white blood cells:

Neutrophils

Lymphocytes

Monocytes

Eosinophils

Basophils

A white blood count measures the total number of these cells in your blood. Another test, called a blood differential, measures the amount of each type of white blood cell.

Other names: WBC count, white cell count, white blood cell count, Leukocyte Count, WBC

WHAT IS IT USED FOR?

A white blood count is most often used to help diagnose or monitor disorders related to having a high white blood cell count or low white blood cell count.

Disorders related to having a high white blood count include:

Autoimmune and inflammatory diseases, conditions that cause the immune system to attack healthy tissues

Bacterial or viral infections

Cancers such as leukemia and Hodgkin disease

Allergic reactions

Disorders related to having a low white blood count include:

Diseases of the immune system, such as HIV

Lymphoma, a cancer of the bone marrow

Diseases of the liver or spleen

A white blood count can show if the number of your white blood cells is too high or too low, but it can't confirm a diagnosis. So it is usually done along with other tests to help confirm your diagnosis. These other tests could include a complete blood count, blood differential, blood smear, and/or bone marrow test.

WHY DO I NEED A WHITE BLOOD COUNT?

You may need this test if you have signs of an infection, inflammation, or autoimmune disease. The symptoms of an infection may include:

Fever

Chills

Body aches

Headache

Wound that is red, has pus, or won't heal

Ongoing cough

The symptoms of inflammation and autoimmune diseases will be different, depending on where in the body the inflammation is and the type of disease you have.

You may also need this test if you have a disease that weakens your immune system or are taking medicine that lowers your immune response. If the test shows your white blood count is getting too low, your provider may want to adjust your treatment.

Your newborn or older child may also be tested as part of a routine screening or if they have symptoms of a white blood cell disorder.

WHAT HAPPENS DURING A WHITE BLOOD COUNT?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out.

To test children, a health care provider will take a sample from the heel (newborns and young babies) or the fingertip or arm (older babies and children). The provider will clean the heel, fingertip or arm with alcohol and poke the site with a small needle. The provider will collect a blood sample and put a bandage on the site.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a white blood count. You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

ARE THERE ANY RISKS TO THE TEST?

After a blood test, you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There is very little risk to your baby or child with a needle stick test. Your child may feel a little pinch when the site is poked, and a small bruise may form at the site. This should go away quickly.

WHAT DO THE RESULTS MEAN?

Conditions that may cause a high white blood count, also called leukocytosis, include:

An infection

An inflammatory disease such as rheumatoid arthritis

An allergy

Leukemia or Hodgkin disease

Tissue damage from a burn injury or surgery

Smoking, stress, a reaction to a medicine, or pregnancy may also cause your body to make more white blood cells, which leads to a high white blood count.

Conditions that may cause a low white blood count, also called leukopenia, include:

Bone marrow damage. This may be caused by infection, disease, or treatments such as chemotherapy.

Cancers that affect the bone marrow.

An autoimmune disorder, such as lupus.

HIV.

If you are already being treated for a white blood cell disorder, your results may show if your treatment is working or

whether your condition has improved.

If you have questions about your results, talk to your provider. Your provider will consider your symptoms, medical history, and the results of other blood tests to understand your white blood count results.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A WHITE BLOOD COUNT?

White blood count results are often compared with results of other blood tests, including a blood differential. A blood differential test shows the amount of each type of white blood cell, such as neutrophils or lymphocytes. Neutrophils mostly target bacterial infections. Lymphocytes mostly target viral infections.

A higher-than-normal amount of neutrophils is known as neutrophilia.

A lower-than-normal amount is known as neutropenia.

A higher-than-normal amount of lymphocytes is known as lymphocytosis.

A lower-than-normal amount is known as lymphopenia.

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- Autoimmune Diseases
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Whooping Cough Tests

<https://medlineplus.gov/lab-tests/whooping-cough-diagnosis/>

WHOOPING COUGH TESTS

WHAT ARE WHOOPING COUGH TESTS?

A whooping cough test usually uses fluid from your nose to help diagnose whooping cough, also known as pertussis. Whooping cough is a serious bacterial infection in your respiratory system that may cause severe fits of rapid coughing. In serious cases, the coughing may last until all the air is gone from your lungs. This results in the "whooping" sound that the infection is named after.

Whooping cough is easily spread from person to person, usually by coughing or sneezing. You can get it at any age, but the illness is most serious in babies and children who are too young to get their childhood vaccines against whooping cough. Babies who are less than a year old who get whooping cough often need hospital care.

The best way to protect against whooping cough is with the pertussis vaccination. Vaccinated people may still get the infection, but their illness is usually milder and shorter. Ask your provider when you and your child need to be vaccinated. Your provider can also answer any questions you have about possible vaccine side effects.

If you're pregnant, you should be vaccinated between the 27th and 36th weeks of your pregnancy. If you're vaccinated during each pregnancy, you will pass some protection against whooping cough to your baby for the first months of life before the baby can be vaccinated.

Other names: pertussis test, bordetella pertussis culture, PCR, antibodies (IgA, IgG, IgM)

WHAT ARE THE TESTS USED FOR?

Whooping cough tests are used to find out whether you or your child has whooping cough. Getting diagnosed and treated in the early stages of infection may make your symptoms less severe and help prevent the spread of the disease.

WHY DO I NEED A WHOOPING COUGH TEST?

Your health care provider may order a whooping cough test if you or your child has been exposed to someone with whooping cough or has symptoms of the infection. Symptoms usually happen in three stages. In the first stage, symptoms are like those of a common cold. They can last for one or two weeks and usually include:

Runny nose.

Mild fever.

Mild cough (babies may not cough).

Pauses in breathing, called apnea, especially in babies.

In the second stage, which can last 10 weeks or more, the symptoms are more serious and may include:

Severe coughing fits followed by a high-pitch "whoop" sound.

Coughing so hard it causes vomiting.

Feeling very tired after coughing fits.

The third stage marks the beginning of the recovery. Your cough may be milder, and you may cough less often.

A provider may order a test during any one of these stages. They may order it even if you don't have the "whooping" sound when you cough, especially if you've been exposed to someone with this infection.

To avoid getting or spreading whooping cough:

Avoid close contact with those who have whooping cough.

Stay at home if you or your child has whooping cough.

Cover your mouth and nose when coughing or sneezing.

Thoroughly wash your hands, especially before and after handling food or going to the bathroom.

The infection is usually milder in teens and adults, especially those who have had the vaccine.

WHAT HAPPENS DURING A WHOOPING COUGH TEST?

Two types of tests are commonly used to diagnose whooping cough. Both tests use a sample of fluid and cells from the back of your nose. Your provider may order one or both of these tests:

Bacteria culture test. This test is most accurate when it's done during the first two weeks after coughing begins. Your fluid sample is sent to a lab. Cells in the sample are grown until there are enough to test. The sample is then checked for whooping cough bacteria. Test results may take up to a week.

PCR (polymerase chain reaction) test. This test is most accurate when it's done in the first three weeks after coughing begins. A PCR test checks your fluid sample for DNA left behind by the whooping cough bacteria. Results may be ready in as soon as two hours.

For a bacteria culture or a PCR test, your provider will use one of these ways to gather a sample of fluid and cells from your nose:

Nasopharyngeal swab test. Your provider will insert a long swab into your nostril until it reaches your nasopharynx (the upper part of your throat). They will then rotate the swab for 10 to 15 seconds, remove it, and test that sample for signs of the whooping cough bacteria.

Nasal aspirate or wash. Your provider will squirt a saline solution (salt water) into your nose and remove the sample with gentle suction.

Blood tests that look for antibodies to whooping cough are generally not used to diagnose infections. Researchers use these tests to study the spread of whooping cough in the community. Whooping cough antibodies are proteins in your blood that your immune system makes to fight the infection. A blood antibody test may be done from 2 to 12 weeks after coughing begins. So, the test may be used in later stages when culture and PCR tests may not give accurate results.

If you have a blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

WILL I NEED TO DO ANYTHING TO PREPARE FOR A WHOOPING COUGH TEST?

You don't need any special preparations for a whooping cough test. However, if you have taken antibiotics in the days before your test, let your provider know. This may affect your test results.

ARE THERE ANY RISKS TO THE TESTS?

There is very little risk to whooping cough tests.

A nasal aspirate or wash may feel uncomfortable.

A nasopharyngeal swab test may tickle, feel uncomfortable, or cause you to cough or gag.

For a blood test you may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

WHAT DO THE RESULTS MEAN?

For a bacteria culture test:

A positive result means that you have whooping cough.

A negative result may mean that you don't have whooping cough, but it doesn't rule it out. If your test sample was taken too late or if you took antibiotics before your test, the results may not be accurate.

For a PCR test:

A positive result means that genetic material from the whooping cough bacteria was found in your test sample, and you probably have whooping cough. But the PCR test may have found material from other related bacteria and mistaken it for whooping cough bacteria.

A negative result means that you probably don't have whooping cough, but it doesn't rule it out. It's possible that your sample didn't have enough bacteria for the test to be accurate.

To make a diagnosis for whooping cough, your provider can use your:

Test results

Symptoms

Medical history

Potential exposures to whooping cough

Blood testing for whooping cough antibodies is not recommended for diagnosing a whooping cough infection. If you had this test and you were vaccinated, certain antibodies will show up in your test results. If a blood test finds certain other antibodies, it may mean you had a recent whooping cough infection.

Whooping cough is treated with antibiotic medicine. Antibiotics can make your infection less serious if you start treatment before your cough gets bad. Treatment may also help prevent you from spreading the disease to others. If you have questions about your test results or treatment, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT WHOOPING COUGH TESTS?

In most cases, whooping cough is diagnosed with a simple swab from the back of the nose or throat. But in rare cases, your provider may use a procedure called a bronchoalveolar lavage (BAL). This is a procedure that can be done during a bronchoscopy (a procedure that's used to check for the cause of a lung problem).

A bronchoalveolar lavage collects a sample from your lungs and checks it for the pertussis and parapertussis bacteria. Parapertussis is very similar to pertussis and can cause the same symptoms as those of whooping cough. The whooping cough vaccine does not protect you from parapertussis.

Your provider might order this test if:

You aren't improving with treatment for whooping cough.

They suspect another respiratory illness, such as pneumonia.

They need a tissue sample from the lower respiratory tract to see if your symptoms are being caused by a lower respiratory infection (LRTI).

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Xylose Testing

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XYLOSE TESTING

WHAT IS A XYLOSE TEST?

Xylose, also known as D-xylose, is a type of sugar (carbohydrate) that is normally easily absorbed by your intestines. A xylose test checks the level of xylose in both your blood and urine to see how well you absorb carbohydrates.

Usually, your body digests in stages. The proteins, fats, and carbohydrates in the food you eat are:

Broken down in the stomach by acid and enzymes (proteins that speed up certain chemical reactions in your body)

Broken down further in your small intestines by pancreatic enzymes and bile (liquid made in your liver to help digest fats)

Absorbed into your intestines to be used or stored

If you don't have enough bile or pancreatic enzymes, the carbohydrates and other foods you eat can't be digested well.

Lower-than-normal xylose levels can mean there is a problem with your body's ability to absorb nutrients.

Xylose testing is no longer routinely ordered to check for malabsorption, and it may not be available everywhere. This test may be ordered as a follow-up test to other tests, such as a fecal fat test.

Other names: xylose tolerance test, xylose absorption test, D-xylose tolerance test, D-xylose absorption test

WHAT IS IT USED FOR?

A xylose test is most often used to:

Help diagnose malabsorption disorders, conditions that affect your ability to digest and absorb nutrients from food

Find out why a child is not gaining weight, especially if the child seems to be eating enough food

WHY DO I NEED A XYLOSE TEST?

You may need this test if you have symptoms of a malabsorption disorder, which include:

Chronic (long-lasting) diarrhea

Abdominal (belly) pain

Bloating (a feeling of fullness or swelling in your belly)

Gas

Weight loss for no known reason, or in children, not being able to gain weight

WHAT HAPPENS DURING A XYLOSE TEST?

A xylose test involves taking timed samples of your blood and urine. You will be tested before and after drinking a solution containing eight ounces of water mixed with a small amount of xylose.

For the blood tests:

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Next, you will drink the xylose solution.

You will be asked to rest quietly.

Your provider will take another blood sample two hours later. For children, it may be one hour later.

For the urine tests, you will need to:

Go to the bathroom to empty your bladder before starting the urine (pee) test or drinking the xylose solution. You won't save that urine.

After you drink the xylose solution, you will collect all your urine for the next five hours.

Your provider will give you instructions on how to collect your urine during the five-hour period. Each of your samples will be tested for xylose. This includes the first fasting blood sample, the blood sample taken two hours later, and the five-hour urine sample.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You will need to fast (not eat or drink) for eight hours before the test. Children younger than nine years old should fast for four hours before the test.

For 24 hours before the test, you must avoid foods high in a type of sugar known as pentose, which is similar to xylose. These foods include jams, pastries, and fruits.

Some medicines may affect your test results. You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to.

Your provider will let you know if you need to do any other preparations.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

The xylose solution may make you feel nauseous.

There is no risk to having a urine test.

WHAT DO THE RESULTS MEAN?

High levels of xylose are normal. It means that your intestines are absorbing the xylose. If your results are high, that may mean that your symptoms are not caused by malabsorption.

If your results showed lower-than-normal levels of xylose in the blood or urine, it may mean you have a malabsorption disorder, such as:

Celiac disease, a chronic (long-lasting) digestive and immune disorder that damages your small intestine. It is triggered by foods that contain gluten. Gluten is a protein found in wheat, barley, rye, and other grains.

Crohn's disease, a chronic condition that causes inflammation in your digestive tract.

Whipple disease, a rare condition that prevents the small intestine from allowing nutrients to pass into the rest of the body.

Lower-than-normal levels may also be caused by an infection from a parasite, such as:

Hookworm

Giardiasis

If your xylose blood levels were normal but urine levels were low, it may be a sign of kidney disease and/or malabsorption.

You may need more tests before your provider can make a diagnosis. To understand the results of your xylose testing, your provider may consider your symptoms, medical history, and the results of other tests.

Talk to your provider if you have questions about your or your child's results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT XYLOSE TESTING?

A xylose test takes a long time. You may want to bring a book, game, or other activity to keep yourself or your child busy while you wait.

Learn more about laboratory tests, reference ranges, and understanding results.

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Yeast Infection Tests

<https://medlineplus.gov/lab-tests/yeast-infection-tests/>

YEAST INFECTION TESTS

WHAT IS A YEAST INFECTION TEST?

A yeast test can help find out whether you have a yeast infection. Candidiasis, candidosis, and moniliasis are other names for a yeast infection.

Yeast is a type of fungus. Fungi (more than one fungus) are plant-like life forms that live everywhere, including on your skin, genitals, mouth, and intestines (gut). Usually, yeast doesn't make you sick. But if yeast grows out of control, it can cause an infection that needs treatment.

If you're generally healthy, a yeast infection is rarely serious. But for some people, a yeast infection may become very serious. It may affect the blood, heart, brain, eyes, bones, or other parts of your body. This is called an invasive yeast infection.

Invasive infections are usually not a risk for healthy people. But you may be at higher risk of an invasive infection, for example if you have a weakened immune system or a central venous catheter, or if you've had surgery or chemotherapy.

Other names: potassium hydroxide preparation, fungal culture; fungal antigen and antibody tests, calcofluor white stain, fungal smear

WHAT IS IT USED FOR?

A yeast test is used to diagnose yeast infections. There are different methods of yeast testing, depending on where you have symptoms.

WHY DO I NEED A YEAST TEST?

Your health care provider may order a test if you have symptoms of a yeast infection. Symptoms depend on where the yeast is growing. Yeast infections tend to grow in moist areas of your skin and mucous membranes.

These are the typical symptoms of some common types of yeast infections, but your symptoms may vary:

Yeast infections on the folds of your skin include conditions such as athlete's foot and diaper rash. Symptoms include:

Bright red rash, often with ulcers (open sores)

Itching and/or burning

Pimples

Vaginal yeast infections are very common. Symptoms include:

Itching and/or burning in your vagina and vulva (the external female genital organs)

A white discharge that looks like cottage cheese

Pain when urinating (peeing) and during sex

Redness and swelling of your vagina and labia (folds of skin around the vagina)

Yeast infection of the penis is more common if you have diabetes or a foreskin (you are not circumcised). Symptoms include:

Redness

Itching and/or burning

Rash on the head of the penis that may be painful

Yeast infection of the mouth is called thrush. It is common in young children. Thrush in adults may be a sign of a weakened immune system. Symptoms include:

White patches on the:

Tongue

Inside of cheeks
Roof of the mouth
Throat
Tongue
Inside of cheeks
Roof of the mouth
Throat
Redness or soreness
Cotton-like feeling in the mouth
Loss of taste
Pain while eating
Yeast infection at the corners of the mouth may be caused by thumb sucking, ill-fitting dentures, or frequent licking of the lips. Symptoms include:
Cracks and redness at the corners of the mouth
Yeast infection in the nail beds can happen in the fingernails or toenails, but it's more common in toenails. Symptoms include:
Discolored nail (yellow, brown, or white)
Thick nail
Cracks in the nail
Pain and redness if the infection is severe

WHAT HAPPENS DURING A YEAST TEST?

The type of test depends on the location of your symptoms.

A vaginal yeast infection - Your provider will perform a pelvic exam and take a sample of the discharge from your vagina.

Thrush - Your provider will look at the infected area in your mouth and may scrape some cells to examine under a microscope.

A yeast infection on the skin or nails - Your provider may use a blunt-edged instrument to scrape off a small bit of skin or part of a nail to examine. During this type of test, you may feel some pressure and a little discomfort.

Your provider may be able to tell if you have a yeast infection just by looking at the infected area and examining a sample of the cells under a microscope. If there are not enough cells in the sample to find an infection, you may need a fungal culture test.

During a fungal culture test, the cells in your sample are sent to a lab. Your sample is put in a dish with a special substance that allows cells to grow until there are enough to test. Results are often available within a few days. But some yeast grows slowly, so it may take weeks to get a result.

If your provider thinks you may have an invasive yeast infection in your blood or organs, you may have a fungal culture test or other tests on a sample of blood, other fluid, or tissue from the part of your body that may be infected.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a yeast test.

ARE THERE ANY RISKS TO THE TEST?

There is no known risk to having a yeast test.

WHAT DO THE RESULTS MEAN?

If your results show that you have a yeast infection, your provider may recommend a prescription or over-the-counter

antifungal medicine. Antifungal medicines come in many forms for different types of infections, including pills, vaginal creams and suppositories, skin creams, powders, and lotions. Your provider will tell you which treatment is best for you. It's important to take all your medicine as prescribed, even if you feel better before you finish it. Many yeast infections get better after a few days or weeks of treatment, but certain fungal infections may need treatment for several months or longer before they clear up completely.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A YEAST TEST?

You may have a higher risk of a yeast infection if you:

Are taking antibiotics. Certain antibiotics can cause yeast to grow out of control. So, be sure to tell your provider about any medicines you are taking.

Are pregnant. Hormonal changes may increase your risk of yeast infections.

Have HIV. People with HIV have a higher risk of esophageal candidiasis.

At-home tests for vaginal yeast infections can tell you if your symptoms are likely to be from a yeast infection. But they may not be able to confirm the diagnosis. It's best to talk with your provider about your symptoms and which test is best for you. This is important, because many symptoms of a vaginal yeast infection are the same as the symptoms of certain sexually transmitted infections (STIs).

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Zika Virus Test

<https://medlineplus.gov/lab-tests/zika-virus-test/>

ZIKA VIRUS TEST

WHAT IS A ZIKA VIRUS TEST?

A Zika virus test looks for signs of the Zika infection in your blood or urine. Zika is a type of flavivirus, as is dengue fever and West Nile. Like those viral infections, Zika is usually spread by mosquitoes, especially in areas with tropical climates, such as parts of Central and South America, Africa, Mexico, and islands in the Caribbean and Pacific. Because Zika can enter bodily fluids such as blood, urine (pee), and semen, it can also spread through sex with an infected person or from a pregnant woman to her fetus.

There are two types of tests used to check for Zika infections:

Molecular tests. These tests look for genetic material from the Zika virus. The tests are also called nucleic acid amplification tests (NAAT or NAT). A PCR test (polymerase chain reaction) is a common type of molecular test that checks for the Zika virus in a sample of your blood or urine. This test is recommended if you're pregnant.

Antibody tests. Antibodies are proteins that your immune system makes to fight viruses and other germs. A Zika antibody test checks a sample of your blood for the antibodies that fight Zika. Your body needs time to make antibodies. So, to be accurate, an antibody test can be done starting within a week after symptoms begin.

Zika virus outbreaks tend to come and go. The recommendations for Zika testing are likely to change as outbreaks develop and disappear. Even without confirmed cases of Zika in the U.S., it is still possible to get the infection.

Other names: Zika Antibody Test, Zika RT-PCR Test, Zika test

WHAT IS IT USED FOR?

A Zika virus test is used to check for a Zika virus infection. It's mostly used on pregnant women who have recently traveled to an area where there is a risk of Zika infection. This is because a Zika infection can cause serious health problems for a fetus, including:

Microcephaly, a birth defect that severely affects the development of a fetus' brain.

Cerebral palsy, a condition that affects coordination and the ability to control one's muscles.

Brain development issues.

Vision or hearing problems.

Low birth weight.

Congenital Zika syndrome, a combination of conditions at birth, including microcephaly, a partially collapsed skull, and eye damage.

Zika infections during pregnancy have also been linked to an increased risk of miscarriage and stillbirth.

In rare cases, children and adults infected with Zika may get a disease called Guillain-Barré syndrome (GBS). GBS is a disorder that causes the body's immune system to attack part of the nervous system. GBS is serious, but treatable. With proper care, most people recover from GBS over time.

WHY DO I NEED A ZIKA VIRUS TEST?

You may need a Zika virus test if you are pregnant and have recently traveled to an area where there is a risk of Zika infection. You may also need a Zika test if you are pregnant and have had sex with a partner who traveled to one of these areas.

A Zika test might be ordered if you have symptoms of Zika. Most people with Zika don't have symptoms, but when there are symptoms, they often include:

Fever.

Rash.

Joint pain.

Muscle pain.
Headache.
Red eyes (pink eye, also called conjunctivitis).

WHAT HAPPENS DURING A ZIKA VIRUS TEST?

A Zika virus test is usually a blood test or a urine test.

If you are getting a Zika blood test, a health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

If you are getting a Zika urine test, a health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

Wash your hands with soap and water and dry them.

Open the container without touching the inside.

Clean your genital area with the cleansing wipe:

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.

For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.

Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.

Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.

Finish urinating into the toilet.

Put the cap on the container and return it as instructed.

If you have hemorrhoids that bleed or are having your menstrual period, tell your provider before your test.

If you are pregnant and your prenatal ultrasound shows the possibility of microcephaly, your health care provider may recommend a procedure called amniocentesis to check for Zika. Amniocentesis is a test that looks at the fluid that surrounds the fetus (amniotic fluid). For this test, your provider will insert a special hollow needle into your belly and withdraw a small sample of fluid for testing.

WILL I NEED TO DO ANYTHING TO PREPARE FOR THE TEST?

You don't need any special preparations for a Zika virus test.

ARE THERE ANY RISKS TO THE TEST?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

There are no known risks to a urine test.

Amniocentesis may cause some cramping or pain in your belly. There is a small chance the procedure will cause a miscarriage. Talk to your provider about the benefits and risks of this test.

WHAT DO THE RESULTS MEAN?

A positive Zika test result means you probably have a Zika infection.

A negative result can mean that you aren't infected or that you were tested too soon for the virus to show up in testing. If you think you were exposed to the virus, talk to your provider about when or if you need to be retested.

If you are diagnosed with Zika and are pregnant, you can start to prepare for your baby's possible health problems before they are born. While Zika exposure doesn't always cause birth defects or health problems, many children born with Zika have long-lasting special needs. Talk to your provider about how to get support and health care services should you need them. Early intervention may make a difference in your child's health and quality of life.

If you are diagnosed with Zika and are not pregnant, but would like to become pregnant in the future, talk to your provider. They can tell you how long you should wait before trying to have a baby and if you need to be retested.

Learn more about laboratory tests, reference ranges, and understanding results.

IS THERE ANYTHING ELSE I NEED TO KNOW ABOUT A ZIKA VIRUS TEST?

If you are pregnant or planning to become pregnant, you should take steps to reduce your risk of getting a Zika infection. The Centers for Disease Control and Prevention (CDC) recommends that pregnant women and their partners avoid traveling in areas that may put them at risk for Zika infection. If you can't avoid travel or if you live in one of these areas, you should:

Apply an insect repellent containing DEET on your skin and clothing. DEET is safe and effective for pregnant women.

Wear long-sleeved shirts and pants.

Use screens on windows and doors.

Sleep under a mosquito net.

If you or your partner think either of you have been exposed to Zika, use a condom during vaginal, anal, and oral sex. This step can greatly reduce your chance of getting or spreading the infection through sexual contact.

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